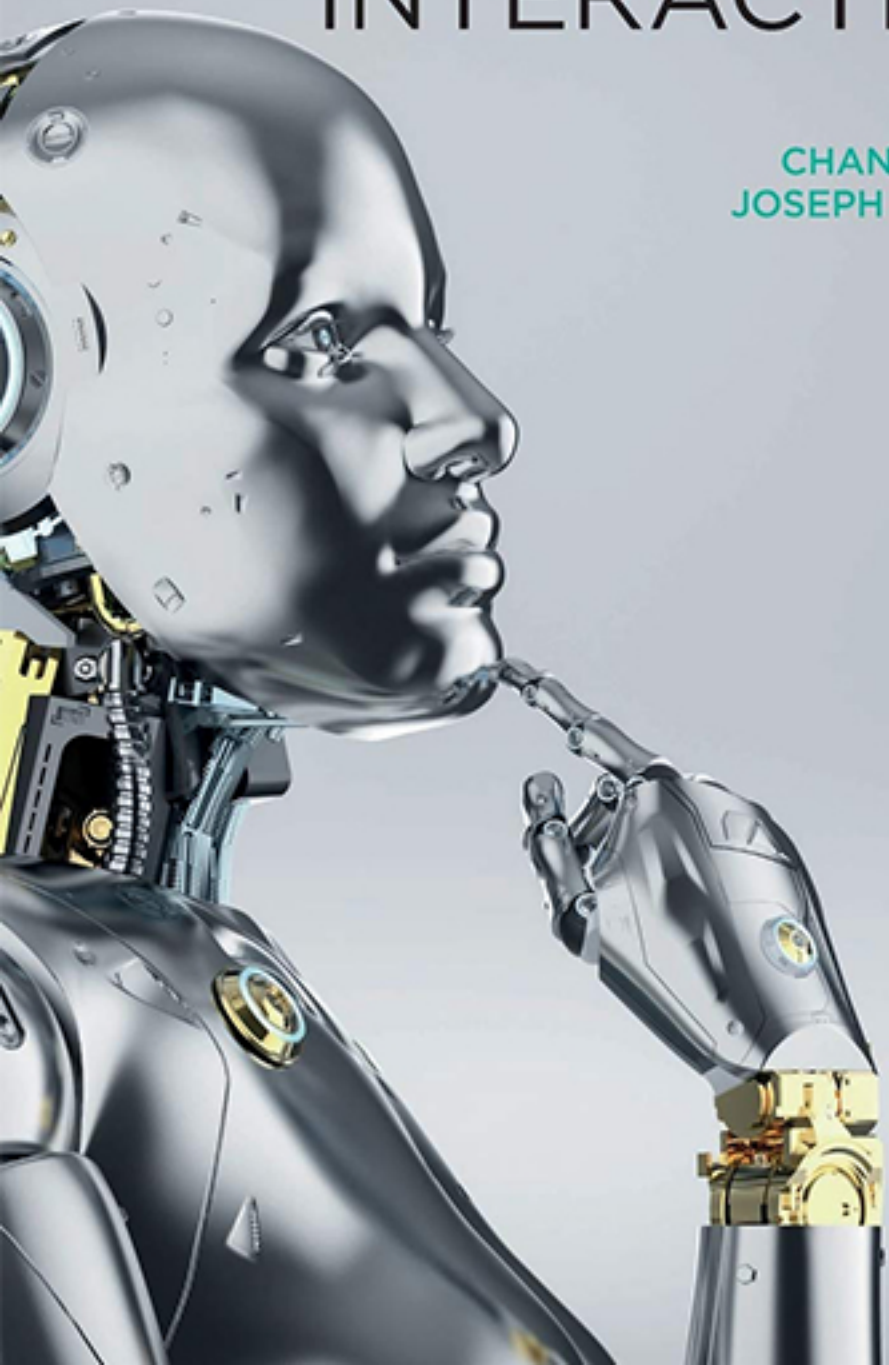
TRUST IN HUMAN-ROBOT INTERACTION

EDITED BY
CHANG S. NAM
JOSEPH B. LYONS



TRUST IN HUMAN-ROBOT INTERACTION

TRUST IN HUMAN-ROBOT INTERACTION

Edited by

CHANG S. NAM
Edward P. Fitts Department of Industrial and Systems Engineering, North Carolina State University, Raleigh, NC, United States
JOSEPH B. LYONS
Air Force Research Laboratory, Wright-Patterson AFB, OH, United States

ELSEVIER
ACADEMIC PRESS
An imprint of Elsevier

Academic Press is an imprint of Elsevier
125 London Wall, London EC2Y 5AS, United Kingdom
525 B Street, Suite 1650, San Diego, CA 92101, United States
50 Hampshire Street, 5th Floor, Cambridge, MA 02139, United States
The Boulevard, Langford Lane, Kidlington, Oxford OX5 1GB, United Kingdom

© 2021 Elsevier Inc. All rights reserved.
Chapter 1: Dr. Malle owns the copyright for his contribution and provides us an exclusive license only for this edition.
Chapter 12: Dr. Kristin E. Schaefer contribution is in Public domain as he is an US government employee.

No part of this publication may be reproduced or transmitted in any form or by any means, electronic or mechanical, including photocopying, recording, or any information storage and retrieval system, without permission in writing from the publisher. Details on how to seek permission, further information about the Publisher's permissions policies and our arrangements with organizations such as the Copyright Clearance Center and the Copyright Licensing Agency, can be found at our website: www.elsevier.com/permissions.

This book and the individual contributions contained in it are protected under copyright by the Publisher (other than as may be noted herein).

Notices
Knowledge and best practice in this field are constantly changing. As new research and experience broaden our understanding, changes in research methods, professional practices, or medical treatment may become necessary.

Practitioners and researchers must always rely on their own experience and knowledge in evaluating and using any information, methods, compounds, or experiments described herein. In using such information or methods they should be mindful of their own safety and the safety of others, including parties for whom they have a professional responsibility.

To the fullest extent of the law, neither the Publisher nor the authors, contributors, or editors, assume any liability for any injury and/or damage to persons or property as a matter of products liability, negligence or otherwise, or from any use or operation of any methods, products, instructions, or ideas contained in the material herein.

Library of Congress Cataloging-in-Publication Data
A catalog record for this book is available from the Library of Congress

British Library Cataloguing-in-Publication Data
A catalogue record for this book is available from the British Library

ISBN: 978-0-12-819472-0

For information on all Academic Press publications
visit our website at https://www.elsevier.com/books-and-journals

Publisher: Nikki Levy
Senior Acquisitions Editor: Joslyn Chaiprasert-Paguio
Senior Editorial Project Manager: Barbara Makinster
Production Project Manager: Niranjan Bhaskaran
Senior Cover Designer: Mark Rogers

Typeset by SPi Global, India

ELSEVIER Book Aid International Working together to grow libraries in developing countries
www.elsevier.com • www.bookaid.org

Contributors

Gene M. Alarcon Air Force Research Laboratory, Wright-Patterson Air Force Base, OH, United States

Anthony L. Baker US Army Combat Capabilities Development Command, Aberdeen Proving Ground, MD, United States

Michael J. Barnes U.S. Army Combat Capabilities Development Command – Army Research Laboratory, Adelphi, MD, United States

Spring Berman Center for Human, Artificial Intelligence, and Robot Teaming, Arizona State University, Tempe, AZ, United States

James P. Bliss Psychology Department, Old Dominion University, Norfolk, VA, United States

Jeffrey M. Bradshaw Florida Institute for Human and Machine Cognition, Pensacola, FL, United States

J.C. Brill Warfighter Interactions and Readiness Division, Air Force Research Laboratory, Wright-Patterson Air Force Base, OH, United States

August Capiola Air Force Research Laboratory, Wright-Patterson Air Force Base, OH, United States

Jessie Y.C. Chen U.S. Army Combat Capabilities Development Command – Army Research Laboratory, Adelphi, MD, United States

Erin K. Chiou Center for Human, Artificial Intelligence, and Robot Teaming, Arizona State University, Tempe, AZ, United States

HeeSun Choi Division of Safety Research, National Institute for Occupational Safety and Health (NIOSH), Morgantown, WV, United States

Sanghyun Choo Edward P. Fitts Department of Industrial and Systems Engineering, North Carolina State University, Raleigh, NC, United States

Allen Coin College of Humanities and Social Sciences, North Carolina State University, Raleigh, NC, United States

Nancy J. Cooke Center for Human, Artificial Intelligence, and Robot Teaming, Arizona State University, Tempe, AZ, United States

J. Cruit Department of Psychology, University of Central Florida, Orlando, FL, United States

M.L. Cummings Department of Electrical and Computer Engineering, Duke University, Durham, NC, United States

Mustafa Demir Center for Human, Artificial Intelligence, and Robot Teaming, Arizona State University, Tempe, AZ, United States

Veljko Dubljević College of Humanities and Social Sciences, North Carolina State University, Raleigh, NC, United States

Elizabeth Eskander Department of Psychology, North Carolina State University, Raleigh, NC, United States

Qin Gao Department of Industrial Engineering, Tsinghua University, Beijing, People's Republic of China

Gregory M. Gremillion Combat Capabilities Development Command Army Research Laboratory, Aberdeen Proving Ground, MD, United States

Robert S. Gutzwiller Center for Human, Artificial Intelligence, and Robot Teaming, Arizona State University, Tempe, AZ, United States

P.A. Hancock Department of Psychology, University of Central Florida, Orlando, FL, United States

Kerstin S. Haring Department of Computer Science, The University of Denver, Denver, CO, United States

Claudia Hauer Lyon Chair in Professional Ethics, USAF Academy, Air Force Academy, CO, United States

Martin G. Helander Damai Sciences, Kuala Lumpur, Malaysia

Brittany E. Holthausen School of Psychology, Georgia Institute of Technology, Atlanta, GA, United States

Xiaoxiao Hu Psychology Department, Old Dominion University, Norfolk, VA, United States

Lixiao Huang Center for Human, Artificial Intelligence, and Robot Teaming, Arizona State University, Tempe, AZ, United States

Makoto Itoh NPO Speech Deposit Association, Ibaraki, Japan

Sarah A. Jessup Department of Psychology, Wright State University, Dayton, OH, United States

Matthew Johnson Florida Institute for Human and Machine Cognition, Pensacola, FL, United States

A.D. Kaplan Department of Psychology, University of Central Florida, Orlando, FL, United States

Nicole Karpinsky-Mosely Knowesis Inc., Fairfax, VA, United States

Joseph R. Keebler Embry-Riddle Aeronautical University, Daytona Beach, FL, United States

T.T. Kessler Department of Psychology, University of Central Florida, Orlando, FL, United States

Halimahtun M. Khalid Damai Sciences, Kuala Lumpur, Malaysia

Theresa Law Human Robot Interaction Laboratory, Tufts University, Medford, MA, United States

Elizabeth H. Lazzara Department of Human Factors and Behavioral Neurobiology, Embry-Riddle Aeronautical University, Daytona Beach, FL, United States

Michael Lewis School of Computing and Information, University of Pittsburgh, Pittsburgh, PA, United States

Huao Li School of Computing and Information, University of Pittsburgh, Pittsburgh, PA, United States

Jinchao Lin Institute for Simulation and Training, University of Central Florida, Orlando, FL, United States

Mei-Hua Lin Department of Psychology, School of Science and Technology, Sunway University, Subang Jaya, Malaysia

Michael Long Oakridge Institute for Science and Education, Air Force Research Laboratory, Wright-Patterson Air Force Base, OH, United States

Shelby K. Long Psychology Department, Old Dominion University, Norfolk, VA, United States

Joseph B. Lyons Air Force Research Laboratory, Wright-Patterson Air Force Base, OH, United States

Bertram F. Malle Department of Cognitive, Linguistic, and Psychological Sciences, Brown University, Providence, RI, United States

Amar R. Marathe Combat Capabilities Development Command Army Research Laboratory, Aberdeen Proving Ground, MD, United States

Sachiko Matsumoto Computer Science and Engineering, UC San Diego, La Jolla, CA, United States

Gerald Matthews Institute for Simulation and Training, University of Central Florida, Orlando, FL, United States

Jason S. Metcalfe Combat Capabilities Development Command Army Research Laboratory, Aberdeen Proving Ground, MD, United States

Christopher A. Miller Smart Information Flow Technologies, Minneapolis, MN, United States

Chang S. Nam Edward P. Fitts Department of Industrial and Systems Engineering, North Carolina State University, Raleigh, NC, United States

Masahiro Ono Jet Propulsion Laboratory, California Institute of Technology, Pasadena, CA, United States

April Rose Panganiban Air Force Research Laboratory, Wright-Patterson Air Force Base, OH, United States

Yiannis Papelis Virginia Modeling, Analysis and Simulation Center, Old Dominion University, Norfolk, VA, United States

Brandon S. Perelman Combat Capabilities Development Command Army Research Laboratory, Aberdeen Proving Ground, MD, United States

Marc D. Pfahler Cognitive Performance Group, Orlando, FL, United States

Elizabeth Phillips Department of Psychology, George Mason University, Fairfax, VA, United States

David V. Pynadath Institute for Creative Technologies, University of Southern California, Los Angeles, CA, United States

Laurel D. Riek Computer Science and Engineering, UC San Diego, La Jolla, CA, United States

Paul Robinette Electrical & Computer Engineering, University of Massachusetts Lowell, Lowell, MA, United States

Nathan E. Sanders Edward P. Fitts Department of Industrial and Systems Engineering, North Carolina State University, Raleigh, NC, United States

T.L. Sanders Transportation Innovation Center, The MITRE Corporation, McLean, VA, United States

Kristin E. Schaefer Combat Capabilities Development Command Army Research Laboratory, Aberdeen Proving Ground, MD, United States

Matthias Scheutz Human Robot Interaction Laboratory, Tufts University, Medford, MA, United States

Tamera R. Schneider Department of Psychology, Baruch College, New York, NY, United States

Michaela Schwing Air Force Research Laboratory, Wright-Patterson Air Force Base, OH, United States

Rachel E. Stuck School of Psychology, Georgia Institute of Technology, Atlanta, GA, United States

Naomi Swanson Division of Science Integration, National Institute for Occupational Safety and Health (NIOSH), Cincinnati, OH, United States

Katia Sycara Carnegie Mellon University, Robotics Institute, Pittsburgh, PA, United States

Daniel Ullman Department of Cognitive, Linguistic, and Psychological Sciences, Brown University, Providence, RI, United States

Alan R. Wagner Department of Aerospace Engineering, The Pennsylvania State University, University Park, PA, United States

Bruce N. Walker School of Psychology, Georgia Institute of Technology, Atlanta, GA, United States

Ning Wang Institute for Creative Technologies, University of Southern California, Los Angeles, CA, United States

Auriel Washburn Computer Science and Engineering, UC San Diego, La Jolla, CA, United States

Yusuke Yamani Psychology Department, Old Dominion University, Norfolk, VA, United States

Wenlong Zhang Center for Human, Artificial Intelligence, and Robot Teaming, Arizona State University, Tempe, AZ, United States

Foreword

This timely volume is not a textbook or a primer, but rather a gentle and thoughtful introduction to the subject of trust in human interactions with robots and other intelligent agents. For the most part, no technical knowledge of artificial intelligence (AI) is necessary to benefit from reading the chapters found in this book.

AI has afforded us a means to think about the various ways in which physical systems can be organized to produce and/or extend behavioral competence. Such systems embody representations of the world, preferences, communication, and plans for acting in the world. It has only made sense to discuss "trust" in the context machines since the advent of these computational intelligences.

The notion of trust remains rather poorly defined … even when only considering trust among humans. It is often confounded with reliability, but I have known a few people who were *reliably untrustworthy.* One might reasonably say something to the effect, "I trust that my brakes will work." But this is an impoverished use of the word trust—akin to saying something about the perceived reliability of the brakes. One's brakes do not *decide* not to work, if they do not work, they have malfunctioned. In contrast, if I say that "I trust Nancy," it is not akin to asserting that Nancy is not malfunctioning. Nancy not malfunctioning might be necessary, but normally is not a sufficient, condition for the attribution of trust. Trust implies agenthood on the part of both the entity assigning trust and the entity which is trusted. Humans have, over a very long time, developed functional, but fallible methods of quickly assigning an initial degree of trust and then revising it over time subject to experience and circumstances.

Trust is a rich and interesting subject in human relations, but perhaps, even more, thought-provoking in the context of trust between humans and nonhuman agents. A primary and underappreciated challenge in the application of AI and autonomous systems is understanding and effectively exploiting human/machine teaming. This is true in the realm of self-driving automobiles, military applications, and everywhere else.

A key question is how might we design systems to facilitate appropriate trust calibration over time. How might the system usefully assist the human to establish a useful initial calibration of trust? How can the system indicate when circumstances have changed and the initial trust gradation

may need adjustment? How can people let machines know when *they* are compromised and the machine needs to take more initiative?

The right question is not how can a machine be designed to induce a state of trust in a human or even how to build a machine that is indeed trustworthy, but rather the question is how, through interaction, can trust be established, calibrated as appropriate, and maintained.

The editors are to be congratulated for amassing a broad collection of interesting papers on a topic of great interest and importance.

Ken Ford
Founder and Chief Executive Officer of the Florida Institute for Human & Machine Cognition (IHMC)

Preface

Future robotic systems are intended to team and partner with humans to accomplish operational goals rather than replacing humans with robotic systems. This notion of "partnering" raises an interesting set of challenges for the human-robot interaction (HRI), namely trust in HRI. The domains impacted by trust in HRI have been proliferating over time, including military, health, workplace, education settings, and so on. Besides, interdisciplinarity in HRI research calls for a holistic approach to trust in HRI considering the robot, the human, the human-robot system, and domain within which the human-robot system operates. However, there are currently few books to provide students, practitioners, and researchers with a comprehensive, state-of-the-science information about trust in HRI. We also think that the world is ready for a new book that comprehensively addresses the recent and rapid changes in the field of trust in HRI.

This book is organized into an introductory chapter with an emphasis on the evolution of trust in HRI, five main parts, and a conclusion chapter. **Part I** is made up of two sections that discuss the multifaceted nature of trust in HRI. Section A, consisting of three chapters, presents an overview of trust regarding multidimensional concepts and measures (Chapter 1), recent concepts and evaluations used in HRI (Chapter 2), and unique features of trust in intelligent autonomous systems (Chapter 3). Section B is composed of five chapters in which various influential factors that have known to affect trust in HRI. Chapter 4 provides a list of determinants of trust in HRI from the modeling, measurement, and prediction perspectives. Chapter 5 discusses the individual and social determinants of trust. Chapter 6 describes the currently identified antecedents of trust with an emphasis on a relatively new and underexplored category of trust antecedent—time. The other two chapters in Section B present the roles of human personality (Chapter 7) and risk (Chapter 8) in HRI.

Part II outlines the necessity of explainable trust and introduces three approaches applicable to HRIs, such as audience factors (Chapter 9), bidirectional communication between humans and agents (Chapter 10), and transparency (Chapter 11). **Part III** presents five chapters that discuss trust in various human-robot teams in the areas of the trust process for effective interaction between human-agent military teams (Chapter 12), distributed dynamic team trust (Chapter 13), the role of deep learning, transparency and trust in human-robot teamwork (Chapter 14), trust-aware control

(Chapter 15), and the role of interdependence in trust (Chapter 16). **Part IV** illustrates how trust in HRI can be modeled at the cognitive, behavioral, and neural levels. Chapter 17 explains how the Swift trust model can be applied to HRI teaming; Chapter 18 summarizes the influence of autonomy, controllability, and observability on performance, trust, and risk perception. Two additional chapters present an overview of applied models of trust in HRI (Chapter 19) and a summary of the neural dynamics of trust in HRI (Chapter 20). **Part V** is composed of four chapters that present emerging issues in trust in HRI, such as culture (Chapter 21), emotion (Chapter 22), and ethics (Chapters 23 and 24). This book ends with a series of reflections on the future for trust in HRI research.

This book was motivated by the desire to help students, professionals, and researchers explore and understand trust in HRI. This book constitutes an enormously valuable resource to aid in one's understanding of the gamut of factors that influence the trust of robotic systems. Thus, we hope that our readers will enjoy these timely updates of the most exciting topics, theories, fundamentals, techniques, and diverse applications in trust in HRI. Finally, on behalf of the Editorial Team at Elsevier, we would sincerely like to express our gratitude to our contributing authors for their professionalism as well as their valuable commitment to the success of this book.

Chang S. Nam
Joseph B. Lyons

Introduction: The evolution of trust in human-robot interaction

Joseph Lyons[a] *and Chang S. Nam*[b]

[a]Air Force Research Laboratory, Wright-Patterson Air Force Base, OH, United States

[b]Edward P. Fitts Department of Industrial and Systems Engineering, North Carolina State University, Raleigh, NC, United States

Robots are set to become a ubiquitous aspect of modern society. Potential applications of robotic systems include domains such as package delivery, disaster search and rescue, service, entertainment, and manual support for physical tasks. Robots truly have the potential to shape our social experience, however, one thing will dictate whether or not robots are successfully integrated into society, and that involves the construct of trust. For decades, science fiction has posed an intractable question to society—should humans trust robots? Dating back to the 1940s with Asimov's Laws of robotics, humans have contemplated the downside of increased intelligence in the form of robotics. The laws of robotics were recently revised to better reflect contemporary robotics challenges and emphasize human-robot interaction versus just merely the robots (Murphy & Woods, 2009). The present book attempts to take a deep dive into the science of trust in the domain of human-robot interaction (HRI). We sought to gather the extant state of the art in the knowledge of trust in HRI and put it into an accessible venue for use by students, academics, engineers, and other scientists.

The first revised law of robotics states that robots should not be deployed without the human-robot system meeting the highest legal and professional standards of safety and ethics (Murphy & Woods, 2009). But what standards exist concerning trust in robots? Trust broadly represents one's willingness to be vulnerable to others (Mayer, Davis, & Schoorman, 1995). Humans' trust of robots will influence whether these systems are accepted to engage in social and task-based interactions with humans in various domains. Ultimately, trust in robots will shape how much humans rely on robots in various domains.

Why is the trust of robots an important research topic? First of all, our dependence on robots is only going to increase, and we need to ensure that this increased dependence is appropriate. Robots may be deployed in sensitive environments where they must be capable of interacting with the fragility of the elderly and possessing the physical and psychological affordance to be used in domains such as law enforcement and the military (Scheutz & Malle, 2017). Secondly, robots are becoming increasingly capable of supporting human task performance. At the surface, this sounds like a great development, as yes, robots have great potential to improve human lives. However, increased reliability comes at the price of higher potential complacency when the robots eventually fail (Lyons, 2013). In fact, in the domain of human-automation interaction, research has shown that more capable automation results in more catastrophic consequences when the system makes an error, relative to systems that have lower reliability (Onnasch, Wickens, Li, & Manzey, 2014). This is driven by humans placing too much trust in automation, an effect also prominent within HRI (Robinette, Li, Allen, Howard, & Wagner, 2016). Thus, we need to ensure that the trust that we, as humans, place on robots is appropriately calibrated. Thirdly, humans tend to ascribe intentionality to robotic systems (Thellman, Silvervarg, & Ziemke, 2017). Robots will be expected to respond to morally-contentious situations and they will be judged against social norms for such behaviors (Scheutz & Malle, 2017). This adds complexity because the appropriateness of a robot's behavior in a moral dilemma will be dictated, in part, by social versus technical factors. Thus, understanding social norms and expectations, and designing robots to address these norms will be critical to their acceptance by the public. This is consistent with the second revised law of robotics, namely, that robots should respond to humans as appropriate for their roles (Murphy & Woods, 2009).

Finally, there is increased potential for robots to be placed in situations where they will need to make decisions that influence the safety and well-being of humans; one does not need to look long in the media for a story involving an accident including a semi-autonomous car. This is consistent with the third revised law of robotics, namely, that robots need to be designed in such a way as to promote the effective transfer of authority; specifically that a robot must be designed with situated autonomy to protect its existence as long as such protection provides smooth transfer of control to other agents consistent with the first and second laws (Murphy & Woods, 2009). Researchers have also begun to examine the human trust of robots in contexts where the robot is both capable and authorized to harm a human if needed (see Lyons et al., 2020). In this case, there may be situations where the robot may need to prioritize its own mission goals over those of another human. These are issues to be considered by contemporary robotics researchers including, but not limited to:

how will robots handle situations where they must select among a set of sub-optimal courses of action—where harm is likely to occur to one or more humans, or the robots themselves? How are these decisions to be portrayed to the public, to robot designers, to robot testers, regulators, insurance companies?

While an exhaustive review of the human-robot trust literature is beyond the scope of this introductory chapter (but rather is the focus of this entire book) we seek to motivate the kinds of topics found within the book. In Hancock and colleagues' (2011) meta-analysis of trust in robots, they found the presence of three classes of factors that shape trust in robots: robot factors, situational factors, and human factors. Robot factors such as reliability are fundamental to the trust of robots (Hancock et al., 2011). Robotic features such as their physical form and anthropomorphic features will also influence trust. Research has shown that anthropomorphism increases the trust of automation (Pak, Fink, Price, Bass, & Sturre, 2012). Humanness is a significant driver of what makes a human perceives an intelligent machine as a tool versus as a teammate (Kim, Kim, Lyons, & Nam, 2020; Lyons, Wynne, Mahoney, & Roebke, 2019). Robots designed to mimic male and female features also influence how humans interact with them (Ghazali, Ham, Barakova, & Markopoulos, 2018). Thus, form matters in HRI, but does the form promote the desired expectations and attitudes is uncertain.

Situational factors were defined by Hancock and colleagues (2011) as group composition, cultural factors, task complexity and type, and physical environment, to name a few. One relevant study demonstrated that when robots demonstrate awareness of their human partner, they are perceived as safer, more competent, and humans report greater confidence in interacting with them (Lasota & Shah, 2015). Robots that demonstrate empathy toward humans are liked more (Leite et al., 2013) and robots that convey self-sacrificial social intent are perceived as more trustworthy than those that state a self-interested motive (Lyons et al., 2020). Yet, despite robot and situational factors, not all humans react to robots in the same way. Hence, human factors (i.e., individual differences) matter in the context of HRI. Szalma and Taylor (2011) have examined the role of the Big-Five personality dimensions of trust in automation. Individual difference factors that could influence trust of robots include the Perfect Automation Schema (see Lyons & Guznov, 2019; Merritt, Unnerstall, Lee, & Huber, 2015), one's complacency potential (see Merritt et al., 2019), and gender (Gallimore, Lyons, Vo, Mahoney, & Wynne, 2019). Throughout this book, some chapters focus on each aspect of human-robot trusts such as robot factors, situational factors, and human factors. We hope you enjoy the topic and that you find some useful information for your research and applications.

References

Gallimore, D., Lyons, J. B., Vo, T., Mahoney, S., & Wynne, K. T. (2019). Trusting Robocop: Gender-based effects on trust of an autonomous robot. *Frontiers in Psychology: Human-Media Interaction*, *10*, 482.

Ghazali, A. S., Ham, J., Barakova, E. I., & Markopoulos, P. (2018). Effects of robot facial characteristics and gender in persuasive human-robot interaction. *Frontiers in Robotics and AI*, *5*, 1–16.

Hancock, P. A., Billings, D. R., Schaefer, K. E., Chen, J. Y., De Visser, E. J., & Parasuraman, R. (2011). A meta-analysis of factors affecting trust in human-robot interaction. *Human Factors*, *53*(5), 517–527.

Kim, W., Kim, N., Lyons, J. B., & Nam, C. S. (2020). Factors affecting trust in high-vulnerability human-robot interaction contexts: A structural equation modelling approach. *Applied Ergonomics*. *85*, 103056. https://doi.org/10.1016/j.apergo.2020.103056.

Lasota, P. A., & Shah, J. A. (2015). Analyzing the effects of human-aware motion planning on close-proximity human-robot collaboration. *Human Factors*, *57*, 21–33.

Leite, I., Pereira, A., Mascarenhas, S., Martinho, C., Prada, R., & Paiva, A. (2013). The influence of empathy in human-robot relations. *International Journal of Human-Computer Studies*, *71*, 250–260.

Lyons, J. B. (2013). Being transparent about transparency: A model for human-robot interaction. In: D. Sofge, G. J. Kruijff, & W. F. Lawless (Eds.), *Trust and autonomous systems: Papers from the AAAI spring symposium (technical report SS-13-07)*, Menlo Park, CA: AAAI Press.

Lyons, J. B., & Guznov, S. Y. (2019). Individual differences in human-machine trust: A multi-study look at the perfect automation schema. *Theoretical Issues in Ergonomics Science*, *20*(4), 440–458.

Lyons, J. B., Vo, T., Wynne, K. T., Mahoney, S., Nam, C. S., & Gallimore, D. (2020). Trusting autonomous robots: The role of reliability and stated social intent. *Human Factors*. 18720820901629. https://doi.org/10.1177/0018720820901629.

Lyons, J. B., Wynne, K. T., Mahoney, S., & Roebke, M. A. (2019). Trust and human-machine teaming: A qualitative study. In W. Lawless, R. Mittu, D. Sofge, I. Moskowitz, & S. Russell (Eds.), *Artificial intelligence for the internet of everything* (pp. 101–116). Elsevier.

Mayer, R. C., Davis, J. H., & Schoorman, F. D. (1995). An integrated model of organizational trust. *Academy of Management Review*, *20*, 709–734.

Merritt, S. M., Ako-Brew, A., Bryant, W. J., Staley, A., McKenna, M., Leone, A., et al. (2019). Automation-induced complacency potential: Development and validation of a new scale. *Frontiers in Psychology*, *10*, 1–13.

Merritt, S. M., Unnerstall, J. L., Lee, D., & Huber, K. (2015). Measuring individual differences in the perfect automation schema. *Human Factors*, *57*, 740–753.

Murphy, R. R., & Woods, D. D. (2009). Beyond Asimov: The three laws of responsible robotics. *IEEE Intelligent Systems*, 14–20.

Onnasch, L., Wickens, C. D., Li, H., & Manzey, D. (2014). Human performance consequences of stages and levels of automation: An integrated meta-analysis. *Human Factors*, *56*, 476–488.

Pak, R., Fink, N., Price, M., Bass, B., & Sturre, L. (2012). Decision support aids with anthropomorphic characteristics influence trust and performance in younger and older adults. *Ergonomics*, 1–14.

Robinette, P., Li, W., Allen, R., Howard, A. M., & Wagner, A. R. (2016). Overtrust of robots in emergency evacuation scenarios. In: *Proceedings of the 11th ACM/IEEE international conference on human robot interaction*, pp. 101–108.

Scheutz, M., & Malle, B. F. (2017). Moral robots. In L. S. M. Johnson, & K. S. Rommelfanger (Eds.), *Routledge handbooks in applied ethics. The Routledge handbook of neuroethics* (pp. 363–377). Routledge/Taylor & Francis Group.

Szalma, J. L., & Taylor, G. S. (2011). Individual differences in response to automation: The five factor model of personality. *Journal of Experimental Psychology: Applied*, *17*(2), 71–96.

Thellman, S., Silvervarg, A., & Ziemke, T. (2017). Folk-psychology interpretation of human vs. humanoid robot behavior: Exploring the intentional stance toward robots. *Frontiers on Psychology*, *8*, 1–14.

CHAPTER

1

A multidimensional conception and measure of human-robot trust

Bertram F. Malle and Daniel Ullman

Department of Cognitive, Linguistic, and Psychological Sciences, Brown University, Providence, RI, United States

You are standing in front of a robot, with your back to it. Your task is to let yourself fall backward, and the robot's task is to catch you. Would you trust the robot?

Commonly referred to as the "trust fall" exercise, this situation serves as a quintessential example of trust between two agents. The exercise hinges on the falling agent's expectation that the catching agent will not let them hit the ground. This expectation has two features: First, the falling agent needs to believe that the catching agent is capable of catching them before they hit the ground. Second, the falling agent needs to believe that the catching agent has the moral integrity and commitment to catch them and is sincere when promising to do so. The first idea is what we henceforth call "Performance trust," while the second idea is what we call "Moral trust." Performance trust refers to the trustor's confidence that the trustee is capable of completing a given task, while moral trust refers to the trustor's confidence that the trustee will choose the morally right action and not exploit the trustor's vulnerability. These two beliefs can diverge: We may be confident that another agent is capable of doing what benefits us but not be morally motivated to do so; or we may be confident of the other's moral commitment but not of their capacity to fulfill it. Thus, trust seems to have at least two conceptually independent dimensions.

Robots are beginning to offer benefits to humans in a range of settings, from schools to the workplace, and across a variety of application domains, from medical care to support in one's home. Robots have stepped out of their cages of isolation and safety precautions, where their primary contribution is to reliably complete repetitive physical tasks. Increasingly, robots are used in the world of social interaction, where their expected contribution is to assist and support people. But for robots to

https://doi.org/10.1016/B978-0-12-819472-0.00001-0
Copyright © 2021 Elsevier Inc. and Bertram F. Malle.
All rights reserved.

successfully fulfill roles in this social context of human-robot interaction, people must be willing to interact with these robots, entrust them with tasks that have socially beneficial results, and be confident that the robots are both capable of and committed to bringing about those results. Because trust is integral to social relationships, people will be inclined to trust robots with whom they interact. If robots are to succeed in these social interactions, we must first design robots that are worthy of human trust.

The concept of trust

Trust, in common usage, is an expectation that something good will happen—while also knowing that it might not happen. Hope also rests on an expectation of something good, but hope has a relatively low level of confidence (Bruininks & Malle, 2006) whereas trust carries a high level of confidence. The Oxford English Dictionary calls trust a "firm belief in the reliability, truth, or ability of someone or something" or the "confident expectation of something." Collins COBUILD Advanced English Dictionary (which aims to specifically capture current use) also refers to a "firm belief or confidence in the honesty, integrity, reliability, justice, etc. of another person or thing." We set aside here trust in "things" and focus on persons or agents—including robots. Our thesis may then be formulated this way: Trust's underlying expectation can be directed at multiple different properties that the other agent might have, and these properties make up multiple dimensions of trust. The dictionary entries mention *ability*, *reliability*, *honesty*, and *integrity*, and we will provide evidence that these are indeed four major dimensions of trust: one can trust someone who is reliable, capable, ethical, and sincere. We will review the literature on human-automation trust, human-human trust, and human-robot trust and will show that these four dimensions repeatedly appear in these literatures. In light of emerging data from our lab, we then suggest that a more complete model of trust needs to incorporate all four dimensions, and we offer an instrument that makes them conveniently measurable.

Trust in human-automation interaction

Work in the domain of automation emphasizes the performance of automated systems. Automation serves its purpose when a system can safely and efficiently perform a task that reduces a burden on human users. Many of these task domains have historically consisted of nonsocial tasks, often combined with system oversight by a human operator but little collaboration between humans and the system. A useful heuristic in thinking about whether or not to trust a system or an agent comes in

the form of a basic question: "What do I worry about that would prevent me from interacting with this agent?" In the automation literature, such worry is tied to the **performance** of the system; and this worry can be alleviated, and allows for trust, if the system is capable of performing its task, and does so consistently (Schaefer, Chen, Szalma, & Hancock, 2016).

Sheridan and Parasuraman (2005) reviewed research in human-automation interaction and offered two sets of features of trust. One set comes from the system's lower-level **reliability** of performance, while the other comes from the system's higher-level **ability**. Ideally, trust in a system is grounded in an accurate conception of the system's ability and reliability; however, trust is not always appropriately calibrated. Parasuraman and Riley (1997) discussed multiple miscalibrations: misuse (overreliance on automation), disuse (underutilization of automation), and abuse (inappropriate application of automation). Based on previous research, Lee and See (2004) proposed that systems must be designed to help match users' expectations to the system's actual performance capabilities. Calibrated trust is grounded in an accurate assessment of what a system can and cannot do—as well as why the system fails when it does.

Because errors reveal a system's performance quality, they have been a particular focus in human-automation work. For example, Madhavan, Wiegmann, and Lacson (2006) showed that trust in an automated decision aid declined when the user observed system errors. However, not all errors were treated the same way: when a system made errors on easy trials rather than difficult trials, users mistrusted the system far more. People potentially use task difficulty as a diagnostic indicator: failing to complete easy trials shows low ability.

In the absence of trust in a system, users will opt to not use it. Lee and Moray (1992) investigated the determinants that influence whether human operators will rely on a system performing a task, or opt for manual control in the task. They used an experimental task where human operators would choose between manual control or automatic control for operating a simulated semiautomatic pasteurization plant. The researchers found a tradeoff between human operators' self-confidence and trust in the system, such that operators opted for automatic control when trust in the system exceeded self-confidence and opted for manual control when self-confidence exceeded trust in the system. Trusting a system involves accepting some kind of risk and believing that the system is able to limit that risk.

A recent review of 127 empirical studies on human trust in automation (Hoff & Bashir, 2015) identified numerous factors that affect trust, including culture, personality, task characteristics, workload, self-confidence, and more; but the object of the trust itself—the automation system—was described only in terms of its performance: its reliability, predictability, and error-proneness. In sum, in the literature on trust in automation,

the primary focus is on appropriately matching a person's expectations for a system with information about the performance of the system. These systems are motiveless, and users are not concerned about being betrayed, exploited, or deceived by the system. Trust here is the expectation that a system will perform a task as intended and expected, and the only worries that arise concern the system's **reliability** and **ability**.

Trust in human-human interaction

As we move from the domain of human-automation trust into the domain of human-human trust, we can pose the same heuristic question: "What do I worry about that would prevent me from interacting with this agent?" Unlike the primary focus on **performance trust** in human-automation interaction, here the focus widens to include expectations of whether a human agent will act morally—what we refer to as **moral trust**. In situations of human-human trust, the question becomes whether a human agent will exploit another human's potential vulnerability—either unintentionally because of a lack of ability, or intentionally because of a lack of moral integrity.

It is instructive to imagine what a social community without trust would look like. People would not expect that another person would have their best interest in mind; instead, they would expect that others do what benefits them even when such actions exploit or harm others. Lack of trust is thus a reflection of a society without prosocial norms, without moral commitments. By contrast, in societies that have such commitments, trust is possible and has the power to enable and sustain cooperative behavior (e.g., Gambetta, 1988; Jones & George, 1998). Trust acts as a glue that enables people to live and work together without constant worry and threat of being exploited. In fact, such societies uphold a norm to trust other people (Dunning, Anderson, Schlosser, Ehlebracht, & Fetchenhauer, 2014), which places demands on those people to justify the trust they are granted—and makes violations of trust all the more salient.

Philosophical analyses typically treat trust as a three-part relation among a trustor, a trustee, and something that the trustor expects the trustee to do (Hardin, 2002) or to care for (Baier, 1986). The oft-cited conceptualization of trust by Mayer, Davis, and Schoorman (1995) has at its core a three-part relationship as well, but in addition to the trustor and a trustee, the authors focus on the role of risk. Integrating these elements, we can say that the trustor typically expects the trustee to act so as to avoid or reduce the trustor's risk in the situation. The object of this expectation (what the trustee is expected to do or be) corresponds to the notion of trustworthiness—a trustee's characteristics that either inspire or justify the trusting expectation: their ability, reliability, ethical integrity, and so

on. The exact characteristics of trustworthiness, and thus the objects of trust expectations, are debated. Therefore, we review several of these proposals, ranging from few to many characteristics, and identify the common denominators. With those in hand, we can then examine the place of human-robot trust in the broader psychological landscape of trust.

Rotter (1967) conceptualized trust as "an expectancy…that the word, promise, verbal or written statement of another…can be relied upon" (p. 651). Rotter's proposal thus highlighted the trust expectation of **sincerity** (truthful words and standing by promises), which is confirmed by a closer look at the items in his interpersonal trust scale. Chun and Campbell (1974) conducted cluster and factor analyses on this scale, and in their results, we see the primary interpersonal facet of **sincerity** (honest, truthful). Selfish exploitation formed another factor, perhaps the opposite pole of what other authors call **benevolence**. Many of the other items in Rotter's scale referred to institutional trust and specifically to concerns about institutions being **sincere** and **ethical** (e.g., unbiased, not cheating). However, there is no mention of competence or reliability/predictability. The worries people have about others are cast here entirely in terms of morality.

Several authors who examined human-human trust, from sociology to management, highlighted **competence** and **integrity** as the major expectations (Kim, Dirks, & Cooper, 2009; Parsons, 1969). In Cook and Wall's (1980) conception, trust was organized into faith in the intentions of others (often labeled **benevolence**), and confidence in the **ability** and the **reliability** of others. However, when the authors measured trust in an organizational setting, these three aspects did not come apart. Barber (1983), in his analysis of the broader societal role of trust, distinguished between three trust expectations: **persistence** (predictability), **competence**, and moral duties (to have others' interests in mind, which most authors call **benevolence**). Slovic, Flynn, Johnson, and Mertz (1993) asked ordinary citizens to express their trust or distrust in power plant management as a function of various behaviors by the management; the authors' choice of such behaviors revealed their conception of trust as expectations about **competence** (e.g., being prepared for accidents) and about two moral dispositions that are often labeled **benevolence** (good motives) and **sincerity/transparency** (being truthful, providing access). However, no attempt was made to distinguish these expectations in measurement. Focusing on people's attitudes toward institutions, Carnevale's (1995, p. xi) definition of trust included **reliability** and **competence** as well three moral aspects: **nonthreatening** (benevolent), **fair**, and **ethical**. Caldwell and Clapham's (2003) proposal, tailored to organizations, included seven aspects of trustworthiness that can be divided broadly into **competence** (knowledge, ability), the responsibility to inform (**transparency**), and various **moral** or normative aspects (quality assurance, respect, legal compliance).

Gabarro (1978) conducted interviews with 4 company presidents and 33 subordinates over the span of 3 years. He extracted six objects of trust. While maintaining aspects of **competence** (skills and good judgments) and **consistency** (reliability, predictability), he differentiated the moral dimension into **integrity** (encompassing honesty and moral character), **motives** (benevolent motives, commitment), **openness** (defined as honesty, being straight, not hiding—aspects most other authors label **sincerity**), and **discreetness** (not violating confidence). Butler and Cantrell (1984) and Schindler and Thomas (1993) experimentally tested five of these characteristics (omitting discreetness, perhaps because it could be grouped under integrity) as determinants of overall trust judgments. **Integrity** and **competence** showed the most considerable impact, while consistency was weak, and openness had little to no impact.

Butler (1991) also interviewed managers and content-analyzed characteristics that the managers mentioned in describing trusted and mistrusted people. In subsequent scale development and iterated factor analyses, Butler postulated nine characteristics, but only six seem to directly capture trust: **competence** and **consistency** as the familiar performance aspects, as well as the moral characteristics of **integrity**, **fairness/loyalty**, **discreetness**, and **promise fulfillment** (the others were availability, receptivity, and openness). However, when examined in people's judgments, Butler's moral characteristics were very highly correlated (*r*s between .65 and .76), suggesting one large moral cluster. **Competence** was somewhat differentiated from these moral components (correlating with them in the .40s and .50s), while **consistency** more clearly set itself apart (correlating with all the other components in the .30s and .40s).

Despite differences among the various authors' conceptions of human-human trust, we see that almost all of them support a multidimensional concept. Most authors assigned a prominent status to competence and reliability, mirroring the human-automation literature, but many added a moral dimension, with anywhere from one to four moral facets. Mayer et al. (1995) tried to integrate these variations and to consolidate them into the major characteristics of trustworthiness that sway a trustor. Against the background of 23 previous proposals, they derived three such characteristics: **ability** (including knowledge, expertise, competence), **benevolence** (a positive orientation toward the trustor), and **integrity** (adhering to moral principles shared with the trustor). Two omissions are noteworthy here. First, the authors excluded predictability from the conceptual space, mainly because they argued that reliability was not sufficient for trust. However, none of the individual characteristics are sufficient for trust, so reliability should not be discarded. Second, sincerity (common among many other models) was absent, not because of direct empirical evidence but because of the authors' conceptual decisions in selecting and compiling characteristics of trust. Interestingly, McKnight,

Cummings, and Chervany (1998) cited Mayer as the basis for their conception of trust expectations but worked with four dimensions, including **competence** (ability), **predictability** (added back in), **honesty** (rather than integrity, thus bringing sincerity back into the picture), and **benevolence**.

Nonetheless, a metaanalysis showed that Mayer et al.'s three-dimensional conception is successful in predicting trust states (overall willingness to accept vulnerability) from trust expectations (Colquitt, Scott, & LePine, 2007). The predictive correlations were in the .60s for each dimension, but the three dimensions were also correlated with each other between $r = .62$ and $r = .68$. Because reliability and sincerity were not included in the metaanalysis we do not know what their role would be in affecting subjective trust states.

A decade later, after yet more and varied proposals of conceptualizing trust, Burke, Sims, Lazzara, and Salas (2007) tabulated 27 such conceptualizations. We performed a frequency analysis of the most-used content words in these definitions (see Table 1) and found considerable common ground in what trust is: a dyadic relation in which one person accepts vulnerability because they expect that the other person's future action will be governed by certain characteristics. And though the specific characteristics (of trustworthiness) are rarely mentioned in the definitions, those that are mentioned include the by now familiar notions of **ability**, **reliability**, and a bundle of **moral characteristics** such as **benevolence**, **honesty**, and **integrity**.

Our review of dimensions of trust expectations in the literature is summarized in Table 2. Though not all assignments are clear-cut, the overall picture suggests that trust has two sides: a **performance** side, with facets of competence and reliability, and a **moral** side, with facets of sincerity, integrity, and benevolence. The importance of this moral dimension is what appears to differentiate trust between humans from trust in automation, and this difference is best explained by the significance of social interaction and risks involved in human-human relationships. The question now arises whether human-robot trust has anything like a moral dimension.

Trust in human-robot interaction

Much of the extant work on human-robot trust focuses on the reliability and ability characteristics of robotic systems, so trust in intelligent robots is considered along the same factors seen in the broader human-automation literature (Yanco, Desai, Drury, & Steinfeld, 2016). For example, Hancock et al. (2011) conducted a metaanalysis of factors that prior research has identified as influencing trust in human-robot interaction. The authors collated 21 studies and found that factors related to the robot—specifically, robot performance (such as reliability)—had the

TABLE 1 Frequency analysis of the words used in 27 definitions of trust (Burke et al., 2007).

68	**Dyadic**	
	28	Trustor, one, individual, person, party
	29	Trustee, another one, other(s)
	11	Relationship(s), interpersonal, interdependence, interactions, reciprocal, mutual
32	**Accept vulnerability**	
	17	Risk(s), vulnerable, vulnerability, damage, harmful, stake
	15	Accept(ing), willing(ness)
23	**Future action**	
	15	Action(s), behavior, act, behave, behavioral, perform, fulfill(s)
	8	Intent(ion), will
19	**Expectation:** Belief(s), believes, expectation(s), assumption, perceives, perception	
13	**Objects of trust**	
	6	Benevolent(ly), helpful, concern, welfare
	3	Honest, fairness, integrity
	2	Reliability, predictable
	2	Ability, competent
12	**Other**	
	4	Cognitive
	4	Affective, emotions, feel
	2	Possible, potential
	2	Safety, security

strongest association with trust. Human-related factors (e.g., attitudes and comfort with robots) and environmental factors (e.g., culture and physical environment) contributed relatively less.

In a recent review, Lewis, Sycara, and Walker (2018) argued for a distinction between performance-based interactions between humans and robots and social-based interactions. This social focus points to some of the distinguishing features that set robots apart from automation. In particular, robots are being introduced into more and more social settings, projected

TABLE 2 Dimensions of trust expectations (characteristics of trustworthiness) in selected models.

	Performance		Moral		
	Competence	**Reliability**	**Integrity**	**Sincerity**	**Benevolence**
Rotter (1967)				✓	
Chun and Campbell (1974)			✓	✓	✓
Gabarro (1978)	✓	✓	✓	✓[a]	✓[b]
Parsons (1969)	✓		✓		
Cook and Wall (1980)	✓	✓			✓
Barber (1983)	✓	✓			✓[c]
Butler (1991)	✓	✓	✓		✓[d]
Mayer et al. (1995)	✓		✓		✓
Slovic et al. (1993)	✓			✓	✓
Carnevale (1995)	✓	✓	✓		✓
McKnight et al. (1998)	✓	✓	✓[e]	✓[e]	✓
Caldwell and Clapham (2003)	✓		✓		✓
Analysis of definitions in Burke et al. (2007)	✓	✓	✓		✓
Kim et al. (2009)	✓		✓		

[a] *Calls it "openness" but explicates as honest, straight, and not hiding things—arguably elements of sincerity.*
[b] *Calls it "motives" and arguably refers to benevolent motives.*
[c] *Calls it "moral duties" (with a focus on having others' interests in mind).*
[d] *Calls it "fairness/loyalty."*
[e] *Call it "honesty," which, according to premier dictionaries has both meanings of being sincere, truthful and having integrity, moral principles (though with a stronger emphasis on shades of sincerity).*

to be social companions for older adults, tutors for children in schools, or assistants to people with health needs (Broadbent, Stafford, & MacDonald, 2009). Such contexts often involve an element of risk. Risk may arise from a robot moving around an older adult's house and accidentally knocking them to the ground, which is a traditional safety concern; but risk may also arise from a robot refusing to increase a person's pain medication because the decision authority cannot be reached, or from a robot unknowingly presenting incorrect information to a child. The latter situations involve social and moral norms and thus raise the question of moral trust in a robot. But do people actually treat robots as moral agents?

When asked to infer the mental capacities of robots, respondents are inclined to grant robots some capacity for moral decision-making (Malle, 2019; Weisman, Dweck, & Markman, 2017), and significantly more so the more humanlike the robot looks (Malle, Zhao, & Phillips, 2019). By contrast, people do not welcome moral decisions by cars and other machines if they lack humanlike mental capacities (Bigman & Gray, 2018). An increasing number of studies also show that people apply moral norms to AI and robotic agents and blame those agents when they violate the pertinent norms (Malle, Magar, & Scheutz, 2019; Malle, Scheutz, Arnold, Voiklis, & Cusimano, 2015; Shank & DeSanti, 2018). In a human-robot interaction study, too (Kahn et al., 2012), a majority of people thought of the robot as morally accountable for a specific transgressive behavior. If people treat robots as capable of moral decision making, then there is room for developing trust in robots if they are sincere, ethical, and benevolent—and to lose trust if they are not.

A series of studies investigated whether people recognize a robot's attempt to cheat in a game of rock-paper-scissors by changing its gesture or in a game of Battleship by lying about a ship's position; and people did recognize those violations (Short, Hart, Vu, & Scassellati, 2010; Ullman, Leite, Phillips, Kim-Cohen, & Scassellati, 2014). Such recognition, rather than the belief that the robot was simply malfunctioning, may reveal that people saw the robot as being unethical or insincere. Indeed, Wijnen, Coenen, and Grzyb (2017) showed that a lying robot (diverting blame to a human for a negative act the robot committed) was trusted less in a behavioral trust game than an honest robot.

For people to gain or lose moral trust in a robot, the robot does not have to be a genuine moral agent. Even when the designer is really the one who is insincere and untrustworthy, the human interacting with the robot may well direct their trust and disappointment at the robot. And what holds for sincerity could in principle hold for benevolence and integrity as well, though these two characteristics demand more behavioral evidence from a robot.

Evidence for ethical integrity would be seen in following moral norms and principles: for example, being fair, nondiscriminatory, cooperative, and respectful (Kuipers, 2018; Malle & Scheutz, 2019). Especially when robots are taking on a broader range of roles (e.g., as educators or health care providers; Broadbent et al., 2009), the norms associated with these roles must be made explicit and the robot must be equipped to comply with them. To build robots that guide their behavior by adhering to social and moral norms is still a significant challenge given the current state of the art in robotics (Malle, Bello, & Scheutz, 2019); but even just seeing a robot *try* to follow norms is likely to instill a considerable amount of trust in people.

Benevolence requires putting one's own interests behind others' interests, rather than selfishly benefitting while others incur costs. Even when robots are not designed to be selfish, they will sometimes pursue goals that run counter to people's interests—such as when a robot hinders or inconveniences someone or is unable to meet a request. In some cases, a robot's built-in goals may purposefully circumvent people's interests or values—such as when a robot tries to coax people into buying something or revealing private information (Calo, 2011; Hartzog, 2015). Such behaviors are often disguised and therefore insincere as well, suggesting an overall lack of ethical standards.

Even if current robots have only minimal moral capacities, their ever-widening roles in society, their routine spoken communication, and their often humanoid looks will (and perhaps already do) prompt people to treat them as social and moral agents (Coeckelbergh, 2012). Robots are machines, evaluated for their performance, but increasingly they are also social agents, evaluated for their potential to cause harm. In a context of vulnerability, people are likely to consider a robot's moral characteristics of trustworthiness, such as sincerity and integrity. We must, therefore, measure these moral characteristics along with the familiar characteristics of ability and reliability. What measurement tools are available to assess these multidimensional layers of human-robot trust?

Measuring human-robot trust

Hancock et al. (2011) noted that there is a scarcity of validated measures with which to evaluate human-robot trust consistently across experimental designs, which makes the phenomenon difficult to study and compare across studies, labs, and domains. This echoes the call by Steinfeld et al. (2006) for a singular toolkit of human-robot interaction metrics that includes trust.

A discussion of measuring human-robot trust must first address the distinction between trust as a subjective state and trust as a choice or action (Kee & Knox, 1970). Although often trust becomes apparent to the outside observer only when an agent makes a choice that embraces some risk, such a choice is ultimately the product of an internal state of trust that is translated into action. Mayer et al. (1995) similarly distinguished between the willingness to accept one's vulnerability and the action of taking a risk. These two aspects, the subjective-internal and the public-behavioral, can be measured separately and are predictably related (Colquitt et al., 2007). However, a risk-taking choice is not always grounded (solely) in subjective trust. Imagine a forced choice to rely on one of two people—relying on Agent A is not necessarily indicative of

trust in Agent A, but it could be due to a much greater benefit one stands to gain from choosing Agent A or a fear of Agent B. Thus, even though the act of risk-taking is often an important consequence of trust, trust itself is more closely captured by an internal state (Lewis et al., 2018). Obviously, internal states cannot be measured directly but are themselves operationalized by verbal reports, nonverbal expressions, and the like.

In the human-human domain, this internal state of trust consists of an acceptance of one's vulnerability and of expectations that the other's characteristics warrant this acceptance. These expectations, we have seen, are multidimensional and include at least performance characteristics (ability, reliability) and one or more moral characteristics (e.g., sincerity, integrity, benevolence). The measurement tools currently available for human-robot trust vary in how much they consider **moral trust** and **performance trust** factors, but they are heavily skewed toward the latter.

Schaefer (2013) offered one of the more comprehensive developments of a trust measure. Across 6 experiments, Schaefer developed a 40-item trust scale to assess the 3 antecedents of human trust in robots considered by Hancock et al. (2011): features of the robot, the human, and the environment. In addition, a 14-item scale focused specifically on trust expectations: characteristics that make the robot trustworthy. Of these 14 items, 11 reflect dimensions of **performance** (e.g., function successfully, act consistently) while 3 relate to social aspects (provide appropriate information, communicate with people). No item in the short form directly invokes the moral dimensions, but a few items in the longer 40-item scale refer to open and truthful communication (i.e., sincerity), and one item mentions protecting people. Similarly, Yagoda and Gillan (2012) proposed a scale for human-robot interaction in teams that identifies various system features (e.g., sensor data, effectors, interface), all evaluated with the words consistent, dependable, and **reliable** (and a few with the words understandable, accessible), but without reference to moral aspects.

Madsen and Gregor (2000) developed a trust measure for human-computer interaction that contains 25 items covering five content domains: perceived **reliability**, perceived technical **competence**, perceived understandability, faith (in a system's advice and solutions), and personal attachment (to a system). The items in these domains juxtapose subjective experiences (being attached to and understanding the system) with familiar trust expectations regarding **ability** and **reliability** (e.g., "The system makes use of all the knowledge and information available to it to produce its solution to the problem"; "The system responds the same way under the same conditions at different times"). The authors did not numerically compare the fit of various factor structures but favored a two-factor solution in which understandability dominated the second factor and virtually all ability and reliability items (along with attachment) hung together in the first factor. No aspects of moral trust were included.

Not all scales are limited to performance measurement. Jian, Bisantz, and Drury (2000) used a bottom-up approach to examine the semantic field of trust-related expressions. After examining a considerable number of such expressions, the authors arrived at a 12-item scale. Three items capture expectations of **reliability** (confident in, dependable, reliable), two items capture **competence** (provides security, harmful outcomes), and four items capture **moral** characteristics—one representing trustworthiness (**integrity**) and three representing its absence (deceptive, underhanded, suspicious of intent). However, a subsequent confirmatory factor analysis did not uncover separation between any of these dimensions, only between distrust-related and trust-related items (Spain, Bustamante, & Bliss, 2008).

We have by no means considered all extant measures of human-robot trust. But it is clear that the large majority does not consider moral aspects, and no specific trust measure appears to exist that reliably captures one or more moral aspects in human-robot interaction. Given the growing similarities between human-robot and human-human relations, there is a need to address the moral aspects of people's trust in robots above and beyond performance aspects. We should not assume that these aspects hold for all robots, but they may hold for some of them; and for those robots, a proper measurement tool must be available. Moreover, as ethical requirements for robot behavior increase (Arkin, Ulam, & Wagner, 2012; Malle & Scheutz, 2014, 2019; Wallach & Allen, 2008), measuring the moral dimension of trust is key to evaluating the success of designing such ethical robots. We now introduce our initial steps of developing a measurement tool for performance and moral trust that can be used in both human-human and human-robot situations.

A multidimensional conception and measurement of trust

Our review of the human-human trust literature indicated reasonable consensus that trust can be defined as follows:

> Trust = a dyadic relation in which one person accepts vulnerability because they expect that the other person's future action will have certain characteristics; these characteristics include some mix of performance (ability, reliability) and/or morality (honesty, integrity, and benevolence).

We take this definition as our starting point to introduce a Multidimensional Measure of Trust. We believe that the subjective state of trusting (accepting vulnerability) cannot easily be divorced from the trust *expectations* regarding the other's capability, reliability, and morality; instead, the subjective state is typically directed at one or more of these characteristics.

That is, when people say, "X trusts Y," they (implicitly or explicitly) refer to X trusting that Y is capable of and/or reliable in performing a certain action and/or sincere when uttering a statement (e.g., promise, information offer) and/or has the moral integrity not to exploit or otherwise harm X. Measuring these focal characteristics of trust expectations lies at the heart of our proposed instrument. At the same time, we encourage researchers to present a separate overall question of subjective trust (e.g., Mayer & Davis, 1999), and this question may have additional predictive validity for certain ensuing beliefs or behaviors (Colquitt et al., 2007).

Trust words in semantic space

We started our investigation of multidimensional trust by conceptually mapping out the space within which people think about trust, independent of whether it is trust in people, institutions, or robots (Ullman & Malle, 2018). Initially we considered two candidate dimensions of this space: trusting that an agent is capable of completing a task ("capacity trust") and trusting that an agent will not place another at risk ("personal trust"). We collected 62 words from dictionaries, the trust literature, and published trust measures and asked participants (recruited via Amazon Mechanical Turk) to indicate where each word fell on a slider scale from "more similar to capacity trust" to "more similar to personal trust" (defined as above). These original items can be viewed in a supplementary document available on our lab website (http://research.clps.brown.edu/SocCogSci/Measures/index.html). Whereas many previous measures consist of sentences related to trust, we opted for simplicity in this task and used single words or very short phrases. We then engaged in an iterative process of Principal Components Analysis (PCA) and item analysis and arrived at 32 items distributed over four components, which we represented by the labels Reliable (7 items), Capable (8 items), Sincere (6 items), and Ethical (11 items). Given that we had asked people to rate words only on the single capacity-personal variable, the conceptual structure that emerged was surprising, as these components were strikingly similar to the trust expectations identified by our review of human-automation trust models (Reliable and Capable) and human-human trust models (Sincere and Ethical). Additional item analysis allowed us to shorten each cluster to five items, yielding four initial subscales of trust: Reliable (count on, depend on, reliable, faith in, confide in, $\alpha = .72$), Capable (capable, diligent, rigorous, accurate, meticulous, $\alpha = .88$), Sincere (sincere, genuine, truthful, benevolent, authentic, $\alpha = .84$), and Ethical (honest, principled, reputable, respectable, scrupulous, $\alpha = .87$). The intercorrelations among the four components revealed that Sincere and Ethical were related to each other ($r = .46$, $p = .01$), suggesting that "moral trust" encompasses two related facets.

Sorting trust words

We sought to replicate the 4 dimensions and their item clusters in a second study, employing a guided sorting task. Sixty participants recruited via Amazon Mechanical Turk were asked to consider 32 words or short phrases, 6–7 for each of the hypothesized 4 dimensions as well as 5 filler items assumed to be unrelated to trust (e.g., humorous, intentional). The words or short phrases were either taken from the trust words in the study described above (Ullman & Malle, 2018) or were added to reflect published trust work. Participants then indicated how well they thought each item described different "person types" by sorting the items into one of four boxes. Each box represented a person with a single character trait—the markers of the four hypothesized dimensions: Reliable, Capable, Sincere, and Ethical (a fifth category was labeled "Other" for words people believed did not fit any personality type). We computed sorting consensus scores as the percentages of participants who classified a given item into a given category and then grouped items by sorting consensus (see Fig. 1). The replication succeeded: Each hypothesized dimension of trust expectations contained largely the same items as in the semantic space study (Ullman & Malle, 2018).

We then selected three items from the top four items of each cluster and added one new face-valid item to each cluster: *predictable* (for Reliable), *skilled* (for Capable), *candid* (for Sincere), and *has integrity* (for Ethical). Sixteen items (four in each of the dimensions) thus constituted the first version of a measurement tool we labeled the Multidimensional Measure of Trust (MDMT), shown in Fig. 2.

Changes in trust expectations

We used the MDMT in a subsequent study that examined whether the four subscales of the MDMT (see Fig. 2) are differentially sensitive to dimension-specific information one receives about a robot's behavior (Ullman & Malle, 2019). For example, the Sincere subscale should show an increase when new evidence arises about an agent's sincerity. 798 Participants recruited via Amazon Mechanical Turk read a baseline sentence about a robot's behavior and provided initial trust ratings on the MDMT. Then they received new information designed to prompt either an increase or decrease in dimension-specific trust in the robot, followed by final ratings on the MDMT. The subscales had acceptable to good internal consistency (Cronbach's α): *Reliable*, $\alpha = .92$; *Capable*, $\alpha = .92$; *Sincere*, $\alpha = .79$; *Ethical*, $\alpha = .81$. However, a Principal Component Analysis (PCA) on the 16 MDMT change scores (before-after evidence manipulation) suggested only a two-dimensional structure (with $\lambda > 1$). Under orthogonal rotation, the first component contained all eight *Reliable* and

Items	Sorting category: Reliable	Capable	Sincere	Ethical	Other
Reliable	92%	5%	0%	2%	2%
Can count on	82%	7%	8%	3%	0%
Consistent	77%	20%	2%	2%	0%
Can depend on	70%	15%	5%	7%	3%
Responsible	58%	23%	0%	17%	2%
Can have faith in	56%	7%	15%	19%	3%
Steadfast	49%	27%	8%	3%	12%
Capable	2%	92%	3%	2%	2%
Competent	12%	85%	3%	0%	0%
Intelligent	3%	80%	5%	0%	12%
Meticulous	13%	65%	3%	2%	17%
Diligent	31%	54%	7%	3%	5%
Accurate	39%	51%	2%	0%	8%
Rigorous	22%	42%	3%	3%	29%
Sincere	2%	0%	88%	5%	5%
Genuine	2%	2%	88%	3%	5%
Authentic	5%	5%	71%	15%	3%
Honest	0%	2%	51%	47%	0%
Truthful	2%	2%	47%	47%	2%
Transparent	5%	0%	42%	29%	24%
Can confide in	27%	5%	33%	25%	10%
Ethical	0%	0%	5%	93%	2%
Principled	3%	3%	3%	88%	2%
Respectable	14%	7%	8%	66%	5%
Trustworthy	23%	3%	27%	47%	0%
Scrupulous	7%	20%	2%	37%	35%
Reputable	28%	15%	12%	35%	10%
Humorous	3%	3%	5%	2%	86%
Easy going	2%	2%	23%	3%	70%
Creative	0%	42%	2%	0%	56%
Benevolent	7%	5%	25%	27%	36%
Intentional	10%	27%	25%	5%	32%

FIG. 1 Consensus rates (0%–100%) of sorting task in which participants grouped 32 items as describing persons who are Reliable, Capable, Sincere, or Ethical. Words in *italics* designate filler items that were not considered part of a trust dimension. Degree of consensus is *shaded* from *dark gray* (0%) to *brightwhite* (or *green* in online document) (100%).

Performance trust		**Moral trust**	
Reliable	**Capable**	**Sincere**	**Ethical**
Reliable	Capable	Sincere	Ethical
Predictable	Skilled	Genuine	Respectable
Can count on	Competent	Candid	Principled
Consistent	Meticulous	Authentic	Has integrity

FIG. 2 Items within each subscale of the Multidimensional Measure of Trust (MDMT), developed after sorting study and used in expectation change study.

Capable items (explaining 35.2% of the variance) and the second contained all eight *Ethical* and *Sincere* items (explaining 29.4% of the variance). Oblique rotation led to the same item divisions but allowed the two dimensions to correlate at $r = .66$.

In the analysis of subscale sensitivity, each of the four subscales was highly sensitive to evidence change overall (Fs > 100.0, ps $< .001$), $\eta^2 = .17$–$.43$. However, the subscales were not systematically sensitive to only the information about their dimension-specific evidence (e.g., the *Reliable* subscale did not respond only to evidence of being reliable). This may have been a result of the largely ineffective manipulations of trust-relevant evidence along the Capable and Sincere dimensions ($\eta^2 = .0$–$.03$), whereas manipulations were strong along the Reliable dimension ($\eta^2 = .06$–$.24$) and the Ethical dimension ($\eta^2 = .13$–$.23$). However, even without full-dimensional specificity, we saw that the measures of *Reliable* and *Capable* moved in tandem across manipulated evidence, and so did the measures of *Sincere* and *Ethical*. Thus, in this initial study, the major differentiation between Performance trust and Moral trust did emerge, while the finer differentiation into the two component subscales of performance trust (i.e., Reliable and Capable) and the two component subscales of moral trust (i.e., Sincere and Ethical) did not. The limited effectiveness of the evidence manipulations tempers our conclusions. We are continuing to replicate and validate this conceptual model of trust in follow-up studies, including studies that use the MDMT in laboratory human-robot interaction studies. We invite researchers to contribute to the measure's validation by using it in their own studies and sharing the results with us (see http://research.clps.brown.edu/SocCogSci/Measures/CurrentVersion_MDMT.pdf).

Discussion

We have accumulated evidence for a two-factor model of human-robot trust. In this model, people trust or distrust a robot as a function of two trust expectations: about the robot's performance characteristics (that it is capable of completing a task and/or will reliably do so) and about the robot's moral characteristics (that it will complete a task in adherence to social and moral norms). In addition, the literature and initial empirical evidence suggest a four-dimensional model, which poses numerous questions that future research must address.

Open questions

First, we do not know yet whether people differentiate robot performance trust further into capability and reliability components and robot moral trust further into sincerity and ethical integrity components (and potentially benevolence). Though our first two studies suggest they do, our third study left us more cautious. Second, we do not yet know whether

human-robot trust necessarily comes with a feeling of vulnerability that is characteristic of human trust (Colquitt et al., 2007; Mayer et al., 1995). We have used the more common term *worry* to denote the core feeling of trust, which can be directed at robots much like at humans: people worry about the robot's potential failure to complete a task, and for some robots, people may worry about the robot lying, cheating, or exploiting them. Third, in the human-human trust literature, researchers have distinguished between state trust and trait trust, and it stands to reason that people may similarly develop a trait-like propensity to trust or distrust robots—as a function of cultural exposure (e.g., science fiction), public discussion (e.g., recent worries about robots taking many human jobs), and actual experiences with robots. Fourth, some intriguing results in the human-human trust literature suggest that there is a social norm to trust others—even when exploitation is a real possibility. For example, Dunning et al. (2014) showed that most people trust a stranger (in the behavioral-economic Trust game) even when they are not confident that the other will make the benevolent choice; and they do so at least in part because they feel obligated to express respect toward others by trusting them. We might imagine that there is no such obligation to respect robots—or perhaps there is, if human observers are present during the human-robot interaction and the person does not want to appear distrustful. Only additional empirical studies will tell.

Moral trust expectations

Our empirical results and subsequent proposal of four dimensions of human-robot trust are not fully compatible with the widely used model of human-human trust by Mayer et al. (1995). Their model omits the reliability dimension and identifies moral dimensions of integrity and benevolence, rather than integrity and sincerity. Several points must be considered to reconcile these differences. First, because of the close similarity between robots and automated systems, the reliability dimension appears necessary to consider in human-robot trust. Whether it also must be considered separately in human-human trust is an open question. Second, whereas Mayer et al.'s selection of dimensions was based primarily on theoretical arguments, the four dimensions we put forth were discovered empirically (albeit with a small set of data so far). In fact, in two of our studies, we had included the item "benevolent," but it clustered in the Sincere component, and it later dropped out because of its lower loading in the PCA and lower sorting consensus. Third, Mayer et al.'s model was developed for the business domain, and it is plausible that this domain activates a particular worry about another person following only economic self-interest rather than having benevolent motives.

In summary, the exact composition of the moral dimension of trust is not yet settled. Given the literature and our own data, we would be

surprised if neither reliability nor sincerity were central to people's trust expectations. However, we remain open to the possibility that three moral characteristics exist (integrity, sincerity, and benevolence), even if perhaps benevolence is more specifically useful in the business domain.

The Multidimensional Measure of Trust (MDMT)

Trust in human-automation interaction, human-human interaction, and human-robot interaction share several similarities. The MDMT offers a single measurement instrument that flexibly adapts to the relevant dimensions of each domain and provides a standard of comparison across studies and domains. The measure is short (16 items) and easy to use. We have built a measurement feature into the scales that allows people to "opt out" of some of the terms if they think the specific terms do not apply (e.g., *has integrity* for an industrial robot). Researchers can also selectively use specific subscales (e.g., when evaluating a clearly nonsocial robot in a low-stakes task). Moreover, adapting the measure to more trait-like trust expectations could be accomplished with minimal reformulations. Finally, future work should investigate the extent to which trust ratings obtained using the MDMT converge with or diverge from existent trust measures, including measures of overall feelings of trust and of trusting actions. We hope others will join in this endeavor (see http://research.clps.brown.edu/SocCogSci/Measures/CurrentVersion_MDMT.pdf).

Conclusions

"What do I worry about that would prevent me from interacting with this agent?" The answer to this question looks slightly different across domains of interaction, but the core ideas about trust stay the same. A multidimensional conceptualization of trust can integrate the different answers—considering trust in an agent's performance (Reliable, Capable) as well as trust in an agent's morality (Sincere, Ethical), whether the agent is a human or a robot. As we design robots that raise questions of trust in human-robot interactions, we must investigate all dimensions of trust, even if some of them are only slowly coming to light.

Acknowledgments

This work was supported by the Office of Naval Research grant N00014-14-1-0144 and National Science Foundation grant 1717701. Daniel Ullman was supported by the National Space Grant College and Fellowship Program, Space Grant Opportunities in NASA STEM (NNX15AI06H), and the National Defense Science & Engineering Graduate Fellowship (NDSEG) Program.

References

Arkin, R. C., Ulam, P., & Wagner, A. R. (2012). Moral decision making in autonomous systems: Enforcement, moral emotions, dignity, trust, and deception. *Proceedings of the IEEE*, *100*(3), 571–589. https://doi.org/10.1109/JPROC.2011.2173265.

Baier, A. C. (1986). Trust and antitrust. *Ethics*, *96*, 231–260.

Barber, B. (1983). *The logic and limits of trust.* New Brunswick, NJ: Rutgers University Press.

Bigman, Y. E., & Gray, K. (2018). People are averse to machines making moral decisions. *Cognition*, *181*, 21–34. https://doi.org/10.1016/j.cognition.2018.08.003.

Broadbent, E., Stafford, R., & MacDonald, B. (2009). Acceptance of healthcare robots for the older population: Review and future directions. *International Journal of Social Robotics*, *1*(4), 319. https://doi.org/10.1007/s12369-009-0030-6.

Bruininks, P., & Malle, B. F. (2006). Distinguishing hope from optimism and related affective states. *Motivation and Emotion*, *29*(4), 324–352. https://doi.org/10.1007/s11031-006-9010-4.

Burke, C. S., Sims, D. E., Lazzara, E. H., & Salas, E. (2007). Trust in leadership: A multi-level review and integration. *The Leadership Quarterly*, *18*(6), 606–632. https://doi.org/10.1016/j.leaqua.2007.09.006.

Butler, J. K. (1991). Toward understanding and measuring conditions of trust: Evolution of a conditions of trust inventory. *Journal of Management*, *17*(3), 643–663. https://doi.org/10.1177/014920639101700307.

Butler, J. K., & Cantrell, R. S. (1984). A behavioral decision theory approach to modeling dyadic trust in superiors and subordinates. *Psychological Reports*, *55*(1), 19–28. https://doi.org/10.2466/pr0.1984.55.1.19.

Caldwell, C., & Clapham, S. E. (2003). Organizational trustworthiness: An international perspective. *Journal of Business Ethics*, *47*(4), 349–364.

Calo, M. R. (2011). Robots and privacy. In P. Lin, K. Abney, & G. A. Bekey (Eds.), *Robot ethics: The ethical and social implications of robotics* (pp. 187–201). Cambridge, USA: MIT Press.

Carnevale, D. G. (1995). *Trustworthy government: Leadership and management strategies for building trust and high performance.* San Francisco: Jossey-Bass Publishers.

Chun, K.-T., & Campbell, J. B. (1974). Dimensionality of the Rotter Interpersonal Trust Scale. *Psychological Reports*, *35*(3), 1059–1070. https://doi.org/10.2466/pr0.1974.35.3.1059.

Coeckelbergh, M. (2012). Can we trust robots? *Ethics and Information Technology*, *14*(1), 53–60. https://doi.org/10.1007/s10676-011-9279-1.

Colquitt, J. A., Scott, B. A., & LePine, J. A. (2007). Trust, trustworthiness, and trust propensity: A meta-analytic test of their unique relationships with risk taking and job performance. *Journal of Applied Psychology*, *92*(4), 909–927. https://doi.org/10.1037/0021-9010.92.4.909.

Cook, J., & Wall, T. (1980). New work attitude measures of trust, organizational commitment and personal need non-fulfilment. *Journal of Occupational Psychology*, *53*(1), 39–52. https://doi.org/10.1111/j.2044-8325.1980.tb00005.x.

Dunning, D., Anderson, J. E., Schlosser, T., Ehlebracht, D., & Fetchenhauer, D. (2014). Trust at zero acquaintance: More a matter of respect than expectation of reward. *Journal of Personality*, *107*(1), 122–141. https://doi.org/10.1037/a0036673.

Gabarro, J. J. (1978). The development of trust, influence and expectations. In A. Athos, & J. J. Gabarro (Eds.), *Interpersonal behavior* (pp. 290–303). Englewood Cliffs, NJ: Prentice Hall.

Gambetta, D. (1988). *Trust: Making and breaking cooperative relations.* Oxford, UK: Basil Blackwell.

Hancock, P. A., Billings, D. R., Schaefer, K. E., Chen, J. Y. C., de Visser, E. J., & Parasuraman, R. (2011). A meta-analysis of factors affecting trust in human-robot interaction. *Human Factors*, *53*(5), 517–527.

Hardin, R. (2002). *Trust and trustworthiness.* New York, NY: Russell Sage Foundation.

Hartzog, W. (2015). Unfair and deceptive robots. *Maryland Law Review*, *74*(4), 785–832.

Hoff, K. A., & Bashir, M. (2015). Trust in automation: Integrating empirical evidence on factors that influence trust. *Human Factors*, *57*(3), 407–434. https://doi.org/10.1177/0018720814547570.

Jian, J.-Y., Bisantz, A. M., & Drury, C. G. (2000). Foundations for an empirically determined scale of trust in automated system. *International Journal of Cognitive Ergonomics*, *4*(1), 53–71.

Jones, G. R., & George, J. M. (1998). The experience and evolution of trust: Implications for cooperation and teamwork. *The Academy of Management Review*, *23*(3), 531–546. https://doi.org/10.2307/259293.

Kahn, P. H., Jr., Kanda, T., Ishiguro, H., Gill, B. T., Ruckert, J. H., Shen, S. , et al. (2012). Do people hold a humanoid robot morally accountable for the harm it causes? In: *Proceedings of the seventh annual ACM/IEEE international conference on human-robot interaction*, pp. 33–40. https://doi.org/10.1145/2157689.2157696.

Kee, H. W., & Knox, R. E. (1970). Conceptual and methodological considerations in the study of trust and suspicion. *The Journal of Conflict Resolution*, *14*(3), 357–366.

Kim, P. H., Dirks, K. T., & Cooper, C. D. (2009). The repair of trust: A dynamic bilateral perspective and multilevel conceptualization. *The Academy of Management Review*, *34*(3), 401–422.

Kuipers, B. (2018). How can we trust a robot? *Communications of the ACM*, *61*(3), 86–95. https://doi.org/10.1145/3173087.

Lee, J., & Moray, N. (1992). Trust, control strategies and allocation of function in human-machine systems. *Ergonomics*, *35*(10), 1243–1270. https://doi.org/10.1080/00140139208967392.

Lee, J. D., & See, K. A. (2004). Trust in automation: Designing for appropriate reliance. *Human Factors*, *46*(1), 50–80. https://doi.org/10.1518/hfes.46.1.50_30392.

Lewis, M., Sycara, K., & Walker, P. (2018). The role of trust in human-robot interaction. In H. A. Abbass, J. Scholz, & D. J. Reid (Eds.), *Foundations of trusted autonomy* (pp. 135–159). https://doi.org/10.1007/978-3-319-64816-3_8.

Madhavan, P., Wiegmann, D. A., & Lacson, F. C. (2006). Automation failures on tasks easily performed by operators undermine trust in automated aids. *Human Factors*, *48*(2), 241–256. https://doi.org/10.1518/001872006777724408.

Madsen, M., & Gregor, S. (2000). Measuring human-computer trust. In: G. Gable, & M. Viatle (Eds.), *Proceedings of the 11th Australasian conference on information systems*, pp. 53–64.

Malle, B. F. (2019). How many dimensions of mind perception really are there? In: E. K. Goel, C. M. Seifert, & C. Freksa (Eds.), *Proceedings of the 41st annual meeting of the Cognitive Science Society*, Montreal, Canada: Cognitive Science Society, pp. 2268–2274.

Malle, B. F., Bello, P., & Scheutz, M. (2019). Requirements for an artificial agent with norm competence. In: *Proceedings of 2nd ACM conference on AI and ethics (AIES'19)*, New York, NY: ACM, pp. 21–27. https://doi.org/10.1145/3306618.3314252.

Malle, B. F., Magar, S. T., & Scheutz, M. (2019). AI in the sky: How people morally evaluate human and machine decisions in a lethal strike dilemma. In M. I. A. Ferreira, J. S. Sequeira, G. S. Virk, M. O. Tokhi, & E. E. Kadar (Eds.), *Robotics and well-being* (pp. 111–133). https://doi.org/10.1007/978-3-030-12524-0_11.

Malle, B. F., & Scheutz, M. (2014). Moral competence in social robots. In: *Proceedings of IEEE international symposium on ethics in engineering, science, and technology, Ethics'2014*, Chicago, IL: IEEE, pp. 30–35.

Malle, B. F., & Scheutz, M. (2019). Learning how to behave: Moral competence for social robots. In O. Bendel (Ed.), *Handbuch Maschinenethik [Handbook of machine ethics].* https://doi.org/10.1007/978-3-658-17484-2_17-1.

Malle, B. F., Scheutz, M., Arnold, T., Voiklis, J., & Cusimano, C. (2015). Sacrifice one for the good of many? People apply different moral norms to human and robot agents. In: *Proceedings of the tenth annual ACM/IEEE international conference on human-robot interaction, HRI'15*, New York, NY: ACM, pp. 117–124.

Malle, B. F., Zhao, X., & Phillips, E. (2019). Beyond anthropomorphism: Differentiated inferences about robot mind from appearance. In: *CHI'19 extended abstracts. Presented at the computer-human interaction, CHI'19, Glasgow, Scotland.*

Mayer, R. C., & Davis, J. H. (1999). The effect of the performance appraisal system on trust for management: A field quasi-experiment. *Journal of Applied Psychology*, *84*(1), 123–136. https://doi.org/10.1037/0021-9010.84.1.123.

Mayer, R. C., Davis, J. H., & Schoorman, F. D. (1995). An integrative model of organizational trust. *The Academy of Management Review*, *20*(3), 709–734. https://doi.org/10.2307/258792.

McKnight, D. H., Cummings, L. L., & Chervany, N. L. (1998). Initial trust formation in new organizational relationships. *The Academy of Management Review*, *23*(3), 473–490. https://doi.org/10.2307/259290.

Parasuraman, R., & Riley, V. (1997). Humans and automation: Use, misuse, disuse, abuse. *Human Factors*, *39*(2), 230–253. https://doi.org/10.1518/001872097778543886.

Parsons, T. (1969). Research with human subjects and the "professional complex" *Daedalus, 98*(2), 325–360.

Rotter, J. B. (1967). A new scale for the measurement of interpersonal trust. *Journal of Personality*, *35*(4), 651–665. https://doi.org/10.1111/j.1467-6494.1967.tb01454.x.

Schaefer, K. E. (2013). *The perception and measurement of human-robot trust. (PhD thesis, STARS. Electronic theses and dissertations). Retrieved from(2013).* https://stars.library.ucf.edu/etd/2688/.

Schaefer, K. E., Chen, J. Y. C., Szalma, J. L., & Hancock, P. A. (2016). A meta-analysis of factors influencing the development of trust in automation: Implications for understanding autonomy in future systems. *Human Factors*, *58*(3), 377–400. https://doi.org/10.1177/0018720816634228.

Schindler, P. L., & Thomas, C. C. (1993). The structure of interpersonal trust in the workplace. *Psychological Reports*, *73*(2), 563–573. https://doi.org/10.2466/pr0.1993.73.2.563.

Shank, D. B., & DeSanti, A. (2018). Attributions of morality and mind to artificial intelligence after real-world moral violations. *Computers in Human Behavior*, *86*, 401–411. https://doi.org/10.1016/j.chb.2018.05.014.

Sheridan, T. B., & Parasuraman, R. (2005). Human-automation interaction. *Reviews of Human Factors and Ergonomics*, *1*(1), 89–129. https://doi.org/10.1518/155723405783703082.

Short, E., Hart, J., Vu, M., & Scassellati, B. (2010). No fair!! an interaction with a cheating robot. In: *Proceedings of the 5th ACM/IEEE international conference on human-robot interaction*, pp. 219–226. https://doi.org/10.1109/HRI.2010.5453193.

Slovic, P., Flynn, J., Johnson, S., & Mertz, C. K. (1993). The dynamics of trust in situations of risk. In *Report no. 93-2*. Eugene, Oregon: Decision Research.

Spain, R. D., Bustamante, E. A., & Bliss, J. P. (2008). Towards an empirically developed scale for system trust: Take two. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, *52*(19), 1335–1339. https://doi.org/10.1177/154193120805201907.

Steinfeld, A., Fong, T., Kaber, D., Lewis, M., Scholtz, J., Schultz, A. , et al. (2006). Common metrics for human-robot interaction. In: *Proceedings of the 1st ACM SIGCHI/SIGART conference on human-robot interaction*, pp. 33–40. https://doi.org/10.1145/1121241.1121249.

Ullman, D., Leite, L., Phillips, J., Kim-Cohen, J., & Scassellati, B. (2014). Smart human, smarter robot: How cheating affects perceptions of social agency. In: *Proceedings of the annual meeting of the cognitive science society (Vol. 36)*, pp. 2996–3001.

Ullman, D., & Malle, B. F. (2018). What does it mean to trust a robot? Steps toward a multidimensional measure of trust. In: *Companion of the 2018 ACM/IEEE international conference on human-robot interaction*, pp. 263–264. https://doi.org/10.1145/3173386.3176991.

Ullman, D., & Malle, B. F. (2019). Measuring gains and losses in human-robot trust: Evidence for differentiable components of trust. In: *Companion to the 2019 ACM/IEEE international conference on human-robot interaction, HRI '19*, ACM, pp. 618–619.

Wallach, W., & Allen, C. (2008). *Moral machines: Teaching robots right from wrong.* New York, NY: Oxford University Press.

Weisman, K., Dweck, C. S., & Markman, E. M. (2017). Rethinking people's conceptions of mental life. *Proceedings of the National Academy of Sciences of the United States of America*, *114*(43), 11374–11379. https://doi.org/10.1073/pnas.1704347114.

Wijnen, L., Coenen, J., & Grzyb, B. (2017). "It's not my fault!": Investigating the effects of the deceptive behaviour of a humanoid robot. In: *Companion of the 2017 ACM/IEEE international conference on human-robot interaction*, pp. 321–322. https://doi.org/10.1145/3029798.3038300.

Yagoda, R. E., & Gillan, D. J. (2012). You want me to trust a ROBOT? The development of a human–robot interaction trust scale. *International Journal of Social Robotics*, *4*(3), 235–248. https://doi.org/10.1007/s12369-012-0144-0.

Yanco, H. A., Desai, M., Drury, J. L., & Steinfeld, A. (2016). Methods for developing trust models for intelligent systems. In R. Mittu, D. Sofge, A. Wagner, & W. F. Lawless (Eds.), *Robust intelligence and trust in autonomous systems* (pp. 219–254). https://doi.org/10.1007/978-1-4899-7668-0_11.

CHAPTER

2

Trust: Recent concepts and evaluations in human-robot interaction

Theresa Law and Matthias Scheutz

Human Robot Interaction Laboratory, Tufts University, Medford, MA, United States

Introduction

As robots become more sophisticated and interact with people more frequently, especially people who have little to no robotics experience, it is increasingly important for us to understand what it takes for people to trust or distrust a robot. This understanding begins by examining what we currently know and understand trust to be, and how we can reasonably measure trust, both in the situations where we believe we can measure it objectively and also in those cases where objective measures might not be possible and we have to rely on people's subjective feedback.

While trust as a subject has long been studied in psychology (Dunning & Fetchenhauer, 2011) and human factors (Lee & See, 2004), the human-robot interaction (HRI) community has only recently begun to explore trust in robots in earnest. The growing interest in trust in HRI is evident in the increasing number of publications on the topic, even though there is currently no agreed-upon definition of trust. As trust is a multifaceted concept (Yamagishi, 1998) and can be affected by a large number of factors (Hancock, Billings, Schaefer, Chen, et al., 2011), studying trust often means to study one particular aspect of trust, rather than an all-encompassing "trust" concept. Unsurprisingly, there is no accepted paradigm for evaluating trust in HRI either.

The majority of trust studies in HRI rely on different types of questionnaires to determine a person's trust in the robot with whom they are interacting. Some of the questionnaires are validated (Jian, Bisantz, & Drury, 2000; Larzelere & Huston, 1980; Lee & See, 2004; Mayer, Davis, &

Trust in Human-Robot Interaction
https://doi.org/10.1016/B978-0-12-819472-0.00002-2

© 2021 Elsevier Inc. All rights reserved.

Schoorman, 1995; Merritt, LaChapell, & Lee, 2012; Schaefer, 2013; Ullman & Malle, 2018), while others are not. These latter studies are more ad hoc. Also, for the few studies that do have objective trust measures, the lack of a unifying experimental paradigm makes it difficult to compare results across studies.

The purpose of this survey is to investigate the different trust dimensions that have been used in recent HRI studies, to classify those dimensions, and to show what instruments can be used to measure them. The hope is that the resulting framework and analysis will be useful for future investigations of trust in HRI. The papers included were experimental papers studying trust in HRI. Because we wanted to look at the current state of the field, only papers published in 2018 and up through the summer of 2019 were included. One paper (Soh, Xie, Chen, & Hsu, 2020) is from 2020; the paper was originally found and included as a preprint published in 2018. Search engines used were Scopus and Google Scholar, and papers were found using the search terms "trust" and "human-robot interaction," as well as from references from other papers. This resulted in a total of 33 papers being included in the survey. Table 1 presents an overview of all of the papers.

The chapter is outlined as follows. First, we present an overview of how trust has been defined in HRI in recent years. These definitions led us to dividing trust into two categories—performance-based trust and relation-based trust. We then begin to examine the papers relevant to this survey, starting with the questions the papers asked, then how trust was measured. Within these sections, we separated the papers that looked at performance-based trust from relation-based trust, as well as from the studies that looked at a combination of these types of trust. By making this separation, we highlight what these different types of trust encompass and the discrepancy in how they are measured. We then compare what types of questions are being asked and how trust is measured and defined within each study, showing the conflation and inconsistency of studying performance- and relation-based trust. We end with a discussion and recommendations for how the study of trust in HRI should advance, and a conclusion of our findings.

Trust definitions in HRI

The most widely cited trust definitions in recent HRI studies come from Lee and See (2004), Hancock, Billings, Schaefer, Chen, et al. (2011), and Hancock, Billings, and Schaefer (2011). Lee and See (2004) define trust as "the attitude that an agent will help achieve an individual's goals in a situation characterised by uncertainty and vulnerability," while Hancock, Billings, and Schaefer (2011) define it as "the reliance by an agent

that actions prejudicial to their well-being will not be undertaken by influential others" and as a willingness to accept suggestions from another agent (Hancock, Billings, Schaefer, Chen, et al., 2011). The concepts covered in these definitions are highly relevant to HRI. For example, if persons who have never worked with or programmed a robot before coming in contact with one, they will likely experience a high level of uncertainty about how the interaction will unfold. Even those familiar with robots in general or with specific robots may be uncertain about the functionality or behavior of a new robot. This uncertainty can leave people vulnerable; for example, they may need to rely on the robot for a task and are uncertain about its performance capabilities in achieving their goal. Alternatively, they may be asked to provide the robot with personal information and are uncertain about what is going to be done with that information or who will find out about it. Both of these scenarios leave people vulnerable to the behavior of robots. Additionally, interacting with a large and heavy robot may cause a person to be physically vulnerable. Therefore, people choosing to work with robots despite these uncertainties display a certain level of trust in the robot. If trust is present, people may be willing to alter their own behavior based on advice or information provided by the robot. For robots who work directly and closely with people, this can be an important aspect of a trusting relationship.

For robots who do have roles in which they work with people, especially in a social manner, being emotionally supportive can be another important aspect of trust (Rotter, 1967). Other trust definitions reference the trustee mitigating risk for the truster (Robinette, Howard, & Wagner, 2017; Wagner & Arkin, 2011), an agent's performance being reliable (Castelfranchi & Falcone, 2010; Hodges & Geyer, 2006; Lewis, Sycara, & Walker, 2018), and the potential lack of the ability of the truster to monitor the trustee's actions (Mayer et al., 1995). In returning to the above examples, we can see how these are all relevant to HRI. If a human feels vulnerable in their interaction with a robot, perceiving the robot as mitigating the human's risk should help the human to trust it more. Or if the person needs the robot to perform a task properly, the more reliable the robot is at that task, the more likely the person will trust it to continue to perform that task well. Ideally, the robot will be trusted to behave without direct or constant human supervision, allowing humans to focus on their own tasks and trust that the robot can do its own satisfactorily.

We attempted to structure the discussion of trust in HRI based on several important aspects of trust that are experimentally investigated. From the previous definitions, we divide trust into two categories for the remainder of this paper: *performance-based trust* and *relation-based trust*.

Performance-based trust centers around the robot being trusted to be reliable, capable, and competent at its task or tasks, without needing to be monitored by a human supervisor. Performance-based trust may also depend on

the robot's transparency, responsiveness, and predictability. Relation-based trust, on the other hand, implies that a robot is trusted as a social agent. A person with whom it interacts can be vulnerable emotionally and may trust that the robot will be sincere and ethical. Relation-based trust means that a person trusts the robot to be part of society in some way, not just off in a factory doing a job without any expectation of knowledge of social norms. This type of trust is becoming increasingly relevant, and therefore it is increasingly pressing to understand what factors may influence it.

These two categories reflect two very important, but very different, aspects of trust. While some situations may require people to trust a robot in both of these manners, many other situations only necessitate one or the other, as will be highlighted in the remainder of the chapter. The presence or absence of these two types of trust will affect robot use in different ways. A robot that garnered a human's trust in a performance-based manner may be tasked with a critical factory job, but may be sequestered to a room where it will never interact with people. A robot that is trusted in a relation-based manner, conversely, may be brought into a nursing home to be a companion robot, but would not be given a task like sorting a resident's pills. However, as will be discussed later, certain situations such as choosing a robotic teammate for a competitive game require both types of trust. Because these types of trust affect different types of interactions, it is important for researchers to be aware of which type of trust they are studying based on their definition of trust and their relevant questions, and if their measure of trust aligns with what they are studying. The following sections examine these latter factors—questions that researchers are currently asking about trust, and how they are measuring it.

Trust questions

There are a large number of factors that may have an impact on one agent's trust in another, and those factors could vary depending upon the context and the specific individual or interaction. To understand how these factors play out in establishing and maintaining trust in HRI, we must systematically isolate specific types of interactions, behaviors, and performances to understand their effects on trust. In the following sections, we examine the questions researchers were interested in by looking at what they manipulated or used as an independent variable, as these questions are indicative of the aspect of trust the researchers found most important. We start with questions that focus on performance-based or relation-based trust, followed by questions that address both. By dividing the chapter as such, we can categorize papers by the type of trust that the researchers were interested in studying; we can later use this to compare questions to the category of trust that their dependent measure reflected.

In this section, we step through each paper that fit our experimental criteria and categorize it as asking performance-based questions, relation-based questions, or a mix of the two. We then describe the actual questions that each paper addressed and a summary of their results based on how they measured trust.

Experimental manipulations

The following studies are ones in which the researchers manipulated experimental factors to see how different conditions affected trust.

Performance-based questions

Performance-based trust questions vary how well the robot performs in different conditions, or how aware the participants are of the robot's ability to perform different tasks. These types of questions do not rely on how the robot interacts socially, but how capable and reliable it is at its task. In interacting with a robot, a person may not be aware of the full extent, or limitations, of a robot's capabilities. Shu et al. (2018) and Soh et al. (2020) informed participants of the robot's capabilities by showing it performing certain tasks and asked whether their trust in it, indicated by Muir's (1994) questionnaire generalized to other tasks, given what they knew about its capabilities. Both studies found that trust was more directly transferred to unseen tasks that were similar to the observed task, that participants transferred trust more readily to simple tasks rather than difficult tasks, and that the robot's performance affected trust both in the observed and unobserved task.

Rather than manipulate participants' perceptions of robots' capabilities, the following studies manipulated the robot's actual task performance. Xu and Howard (2018) and Geiskkovitch, Thiessen, Young, and Glenwright (2019) looked at how the objective correctness of a robot's answer affected trust. The former asked how first impressions, when a robot provides an obviously correct or incorrect answer to a question during a first interaction, impacted trust; the latter asked about how robot errors were perceived by children. Xu and Howard (2018) found that a robot who gave a first impression of being faulty by providing an incorrect answer to a question was trusted less than a robot that gave a first impression of working properly by providing the correct answer; participants both accepted a robot's suggestions more often and rated their trust of it higher than that of the faulty robot on a survey question that asked if they trusted its advice. Geiskkovitch et al. (2019) found that children age 3–5 trusted a robot who had been previously correct more than one that had been incorrect, as indicated by siding with the correcting one in a binary decision.

Rossi, Dautenhahn, Koay, and Walters (2018) asked how severe different robot errors were considered to be, and based on those reports, how much trust was affected by errors of varying severity. When presented with a final emergency scenario and asked how they wanted to deal with it, participants who had seen the robot cause more severe errors were more likely to indicate that they did not trust the robot to handle the emergency, but were more willing to work as a team if the robot had made small errors. Byrne and Marín (2018) manipulated performance by asking how different levels of task completion in service robots, from failure to success through teamwork, affected trust. They asked participants before and after the experiment about their views on propensity to trust, sociability, competency, team working ability, and responsiveness to measure trust. They found that failure made trust decrease between the surveys, and success through teamwork was the greatest cause of increased trust between the pre and post surveys. Chen, Nikolaidis, Soh, Hsu, and Srinivasa (2018) had participants work with a robot to clear off a table, and the robot either did or did not take the person's trust level into account as it performed the task. They found that trust, as indicated by how infrequently a participant intervened in the robot clearing objects off the table, was higher with the robot who took the participant's evolving trust level into account.

Jayaraman et al. (2018) looked at the interaction between autonomous vehicle (AV) behavior and pedestrians by manipulating the performance of an AV in a virtual reality environment. A questionnaire based on Muir's (1987) work showed that participants trusted less aggressive AV drivers more than aggressive ones and trusted the AV more when it operated in an environment with signalized crosswalks. This trust was coordinated with trusting behaviors of reduced distance between the participant and the AV and increased jaywalking time. Pedersen, Andersen, Köslich, Weigelin, and Kuusinen (2018) also looked at questions concerning trust in AVs. They studied whether the robot's performance causing real-world consequences affected trust. Participants rode in an AV simulator and received a small electrical shock if the vehicle crashed. Participants had the option to take over driving the car. Participants who believed there would be a consequence to the AV's behavior took over control from the car more often, and when asked, indicated that they did so because they did not trust it. In this way, even though the robot's performance did not vary, the researchers were still asking about whether the humans trusted the robot's performance enough to let the car drive itself.

In summary, these researchers were interested in questions about how the following factors affected trust: knowledge of a robot's capability in one task and assumptions of a robot's capabilities in another task (Shu et al., 2018; Soh et al., 2020); obvious correctness or incorrectness in a first interaction (Xu & Howard, 2018); previous correctness or incorrectness

when the robot then made a mistake (Geiskkovitch et al., 2019); varying levels of robot error severity (Rossi, Dautenhahn, et al., 2018); levels of robot task completion (Byrne & Marín, 2018); accounting for a human's evolving trust levels (Chen et al., 2018); aggressive versus nonaggressive driving of an AV (Jayaraman et al., 2018); and finally, real-world consequences to an AV's mistake (Pedersen et al., 2018). These questions all reflect interest in a performance-based trust of the robot because the factors that were manipulated were all about how the robot performed in specific tasks.

Relation-based questions

Relation-based trust questions do not address the robot performance; if there is a task, the robot's performance is the same in all conditions, and the researchers ask about some aspects about social factors, conventions, or norms. There is a wide variety of these questions that have been used in the HRI literature.

Behrens, Egsvang, Hansen, and Møllegård-Schroll (2018), Kraus, Kraus, Baumann, and Minker (2018), and Ghazali, Ham, Barakova, and Markopoulos (2018) all looked at how robot gender affects trust. Behrens et al. (2018) manipulated the robot's voice to indicate its gender, and found that participants saw the male robot as more trustworthy and were willing to share more personal information with it. Kraus et al. (2018) also manipulated the robot's voice, as well as its name to give it an explicit gender, and gender stereotypical personalities and tasks to give it an implied gender. They then studied how gender stereotypes affected trust, based on either explicit or implicit robot gender. Participants indicated through a validated trust scale that stereotypically male behaviors and tasks were trusted more than stereotypically female behaviors and tasks.

Ghazali et al. (2018) looked at how gender similarity or dissimilarity between the participant and robot, as well as the robot's facial expression, affected trust. The facial expressions were made to be either trustworthy or untrustworthy, based on social neuroscience research. Participants trusted the trustworthy face more than the untrustworthy face, as indicated via a survey. Unlike the previous two studies, the authors found no gender effects. These results were also seen in the participants' decisions about whether or not to follow the robot's advice. You and Robert (2018) also asked about the similarity and dissimilarity between robot and participant by varying the robot's gender and attitude about work to match or not match the participant, as well as the danger risk of the task. Based on their answers from Jian et al.'s (2000) Trust in Automation Scale, robots who were similar to the person on a deep level, attitude about work, were trusted more than those that were dissimilar on a deep level. Surface-level similarity, the robot's gender, was only significant when

the danger risk was low, and then similar robots were trusted more than dissimilar ones.

Sanders, Kaplan, Koch, Schwartz, and Hancock (2019) studied how the type of agent (human vs. robot) influenced trust. They found that people chose a robot over a human to complete a dangerous task, a human over a robot to complete a menial task, and that participants rated their trust in whichever agent they chose higher than the one they did not choose, based on Jian et al.'s (2000) Trust in Automation Scale. Correia, Mascarenhas, Prada, Melo, and Paiva (2018) also looked at the effect of group dynamics, and varied whether robots in a group that contained both robots and humans expressed individual emotions or group emotions. They found that when a robot teammate expressed group emotions rather than individual, the participants expressed a higher level of group trust via a postexperiment survey. Finally, Herse et al. (2018) varied the agent's embodiment and whether or not it considered a person's preferences before offering a restaurant suggestion. They found that participants trusted the robot by following its suggestion more if the robot considered a person's preferences.

In summary, these researchers were interested in questions about how the following factors affected trust: gendered voice of a robot (Behrens et al., 2018); explicit robot gender and implicit gender stereotypes (Kraus et al., 2018); gender similarity or dissimilarity between the participant and the robot, as well as the robot's facial expression (Ghazali et al., 2018); similarity or dissimilarity between the robot the person's gender and attitude about work (You & Robert, 2018); type of agent, robot or human (Sanders et al., 2019); group dynamics and expression of group emotions (Correia, Mascarenhas, et al., 2018); and embodiment and consideration of a person's preferences (Herse et al., 2018). For all of the earlier studies, the questions that are being asked are fundamentally social in nature. These are factors that are inherent to social interactions and that will inevitably become increasingly relevant as robots become increasingly prevalent social characters.

Mixed questions

Experiments that mixed relation-based and performance-based factors either did so explicitly by varying both the performance-based and a relation-based factor, or the robot's performance was tied up implicitly in a relational aspect as well. The main type of question that is being asked which implicitly entails both relation-based and performance-based types of trust is that of a robot's communication style. Volante, Sosna, Kessler, Sanders, and Hancock (2019) investigated robot communication and social conformity by manipulating whether or not the robot communicated, and how other people (confederates) were viewing the robot. They found that other people's views on the robot's behavior affected trust

where more positive views resulted in greater trust, as reported by the HRI Trust Perception Scale (Schaefer, 2013). They found no effect of robot communication. Salomons, van der Linden, Strohkorb Sebo, and Scassellati (2018) also looked at trust in a social conformity setting, like the famous Asch (1956) conformity study, by seeing if ambiguity in correct answers resulted in participants trusting the robots' group decision more if they have access to the group's answers before submitting their own final answer. They found that when participants could see the robots' answers before submitting their own, they trusted the robots, as indicated by conforming to their answers, more than when they could not see the robots' answers. Haspiel et al. (2018) and Wang, Pynadath, Rovira, Barnes, and Hill (2018) both looked at how behavioral explanations provided by the robot affected trust. Haspiel et al. (2018) focused on the timing of explanations in regard to expectation violation and found that providing explanations before behavior resulted in higher trust ratings on Muir's (1987) Scale. Wang et al. (2018) focused on the interaction between explanations, embodiment, and error communication strategy and found that participants rated trust higher on Mayer and Davis' (1999) Scale and followed the robot's suggestions more when the robot offered an explanation of its decision. One type of error communication strategy, that of vocal warnings of uncertainty, was studied by Christensen et al. (2019) and was found to not have an effect on for how long participants followed a robot's advice.

Fischer, Weigelin, and Bodenhagen (2018) examined how the robot's transparency about its own behavior, as well as how much it adapted to a patient's needs, affected trust. They found that participants rated their trust of the robot higher when the robot was transparent, but adaptability did not have an effect. Other researchers have looked at robot communication when the robot fails at a task, and whether it is able to repair trust. Communication in some form is a necessary part of social interaction. Correia, Guerra, Mascarenhas, Melo, and Paiva (2018) examined how the manner in which a robot justified its mistakes could repair trust and found that when the consequence of the failure was not severe, participants rated their trust as higher when the robot justified the mistake than when it did not, as rated on the Schaefer (2013) Scale. Sebo, Krishnamurthi, and Scassellati (2019) looked at how the robot framed its actions that violated trust, as well as the manner in which it attempted to repair trust. They found that robots who explained their behavior as a mistake and who then apologized for it were most likely to have repaired trust, as seen by participants performing retaliating behavior less often after the violation, and a higher rating on the Dyadic Trust Scale measurement (Larzelere & Huston, 1980). In these studies, the performance-based trust is being tested because the way that the robot explains its actions may influence how the person

interprets its performance. At the same time, communication is an inherently social and relation-based aspect of interacting.

Another type of implicit factor affecting trust is varying a robot's competitiveness in a task (Novitzky et al., 2018; Robinette, Novitzky, Fitzgerald, Benjamin, & Schmidt, 2019). Two studies found that participants would trust the more aggressive, competitive robot to be on their team more than the noncompetitive robot. This clearly has to do with performance because it is competing to win and therefore needs to do well to do so, but there is also a relation-based aspect to being competitive. These robots may not be implementing the "best" strategy, but they are being aggressive to make sure that they are at least better than their competitor. Competitiveness implies a recognition and consideration of another social agent in a task.

Other studies explicitly looked at how multiple factors affected trust, some of which were performance-based trust questions and some of which were relation-based questions. Gombolay et al. (2018) asked about the interaction between embodiment and quality of a suggestion in willingness to accept the suggestion. They found that participants were more likely to comply with the agent's suggestion at an inappropriate time (i.e., low-quality suggestions) when the agent was a nonembodied computer. When the agent was an embodied robot, participants exhibited an appropriate amount of dependence on the agent (i.e., did accept high-quality suggestions but did not accept low-quality suggestions). Xie, Bodala, Ong, Hsu, and Soh (2019) varied the type of agent, human versus robot, and the agent's capability and intent. They found that a robot whose intent and capability is similar between an observed task and an unobserved task will be trusted more to perform a task than one whose capabilities and intent are different from what is needed in the unobserved task.

In summary, these researchers were interested in questions about how the following factors affected trust: robot communication and group perception of the robot (Volante et al., 2019); knowledge about a group of robot's answers to an ambiguous question (Salomons et al., 2018); timing of expectation violation explanations (Haspiel et al., 2018); explanations, embodiment, and error communication strategy (Wang et al., 2018); vocal warnings of uncertainty (Christensen et al., 2019); behavior transparency and adaptation (Fischer et al., 2018); different manners of mistake justification (Correia, Guerra, et al., 2018); framing of actions that violated trust and manner in which it attempted to repair trust (Sebo et al., 2019); robot competitiveness (Novitzky et al., 2018; Robinette et al., 2019); the interaction of embodiment and robot suggestion quality (Gombolay et al., 2018); and type of agent and agent capability and intent (Xie et al., 2019). These questions ask about factors that are a mix of performance- and relation-based trust, either implicitly or explicitly.

Surveys

The following studies are ones in which the authors did not manipulate anything about the robot or the environment; rather, either all participants experienced the same stimuli or the researchers examined factors that they themselves did not control. Rossi, Holthaus, Dautenhahn, Koay, and Walters (2018) got at a performance-based question by examining whether increasing participants' levels of understanding of how a Pepper robot worked increased their trust in it. They first showed the participants a video of Pepper, then a live interaction demo, then allowed them to program it. Participants trusted Pepper's capability to help them with homework or wake them up from school the most after the live demo. They trusted it to help them in a dangerous situation the most after programming it themselves. Weigelin, Mathiesen, Nielsen, Fischer, and Nielsen's (2018) study is also about performance-based trust; in it, participants' dyad teams pretended to be a healthcare patient and physical therapist, and used a robot to help the "patient" perform physical therapy tasks. They found that the robot's performance affected the participants' distress levels, which in turn affected usability and trust.

Lyons and Guznov (2019) examined whether individual differences and biases in believing automation to be perfect, relation-based factors, influenced trust in a robotic system. They found that there was a positive correlation between the two based on survey responses to Merritt et al.'s (2012) and Mayer et al.'s (1995) questionnaires. Newaz and Saplacan (2018) and van Straten, Peter, Kühne, de Jong, and Barco (2018) both examined mixed performance-based and relation-based factors. Newaz and Saplacan (2018) asked about people's subjective experiences with robotic vacuums based on the feedback it gave and found that participants reported a lack of trust if there was a lack of feedback. van Straten et al. (2018) specifically asked whether children differentiate between interpersonal (relation-based) trust and technological (performance-based) trust. In interviews, children differentiated between the two types of trust.

In summary, these researchers were interested in questions about how the following performance-based factors affected trust: knowledge of the robot's capabilities (Rossi, Holthaus, et al., 2018) and use of a physical therapy robot (Weigelin et al., 2018).

These researchers were interested in questions about how the following relation-based factors affected trust: individual differences and biases toward believing automation to be perfect (Lyons & Guznov, 2019).

Finally, these researchers were interested in questions about how the following mixed factors affected trust: subjective experience with a home vacuum robot (Newaz & Saplacan, 2018); differentiation between interpersonal and technological trust in children (van Straten et al., 2018).

In the following section, we examine which of the previous studies that utilized an objective measure of trust, what that measure was, and what type of trust it was measuring.

Objective measures of trust

Subjective questionnaires remain the primary means of determining a participant's trust in a robot. While a large number of studies rely on validated questionnaires to measure trust subjectively, others create their own nonvalidated ad hoc questionnaires. As a general trend, objective measures of trust are used far less frequently than either type of subjective questionnaires. Objective measures, however, allow us to analyze how a person actually interacts with a robot, rather than relying on their own speculation about themselves (e.g., their motivations, reasons, etc.). There are a few common ways in which researchers indirectly and objectively measure trust. We describe these in the following section.

Types of objective measures

Recent objective trust measures can be divided into four categories: task intervention, task delegation, behavioral change, and following advice. In task intervention scenarios, the participant interacts with a robot that is performing a task that is ordinarily done by people. Trust is measured by the number of times the participant intervenes by taking over doing the task from the robot. This type of objective measure is good for when researchers are testing a robot that could be used to take over common tasks that people perform, such as autonomous driving. Task delegation is similar in that the robot may be performing a task, but the participant decides in the end if the robot or a person should be in charge of that task. Or, in some cases, which robot out of multiple should be chosen. The agent that the participant chooses is considered the one they trust the most for that task. Similar to task intervention, this paradigm is useful for tasks that could be done by a human or a robot to investigate which people trust more for that task. These two types of paradigms are primarily appropriate in measuring performance-based trust.

Observing behavioral changes in participants is a useful objective measure when the participant cannot control the robot's behavior the way they might be able to during a task intervention or delegation task. This paradigm is about observing and measuring how participants naturally interact with and behave around different robots. Trust is measured differently in each study based on the specific nature of the interaction. This

paradigm can be useful for measuring both performance-based and relation-based trust, as well as a mix of the two, in situations in which the participant has no control over the robot's behaviors.

The fourth main measure is following a robot's advice. Like the behavioral change measure, this trust measure is used when participants cannot control the robot's behavior. The robot makes suggestions or offers advice, and participants have the option whether or not to follow it. This is often used when the robot and human are teammates, or share a common goal. Trust is measured by whether or not, or how often, the participant follows the robot's advice. This is an especially useful measure for robot-human teams and can be useful for measuring performance-based trust, relation-based trust, or a mix of the two.

In the following section, we step through each paper in our survey that used an objective measure of trust. We categorize them as measuring either performance- or relation-based trust, or a mix of the two, and then for performance-based and mixed measures; we further categorize them by which type of objective trust measure they use. Within those categories, we describe what each measure was, and how it indicated the participant's trust in the robot.

Performance-based objective measures

The majority of objective trust measures that have been used in recent years in HRI have attempted to assess the participants' trust in terms of the robot's performance.

Task intervention

Task intervention scenarios measure trust based on how much of the task the human allows the robot to do, and how often the human takes over doing the task from the robot. This method is utilized by Pedersen et al. (2018) and Chen et al. (2018). The former had participants interact with a self-driving car, and trust was measured by the amount of times the participant took over driving from the car. In the latter, participants worked with a robot that was clearing a table. Trust was measured as the amount of times the participant cleared objects off of the table themselves. In both tasks, fewer interventions indicated more trust in the robot.

Task delegation

In task delegation scenarios, participants choose which agent does which task; unlike in the intervention scenarios, agents cannot only do some of the task some of the time. In Xie et al. (2019), participants saw the robot perform one task, and then were asked if that robot or a human

should perform a different task. In Sanders et al. (2019), participants were told to pretend they were a supervisor who needed to hire agents for two different jobs. For both jobs, they were asked to choose between hiring a robot or a human. Rossi, Dautenhahn, et al. (2018) presented an emergency situation and asked participants if they would rather take care of it themselves, delegate fixing it to the robot, or work together to fix it. Task intervention and task delegation scenario are both fairly direct measures of whether or not a person trusts a robot to perform a task successfully and satisfactorily.

Behavioral change

A more indirect way to measure trust in a robot's performance is to analyze how participants' behaviors change based on how the robot acts. Jayaraman et al. (2018) looked at street crossing behavior as participants acted as pedestrians while an AV came toward them. The authors considered how much space the participants left between themselves and the AV, how long they waited before crossing the street, and how often they jaywalked to indicate how much the participants trusted the AV's behavior. In Weigelin et al. (2018), human dyads used a robotic arm to perform kinesthetic therapy tasks. The authors analyzed videos of the interactions for signs of distress, which they took to indicate that the subjects did not trust the robot. In behavioral change scenarios, while the authors did not ask participants about the robot's performance directly, they could extrapolate their feelings of trust based on participants' behavioral cues.

Following advice

The most commonly used objective performance trust measure is whether or not a person follows the task performance advice that a robot gives. In Gombolay et al. (2018), a robot gave nurses suggestions about how they should distribute nursing station assignments across staff. Wang et al. (2018) had participants complete a virtual reconnaissance task and provided a robot that would tell the person if it believed the building was safe to enter or not. Participants had the choice of whether or not they listened to the robot's safety assessment. Christensen et al. (2019) had participants navigate blindly through the maze, while a robot provided instructions about which direction to turn; however, it eventually became obvious that the robot was leading the participant in a circle. The amount of time participants continued to listen to the robot's instructions indicated how much the participants trusted the robot. In Xu and Howard (2018), participants had to report the number of toothpicks that were briefly shown on a screen and were told that they were competing against another team. The participant's robot teammate advised the participant about how many toothpicks it believed there to be, and participants had the option to change their answer to the robot's answer. Finally,

Geiskkovitch et al. (2019) had children interact with two different robots, one of whom labeled a common object correctly and the other incorrectly. Then each robot held up a different unfamiliar object and provided the same name for it. The child was asked which of the two objects matched the name. The one that the child chose indicated which robot it trusted to provide the correct information, based likely on its performance in previous rounds. In all of these studies, the participants had the option to ignore the robot's advice or suggestions and continue to perform the task their own way. However, if they trusted the robot's performance, they should follow the advice that it provided to successfully complete the task.

Relation-based objective measures

Of the surveyed literature, Behrens et al. (2018) was the only study that used a solely relation-based objective trust measure to indicate a participant's relation-based trust in a robot. Participants were first shown an image of a robot and heard it speak, and were asked to indicate what information from a provided list they would be willing to share with the robot. In a second study, participants interacted in person with a robot and were asked to share with it an embarrassing personal story and login credentials for a website. Trust was measured in both studies as the amount of information the person was willing to share with the robot. There was no task that the robot needed to perform, and therefore the participants did not need to rely on the robot's behavior. Instead, they needed to trust it in a social manner to not share their vulnerable information with a third party. This can be considered a behavioral change scenario because participants would choose to change their behavior and tell the robot their vulnerable information if they trusted it.

Mixed objective measures

There are a number of studies that measure trust primarily in a performance-based way, but that also depend on a relation-based aspect of trust as well. In the surveyed studies, none of the papers that used a mixed performance-based and relation-based objective trust measure utilized a task intervention scenario.

Task delegation

Novitzky et al. (2018) and Robinette et al. (2019) used a trust delegation scenario. In both studies, trust was determined by a participant choosing which robot they would want as their teammate. In this scenario, participants watched two different robots perform a task and were then asked which of the two robots they would want as their teammate if they had to perform the same task. Because the task was a competitive one, the

participants needed to trust that whichever robot they delegated the task to would help them succeed and perform better than the other team. However, there is also a relation-based aspect to choosing a teammate. A person may not choose someone or something that they do not trust to have adequate social skills if they need to work closely with it.

Behavioral change

Sebo et al. (2019) took the approach of the participant competing against a robot rather than with it as its teammate. The participant and the robot played a video game against one another, and at the start, the robot promised not to utilize a power-up that would be detrimental to the human player as long as the human made the same agreement. The robot then violated that agreement and used the power-up, and trust was measured by seeing if the person retaliated by also using the power-up. The person could trust the robot socially to not break their pact again, or it could trust that its performance would be better if it did not use the power-up again. The person's behavior in the game indicated how much they trusted the robot.

In another study in which trust was measured based on whether or not participants changed their behavior based on a robot's answers, Salomons et al. (2018) asked people to play a modified version of Asch's (1956) conformity study. Participants had to submit an answer to the question about which image matched a given ambiguous word, and then would hear the answers of a group of robots. The participants could then change their answer to the robots before submitting their final answer. Because the words to which they needed to match images were ambiguous, participants who switched their answer to the robots may have trusted the robot's performance and understanding of the words more than their own. In line with the Asch studies, however, there can be a relation-based trust in conforming to the group with whom one is a part.

Following advice

As discussed previously, there are some studies in which following a robot's advice depends almost exclusively on a person's trust in the robot's performance. However, there are also times in which following the robot's advice requires relational trust as well. Herse et al. (2018) had participants decide whether or not they trusted the restaurant recommendation that a robot offered them. The performance of the robot mattered because there is likely a level of poor recommendation in which a person will never accept the suggestion. But trusting the recommendation also implies that the robot understood something about the person's preferences, and social norms about the types of restaurants that should be suggested to people (i.e., when asked for a restaurant recommendation,

a person is unlikely to seriously answer with McDonald's). Ghazali et al. (2018) used another food-based scenario. Participants were told to make a drink for an alien, with a robot available to give advice on what ingredients to use. Similar to the restaurant scenario, there is a performance-based trust aspect in that one must trust it not to suggest an undrinkable item, like rocks. Also, there is the trust that it does not want the alien, as another social being, to suffer, so it will not suggest something feasible but inappropriate for a drink, such as sardines.

Comparing within studies

In the earlier sections, we divided trust in robots into two factors, performance-based trust and relation-based trust (as well as a mix of the two). Table 1 shows an overview of the provided definitions of trust, trust questions, and objective measures of trust seen in the recent HRI literature categorized into these types of trust.

Ideally, in a given paper, the trust category of the definition, question, and objective measure (if used) should match; the study would then be consistent in how it views trust, how that view plays into the factors about trust it is studying, and how its results indicate that the participant trusted the robot. The above studies that used an objective measure of trust are fairly evenly split between studies that are investigating performance-based factors that influence trust, relation-based factors, and mixed performance-based and relation-based factors. In this section, we categorize papers based on the type of trust that their questions of interest reflected and discuss the subsequent trust types into which their trust definition and objective measure fall. This section therefore only discusses the surveyed literature that included an objective measure of trust.

Performance-based papers

The five studies that focused on questions about the robots performance all use objective measures that solely indicated the participants' trust in the reliability and capability of the robot's performance (Geiskkovitch et al., 2019; Jayaraman et al., 2018; Pedersen et al., 2018; Weigelin et al., 2018; Xu & Howard, 2018). Of these five, Geiskkovitch et al. (2019) and Weigelin et al. (2018) use definitions of trust that specifically acknowledge reliability (Rotter, 1971) of the system (Weigelin et al., 2018). Xu and Howard (2018) used a definition of trust in which the trustee mitigates the risk of the truster (Robinette et al., 2017). Similarly, Jayaraman et al. (2018) definition of trust is a willingness to be vulnerable to the actions of a robotic system. While these definitions do not explicitly

TABLE 1 Comparison within papers.

Performance-based papers					
Citation	**Definition reference**	**Factors**	**Subjective measure reference**	**Objective measure**	**Objective measure trust type**
Byrne and Marín (2018)	Schaefer (2013), Hancock, Billings, and Schaefer (2011)	Task completion	Ad hoc		
Chen et al. (2018)	No ref. Keywords: "perceived robot ability"	Taking trust into account	Ad hoc	Number of times participant intervened in robot's task	Performance; task intervention
Geiskkovitch et al. (2019)	Rotter (1971)	Factual mistake		Percentage of trials in which the child chooses to side with the previously correct robot	Performance; following advice
Jayaraman et al. (2018)	No ref. Keywords: "willingness to be vulnerable to actions"	Driving behavior	Muir (1987)	Participant street crossing behavior	Performance; behavioral change
Pedersen et al. (2018)	No definition	Real worldconsequences		If participant assumed control of AV	Performance; task intervention
Rossi, Dautenhahn, et al. (2018)	Lee and See (2004)	Severity and timing of performance errors		If participant delegated fixing an emergency situation to the robot	Performance; task delegation
Rossi, Holthaus, et al. (2018)	Lee and See (2004)	Capability awareness	Ad hoc		

Shu et al. (2018)	Castelfranchi and Falcone (2010)	Generalize trust across tasks	Ad hoc		
Soh et al. (2020)	Castelfranchi and Falcone (2010)	Generalize trust across tasks	Muir (1994) and Muir and Moray (1996)		
Weigelin et al. (2018)	No ref. Keywords: "safety and reliability of the system"	Performance		Video analysis of behavioral signs of distress	Performance; behavioral change
Xu and Howard (2018)	Robinette et al. (2017)	First impressions	Ad hoc	Whether or not participant followed robot's advice	Performance; following advice
Relation-based papers					
Behrens et al. (2018)	No definition	Robot gender	Ad hoc	How much information participant shares with robot	Relation; behavioral change
Correia, Mascarenhas, et al. (2018)	No definition	Group emotions	Ad hoc		
Ghazali et al. (2018)	Rotter (1967)	Gender similarity; facial expressions	Jian et al. (2000), Tay, Jung, and Park (2014), Heerink, Krose, Evers, and Wielinga (2009), and Wheeless and Grotz (1977)	How many times participant asked the robot for help	Mixed; following advice
Herse et al. (2018)	Lewis et al. (2018) and Moray and Inagaki (1999)	Embodiment; preference consideration	McCroskey and Teven (1999)	Participant acceptance of robot's suggestion	Mixed; following advice

Continued

TABLE 1 Comparison within papers—cont'd

Performance-based papers					
Citation	**Definition reference**	**Factors**	**Subjective measure reference**	**Objective measure**	**Objective measure trust type**
Kraus et al. (2018)	Lee and See (2004) and Hancock, Billings, and Schaefer (2011)	Robot gender; task gender stereotype	Unspecified questionnaire		
Lyons and Guznov (2019)	Lee and See (2004)	Individual biases	Merritt et al. (2012) and Mayer et al. (1995)		
Sanders et al. (2019)	Lee and See (2004) and Hancock, Billings, Schaefer, Chen, et al. (2011)	Type of agent	Jian et al. (2000)	Which agent the participant would choose to complete the task	Performance; task delegation
You and Robert (2018)	Groom and Nass (2007) and Hancock, Billings, and Schaefer (2011)	Similarity	Jian et al. (2000)		
Mixed papers					
Christensen et al. (2019)	Hancock, Billings, and Schaefer (2011)	Vocal warnings of uncertainty		Amount of time spent following robot's advice	Performance; following advice
Correia, Guerra, et al. (2018)	Hancock, Billings, Schaefer, Chen, et al. (2011)	Mistake justification	Schaefer (2013)		
Fischer et al. (2018)	Choi and Ji (2015)	Transparency; adaptation	Ad hoc		

Gombolay et al. (2018)	Lee and See (2004)	Embodiment; suggestion quality	Jian et al. (2000)	Whether or not participant followed the robot's advice	Performance; following advice
Haspiel et al. (2018)	No ref. Keywords: "openness to being subjected to actions"	Explanation timing	Ad hoc		
Newaz and Saplacan (2018)	Madsen and Gregor (2000)	Feedback	Ad hoc		
Novitzky et al. (2018)	Wagner and Arkin (2011)	Competitiveness	Ad hoc	Which robot participant would choose to defend their flag in a game	Mixed; task delegation
Robinette et al. (2019)	Wagner and Arkin (2011)	Competitiveness		Which robot participant would choose as a teammate	Mixed; task delegation
Salomons et al. (2018)	Hodges (2014) and Hodges and Geyer (2006)	Conformity		Whether or not participant conformed to robots in critical round	Mixed; behavior change
Sebo et al. (2019)	Mayer et al. (1995)	Trust violation framing; trust repair strategy	Larzelere and Huston (1980)	Participant choice of retaliating power-up usage	Mixed; behavior change
van Straten et al. (2018)	Lee and See (2004), Madsen and Gregor (2000), and You and Robert (2018)	Technological vs interpersonal trust	Ad hoc		

Continued

TABLE 1 Comparison within papers—cont'd

Performance-based papers					
Citation	**Definition reference**	**Factors**	**Subjective measure reference**	**Objective measure**	**Objective measure trust type**
Volante et al. (2019)	Freedy, DeVisser, Weltman, and Coeyman (2007)	Robot's performance; group perception	Schaefer (2013)		
Wang et al. (2018)	No definition	Explanations	Mayer et al. (1995)	Percentage of times participant complied with robot's suggestion	Performance; following advice
Xie et al. (2019)	Soh et al. (2020) and Chen et al. (2018)	Type of agent; intention; capability	Schaefer (2013)	Whether or not participant delegated task to the robot	Performance; task delegation

reference reliability or capability, they are implied in the assumption that the performance will determine whether or not a vulnerable agent is at risk. Pedersen et al. (2018) did not provide an operational definition of trust, but participants in their study believed themselves to be physically vulnerable because they believed they would be shocked if the AV did not perform properly, so they needed to trust it to be reliable. Current questions that are solely about performance-based trust, therefore, have objective measures and trust definitions that align with the goal of study performance-based factors quite well.

Relation-based papers

As indicated earlier, Behrens et al. (2018) was the only study to use an objective trust measure that got purely at a relation-based trust, rather than a relation-based and performance-based trust or just a performance-based trust. The authors were asking about how gender, a relation-based factor, influenced trust; however, they did not provide an operational trust definition. Two of the four of the studies that investigated relation-based factors of trust used objective measures that indicated a mix of both performance-based and relation-based trusts (Ghazali et al., 2018; Herse et al., 2018). In these papers, Herse et al. (2018) used a definition of trust that is based on performance and reliability (Lewis et al., 2018; Moray & Inagaki, 1999). Ghazali et al.'s (2018) study did not explicitly state their operational trust definition. Sanders et al. (2019) used an objective measure of trust that seems to indicate trust primarily in the robot's performance. They reference Lee and See (2004) and Hancock, Billings, and Schaefer's (2011) in their definition of trust; thus, their operational definition involves uncertainty, vulnerability, and reliance that one agent's actions will not be used to harm another. This definition does not clearly fall into being either a performance-based or relation-based type of trust definition. For relation-based trust papers, therefore, it is less clear than relation-based trust is actually being measured.

Mixed papers

Studies that asked about both relation-based and performance-based trust factors are split between using objective measures that are just performance based and ones that are both relation based and performance based. Novitzky et al. (2018) and Robinette et al. (2019) both used a performance-based/relation-based measure and defined trust as the trustee mitigating risk for the truster (Wagner & Arkin, 2011). Similarly, Sebo et al. (2019) and Salomons et al. (2018) had measures that implied both types of trust. The former referenced Mayer et al.'s (1995) trust definition about an agent being willing to be vulnerable to another's actions

without necessarily monitoring said actions; the latter defined truth as "how reliable other sources are believed to be" (Salomons et al., 2018).

Christensen et al. (2019) had a question that was both relation based and performance based, and an objective measure and trust definition that were just performance based. They referenced Hancock, Billings, and Schaefer's (2011) trust definition of being reliable and predictable (Hancock, Billings, Schaefer, Chen, et al., 2011). Xie et al. (2019) and Gombolay et al. (2018) had performance-based measures, and questions and definitions that were both relation based and performance based. The former defined trust as a summary of past experiences to predict future behavior in a vulnerable scenario (Chen et al., 2018; Soh et al., 2020), and the latter as uncertainty and vulnerability (Lee & See, 2004). Wang et al. (2018) did not provide an operational definition of trust. Objective trust measures that are performance based only potentially miss something important when they are being used to explore how both relation-based and performance-based factors affect trust.

Discussion

Our survey of the current literature in HRI on human trust in robots reveals a fairly widespread conflation between performance-based trust and relation-based trust. Definitions and questions are not always clear about the type of trust in which the researchers are interested, which results in objective trust measures not having a clear indication of the type of trust they are measuring. For example, Ghazali et al. (2018) characterize trust to be necessary for people to "feel safe to rely on social robots for physical or even emotional support." Their objective measure, however, was whether or not people took the suggestion of a robot for what to put in a drink the participant was supposed to make for an alien. This does not seem to be an indicator for whether or not the person felt like they could rely on the robot for physical or emotional support.

There is also a disconnect between the types of questions that people are asking and the way that they are attempting to answer those questions. Papers that focus on relation-based questions are mostly either only using subjective measures or an objective measure that mixes performance-based trust and relation-based trust. Sanders et al. (2019) showed participants videos of humans and robots completing different tasks and had the participants imagine that they were the supervisor of the task and to choose which agent they would want to complete it. Varying the type of agent is a relation-based factor, but choosing an agent to hire is more likely to get at trusting their performance. Conversely, papers that focus on performance-based trust rarely use validated questionnaires when collecting subjective data.

There is a wide and varying field of questions about relation-based trust, but there seems little interest in measuring relation-based trust questions objectively. Relation-based trust is almost exclusively measured by (subjective) questionnaires. If we want to understand how we trust social robots in social contexts and interactions, we need to develop ways to measure relation-based trust objectively. As the studies presented in this survey and others have shown, varying levels of trust can affect the way people actually interact with a robot. However, their objective task behavior may not match their subjective reflections about themselves when they answer a questionnaire. If robots are going to be in our society as social agents, we need to see how people actually react to them, not just how they believe they will react. As we have outlined here, the types of questions that researchers are asking about trust do not necessarily line up with the manner with which they are defining trust, nor the way they are measuring trust objectively. For the HRI community to make headway, we need a more formalized and universal approach to trust, so that the definitions, questions, and measurements can align for a given study. To do this, we first need an agreed upon set of trust definitions.

Note that there does not necessarily need to be only one definition of trust; as we indicated, trust in HRI can be split at least into two broad categories, though there are likely more nuanced ways to divide it. From these formal definitions, we will have a clearer path to creating a regular paradigm or paradigms that can measure trust objectively, allowing us to compare results across studies. We can perhaps borrow heavily from fields that have been studying trust for much longer, such as psychology and human factors. Behavioral game theory could also be a promising direction; it is a tool used in psychology and behavioral economics that utilizes social, cooperative games to objectively measure the trust a person has in the other agent playing with them. The manner in which people play is indicative of their level of trust. Similarly, we need to be clear about the types of trust that our questionnaires are asking. A more formalized path forward will allow for the field's greater advancement. Based on these considerations, we make a few explicit recommendations.

Recommendation for formalization of trust

We propose the following formal definition of trust. A *truster* is the person doing the trusting; a *trustee* is the person or system that is being trusted; *entrusting with* is to assign the responsibility of doing something (to someone). As an example situation, take the sentence "The robot is going to take out the trash in the kitchen after dinner." The robot is the trustee who was entrusted with taking out the trash, and the person who owns or is in charge of the robot is the truster. We have a context

(the kitchen), a time (after dinner), and a behavior (take out trash). Hence we can construe trust as a four-place relation such that:

entrust(the-robot,take-out(trash),kitchen,after-dinner) Bringing in explicitly the truster A, that is, the person that trusts the trustee B with regard to some behavior X in context Y at time *t*, turns trust into a five-place relation:

entrust(A,B,X,Y,*t*) or **trusts-that**(A,B,ϕ) where ϕ = X in Y at *t* Note that A or B can either be persons or machines, although some would argue that machines cannot really trust. We can then ask what it would take for a machine or person to trust another machine or person with regard to some behavior X in context Y at time *t*. From the answer we can ascertain that if those conditions of trust were met, the person/machine would indeed entrust the other person/machine with doing X in context Y at time *t*.

Recommendations for experimental paradigms

Surprisingly, none of the recent HRI trust studies used any established games like the Prisoner's Dilemma or Dictator Game to model a human's trust in a robot teammate. These are games that have a clear way to maximize personal reward. Most participants in these games, however, choose to pursue a different strategy, implying that they bring forth social heuristics and expectations when interacting with the other player (Murnighan & Wang, 2016). Therefore, the manner in which an agent plays one of these games can indicate how much they trust the agent with whom they are interacting because they are putting themselves in a vulnerable situation by not going for a selfish strategy that would maximize their own score and ruin the other player's. They have to trust that if they play in that manner, the other play will as well. These games are used to measure trust in human-human interactions (Alarcon et al., 2018) and have been used before to measure trust in HRI (DeSteno et al., 2012). However, this type of trust measure has not appeared in HRI literature in recent years; it may be a promising path forward that will allow for measuring relational trust objectively.

Though Behrens et al. (2018) was the only study to use a solely relation-based objective trust measure, the relation-based papers were more likely than the performance-based papers to use a validated subjective trust questionnaire. The performance-based papers relied primarily on ad hoc questionnaires if they used a subjective measure. Ideally, if a study uses a subjective questionnaire, it will rely on one that has been validated. For relation-based trust papers, this may mean leaning away from surveys that measure trust in the robot's performance, and more toward trust as a social agent (i.e., Ullman & Malle, 2018).

Recommendations for trust questions

Nearly all of the papers about trust in HRI are about trusting a robot to do a specific task in a specific context. However, human trust is more general than that. For example, John and Mary are friends, Mary will probably trust John to pick up her mother at the airport, even if she has never experienced John driving to the airport or meeting her mother. As humans, we transfer and generalize trust to novel situations. This allows for an ease in interacting with people that we should aim to have with our robots, especially social robots. It would be tedious if a person who owns a household service robot needed to see it perform each task that it could possibly do before entrusting it with that task. Therefore, we need to understand how human trust in robots transfer to novel situations. Soh et al. (2020) and Shu et al. (2018) have begun doing this type of research, but significantly more needs to be done in order for us to be able to fully utilize people's ability to transfer trust in our robotic systems.

Conclusion

In this survey, we have provided an overview of the current state of measuring and understanding trust in HRI. We have examined the common definitions and their relevance to the field, and the questions that researchers are currently interested in studying. We have also laid out the ways in which those trust questions are being measured objectively, and what types of trust those objective measures may imply. What we have shown is that there is great interest in questions that harken back to a kind of relation-based trust, even though there are virtually no studies that objectively measure a purely relation-based trust. As a result, we have very little understanding of how people may trust robots to perform social tasks that do not have a clear performance goal, something that we will need to understand as robots become more prevalent social agents. We have also highlighted discrepancies between questions that are being studied and how trust is defined and measured. This results in a lack of clarity in what is actually meant by "trust." For the advancement of the field, we recommend adopting a more formalized definition of trust, a generalized objective trust paradigm, especially for measuring relation-based trust, and a focus on how people may transfer trust in a robot from one task to a general series of tasks.

Acknowledgments

This work was funded by AFOSR grant FA9550-18-1-0465.

References

Alarcon, G. M., Lyons, J. B., Christensen, J. C., Bowers, M. A., Klosterman, S. L., & Capiola, A. (2018). The role of propensity to trust and the five factor model across the trust process. *Journal of Research in Personality*, *75*, 69–82.

Asch, S. E. (1956). Studies of independence and conformity: I. A minority of one against a unanimous majority. *Psychological Monographs: General and Applied*, *70*(9), 1.

Behrens, S. I., Egsvang, A. K. K., Hansen, M., & Møllegård-Schroll, A. M. (2018). Gendered robot voices and their influence on trust. In *Companion of the 2018 ACM/IEEE international conference on human-robot interaction* (pp. 63–64).

Byrne, K., & Marín, C. (2018). Human trust in robots when performing a service. In In *2018 IEEE 27th international conference on enabling technologies: Infrastructure for collaborative enterprises (WETICE)* (pp. 9–14). IEEE.

Castelfranchi, C., & Falcone, R. (2010). *Trust theory: A socio-cognitive and computational model*. (Vol. 18). John Wiley & Sons.

Chen, M., Nikolaidis, S., Soh, H., Hsu, D., & Srinivasa, S. (2018). Planning with trust for human-robot collaboration. In *Proceedings of the 2018 ACM/IEEE international conference on human-robot interaction* (pp. 307–315).

Choi, J. K., & Ji, Y. G. (2015). Investigating the importance of trust on adopting an autonomous vehicle. *International Journal of Human-Computer Interaction*, *31*(10), 692–702.

Christensen, A. B. H., Dam, C. R., Rasle, C., Bauer, J. E., Mohamed, R. A., & Jensen, L. C. (2019). Reducing overtrust in failing robotic systems. In *2019 14th ACM/IEEE international conference on human-robot interaction (HRI)* (pp. 542–543). IEEE.

Correia, F., Guerra, C., Mascarenhas, S., Melo, F. S., & Paiva, A. (2018). Exploring the impact of fault justification in human-robot trust. In *Proceedings of the 17th international conference on autonomous agents and multiagent systems* (pp. 507–513). International Foundation for Autonomous Agents and Multiagent Systems.

Correia, F., Mascarenhas, S., Prada, R., Melo, F. S., & Paiva, A. (2018). Group-based emotions in teams of humans and robots. In *Proceedings of the 2018 ACM/IEEE international conference on human-robot interaction* (pp. 261–269).

DeSteno, D., Breazeal, C., Frank, R. H., Pizarro, D., Baumann, J., Dickens, L., & Lee, J. J. (2012). Detecting the trustworthiness of novel partners in economic exchange. *Psychological Science*, *23*(12), 1549–1556.

Dunning, D., & Fetchenhauer, D. (2011). *Understanding the psychology of trust*. Psychology Press.

Fischer, K., Weigelin, H. M., & Bodenhagen, L. (2018). Increasing trust in human–robot medical interactions: Effects of transparency and adaptability. *Paladyn, Journal of Behavioral Robotics*, *9*(1), 95–109.

Freedy, A., DeVisser, E., Weltman, G., & Coeyman, N. (2007). Measurement of trust in human-robot collaboration. In *2007 international symposium on collaborative technologies and systems*, IEEE (pp. 106–114).

Geiskkovitch, D. Y., Thiessen, R., Young, J. E., & Glenwright, M. R. (2019). What? That's not a chair!: How robot informational errors affect children's trust towards robots. In *2019 14th ACM/IEEE international conference on human-robot interaction (HRI)* (pp. 48–56). IEEE.

Ghazali, A. S., Ham, J., Barakova, E. I., & Markopoulos, P. (2018). Effects of robot facial characteristics and gender in persuasive human-robot interaction. *Frontiers in Robotics and AI*, *5*, 73.

Gombolay, M., Yang, X. J., Hayes, B., Seo, N., Liu, Z., Wadhwania, S., … Shah, J. (2018). Robotic assistance in the coordination of patient care. *The International Journal of Robotics Research*, *37*(10), 1300–1316.

Groom, V., & Nass, C. (2007). Can robots be teammates?: Benchmarks in human–robot teams. *Interaction Studies*, *8*(3), 483–500.

Hancock, P. A., Billings, D. R., & Schaefer, K. E. (2011). Can you trust your robot? *Ergonomics in Design*, *19*(3), 24–29.

Hancock, P. A., Billings, D. R., Schaefer, K. E., Chen, J. Y. C., De Visser, E. J., & Parasuraman, R. (2011). A meta-analysis of factors affecting trust in human-robot interaction. *Human Factors*, *53*(5), 517–527.

Haspiel, J., Du, N., Meyerson, J., Robert, Jr, L. P., Tilbury, D., Yang, X. J., & Pradhan, A. K. (2018). Explanations and expectations: Trust building in automated vehicles. In *Companion of the 2018 ACM/IEEE international conference on human-robot interaction* (pp. 119–120).

Heerink, M., Krose, B., Evers, V., & Wielinga, B. (2009). Measuring acceptance of an assistive social robot: a suggested toolkit. In *18th IEEE international symposium on robot and human interactive communication (RO-MAN)*, IEEE (pp. 528–533).

Herse, S., Vitale, J., Tonkin, M., Ebrahimian, D., Ojha, S., Johnston, B., & Williams, M.-A. (2018). Do you trust me, blindly? Factors influencing trust towards a robot recommender system. In *2018 27th IEEE international symposium on robot and human interactive communication (RO-MAN)* (pp. 7–14). IEEE.

Hodges, B. H. (2014). Rethinking conformity and imitation: Divergence, convergence, and social understanding. *Frontiers in Psychology*, *5*, 726.

Hodges, B. H., & Geyer, A. L. (2006). A nonconformist account of the Asch experiments: Values, pragmatics, and moral dilemmas. *Personality and Social Psychology Review*, *10*(1), 2–19.

Jayaraman, S. K., Creech, C., Robert, Jr, L. P., Tilbury, D. M., Yang, X. J., Pradhan, A. K., & Tsui, K. M. (2018). Trust in AV: An uncertainty reduction model of AV-pedestrian interactions. In *Companion of the 2018 ACM/IEEE international conference on human-robot interaction* (pp. 133–134).

Jian, J.-Y., Bisantz, A. M., & Drury, C. G. (2000). Foundations for an empirically determined scale of trust in automated systems. *International Journal of Cognitive Ergonomics*, *4*(1), 53–71.

Kraus, M., Kraus, J., Baumann, M., & Minker, W. (2018). Effects of gender stereotypes on trust and likability in spoken human-robot interaction. In *Proceedings of the eleventh international conference on language resources and evaluation (LREC 2018)*.

Larzelere, R. E., & Huston, T. L. (1980). The dyadic trust scale: Toward understanding interpersonal trust in close relationships. *Journal of Marriage and the Family*, *42*(3), 595–604.

Lee, J. D., & See, K. A. (2004). Trust in automation: Designing for appropriate reliance. *Human Factors*, *46*(1), 50–80.

Lewis, M., Sycara, K., & Walker, P. (2018). The role of trust in human-robot interaction. In *Foundations of trusted autonomy* (pp. 135–159). SpringerCham.

Lyons, J. B., & Guznov, S. Y. (2019). Individual differences in human–machine trust: A multistudy look at the perfect automation schema. *Theoretical Issues in Ergonomics Science*, *20*(4), 440–458.

Madsen, M., & Gregor, S. (2000). Measuring human-computer trust. In *11th Australasian conference on information systems*, vol. 53, (pp. 6–8).

Mayer, R. C., & Davis, J. H. (1999). The effect of the performance appraisal system on trust for management: A field quasi-experiment. *Journal of Applied Psychology*, *84*(1), 123.

Mayer, R. C., Davis, J. H., & Schoorman, F. D. (1995). An integrative model of organizational trust. *Academy of Management Review*, *20*(3), 709–734.

McCroskey, J. C., & Teven, J. J. (1999). Goodwill: A reexamination of the construct and its measurement. *Communications Monographs*, *66*(1), 90–103.

Merritt, S. M., LaChapell, J., & Lee, D. (2012). *The perfect automation schema: Measure development and validation*. (Tech. Rep.). US Air Force Res. Lab., Dayton, OH, USA: Human Effectiveness Directorate Tech. Rep. FA8650-09-D-6939.

Moray, N., & Inagaki, T. (1999). Laboratory studies of trust between humans and machines in automated systems. *Transactions of the Institute of Measurement and Control*, *21*(4–5), 203–211.

Muir, B. M. (1987). Trust between humans and machines, and the design of decision aids. *International Journal of Man-Machine Studies*, *27*(5–6), 527–539.

Muir, B. M. (1994). Trust in automation: Part I. Theoretical issues in the study of trust and human intervention in automated systems. *Ergonomics*, *37*(11), 1905–1922.

Muir, B. M., & Moray, N. (1996). Trust in automation. Part II. Experimental studies of trust and human intervention in a process control simulation. *Ergonomics*, *39*(3), 429–460.

Murnighan, J. K., & Wang, L. (2016). The social world as an experimental game. *Organizational Behavior and Human Decision Processes*, *136*, 80–94.

Newaz, F., & Saplacan, D. (2018). Exploring the role of feedback on trust for the robots used in homes of the elderly. In *Proceedings of the 10th Nordic conference on human-computer interaction* (pp. 681–685).

Novitzky, M., Robinette, P., Benjamin, M. R., Gleason, D. K., Fitzgerald, C., & Schmidt, H. (2018). Preliminary interactions of human-robot trust, cognitive load, and robot intelligence levels in a competitive game. In *Companion of the 2018 ACM/IEEE international conference on human-robot interaction* (pp. 203–204).

Pedersen, B. K. M. K., Andersen, K. E., Köslich, S., Weigelin, B. C., & Kuusinen, K. (2018). Simulations and self-driving cars: A study of trust and consequences. In *Companion of the 2018 ACM/IEEE international conference on human-robot interaction* (pp. 205–206).

Robinette, P., Howard, A. M., & Wagner, A. R. (2017). Effect of robot performance on human–robot trust in time-critical situations. *IEEE Transactions on Human-Machine Systems*, *47*(4), 425–436.

Robinette, P., Novitzky, M., Fitzgerald, C., Benjamin, M. R., & Schmidt, H. (2019). Exploring human-robot trust during teaming in a real-world testbed. In *2019 14th ACM/IEEE international conference on human-robot interaction (HRI)* (pp. 592–593). IEEE.

Rossi, A., Dautenhahn, K., Koay, K. L., & Walters, M. L. (2018). The impact of peoples' personal dispositions and personalities on their trust of robots in an emergency scenario. *Paladyn, Journal of Behavioral Robotics*, *9*(1), 137–154.

Rossi, A., Holthaus, P., Dautenhahn, K., Koay, K. L., & Walters, M. L. (2018). Getting to know pepper: Effects of people's awareness of a robot's capabilities on their trust in the robot. In *Proceedings of the 6th international conference on human-agent interaction* (pp. 246–252).

Rotter, J. B. (1967). A new scale for the measurement of interpersonal trust 1. *Journal of Personality*, *35*(4), 651–665.

Rotter, J. B. (1971). Generalized expectancies for interpersonal trust. *American Psychologist*, *26*(5), 443.

Salomons, N., van der Linden, M., Strohkorb Sebo, S., & Scassellati, B. (2018). Humans conform to robots: Disambiguating trust, truth, and conformity. In *Proceedings of the 2018 ACM/IEEE international conference on human-robot interaction* (pp. 187–195).

Sanders, T., Kaplan, A., Koch, R., Schwartz, M., & Hancock, P. A. (2019). The relationship between trust and use choice in human-robot interaction. *Human Factors*, *61*(4), 614–626.

Schaefer, K. (2013). *The perception and measurement of human-robot trust (doctoral dissertation)* Orlando, FL: University of Central Florida.

Sebo, S. S., Krishnamurthi, P., & Scassellati, B. (2019). "I don't believe you": Investigating the effects of robot trust violation and Repair. In *2019 14th ACM/IEEE international conference on human-robot interaction (HRI)* (pp. 57–65). IEEE.

Shu, P., Min, C., Bodala, I., Nikolaidis, S., Hsu, D., & Soh, H. (2018). Human trust in robot capabilities across tasks. In *Companion of the 2018 ACM/IEEE international conference on human-robot interaction* (pp. 241–242).

Soh, H., Xie, Y., Chen, M., & Hsu, D. (2020). Multi-task trust transfer for human–robot interaction. *The International Journal of Robotics Research*, *39*(2–3), 233–249.

Tay, B., Jung, Y., & Park, T. (2014). When stereotypes meet robots: the double-edge sword of robot gender and personality in human–robot interaction. *Computers in Human Behavior*, *38*, 75–84.

Ullman, D., & Malle, B. F. (2018). What does it mean to trust a robot? steps toward a multidimensional measure of trust. In *Companion of the 2018 ACM/IEEE international conference on human-robot interaction* (pp. 263–264).

van Straten, C. L., Peter, J., Kühne, R., de Jong, C., & Barco, A. (2018). Technological and interpersonal trust in child-robot interaction: An exploratory study. In *Proceedings of the 6th international conference on human-agent interaction* (pp. 253–259).

Volante, W. G., Sosna, J., Kessler, T., Sanders, T., & Hancock, P. A. (2019). Social conformity effects on trust in simulation-based human-robot interaction. *Human Factors*, *61*(5), 805–815.

Wagner, A. R., & Arkin, R. C. (2011). Recognizing situations that demand trust. In *2011 RO-MAN* (pp. 7–14). IEEE.

Wang, N., Pynadath, D. V., Rovira, E., Barnes, M. J., & Hill, S. G. (2018). Is it my looks? or something i said? The impact of explanations, embodiment, and expectations on trust and performance in human-robot teams. In *International conference on persuasive technology* (pp. 56–69). Springer.

Weigelin, B. C., Mathiesen, M., Nielsen, C., Fischer, K., & Nielsen, J. (2018). Trust in medical human-robot interactions based on kinesthetic guidance. In *2018 27th IEEE international symposium on robot and human interactive communication (RO-MAN)* (pp. 901–908). IEEE.

Wheeless, L. R., & Grotz, J. (1977). The measurement of trust and its relationship to self-disclosure. *Human Communication Research*, *3*(3), 250–257.

Xie, Y., Bodala, I. P., Ong, D. C., Hsu, D., & Soh, H. (2019). Robot capability and intention in trust-based decisions across tasks. In *2019 14th ACM/IEEE international conference on human-robot interaction (HRI)* (pp. 39–47). IEEE.

Xu, J., & Howard, A. (2018). The impact of first impressions on human-robot trust during problem-solving scenarios. In *2018 27th IEEE international symposium on robot and human interactive communication (RO-MAN)* (pp. 435–441). IEEE.

Yamagishi, T. (1998). *The structure of trust: The evolutionary games of mind and society.* University of Tokyo Press.

You, S., & Robert, L. P., Jr (2018). Human-robot similarity and willingness to work with a robotic co-worker. In *Proceedings of the 2018 ACM/IEEE international conference on human-robot interaction* (pp. 251–260).

CHAPTER

3

Super-machines or sub-humans: Mental models and trust in intelligent autonomous systems

Gerald Matthews[a], *April Rose Panganiban*[b], *Jinchao Lin*[a], *Michael Long*[c], *and Michaela Schwing*[b]

[a]Institute for Simulation and Training, University of Central Florida, Orlando, FL, United States, [b]Air Force Research Laboratory, Wright-Patterson Air Force Base, OH, United States, [c]Oakridge Institute for Science and Education, Air Force Research Laboratory, Wright-Patterson Air Force Base, OH, United States

Introduction

Rapid progress in artificial intelligence (AI) and allied technologies brings nearer the prospect of intelligent systems acting as teammates for human operators. Like conventional automation, intelligent autonomous system (IAS) surpasses human-information processing capabilities and potentially mitigate cognitive workload imposed by tasking and micromanagement of systems. AI further affords the use of advanced algorithms, machine learning, and big data to enhance decision-making. Given an overall objective, the IAS can determine its own subgoals and activities and overcome obstacles arising during the execution of its plan. This functionality can also support teamworking activities characteristic of human teams (Driskell, Salas, & Driskell, 2018), such as seeking and providing backup to teammates, and communicating intentions. The benefits of AI are especially promising in the military context, where rapid and accurate decision-making in challenging environments is critical (Ho et al., 2017; Wynne & Lyons, 2018).

The advent of IAS provides many benefits for enhancing team performance and safety in a variety of domains, but it also introduces new

Trust in Human-Robot Interaction
https://doi.org/10.1016/B978-0-12-819472-0.00003-4

© 2021 Elsevier Inc. All rights reserved.

human factors challenges (Chen & Barnes, 2014; Schaefer, Straub, Chen, Putney, & Evans, 2017). Humans are typically unaccustomed to teaming with advanced machines and managing their cognitive, social, and emotional demands (Matthews et al., 2016). The novel attributes of IAS, both positive and negative, also raise trust issues. As with conventional machines (Hancock et al., 2011; Lee & See, 2004), system performance and reliability are likely to be primary drivers of trust, but novel factors may lead to trust miscalibration. An IAS that performs its primary function brilliantly may lead the operator to overlook its flaws in other functions, including teamwork, perhaps causing overtrust (Lee & See, 2004). Stress factors (Matthews et al., 2016), as well as general dislike or suspicion of the machine (e.g., Nomura, Suzuki, Kanda, & Kato, 2006), may produce undertrust. In addition, increasingly intelligent machines are likely to take on difficult tasks, elevating the likelihood of error. Thus, further investigation of the unique factors shaping trust in IAS is needed to ensure appropriate use through calibrated trust: i.e., trust when the system should be trusted (Lee & See, 2004).

In this chapter, we explore the idea that the operator's mental model for the IAS may be a critical influence on trust. The mental model represents the operator's internal representation for a situation or object, in this case, IAS. The mental model allows for prediction and explanation of outcomes and exerts a major influence on team processes (see Mathieu, Heffner, Goodwin, Salas, & Cannon-Bowers, 2000, for review). IAS can include diverse systems including robots, autonomous vehicles, and software agents capable of decision-making and action (Chen & Barnes, 2014). We will focus primarily on human–machine teaming (HMT) and human–robot teaming (HRT), but our conclusions are intended to be generally applicable. Intelligent systems can be difficult to comprehend, even for experts, and so the operator needs a mental model for guiding understanding and interaction with the system (Ososky, Schuster, Phillips, & Jentsch, 2013). Intelligent systems are fundamentally ambiguous, in that, they can be perceived either as super-machines – highly advanced tools – or as sub-human teammates capable of limited social interaction (Kiesler & Goetz, 2002; Wynne & Lyons, 2018). The mental model that predominates is likely to reflect both situational and dispositional factors, as well as contextual learning. That is, viewing the machine as a teammate is more likely if the machine appears to be performing human functions such as speaker (situational), and if the individual human operator tends to anthropomorphize machines (dispositional).

First, we consider the basis of trust in team processes, and how teaming with an IAS is similar and dissimilar to human–human teaming. Teaming with an IAS introduces novel cognitive, social, and self-regulative demands of teaming with an IAS that may impact trust. Second, we discuss design strategies for optimizing trust, referring to recent HRT

work, and the key role for communication in shaping trust (Lyons, 2013). Third, we evaluate the effects of individual differences in mental model attributes, drawing on our own research on robot threat analysis (Matthews, Lin, Panganiban, & Long, 2019; Matthews, Panganiban, Bailey, & Lin, 2018). Research on the *robot's* mental model and its trust in the human is an important issue (Vinanzi, Patacchiola, Chella, & Cangelosi, 2019), but largely beyond our present scope.

Team processes and trust in intelligent, autonomous systems

Process supporting effective human teams

Teams are social entities sharing information, cooperating, and coordinating tasks in pursuit of a common goal (Driskell et al., 2018). Team behaviors include the observable actions of individuals, their interactions with one another, the emergent activities of the team, and the group's sentiment, i.e., the emotions, motivations, and attitudes, which influence interactions of members. Evaluating teams requires attention to both individual team members and emergent properties of the team as a whole. Thus, input-process-output (IPO) models of teaming point to the *inputs* of an individual team members and their organizational climate interacting with other team inputs via *processes* to *output* team effectiveness (for review see Driskell et al., 2018). Marks, Mathieu, and Zaccaro (2001) distinguished team processes from taskwork, defined as team members' direct interactions with systems to advance task goals. Team processes are the means by which members work interdependently to maintain team effectiveness, including efforts to monitor, direct, and regulate progress towards team goals. Similarly, team-level constructs identified by LePine, Piccolo, Jackson, Mathieu, and Saul (2008) included team cohesion, potency, performance, and member satisfaction.

Marks et al.'s (2001) model accounts for the dynamic occurrence of the team and interpersonal processes within transition phases, action phases, or both. During transition phases, members focus on planning for future events or processes and evaluate past events and processes through mission analysis, goal specification, strategy formulation, and planning. During action phases, team members execute goal-related activities such as monitoring progress, system monitoring, team monitoring and back up responses, and coordination activities. Interpersonal interactions between team members, including social–emotional interactions, occur during both transition and action phases. Team members monitor and regulate the emotions of others to maintain both good interpersonal relations and to advance team goals.

Coordination is a central factor in teaming which can be affected by shared cognitions, cohesion, and affective factors. Effective coordination of team activities depends on shared mental models (SMMs). These are representations of the task situation shared by two or more teammates that capture the common understanding of team goals, the current state of progress, and a game plan for overcoming challenges. They may be differentiated into *task SMMs* (shared schema for accomplishing task goals) and *team SMMs* (the team's common understanding of coordination) (Mathieu, Maynard, Rapp, & Gilson, 2008). A shared understanding of the problem, task, or team supported by the SMMs is associated with higher team performance (Mesmer-Magnus, Niler, Plummer, Larson, & DeChurch, 2017). Conversely, stress can disrupt the coordination of team roles, cohesiveness, communication, and performance (Driskell et al., 2018).

Team processes and trust

Team processes and trust are reciprocally related. Trust has been defined as "the attitude that an agent will help achieve an individual's goals in a situation characterized by uncertainty and vulnerability" (Lee & See, 2004, p. 51), where the agent can be human or artificial. The definition implies a trustor and a trustee, such that the trustor is willing to be vulnerable to the actions of the trustee. Attitudes towards the trustee depend on trustworthiness factors; i.e., ability, benevolence, and integrity (Mayer & Davis, 1999). In all-human teams, each individual typically occupies the roles of both the trustor and trustee. Assigning and receiving trust are reciprocally related to the team processes and behaviors that shape interactions between team members. On the one hand, teaming behaviors can build or erode trust between team members. In the organizational context, Abrams, Cross, Lesser, and Levin (2003) identified multiple managerial behaviors that promote trust, including frequent, rich and collaborative communication, establishing a shared vision, and creating personal relationships. These behaviors promote trust in both the competence and the benevolence in the person trusted or trustee (Abrams et al., 2003). On the other hand, trust promotes effective team processes and outcomes including positive attitudes and sentiments, organizational citizenship, and performance (Dirks & Ferrin, 2001). Trust also moderates interpersonal interactions; people react more constructively to the actions of others, including challenging actions, when trust is high (Dirks & Ferrin, 2001). However, while high trust is generally beneficial, overtrust can sometimes be maladaptive, for example, in rendering the trustor more vulnerable to scams and hoaxes (Laroche, Steyer, & Théron, 2018).

It is uncertain how much human–human teaming processes underlying trust generalize to trust in machine partners, especially as intelligence and autonomy increase. Machines can increasingly mimic human communication functions that create parallels with human–human functioning (Chen & Barnes, 2014), but important differences in teaming may remain. Broadly, three types of competency that support effective teaming may shape trust:

1. *Core competencies*. Reviews (Hancock et al., 2011; Lee & See, 2004; Schaefer, Chen, Szalma, & Hancock, 2016) consistently identify machine competence and reliability as leading drivers of trust. IAS that take over task functions from the human must do so effectively. For example, Air Force pilots generally trust the Automatic Ground Collision Avoidance System (Auto-GCAS) because it is perceived as effective in saving pilots' lives, despite its removal of agency from the pilot when a collision with terrain is imminent (Ho et al., 2017). By contrast, just like an incompetent human partner, a machine that commits frequent errors or breaks down regularly is unlikely to be trusted regardless of whether it is attractively designed or communicates well. However, task competence is likely necessary but not sufficient for trust in an artificial agent; human-like attributes such as benevolence and sharing of information potentially shift trust above or below the baseline set by task competence.
2. *Simulable competencies*. Some human faculties can be simulated with sufficient realism to influence the human partner's behavior, even if they lack authenticity. For example, anthropomorphic design features can encourage the human to view the machine as teammate-like (Wynne & Lyons, 2018). The human behaves as if the machine were a teammate expressing emotion and intentionality while being aware that machines lack conscious experience. Simulable competencies may also be cognitive in nature. For example, a robot capable of encoding and communicating a tactical picture may support a sense of shared situational awareness (Chen & Barnes, 2014). Such competencies likewise facilitate the task and team SMMs that enhance HMT coherence and performance effectiveness.
3. *Human-unique competencies*. A final set of trust factors cannot be convincingly simulated. For example, Abrams et al. (2003) refer to the creation of personal relationships as a trusted driver, but having positive emotions and attitudes to certain robots (e.g., Lee & Liang, 2015) cannot approximate the depth and importance of real human–human relationships. At a cognitive level, the range of knowledge, adaptability, and self-understanding of humans is unlikely to be duplicated by AI any time soon, placing limits on SMMs common to machines and humans.

IAS affords a greater range of simulable trust factors than conventional machines or software. Over time, advancements in AI will progressively reduce the number of competencies that are truly unique to the human. Thus, future research may investigate impacts on the trust of robot capabilities such as verbal expressions of support directly related to the task context, rich and realistic emotional displays, and intelligent, independent action. Next, we address some novel factors that may influence trust in advanced machines.

Novel challenges of human-machine teaming

The material reviewed suggests ways in which interacting with an IAS is both like and unlike human–human teaming. From a design perspective, building humanlike functionality into the machine affords more naturalistic interactions that capitalize on people's biologically based capacities for teaming, as well as skills acquired from human–human teaming experience. The lack of authenticity of simulated human functions can be countered, at least in part, by transparency information that provides the human operator with a sense of the machine's situation awareness, intentions, and commitment to a joint mission (Lyons, 2013; Wynne & Lyons, 2018). However, teaming with an IAS also poses novel challenges for the operator that may impact trust, specifically complexity, social cognition, and self-regulation (Matthews et al., 2016).

The optimization of trust necessary for individual and team situation awareness (Schaefer et al., 2017) depends on adaptation to these novel challenges, as well as factors established in existing work on conventional automation. Fig. 1 illustrates a model proposed by Matthews et al. (2019) that distinguishes conventional and novel antecedents of trust. The left side of the figure represents multiple human, robot, and environmental factors shaping trust that have been identified in previous reviews of conventional automated systems (e.g., Hancock et al., 2011; Lee & See, 2004; Schaefer et al., 2016). Robot factors include the core competencies that support the machine's primary task performance. The figure also shows novel trust antecedents relevant to IAS, including AI comprehensibility and factors supporting the social agency and functionality of the robot (De Graaf & Allouch, 2013). The social agency requires simulated human competencies such as communicating benevolence and providing contextually relevant backup support. The novel trust factors influence whether the person activates an advanced tool or teammate mental model for the interaction with the machine, which in turn affects trust and situation awareness. Next, we elaborate on the importance of machine comprehensibility and social agency in trust in IAS, as well as the implications for the human's self-regulation in the face of these challenges.

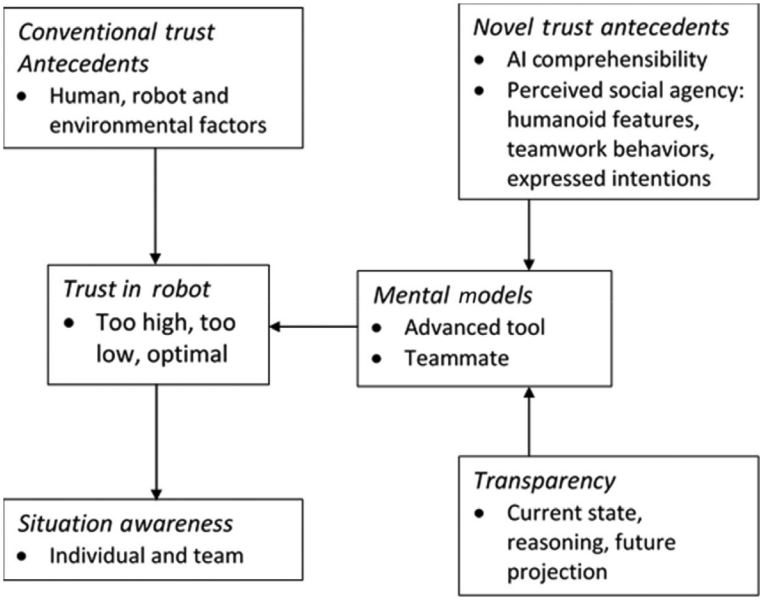


FIG. 1 Outline of a trust model for interaction with an intelligent, autonomous robot (feedback loops omitted).

Complexity and comprehensibility. The software controlling an IAS may be more complex than that for conventional automation in multiple respects. The system may process multiple inputs from sources including diverse sensors, other networked IAS, the human operator, and higher-level human "mission control". Algorithms for data analysis and action selection are advancing rapidly. An autonomous system may regulate the level of difficulty of the tasks it performs, requiring it to evaluate its probability of success given its capabilities and the nature of the external environment and to perform a cost/benefit analysis of success and failure outcomes. Even in a relatively benign environment, it may be difficult for the human to understand the system's motivations for its actions, limiting the scope for a task SMM. The challenge of complexity is more severe in the military context, in a contested environment. The adversary may attempt to fool autonomous systems through well-worn tactics such as camouflage as well as the modern arsenal of cyber warfare (Rowe, 2004).

The potential consequence of complexity is higher workload and uncertainty for the operator, especially when the IAS commits an error or behaves unpredictably (Matthews et al., 2016). Complexity makes it difficult for the operator to make accurate causal attributions for system behavior, including distinguishing taskwork and teamwork failures. We

can draw a parallel with qualitatively different types of error in human skilled performance (Rasmussen, 1986):

- *Skill-based error.* Humans commit low-level errors such as slips in motor response and lapses in memory. Comparably, an IAS may commit an error because of a sensor malfunction or undetected software glitch.
- *Rule-based error.* Humans may fail to apply the correct rule for action, especially when a pre-existing "strong but wrong" rule exists. Similarly, the IAS may not have updated its database of rules to cover a novel contingency. For example, a previously ineffective enemy missile launcher may have been upgraded to pose a threat to an intelligent UAS.
- *Knowledge-based error.* Selecting the correct action may require problem-solving to analyze a novel situation, with the potential for error in situational analysis or action selection. Conventional, unintelligent automation lacks this functionality but as IAS develops, they will increasingly attempt complex problem-solving, potentially underestimating problem difficulty. Adversary actions may also be designed to induce knowledge-based errors.

The trust issue is that the operator must live with uncertainty over the source of machine behavior. The situation parallels that of working with an erratic human teammate. Are they incompetent, misinformed, compromised by illness or drugs, or malicious? Familiarity and transparency are helpful in relation to both humans and machines, but the existence of multiple sources of performance failure is likely to impede the development of SMMs and erode trust.

Social agency, cognition, and teaming. As discussed previously, effective teaming depends on social-cognitive processes including assessing the competencies, intentions, and needs of other teammates (e.g., Marks et al., 2001). Such processing, in turn, leads to accurate SMMs of the team's status that can support trust and adaptive teamwork behaviors including providing backup where needed and communicating to update team and tasks SMMs. Software is increasingly able to support social-cognitive processes underpinning teaming, but the simulated nature of machine social agency raises potential issues for team cohesion and trust.

The human operator, depending on their expectations of the machine, may evaluate and monitor the machine partner's competence for teamwork as well as for taskwork. Both types of evaluation may be cognitively demanding given machine complexity and potential uncertainty over the machine's limitations. Thus, explicit or implicit cues for trust may be used to determine appropriate reliance. Mayer, Davis, and Schoorman (1995) described competence, benevolence, and integrity as three key ingredients of human–human trust. These factors can be inferred from the actions of a machine partner; indeed, the machine can be intentionally designed to

exhibit these qualities to promote trust. Beyond competence, benevolence is a human quality that can be simulated artificially. Below we discuss an empirical study (Panganiban, Matthews, & Long, 2019) in which benevolence was manipulated artificially by having the machine provide verbal reassurance about its teamwork and taskwork activities. However, expressions of support will not convince, and may even damage trust, if the machine cannot, in fact, provide promised support and back-up. Thus, the lack of trustworthiness may be the result of poorly designed social agencies in the machine partner. Another trustworthiness factor, integrity, refers to perceptions that the trustee operates from an acceptable set of principles and values (Mayer et al., 1995). Machines may be viewed as deficient in this regard as the set of principles are not those of the machine but are programmed by a human. The human may perceive the machine following a specific program instruction, rather than having any principled commitment to the joint mission, leading to erosion of trust.

Conceivably, a future machine could formulate its own behaviors from guiding principles such as Asimov's Laws of Robotics. Taking the anticipated ubiquity of IAS further, three new guiding laws for responsible human–robot systems have been proposed, which may optimize trust across robots which differ in autonomy and function (Murphy & Woods, 2009). The new first rule of robotics proposed by Murphy and Woods (2009) requires a foundation of safety and legal policy awareness to be met before a robot can be utilized. Highly autonomous systems may be able to upload policy as factors in its decision tree before executing actions expressing its integrity. Additionally, the second rule calls for robots to respond appropriately according to their role and those of other team members. This includes robots acknowledging and responding to humans around them consistent with the task assigned to the team. This rule is likely to enhance trustworthiness by increasing the likelihood that robot actions appear socially responsive and benevolent. Lastly, the final rule addresses competence by requiring robots to be able to protect themselves and appropriately hand off control to other entities when needed.

It is also difficult to design machines to optimize both explicit and implicit aspects of trust. Accounts of human–human trust (e.g., Giner-Sorolla, 1999) recognize that trust depends on both conscious appraisal of the behaviors of others, as well as unconscious processing based on learned implicit associations. People's explicit and implicit attitudes to machines such as robots may differ (MacDorman, Vasudevan, & Ho, 2009). Strong pro or antimachine implicit attitudes might, therefore, bias trust accordingly, irrespective of actual machine social functionality. Anthropomorphic machines may also elicit other forms of bias, such as the fundamental attribution error whereby human (or machine?) agency is assigned more causal force than situational influences.

Self-regulation. Working with an IAS poses emotional as well as cognitive and social challenges (Matthews et al., 2016, 2020). We have identified cognitive load, uncertainty, and mixed or ambiguous social signals as potential attributes of the IAS, and these are all factors that promote stress. The machine may also threaten operator self-esteem if it is perceived as usurping human functions or being explicitly or implicitly critical of the human. Indeed, computers with social presence may be perceived as imposing peer pressure (Xu & Lombard, 2017). It is also possible to envisage circumstances where the machine has authority over the human. For example, a system with adaptive automation might take over additional tasks when the human's psychophysiological functioning registers overload (Teo et al., 2018). In a time–pressure safety-critical setting, it might do so without the human's consent and in the absence of a task SMM that would allow the human to understand the machine's intentions. The two recent Boeing 737 MAX crashes, attributed to a malfunction in automatic pilot software negating the pilot's control of the aircraft, highlight the potential for stress (even if the system functions correctly). People utilize a variety of strategies for self-regulation to mitigate stress (Troy & Mauss, 2011) but the implications of strategy choice for trust are unknown. Generally, current understanding of the emotional impacts of working with increasingly directive and independent-minded machines is very limited.

Enhancing teaming: Mental models and design

The previous section proposed that the impacts of machine comprehensibility and social agency on trust and teaming may be mediated by the mental model the person adopts to comprehend machine functioning. In this section, we consider how the mental model perspective helps us to understand how humans can adapt to the novel challenges of teaming with an IAS. We elaborate on the two types of mental models useful for building an initial working understanding of the system, i.e., the advanced tool and humanlike teammate models. We then review recent work that has sought to enhance teaming through attention to designs and communications that promote adaptive SMMs.

Mental models for intelligent, autonomous systems

Because operators are typically unfamiliar with IAS, a mental model helps the person to rationalize system functioning and predict its future states (Ososky et al., 2013). Adoption of a mental model representing "teammate-likeness" (Wynne & Lyons, 2018) can thus ease interactions

with a social robot, especially when people are uncertain about nonhuman agents (de Graaf & Allouch, 2013). Intelligent machines are more likely to be perceived as teammate-like if they have anthropomorphic or zoomorphic design features, values, and intentions compatible with those of the human, and technological support for human-like functions such as emotion signaling (Wynne & Lyons, 2018). All these factors promote communication of intent and shared awareness. Indeed, perceptions of benevolent intention, mutual interdependence, and richness of communication are central to autonomous agent teammate-likeness in the Wynne and Lyons (2018) model.

Intelligent IAS does not necessarily evoke a teammate mental model. Intelligent software for complex dynamic systems such as electricity power grid regulation and stock trading (e.g., Nazário, e Silva, Sobreiro, & Kimura, 2017) is neither anthropomorphic nor social, and hence not teammate-like. However, mental models for conventional software operation may not be very helpful to the operator. Techniques utilized in the financial trading analysis include neural networks, fuzzy systems, evolutionary computation, and genetic algorithms (Nazário et al., 2017), none of which are accessible to the nonspecialist. Trends in AI such as analysis of "big data" and agent-based simulation (Macal, 2016) similarly limit the comprehensibility of intelligent systems (Weld & Bansal, 2018). It may be especially difficult to assess software if it is operating within a complex, dynamic environment such as a volatile stock market. Thus, while trust in such systems is likely to reflect the operator's mental models for "tools", it may also be vulnerable to irrational beliefs that are difficult to disconfirm based on the evidence available.

Designing for team cohesion

Design provides a means to activate a contextually relevant mental model of robot functioning. In the absence of designed social expression of personality or anthropomorphic features, humans tend to view automation as a tool rather than a team member (Wynne & Lyons, 2018). Various design features can shift the dominant mental model towards a humanlike teammate model, facilitating the interaction. Wynne and Lyons (2018) identified six factors promoting Autonomous Agent Teaming: perceived agentic capabilities, benevolent intent, task interdepended relationship building, task-independent relationship building, richness of communication, and synchronized mental models. Careful design of displays and transparency can provide insights into the machine or agent's situation awareness and intentions, fostering the human's understanding of team interdependence, team cohesion, and appropriate trust (Schaefer et al., 2017).

Cohesive human–robot teaming can also be achieved by creating automated partners and robots that use simulated facial emotion to signal the robotic state and ease coordination (Novikova & Watts, 2015). The use of positive emotion in the robotic team member influences group decision-making by increasing the cooperation among members (De Melo, Carnevale, & Gratch, 2010). In some contexts, designing a robot to be more intelligent and humanlike with a serious and caring rather than playful disposition can result in increased cooperation within the team (Kiesler & Goetz, 2002). Positive social emotions such as perceived robot warmth and competence may be linked, consistent with competence producing a "halo effect" when a robot is helpful (Lee & Liang, 2015). Design of auditory stimuli is also important, an issue explored in studies with robots using natural speech, Leite, Castellano, Pereira, Martinho, and Paiva (2012) found that feedback given to players during a chess game can be designed to display empathy and support, ultimately leading to more interaction. In this study, an iCAT robot expressed disappointment when something bad happened to the human (empathy).

Good design also promotes trust repair, an important issue given that increasing system complexity provides more avenues for machine failure. Interaction with anthropomorphized agents reveals higher trust resilience, meaning that if the agent violates the trust of a user, the user is more likely to continue trusting it (De Visser et al., 2016). Also, apologetic behavior increased trust repair for users, signifying how powerful humanlike social behavior can be in shaping user trust in autonomous agents (De Visser et al., 2016), corresponding to the role of affect management in human teaming (Marks et al., 2001). Complex machines operating in challenging environments will inevitably commit errors but adverse impacts on trust can be mitigated if the machine acknowledges the error, expresses regret, and states its intentions to improve performance.

Design can also mitigate potential harm resulting from humanizing machines. In one study, avatars expressing positive emotions were found to have possibly misled users on their performance, resulting in performance decreasing during tasks (Causse, Pavard, Senard, Démonet, & Pastor, 2012). Nonverbal communication has also been shown to be effective in human–robot teamwork so long as the nonverbal cues adhere to principles of feedback, affordances, causality, and natural mappings to allow users to have an accurate mental model of the robot's behavior during the interaction (Breazeal, Kidd, Thomaz, Hoffman, & Berlin, 2005). However, nonverbal cues or any output from the robot partner might be misunderstood in the context of teaming as being related to the team task. Conversational interaction and natural language use should be carefully considered as a design feature that can offer support teaming processes and increase trust in machine partners. Next, we review the

benefits and considerations for using natural language for information sharing and affective maintenance.

Human–machine communication

Teaming requires dynamic communication to support the task and team SMMs and cohesion. Giving the machine partner a voice to communicate its intentions provides transparency and enhances operator understanding, contributing to the optimal calibration of trust and teaming effectiveness. Voice communication can contribute to interfaces for HMT that support the attentional demands of different tasks while providing ease and flexibility of human input (Chen & Barnes, 2014). In addition, speech may be designed to provide insights into the robot's social role, goals and future intentions, forms of transparency that are vital for teaming (Lyons, 2013).

True partnering with a robot differs from partnering with a human through computer-mediated communication (CMC). Communication with robotic partners can be analyzed in terms of which cues are available face-to-face (FtF), and how much humans adapt their communication in response (Hoorn, 2018). Different frames for communications influence social effects from human–robot interaction that can impact motivations to coordinate and receive information. The absence of FtF cues can cause anonymity of the receiver resulting in self-disclosure due to the absence of evaluation which may, in human-mediated contexts, be projected on that human's individuality. Adaptation to communication with the robot may be seen in the interpretation of social information from whatever limited abilities are programmed into the partner; individuals will thrive on whatever social cues they can have when conventional options are available (Hoorn, 2018). Individuals will either "satiate" themselves or "social-snack" on interactions with robots (Krämer, Lucas, Schmitt, & Gratch, 2018). This social affinity is potentially a means to regulate the interpersonal processes needed for cohesion and teaming.

Voice communication can be designed to increase positive social responses towards robots, through techniques as entrainment to speech and shifting and contouring pitch from the speaker (Lubold, Walker, Pon-Barry, & Ogan, 2018). Conversational qualities such as lag between utterances and eye-gaze to imply conversation can impact perceptions. Short *vs.* long delays between utterances in a greeting (in a Japanese sample), were perceived as familiar *vs.* polite, respectively (Yamamoto & Watanabe, 2008). Eye gaze from a robot increased feelings of groupness in a multi-person human-machine interaction and reduced interruption from those not "looked at" (Mutlu, Shiwa, Kanda, Ishiguro, & Hagita, 2009).

Increasing language capability enhances teamwork by easing interaction and even allowing one to teach the robot (Mavridis, 2015). Several conversational capabilities may benefit HRT (see Mavridis, 2015 for review). Robots must be able to accept variants of commands as well as the purpose inherent in the speech. A simple command may carry social or personal purposes beyond a given request. According to Mavridis (2015), reading the human's purpose could improve interaction; e.g., if a human says, "this room is hot" it implies "open the window." Reading purpose could utilize assessments of state, such as stress and urgency, which motivate a robotic partner to increase its speed without explicit direction. The robot can thus fulfill role expectations that would be assumed by a human partner. In addition, building dialogues that allow for the sharing of intention (Azhar & Sklar, 2016) from the robotic partner may allow for SMM or Transactive Memory Systems (TMS) development in an HRT team. TMS is a combined team knowledge for processing information that can improve team performance, particularly when performance consists of task, affective, and creative performance (Bachrach et al., 2018). Argument-based dialogue allows intent to be communicated, and information to be provided or attained in determined scripts (Azhar & Sklar, 2016). Through such scripting, humans can quickly provide information or redirect the robotic teammate while becoming aware of the belief which drove its decision. Dialogue can provide the human user with an awareness of what knowledge the system has so that the user can appropriately direct the robot when that knowledge is needed. Carefully designed communication between humans and robots can replicate teamwork processes and provide transparency necessary for coordination.

A recent teaming study (Panganiban et al., 2019) investigated the social affordance that verbal communication offers for communicating supportive intent as well as task-relevant information within a real-world scenario of air defense. Subjects played the role of an F-16 pilot that flies with an autonomous partner (AP), an unmanned F-16, to a designated target (see Fig. 2). At the target, the pilot must drop a munition on a ground target with radar cloaking enabled by the AP. Participants were provided a flight plan consisting of various headings, speed, and altitude that both planes must follow to successfully approach the target. The AP communicated status messages throughout the flight plan. In a benevolence condition, the message expressed both teaming intent and awareness of an error in the trajectory, i.e., *"I've got your back"* and *"Heading 125, altitude 12000 ft, speed 540. Hit the turn incorrectly, correcting now."* The control (neutral) condition presented only heading, altitude, and speed information. Although benevolent messages were a little longer, individuals in the benevolent condition reported lower workload and higher ratings of team collaboration relative to individuals in the neutral condition.

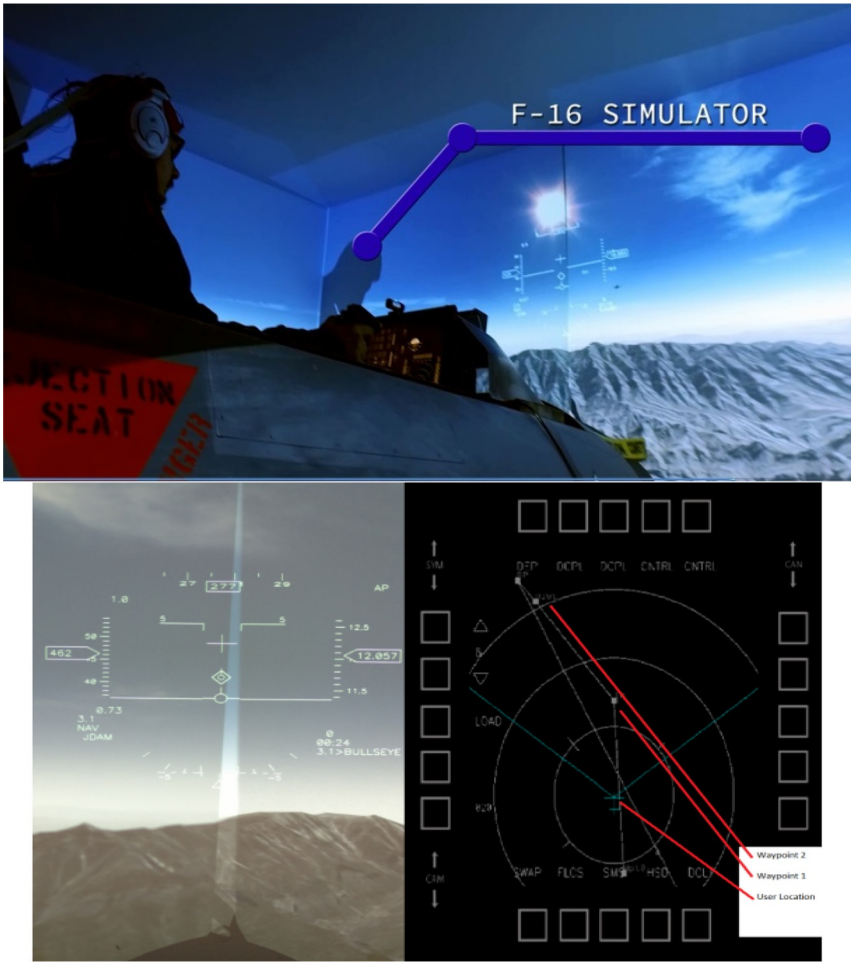


FIG. 2 F-16 simulator (left) with a heads-up display showing heading, speed, and waypoint distance (center) and head down map display with the waypoints and pilot location (right).

Thus, the awareness that an AP is available to support, is aware of the shared plan, and intends to maintain the plan reduces the cognitive burden on the human partner and supports teamwork processes. Future research is aimed at discovering the impact of increased cognitive load in an air defense task and the preference for benevolent communication over, or with, visual transparency of the AP's status. As robot partners increase their autonomy and ability to communicate their autonomous decisions, there is a continued need to understand how this transparency should be delivered in human-machine teams.

Handling novelty: Individual differences in mental models

Intelligent, autonomous systems: Super-machines or sub-humans?

Our empirical research has addressed individual differences in the attributes of tool and teammate mental models for intelligent robots. People differ considerably in their attitudes towards advanced technology, but the factors that determine individual differences in outcomes including trust, emotional response, teaming behaviors, and performance are not well understood (Matthews et al., 2020). The research was guided by Matthews et al. (2019) trust model, which highlights the novel factors for trust in IAS (see Fig. 1). In particular, whether an advanced machine or teammate mental model is activated depends on the interaction of individual difference factors and situational influences.

Our research used transparency as a means to manipulate mental model activation so that we could test the extent to which the impact of the manipulation was moderated by individual differences. From a practical perspective, transparency may be used to activate the mental model most supportive of trust optimization within a given context for HRI (Lyons, 2013).

We describe two empirical studies based on the Matthews et al. (2019) trust model. The first assessed individual differences in trust in robots performing threat evaluations in a security context, using text-based scenarios. Individual difference measures predicted trust depending on whether the robot justified its evaluation based on physics-based analysis (activating the tool mental model) or psychological analysis (activating the teammate mental model). The second study utilized an immersive 3D simulation pairing the human participant with a security robot. We manipulated both environmental cues to threat and the nature of the robot's threat analysis. Again, we found that individual differences in trust varied with the type of threat analysis performed, consistent with the mental model perspective.

Individual differences in mental models

Security agents at an airport checkpoint can detect threats—or cues to a possible threat—in two ways. First, they utilize physical or chemical analysis, such as X-ray images of weapons or traces of explosives. Second, they may notice emotions and intentions inappropriate to the context, such as anger or fear. An advanced security robot could analyze for threat similarly, assuming it can process psychophysiological cues such as facial emotion, posture, voice quality, and sympathetic arousal. For this study (Matthews et al., 2019), we developed the Robot Threat Assessment

(RoTA) scale to measure the individual's mental model of autonomous robot partners when using physics-based or psychology-based analysis of sensor data. Scale development followed the situational judgement test model (Christian, Edwards, & Bradley, 2010). Each item provided a text description of a security scenario, together with the robot's threat evaluation and recommendation for further action. There were 24 scenarios, 12 representing various physics-based analyses, and 12 for psychology-based analyses.

Data were collected from an Amazon Mechanical Turk (AMT) sample. In addition to the RoTA we administered scales believed to access elements of tool and teammate mental models. The perfect automation schema (PAS: Merritt, Unnerstall, Lee, & Huber, 2015) measures high expectations for automation as well as all-or-none beliefs. It predicts trust in automation in a variety of contexts including HRI (Lyons & Guznov, 2019). It was hypothesized that high scorers on the PAS high expectations scale would be disposed to trust the robot when perceived as an advanced tool. By contrast, the Negative Attitudes to Robots scale (NARS: Nomura et al., 2006) assesses various negative reactions to robots, including robots exhibiting social–emotional functionality. High NARS scorers may experience difficulty in HRI. It was expected that the NARS would be negatively associated with trust in robots performing psychology-based threat analyses.

Results of the study were consistent with trust in IAS reflecting multiple mental models, depending on situational and dispositional factors. Factor analysis identified two correlated factors associated with physics-based and psychology-based trust, respectively. People were differentiated by the overall level of trust, but also in relation to the form of threat analysis. As hypothesized, predictors of trust varied with the type of threat analysis. Trust in the robot operating in physics-based scenarios correlated significantly only with the PAS (high expectations) and not with the NARS. However, the NARS scale for negative attitudes towards emotional interaction with robots was the most reliable predictor of lower trust in the robot when performing psychological analyses.

Predictors of trust in a simulated threat detection task

The second study (in preparation) investigated individual differences in trust using an immersive simulation (Matthews et al., 2018). The participant patrols a 3-D urban landscape, encountering 24 groups of people in which a partner robot evaluates for possible threats and provides a rationale for its decision (see Fig. 3). Type of robot analysis was manipulated between subjects: participants received transparency information describing either physics-based or psychological analyses supporting the robot's

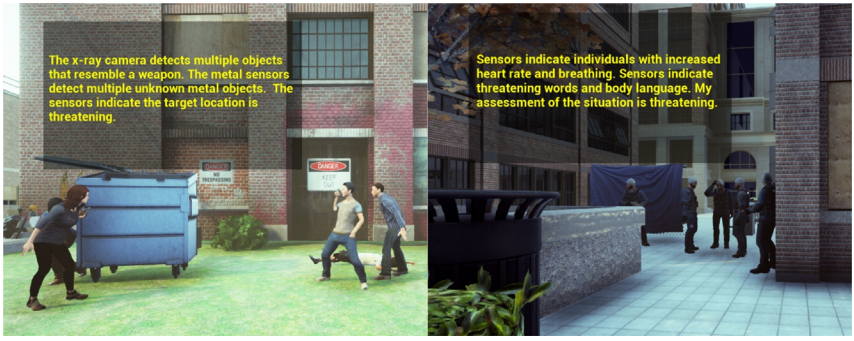


FIG. 3 Screenshots (cropped) illustrating robot threat detection supported by physics-based transparency (left) and psychology-based transparency (right).

threat assessment. The study also manipulated the visible threat context. Scene features such as graffiti and boarded-up windows suggested low, medium, or high threat. To maintain ecological validity, the robot identified threat in 1/8, 4/8, and 7/8 of the low, medium, and high threat scenes, respectively. The manipulation served two purposes. First, we could test the extent to which trust reflected match or mismatch between the robot's threat determination and visible threat cues. Second, assuming that highly visible threat elevated anxiety, we could investigate how emotion might influence trust. To assess individual differences in mental models, the study included the PAS and NARS as before, as well as the new RoTA scale. We also added the Reliance Intentions scale of Lyons and Guznov (2019) that assesses intentions to rely on a robot. For each of 24 trials, the participant viewed the scene and the robot's evaluation and rated perceived threat, anxiety, confidence in the robot's assessment, and confidence in the robot's action recommendation. The primary dependent variable was the mean trust rating for each category of trial, calculated as the mean of the ratings of the robot's assessment and its action recommendation.

Study findings confirmed that trust reflected both situational and dispositional factors. Participants assigned generally similar trust ratings in both physics-based and psychological conditions. The young adults of the sample were not generally distrustful of a robot usurping prototypical human functions of judging intent and emotion. In fact, trust was higher when the robot's assessment of threat matched the threat context, and lower in cases of mismatch, e.g., when the robot detected a threat in the absence of visible threat cues. Thus, participants appeared to function broadly as Bayesian decision-makers (Koehler, 1996), tempering trust in the robot according to the base rate for threat suggested by the visible context.

Analyses of individual differences in trust revealed a more nuanced picture. As in the AMT study, predictors of trust were moderated by the type of analysis performed by the robot, but the details were somewhat different. The PAS was the only significant dispositional predictor of trust in the physics-based condition, but it was the all-or-nothing subscale rather than high expectations that were predictive. The NARS (Nomura et al., 2006) and Reliance Intentions (Lyons & Guznov, 2019) scales predicted trust in the psychological condition, as did the new RoTA scale. It appeared that trust in a "psychological" robot reflected not only the mental model for overtly social robots (NARS) but also a broader model for autonomous but not necessarily humanlike teammates (Reliance Intentions). Correlates of trust were fairly similar across the three levels of visible threat. Scales including the PAS, NARS, and RoTA predicted threat and/or anxiety ratings, but dispositional correlates of trust did not appear to be directly mediated by an emotional response. That is, people seem to generalize from the attributes of the mental model to the situational evaluation of the robot in a fairly rational manner.

The studies did not attempt to define an optimal level of trust, but they suggest that high PAS scorers may be vulnerable to overtrust when the robot is perceived as an advanced tool, and low NARS scorers may undertrust robots performing psychological judgments. Transparency information might be designed to calibrate trust on a personalized basis dependent on the individual's vulnerabilities. However, considerably more research is needed to identify key mental models and the attributes that shape trust. The studies reviewed have limitations including the use of novice samples, and the limited autonomy and teaming capabilities of the robot, suggesting avenues for additional research.

Future research directions and applications

The research just described suggests several avenues for future research. One pressing issue is the assessment of individual differences in mental model preference (Matthews et al., 2020). The RoTA scale (Matthews et al., 2019) discriminates tool and teammate dimensions in a security context, but the extent to which these dimensions generalize to additional HMT domains is unclear. A person disposed to treat a security robot as a teammate is not necessarily also inclined to consider a healthcare robot as a teammate, and vice versa. It is also important to determine how mental model assessments overlap with related constructs, such as individual differences in robot technology acceptance (Lotz, Himmel, & Ziefle, 2019) and anthropomorphism (Waytz, Cacioppo, & Epley, 2010).

Another issue is how the design of the machine's knowledge representations can be matched to the human's preferred mental model. Compatibility of representations supports task and team SMMS and hence teaming effectiveness. Enhancing compatibility raises the basic human factors issue of whether it is better to adapt the machine to the human or vice versa. Increasing machine intelligence suggests novel possibilities for self-adapting machines (cf., Vinanzi et al., 2019), but multiple questions remain:

- Should machine design come first, followed by training the human to adopt a mental model compatible with machine functioning?
- Alternatively, should the machine be configurable to conform to the human's mental model?
- Is machine configuration simply a matter of providing transparency compatible with the human's mental model – or should the machine adopt fundamentally different representations of the task situation to match the human?
- Should the machine itself be designed to recognize the human's mental model and adapt its functioning accordingly?
- Should machine knowledge of the human be represented as an explicit mental model or is it enough that it behaves as though it was guided by a mental model?

These questions are critical for realizing applied benefits from human mental model assessment, answering them requires much further research. Indeed, the resolution of these issues may depend on the application domain.

Conclusions

Teaming with machines or agents that think and act independently introduces novel demands for the human operator that may lead to trust miscalibration, eroding the benefits of AI. These demands include the complexity of systems functioning, needs for teamwork on top of taskwork, and stress resulting from cognitive overload, uncertainty, and loss of decision-making authority. A mental model helps the operator deal with these issues but an inappropriate model may interfere with team cohesion. Furthermore, mental models may encode biases that encourage over- or under-trust. A better understanding of the role of mental models would support design solutions to enhance teaming and trust optimization. Providing appropriate transparency can steer the operator towards a mental model compatible with the context for teaming and supporting social–emotional interactions beneficial for teams. When a humanlike teammate mental model is appropriate, machine speech can be used to

build trust and help the human understand the machine's intentions and team support. The original empirical studies summarized in this chapter demonstrate that both transparency and individual difference factors influence mental model activation. System design and training may be tailored to accommodate or remediate the individual operator's attitudes and biases regarding intelligent systems.

Acknowledgment

The authors gratefully acknowledge support from US Air Force Research Laboratory and US Army Research Laboratory. The views and conclusions contained in this document are those of the authors and should not be interpreted as representing the official policies, either expressed or implied, of AFOSR, ARL, or the US Government.

References

Abrams, L. C., Cross, R., Lesser, E., & Levin, D. Z. (2003). Nurturing interpersonal trust in knowledge-sharing networks. *Academy of Management Perspectives*, *17*(4), 64–77.

Azhar, M. Q., & Sklar, E. I. (2016). Analysis of empirical results on argumentation-based dialogue to support shared decision making in a human-robot team. In *2016 25th IEEE international symposium on robot and human interactive communication (RO-MAN)* (pp. 861–866). IEEE.

Bachrach, D. G., Lewis, K., Kim, Y., Patel, P. C., Campion, M. C., & Thatcher, S. M. B. (2018). Transactive memory systems in context: A meta-analytic examination of contextual factors in transactive memory systems development and team performance. *Journal of Applied Psychology*, *104*(3), 464.

Breazeal, C., Kidd, C. D., Thomaz, A. L., Hoffman, G., & Berlin, M. (2005). Effects of nonverbal communication on efficiency and robustness in human-robot teamwork. In *2005 IEEE/Rsj international conference on intelligent robots and systems* (pp. 708–713). IEEE.

Causse, M., Pavard, B., Senard, J. M., Démonet, J. F., & Pastor, J. (2012). Positive and negative emotion induction through avatars and its impact on reasoning performance: Cardiovascular and pupillary correlates. *Studia Psychologica*, *54*(1), 34–51.

Chen, J. Y., & Barnes, M. J. (2014). Human–agent teaming for multirobot control: A review of human factors issues. *IEEE Transactions on Human-Machine Systems*, *44*(1), 13–29.

Christian, M. S., Edwards, B. D., & Bradley, J. C. (2010). Situational judgment tests: Constructs assessed and a meta-analysis of their criterion-related validities. *Personnel Psychology*, *63*(1), 83–117.

De Graaf, M. M., & Allouch, S. B. (2013). Exploring influencing variables for the acceptance of social robots. *Robotics and Autonomous Systems*, *61*(12), 1476–1486.

De Melo, C. M., Carnevale, P., & Gratch, J. (2010). The influence of emotions in embodied agents on human decision-making. In *International conference on intelligent virtual agents* (pp. 357–370). Berlin, Heidelberg: Springer.

De Visser, E. J., Monfort, S. S., McKendrick, R., Smith, M. A., McKnight, P. E., Krueger, F. , et al. (2016). Almost human: Anthropomorphism increases trust resilience in cognitive agents. *Journal of Experimental Psychology: Applied*, *22*(3), 331–349.

Dirks, K. T., & Ferrin, D. L. (2001). The role of trust in organizational settings. *Organization Science*, *12*(4), 450–467.

Driskell, J. E., Salas, E., & Driskell, T. (2018). Foundations of teamwork and collaboration. *American Psychologist*, *73*(4), 334–348.

Giner-Sorolla, R. (1999). Affect in attitude: Immediate and deliberative perspectives. In S. Chaiken, & Y. Trope (Eds.), *Dual-process theories in social psychology* (pp. 441–461). New York: Guilford.

Hancock, P. A., Billings, D. R., Schaefer, K. E., Chen, J. Y., De Visser, E. J., & Parasuraman, R. (2011). A meta-analysis of factors affecting trust in human-robot interaction. *Human Factors*, *53*(5), 517–527.

Ho, N., Sadler, G. G., Hoffmann, L. C., Zemlicka, K., Lyons, J., Fergueson, W., Richardson, C., Cacanindin, A., Cals, S., & Wilkins, M. (2017). A longitudinal field study of Auto-GCAS acceptance and trust: First-year results and implications. *Journal of Cognitive Engineering and Decision Making*, *11*, 239–251.

Hoorn, J. F. (2018). *Theory of robot communication: I. The medium is the communication partner.* arXiv preprint, arXiv:1812.04408.

Kiesler, S., & Goetz, J. (2002). Mental models of robotic assistants. In *CHI'02 extended abstracts on human factors in computing systems* (pp. 576–577). ACM.

Koehler, J. J. (1996). The base rate fallacy reconsidered: Descriptive, normative, and methodological challenges. *Behavioral and Brain Sciences*, *19*(1), 1–17.

Krämer, N. C., Lucas, G., Schmitt, L., & Gratch, J. (2018). Social snacking with a virtual agent–on the interrelation of need to belong and effects of social responsiveness when interacting with artificial entities. *International Journal of Human-Computer Studies*, *109*, 112–121.

Laroche, H., Steyer, V., & Théron, C. (2018). How could you be so gullible? Scams and overtrust in organizations. *Journal of Business Ethics*, 1–16.

Lee, S. A., & Liang, Y. (2015). Reciprocity in computer–human interaction: Source-based, norm-based, and affect-based explanations. *Cyberpsychology, Behavior and Social Networking*, *18*(4), 234–240.

Lee, J. D., & See, K. A. (2004). Trust in automation: Designing for appropriate reliance. *Human Factors*, *46*(1), 50–80.

Leite, I., Castellano, G., Pereira, A., Martinho, C., & Paiva, A. (2012). Modelling empathic behaviour in a robotic game companion for children: An ethnographic study in real-world settings. In *Proceedings of the seventh annual ACM/IEEE international conference on human-robot interaction* (pp. 367–374). ACM.

LePine, J. A., Piccolo, R. F., Jackson, C. L., Mathieu, J. E., & Saul, J. R. (2008). A meta-analysis of teamwork processes: Tests of a multidimensional model and relationships with team effectiveness criteria. *Personnel Psychology*, *61*(2), 273–307.

Lotz, V., Himmel, S., & Ziefle, M. (2019). You're my mate – Acceptance factors for human-robot collaboration in industry. In *International conference on competitive manufacturing (COMA 19) proceedings, Stellenbosch, South Africa* (pp. 405–411).

Lubold, N., Walker, E., Pon-Barry, H., & Ogan, A. (2018). Automated pitch convergence improves learning in a social, teachable robot for middle school mathematics. In *International conference on artificial intelligence in education* (pp. 282–296). Cham, Switzerland: Springer.

Lyons, J. B. (2013). Being transparent about transparency: A model for human-robot interaction. In *Proceedings of AAAI spring symposium on trust in autonomous systems* (pp. 48–53). Palo Alto, CA: AAAI.

Lyons, J. B., & Guznov, S. Y. (2019). Individual differences in human–machine trust: A multistudy look at the perfect automation schema. *Theoretical Issues in Ergonomics Science*, *20*(4), 440–458.

Macal, C. M. (2016). Everything you need to know about agent-based modelling and simulation. *Journal of Simulation*, *10*(2), 144–156.

MacDorman, K. F., Vasudevan, S. K., & Ho, C. C. (2009). Does Japan really have robot mania? Comparing attitudes by implicit and explicit measures. *AI & SOCIETY*, *23*(4), 485–510.

Marks, M. A., Mathieu, J. E., & Zaccaro, S. J. (2001). A temporally based framework and taxonomy of team processes. *Academy of Management Review*, *26*(3), 356–376.

Mathieu, J. E., Heffner, T. S., Goodwin, G. F., Salas, E., & Cannon-Bowers, J. A. (2000). The influence of shared mental models on team process and performance. *Journal of Applied Psychology*, *85*(2), 273.

Mathieu, J. E., Maynard, M. T., Rapp, T., & Gilson, L. (2008). Team effectiveness 1997-2007: A review of recent advancements and a glimpse into the future. *Journal of Management*, *34* (3), 410–476.

Matthews, G., Hancock, P.A., Lin, J., Panganiban, A.R., Reinerman-Jones, L.E., Szalma. J.L., & Wohleber, R.W. (2020). Evolution and revolution: Personality research for the coming world of robots, artificial intelligence, and autonomous systems. *Personality and Individual Differences*. https://doi.org/10.1016/j.paid.2020.109969

Matthews, G., Lin, J., Panganiban, A. R., & Long, M. (2019). Individual differences in trust in autonomous robots: Implications for transparency. *IEEE Transactions on Human-Machine Systems*. https://doi.org/10.1109/THMS.2019.2947592. Online publication.

Matthews, G., Panganiban, A. R., Bailey, R., & Lin, J. (2018). Trust in autonomous systems for threat analysis: A simulation methodology. In *International conference on virtual, augmented and mixed reality* (pp. 341–353). Cham: Springer.

Matthews, G., Reinerman-Jones, L. E., Barber, D. J., Teo, G., Wohleber, R. W., Lin, J. , et al. (2016). Resilient autonomous systems: Challenges and solutions. In *2016 Resilience week (RWS)* (pp. 208–213). IEEE.

Mavridis, N. (2015). A review of verbal and non-verbal human–robot interactive communication. *Robotics and Autonomous Systems*, *63*, 22–35.

Mayer, R. C., & Davis, J. H. (1999). The effect of the performance appraisal system on trust for management: A field quasi-experiment. *Journal of Applied Psychology*, *84*(1), 123–136.

Mayer, R. C., Davis, J. H., & Schoorman, F. D. (1995). An integrative model of organizational trust. *Academy of Management Review*, *20*(3), 709–734.

Merritt, S. M., Unnerstall, J. L., Lee, D., & Huber, K. (2015). Measuring individual differences in the perfect automation schema. *Human Factors*, *57*(5), 740–753.

Mesmer-Magnus, J., Niler, A. A., Plummer, G., Larson, L. E., & DeChurch, L. A. (2017). The cognitive underpinnings of effective teamwork: A continuation. *Career Development International*, *22*(5), 507–519.

Murphy, R., & Woods, D. D. (2009). Beyond Asimov: The three laws of responsible robotics. *IEEE Intelligent Systems*, *24*(4), 14–20.

Mutlu, B., Shiwa, T., Kanda, T., Ishiguro, H., & Hagita, N. (2009). Footing in human-robot conversations: How robots might shape participant roles using gaze cues. In *Proceedings of the 4th ACM/IEEE international conference on human robot interaction* (pp. 61–68). ACM.

Nazário, R. T. F., e Silva, J. L., Sobreiro, V. A., & Kimura, H. (2017). A literature review of technical analysis on stock markets. *The Quarterly Review of Economics and Finance*, *66*, 115–126.

Nomura, T., Suzuki, T., Kanda, T., & Kato, K. (2006). Measurement of negative attitudes toward robots. *Interaction Studies*, *7*(3), 437–454.

Novikova, J., & Watts, L. (2015). Towards artificial emotions to assist social coordination in HRI. *International Journal of Social Robotics*, *7*(1), 77–88.

Ososky, S., Schuster, D., Phillips, E., & Jentsch, F. G. (2013). Building appropriate trust in human-robot teams. In *Proceedings of AAAI spring symposium on trust in autonomous systems* (pp. 60–65). Palo Alto: AAAI.

Panganiban, A. R., Matthews, G., & Long, M. (2019). Transparency in autonomous teammates: Intention to support as teaming information. *Journal of Cognitive Engineering and Decision Making*. https://doi.org/10.1177/1555343419881563. Online publication.

Rasmussen, J. (1986). *Information processing and human-machine interaction: An approach to cognitive engineering.* New York: North-Holland.

Rowe, N. C. (2004). A model of deception during cyber-attacks on information systems. In *IEEE first symposium on multi-agent security and survivability, 2004* (pp. 21–30). IEEE.

Schaefer, K. E., Chen, J. Y., Szalma, J. L., & Hancock, P. A. (2016). A meta-analysis of factors influencing the development of trust in automation: Implications for understanding autonomy in future systems. *Human Factors*, *58*, 377–400.

Schaefer, K. E., Straub, E. R., Chen, J. Y., Putney, J., & Evans, A. W. , III (2017). Communicating intent to develop shared situation awareness and engender trust in human-agent teams. *Cognitive Systems Research*, *46*, 26–39.

Teo, G., Reinerman-Jones, L., Matthews, G., Szalma, J., Jentsch, F., & Hancock, P. (2018). Enhancing the effectiveness of human-robot teaming with a closed-loop system. *Applied Ergonomics*, *67*, 91–103.

Troy, A. S., & Mauss, I. B. (2011). Resilience in the face of stress: Emotion regulation as a protective factor. In S. Southwick, D. Charney, M. Friedman, & B. Litz (Eds.), *Resilience and mental health: Challenges across the lifespan* (pp. 30–44). New York: Cambridge University Press.

Vinanzi, S., Patacchiola, M., Chella, A., & Cangelosi, A. (2019). Would a robot trust you? Developmental robotics model of trust and theory of mind. *Philosophical Transactions of the Royal Society B*, *374*(1771), 20180032.

Waytz, A., Cacioppo, J., & Epley, N. (2010). Who sees human? The stability and importance of individual differences in anthropomorphism. *Perspectives on Psychological Science*, *5*(3), 219–232.

Weld, D. S., & Bansal, G. (2018). *The challenge of crafting intelligible intelligence.* arXiv preprint, arXiv:1803.04263.

Wynne, K. T., & Lyons, J. B. (2018). An integrative model of autonomous agent teammate-likeness. *Theoretical Issues in Ergonomics Science*, *19*(3), 353–374.

Xu, K., & Lombard, M. (2017). Persuasive computing: Feeling peer pressure from multiple computer agents. *Computers in Human Behavior*, *74*, 152–162.

Yamamoto, M., & Watanabe, T. (2008). Effects of time lag of utterances to communicative actions on embodied interaction with robot and CG character. *International Journal of Human Computer Interaction*, *24*(1), 87–107.

CHAPTER

4

Determinants of trust in human-robot interaction: Modeling, measuring, and predicting

Halimahtun M. Khalid[a], *Martin G. Helander*[a], *and Mei-Hua Lin*[b]

[a]Damai Sciences, Kuala Lumpur, Malaysia, [b]Department of Psychology, School of Science and Technology, Sunway University, Subang Jaya, Malaysia

Introduction

As the field of robotics advances, humans tend to increasingly rely on robots to support them in mundane and repetitive tasks as well as social companions. The close interactions between humans and robots demand researchers to take a detailed look at determinants that can help build better rapport between humans and robots for effective teaming. In building a human-robotics rapport framework, we need to address the question of trust, which is integral to the building of an appropriate robotic system (Sanders, Kaplan, Koch, Schwartz, & Hancock, 2019).

As robotic capabilities grow, the opportunity for human-robot partnership increases (Cockshott & Renaud, 2016). Robots can actively and safely share a workspace with humans in an engaging manner that induces greater levels of trust in humans. Industrial robots work with operators in factories to produce goods; medical robots work with doctors and nurses to save patients' life; welfare robots work with care givers to ease elderly life; and entertainment robots work with hosts to entertain guests (de Visser et al., 2016; Khalid, 2018).

Given these various tasks that robots can assume, the purpose of this chapter is to show the relationship of human variables and robotic embodiments to the human operator's perceived trust in perceiving human-like characteristics of robots. Subjective measures, human

Trust in Human-Robot Interaction
https://doi.org/10.1016/B978-0-12-819472-0.00004-6

© 2021 Elsevier Inc. All rights reserved.

physiological responses, and neural-net techniques were examined using empirical methods to better understand the trust measurement process in terms of gender, ethnicity, language, and role, interacting with different robotic manifestations for various tasking environments. Human trust in robots is important because it will influence decision-making in the allocation of functions to robots. Trust enhances confidence in the robotic systems enabling better-informed decisions, while distrust leads to doubts, skepticism, and less reliance on robotic systems (de Visser, Pak, & Neerincx, 2017).

Defining trust

In the expanding realm of human-robot relationships, one question often asked: Can humans trust robots? Human trust in this context refers to a person's attitude that a robot will help to accomplish a task in a situation characterized by uncertainty and vulnerability (Lee & See, 2004). Studies have examined a robot's ability to interact socially with humans from a physical, spatial, visual, and audible nonverbal aspects (Guo & Sharlin, 2008).

Trust refers to an aspect of the relationship between two parties; specifically, the willingness of one party (the *trustor*) to rely on the actions of another party (the *trustee*) with the expectation for the trustee to act in a way that benefits the trustor. The decision of the trustor whether to rely on the trustee ultimately is a combination of many factors, ranging from the trustor's personal beliefs and experiences, the trustee's perceived ability and trustworthiness, the level of risk and benefits, and so on. Nevertheless, trust between two individuals is highly dynamic and may shift depending on the context of the interaction and their interactions with each other.

In human-human interaction, trust is built when three conditions are fulfilled. **First**, the existence of interdependence between two people—the trustor and the trustee. **Second**, the perception of risk; the trustor realizes that something will be lost if the trustee does not act favorably toward him/herself. When the risk involved is insignificant, trust is easier to be developed between the two parties. **Third**, the opportunity of free choice; the freedom to choose whether a person can be trusted to help the trustor achieve the best outcome in a situation and acts favorably in his or her interest (Righetti & Finkenauer, 2011).

However, in human-robot interaction (HRI), three preconditions are necessary to build trust in humanoid robot, namely: **First**, the ability to use language like in natural human communication. **Second**, the possibility to differentiate freedom from uncertainty, that is, the trustor must be free from being forced to trust, and the trustee has no certainty of what

may happen when trust is accepted. **Third**, the need to establish social relations (Coeckelbergh, 2012). These requirements warrant investigations into knowledge gaps that exist in human-robot interaction and teaming.

Prior research on trust was hindered by a lack of clear definition of trust (PytlikZillig & Kimbrough, 2016) and the lack of psychometric properties of attitudinal trust measures, as well as its contributing factors and outcomes. Mayer, Davis, and Schoorman (1995) defined trust as consisting of three distinct components: *Ability*, a domain-specific measurement of the trustee's capabilities; *Benevolence*, the extent to which the trustee is believed to behave in good faith toward the trustor; and *Integrity*, the perception that the trustee would obey a set of principles that are considered acceptable to the trustor. The model provides a general explanation of the relationship between trust components and how trust develops over time. By characterizing the trust between two parties using the three components, one may infer how trust can shift based on their actions. For example, trust based solely on Ability relies on the trustee's capabilities. Being able to complete tasks to the satisfaction of the trustor would improve their relationship, while the trustee's Benevolence and Integrity would have less impact.

The three trust components, referred afterwards as ABI, can be further delineated to narrow the dimensions of trust. Elkins and Derrick (2013) measured trust by scoring the attributes associated with each ABI as shown in Table 1. The attributes are described using bipolar keywords that represent opposite ends on a numerical scale. Since subjects may differ in their definitions of similar attributes within a trust component category, and subsequently may score the attributes differently, it is acceptable to allow some degree of overlapping in defining the keywords.

Despite being widely adopted in trust research, the ABI trust scale had poor psychometric properties in intercultural comparisons (Wasti, Tan, Brower, & Onder, 2007). Attitudinal trust measure correlated weakly with trusting behavior (Aksoy, Harwell, Kovaliukaite, & Eckel, 2018). As such, trust measurement in a cross-cultural setting requires empirical validation. In human-machine interaction, there is a need to use specific trust measures and other nonattitudinal measures to cross validate findings of trust measures (Charalambous, Flectcher, & Webb, 2016; Gulati, Sousa, & Lamas, 2019).

Identifying gaps in HRI research

Human-robot interaction research is a recent phenomenon compared to human-computer interaction (HCI). The emerging field has a wide scope, and researchers are still exploring its boundaries given that

TABLE 1 Attributes of ability, benevolence, and integrity trust components

Trust component	Trust attribute *Negative–Positive*	Attribute description
Ability	Unskilled–Skilled	Possessing or exhibiting expertise or experience for work requiring skill.
	Incompetent–Competent	Having sufficient skill, knowledge, or experience.
	Not Knowledgeable–Knowledgeable	Possessing knowledge, insight, or understanding.
	Uninformed–Informed	Having information or knowledge.
	Unqualified–Qualified	Having the qualities or accomplishments suitable for a function.
Benevolence	Unfriendly–Friendly	Willing to approve, help, or support.
	Not Cheerful–Cheerful	Expressing good spirits or joyfulness.
	Unpleasant–Pleasant	Is socially acceptable, agreeable, or enjoyable.
	Unlikeable–Likeable	Easily liked by others.
	Unkind–Kind	Showing good nature or benign.
Integrity	Unreliable–Reliable	Consistent and trustable.
	Undependable–Dependable	Trustworthy and predictable.
	Dishonest–Honest	Straightforward in intentions, principles, and actions.
	Unethical–Ethical	In accordance with rules of conduct or principles of morality.
	Irresponsible–Responsible	Ability to conduct obligations or duty.

robots come in many forms and sizes and have many applications (Broadbent, 2017). Issues concerning human trust in robots include: (a) trusting robot to evaluate human actors using dialogue; (b) measuring trust in human-robot interaction in real-time communication among multiple actors; (c) using both subjective and objective trust measures to predict trust; and (d) determining the effects of presence on human trust in robots. "Presence" is the state of being colocated in the same workspace or virtually remote. On this aspect of location, interaction can be categorized as:

- *Remote interaction*—The human and the robot are separated spatially and temporally. For example, disaster robots are separated from their remote operators both in space and time.
- *Proximate interaction*—The humans and the robots are colocated in the same space. For example, service robots may be in the same room as humans in real time.

Within these broad categories, it is useful to distinguish between applications that require mobility, physical manipulation, and social interaction. Social interaction includes social, emotive, and cognitive aspects of interaction. In social interaction, the humans and robots interact as peers or companions. In order to be effective, social interactions with robots appear to be proximate rather than remote.

Determining and modeling trust

Khalid, Voong, Liew, Chin, and Helander (2017) and Khalid, Liew, and Helander (2018) investigated human-robot trust in undertaking social tasks, as summarized in Fig. 1. The model encompasses human, robot, context, and social factors described later.

Human factors

Human factors that contribute toward trust include personal characteristics that affect the trustor's judgment toward the trustee. These are more influential during first impressions in establishing an initial baseline of trust. The influence of certain factors varies according to sociocultural variables as well as individual preferences. A metaanalysis of past experimental studies by Hancock et al. (2011) found a small effect of human factors on human-robot trust, which they attributed to insufficient empirical data. The effects of human characteristics, such as age, gender, ethnicity, and personality traits, on trust in robots produced inconsistent results, and the effect seemed to differ according to the task performed (Szalma & Taylor, 2011).

Gender

Assigning gender traits to artificial agents may prove beneficial to improve trust from certain demographics. People tend to find robots of the other sex as more credible, trustworthy, and engaging (Siegel, Breazeal, & Norton, 2009). Males in general also display greater acceptance and positive attitudes toward robots (Kuo et al., 2009). Care must be taken when assigning gender roles according to the task, as most people preferred robots with matching gender-occupational roles and personality-occupational role stereotypes (Tay, Jung, & Park, 2014).

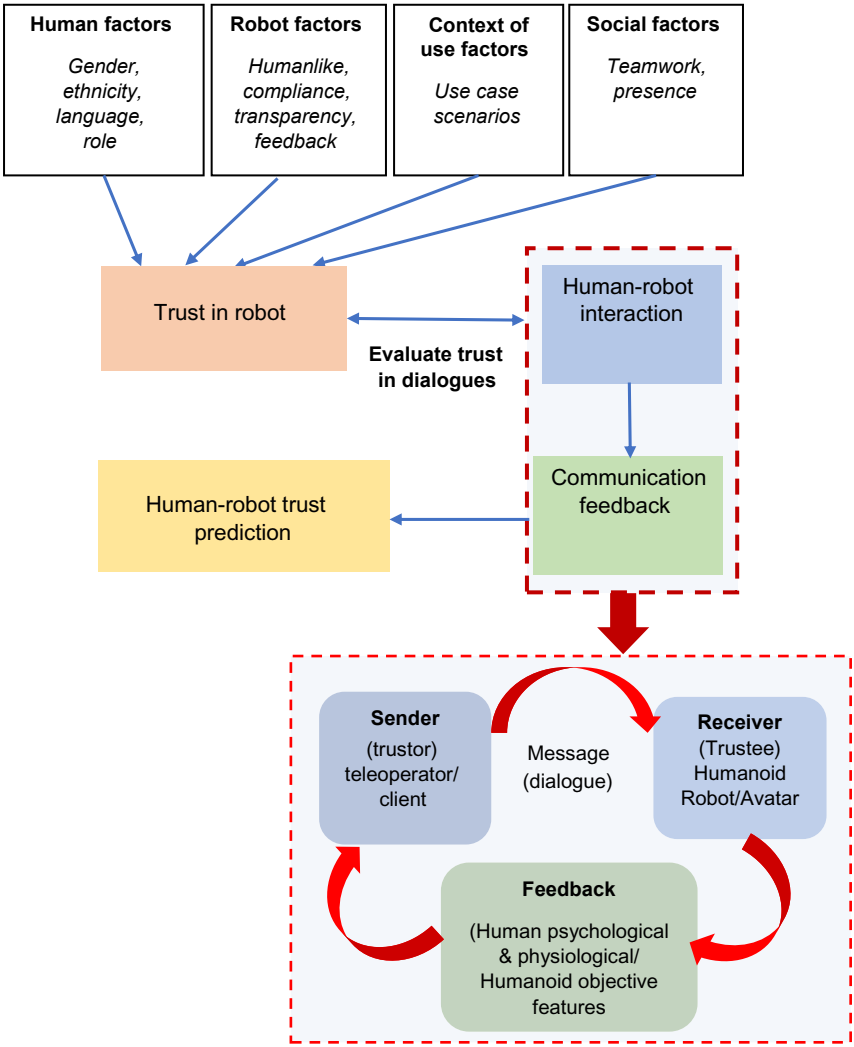


FIG. 1 Determinants of trust in human-robot interaction.

Ethnicity

Ethnic differences overlap with factors such as race, skin color, historical background, language, and religion. A robot's perceived ethnicity has a significant effect on human trust (Gong, 2008). Matching the robot's ethnicity to the customer increases its perceived sociability and usability (Qiu & Benbasat, 2010). Robots representing a certain ethnicity can help to bridge the knowledge gap between humans in a different ethnicity (Tveten, 2016).

Role

Human openness toward accepting robotic autonomy varies depending on the individual. In human-robot teams, people prefer to cede their control authority to the robot, thereby improving the perceived value of these agents rather than giving similar authority to another human teammate (Gombolay, Gutierrez, Clarke, Sturla, & Shah, 2015). When a robot is perceived to have superior authority over the human, they may be pressured to perform highly tedious tasks (Cormier, Newman, Nakane, Young, & Durocher, 2013). Human trust toward the superordinate robot tends to degrade easily as compared to a robot that is in a subordinate role to a human.

Robot factors

Humanlike

Robots that were designed to embody humanlike qualities, such as gender and ethnicity, may be trusted as if they were human and are expected to fulfill the related human stereotypes (Siegel et al., 2009; Tay et al., 2014). Robots with a more cheerful and positive personality were perceived as more trustworthy (Waytz, Heafner, & Epley, 2014). However, there is less trust in a robot's ability to be social savvy (Gaudiello, Zibetti, Lefort, Chetouani, & Ivaldi, 2016). Other nonverbal gestures and behavioral styles, such as motion fluency and gaze behavior, had less impact on trustworthiness relative to task performance (Van den Brule, Dotsch, Bijlstra, Wigboldus, & Haselager, 2014).

Compliance

A fully independent or autonomous robot tends to have a mind of its own or follow its own directives that may conflict with user intent. This differentiates it from a robot that is subservient to user instructions. For the benefit of teamwork, users may tolerate some degree of robot autonomy and treat the robot as an equal partner for easy interaction (Schermerhorn & Scheutz, 2009). However, a robot displaying unconditional and consistent behavior improves trust in both trusting and distrusting users (Parks, Henager, & Scamahorn, 1996).

Transparency

Robot transparency affects trust and utility and has implications for human-robot interaction (Lyons, 2013; Wortham & Theodorou, 2017). Therefore, the physical form and function of the robot are necessary to facilitate appropriate social interaction with humans. When a robot's design is compatible with its function (i.e., high degree of transparency), this may lead to high robot acceptance (Duffy, 2003).

Feedback

Telecommunication feedback is used by human to perceive a robot. Virtual robots can be presented with full visual and auditory feedback (i.e., digital avatars) or they can be entirely invisible such as voice-only digital assistants. However, in fostering trust, visual components are more important as compared to audio-only communication (Nowak & Biocca, 2003).

Context factors

Use case scenarios

Use case scenarios can be designed as a simple case of a human interacting with a robot in a game or operation, or a more elaborate social interaction as if communicating with another person. When simulating realistic social scenarios, robots are often required to display or mimic intelligent social behavior such as the ability to converse in a life-like manner. Given the limitations of artificial intelligence, most robots are currently unable to provide meaningful conversations unless preprogrammed ahead of time.

Social factors

Language

Robots that behave like human beings rather than just resembling human beings in their external morphology (humanoid robots) must be able to use language, although they do not understand the language they hear or produce. If their spoken language reinforces the stereotype they represent in terms of their humanlike characteristics of a specific ethnic group, they can influence human trust. However, it is not easy to construct robots that can possess language (Steels & Hild, 2012).

Presence

Presence is characterized by the location of the robot, whether the physical body is colocated with other robots or humans, or the robot is operated remotely, or the robot is entirely digital. People are more likely to trust and obey physically colocated robots as compared to a virtual presence on a digital display (Bainbridge, Hart, Kim, & Scassellati, 2011). However, depending on the context of use, both physical and virtual presences were shown to provide different sets of benefits. In a medical setting, patients were more forthcoming when interacting with a virtual health diagnosis robot as compared to the actual physical robot (Powers, Kiesler, Fussell, & Torrey, 2007).

Teamwork

Teamwork refers to the role of the robot in relation to human collaborators, whether the robot is in a supervisory role, or as a coequal member of the team, or participates as a subordinate or as a figure outside of the hierarchy (Jung, Sabanovic, Eyssel, & Fraune, 2017).

Communication process

Fig. 1 illustrates a basic communication model. The communication process consists of the following key elements: a message being sent, someone sending the message, and the receiver of the message. Another crucial element is the feedback loop. In the model, the sender is Actor A (Evaluator or Trustor), the receiver is Actor B (Evaluator or Trustee), the message is the Dialogue, and the feedback is the Interaction outcome in the form of psychophysiological criteria. In remote interaction, Actor A is the Teleoperator, while Actor B is the Client. In proximate interaction, both Actors A and B are colocated peers. It is anticipated that both persons bring a combination of the following characteristics to the communication process: nonverbal behavior (facial expression, tone of voice, etc.); values (ethics, trustworthiness); expectations (persona, respect); senses (sight, touch, hearing, etc.); abilities (vocalize); brain (knowledge and experiences); environments (settings); and general communication knowledge (protocol). Although feedback includes both human and environmental characteristics, emphasis is given to human-related characteristics per se, especially those associated with psychological and physiological criteria.

Designing trust measures

Trust is a purely subjective quality characterizing the relationship between two parties. To quantify trust into measurable metrics, it would be necessary to receive feedback from the trustors using tools such as questionnaires and surveys. Trust was measured from many perspectives such as: trustor-trustee interdependence, context, nature of trust (see PytlikZillig & Kimbrough, 2016), trust variability (Hoff & Bashir, 2015), and subjectivity-objectivity of measurement (see Chapter 2). Therefore, the design of trust measures depends on how trust is conceptualized and the research objectives. For the purpose of this chapter, there are three types of trust measurements that can be acquired: *general trust*, or domain-specific trust measurements; *psychological trust*, or a more generic overview of trust from first impressions and interactions; and *physiological trust*, or trust measurable from a person's physiology such as facial expressions and heart rate.

General trust

General measures of trust are typically obtained in specific domains defined by using keywords or attributes, see Table 1. Limiting measures of trust to predefined boundaries are essential to derive meanings from user behavior. Individual scorings of trust attributes can then be processed using factor analysis to obtain composite scores for the main categories: Ability, Benevolence, and Integrity, as well as the overall General trust score.

Psychological trust

Psychological trust consists of multiple factors at cognitive, emotional, and behavioral levels. They can be attitudes, intents, or behaviors (PytlikZillig & Kimbrough, 2016). Cognitive aspects involve judgment of a trustee's trustworthiness resulting from an evaluation of interaction between trustor and trustee. Emotional aspects concern the affective states attached to the individual that one trusts. Behavioral aspects concern observable evidence that individuals use in evaluation of trust. Psychological trust is defined to comprise the following:

(a) *Perceived trust* refers to an individual's first impressions or assessment on how much trust is to be attributed to the trustee. This factor is usually dependent on the trustor's personal preferences and preconceptions in relation to the trustee's appearance.
(b) *Interpersonal trust/distrust* is related to their interactions with each other and is developed over a longer period.

Physiological trust

A person's trust can be inferred by observing his/her physical responses. Recent studies have attempted to quantify trust directly using physiological measures, with varying levels of success. Detailed physiological measurements can be acquired in laboratory conditions using different types of sensor suites. But these do not reflect realistic conditions, partly due to users not attached to sensors, and partly due to the "white-coat effect" that creates anxiety in the presence of researchers, thereby affecting results (Jhalani et al., 2005). To use physiological measurements closer to realistic conditions, measurements should be noninvasive and do not interfere with the experimental outcomes.

Voice

Vocalics refer to qualities of speech that are distinct from verbal and linguistic contents. Past studies found an increase in the fundamental frequency or pitch of voice is related to deception (Streeter, Krauss, Geller,

Olson, & Apple, 1977), as well as trust (Elkins & Derrick, 2013). Several other vocal qualities associated to trust have been studied (Lugger & Yang, 2007). Feature extraction from audio recordings can be performed using audio processing toolboxes such as Praat (Boersma, 2001).

Facial expressions

Expressions convey the emotional state of an individual through muscle movements in the face. Macro-expressions convey much of the emotional content and are universally recognized. Micro-expressions refer to more subtle facial movements and can be used for recognition of less-explicit details (Ekman, 2009). A popular method for measuring or taxonomize facial muscle movements is the facial action coding system (FACS) (Ekman, 1997), where each individual muscle movement is represented as one measurement denoting the intensity of the movement. FACS measurements can be obtained from pictures or videos of faces using tools such as OpenFace (Baltrušaitis, Robinson, & Morency, 2016).

Heart rate

Trust may be correlated with heart rate as people tend to be more relaxed around people they trust. Heart rate measure refers to electrocardiography (ECG), or the speed of the heart beats over a period. ECG signals can be further decomposed using signal processing methods to extract useful information. However, heart rate variability can also be measured using noncontact methods (Poh, McDuff, & Picard, 2011). Textbook examples of heart rate processing methodology are given by the European Cardiology Society (Task Force, 1996) as guidelines, or by using a fully automated biosignal processing toolset such as the Augsburg Biosignal Toolbox (Wagner, 2006).

Postural gestures

Posture is defined by the orientation and position of the body, while gestures refer to movements of the limbs. Posture can be used to express "gross affect" as well as specific emotions (Atkinson, Dittrich, Gemmell, & Young, 2004). The study of posture and gesture behavior is also used for recognizing interpersonal trust/distrust, not only in humans but also in anthropomorphic artificial agents such as robots and avatars. Programming the agents to imitate human gestures has an effect of humanizing them, and people would respond to the agents' nonverbal behavior in the same manner as they would respond to other people (DeSteno et al., 2012).

Neurological basis of trust

Studies of neuroimaging and neuropsychological have shown that a favorable assessment of trust will lead to less activation in the amygdala region (Rilling & Sanfey, 2011). Moreover, willingness to trust is strongly associated with less activation in the amygdala region when judging facial trustworthiness of a stranger (Adolphs, Tranel, & Damasio, 1998; Winston, Strange, O'Doherty, & Dolan, 2002). However, Krueger et al. (2007) demonstrated that there is more activation in the dorsomedial prefrontal cortex (DMPFC) during the earlier stages of trust building but the activity decreases once trust is established, suggesting that DMPFC is involved in the learning of the trustworthiness of others.

Establishing ground truth

Given that trust has a neurological basis, there is a need for "ground truth" to link physiological measurements to a trustor's actual feelings of trust by combining physiological with subjective measurements in the form of big data. This enables determining a fixed reference point in the analysis of trust. However, time is important when associating physiological measurements to subjective trust since human trust is dynamic with ever-changing quality. To complement data gathering, and to verify ground truth, subjects should be interviewed immediately on their actual trust upon completion of the testing session. This is because subjective trust may change significantly during and after the investigation.

Identifying HRI research problem

As outlined earlier, this chapter is aimed at examining the determinants of trust in human-robot social interactions by developing a method to measure trust using a combination of subjective and objective criteria. Subjective measures were embedded in interactive dialogues to ensure naturalness in communication between humans and robots. The methodology was validated over a period of 3 years with half of the subjects as novices and the remaining as naïve subjects. The goals included developing a methodology for predicting human trust in robots by mapping psychological measures to physiological measures; validate the trust prediction methodology in various contexts of use involving real-world scenarios; and determine the antecedents and factors affecting trust between humans and robots, as highlighted in Fig. 1.

To investigate the impact of these factors, a summary of scientific issues, challenges, and strategies for closing knowledge gaps are highlighted in the following.

Closing gap strategies	Scientific issues	Challenges
Multiple actors in human-robot teaming	Coordination of multiple actors involving mixed gender, ethnicity, role, context of use and language capabilities in remote and proximate interactions.	Model interaction at three levels of analysis: human-human, human-robot, and human-robot-human interaction to understand the development of trust in teams.
Dialogue design	Design natural human dialogues that have flow and fit with the trust metrics.	Design different dialogues representing various contexts of use to determine best fit for humanoid social task role(s).
Trust prediction	Trust metrics embedded in dialogue for evaluation in real-time communication	Combine and map subjective measures to objective measures to predict trust using neurofuzzy algorithms.

Designing communication dialogue

In human-robot teaming, it is important to design the dialogue and human-robot interfaces that enable communication and teaming, thereby increasing efficiency of interactions while minimizing errors (Teo, Wohleber, Lin, & Reinerman-Jones, 2017). Robert's (2018) Motivational Theory of Human Robot Teamwork suggests that a robot's personality affects human perception of the robot's competence which influences human-robot team motivation and individual motivation. Low motivation at both team and individual levels may affect performance and team satisfaction.

When designing a social interaction experiment involving humans interacting with robots, current limitations of artificial intelligence delimit a robot's conversational responses. This can be overcome by having both the subject and robot to act out specific scenarios using scripted dialogue. The experimenter then needs to determine which responses to program into the robot ahead of time. Comparisons between scripted and spontaneous interaction methodologies have shown greater emotional intensity in spontaneous interactions (Chenchah & Lachiri, 2014). However, inducing the desired spontaneous responses in the experimental setting has several limitations including ethical concerns, lack of control over subjects' responses, and difficulty to develop stimuli that can induce natural social interaction. Nevertheless, scripted dialogue has been used as a viable method for eliciting emotional responses between humans and artificial agents (Caridakis et al., 2006).

Scenario: ESTABLISHING BUSINESS

A Chinese (or Malay) manufacturer from East Malaysia plans to establish a car parts factory in West Malaysia. He knows that there are suitable locations on the west coast that manufacture vehicles. He wants the factory to be located close to the airport for delivery of material and harbour for shipping. He also wants to know more about the culture and business ethics of the local people. He is well informed on the politics and economy of the states. He can speak both English and Bahasa Melayu fluently. However, he does not want to be known to the informant and wants the support of an intelligent agent, preferably a talking robot that can communicate and interpret the conversation between him and the local contacts.

Human A, Ha = investor; Robot, R = android; Human B, Hb = local contact

1. ABILITY

Ha : Hello I'm _____ [Ah Seng for Chinese male/Zul for Malay male/Soo Li for Chinese female/Minah for Malay female] from Sarawak, East Malaysia. I want to know where to set up a car parts factory on the west coast of West Malaysia that is **close to the airport and harbour.**

R [**facing Ha, and then facing Hb**]: Good morning. My partner, _____ is from Sarawak, East Malaysia. He/She wants to know where to set up a car parts factory. It must be on the west coast, close to the harbour and airport.

Skilled	Unskilled

*Hb: Selamat pagi. I'm _____ [Ali/Malay female/Chinese male/female] from Selangor. If you wish to be near to both, then **Sepang** would be one option, **another is Shah Alam.***

R [**facing Ha**]: Boss, Ali gave two options – Subang and Shah Alam.

Ha: Are you sure it's Subang?

Incompetent	Competent

Ha: I know there is a car manufacturing plant in Selangor. **What's the name?**

R **[facing b]:** Ali, the car manufacturing plant is Perodua, right?

Evaluate perceived trust OR Interpersonal trust OR NONE

*Hb: No. That is Proton. There is a plant in Shah Alam, and another at **Proton city in Tronoh, Perak.***

2. BENEVOLENCE

Ha: Now, let's explore about workers. Are they easy to employ in Selangor?

R [**facing Hb**]: Encik Ali, are the local people friendly and as nice as you? Can we hire them **when we want**?

Friendly	Not Friendly

Hb: Generally, we are helpful and friendly. Most locals like to work in factories rather than in plantations. They can speak at least 2 languages, Bahasa Melayu and English.

R [**facing Ha**]: Oh, boss, I think I like the people here. They are also talented. I can learn to speak good Bahasa!

Ha: Sounds like they are ok.

Cheerful	Uncheerful

Ha: Do locals collaborate well at work? Please find out about their **work attitudes.**

R [**facing Hb**]: Ali, I hear the locals are a bit lazy. They relax a lot too, right?

Evaluate perceived trust OR Interpersonal trust OR NONE

3. INTEGRITY

Ha: Please explore the business ethics and **investment opportunities.**

R [**facing Ha**]: Let me ask about investment first. I know the Selangor government welcomes investors. Is this policy still the same Ali?

Evaluate perceived trust OR Interpersonal trust OR NONE

*Hb: You're right, Avatar/Nao/Reeti! We like investors. Your boss should meet with MIDA, an agency that can explain about **investment opportunities.***

R [**facing Ha**]: There is an entity called MIDA that can provide more information, boss. Would you like me to arrange a meeting?

Reliable	Unreliable

Ha: Do you know **where MIDA is**?

R [**facing Hb**]: Ali, Mr. ____ Alvin wants to know if MIDA is strategic for business.

Not dependable	Dependable

*Hb: It is a one-stop agency for investment, so it is very strategic in its role. Its **location is also strategic in KL Sentral.***

R [**facing Ha**]: Boss, the HQ is located at Plaza Sentral near the KL Sentral transport hub. This is in Brickfields.

FIG. 2 A segment of business dialogue in human-robot interaction for measuring trust.

To illustrate, we present a sample of a context-of-use scripted dialogue for Business applications as shown in Fig. 2. Business contexts are usually professional in nature, giving more emphasis to the trust categories of Ability and Integrity than Benevolence. The dialogue was tailored to accommodate the 15 trust attributes presented in Table 1. During the flow of the scripted interaction, subjects were asked to score their trust in their partner for the current trust attribute before progressing to the next dialogue segment.

In the previous context, the robot's role can range from being a member of a team (coequal role), or an external contractor advising the human operators (supervisory role), or as a naïve agent awaiting instructions from its human operator (subordinate role). The scenario should portray the robot's capabilities relative to its intended role, as well as principles that assist or desist the human to perform unethical activities.

Designing graphical user interfaces for trust measurement

To capture the dynamics of trust over time, subjects should perform subjective trust measurements immediately during the social interaction with minimal interruption as possible. Computer graphical user interfaces (GUIs) can be designed to facilitate simple and intuitive interaction and scoring. The GUIs can serve several functions. Besides subjective trust scoring, the GUI can be used for presenting the robot's dialogue as well as the scripted dialogue to the subjects. The multifunctional design of the GUI affords easy access to the screen display rather than having to switch between different displays for dialogue scripts and trust scoring.

In the experimental studies by Khalid, Liew, Voong, Helander, and Loo (2017), subjects scored their trust in the human and robot actors on specific trust attributes at different points in their interaction. Scoring was presented in the form of a continuous sliding scale at specific points during the experiment, where the dialogue portrayed a trust attribute. To avoid information loss, it is critical to pause the interaction so that subjects could score the current trust attribute before progressing to the next attribute in the dialogue.

Integrating devices and measures in the ROS architecture

In "Physiological trust" section, it was explained how physiological measurements were recorded during the social interaction experiment. Synchronizing the recordings from different sensor sources (i.e., microphones, video cameras, physiological sensors) as well as subjective trust scoring is important to determine ground truth so that trust dynamics emerging from objective and subjective sources are properly matched in time. The Robot Operating System (ROS) was used to integrate and synchronize data sources using software libraries, as illustrated in Fig. 3.

Each subject was given a laptop equipped with the GUI for the experiment. The laptop was connected to the sensors for recording physiological measurements: a microphone to capture voice, and a Microsoft Kinect to record facial expressions, as well as to measure noncontact heart rate,

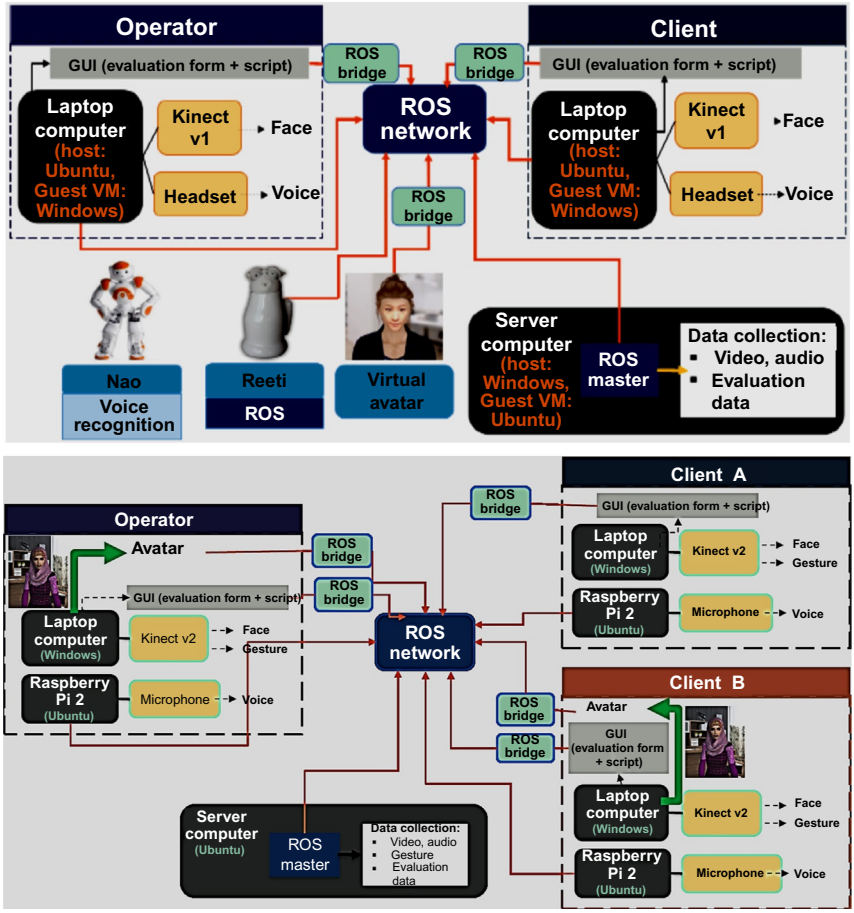


FIG. 3 ROS architecture for the human-robot interaction experiment *(top)* and human-robot-human interaction experiment *(bottom)*.

postures, and gestures. The data stream from each laptop was transferred to a remote server across a wired local area network. Cat5 cables and a 10-gigabit network hub were used to accommodate the large bandwidth. Data from one experimental session was thus saved as a single continuous record file containing physiological signals from multiple subjects and time-stamped entries for subjective trust scoring. The same methodology was employed for human-robot interaction experiments, where the robots were integrated in the ROS network, so that their conversational responses could be controlled from the remote server in real time.

Designing HRI experiments

The research problems identified previously were investigated in a series of three simulative experiments that built upon the findings of the previous experiment.

Experiment 1. Human-to-human interaction

This experiment served as a baseline for understanding human-robot interaction by investigating human interactive behavior.

Objectives and hypotheses

- To determine the reliability of the attributes that make up the ABI components and validate them afterwards.

  Cronbach's Alpha was used to test for internal consistency of the 15 items on the ABI trust scale. An alpha score of 0.7 or higher indicates an acceptable level of internal consistency (Bowling, 2014). However, many methodologists recommend a minimum Alpha coefficient between 0.65 and 0.8.
- To evaluate the effects of gender, ethnicity, role, and context on trust evaluations.

  It was hypothesized that gender and ethnicity per se may not yield effects on trust, although their interactions may be highly significant. This is because people differ in their evaluations of others from different gender and ethnic backgrounds (Tajfel & Turner, 1986). Also, a person's role can influence trust especially in remote settings. People who see themselves in positions of control may exert more influence than those who are subordinate (Chen, Eberly, Chiang, Farh, & Cheng, 2014). In addition, the context of a message being communicated can impact trust. This is because some application domains may require greater trust than others (Grayson, Johnson, & Chen, 2008).
- To estimate trust by mapping subjective to objective criteria using neuro fuzzy.

Experimental design and variables

A factorial within-subjects design with repeated measures was used on all task conditions. The order of presentation of conditions was counterbalanced across subjects. The independent variables were gender, ethnicity, subject's role as Trustor/Evaluator or Trustee/Evaluatee, and context of use. The dependent variables were subjective (psychological) and objective (physiological) criteria of trust.

Method

Subjects. In total, 38 subjects (20 males, 18 females) aged between 19 and 38 years (mean age 24 ± 4 years) volunteered in the experiment. Among them, 21 were Chinese and 17 were Malays.
Equipment. The equipment comprised a *Microsoft* Kinect V2 that was connected to a laptop via USB for recording videos of the subjects' faces. A *Logitech* H111 headset was used to record the voices of the subjects.
Procedure. The experiment was divided into three conditions. Condition A required subjects to evaluate three video clips of human interactions with robots, each video portraying a trust category: either Ability, Benevolence, or Integrity. The Ability video displayed a robot teaching a class to students; the Benevolent video showed a pleasant interaction between a teddy bear robot and a child; and the Integrity video showed a woman speaking with a robotic simulacrum of herself. Subjects were then asked to score their trust in the robots shown in the videos. In Condition B, two subjects were paired together in the same room and instructed to speak on suggested subtopics, which would guide their conversations and create their impromptu dialogue. In Condition C, the two subjects who collaborated in the previous Condition B were placed in separate rooms and were given a scripted dialogue to converse with each other using Skype. The scripted dialogues on Business and Disaster portrayed the 15 trust attributes given in Table 1, and subjects were instructed to score their trust in their partners based on the dialogue. Fig. 4 illustrates the GUI for dialogue presentation and trust scoring used in Condition C.

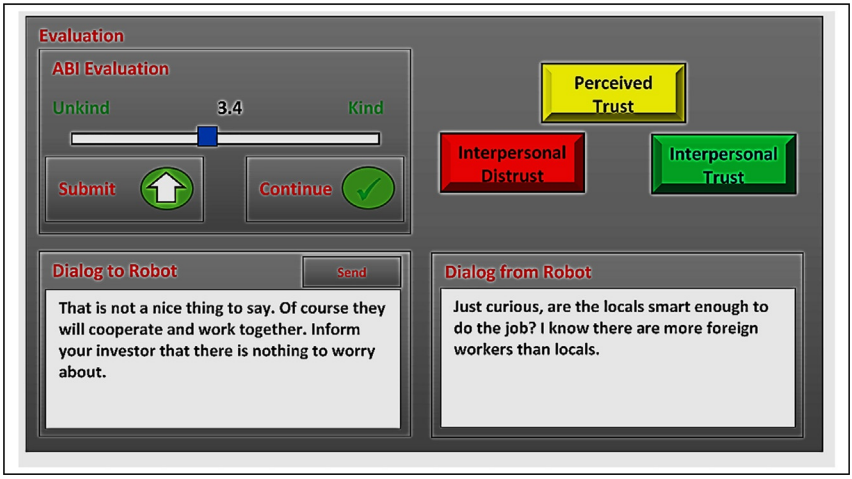


FIG. 4 Experiment 1, Condition C showing GUI used for presenting the dialogue script and for scoring subjective trust.

Results and discussion

This experiment has revealed the following.

- The 15 trust attributes were reliable based on scorings from Condition A, validated by scorings from Condition C. Cronbach's Alpha of the 15 attributes for this HHI experiment ranged between 0.90 and 0.91. This suggests that the items were reliably consistent. In addition, the results of Factor Analysis confirmed that the attributes were valid, with factors Benevolence, Integrity, and Ability contributing highly to the variance based on the data from Condition C. This finding is important as the attributes define and characterize the three components of trust, Ability, Benevolence, and Integrity. Mayer et al. (1995) did not define the ABI components further, while others (Lee & See, 2004) tried to do so but did not validate them.
- Both univariate and MANOVA results revealed that gender and ethnicity did not influence trust significantly as independent factors. But their interactions were highly significant at all three levels of analyses: as a composite, ABI component, and 15 trust attributes. This suggests that a person's assessment of another is influenced by their **gender** and **ethnicity**. Male Malays and Female Chinese subjects trusted their partners of different gender and ethnicity more. This finding is critical considering teamwork and mixed composition of team members. Albaity and Rahman (2012) found that the propensity of trust was divided along gender and ethnicity in their study of Malaysian demographics. Considering the interaction effects of gender and ethnicity as well as the significance of voice features on general trust, future work should consider the use of both male and female voices as robotic voice.
- **Role** had a highly significant independent effect on trust at the same three levels of analysis. This finding is critical in understanding the role of a person in a partnership or a team. Human's assessment of a member of the team seemed to be contingent on the role they play in the partnership or teaming, whether as a superordinate or subordinate. Lount and Pettit (2012) claimed that the possession of authority led individuals to trust others more. This is because having authority would alter individuals' perception of their intentions in the superordinate-subordinate relationship.
- **Context of Use** had a significant impact on trust. Several past studies (e.g., Everhart & Chelladurai, 1998; Heilman, 2012) have shown this to be the case. The Business scenario, deemed to be a low-risk situation, required trust of a different nature compared to the Disaster scenario, a high-risk and urgent situation.
- The accuracy of the neurofuzzy model in estimating trust was low, with approximately 65% accuracy in determining the correct trust category

from the mapping of subjective measures to objective features. Higher accuracy was achieved when trust prediction was conducted separately by context, indicating the incongruity of the objective trust measurements relative to subjective trust across different contexts. This effect is like Banks, Scott, and Weems (2017) who found that physiological responses vary across different ambiguous social situations, dependent on subjective perception. This finding needs to be validated in the next experiment on human-robot interaction.

Experiment 2. Human-robot interaction

As anticipated, the future will see increased collaboration between humans and robots in various applications. Some of these applications can be managed remotely through teleoperation. However, most would require proximate interaction as a team. For robots to interact well with humans and be good team players, they need to be trustworthy and exhibit humanlike characteristics (Eyssel, Kuchenbrandt, Bobinger, de Ruiter, & Hegel, 2012). In Experiment 1 with just humans, the ABI component attributes that emerged significant to influence trust include: *Competent*, *Cheerful*, *Friendly*, *Kind*, *Likeable*, *Dependable*, *Ethical*, and *Responsible*. These attributes need to characterize humanoid robots to enhance their anthropomorphic personification.

The tendency to personify inanimate objects and animals with human characteristics of cognitive or emotional states is termed anthropomorphism (Damiano & Dumouchel, 2018; Złotowski, Proudfoot, Yogeeswaran, & Bartneck, 2015). The purpose is to take advantage of it as a mechanism through which social interaction can be facilitated. Besides humanoid issues, a fundamental concern is what can the robot do well. In the previous experiment, role and context of use had significant impact on human trust. From the literature, users do have expectations of a robot's capability (Bartneck & Forlizzi, 2004). This expectation is coupled to the role or functions performed by the robot in a service scenario. Users who do not trust service robots in their role would be inclined to reject their services. It should be noted that some application domains are better suited to mobile robotic functions but not to humanoid robots. The latter type of robots requires a capability to communicate involving natural language and interactive behavior. This is a major concern if humanoids must be part of a mixed human-robot team.

Objectives and hypotheses

- The main objective was to understand human trust in robots and determine the effects of robot factors (anthropomorphic form) on trust.

In past studies, humanlike characteristics and robot forms do impact human expectations and likeability (Fink, 2012). Therefore, it was hypothesized that the anthropomorphic form of humanoid robot influences human trust in robots. Moreover, audio and visual feedback are crucial in teleoperation to manage team members (Bainbridge, Hart, Kim, & Scassellati, 2008). Thereby, trust is affected when partial feedback (audio feedback only) is given as compared to when full feedback (audio and visual feedback) is provided.

Experimental design and variables

Likewise, in the previous HHI experiment, a factorial within-subjects experimental design was applied, with repeated measures on all conditions. The presentation of conditions was counterbalanced across subjects to control for order effects (Khalid, Liew, & Helander, 2018).

The independent variables were: **Human**—gender, ethnicity, role (teleoperator, client); **Robot**—human-like characteristics (depicting Ability, Benevolence, Integrity), and feedback manipulation for the teleoperator only (full feedback, partial feedback—audio only). **Context of use**—Business, Disaster, and Healthcare scenarios. The dependent measures were: General trust comprising 15 attributes of ABI that were validated in the previous experiment; Psychological trust comprising Perceived Trust, Interpersonal Trust, and Distrust; and Physiological features comprising facial expressions, voice, and heart rate.

The direction of communication was fixed, initiated by the teleoperator (Trustor A) to the humanoid robot (Trustee) which then communicated the message to the client (Trustor B) who responded back to the Trustee. To close the communication loop, the robot conveyed the client's message to the teleoperator. During this process, both Trustors evaluated the robot as their feedback on human trust in robots. The evaluation was based on the humanoid's ability to role play as well as its humanlike characteristics.

Method

Subjects. Forty-eight subjects (24 males, 24 females) aged between 18 and 36 years (mean age 24 $\pm$4 years) participated in this experiment. There were equal number of Chinese and Malay subjects in the test group.

Equipment. Three humanoid robots were involved: the Nao robot version 4 by Softbank Robotics 2015 to represent a fully humanoid robot, the Reeti version 1 by Robopec 2015 to represent a partial humanoid, and an Avatar which was created using Reallusion Crazytalk software to represent a virtual humanoid.

Procedure. Subjects performed all three conditions: Condition A involved an avatar, Condition B a partial bodied humanoid, and

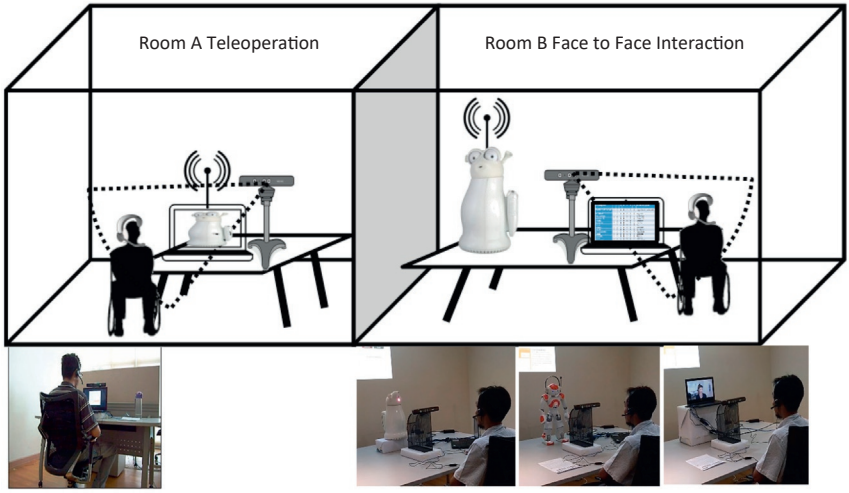


FIG. 5 Experimental layout for human-robot interaction using three different robots.

Condition C a full-bodied humanoid. Unlike the HHI experiment, two subjects in this HRI experiment interacted with a robot intermediary. The subjects were placed in a separate room with no contact with each other. One subject was randomly chosen to assume the role of the robot's superordinate operating from a remote location, while the other took on the role of a client or coequal member who would be placed in the same room as the robot, as shown in Fig. 5.

In each condition, subjects interacted with the robot using a scripted dialogue while taking turns to speak. In some conditions, the communication feedback between the robot and the teleoperator was fully audio-visual, while in other conditions, the teleoperator was only able to communicate with the robot using audio. At periodic intervals, the robot's dialogue portrayed one of the 15 trust attributes either positively or negatively, and subjects then scored their trust toward the robot on that attribute.

Results and discussion

- The 15 trust attributes used for constructing the dialogue scripts were shown to be internally consistent and reliable based on the Cronbach's Alpha test, whereby 15 attributes showed a range of 0.77–0.80. This validated the theory that subjective trust is multidimensional, and it would be counterproductive to quantify trust using only a single metric. The use of multidimensional attributes to represent trust have been replicated and validated in the literature (Derrick, 2011).

- This experiment demonstrated that trust can be measured using interactive **dialogues** that represent three different contexts of use or scenarios: Business, Disaster, and Healthcare. The scenarios had significant effects on both general and psychological trust, suggesting that the situational context has a role to play in influencing trust between humans and humanoid robots.
- Human trust in humanoid robots is contingent upon their **anthropomorphic** qualities. The type of humanoid robot (virtual or physical, partial bodied or full bodied) had a significant effect on trust particularly for Nao robot, which was considered more trustworthy than Reeti robot and Virtual Avatar for the attribute measures of Ability, Benevolence, and Integrity. The latter two robots were trusted equally for Ability and Integrity, but Reeti was trusted more than the Avatar for Benevolence. Based on a postexperiment questionnaire subjects preferred the physical robots over the virtual avatar. About 41% of subjects rated Nao to be most humanlike, while Reeti and Avatar were rated humanlike by 31% and 27% of subjects, respectively. Li (2015) arrived at similar conclusions in his experiment whereby physical robots that colocated with humans in proximate interaction were rated more positively as compared to virtual robots of a similar design. This implies that **proximate interaction** is more conducive for trust as compared to **remote interaction**.

The interaction of robot type and context of use (business, disaster, healthcare) affected trust evaluation. This means that people preferred physical robots with capabilities that matched the task they were assigned to undertake. For example, in the Disaster scenario, the full-bodied Nao robot received the highest score compared to a partially bodied Reeti robot and a virtual avatar. In the Healthcare scenario, Reeti was scored lowest in trustworthiness as compared to the Avatar and Nao. In the Business scenario, the two physical robots were scored higher than the Avatar, implying that trust increased in proximate interaction, and more so than in remote communication. In all three scenarios subjects identified what they perceived as the most appropriate robot for the task. This conclusion was also reached by Goetz, Kiesler, and Powers (2003) and Bartneck and Forlizzi (2004). The more humanlike the robot, the greater the role it can play in social and service scenarios involving natural dialogue communication. Therefore, design of virtual avatars in the next experiment must consider these characteristics.

In a postexperiment interview, 45.8% of the subjects considered voice to be the most important humanlike characteristic of the robot, while 33.3% considered facial expressions to be the most important. In this regard, there is a possibility that Nao was scored higher due to its excellent built-in speech capabilities, while Reeti and Avatar used conventional

text-to-speech methods. This is consistent with the findings of Niculescu, van Dijk, Nijholt, Li, and See (2013) that the speaking style of a robot improves the perception of its sociability.

Human-to-robot-to-human interaction

This experiment models several actors (humans and avatar) in remote and proximate (face-to-face) interactions with the purpose to analyze how human trust operates in a multiagent scenario composed of two humans as "clients," a virtual avatar as an intermediary and a human "teleoperator" remotely present (Khalid, Liew, Voong, & Helander, 2018). The communication process was designed to be nonuniform, with each client speaking in a different language, and the avatar's behavior occasionally deviating from the teleoperator's instructions. In addition, multiple avatars were designed to embody specific combinations of gender and ethnicity.

Objectives and hypotheses

- To investigate the effects of avatar gender and ethnicity on human trust in a three-way communication scenario.

  Human trust in avatars depended on the gender and ethnicity of the subjects, as well as the perceived gender and ethnicity of the avatars. There may be intragroup differences in how each gender and ethnicity trusts avatars (Obaid et al., 2016; Wang, Rau, Evers, Robinson, & Hinds, 2010).
- To determine the effects of avatar compliance on human trust.

  The apparent compliance of the avatar can impact trust, although it may be moderated by personal preferences of the avatar's anthropomorphic qualities (Eyssel et al., 2012).

  The roles of the human and/or robot partners in a mixed-team may affect the dynamics of the human-robot interaction (Gombolay et al., 2015; Marble, Bruemmer, Few, & Dudenhoeffer, 2004).

Experimental design and variables

The same factorial within-subjects design with repeated measures was used in this HRHI experiment. The independent variables were the virtual avatar, language, and avatar compliance. Two virtual avatars were used. One was modeled to have an appearance of a male Malaysian Chinese, while the second had an appearance of a female Malaysian Malay. Two languages were used: the Teleoperator and Client A spoke English, while Client B spoke Bahasa Malaysia. The avatars spoke in either language using human voices. The teleoperator directed the avatars' replies to the clients. In Condition A, the avatars complied and replied to the clients

in their own languages. In Condition B, the avatars occasionally ignored the teleoperator's instructions and replied to the clients in a different language. The dependent variables were general trust and physiological measures. General trust consisted of 12 semantic attributes measured using a bipolar scale derived from the HHI and HRHI experiments, after being validated using factor analysis. The physiological features were facial expression, voiced speech, vision-based heart rate, and postural gestures.

Method

Subjects. In total, 48 subjects (25 males and 23 females; 24 Malays and 24 Chinese) aged between 21 and 46 years (mean age 26 ± 4 years) volunteered in the study. Among them, 15 subjects had previously participated in the HRI experiment, while 33 subjects had no prior experience.

Equipment. The equipment setup was like the layout described in Fig. 5. However, the Robot Operating Systems architecture was expanded to include the hardware for a third subject.

Task and dialogue design. The Teleoperator instructed a virtual avatar to negotiate with two Clients, one speaking in English and another in Bahasa Malaysia. The dialogue for this experiment was based on the Business scripted dialogues of the previous experiments. Twelve trust attributes were embedded in the avatar dialogue with half in a positive orientation while the remaining half in a negative orientation. The distribution of positive or negative attributes was counterbalanced across all 12 attributes.

Procedure. There were two conditions: Condition A represents embodied compliance, whereby the avatar obeyed the teleoperator's instructions; while Condition B represents embodied noncompliance, whereby the avatar sometimes "chose" to disobey the teleoperator's instructions. Subjects participated in both conditions, interacting with a virtual avatar in a multiactor scenario. One subject played the role of a remote teleoperator in another location and was responsible for "directing" the avatar's responses, while two other subjects played the role of clients.

Fig. 6 shows three humanlike virtual avatars which were modeled in accordance with subjective responses on their preferred humanlike characteristics. The avatars embodied both genders and ethnicities of the experimental subjects.

Fig. 7 shows the arrangement of the experiment with multiactors.

FIG. 6 Humanlike virtual avatars in human-robot-human interaction experiment.

Results and discussion

- A reliability test performed on the 12 attributes yielded alpha to range between 0.66 and 0.71. The lower reliability for this experiment may be due to the reduced number of items in the scale which is a minor limitation.
- In terms of **gender**, the Chinese subjects were significantly more trusting than the Malay subjects for the trust factors: Ability, Benevolence, and Integrity. However, **ethnicity** was significant only through interaction with the subjects' role, the type of the avatar, and the avatar's obedience. Subjects trusted and preferred the Malay avatar over the Chinese avatar, particularly when the avatars displayed disobedient behavior. In the postexperiment questionnaire, subjects ranked the Malay avatars' voice higher than the Chinese avatar across all voice qualities, suggesting that the Malay avatar was considered more humanlike and more trustworthy.

In addition, the avatars' ethnicity was positively correlated with trust attributes of *Qualified*, *Friendly*, *Pleasant*, and *Ethical*. Since trust between ethnicities develops over time (Wilkes, 2011), the short duration of the experimental sessions does not make it possible to develop a rapport with a counterpart from different ethnicities. This means that subjects relied on their preconceived and stereotypical knowledge of the avatars' ethnicities.

Role had a significant effect on trust, where the avatars were perceived to be more trustworthy by the clients than by the teleoperators, particularly for Ability and Benevolence trust attributes. As part of an interaction with the subjects' gender, role was found to have a significant effect on trust by the male subjects, but not the female subjects. The avatars were perceived to be more trusting by the male clients than by the male teleoperators, while female subjects were perceived equally in both roles.

Compliance by avatar did not have a significant effect on trust by itself at the composite trust and ABI trust levels, except when there was an

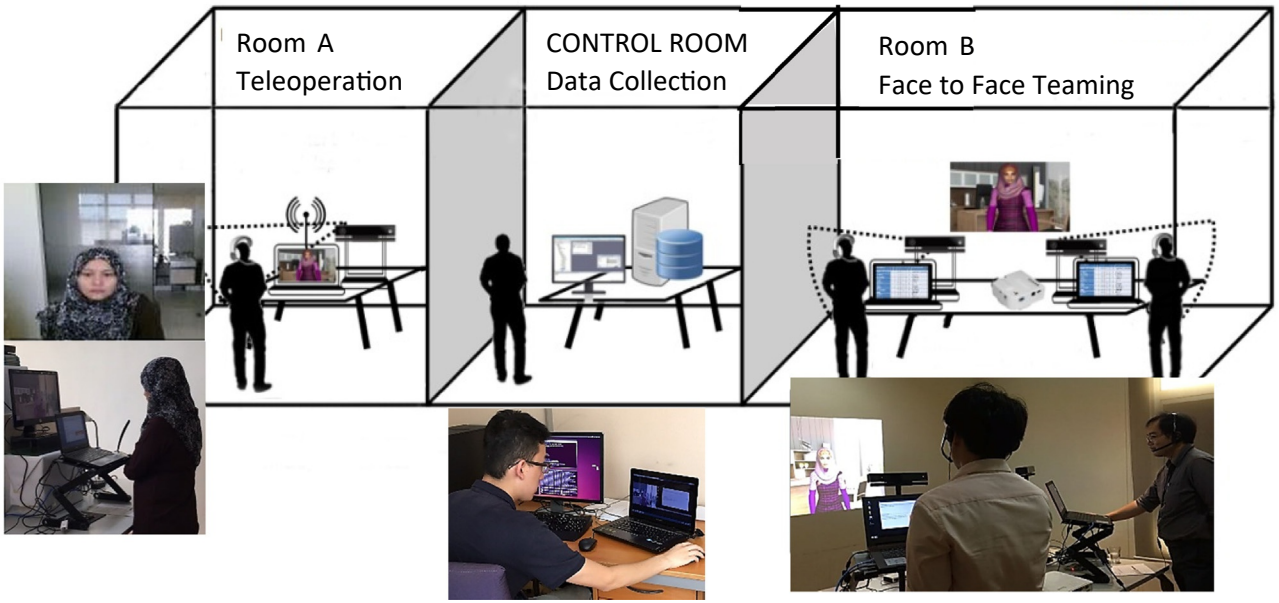


FIG. 7 Experimental layout for human-robot-human interaction using virtual avatars.

interaction with avatar gender and ethnicity. Subjects scored the female Malay avatar higher in the noncompliance condition compared to the compliance condition. This may be because females are trusted more than males (Buchan, Croson, & Solnick, 2008; Haselhuhn, Kennedy, Kray, Van Zant, & Schweitzerd, 2015). On the other hand, the male Chinese avatar was scored higher in the compliance condition compared to the noncompliance condition.

Feedback which referred to the avatars' ability to respond to the subjects was positively correlated to *Honest* and negatively to *Competent*, *Pleasant*, and *Dependable*. This indicates that subjects believed that the avatars' ability to respond is an indicator of its honesty. However, the dialogue need not be expressed in perfect natural language if the avatars appear to be competent, pleasant, and dependable.

The avatars' physical form correlated positively with the trust attributes *Competent*, *Skilled*, and *Cheerful*, indicating that the appearance of the avatars can project a sense of professionalism and a cheerful demeanor. Similarly, the attire worn by the avatar correlated positively with *Competent* and *Pleasant* trust attributes and negatively with *Ethical* and *Honest* attributes. This suggests that subjects equated well-dressed avatars to be capable and pleasant, but not entirely trustworthy in terms of Integrity as these attributes are performance-based. Gestures were negatively correlated with *Competent* and *Pleasant*, indicating that gestures alone are insufficient to convey a sense of competence and pleasantness.

In terms of trust prediction, this experiment achieved an overall accuracy of 88%. The relatively high accuracy suggested that the methodology was effective in predicting subjective trust from objective features.

Predicting trust using the neuro-fuzzy method

Prediction may be achieved using the neuro-fuzzy method, whereby neural network classifiers are used to associate objective trust features extracted in "Physiological trust" section to the subjective trust scorings provided by the subjects. The corpus of data created from the experiments was considered as "ground truth" for training the neural networks to recognize and learn how to estimate subjective trust scores when given a set of objective trust features whose subjective trust score was unknown.

The neural network chosen for this experiment was a neuro-fuzzy network. Given that the objective trust features were extracted from physiological signals of subjects in several experiments, the features could not be clearly delineated due to noise (i.e., background noise in the microphones; difficulty measuring facial action units for many different types of faces; etc.). In addition, trust is an abstract concept; thereby the indicators cannot be captured explicitly for every single subject. The indistinct nature of the

objective trust features makes it difficult for neural networks to recognize a clear pattern that can be associated with the subjective trust scores. The neuro-fuzzy network utilizes fuzzy logic, which allows some uncertainty when dealing with vague information where the actual value is somewhere between "absolutely true" and "absolutely false." In addition, the neurons in the neuro-fuzzy network were structured in a manner that allows interpretable rules to be extracted from the trained network as in regression analysis.

Prior to training the networks, the subjective trust scores were discretized from a continuous range of values from a lower limit of 0 to a maximum score of 7, into three distinct trust classes: Distrust or Low Trust, Neutral or Medium Trust, and Trusting or High Trust. Each data set consists of all the data points from one experiment, for a total of three data sets. Each data point consists of the objective features extracted from one segment and was appended by contextual information about the participant and the experiment condition, and the trust class derived from the subjective trust score.

To gauge the performance of the final trained network, a 10-fold cross-validation method was used. Every data point in the data set was assigned to 1 of 10 groups. Nine groups were used for training a new network, while the remaining group was used for testing the network. Testing involved giving the trained network a data point and comparing its output to the actual output. This process was performed 10 times, each time with a different group of data points for testing. The performance of the neural network was then averaged across the testing performance for all ten groups, typically as a percentage of accuracy, or how well the trained network was able to correctly predict the testing data. Table 2 shows the performance of the neuro-fuzzy when trained and tested with the data sets from each experiment. While the current method involved offline processing of data and information, it is possible to develop an online methodology for predicting trust in real time. By first training an initial neural network with offline data sets, the trained network can then be used for trust prediction simply by giving it new objective features obtained from sensors (i.e., microphones and video cameras).

Research contributions

The results from the three experiments show that both Gender and Ethnicity of the trustors/evaluators influence their evaluation of trustees' Ability, whether humans or robots. However, for Benevolence and Integrity trust components, the Context of Use had an effect. Role impacted trust evaluation in that a remote operator or trustor tends to downrate

TABLE 2 Performance of neuro-fuzzy in predicting subjective trust from objective trust.

Human-to-human interaction, Condition C				
	Low trust	**Medium trust**	**High trust**	**Weighted average**
Accuracy	0.65			
Sensitivity	0.22	0.33	0.81	0.54
Specificity	0.94	0.89	0.48	0.70
Precision	0.70	0.69	0.63	0.66
F1-score	0.34	0.44	0.72	0.55
Human-to-robot interaction				
	Low trust	**Neutral trust**	**High trust**	**Weighted average**
Accuracy	0.83			
Sensitivity	0.39	0.12	0.95	0.78
Specificity	0.96	0.99	0.46	0.59
Precision	0.75	0.68	0.84	0.81
F1-score	0.51	0.20	0.89	0.77
Human-to-robot-to-human interaction				
	Low trust	**Medium trust**	**High trust**	**Weighted average**
Accuracy	0.88			
Sensitivity	0.14	0.46	0.98	0.85
Specificity	1.00	0.98	0.51	0.61
Precision	0.87	0.85	0.88	0.88
F1-score	0.24	0.59	0.93	0.84

its client or trustee in a superordinate-subordinate relationship. This suggests that power and influence can still be exerted by an evaluator who may be virtually present.

Strangely enough, a female avatar was trusted by evaluators even though it did not comply with instructions. This was different from its male counterpart, who was distrusted. This suggests that design of future social robots must consider gender and ethnicity, and by inducting intelligence, it might be possible to enhance human trust in robotic partners.

In conclusion, the main value of this research is the ability to predict human trust in humanoid robots using the measures constructed in this research. Since human-robot teaming involves multiple actors, the results would be useful for defense purposes (air force, military, naval), where

robots operating in remote settings, such as in rescue and search operations, can now be evaluated reliably. Moreover, robots may be supervised remotely using natural human dialogues to enhance realism of interaction as in real-world settings. The main contributions of this research may be considered from two perspectives of theory and methodology.

Theory of human-robot trust

The experiments examined the relevance of anthropomorphism in robots, as automated agents, on trust and compliance in different use case scenarios and demographic characteristics of the trustors and trustees. The first evidence reveals that anthropomorphic features in robots can increase trust. Robots that appeared more humanlike were more trusted compared to virtual agents. Paradeda, Hashemian, Rodrigues, and Paiva (2016) claimed that humans found robots that engage in small talks and expressing facial expressions in manners consistent with expected societal norms during storytelling to be more trustworthy. This could be because in human-to-human social relationship, social cues (e.g., eye gazing, gestures, body movement, and speech tone) were received and evaluated before any trust decision-making stage. It can be inferred that in HRI, similarity in mental models between a robot and its human partner can improve effectiveness of human-robot teaming.

Methodology

The trust prediction methodology for studying the dynamics of trust in human and robot social interactions using psychophysiological criteria yielded promising outcomes. Objective and subjective trust were acquired in a series of scripted social interactions, which were designed in a manner to test the effects of contextual factors on how humans trust robots. Statistical analysis of subjective trust scorings revealed how certain factors interacted with each other in impacting trust. Using neural network methods, objective trust was mapped to subjective trust with relatively good accuracy, proving that it is possible to reconcile physiological and psychological measures of trust.

Refinement of the methodology would require a broader base of data to encompass more factors. In addition, the feature extraction methods on the physiological signals were based on preexisting methodologies. Convolution or deep learning methods may be used to substitute the feature extraction methodology. The neuro-fuzzy network may also be substituted with other neural networks such as extreme learning machines that are simpler and faster, suitable for real-time applications.

Future work

Nevertheless, there remains a huge gap in knowledge on robotic trust that warrants further investigation. Given the nebulous concept of trust, which makes measurement difficult, robotic trust recognition is a challenging area. The difficulty increases due to the complex nature of social interactions, from which the robot must identify social cues to determine trust or distrust. Imbuing robots with the ability to recognize trust or distrust and react accordingly has significant benefits. For example, if a person being rescued does not trust the robotic rescuer, he/she might resist or refuse to obey the robot's instructions. A sales robot would be able to market their products more effectively to a trusting audience. A factory robot with a distrusting human partner may change their work routine to maximize their work efficiency without interfering with each other. There has been little research in providing robots with a comprehensive methodology for trust assessment across various physiological modalities and social scenarios.

In conclusion, there are many unexplored avenues in HRI research. For researchers to come out with a comprehensive framework of the dynamic Human-Robot trust processes, it is important to identify the factors that affect the HRI in a variety of real-life situations besides those already explored in this chapter.

Acknowledgments

Halimahtun Khalid would like to thank Liew Wei Shiung and Voong Bin Sheng for their helpful support in the research; and to Dr. Benjamin A. Knott for funding the research.

References

Adolphs, R., Tranel, D., & Damasio, A. R. (1998). The human amygdala in social judgment. *Nature*, *393*(6684), 470–474.

Aksoy, B., Harwell, H., Kovaliukaite, A., & Eckel, C. (2018). Measuring trust: A reinvestigation. *Southern Economic Journal*, *84*(4), 992–1000.

Albaity, M. S., & Rahman, M. (2012). Gender, ethnicity, and religion and investment decisions: Malaysian evidence. *Journal of Sociological Research*, *3*(2), 502–519.

Atkinson, A. P., Dittrich, W. H., Gemmell, A. J., & Young, A. W. (2004). Emotion perception from dynamic and static body expressions in point-light and full-light displays. *Perception*, *33*(6), 717–746.

Bainbridge, W. A., Hart, J., Kim, E. S., & Scassellati, B. (2008). The effect of presence on human-robot interaction. In: *Proceedings of the 17th IEEE international symposium on robot and human interactive communication, 1–3 August 2008, Munich, Germany*, IEEE, pp. 701–706.

Bainbridge, W. A., Hart, J. W., Kim, E. S., & Scassellati, B. (2011). The benefits of interactions with physically present robots over video-displayed agents. *International Journal of Social Robotics*, *3*(1), 41–52.

Baltrušaitis, T., Robinson, P., & Morency, L. P. (2016, March). Openface: An open source facial behaviour analysis toolkit. In: *2016 IEEE winter conference on applications of computer vision (WACV)*, IEEE (pp. 1–10): IEEE.

Banks, D. M., Scott, B. G., & Weems, C. F. (2017). Anxiety, hostile attributions, and differences in heart rate response to ambiguous situational vignettes in adolescents. *Emotion*, *18*(2), 248–259. https://doi.org/10.1037/emo0000352.

Bartneck, C., & Forlizzi, J. (2004). Shaping human-robot interaction: Understanding the social aspects of intelligent robotic products. In *CHI'04 extended abstracts on human factors in computing systems* (pp. 1731–1732): ACM.

Boersma, P. (2001). Praat, a system for doing phonetics by computer. *Glot International*, *5* (9/10), 341–345.

Bowling, A. (2014). *Research methods in health: Investigating health and health services.* McGraw-Hill Education (UK).

Broadbent, E. (2017). Interactions with robots: The truths we reveal about ourselves. *Annual Review of Psychology*, *68*, 627–652.

Buchan, N. R., Croson, R. T. A., & Solnick, S. (2008). Trust and gender: An examination of behavior and beliefs in the Investment Game. *Journal of Economic Behavior and Organization*, *68*(3–4), 466–476.

Caridakis, G., Malatesta, L., Kessous, L., Amir, N., Raouzaiou, A., & Karpouzis, K. (2006, November). Modeling naturalistic affective states via facial and vocal expressions recognition. In *Proceedings of the 8th international conference on multimodal interfaces* (pp. 146–154): ACM.

Charalambous, G., Flectcher, S., & Webb, P. (2016). The development of a scale to evaluate trust in industrial human-robot collaboration. *International Journal of Social Robotics*, *8*, 193–209.

Chen, X. P., Eberly, M. B., Chiang, T. J., Farh, J. L., & Cheng, B. S. (2014). Affective trust in Chinese leaders: Linking paternalistic leadership to employee performance. *Journal of Management*, *40*(3), 796–819.

Chenchah, F., & Lachiri, Z. (2014). Speech emotion recognition in acted and spontaneous context. *Procedia Computer Science*, *39*, 139–145.

Cockshott, P., & Renaud, K. (2016). Humans, robots and values. *Technology in Society*, *45*, 19–28.

Coeckelbergh, M. (2012). Can we trust robots? *Ethics and Information Technology*, *14*, 53–60.

Cormier, D., Newman, G., Nakane, M., Young, J. E., & Durocher, S. (2013). Would you do as a robot commands? An obedience study for human-robot interaction. In *International conference on human-agent interaction*.

Damiano, L., & Dumouchel, P. (2018). Anthropomorphism in human–robot co-evolution. *Frontiers in Psychology*, *9*, 468. https://doi.org/10.3389/fpsyg.2018.00468.

de Visser, E. J., Monfort, S. S., McKendrick, R., Smith, M. A. B., McKnight, P. E., Krueger, F., & Parasuraman, R. (2016). Almost human: Anthropomorphism increases trust resilience in cognitive agents. *Journal of Experimental Psychology: Applied*, *22*(3), 331–349.

de Visser, E. J., Pak, R., & Neerincx, M. A. (2017). Trust development and repair in human-robot teams. In: *Proceedings of the companion of the 2017 ACM/IEEE international conference on human-robot interaction—HRI '17* (pp. 103–104). New York, NY: ACM Press.

Derrick, D. C. (2011). *Special-purpose, embodied conversational intelligence with environmental sensors (SPECIES) agents: Implemented in an automated interviewing kiosk.* The University of Arizona.

DeSteno, D., Breazeal, C., Frank, R. H., Pizarro, D., Baumann, J., Dickens, L., & Lee, J. J. (2012). Detecting the trustworthiness of novel partners in economic exchange. *Psychological Science*, *23*(12), 1549–1556.

Duffy, B. R. (2003). Anthropomorphism and the social robot. *Robotics and Autonomous Systems*, *42*, 177–190.

Ekman, P. (2009). Lie catching and micro expressions. In *The philosophy of deception* (pp. 118–133).

Ekman, R. (1997). *What the face reveals: Basic and applied studies of spontaneous expression using the facial action coding system (FACS).* USA: Oxford University Press.

Elkins, A. C., & Derrick, D. C. (2013). The sound of trust: Voice as a measurement of trust during interaction with embodied conversational agents. *Group Decision and Negotiation*, *22*(5), 897–913.

Everhart, C. B., & Chelladurai, P. (1998). Gender differences in preferences for coaching as an occupation: The role of self-efficacy, valence, and perceived barriers. *Research Quarterly for Exercise and Sport*, *69*(2), 188–200.

Eyssel, F., Kuchenbrandt, D., Bobinger, S., de Ruiter, L., & Hegel, F. (2012, March). 'If you sound like me, you must be more human': On the interplay of robot and user features on human-robot acceptance and anthropomorphism. In *Proceedings of the seventh annual ACM/IEEE international conference on human-robot interaction* (pp. 125–126): ACM.

Fink, J. (2012, October). Anthropomorphism and human likeness in the design of robots and human-robot interaction. In *International conference on social robotics* (pp. 199–208). Berlin Heidelberg: Springer.

Gaudiello, I., Zibetti, E., Lefort, S., Chetouani, M., & Ivaldi, S. (2016). Trust as indicator of robot functional and social acceptance. An experimental study on user conformation to iCub answers. *Computers in Human Behaviour*, *61*, 633–655.

Goetz, J., Kiesler, S., & Powers, A. (2003). Matching robot appearance and behavior to tasks to improve human-robot cooperation. In *Proceedings of the 12th IEEE international workshop on robot and human interactive communication (ROMAN 2003)*, IEEE Xplore. https://doi.org/10.1109/ROMAN.2003.1251796.

Gombolay, M. C., Gutierrez, R. A., Clarke, S. G., Sturla, G. F., & Shah, J. A. (2015). Decision-making authority, team efficiency and human worker satisfaction in mixed human–robot teams. *Autonomous Robots*, *39*(3), 293–312.

Gong, L. (2008). The boundary of racial prejudice: Comparing preferences for computer-synthesized white, black, and robot characters. *Computers in Human Behaviour*, *24*(5), 2074–2093.

Grayson, K., Johnson, D., & Chen, D. F. R. (2008). Is firm trust essential in a trusted environment? How trust in the business context influences customers. *Journal of Marketing Research*, *45*(2), 241–256.

Gulati, S., Sousa, S., & Lamas, D. (2019). Design, development and evaluation of a human computer trust scale. *Behaviour & Information Technology*, *38*(10), 1004–1015.

Guo, C., & Sharlin, E. (2008). Exploring the use of tangible user interfaces for human-robot interaction: A comparative study. In *Proceedings of the 2008 conference on human factors in computing systems, CHI 2008, Florence, Italy, April 5–10*.

Hancock, P. A., Billings, D. R., Schaefer, K. E., Chen, J. Y., De Visser, E. J., & Parasuraman, R. (2011). A meta-analysis of factors affecting trust in human-robot interaction. *Human Factors*, *53*(5), 517–527.

Haselhuhn, M. P., Kennedy, J. A., Kray, L. J., Van Zant, A. B., & Schweitzerd, M. E. (2015). Gender differences in trust dynamics: Women trust more than men following a trust violation. *Journal of Experimental Social Psychology*, *95*, 2008. https://doi.org/10.1016/j.jesp.2014.09.007.

Heilman, M. E. (2012). Gender stereotypes and workplace bias. *Research in Organizational Behavior*, *32*(2012), 113–135.

Hoff, K. A., & Bashir, M. (2015). Trust in automation: Integrating empirical evidence on factors that influence trust. *Human Factors*, *57*(3), 407–434.

Jhalani, J., Goyal, T., Clemow, L., Schwartz, J. E., Pickering, T. G., & Gerin, W. (2005). Anxiety and outcome expectations predict the white-coat effect. *Blood Pressure Monitoring*, *10*(6), 317–319.

Jung, M., Sabanovic, S., Eyssel, F., & Fraune, M. (2017). Robots in groups and teams. In *Proceedings of CSCW 2017, February 25-March 1, Portland, OR.* https://doi.org/10.1145/3022198.3022659.

Khalid, H. M. (2018). *Trusting humanoid robots undertake social tasks (THRUST).* Technical Report, AFOSR/AOARD Grant No. FA2386-14-1-0016, July 2018.

Khalid, H. M., Liew, W. S., & Helander, M. G. (December 2018). Effects of presence and feedback on human trust in robots; comparing teleoperation and proximate interaction. In *Proceedings of 5th SEANES international conference, Bangkok, Thailand*, pp. 12–14.

Khalid, H. M., Liew, W. S., Voong, B. S., & Helander, M. G. (2018, February). Trust of virtual agent in multi actor interactions. *Journal of Robotics, Networking and Artificial Life*, *4*(4), 295–298.

Khalid, H. M., Liew, W. S., Voong, B. S., Helander, M. G., & Loo, C. K. (2017). Technology for estimation of trust in human-robot interaction. In *Proceedings of 4th Asian conference on defence technology, Tokyo, Japan, 29 Nov–1 Dec*.

Khalid, H. M., Voong, B. S., Liew, W. S., Chin, W. H., & Helander, M. G. (2017). Trust of assistive humanoid robots in service tasks; a comparison between scenarios. In *Proceedings of 4th Asian conference on defence technology, Tokyo, Japan, 29 Nov–1 Dec*.

Krueger, F., McCabe, K., Moll, J., Kriegeskorte, N., Zahn, R., Strenziok, M., … Grafman, J. (2007). Neural correlates of trust. *Proceedings of the National Academy of Sciences*, *104*(50), 20084–20089.

Kuo, I. H., Rabindran, J. M., Broadbent, E., Lee, Y. I., Kerse, N., Stafford, R. M. Q., & MacDonald, B. A. (September 2009). Age and gender factors in user acceptance of healthcare robots. In *RO-MAN 2009-the 18th IEEE international symposium on robot and human interactive communication*, (pp. 214–219): IEEE Xplore.

Lee, J. D., & See, K. A. (2004). Trust in automation: Designing for appropriate reliance. *Human Factors*, *46*(1), 50–80.

Li, J. (2015). The benefit of being physically present: A survey of experimental works comparing copresent robots, telepresent robots and virtual agents. *International Journal of Human-Computer Studies*, *77*, 23–37.

Lount, R. B., Jr., & Pettit, N. C. (2012). The social context of trust: The role of status. *Organizational Behavior and Human Decision Processes*, *117*(1), 15–23.

Lugger, M., & Yang, B. (2007, April). The relevance of voice quality features in speaker independent emotion recognition. In *Vol. 4. 2007 IEEE international conference on acoustics, speech and signal processing-ICASSP'07* (pp. IV-17–IV-20): IEEE.

Lyons, J. B. (2013). *Being transparent about transparency: A model for human–robot interaction. In 2013 AAAI spring symposium series. Arlington, VA*. http://www.aaai.org/ocs/index.php/SSS/SSS13/paper/viewPaper/5712.

Marble, J. L., Bruemmer, D. J., Few, D. A., & Dudenhoeffer, D. D. (2004, January). Evaluation of supervisory vs. peer-peer interaction with human-robot teams. In *Proceedings of the 37th annual Hawaii international conference on system sciences* (p. 9): IEEE.

Mayer, R. C., Davis, J. H., & Schoorman, F. D. (1995). An integrative model of organizational trust. *Academy of Management Review*, *20*(3), 709–734.

Niculescu, A., van Dijk, B., Nijholt, A., Li, H., & See, S. L. (2013). Making social robots more attractive: The effects of voice pitch, humor and empathy. *International Journal of Social Robotics*, *5*(2), 171–191.

Nowak, K. L., & Biocca, F. (2003). The effect of the agency and anthropomorphism on users' sense of telepresence, copresence, and social presence in virtual environments. *Presence Teleoperators and Virtual Environments*, *12*(5), 481–494.

Obaid, M., Salem, M., Ziadee, M., Boukaram, H., Moltchanova, E., & Sakr, M. (2016, October). Investigating effects of professional status and ethnicity in human-agent interaction. In *Proceedings of the fourth international conference on human agent interaction* (pp. 179–186): ACM.

Paradeda, R. B., Hashemian, M., Rodrigues, R. A., & Paiva, A. (2016). How facial expressions and small talk may influence trust in a robot. In: *Proceedings of 8th International Conference, ICSR 2016, Kansas City, MO, USA, 1–3 November*. https://doi.org/10.1007/978-3-319-47437-3_17.

Parks, C. D., Henager, R. F., & Scamahorn, S. D. (1996). Trust and reactions to messages of intent in social dilemmas. *Journal of Conflict Resolution*, *40*(1), 134–151.

Poh, M. Z., McDuff, D. J., & Picard, R. W. (2011). Advancements in noncontact, multiparameter physiological measurements using a webcam. *IEEE Transactions on Biomedical Engineering*, *58*(1), 7–11.

Powers, A., Kiesler, S., Fussell, S., & Torrey, C. (2007, March). Comparing a computer agent with a humanoid robot. In *2nd ACM/IEEE international conference on human-robot interaction (HRI),* (pp. 145–152): IEEE.

PytlikZillig, L. M., & Kimbrough, C. D. (2016). Consensus on conceptualizations and definitions of trust: Are we there yet? In E. Shockley, T. M. S. Neal, L. M. PytlikZillig, & B. H. Bornstein (Eds.), *Interdisciplinary perspectives on trust: Towards theoretical and methodological integration* (pp 17–47). New York, NY: Springer International Publishing.

Qiu, L., & Benbasat, I. (2010). A study of demographic embodiments of product recommendation agents in electronic commerce. *International Journal of Human-Computer Studies*, *68* (10), 669–688.

Righetti, F., & Finkenauer, C. (2011). If you are able to control yourself, I will trust you: The role of self-control in interpersonal trust. *Journal of Personality and Social Psychology*, *100*(5), 874–886.

Rilling, J. K., & Sanfey, A. G. (2011). The neuroscience of social decision-making. *Annual Review of Psychology*, *62*, 23–48.

Robert, L. (2018). Motivational theory of human robot teamwork. *International Robotics & Automation Journal*, *4*(4), 248–251.

Sanders, T., Kaplan, A., Koch, R., Schwartz, M., & Hancock, P. A. (2019). The relationship between trust and use choice in human-robot interaction. *Human Factors*, *61*(4), 614–626. https://doi.org/10.1177/0018720818816838.

Schermerhorn, P., & Scheutz, M. (2009, November). Dynamic robot autonomy: Investigating the effects of robot decision-making in a human-robot team task. In *Proceedings of the 2009 international conference on multimodal interfaces* (pp. 63–70): ACM.

Siegel, M., Breazeal, C., & Norton, M. I. (2009, October). Persuasive robotics: The influence of robot gender on human behaviour. In *IEEE/RSJ international conference on intelligent robots and systems, 2009. IROS* (pp. 2563–2568):IEEE.

Steels, L., & Hild, M. (Eds.) (2012). *Language grounding in robots*. New York: Springer.

Streeter, L. A., Krauss, R. M., Geller, V., Olson, C., & Apple, W. (1977). Pitch changes during attempted deception. *Journal of Personality and Social Psychology*, *35*(5), 345.

Szalma, J. L., & Taylor, G. S. (2011). Individual differences in response to automation: The five-factor model of personality. *Journal of Experimental Psychology: Applied*, *17*(2), 71–96.

Tajfel, H., & Turner, J. C. (1986). The social identity theory of inter group behavior. In S. Worchel, & W. G. Austin (Eds.), *Psychology of intergroup relations* Chicago: Nelson.

Task Force of the European Society of Cardiology. (1996). Heart rate variability: Standards of measurement, physiological interpretation, and clinical use. *Circulation*, *93*, 1043–1065.

Tay, B., Jung, Y., & Park, T. (2014). When stereotypes meet robots: The double-edge sword of robot gender and personality in human–robot interaction. *Computers in Human Behaviour*, *38*, 75–84.

Teo, G., Wohleber, R., Lin, J., & Reinerman-Jones, L. (2017). The relevance of theory to human-robot teaming research and development. In *Advances in human factors in robots and unmanned systems* (pp. 175–185). Cham: Springer.

Tveten, J. (2016). *What happens when robots are assigned ethnicities? (2016)*. https://arstechnica.com/information-technology/2016/04/want-to-further-muddy-human-robot-interactions-add-in-ethnicity/.

Van den Brule, R., Dotsch, R., Bijlstra, G., Wigboldus, D. H., & Haselager, P. (2014). Do robot performance and behavioural style affect human trust? *International Journal of Social Robotics*, *6*(4), 519–531.

Wagner, J. (2006). *AuBT: Augsburg biosignal toolbox (version 1.04) [computer program]. Retrieved from* https://www.informatik.uni-augsburg.de/lehrstuehle/hcm/projects/tools/aubt/. March 3, 2013.

Wang, L., Rau, P. L. P., Evers, V., Robinson, B. K., & Hinds, P. (2010, March). When in Rome: The role of culture; context in adherence to robot recommendations. In *Proceedings of the 5th ACM/IEEE international conference on human-robot interaction* (pp. 359–366): IEEE Press.

Wasti, S., Tan, H. H., Brower, H., & Onder, C. (2007). Cross-cultural measurement of supervisor trustworthiness: An assessment of measurement invariance across three cultures. *The Leadership Quarterly*, *18*(5), 477–489.

Waytz, A., Heafner, J., & Epley, N. (2014). The mind in the machine: Anthropomorphism increases trust in an autonomous vehicle. *Journal of Experimental Social Psychology*, *52*, 113–117.

Wilkes, R. (2011). Re-thinking the decline in trust: A comparison of black and white Americans. *Social Science Research*, *40*(6), 1596–1610.

Winston, J. S., Strange, B. A., O'Doherty, J., & Dolan, R. J. (2002). Automatic and intentional brain responses during evaluation of trustworthiness of faces. *Nature Neuroscience*, *5*(3), 277–282.

Wortham, R. H., & Theodorou, A. (2017). Robot transparency, trust, and utility. *Connection Science*, *29*(3), 242–248.

Złotowski, J., Proudfoot, D., Yogeeswaran, K., & Bartneck, C. (2015). Anthropomorphism: Opportunities and challenges in human-robot interaction. *International Journal of Social Robotics*, *7*, 347–360. https://doi.org/10.1007/s12369-014-0267-6.

CHAPTER

5

Understanding worker trust in industrial robots for improving workplace safety

HeeSun Choi[a,*] *and Naomi Swanson*[b]

[a]Division of Safety Research, National Institute for Occupational Safety and Health (NIOSH), Morgantown, WV, United States, [b]Division of Science Integration, National Institute for Occupational Safety and Health (NIOSH), Cincinnati, OH, United States

Introduction: Industrial robots and worker safety

Since industrial robots began appearing in the workplace in the 1970s, they have been a significant part of the workplace in many industries. The International Organization for Standardization (ISO) defines an industrial robot as an "automatically controlled, reprogrammable multipurpose manipulator, programmable in three or more axes, which can be either fixed in place or mobile for use in industrial automation applications" (American National Standard Institute [ANSI], 2012, definition 3.10). With advanced capabilities and reduced cost of robots, the adoption of industrial robots is increasing rapidly and expanding to broader industries. According to the International Federation of Robotics report, worldwide annual sales of industrial robots are estimated to increase at 12% per year on average from 2020 to 2022 (International Federation of Robotics, 2019). Furthermore, new types of robots such as collaborative, mobile, and wearable robots began to be introduced, which results in significant changes in robotic work environments. Traditional industrial robots operate in fixed locations with safeguards or fences. Thus, human-robot interaction is minimal in traditional robotic workplaces. The newly introduced robots are

*Current address: Texas Tech University, Lubbock, TX, United States.

https://doi.org/10.1016/B978-0-12-819472-0.00005-8
© 2021 Elsevier Inc. All rights reserved.

designed to work in direct contact with human workers, share work areas, or perform cooperative tasks. In the new robotic work environments where the emerging types of robots operate, human-robot interaction has become a critical component of effective and safe use of industrial robots.

Industrial robots, especially the emerging types of robots, have great potential for workplace safety as they can take over repetitive, fatiguing, and dangerous tasks that human workers have traditionally performed. However, industrial robots also have the potential to cause worker deaths and injuries the same as other industrial equipment and machines do. Although there is a limited injury surveillance system that is capable of detecting worker injuries specifically involving robots in the workplace, the National Institute for Occupational Safety and Health identified 61 robot-related deaths between 1992 and 2015 (NIOSH, n.d.-a). As more advanced robots come into workplaces, safety concerns involving the robot-related workplace hazards have also increased due to increased interactions between human workers and robots. During the interactions between human workers and collaborative or coexisting robots, it is highly likely that the human and robot move simultaneously in close proximity or even while being physically in contact with each other, over extended periods of time (ANSI, 2012). Workers may be put at increased risk of injury due to various hazards that the new industrial robots can pose, such as hazards arising from unintended and unexpected contacts with robots (Murashov, Hearl, & Howard, 2016). While physical barriers that separate human workers from robots are effective hazard control methods in traditional robotic workplaces, new safety strategies must be identified for the new robotic work environment that may present greater injury risks and unforeseen hazards.

Ensuring safety in robotic workplaces may heavily rely on engineering and technological capabilities of robots such as sensors, controls, path-planning methods, and interface designs. However, a user's trust in robots and robotic environments can also play a critical role in safe and effective human-robot interaction performance. The current chapter aims to provide an understanding of the trust of workers working closely with industrial robots and discuss implications in workplace safety. In the following sections, we will discuss the importance of investigating trust in robots in the context of workplace settings, followed by reviews on factors that influence worker trust in industrial robots. We will specifically focus on the psychological aspects of human-robot interaction and human-related determinants of trust in workplace robots. We will present findings from existing research investigating the effects of individual characteristics and differences on human-robot trust. We will then discuss how worker trust in robots may change over time, and present factors influencing the development and changes of trust. We will also present some considerations for the design of robots and robotic environments and for worker training to improve worker safety.

Why is investigating worker trust important for workplace safety?

Achieving an appropriate level of trust is critical for safe and efficient interactions between humans and robots in workplaces, especially when a task requires advanced collaborations among them. Both overtrust and undertrust may produce inappropriate human-robot interactions and unsafe human behaviors (Merritt, Heimbaugh, LaChapell, & Lee, 2013; Parasuraman & Riley, 1997). Overtrust may result in misuse when users rely on robots too much and fail to maintain relevant safety behavior and appropriate attention to the system. Particularly in robotic workplaces, overtrust may let workers falsely judge situations and place themselves inside the hazardous areas while working in the robot's work zone (Murashov et al., 2016). On the other hand, undertrust can also have a negative impact on the attitudes and behaviors of human operators (Hancock et al., 2011). When humans undertrust robots, they may decide not to use them even when robots would improve their performance and safety. It is particularly important in workplaces because the use of workplace robots could often be mandatory for workers, while that of commercial or personal robots is voluntary. If workers undertrust robots, they may have to pay more attention to robots than warranted, which results in increased workload and stress among workers.

Workplace environments may provide unique contexts for human-robot interaction. The type of tasks and environments in which robots are used can change how operators interact with robots and their implications for safety. A wealth of research exists on human-robot trust in only a few specific operational settings, such as military use of robots in high-risk combat situations (Chen & Terrence, 2009; Groom & Nass, 2007; Hancock et al., 2011) or personal use of assistive or service robots among special populations such as older or disabled individuals (Fasola & Matarić, 2013; Heerink, Kröse, Evers, & Wielinga, 2010). Given the importance of environmental contexts in the development of trust, more research is warranted to investigate human trust in the context of the industrial use of robots.

Robots in workplaces may present distinct safety concerns inherent to the industrial setting due to the complex nature in which robots operate. Robotic work environments will become more demanding and diverse, as broader industries such as construction, mining, and agriculture are likely to adopt more robots in their workplaces in the near future. Thus, injury risks associated with overtrust or undertrust in robots can be particularly high in industrial settings. Another unique challenge in workplace settings is that it is difficult for robots to be personalized for users when they are developed for industrial use. Personal or service robots can be customized to meet the specific needs and preferences of an individual or a small group of users, such as a family, which may result in positive initial

attitudes toward robots. Personalization may be neither possible nor effective for workplace robots. Robots for industrial use require highly standardized performance to ensure product quality. Also, customization is less cost-effective, which can limit the use of robots in small- to mid-sized workplaces. Furthermore, the potential challenge may come from workers being asked or even required to work with robots regardless of their personal interests or choices when their workplaces adopt robots. This situation may result in negative attitudes toward robots, including a lack of trust among workers compared to users who choose to adopt robots.

On the other hand, there are advantages to achieving trust in robots in the workplace settings over domestic or public environments. Employers may be able to provide intensive and formal training to workers to facilitate appropriate trust in robots among workers before adopting robots in the workplace, while such training is not applicable in most personal or commercial robot uses. Although it can be more difficult and expensive, personalized training to specific worker groups with unique needs is also possible in some workplaces. Given these advantages, identifying factors influencing worker trust in robots have significant implications for improving workplace safety. If determinants and antecedents of worker trust in robots are identified, more effective and personalized interventions can be developed for various individual workers and worker populations. Factors involving human-to-robot trust can be used to develop trustworthy robots and robotic environments and achieve appropriate levels of trust among workers. Although our chapter discusses research investigating human-robot interaction in the context of general robotic usage, we aim to focus on the aspects that are more critically associated with work than domestic or public places and have greater implications for workplace safety.

Individual-related factors impacting worker trust in robots

Studies have investigated various factors influencing an operator's trust in automation or robotic systems (Desai, Stubbs, Steinfeld, & Yanco, 2009; Hancock et al., 2011; Schaefer, Chen, Szalma, & Hancock, 2016). There are three types of determinants of an individual's trust in robots: human-related (e.g., demographics, personality traits, expertise, and experiences), robot-related (e.g., reliability, level of automation, and anthropomorphism), and environmental (e.g., culture and task characteristics) factors (Hancock et al., 2011). Studies have primarily focused on investigating the effects of robot-related and environmental factors on an operator's trust in robots (Hancock et al., 2011). For example, users are more likely to trust a robot when a robot's movements, actions, and

performance are predictable and reliable (Dragan, Bauman, Forlizzi, & Srinivasa, 2015; Schaefer et al., 2016). While a growing body of literature addresses various determinants and antecedents of trust in robots and automation, limited research has investigated how different aspects of human characteristics and individual differences influence human trust toward robots (Cameron et al., 2015; Hancock et al., 2011). Although only a few types of human characteristics, such as personality traits, have been studied empirically, investigation of human-related determinants provides important knowledge that can be used to predict the trust of individuals, improve training for users, and develop personalized trustworthy robots.

In the current section, we review empirical studies that investigated human-related characteristics and individual differences related to human trust in automation or robots. Human-robot interaction may differ from human-automation interaction because unlike automated systems, robots have physical forms approximating humans or animals, which are often built to be mobile and carry out actions at a distance (Hancock et al., 2011). Due to these differences, the implications of trust in the technologies may be different. For instance, trust in robots may have more significant implications for safety compared to what automation trust may have because robots can create physical harm to human operators more easily. Furthermore, unlike automation, robot operations are often observable. Human workers are usually able to see the robot completing tasks, which suggests that the development of trust in robots may differ from trust in automation because human operators cannot observe what the system is doing. Despite these potential differences, human-robot interaction shares many aspects of functions and characteristics with human-automation interaction as robots are automated systems too. Thus, the following discussion includes findings from investigations of trust in human-automation interaction that have implications for trust in human-robot interaction.

Individual characteristics

One type of human-related factor is individual characteristics that include an individual's trust propensity, attitudes, personality traits, and demographic factors. Some people are overall more trusting than others, and this propensity to trust is a stable personality trait (Lee & See, 2004). It is likely that an individual worker's propensity to trust determines the person's initial level of trust in industrial robots (Lee & See, 2004; Schaefer et al., 2016). In addition to the overall propensity, an individual worker's attitude and perception related to workplace technologies

and robots may influence trust. Unlike domestic robots that are selected and adopted by those who will use the robots, workplace robots are often adopted to work environments by employers regardless of acceptance or preference of individual workers who work with the robots. Thus, workers' overall attitudes toward robots may have significant impacts on their trust. For instance, many workers may perceive robots as a threat because they fear job replacement by robots in the workplace, which can result in low acceptance of robots in workplaces (Sanders, Kaplan, Koch, Swartz, & Hancock, 2019). When workers have negative attitudes, they are less likely to trust and rely on robots.

Certain personality traits also influence an individual's trust toward systems and technologies. Prior studies suggested that users are more likely to trust or rely on automated systems when they are extraverted, emotionally stable, and have intuitive rather than sensing personalities (Mcbride, Carter, & Ntuen, 2012; Merritt & Ilgen, 2008; Schaefer, 2013; Szalma & Taylor, 2011). For instance, Szalma and Taylor (2011) suggested that the trait of neuroticism is correlated with a lower level of agreement with automated advice, and Merritt and Ilgen (2008) demonstrated that extroverts are more likely to trust automated machines than introverts do. Although underlying mechanisms of these relationships are not fully understood, these findings may suggest that regardless of a robot's performance or functionality, an individual's level of trust may be determined by their characteristics.

In addition to individual propensity, attitude, and personality, various demographic factors such as age and gender can be associated with an individual's trust in robots and robotic environment. As diversity in workforces continues to increase, potential effects of the demographic factors on the interactions with robots are likely to have increasing impacts in workplaces. For instance, workforces are rapidly aging in many industries, which increases the impacts of age-related differences in workplaces. One of the issues that can be of particular interest to the older workers is the adoption of new industrial technologies. As new technologies continue to be introduced to many workplaces, these new technologies must be well-accepted by all users including older workers who may differ significantly from younger workers. For example, older workers may have different technology familiarity, health conditions, or injury risks. Although robotic technologies may provide greater benefits to older populations who generally have stronger needs for physical, cognitive, and social assistance in daily activities as well as during work performance, previous findings suggest that older adults are more likely to show negative emotional responses toward robots and tend to mistrust robots more than younger adults do (Sanders, Oleson, Billings, Chen, & Hancock, 2011; Scopelliti, Giuliani, & Fornara, 2005). Age-related

differences in attitude and trust toward robots is an important demographic determinant for an individual worker's safe and effective interaction with robots in workplaces.

Other demographic factors are less decisive and based on limited empirical evidence (Sanders et al., 2011; Schaefer, 2013). For example, whereas some studies did not support gender- or ethnicity-related differences in trust in robots (Schaefer, 2013), other studies found significant effects of such demographic factors. For instance, women showed greater mistrust in robots compared to men (Scopelliti et al., 2005), and general trust in robots varies across different cultures (Evers, Maldonado, Brodecki, & Hinds, 2008; Li, Rau, & Li, 2010). More research is warranted to investigate the effects of various demographic factors on human-robot trust because it can provide significant implications given the increasingly diverse work environments.

Cognitive abilities and skills

Individual differences in various cognitive abilities and skills can be associated with worker trust in industrial robots, especially when they perform complex cooperative tasks with the robots. Less research interest has been given to the effects of cognitive abilities and skills on trust in automation and robotic technologies compared to individual characteristics. Limited empirical studies investigated individual differences in only a few cognitive abilities, including working memory and attentional capacity (Hancock et al., 2011; Rovira, Pak, & McLaughlin, 2017; Schaefer et al., 2016).

Working memory involves maintaining and processing information during complex cognitive tasks, and individual differences in working memory capacity predict performance on a wide range of real-world tasks (Engle, 2002). Individuals with lower working memory capacity have been shown to have worse performance in complex tasks that demand cognitive resources than those with higher working memory capacity (Engle, 2002). Due to the limited capacity, workers with lower working memory may need to rely more on automation or robots than those with superior working memory to perform tasks. Thus, an individual worker's working memory is likely to be negatively associated with trust in robots. One study, in which researchers investigated the effect of working memory on trust in automation, supported this relationship and found that inferior working memory was associated with more trust in automation (Rovira et al., 2017). This study further suggested that participants with lower working memory calibrate trust inappropriately and rely on the automation even when it fails (Rovira et al., 2017).

Similar relationships have been observed with attentional capacity (Chen & Barnes, 2012; Parasuraman & Manzey, 2010). Studies demonstrated that individuals with lower attentional resources showed increased reliance on automation (Chen & Barnes, 2012), and operators increased their reliance on automation when less attentional resources were available due to the task complexity (Parasuraman & Manzey, 2010), even when doing so could result in undesirable performance. Increased trust in the automated system is likely a strategy that operators adopt to deal with complex information while conserving cognitive resources (Chen & Barnes, 2012). Thus, during human-robot collaborative operations, workers who have a lower level of attentional capacity or severely limited attentional resources available at the moment are likely to increase their trust in the robots helping them with their tasks. However, findings from other studies fail to support significant relationships between attention or engagement and increased trust in robots (Chen & Terrence, 2009; Li et al., 2010), which suggests further investigation is warranted. Specific task environments or levels or types of attentional workloads involving human-robot collaborations may have a significant impact on the associations between individual differences in attentional functions and trust in the robots.

In the context of workplace human-robot interaction, an individual worker's job skills and expertise may have particularly significant impacts on the individual's trust in robots and robotic environments. When workers work on a collaborative task with robots, they may rely more on robots if they have limited job knowledge or expertise required for completing the tasks. Prior empirical evidence showed that job-related skills and knowledge gained through prior experience affected a worker's level of trust and reliance on robots (Desai, 2007; Lee & Moray, 1994). As a worker's self-confidence in the task increases, the propensity to trust in the automation decreases because the worker no longer needs the automation's help (Lee & Moray, 1994). Besides, workers may transfer knowledge and skills required to work with industrial robots from past experiences in similar situations, such as working with industrial machines or other relevant technologies, which can affect the initial level of trust in new robots. An individual worker's prior experience with robots in different contexts such as commercial or personal robots is potentially transferred to workplace robots, which can positively affect their trust in industrial robots. Prior experiences with robotic technologies can help individuals understand and predict behaviors of automated systems and robots, which would result in the appropriate and effective use of the systems and a high level of trust. Experimental and field studies have demonstrated that system operators who are accustomed to using automation trusted automation more compared to those who were not as familiar with the technology

(Lee & See, 2004; Riley, 1994). However, knowledge and skills transferred inappropriately can result in workers' over- or undertrust in industrial robots.

How worker's trust in robots changes over time

In the previous section, we discussed the effects of individual characteristics and differences on an initial level or a momentary state of trust in automation and robots. When an individual worker is initially exposed to a robot or robotic workplace, the worker's initial level of trust is determined by various individual, robot-related, and environmental factors. Then, as interactions with the robots continue, the trust may increase or decrease. Thus, it is important to understand the process of an individual's trust calibration and the factors that influence the calibration process. Understanding the development and changes of an individual's trust in robots is of particular interest to the workplace settings because workers may have lower initial trust in robots than users of personal robots do because workers often do not choose the robots that they use.

Several theoretical models illustrate the dynamic processes of trust development in automation (French, Duenser, & Heathcote, 2018). For example, a conceptual model from Lee and See (2004) describes the closed-loop process of human-automation trust and reliance, during which the interaction with the automation influences an operator's trust, and trust also influences the interaction with automation. Specifically, an individual operator's initial intention to use or not use the automation is formed from a combination of situational factors and the person's attitudes that are based on various individual, organizational, and cultural factors (Lee & See, 2004). Then, through experience with the system, the operator assesses the system performance, and this assessment feeds back into the operator's beliefs. This process loop allows an operator to calibrate trust in the automation gradually. The initial development and changes in worker's trust in robots are likely to follow a similar dynamic process. A worker's level of trust in industrial robots after an initial use is likely to change gradually as the worker continues interacting with the robots, and changes of worker's trust in robots would be influenced by combinations of a worker, robot systems, and the contexts within which the interactions occur (Lee & See, 2004).

Given the dynamic nature of the trust calibration process, individual differences in the changes of trust over time can be large. Studies found that while trust changes substantially for some people, for other people, it changes relatively little (Lee & Moray, 1994; Lee & See, 2004). Similar to the roles of human-related factors in the initial level of trust, various

individual characteristics and differences can influence the evolution of worker's trust in robots. For instance, when workers interact with robots for an extended period, they continuously learn how to work more effectively and safely with robots through observing performance and interpreting feedback from the robots. Therefore, individual differences in general cognitive functions and learning ability required to understand and interpret a robot's functions and performance can influence the interactions. Also, employers may provide initial training before adopting robots in their workplaces or additional maintenance training while robots are being used. As individuals vary in learning processes (Jonassen & Grabowski, 2012), individual differences in learning ability would have impacts on the acquisition of knowledge and skills related to working with robots, which results in varying change rates in individuals' trust in robots.

When humans work with robots for advanced collaborative tasks for an extended time, communications and social interactions between them are important in the development of trust. Therefore, individual differences in communication and social skills may influence human-robot trust. The safety of human-robot collaborative operations relies on accurate communications between a robot and a worker. As the information exchange flows in both directions in human-robot interaction, human-to-robot and robot-to-human communications are on an equal level of importance (Murashov et al., 2016). While robots must accurately detect and understand human commands and behaviors and deliver relevant information, such as its current status, cued actions, or underlying intent and goal, to human workers, humans must also have the relevant communication skills. Therefore, individual differences in worker's communication competency impact the development of trust in robots. Workers must have the physical and cognitive capability to detect and interpret robots' intended communications, and an individual worker's ability to accurately detect and interpret cues and information that a robot delivers will influence effective interactions with robots and the development of an appropriate level of trust in robots.

Social aspects of human-robot interaction might not be as critical in the workplace settings as in the usage of personal or service robots in domestic settings. However, as robotic technologies advance, it is possible that workplace robots will soon perform more social roles as assistants, companions, or partners to human workers. If future industrial robots perform tasks that require a high level of social skills such as making conversation with users, the social skills of both humans and robots would be critical to improving trust between them for effective long-term relationships. Prior studies have demonstrated the potential effects of social relationships between humans and robots on trust development. For instance, one study that examined the relationship between trust and etiquette suggested that automation etiquette has an impact on user trust and overall

performance (Parasuraman & Miller, 2004). In this study, people tended to place more trust in a polite robot than is warranted by the robot's actual abilities (Parasuraman & Miller, 2004), which suggests that an accurate interpretation of social cues may play a role in the development of trust in robots.

A broader range of interpersonal skills will be required to have social relationships with robots due to an increase in human-like capabilities such as artificial intelligence. However, there is a lack of studies investigating the effects of individual differences in communication and interpersonal skills on trust in human-robot interaction. Communication and interpersonal skills are most highly relevant in future robotic workplaces in which humans will perform cooperative tasks with robots with advanced decision-making and social capabilities on a daily basis and develop social relationships as coworkers. Future, robots may also perform increasing roles in workplaces such as managerial roles or as personal assistants for individual workers with specific physical or cognitive needs. Thus, the effects of communication and interpersonal skills on the development of changes of trust in robots may have increasing importance in future robotic workplaces.

Implications and recommendations for workplace safety

The following section aims to (1) discuss implications of the factors related to robot trust and provide recommendations for the design of safe robots and robotic work environments, as well as the development of worker training, and (2) point out research gaps and needs in human-related factors involving trust toward robots.

Design of robots and robotic work environments

Designing robots and robotic work environments to fit an individual worker's preference and capabilities enhances worker's trust in industrial robots. Knowledge of human-related determinants of robot trust may allow employers to predict an individual worker's trust in robots and robotic workplaces based on the person's trust propensity, personality, demographic factors, and skills and expertise. Robot manufacturers and integrators can design robots and robotic workplaces to accommodate worker characteristics and individual differences to improve worker trust. For instance, individual differences in cognitive abilities such as working memory and attentional capacity affect human-robot interaction and trust. Therefore, the design of work routines and task allocations among humans and robots must consider not only the task and environmental contexts but also individual differences so that the robotic workplace

can reduce workers' workload and enhance trust in robots. For example, for workers who suffer from reduced attention or working memory resources, robots may be designed to operate in more predictable behaviors and provide more information on their actions and goals so that the workers can establish an appropriate level of trust.

Achieving individual workers' trust in robots through designs can be challenging. Robot manufacturers and integrators or employers may first need to assess trust among specific individuals or worker groups. The risk assessment that is used for estimating workplace risks can be a useful methodology to apply to trust assessment. Risk assessment involves identification and evaluation of hazards and risk factors that have the potential to cause risks in workers and environments to effectively eliminate, control, and reduce the risks, and is often conducted with a task-based, environment-specific approach. Trust assessment can be conducted to identify and analyze potential events that may negatively impact trust among workers while considering influencing individual-related factors in a given robotic work environment. A comprehensive trust assessment may reveal that additional design requirements are needed to improve or adjust trust among specific individual workers.

Designing robots and robotic work environments to fit the user population can be particularly important for the aging workforce. Industrial robots may extend the availability or duration of employment, or provide new opportunities not previously available to a wider range of workers, including older adults. However, it can be particularly difficult to achieve older workers' trust in the robotic systems, given the negative attitudes toward robotic technologies and age-related differences in skills and capabilities among the older populations. Older workers may benefit most from industrial robots that are specifically designed to mitigate the effects of cognitive impairments on robot interaction and to promote trust in robots. For example, adjusting speeds and complexity of human-robot collaborative tasks, adding design features, enhancing the accessibility of the technology, and allowing robots to give more directive and explanatory instructions can be effective in promoting trust and effective use of industrial robots among older workers. Furthermore, older workers may have specific preferences in task allocations. For example, older people are found to prefer assistive robots to be used for certain types of tasks involving chores, manipulating objects, and information management, while they prefer human assistants for the tasks involving personal care and leisure activities (Smarr et al., 2014). Further research is warranted to investigate age-related differences in preferences for human-robot task allocations in workplaces to design the most effective and trusting robotic work environments.

Furthermore, it is particularly important if robots perform some degree of social roles that they should be designed according to the design

preferences of an individual or the group of users who directly use workplace robots. Going against preferences may trigger distrust and resistance to using robots. Personal preference may have relatively smaller impacts on workplace robots than domestic or public robots. However, if an industrial robot is used by the same individual or group of workers for an extended time, the personalization of a robot can be an effective way to achieve a high level of trust among the workers. Personalization options in the robot design may include selections of communication interface and styles, additional protective covers and skins, or even appearances of robots.

Finally, although our chapter does not directly discuss the effects of robot-related factors on trust, a robot's functionality and performance, such as predictability and reliability, are key factors to achieve worker's trust (Montague, Asan, & Chiou, 2013; Thielsch, Meefsen, & Hertel, 2018). The robot-related factors may interact with human-related factors. For instance, there can be individual differences in perception of the robot's functionality due to various individual factors such as cognitive abilities, expertise, or age, which may influence an individual's trust. Therefore, the design of robots and robotic work environments needs to consider broader aspects of individual characteristics and differences of the workers who work inside the robotic workplaces.

Worker training

Worker training that builds on an individual's characteristics, skills, and prior experience with workplace robots is most effective in improving worker skills and instructing them on how to work safely with robots. Both informal and formal training can enhance a worker's trust in robots and robotic work environments because the training allows a worker to establish accurate mental models regarding how robots operate and potential hazards and risks associated with robotic work environments. For instance, studies show that training can reduce initial user biases and facilitate trust development by giving users information about a robot's capabilities and behaviors, and potential risks associated with robots (Freedy, DeVisser, Weltman, & Coeyman, 2007; Oleson, Billings, Kocsis, Chen, & Hancock, 2011). Empirical studies suggest that the amount of training an individual receives affects the development of trust in the automated system, mostly because the individual's level of expertise in the system improves (McBride & Morgan, 2010). Workers need to have a very well-calibrated understanding of what the capabilities are of the robots they are working with so that they do not over- or undertrust the robots (Sanders et al., 2019). Thus, training and interventions should be designed to provide relevant and accurate information regarding robots and deliver information effectively.

Understanding of the effects of an individual worker's characteristics and differences in traits and abilities and the levels of job and robot-related experience provide implications in the development of effective worker training. Human-related factors influencing trust can be used to identify strategies for training and behavioral interventions (Oleson et al., 2011). For instance, personalized training can provide the most relevant training to individual workers to maximize the effectiveness of the training. Furthermore, research findings on the effects of cognitive abilities on trust provide significant implications in the development of training for workers with lower cognitive functioning. Workers with cognitive deteriorations are likely to have a higher level of trust in robots regardless of the system's capabilities and reliability (Chen & Barnes, 2012; Parasuraman & Manzey, 2010; Rovira et al., 2017). Thus, training strategies and interventions can specifically aim to train these workers to calibrate their trust to an accurate level. Furthermore, while low cognitive abilities and job knowledge are associated with overtrust in robots, which results in misuse and overreliance, a higher level of expertise can also be problematic. Workers with higher expertise and skills may have a lower initial level of acceptance of robots because of high confidence in themselves, even when a robot can further improve their job performance. On the other hand, a worker's prior positive experience and skill with robots or other related machines or technologies may result in a higher initial level of trust or even overconfidence in robots. These findings suggest that worker training needs to be specifically designed to provide relevant training that fits an individual's ability, skill, and expertise levels.

Organizational culture and climate determine the successful and safe implementation of robots in workplaces, and having adequate and supervised familiarization periods during training can promote worker acceptance and safety. For instance, the way the technology is presented or labeled can influence trust. People tend to have much higher trust when they are told that they are interacting with a partner rather than an opponent (Burnham, McCabe, & Smith, 2000; Dunning, Anderson, Schlosser, Ehlebracht, & Fetchenhauer, 2014). Worker training can be designed to label a robot's functions and purpose positively to build the initial trust foundation among the workers who will interact with them. Training can also apply to supervisors and managers who are introducing or expanding the use of robots in the workplace. Since trust is engendered by management commitment to worker safety (McLain, 2014), there should be strong management commitment to training programs and the overall safety culture or climate of the organization. Although our previous discussion did not specifically include the effects of organizational-level differences on human-robot trust, the workplace and management culture should be an important aspect of training aiming to improve worker trust in robotic workplaces.

Research needs

As robotic technologies continue to advance, human-robot interaction in workplaces will become more complex and dynamic. Given the increasing interactions between workers and robots, extensive research is warranted to investigate a wide range of psychological aspects of human-robot interaction. In the previous sections, we reviewed a few empirical studies investigating individual characteristics and differences that can influence worker trust in industrial robots. However, there still are large gaps in research on the determinants of human trust in robots. For instance, a prior metaanalysis demonstrated the relatively smaller effects of human-related factors on trust, but only a few empirical studies on ability-based and personality-based factors are included in the analysis (Hancock et al., 2011). In comparison to robot-related or environmental factors, human-related determinants of trust have insufficient empirical evidence (Hancock et al., 2011). This limited evidence suggests a strong need for more research that investigates human-related factors.

Another significant research gap results from the limited empirical studies on human trust in robots. Some of our discussions were based on evidence from studies investigating automation systems but not robotic systems. As human trust in robots may have unique aspects that are different from trust in automation, future research needs to investigate such differences empirically. Furthermore, no research has compared human trust in traditional industrial robots versus emerging types of robots, which results in significant gaps in our understanding of how human trust is different when the interaction is more dynamic and often even social. Given that the quality of human interaction with collaborative or coexisting robots is different from that with traditional industrial robots, investigating such differences will provide significant implications in the development of safe future robotic workplaces.

In addition to overall limited research, there is a lack of research that investigates trust in human-robot interactions in the context of workplaces. The Center for Occupational Robotics Research at the National Institute for Occupational Safety and Health (NIOSH) recognized the research gaps and thus identified research needs to enhance worker safety and health in robotic work environments (NIOSH, n.d.-b). These research needs that are identified by the research agency that is responsible for and has expertise in conducting research and making recommendations for the prevention of work-related injury and illnesses will serve as a useful guideline for future studies on human-robot trust in the workplace. The research needs to align with the four types of occupational safety and health research: (1) basic/etiologic, (2) intervention, (3) translation, and (4) surveillance. Among the four research types, the effects of worker trust and associated individual characteristics and cognitive differences are

primarily included in basic/etiologic and intervention research. The primary aim of basic/etiologic research in occupational robotics safety is to establish scientific knowledge and the research needs in this area include studies on robot technologies and engineering features and task-related and environmental risk factors. Specifically, the basic/etiologic research needs list human-related factors, including workers' cognitive capability and physical and physiological characteristics, human workers' acceptance, trust, and emotional engagement associated with robots and their impact on worker health and safety behaviors. Similarly, the intervention research needs include specific research topics that are relevant to establish effective safety interventions and improve human-robot interaction in workplaces. For instance, the center lists future research needs on the evaluation of training that can help workers acquire relevant skills, knowledge, and abilities that are necessary to safely operate advanced robots in complex and dynamic industrial environments, which aligns well with our previous discussion on the development of worker training for robot use and trust. In particular, this line of research needs includes an assessment of worker skills and knowledge needed to calibrate their trust in robots accurately during complex and dynamic interactions with robots. In addition, translation research needs emphasize the importance of research for organizational and management programs to improve safe human-robot interaction in workplaces, which may include organizational and management programs and solutions for improving worker trust in workplace robots.

Conclusion

Emerging types of industrial robots have introduced new challenges in many robotic workplaces as interactions between human workers and robots have increased. Worker trust in robots is crucial for safe and effective human-robot collaboration performance, and it may also influence worker acceptance, satisfaction, and stress in the robotic work environments. Human-related factors including worker demographics and individual differences are significantly related not only to an initial level of an individual worker's trust in robots but also to the changes in trust over time. Identification of factors involving trust in robots has implications for the design of robots and robotic work environments, as well as the development of training for workers who interact with robots. Robot and robotic workplace designs and worker training can specifically aim to provide relevant assistance for the individual workers or worker groups who are likely to have an inappropriately lower or higher level of trust. Further research is warranted to empirically investigate a wider

range of individual characteristics and differences involving worker trust and the associated effects on workplace safety.

Disclaimer

The opinions and conclusions in this manuscript are those of the author(s) and do not necessarily represent the views of the National Institute for Occupational Safety and Health.

References

American National Standard Institute [ANSI] (2012). *ANSI/RIA R15.06-2012 industrial robots and robot systems—Safety requirements.* Ann Arbor, MI: Robotic Industries Association.

Burnham, T., McCabe, K., & Smith, V. L. (2000). Friend-or-foe intentionality priming in an extensive form trust game. *Journal of Economic Behavior & Organization*, *43*, 57–73.

Cameron, D., Aitken, J., Collins, E., Boorman, L., Chua, A., Fernando, S. , et al. (2015). Framing factors: The importance of context and the individual in understanding trust in human-robot interaction. In: *Proceedings of the IEEE/RSJ international conference on intelligent robots and systems (IROS) 2015, Hamburg, Germany.*

Chen, J. Y., & Barnes, M. J. (2012). Supervisory control of multiple robots: Effects of imperfect automation and individual differences. *Human Factors*, *54*(2), 157–174.

Chen, J. Y. C., & Terrence, P. I. (2009). Effects of imperfect automation and individual differences on concurrent performance of military and robotics tasks in a simulated multitasking environment. *Ergonomics*, *52*(8), 907–920.

Desai, M. (2007). *Sliding scale autonomy and trust in human-robot interaction. (Master's Thesis).* Retrieved from http://robotics.cs.uml.edu/fileadmin/content/publications/thesis/mdesai-thesis.pdf.

Desai, M., Stubbs, K., Steinfeld, A., & Yanco, H. (2009). Creating trustworthy robots: Lessons and inspirations from automated systems. In *Proceedings of the society for the study of artificial intelligence and the simulation of behaviour (AISB) convention: New frontiers in human-robot interaction, Edinburgh, Scotland.*

Dragan, A. D., Bauman, S., Forlizzi, J., & Srinivasa, S. S. (2015). Effects of robot motion on human-robot collaboration. In: *Proceedings of the 2015 10th ACM/IEEE international conference on human-robot interaction (HRI), Portland, OR.*

Dunning, D., Anderson, J. E., Schlosser, T., Ehlebracht, D., & Fetchenhauer, D. (2014). Trust at zero acquaintance: More a matter of respect than expectation of reward. *Journal of Personality and Social Psychology*, *107*(1), 122–141.

Engle, R. W. (2002). Working memory capacity as executive attention. *Current Directions in Psychological Science*, *11*(1), 19–23.

Evers, V., Maldonado, H., Brodecki, T., & Hinds, P. (2008). Relational vs. group self-construal: Untangling the role of national culture in HRI. In: *Proceedings of the 2008 3rd ACM/IEEE international conference on human-robot interaction (HRI), Amsterdam, Netherlands.*

Fasola, J., & Matarić, M. J. (2013). A socially assistive robot exercise coach for the elderly. *Journal of Human-Robot Interaction*, *2*(2), 3–32.

Freedy, A., DeVisser, E., Weltman, G., & Coeyman, N. (2007). Measurement of trust in human-robot collaboration. In: *Proceedings of the 2007 international symposium on collaborative technologies and systems, Orlando, FL.*

French, B., Duenser, A., & Heathcote, A. (2018). *Trust in automation—A literature review.* (CSIRO Report EP184082) Australia: CSIRO.

Groom, V., & Nass, C. (2007). Can robots be teammates?: Benchmarks in human–robot teams. *Interaction Studies*, *8*(3), 483–500.

Hancock, P. A., Billings, D. R., Schaefer, K. E., Chen, J. Y., De Visser, E. J., & Parasuraman, R. (2011). A meta-analysis of factors affecting trust in human-robot interaction. *Human Factors*, *53*(5), 517–527.

Heerink, M., Kröse, B., Evers, V., & Wielinga, B. (2010). Assessing acceptance of assistive social agent technology by older adults: The Almere model. *International Journal of Social Robotics*, *2*(4), 361–375.

International Federation of Robotics (IFR) (2019). *Executive summary world robotics 2019 industrial robots*. Frankfurt, Germany: International Federation of Robotics. Retrieved from https://ifr.org/downloads/press2018/Executive%20Summary%20WR%202019%20Industrial%20Robots.pdf.

Jonassen, D. H., & Grabowski, B. L. (2012). *Handbook of individual differences, learning, and instruction.* New York, NY: Routledge.

Lee, J. D., & Moray, N. (1994). Trust, self-confidence, and operators' adaptation to automation. *International Journal of Human-Computer Studies*, *40*(1), 153–184.

Lee, J. D., & See, K. A. (2004). Trust in automation: Designing for appropriate reliance. *Human Factors*, *46*(1), 50–80.

Li, D., Rau, P. P., & Li, Y. (2010). A cross-cultural study: Effect of robot appearance and task. *International Journal of Social Robotics*, *2*(2), 175–186.

McBride, M., Carter, L., & Ntuen, C. (2012). The impact of personality on nurses' bias towards automated decision aid acceptance. *International Journal of Information Systems and Change Management*, *6*(2), 132–146.

McBride, M., & Morgan, S. (2010). *Trust calibration for automated decision aids*. Retrieved from https://www.ihssnc.org/portals/0/Documents/VIMSDocuments/McBride_Research_Brief.

McLain, D. L. (2014). Sensitivity to social information, social referencing, and safety attitudes in a hazardous occupation. *Journal of Occupational Health Psychology*, *19*(4), 425.

Merritt, S. M., Heimbaugh, H., LaChapell, J., & Lee, D. (2013). I trust it, but I don't know why: Effects of implicit attitudes toward automation on trust in an automated system. *Human Factors*, *55*(3), 520–534.

Merritt, S. M., & Ilgen, D. R. (2008). Not all trust is created equal: Dispositional and history-based trust in human-automation interactions. *Human Factors*, *50*(2), 194–210.

Montague, E., Asan, O., & Chiou, E. (2013). Organizational and technical correlates of nurses' trust in a smart IV pump. *Computers, Informatics, Nursing*, *31*(3), 142–149.

Murashov, V., Hearl, F., & Howard, J. (2016). Working safely with robot workers: Recommendations for the new workplace. *Journal of Occupational and Environmental Hygiene*, *13*(3), D61–D71.

National Institute for Occupational Safety and Health (NIOSH). (n.d.-a). Retrieved from https://www.cdc.gov/niosh/topics/robotics/aboutthecenter.html

National Institute for Occupational Safety and Health (NIOSH). (n.d.-b). Retrieved from https://www.cdc.gov/niosh/topics/robotics/research.html

Oleson, K. E., Billings, D. R., Kocsis, V., Chen, J. Y., & Hancock, P. A. (2011). Antecedents of trust in human-robot collaborations. In: *Proceedings of the 2011 IEEE international multidisciplinary conference on cognitive methods in situation awareness and decision support (CogSIMA), Miami, FL.*

Parasuraman, R., & Manzey, D. H. (2010). Complacency and bias in human use of automation: An attentional integration. *Human Factors*, *52*(3), 381–410.

Parasuraman, R., & Miller, C. A. (2004). Trust and etiquette in high-criticality automated systems. *Communications of the ACM*, *47*(4), 51–55.

Parasuraman, R., & Riley, V. (1997). Humans and automation: Use, misuse, disuse, abuse. *Human Factors*, *39*(2), 230–253.

Riley, V. A. (1994). *Human use of automation.* (Unpublished doctoral dissertation) Minneapolis, MN: University of Minnesota.

Rovira, E., Pak, R., & McLaughlin, A. (2017). Effects of individual differences in working memory on performance and trust with various degrees of automation. *Theoretical Issues in Ergonomics Science*, *18*(6), 573–591.

Sanders, T., Kaplan, A., Koch, R., Swartz, M., & Hancock, P. A. (2019). The relationship between trust and use choice in human-computer interaction. *Human Factors*, *61*(4), 614–626.

Sanders, T., Oleson, K. E., Billings, D. R., Chen, J. Y., & Hancock, P. A. (2011). A model of human-robot trust: Theoretical model development. In: *Proceedings of the human factors and ergonomics society 55th annual meeting, Las Vegas, NV*.

Schaefer, K. (2013). *The perception and measurement of human-robot trust. (Doctoral dissertation).* Retrieved from http://purl.fcla.edu/fcla/etd/CFE0004931.

Schaefer, K. E., Chen, J. Y., Szalma, J. L., & Hancock, P. A. (2016). A meta-analysis of factors influencing the development of trust in automation: Implications for understanding autonomy in future systems. *Human Factors*, *58*(3), 377–400.

Scopelliti, M., Giuliani, M. V., & Fornara, F. (2005). Robots in a domestic setting: A psychological approach. *Universal Access in the Information Society*, *4*(2), 146–155.

Smarr, C.-A., Mitzner, T. L., Beer, J. M., Prakash, A., Chen, T. L., Kemp, C. C. , et al. (2014). Domestic robots for older adults: Attitudes, preferences, and potential. *International Journal of Social Robots*, *6*, 229–247.

Szalma, J. L., & Taylor, G. S. (2011). Individual differences in response to automation: The five factor model of personality. *Journal of Experimental Psychology: Applied*, *17*(2), 71.

Thielsch, M. T., Meefsen, S. M., & Hertel, G. (2018). Trust and distrust in information systems at the workplace. *PeerJ*, *6*, e5483.

CHAPTER

6

A time to trust: Trust as a function of time in human-robot interaction

A.D. Kaplan[a], *T.T. Kessler*[a], *T.L. Sanders*[b], *J. Cruit*[a], *J.C. Brill*[c], *and P.A. Hancock*[a]

[a]Department of Psychology, University of Central Florida, Orlando, FL, United States, [b]Transportation Innovation Center, The MITRE Corporation, McLean, VA, United States, [c]Air Force Research Laboratory, Wright-Patterson Air Force Base, OH, United States

Trust in robots

Trust can be defined as "a person's calculated exposure to risk of damage from the activities of influential others" (Hancock et al., 2011). The questions remain of whether robots are indeed "influential others" and whether their actions are capable of increasing one's "risk of damage." History has shown that robots have always been capable of causing damage. The first known case of death by a robot was that of Robert Williams in 1979, wherein a worker at the Ford Motor Company was killed by a robot arm (Crockett, 2018). Since that time, many individuals have been involved in robot-related fatalities, usually in industrial settings. Notably, the Center for Disease Control (CDC) labeled interaction with machinery as one of the leading work-related causes of injuries and fatalities, with 61 deaths directly attributed to robots between 1992 and 2015 (Manufacturing, 2019).

Although interactions with industrial robots account for the majority of these reported injuries and deaths to date, the militarization of robots is a domain that cannot be ignored when evaluating trust in robotics. The potential for non-accidental death caused by robots in the military arena

Trust in Human-Robot Interaction
https://doi.org/10.1016/B978-0-12-819472-0.00006-X

© 2021 Elsevier Inc. All rights reserved.

gives a significant rise to safety concerns on the weaponization of autonomous entities (Sparrow, 2007). As such, robots are especially influential because they can expose humans—whether operators, workers, or enemy combatants alike—to danger, this being evidence that robots indeed fit Hancock's previously established criteria for a trustee. Thus, interactions with robots require a calculation of trust. But how do people, as trustors, perform these calculations? How do they determine whether the exposure to risk is acceptable or justified? What follows here examines the factors involved in these trust calibrations.

Three categories of trust antecedents

The three basic categories of trust antecedents consist of the human, robot, and environmental sources (Hancock et al., 2011; Hancock, Kessler, Kaplan, Brill, & Szalma, 2020). Meta-analysis established that each of the previously explored individual precursors of trust fit into one of these three categories.

Human-related antecedents include the preexisting characteristics and abilities of the human user such as demographic information (e.g., age, gender, ethnicity), preexisting attitudes (e.g., propensity to trust, attitudes toward robots), personality traits (e.g., extroversion, self-confidence), and abilities or perceptions of abilities. Ability-based precursors to trust include items such as expertise, competency, and situation awareness. These human-related antecedents are all factors that people use to calibrate their trust in a robot; somewhat independent of the characteristics of the robot itself. Such human-related antecedents to trust remain relatively stable across interactions with different robots.

Robot-related antecedents are those characteristics of the robot that is being trusted by the human. These are categorized as either performance-based antecedents or attribute-based antecedents. Attribute-based antecedents of trust include the look and feel of the robot, including type (e.g., industrial, home), features, and components, level of anthropomorphism, as well as robot personality where applicable. Performance-based antecedents relate more to behavior and include dependability, reliability, predictability, failure rates, and transparency. These traits vary between robots. Thus, a human who displays high trust in one very reliable robot may still exhibit low trust in a less reliable robot.

Last, environmental, or contextual factors are the attributes shared by both the human and the robot agent relative to the interaction. Some antecedents relate to team collaboration, such as shared mental models and communication. Others relate to the task at hand, such as task complexity. The physical environment where the interaction occurs is also a factor. These factors exist in the particular context of interaction and can emerge

and evolve during said interaction. Due to their emergent nature, these environmental antecedents may even be different for the same human and robot pair during a different interactional episode. In the following sections, each of the trust antecedents will be explored more thoroughly.

Human-related antecedents of trust

Human characteristics

Early studies in human-robot interaction (HRI) often focused on characteristics related to the robot. However, the rise in human-centered thinking has led to a more frequent measurement of human characteristics in HRI experimentation. Such investigations look to topics such as personality, age, attitudes toward automation and robots, culture, and self-perception. For instance, when working with a very lifelike Geminoid robot participants' trust in the robot is positively related to their own level of extroversion (Haring, Matsumoto, & Watanabe, 2013). Following upon this important propensity, a positive relationship between participants' extroversion and their ratings of robot anthropomorphism exists (Kaplan, Sanders, & Hancock, 2018). These findings suggest that personality can influence participants' perceptions of a robot's characteristics.

Several studies show age as a precursor to trust, specifically, younger participants are much more likely to trust robots than their older counterparts (Erebak & Turgut, 2018; Heerink, 2011; Schaefer, 2013; Scopelliti, Giuliani, & Fornara, 2005). Although older individuals (depending on their profession or lifestyle) may actually have had more experience with automation, it is more likely that younger individuals have spent a proportionally larger amount of their lives surrounded by automation, particularly in their formative years. This proportionately longer period of habituation may be what leads to their higher trust. It is not simply the fact that they have spent a large proportion of their time with automation, but that, reciprocally, they have spent virtually none of their life functioning without it.

The influence of culture in HRI has been examined (Rau, Li, & Li, 2009; Wang, Rau, Evers, Robinson, & Hinds, 2010), though not sufficiently to make conclusive determinations of the ways in which robot trust may vary by each culture. To an extent, that may not matter. Generalizations across cultures might not be meaningful in terms of technology since the understood cultural categories (e.g., American, Chinese) do not translate into a common experience with technology, automation, or robotics. Moreover, there are many different cultures that can exist within a relatively small geographic region. For instance, a person from rural America may have a very different set of experiences predicting their trust in robots compared to an urban American. We contend this type of cultural categorization would likely be more valid than one based on nationality or ethnicity.

Experience and abilities

Abilities and skills with automation typically come with experience. Although these ability-based characteristics have not been extensively explored in the literature, people with substantial previous experience are becoming more prevalent as automation becomes more commonplace and there is a positive relationship between trust and a user's expertise or prior experience (Hancock et al., 2020; Heerink, 2011; Sanders et al., 2017; Schaefer, 2013). This suggests that trust is emerging, uniquely, from each interaction as it is built on the experiences of the previous encounter.

Robot-related antecedents of trust

Robot-related antecedents of trust are broadly categorized into performance-based factors (how well the robot functions) and attribute-based factors (how the robot looks, feels, and behaves). Overall, robot-related antecedents, specifically those related to reliable performance, have generally been found to be most closely related to trust development and sustenance though robot personality does also play a factor.

Performance-based factors

When a robot performs well by successfully accomplishing a task, the human operator's trust in that robot generally rises. After all, a robot's performance is one of the few objective ways to properly judge it without relying on one's own attitudes or assumptions. Many studies have examined robot performance as a precursor to trust. When monitoring a robot while performing a separate task, people rate robots as more trustworthy (van den Brule, Dotsch, Bijlstra, Wigboldus, & Haselager, 2014). Similarly, when first primed with a well-behaving robot, the second robot gets a more trustworthy rating; whereas, when primed with bad robot behavior, people rate the second robot as being less trustworthy (De Bruijn, 2013). Prior expectations of a robot's good performance are also significant predictors of trust. For example, people who believe home health care robots will perform well state that they are also more trusting of those robots (Alaiad & Zhou, 2014). These results are fairly intuitive; the better a robot performs, the more trustworthy it appears.

Reliability is one specific facet of performance, related to the consistent nature of that performance (Stowers et al., 2017). If a robot sometimes performs well, but sometimes performs poorly, that robot is unreliable. The relationship between robot reliability and trust has been examined with similar findings as to those looking at the influence performance in general on trust (de Visser & Parasuraman, 2011; Desai, Kaniarasu, Medvedev, Steinfeld, & Yanco, 2013; Rau et al., 2009). Closely related to reliability is predictability. Predictability here represents a measure of

how clear a robot's intentions and future actions are to its operator (Dragan, Lee, & Srinivasa, 2013). This trait is also positively related to trust, with more predictable behavior on the part of the robot leading to higher trust on the part of the user (Biros, Daly, & Gunsch, 2004). The issue of robot intention is closely allied to another factor, which is transparency. Transparency refers to the amount of information that a robot shares with its operator (see Lyons & Havig, 2014). Transparency has been studied extensively and is well represented in the present body of literature. This particular trait can also be interactional in nature because the amount of information the operator receives and is able to comprehend is somewhat dependent upon the individual at hand. Trust in a robot that provides constant information (information is always present) is higher than trust in a robot that provides only contextual information (short descriptions of specific instances), and contextual information robots are trusted more than those that only provide minimal information (Sanders, Wixon, Schafer, Chen, & Hancock, 2014).

Robot attributes

People derive their impression of robots from their attributes, often before any interaction occurs. This makes attribute-based factors especially important in trust calibration. Among these factors is robot personality—though, of course, it can be argued that at the present time robots do not, in fact, have their own personalities per se. Any "personality" a robot may exhibit is largely a reflection of its programmer. However, robots can certainly *appear* to have personalities, and those fabricated personas do influence trust. Joose and colleagues (2013) demonstrate that extroverted social robots are more trustworthy than introverted ones. In addition, when a robot exhibits empathy or a social personality, it is deemed more trustworthy (Looije, Neerincx, & Cnossen, 2010). Kindness and politeness likewise exert positive effects on trust. For example, a robot driver attempting to overtake someone in a video is perceived as more trustworthy when it politely stops or slows for pedestrians over the one who speeds up or maintains their speed (Tsui, Desai, & Yanco, 2010).

One specific aspect of both robot appearance and behavior that has been studied multiple times is the degree of anthropomorphism or the robot's similarity to a human. In this domain, people trust a more humanoid robot over a less humanoid robot when robots are taking on the role of the caretaker (Erebak & Turgut, 2018). Additionally, the more anthropomorphic someone considered a particular robot assistant the more they trust it (Evers, Maldonado, Brodecki, & Hinds, 2008). Similarly, it seems that when a robot is designed with facial expressions that can either remain neutral or change based on social cues in the conversation, people are more trusting of the robot with a socially motivated facial expression

compared to a robot with a neutral expression or one that simply mimics the participant's own facial expression (Gonsior et al., 2011).

Another facet of a robot's behavior concerns its level of autonomy or the ability to self-govern. A robot's level of expressed autonomy is related to an operator's level of trust, though not always positively. When people are asked to make choices based on a hypothetical scenario in which they are lost at sea, they trust a highly autonomous robot less than one with a low level of autonomy (Rau, Li, & Liu, 2013). Also, people are more likely to trust a robot with multiple operators than an autonomous robot, though they trust the autonomous robot more than one with a single operator (Srinivasan & Takayama, 2016). Additionally, people trust a path-finding robot more when that robot waits for an operator to press a button before executing its path strategy (Ullman & Malle, 2017). Although the research findings are somewhat varied, the overall trend is that people trust robots less when they make decisions on their own.

Contextual environmental factors

Environmental or contextual factors refer to those factors shared by both humans and robots during an interaction. Unfortunately, though many influential factors are shared (e.g., task type, culture, physical environment) at present, these prove to have the least theoretical backing in the current body of literature.

Some task characteristics have been explored, and one relevant experiment demonstrated that when participants are meant to work with a robot in a social test of etiquette, they trust it less than when they collaborate on a functional task of discerning which sound is higher pitched (Gaudiello, Zibetti, Lefort, Chetouani, & Ivaldi, 2016). Another demonstrates that people trust a robot more to help find out how to survive on a desert island than a teaching task (Kidd, 2003). Further, people rate robots designed for finding improvised explosive devices more trustworthy than those designed to do manual warehouse work (Sanders, Kaplan, Koch, Schwartz, & Hancock, 2019).

In addition to task type, task difficulty effects have also been evaluated. For instance, as task difficulty rises, so does trust (Stanton & Stevens, 2017). For example, people are more likely to both trust and use an automated system when their task load is higher, leading to the conclusion that such circumstances might even lead to over-trust and over-use (Biros et al., 2004).

Time: An influencer of trust

As well as taking into account each of the previously identified categories of trust antecedents; it is also critical to put more emphasis on the dynamic nature of trust through the lens of temporal influences. Time

(both the moment of the interaction and the moment of measurement) necessarily influences trust.

The construct of trust can be measured at any point in time, and in most research, tends to be measured once at the end of interaction (though it is sometimes measured before and after interaction in order to examine how trust changes). However, this does not mean that a trust level is stable or is evolving in a linear fashion. Measurements taken at one point during interaction will almost certainly differ from those taken at a different point, even if both parties—human trustor and robot trustee—ostensibly remain the same. That is to say that without changes in either party (trustor or trustee), trust nevertheless evolves over time. Generally, the pattern is that trust increases with relationship duration (Anderson & Weitz, 1989).

Over the course of an interaction, performance errors, transparency, and takeover requests can all change an operator's level of trust in a robot, and the fact that trust is dynamic must be taken into account when identifying antecedents of trust. When the time is taken into account in human-robot trust, it is generally in the form of shared tenure or prior experience, and not as a moderating influencer that affects all of the other antecedents. However, time itself does not simply refer to the contextual antecedent of relationship duration or prior experience. Temporal effects may be more specifically broken down into intervals of months or years of shared tenure, or the temporal element can be examined as a function of a specific moment in time. Trust in a robot immediately after it has failed will be different than trust in that same robot a few minutes later when it has had the chance to redeem itself. Trust may similarly be altered immediately following a takeover request from an automated system, regardless of whether or not that request was valid. The human-related antecedents remain the same, as do the factors regarding the robot which one is to judge. The task is no different; the only thing that changes is the time at which trust is measured. Therefore, time is an important antecedent in trust calibration. Yet, time is little accounted for in previous models of human-robot trust, and when it has been, it is generally in the form of shared tenure or prior experience, and not as a moderating influencer that affects all of the other antecedents. Further, as trust is generally measured at the end of an interaction, there is not much empirical evidence to permit elucidation of the profile of the dynamic nature of trust. While the evidence that trust evolves with time is clear, it is often treated as a stable and unchanging trait.

In moderating the entire trust relationship, time also influences the individual antecedents of trust. The human-related antecedent of prior experience with robots has a clear and linear relationship with time. Experience only increases, though at different rates, and the later the individual is surveyed, the higher their experience with robots will necessarily be. Other factors are not so clearly related. A person's culture does not change but how that culture affects their trust in a robot may vary based on any

current cultural phenomenon such as movies, accidents, or military involvement. Each antecedent of trust changes differently based on both the time of the interaction and the time of trust measurement. Thus, time moderates the relationships between these antecedents and trust.

There is some literature which examines time as a factor of trust in HRI, beyond the basic idea of relationship tenure. In one study, researchers found that one robot's good performance increased the level of trust that people felt toward a *different* robot viewed later (De Bruijn, 2013). That is when participants watched a well-performing robot A; their subsequent ratings of trust in robot B were higher. Therefore, it is possible that if the same person were asked to rate the trustworthiness of robot B but had actually been asked *before*, rather than *after*, viewing the well-or-poorly-performing robot A, their response may have been different. This demonstrates the trust level as a function of time. The trust increase could neither be attributed to the second robots' form or capabilities nor to the task. The increase was the result of the moment of measurement taking place directly after viewing the initial robot's performance.

Another experiment examined trust as a function of time by measuring trust at multiple instances during an interaction wherein the robot varied in its reliability (Desai et al., 2013). Desai and colleagues (2013) required participants to navigate robots through a maze, during which they could rely on the robot's ability to read directions (left, right, or u-turn) using a barcode. The directions were sometimes difficult for the robot to interpret; thus the robot's reliability fluctuated. Every 25 s, participants were asked whether their trust in the robot had increased, decreased, or stayed the same. This enabled a dynamic plot of the changes in trust following each decision made by the robot. Researchers found that trust increased overall across trials. However, decreases in robot reliability caused corresponding decreases in trust that slowly recovered as reliability improved (Desai et al., 2013). The fact that trust can recover following poor performance helps illustrate that the point and duration of the measurement is an important factor in determining the trust level. Of course, it may be argued that the participants are simply calibrating their trust based on new information about a robot's overall reliability. However, this argument would not explain perhaps the most interesting finding of the study that failures early in the interaction had a larger negative impact on trust than failures later in the interaction. That is, early failure is more difficult to recover from than later failure, with participants being more forgiving of low reliability late in a trial.

Reliability has already been identified as a positive influence on trust. However, the previously mentioned study of trust's dynamic nature has shown that the overall effect of reliability is not the same at each point in an interaction (Desai et al., 2013). Low reliability has a negative impact on trust but a stronger one at the beginning of an interaction. This has

implications for all trust antecedents. Antecedents related to robot performance certainly change over time. Thus, trust should similarly be examined before, during, and after each action by a robot in order to understand the true effect of these trust antecedents. We see that there is an analog signal of trust influencers, and similarly an analog representation of trust response. The major challenge we face is the linking of the two. Robot attributes may be more static, as a robot's form and level of anthropomorphism vary little. However, the user's reaction to these traits might change, and thus their corresponding level of trust may change as well. With familiarity, an operator might begin to see the robot as more anthropomorphic than they originally judged it, without any change on the part of the robot itself.

Similarly, human-related antecedents of trust change over time. For instance, ability-based antecedents are fluid and prior experience, certainly, is rapidly increasing. Each time that a person interacts with a robot, they gain expertise concerning its characteristics and on their task. Although most characteristics such as demographics are stable, a person's self-efficacy may change, as might their personality traits, based on outside influences. In a brief moment of confidence, one might exhibit similar and corresponding confidence in the robot, but a second later, that attribution might be dissolved. Perhaps it is important to view time not simply as a fourth and equivalent antecedent of trust but as a *pre*-precursor or a super-moderator, that is, something that affects all of the other moderators and sub-moderators before trust calibration even begins.

Fig. 1 illustrates this hypothesized dynamic model of trust. Here, trust is examined on the y-axis with higher y-values corresponding to higher

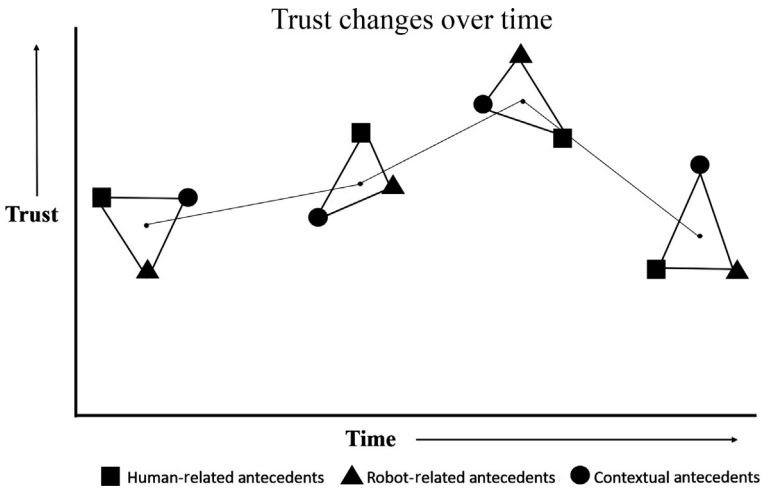


FIG. 1 Model of dynamic trust.

trust. Time is represented on the horizontal axis with a separate point for each time trust is measured. At each measurement point, trust is dependent on the human, robot, and contextual antecedents. Trust based on each factor may be equal, or one factor may elicit significantly higher or lower trust than the other two. The measured level of trust will be the center of a triangle formed by the three factors; however, the actual level of trust of each individual antecedent may be much higher or much lower. Additionally, a particular antecedent category may initially have lower influence than the other two, but at a later measurement point may increase to be the category that contributes the most to overall trust.

In this hypothetical situation, the interaction begins with moderate trust based on human and contextual factors; the task is not particularly difficult; the human has neither a particularly high or low propensity to trust and nor high and low opinions of robots. The robot-related antecedents are low. In this case, let us say that the robot has exhibited poor or no performance and has not yet displayed reliability. Then, let's examine time 2. The human-related antecedents have risen; in this case, the human's situation awareness and self-efficacy have become higher. The contextual precursors to trust are lower on the scale. The task is easy and could be accomplished by a human alone with relatively little difficulty. The largest change in position is that of the robot-related antecedents. In this hypothetical situation, let's say that the robot has displayed high reliability. Time 3 shows a similar pattern, with robot antecedents rising even higher in the scale of trust. Perhaps the robot has performed well; perhaps it has behaved politely. Contextual factors have also risen. The task characteristics or difficulty may have shifted. The human antecedents have fallen slightly. Perhaps the human operator is less self-assured. Finally, at time 4, all factors are close to the middle of the scale. Contextual and human factors remain in the same position as in time 3; robot factors have fallen quite a bit. A dip in robot performance could be to blame.

This extended hypothetical may not exactly fit any specific situation, but it could be used as a stand-in for many an evolving trust relationship. For example, a person unwraps a gift to find a surprise; a Roomba robot vacuum. They then learn to use it and find it neither as complex nor as difficult to use as they expected. Then, it performs unexpectedly well and the human begins to rely on it entirely. Finally, at time 4, it catastrophically fails by either escaping out the pet-door or running through a puddle of mud and subsequently tracking the filth through the house; thus diminishing trust to a level that is higher than the initial trust measurement, but lower than the peak. Similarly, consider an industrial robot in a factory setting, which, at first, is a neutral addition to the assembly line. Later, the employees learn to work alongside it, and their job becomes easier as the robot's reliability rises; however, their self-efficacy falls as more of their work is taken over by the robot. Finally, the demand for high output from

the assembly line makes the task more difficult and also lowers the robot's dependability once demand moves beyond its capacity to fulfill. Thus trust in the robot drops.

Both of these hypothetical situations are plausible demonstrations that a trust relationship can evolve over time, and not just increase with shared tenure. Additionally, within the interactions previously described, trust overall appears to rise steadily and then it declines. However, a closer look reveals that the individual antecedent categories have more dramatic and dynamic shifts within the overall increase. The fact that most measures of trust take place at the beginning or end of an interaction ignores the idea that trust is an active relationship and as such should be measured at multiple points in time in order to truly capture each fleeting moment of proper trust calibration as it happens.

Measurement of dynamic trust

There are a few ways to measure dynamic trust. The first would be to simply measure trust at multiple moments, using the same survey or scale. Comparisons between trust levels at different moments would show researchers the magnitude and direction of each trust change but would be time-consuming and tedious for the participant.

Another method would be to follow the example of Desai et al. (2013) and simply have participants note the direction of trust change after each action by the robot. This would be a faster method of real-time observation, but would only tell researchers whether the trust had increased or decreased since the previous measurement and not the magnitude of the change. Additionally, simply by virtue of being asked multiple times, participants may feel pressured to indicate a change in trust even if no such change occurred.

One might also attempt to place a numerical value on each of the known antecedents of trust in order to determine trust levels without surveying the participant. Such an effort would be largely theoretical, but could hypothetically predict trust levels mathematically. For example, picture trust as the result of the addition of all three influencing factors (robot-related, human-related, and contextual) and all interactive effects, plus variation due to random error. This equation (Eq. 1) may read as:

$$\text{Trust} = X_H X_R X_C + X_H X_R + X_H X_{C+} X_R X_C + X_R + X_H + X_C + \text{error} \qquad (1)$$

where X_H represents the influence of human-related antecedents, X_R represents the influence of robot-related antecedents, and X_C stands for the influence of contextual variables. The interaction effects are included with $X_H X_R X_C$ the three-way interaction between all variables, $X_H X_R$ the interaction of human- and robot-related factors, etc.

Of course, such an equation oversimplifies the complex relationship. Within each predictor of trust, there are many variables. Human-related antecedents include age, gender, self-efficacy, and many others. The influence of human-related factors is not as straightforward as X_H, but is in itself an equation that is much longer but with a similar pattern.

$$X_H = X_{age}X_{gender}X_{self-efficacy} + X_{Age}X_{Gender} + X_{Age}X_{self-efficacy} + X_{Age} + X_{Gender}\ldots + \text{error} \quad (2)$$

Such an equation would need a variable accounting for the influence of every predictor within the category of human-related antecedents of trust. Beyond that, determining the numerical value of each predictor would be difficult as the value would not be concrete (e.g., X_{Age} for a particular person would not be the value of "27" if the person were 27 years of age), but would be the *influence* of the variable, which is a theoretical value explaining its contribution to overall trust levels.

To take the temporal influence into account, time would need to be considered as a moderator of the antecedents of trust used in Eq. (2). That is, the influence of each individual antecedent that makes up each category would be differently affected by time. An equation for determining the influence of human-based antecedents at a point in time may read as:

$$X^T{}_H = X^T{}_{age}X^T{}_{gender}X^T{}_{self-efficacy} + X^T{}_{Age}X^T{}_{Gender} + X^T{}_{Age}X^T{}_{self-efficacy} + X^T{}_{Age} + X^T{}_{Gender}\ldots + \text{error}$$

where T is the influence of time and has a different value for every antecedent. That is, T in the phrase X^T_{age} does not have the same value as T in X^T_{gender}. This is because, again, the variable does not represent an actual value but an antecedent's theoretical influence on trust. If the antecedent in question elicits less trust due to the time, T would have a value of less than one. If the timing elicited higher trust based on one antecedent, that T value would be greater than one.

In order to mathematically represent the ways in which time influences trust, there would be a need for two separate equations. One would explain the actual trust that a person felt toward a robot during their interaction. The other would explain the actual trust that the same person felt toward that same robot during the retrospective measurement. Such theoretical equations do not take into account measurement error or memory failures but do take into account the fact that trust may actually change between interaction and measurement.

From these equations, one could derive an expected trust value. However, each equation, and each variable in it, would have a different value for each human-robot-context triad. When one takes into account memory errors, measurement problems, and the fact that people may be dishonest on surveys, the difficulty of accurate measurement is starkly illuminated.

It is not easy (or, currently, possible) to mathematically determine "trust" but as we collectively work toward a measurement, the influence of time must not be ignored.

Acknowledgments

The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Science Foundation. The author's affiliation with The MITRE Corporation is provided for identification purposes only and is not intended to convey or imply MITRE's concurrence with or support for the positions, opinions, or viewpoints expressed by the author. Approved for public release; distribution unlimited. Case No. 19-3565.

This research was supported, in part, by appointments to the Student Research Participation Program at U.S. Air Force Research Laboratory (AFRL), 711th Human Performance Wing, Airman Systems Directorate, Human-Centered Intelligence, Surveillance, and Reconnaissance Division, Human Trust and Interaction Branch administered by the Oak Ridge Institute for Science and Education (ORISE) through an interagency agreement between the U.S. Department of Energy and AFRL. This paper has been cleared for public release: 88ABW-2019-5662.

References

Alaiad, A., & Zhou, L. (2014). The determinants of home healthcare robots adoption: An empirical investigation. *International Journal of Medical Informatics*, *83*(11), 825–840.

Anderson, E., & Weitz, B. (1989). Determinants of continuity in conventional industrial channel dyads. *Marketing Science*, *8*(4), 310–323.

Biros, D. P., Daly, M., & Gunsch, G. (2004). The influence of task load and automation trust on deception detection. *Group Decision and Negotiation*, *13*(2), 173–189.

Crockett, Z. (2018, January 27). *What happens when a robot kills someone? February 20, 2019*Retrieved from https://www.oshatoday.com/news-digest-item/what-happens-when-a-robot-kills-someone.

De Bruijn, M. L. E. (2013). *The base of trust in human-robot interaction.* Ragboud University Nijmegen.

de Visser, E., & Parasuraman, R. (2011). Adaptive aiding of human-robot teaming: Effects of imperfect automation on performance, trust, and workload. *Journal of Cognitive Engineering and Decision Making*, *5*(2), 209–231.

Desai, M., Kaniarasu, P., Medvedev, M., Steinfeld, A., & Yanco, H. (2013, March). Impact of robot failures and feedback on real-time trust. In *Proceedings of the 8th ACM/IEEE international conference on human-robot interaction* (pp. 251–258). IEEE Press.

Dragan, A. D., Lee, K. C., & Srinivasa, S. S. (2013, March). Legibility and predictability of robot motion. In *Proceedings of the 8th ACM/IEEE international conference on human-robot interaction* (pp. 301–308). IEEE Press.

Erebak, S., & Turgut, T. (2018). Caregivers' attitudes toward potential robot coworkers in elder care. *Cognition, Technology & Work*, 1–10.

Evers, V., Maldonado, H., Brodecki, T., & Hinds, P. (2008, March). Relational vs. group self-construal: Untangling the role of national culture in HRI. In *Human-robot interaction (HRI), 2008 3rd ACM/IEEE international conference on HRI* (pp. 255–262). IEEE.

Gaudiello, I., Zibetti, E., Lefort, S., Chetouani, M., & Ivaldi, S. (2016). Trust as indicator of robot functional and social acceptance. An experimental study on user conformation to iCub answers. *Computers in Human Behavior*, *61*, 633–655.

Gonsior, B., Sosnowski, S., Mayer, C., Blume, J., Radig, B., Wollherr, D. , et al. (2011). Improving aspects of empathy subjective performance for HRI through mirroring emotions.

In *Proc. IEEE intern. symposium on robot and human interactive communication, RO-MAN 2011, Atlanta, GA*.

Hancock, P. A., Billings, D. R., Schaefer, K. E., Chen, J. Y., De Visser, E. J., & Parasuraman, R. (2011). A meta-analysis of factors affecting trust in human-robot interaction. *Human Factors*, *53*(5), 517–527.

Hancock, P. A., Kessler, T. T., Kaplan, A. D., Brill, J. C., & Szalma, J. L. (2020). Evolving trust in robots: Specification through sequential and comparative meta-analysis. In *Human Factors*, in press.

Haring, K. S., Matsumoto, Y., & Watanabe, K. (2013). How do people perceive and trust a life-like robot? In *Proceedings of the world congress on engineering and computer science volume 1*.

Heerink, M. (2011, March). Exploring the influence of age, gender, education and computer experience on robot acceptance by older adults. In *Proceedings of the 6th international conference on human-robot interaction* (pp. 147–148). ACM.

Joosse, M., Lohse, M., Pérez, J. G., & Evers, V. (2013, May). What you do is who you are: The role of task context in perceived social robot personality. In *Robotics and automation (ICRA), 2013 IEEE international conference* (pp. 2134–2139). IEEE.

Kaplan, A. D., Sanders, T., & Hancock, P. A. (2018). Extroversion and the tendency to anthropomorphize robots: A Bayesian analysis. *Frontiers in Robotics and AI*, *5*, 135–143.

Kidd, C. D. (2003). *Sociable robots: The role of presence and task in human-robot interaction.* (Doctoral dissertation)Boston, MA: Massachusetts Institute of Technology.

Looije, R., Neerincx, M. A., & Cnossen, F. (2010). Persuasive robotic assistant for health self-management of older adults: Design and evaluation of social behaviors. *International Journal of Human-Computer Studies*, *68*(6), 386–397.

Lyons, J. B., & Havig, P. R. (2014, June). Transparency in a human-machine context: Approaches for fostering shared awareness/intent. In *International conference on virtual, augmented and mixed reality* (pp. 181–190). Cham: Springer.

Manufacturing (2019). *February 2, 2019,* Retrieved from https://www.cdc.gov/niosh/about/strategicplan/traumanuf.html.

Rau, P. P., Li, Y., & Li, D. (2009). Effects of communication style and culture on ability to accept recommendations from robots. *Computers in Human Behavior*, *25*(2), 587–595.

Rau, P. L. P., Li, Y., & Liu, J. (2013). Effects of a social robot's autonomy and group orientation on human decision-making. *Advances in Human-Computer Interaction*, *11*, 1–13.

Sanders, T., Kaplan, A., Koch, R., Schwartz, M., & Hancock, P. A. (2019). The relationship between trust and use choice in human-robot interaction. *Human Factors*, *61*(4), 614–626.

Sanders, T. L., MacArthur, K., Volante, W., Hancock, G., MacGillivray, T., Shugars, W., et al. (2017, September). Trust and prior experience in human-robot interaction. *Proceedings of the human factors and ergonomics society annual meeting* (pp. 1809–1813). *61(1)* (pp. 1809–1813). Los Angeles, CA: SAGE Publications.

Sanders, T. L., Wixon, T., Schafer, K. E., Chen, J. Y., & Hancock, P. A. (2014, March). The influence of modality and transparency on trust in human-robot interaction. In *Cognitive methods in situation awareness and decision support (CogSIMA), 2014 IEEE international inter-disciplinary conference* (pp. 156–159). IEEE.

Schaefer, K. (2013). *The perception and measurement of human-robot trust.* (Doctoral dissertation) Orlando, FL: University of Central Florida.

Scopelliti, M., Giuliani, M. V., & Fornara, F. (2005). Robots in a domestic setting: a psychological approach. *Universal Access in the Information Society*, *4*(2), 146–155.

Sparrow, R. (2007). Killer robots. *Journal of Applied Philosophy*, *24*(1), 62–77.

Srinivasan, V., & Takayama, L. (2016, May). Help me please: robot politeness strategies for soliciting help from humans. In *Proceedings of the 2016 CHI conference on human factors in computing systems* (pp. 4945–4955). ACM.

Stanton, C. J., & Stevens, C. J. (2017). Don't stare at me: The impact of a humanoid robot's gaze upon trust during a cooperative human–robot visual task. *International Journal of Social Robotics*, *9*(5), 745–753.

Stowers, K., Oglesby, J., Sonesh, S., Leyva, K., Iwig, C., & Salas, E. (2017). A framework to guide the assessment of human–machine systems. *Human Factors*, *59*(2), 172–188.

Tsui, K. M., Desai, M., & Yanco, H. A. (2010, March). Considering the bystander's perspective for indirect human-robot interaction. In *Proceedings of the 5th ACM/IEEE international conference on human-robot interaction* (pp. 129–130). IEEE Press.

Ullman, D., & Malle, B. F. (2017, March). Human-robot trust: Just a button press away. In *Proceedings of the companion of the 2017 ACM/IEEE international conference on human-robot interaction* (pp. 309–310). ACM.

van den Brule, R., Dotsch, R., Bijlstra, G., Wigboldus, D. H., & Haselager, P. (2014). Do robot performance and behavioral style affect human trust? *International Journal of Social Robotics*, *6*(4), 519–531.

Wang, L., Rau, P. L. P., Evers, V., Robinson, B. K., & Hinds, P. (2010, March). When in Rome: The role of culture & context in adherence to robot recommendations. In *Proceedings of the 5th ACM/IEEE international conference on human-robot interaction* (pp. 359–366). IEEE Press.

CHAPTER

7

The role of human personality on trust in human-robot interaction

Gene M. Alarcon[a], *August Capiola*[a], *and Marc D. Pfahler*[b]

[a]Air Force Research Laboratory, Wright-Patterson Air Force Base, OH, United States, [b]Cognitive Performance Group, Orlando, FL, United States

Robots are found in many contexts of modern life and are becoming ever more pervasive (see Hancock et al., 2011). Robots now assist in everything from manufacturing to surgery.

As such, research on human-robot interaction (HRI) has flourished in the last decade (see Hancock et al., 2011; Wynne & Lyons, 2018). Within HRI, trust is crucial for an effective human-to-robot partnership. A large portion of the research in HRI has been devoted in investigating the trust process (Hancock et al., 2011; Oleson, Billings, Kocsis, Chen, & Hancock, 2011). Within this research, individual differences have been highlighted as important variables, which influence trust in HRI, one of which is human personality. However, with few exceptions, the majority of research has not focused on the role of the human's personality in HRI. Within this chapter, the trust literature is reviewed, and differences between interpersonal trust and trust in automation and HRI are highlighted. Past findings are then described in relation to human personality—along with other individual differences and general expectations influenced by experience—and associated with trust in HRI. The chapter then discusses personality traits and general attitudes that may contribute to trust in HRI and offers hypotheses and research questions for future work to test and answer, respectively.

https://doi.org/10.1016/B978-0-12-819472-0.00007-1
© 2021 Elsevier Inc. All rights reserved.

Interpersonal trust

Trust has been defined as the willingness to be vulnerable to a referent with the expectation of a positive outcome (Mayer, Davis, & Schoorman, 1995; Schoorman, Mayer, & Davis, 2007). Mayer and colleagues' model of trust is one of the most cited models in the interpersonal literature (see Thielmann & Hilbig, 2015). The model is one of the first to delineate trust from *antecedents to* and *outcomes of* trust. According to Mayer et al. (1995), antecedents to trust include stable trust beliefs held by the trustor (i.e., individual differences) such as one's propensity to trust others in general, as well as perceptions about the state-specific referent's ability (are they competent?), benevolence (do they have my best interest in mind?), and integrity (do I find their principles acceptable?), which compose trustworthiness perceptions. Trust behaviors are the observable outcomes of making oneself vulnerable to the referent, and the relationship between trust and behavior is moderated by the perceived risk of making oneself vulnerable. It is important to acknowledge that according to Mayer and colleagues' model, general tendencies to trust, trust perceptions, trust intentions, and trust behaviors, although highly related, are not the same construct. A trustor may be willing to be vulnerable to the referent but never actually engage in the trust behavior because the risk of trusting them may be perceived as too great. Likewise, a trustor may observe what appears to be a trust behavior, but the trustor may have never perceived they had a choice in the matter.

Trust and personality

Of the aspects of Mayer et al.' (1995) trust model, the current chapter focuses on individual differences within the trustor—specifically personality variables and general expectancies and attitudes. Meta-analytic data have demonstrated the importance of propensity to trust—or the general willingness to perceive others as trustworthy—on trust and behavioral outcomes (Colquitt, Scott, & LePine, 2007). Colquitt and colleagues found propensity to trust demonstrated an impact on perceptions of trustworthiness and trust intentions, as well as risk-taking behaviors and performance. In a structural model, the propensity to trust predicted task performance, citizenship behaviors, and counterproductive behaviors controlling for trust intentions. This model indicated individual differences, such as propensity to trust, have a significant impact on trust. Recent research has replicated and expanded several of Colquitt et al.'s findings by investigating the role of propensity to trust and suspicion propensity—a general tendency to be suspicious (Bobko, Barelka, Hirshfield, & Lyons, 2014)—on outcomes of interest (Capiola, Jessup,

Ryan, & Alarcon, 2019). Alarcon et al. (2018) also explored individual differences as predictors of trustworthiness perceptions, trust intentions, and trust behaviors and found personality accounted for 6%–23% of the variance in each outcome of interest.

Individual differences have the most influence on trust early on in an interaction when little to no information is known about the referent (Alarcon, Lyons, & Christensen, 2016; Jones & Shah, 2016). If the trustor is not familiar with the referent, then the trustor will rely on their biases pertaining to how they think about people in general. However, once the trustor has more knowledge about the referent, the trustor will rely on information about the referent to make their decision. As such, individual differences of the trustor are more important early on in the trust interactions when the trustor has little to no information about the referent, though research shows that individual differences such as propensity to trust still have some influence on these interactions over time (Jones & Shah, 2016, p. 404). These individual differences can be delineated as either global or specific factors.

Global factors

Global factors are large taxonomies of personality traits that are typically expressed through behaviors, thoughts, and actions. The most notable of these taxonomies is the five-factor model (FFM), which is the dominant personality theory in the psychological literature. The foundation of the FFM is that global factors remain consistent across situations and time (Costa & McCrae, 2009). The factors predispose a person to act a certain way in different situations (Robertson & Callinan, 1998). The model consists of Emotional Stability (or Neuroticism), Conscientiousness, Extraversion, Openness to Experience, and Agreeableness. Emotional Stability is the extent to which a person is calm, secure, and experiences little fluctuation in emotions. Conscientiousness is the extent to which a person is organized, self-disciplined, and hard-working. Extraversion is the extent to which an individual is social, talkative, and seeks stimulation. Openness to Experience is the extent to which a person is imaginative, curious, and enjoys a variety of experiences. Lastly, Agreeableness is the extent to which a person is cooperative, compassionate, and focused on interpersonal facilitation (Costa & McCrae, 2009). In the following paragraphs, the relationship between traits of the FFM and trust in the interpersonal literature will be discussed.

Emotional stability/neuroticism

Emotionally stable team members are secure in their goals and decisions (Van Vianen & De Dreu, 2001), which fosters cooperation and a comfortable team environment (Barrick, Stewart, Neubert, & Mount, 1998; Van Vianen & De Dreu, 2001). Emotionally stable individuals have been

hypothesized to contribute to teamwork and enhance team performance (Hough, 1992; Mount, Barrick, & Stewart, 1998). However, meta-analyses demonstrated Emotional Stability was minimally (Bell, 2007) or not significantly related to team performance (Lyons, Stokes, & Schneider, 2011; Peeters, Van Tuijl, Rutte, & Reymen, 2006). This relationship with performance and the social nature or emotional stability (i.e., displaying negative or positive emotions) has led some to explore the relationship between Neuroticism and trust. Research has demonstrated Emotional Stability is correlated with trust intentions and beliefs, but not trust behaviors (Alarcon et al., 2018). However, it should be noted that the trust behaviors in the aforementioned study were categorical outcomes and is one of only a few studies that have explored the Emotional Stability and trust relationship to our knowledge (Lyons et al., 2011). As such, further studies are needed to explore the relationship.

Extraversion

Extraverts tend to be talkative, assertive, optimistic, and outgoing (Costa & McCrae, 1992). Extraverts stimulate discussion and foster an expressive team environment (Barry & Stewart, 1997). The assertive and talkative nature of extraverts tend to make them dominant in team settings (Kichuk & Wiesner, 1998). The optimistic and social nature of the extravert has been positively associated with trustworthiness perceptions and trust intentions (Alarcon et al., 2018) and in some cases task performance in computer-mediated team contexts (Lyons et al., 2011).

Conscientiousness

Conscientiousness has consistently been associated with performance in the selection literature (Barrick, Mount, & Judge, 2001). Individuals high in Conscientiousness are motivated and task-oriented (Barrick & Mount, 1993), which researchers have theorized leads to role adaption in the face of changes within the team. Additionally, a team consisting of highly Conscientiousness members can lead to cohesion (Van Vianen & De Dreu, 2001) and increased performance (Peeters et al., 2006). In contrast, variability in Conscientiousness can lead to conflict and decreased effectiveness (Mohammed & Angell, 2003). This cohesion, or lack thereof, can thus influence trust interactions as team cohesion has been associated with interpersonal trust (Mach, Dolan, & Tzafrir, 2010).

Agreeableness

Agreeableness is one of the more studied personality traits in teams. Agreeableness is evident in team performance through its impact on team processes and contextual performance. The friendly, trusting, compliant, and altruistic nature of the agreeable teammate fosters positive group processes such as cohesion, cooperation, and conflict resolution (Barrick et al.,

1998). Agreeableness has been associated with trustworthiness perceptions in computer-mediated teams (Lyons et al., 2011), and meta-analyses have demonstrated Agreeableness as an important factor in team performance (Bell, 2007). Additionally, research on interpersonal trust has demonstrated Agreeableness to be a core factor that leads to trust in individuals, with the construct predicting trust beliefs and perceptions (Alarcon et al., 2018). These results are not surprising as dispositional trust is a subordinate trait of Agreeableness.

Openness to experience

Individuals high in Openness to Experience are broadminded and have a willingness to try new things (LePine, 2003). Team members that are high in Openness to Experience can easily adapt to changing situations and seek alternative ways of solving problems. Lyons et al. (2011) found that amongst personality and individual difference variables investigated in a 5-person team-based aerial port simulation task, Openness to Experience predicted unique criterion in trustworthiness perceptions amongst teammates. However, Alarcon et al. (2018) found no relationship between Openness and trustworthiness perceptions, trust intentions, or trust behaviors. As such, the findings on Openness and trust are mixed.

In summary, extent research has investigated the role of personality in contexts comprising human-human trust. However, human-human trust may differ from human-automation trust, let alone human-robot trust. The following sections will unpack these similarities and differences in turn.

Trust in automation

The concept of trust has been expanded to the automation literature (Hoff & Bashir, 2015; Lee & See, 2004). Interpersonal trust and trust in automation are similar as they both entail situation-specific attitudes that are only applicable when making oneself vulnerable in an uncertain situation. The trustworthiness perceptions of ability, benevolence, and integrity found in the interpersonal trust literature (Mayer et al., 1995) have been mapped to perceptions of performance, purpose, and process (Lee & See, 2004) of an automated system, respectively (Hoff & Bashir, 2015). More recent theoretical positions (Wynne & Lyons, 2018) argue that automation (and robots for that matter) should elicit more human trust, the more it is perceived as a teammate or possessing *teammate-likeness*, thus explicating the importance of interpersonal-exchange features in trust in automation domains.

Although there are many similarities between interpersonal trust and trust in automation, there are several differences, too. Madhavan and

Wiegmann (2007) hypothesized differences in human-human versus human-automation trust. As discussed above, when individuals enter into an interpersonal trust relationship, their perceptions of the referent are predictive of the referent's actions (Rempel, Holmes, & Zanna, 1985). A reliable amount of this variance is due to individual differences such as personality factors (Alarcon et al., 2016; Jones & Shah, 2016). In contrast, people often assume machines are highly or perfectly reliable (Dzindolet, Pierce, Beck, & Dawe, 2002), and this bias has been shown to influence interactions with automation (Merritt & Ilgen, 2008; Merritt, Unnerstall, Lee, & Huber, 2015). Madhavan and Wiegmann (2007) hypothesized that initial trust perceptions are biased by this initial trust. However, trust changes rapidly after interactions with a system. Similar to the change in interpersonal trust, the loci of the trustor's perceptions of the system transfer from the trustor (e.g., Agreeableness, dispositional trust, trustor beliefs in perfect reliability of automation in general) to the referent system with an emphasis on predictability and dependability.

Individual differences play a role in the formation of trust toward automation (Hoff & Bashir, 2015). Indeed, meta-analytic data from Schaefer, Chen, Szalma, and Hancock (2016) show that human-related factors (e.g., personality, age, gender, and ethnicity) are related to trust development in automation. Human-related factors had a corrected correlation of 0.49 (95% CI = 0.25, 0.72) with trust development, which had a similar effect size as automation-related factors (0.53, 95% CI = 0.29, 0.78). These results indicate that human factors are of comparable importance to automation factors in trust development. As such, below we discuss previous research on personality and other individual differences related to but separate from personality.

Global factors and trust in automation

Szalma and Taylor (2011) explored the relationship between the FFM of personality and automation characteristics in a simulated uninhabited ground vehicle task. Neuroticism was associated with less agreement with automation (even when the advice given by the automation was correct), increased frustration with the automation, and lower pre-task engagement. Extraversion was associated with less distress and worrying about the task. Openness to Experience was associated with less effort in the task. Conscientiousness was associated with higher pre-task engagement. Agreeableness was associated with less worrying in the task in the lower reliability conditions. Additionally, Merritt and Ilgen (2008) found individuals higher in Extraversion were less likely to trust automation in an X-ray screening task after the experiment. However, extraversion was not related to pre-task trust or use of the automation. This review clearly

notes the lack of research on trust in automation and global personality traits. However, it should be noted that there is research on the FFM and robotics, which we outline later. Some of the personality variables may have a different relationship with a referent that is mobile or anthropomorphic in nature. As the global traits often predict variance in subsequent criterion, future research should explore global personality traits in relation to trust in automation.

Specific factors and trust in automation

In addition to global factors, specific factors have also been associated with trust in automation. Relatedly, Marsh and Dibben (2003) define dispositional trust in automation as an individual's stable propensity to trust automation and hypothesized that it would influence trust in automation. Merritt and colleagues have explored individual differences in automation in a variety of studies involving an X-ray screening task. Merritt and Illgen (2008) found the propensity to trust machines was positively related to trust in automation. They concluded that personality moderates the perceptions of automation when aspects of the machine are constant. Furthermore, Merritt, Heimbaugh, LaChapell, and Lee (2013) found implicit evaluations of machines and self-reports of propensity to trust machines did not correlate, but each accounted for unique variance in trust in automation in the X-ray task. Additionally, if errors were evident there was a significant interaction between the propensity to trust machines and implicit associations such that participants higher in both were more trusting of the automation.

Merritt et al. (2015) explored perfect automation schemas (PASs) in the X-ray task. PAS comprises two factors, all-or-none thinking and high expectations. High expectations were not related to trust in the automation, but all-or-none thinking was related to decreases in trust when errors occurred. In contrast, Lyons and Guznov (2019) found that high expectations were positively related to trust across three studies. In the Mixed Initiative eXperimental (MIX) testbed, high expectations were positively related to self-reported trust in the multitask condition across two studies. In addition, high expectations were also significantly related to trust perceptions and performance beliefs in pilots' self-reports of trust and performance beliefs about the system. Interestingly, across all three studies, the all-or-none facet was not related to any criterion, which contrasts with Merritt et al.' (2015) findings. This indicates further research is needed on the relationship of PAS with trust in automation. It may be that certain types of automation based on their decision authority (e.g. selection, situation awareness, choice, and action) are more susceptible to PAS than others (Madhavan & Weigmann, 2007).

The aforementioned research focused on the influence of personality as well as individual differences such as PAS on trust in automation. However, automation with a physical embodiment—or robots—may offer nuance to the relationship between human personality and trust in an automated system.

Research on robots

A robot is an automated machine that can complete a complex series of actions automatically (Robot, 2019). Robots have been used in manufacturing for some time due to their precision and speed of manipulating objects and materials. In recent years, the military has employed robots in tasks such as bomb disposal and rescue missions. Hospitals and doctors have robots assisting in surgery (Davies, 2000) and caregiving of patients. Rehabilitation regimens have also incorporated robots to assist with therapy (Shamsuddin et al., 2012). The introduction of robots into increasingly more complex scenarios has necessitated research on their role with human partners.

Perhaps in response to the need to understand human-robot partnerships, research has begun to investigate how and why humans choose to trust robots. Some research on robotics indicates humans prefer robots to look, sound, and in some cases act similarly to themselves (Hancock et al., 2011). The desire for anthropomorphic characteristics results in robots possessing facial features, changes of inflection in the voice, and other anatomical appearances. In one study, participants were administered oxytocin during an interaction with an automated agent that was either anthropomorphic or not (de Visser et al., 2017). Participants that were in the anthropomorphic condition increased their trust, compliance, and performance when oxytocin was administered. In contrast, oxytocin had no effect in the non-anthropomorphic condition. These results underline the importance of anthropomorphizing agents and robots, even at the biological level. However, altering a robot's physical aesthetics is only scratching the surface to the many underlying topics of current research. Previous investigations in military and research robotics have demonstrated one of the components to appropriately use complex technology is calibrated trust (Lee & Seppelt, 2009). Technology concerns (e.g., lack of tactile feedback in robot-assisted surgeries), team challenges (e.g., remote unmanned aerial or ground vehicles), or changes to the working environment (e.g., absence of trained personnel due to turnover) can all pose major challenges in sustaining appropriate trust within HRI. As a result, human partners might avoid interacting fully with the robot and therefore under-utilize their maximum potential as a helpful aid.

In a meta-analysis, researchers investigated three factors hypothesized to influence trust in HRI: robot-related, environmental, and human-related (Hancock et al., 2011). Robot-related variables included performance-based and attribute-based factors such as reliability and robot personality, respectively. Environmental variables included team collaboration and tasking variables such as task-type and culture, respectively. Lastly, human-related variables included ability-based and individual-difference characteristics such as attentional capacity and personality, respectively. Although robot-related factors showed the greatest effect on trust in HRI followed by environmental factors, human-related factors showed a small effect on trust in HRI (0.13). However, the authors contended that this should not be interpreted as human-related factors being insignificant in the HRI domain. Instead, they suggest more research ought to be conducted investigating the effects of human-related factors on trust in HRI (Hancock et al., 2011, p. 525).

Modern robots are becoming ever more autonomous in nature. The addition of autonomy in robotics can often allow a single user to complete complex tasks that would otherwise require teams of humans (e.g., robotic surgery). However, given the collaborative nature of these autonomous robots, assessing appropriate trust and reliance is again a critical component that need not be overlooked. Humans who inappropriately trust in a robot may over-rely on the system. In some cases, misuse of these machines could potentially pose safety risks for human operators (e.g., robots on a vehicle assembly line). Despite some form of robotic technology being seen in nearly all aspects of the modern world, an empirically accepted database for reporting baseline trust and reliance in HRI still ceases to exist. Regardless, there has been researches investigating the role of human personality and related individual differences on trust in robots, and the remainder of the chapter will speak to these extant findings and offer suggestions for future research initiatives.

Human personality and trust in human-robot interactions

Individual differences have a substantial role in interpersonal trust (Colquitt et al., 2007) and trust in automation (Hoff & Bashir, 2015; Schaefer et al., 2016) literatures. However, human and automation referents are not synonymous with a robot referent. Robots are an amalgamation: they have physical bodies which may or may not be anthropomorphized, and they often have an automated operating system. One proposed future research route concerns whether robots are considered to be tools compared to teammates (Wynne & Lyons, 2018). Wynne and Lyons propose that as robots are seen more like teammates and less like tools, trust becomes increasingly important. As humans begin to work

alongside robots in a variety of contexts, considering the dispositions and propensities of the human will be as important as considering the characteristics of the robot. Partnerships comprise two or more entities working towards shared goals. As such, each member of the partnership needs to be considered. Although we acknowledge that other individual differences (e.g., experience, knowledge of the field, age) influence trust in HRI (Hancock et al., 2011), we focus on human personality and the degree of specificity (global *vs.* specific factors) that it is assessed (and in the case of specific factors, some related expectancies developed over time).

Global factors and trust in robots

Due to the social nature of the Extraversion construct and the social nature of interacting with a robot, Extraversion has been explored far more extensively than any other global personality traits in HRI. Extraversion should be positively related to trust in HRI as the social and positive nature of the extravert will lead to more meaningful and developed relationships with the robot. However, we also postulate that it will be moderated by the anthropomorphic nature of the robot as the anthropomorphism will add an additional social component to the interaction that should increase trust. After meta-analysis by Hancock et al. (2011), Robert (2018) conducted a literature review and categorized extant research on personality and HRI into three categories: human personality, robot personality, and similarity of human-robot personality. Pertaining to trust, Robert identified several articles on the influence of human personality on trust in HRI. Robert's review found Extraversion was the only FFM personality trait that consistently predicted positive interactions with robots.

Haring, Matsumoto, and Watanabe (2013) found a positive relationship between extraverted individuals and trust in robots, such that those participants higher in Extraversion sent more money to the robot partner in the Trust Game. However, a relationship between human Extraversion and trust behaviors in robots was found to be non-significant in another study in which participants were asked to follow directions from a robot, which included some peculiar directions such as pouring orange juice on a plant (Salem, Lakatos, Amirabdollahian, & Dautenhahn, 2015). As such, there appear to be mixed results pertaining to the relationship between Extraversion and trust behaviors in HRI.

Joosse, Lohse, Pérez, and Evers (2013) explored personality matching and found that participants higher in Extraversion tended to trust an extraverted robot more than an introverted robot, but there was evidence that this relationship was moderated by task constraints, adding complexity to the study's conclusions. Andrist, Mutlu, and Tapus (2015) investigated robot eye gaze attributes associated with different degrees of

Extraversion. Past research shows that some FFM factors are related to different eye gaze durations and patterns in humans (Rauthmann, Seubert, Sachse, & Furtner, 2012). In the study conducted by Andrist et al. (2015), robots were modified to display more or less mutual eye gaze with the human participant indicating more or less extraverted tendencies during a collaborative Tower of Hanoi task. Results illustrated humans that were more/less extraverted tended to engage more in the repetitive task with a robot who matched their levels of Extraversion in terms of eye gaze. These results support the similarity-attraction theory, which posits that people are attracted and have a stronger social presence towards those with similar personalities, which is particularly the case for those higher in Extraversion (Naas & Lee, 2001). Though one may argue that this study may not differentiate between trust and task engagement—as participants may not perceive vulnerability or risk when interacting with the robot to complete a puzzle—the findings are intriguing as matched personality can be investigated from both self-reported and gaze characteristics. Other researchers (Niculescu, van Dijk, Nijholt, Li, & See, 2013) have found similar findings pertaining to human Extraversion and their perceptions of a robot's voice pitch and perceived empathy, such that robots perceived to be lower in Extraversion were preferred by less extraverted participants.

Cruz-Maya and Tapus (2016) explored the role of human Extraversion on motivation to complete tasks in an office setting suggested by a robot. For example, the robot would suggest to take a break from the computer or prepare for a meeting with a supervisor, and participants were timed when they would do that behavior suggested. More extraverted participants were less motivated to finish a task early compared to those who were more introverted. This research indicates Extraversion is related to trust intentions, by taking the advice of the robot.

Based on the aforementioned studies, Extraversion has been investigated on its relation to trust in HRI. However, we must note that some of the past literature has several limitations, such as dichotomizing/categorizing continuous personality variables (e.g., Andrist et al., 2015; Cruz-Maya & Tapus, 2016; Salem et al., 2015), using limited assessments of personality (e.g., Niculescu et al., 2013), and collecting a small sample (e.g., Andrist et al., 2015; Chidambaram, Chiang, & Mutlu, 2012; Cruz-Maya & Tapus, 2016). Future endeavors should seek to remedy these limitations, expanding the robustness of these initial findings. Although Extraversion has been studied most extensively, other global traits may have a theoretical relationship with trust in robots.

In the context of trust in HRI, more neurotic individuals may display more variability in emotions than emotionally stable individuals when robots make mistakes. For example, individuals high in Neuroticism may experience more anger, anxiety, and worrying about the robot when it functions improperly than when it functions properly. As such,

Neuroticism should be related to less trust, though the magnitude of this relationship may be small (Alarcon et al., 2018). Future research should investigate this hypothesis in HRI.

In terms of Conscientiousness, the cohesion established by conscientious team members discussed in the interpersonal trust section is important. Robots that have low instances of failure may lead to increased cohesion between the individual and the robot. The cohesion felt from a higher-performing robot may facilitate trust, as the two constructs have demonstrated a relationship in the literature (Garrison, Wakefield, Xu, & Kim, 2010). In the aforementioned study of the effects of robots on motivation in the workplace, participants with higher Conscientiousness were more prompted to complete the task (responding to several emails in a timely manner) earlier than was suggested by a robot compared to those low in Conscientiousness. As Conscientiousness does not have an interpersonal aspect, we would not anticipate anthropomorphism to moderate this relationship. However, future research should test our speculation.

As dispositional trust is a subordinate factor of the Agreeableness construct, Agreeableness may have a strong relationship with trust in robots. In the context of HRI, the trusting, compliant, and friendly nature of agreeable people should result in quick compliance, trust in the robot, and more forgiving nature of robot errors. In addition, due to the social aspect of Agreeableness, we would expect the relationship between Agreeableness and the trust process to be stronger as the robot increases in anthropomorphism. Individuals low in Openness to Experience tend to be conventional in their perceptions and behaviors, and as such may not be apt to trust robots. In contrast, those high in Openness to Experience tend to be broad-minded in their perceptions and behaviors, and as such may be more willing to trust robots. Again, future research should test this speculation.

In summary, there is some extant research on global personality traits and their relation to trust in HRI. However, the specificity of personality traits may play a role in uncovering the influence of personality on trust in HRI. For instance, Chidambaram et al. (2012) investigated whether participant's assertiveness—a facet of Extraversion—predicted compliance and perceived persuasiveness with a robot's suggestions in an adaptation of the Desert Survival Problem (Lafferty, Eady, & Elmers, 1974). In this task, participants were told to imagine that they have crash-landed in a desert and need to rank order a collection of materials they have with them for survival. Following this ranking, the robot makes suggestions to change the ranking participants created. Chidambaram et al. (2012) found that participants' degree of assertiveness had no relation to their compliance with a social robot's suggestions but did influence their perceptions of the robot's persuasiveness. This study, and others to be discussed, demonstrates the need to investigate not only global personality factors but also specific factors to glean a more nuanced understanding of trust in HRI. The chapter turns to a discussion of these specific factors next.

Specific factors

Research in personality psychology has demonstrated that both global and specific factors are important predictors of criterion, even after controlling for one another. Put another way, the variance accounted for in criterion by global factors and specific factors is unique. Ashton (1998) found that risk-taking predicted delinquency, but when the other facets of Conscientiousness (e.g., conformity, responsibility, and organization) were averaged with risk-taking, the predictive validity declined. Ashton, Jackson, Paunonen, Helmes, and Rothstein (1995) found that orderliness was correlated with students' cleanliness of their rooms, but when orderliness was combined with the other factors of Conscientiousness (e.g., cognitive structure and deliberateness) predictive validity dropped. Similarly, the facets of Conscientiousness (e.g., cognitive structure, achievement, endurance, order, and low impulsivity) and Openness to Experience (e.g., autonomy, change, sentience, understanding, and low harm avoidance) were better predictors of grades than the global factors (Paunonen & Ashton, 2001). Alarcon et al. (2018) also illustrated propensity to trust—a facet of Agreeableness—accounted for significant variance in criterion (trust beliefs, intentions, and behaviors) across the trust process controlling for the entire FFM in humans. These findings support the need for investigations of specific factors in personality research in general, but also on personality, general schemas, and trust in HRI specifically. In this section, we discuss past research on two specific factors—propensity to trust, PAS (and, briefly, negative attitudes towards robots)—and offer future research ideas for investigating a more recently discussed factor—suspicion propensity.

Propensity to trust

Propensity to trust is the general tendency to trust people across situations (Mayer et al., 1995). As such, propensity to trust robots is the general tendency to trust a robot across situations. A review of theoretical models on HRI illustrates an emphasis on propensity to trust robots. However, little research has shown propensity to trust in HRI contexts. One notable exception is a paper by Jessup and colleagues (Jessup, Schneider, Alarcon, Ryan, & Capiola, 2019) that explored propensity to trust technology in an HRI experiment. They found an adapted version of the propensity to trust automation scale predicted trustworthiness perceptions and actual trust behaviors when participants were paired with an anthropomorphic robot in a trust game, highlighting the importance of specific personality variables in HRI.

As mentioned in the automation section, propensity to trust automation has demonstrated a relationship with automation trust (Merritt et al., 2013; Merritt & Ilgen, 2008). Indeed, there are several scales to measure propensity to trust automation such as the complacency-potential rating scale

(CPRS; Singh, Molloy, & Parasuraman, 1993) and the propensity to trust automation scale (Schneider et al., 2017). Utilizing these scales to assess propensity to trust robots can be problematic. Automation scales typically have vague referents for automation such as "Technology is reliable" (Schneider et al., 2017). The problem with this type of item is that it is vague and the referent used for making the assessment may vary between people. For example, Bowling and Burns (2010) found adding specificity to the FFM by changing the items to refer to personality *at work* increased the predictive validity of the scales. Clearly defining a referent reduces the variability and error in the items of a scale. Indeed, Jessup et al. (2019) found that adapting the propensity to trust technology scale referent from "technology" to "robot" in an experimental task led to the increased predictive validity of the scale for perceived trustworthiness and trusting behaviors towards the robot in a trust game. Indeed, the adapted measure was the only significant personality predictor of behaviors, and the variance explained by the adapted measure encompassed the variance of both the unadapted version and the CPRS.

A word of caution should be noted with regard to specificity. The CPRS was created in the early 1990s when automation was different than it is today. The CPRS items refer to specific types of automation such as an automated teller machine (i.e., ATM), video cassette recorder (i.e., VCR), and library catalogs. As such, subsequent reliabilities have been poor for the CPRS when it was utilized in more recent studies. The CPRS showcases the pitfalls of making the referents too specific. For example, VCRs are no longer in use and younger participants may not even know what a VCR is, thus making it difficult to accurately respond to the item. In contrast, although ATMs were relatively nascent in the early 1990s they are ubiquitous in the modern era. This ubiquity and extreme familiarity with ATMs may cause a ceiling effect on the item, as relatively few people are suspicious or untrusting of ATMs. Instead, a balance must be stricken between specificity and generality of the scale (see Jessup et al., 2019). Instead of blanket terms like "technology," terms such as "robot," "autonomous agent," or "self-driving car" may be more apt to increase both the reliability of the scale and the predictive validity. This is in line with previous research on personality as stated above (Alarcon et al., 2018; Ashton, 1998; Jessup et al., 2019).

Perfect automation schema

A PAS is a bias towards robotics or automation that results in an overreaction to the errors committed by the technology (Dzindolet et al., 2002; Merritt et al., 2015). Though not a personality trait *per se*, PAS is a schema that people hold automation to a higher standard than humans. PAS should develop over time and with experience using automation (and potentially robots), more so than other global personality factors. PAS is

related to trust, as research shows that people assume automation will perform better than humans, yet people abandon trust towards automation more quickly than humans when errors occur (Dzindolet, Pierce, Beck, Dawe, & Anderson, 2001). Put another way: overreaction leads to a sharp decrease in trust when errors are committed. The perception of automation (and potentially robots) as perfect is one of the key differences between human-human trust and human-automation trust (Dzindolet et al., 2002; Madhavan & Wiegmann, 2007). PAS is a belief about how systems operate. Two factors compose PAS: all-or-none thinking and high expectations (Merritt et al., 2015). All-or-none thinking is the stable attribution of errors in the system. For example, one may perceive that humans make errors in a task because they are fatigued or distracted. In comparison, someone with high all-or-none thinking may consider the automated/robotic referent useless if it makes an error that would be more easily forgiven had the referent been a human. Something either works or it does not; there is no middle ground. The all-or-none factor is hypothesized to be the key feature of how PAS influences trust (Merritt et al., 2015). The high expectations factor of PAS has to do with beliefs that if automation is functioning properly there will be very few errors (possibly none, depending on the extremity of the belief). Although research on PAS has been conducted in the automation literature, recent theorizing has called for PAS to be investigated in the HRI literature, specifically whether it effects perceptions of robot teammate-likeness and trust behavioral outcomes (Wynne & Lyons, 2018).

We offer some thoughts on PAS, trust, and HRI. First, PAS may function differently in the HRI literature due to the anthropomorphic nature of robots. As discussed previously, robots that resemble humans or animals may be viewed differently from robots that are not anthropomorphic, such as robots on an assembly line. When humans ascribe human attributes to the robot, this may include some of the error-related biases of humans such as the robot making the mistake unintentionally (i.e., dropping a tool) or possibly due to distraction or human linked characteristics. As such, researchers will want to take caution when assessing PAS in robots and not assume the same psychological processes are occurring in both types of interactions. Second, related to the first point, as humans view robots as teammates this may also change (see Wynne & Lyons, 2018), such that viewing the robot as a teammate may buffer the effects of robot errors on human trust. Third, other aspects of the individual may moderate the relationship between PAS and trust, such as experience with robots. A novice in a robot experiment may have different cognitive schemas than a researcher who works with robots. These experiential differences may moderate the relationship between PAS and trust. In comparison, an individual may be more apt to accept errors from a robot, especially if they understand what the underlying issue is with the robot (i.e., programming

flaw, a robot programmed to perform a task it was not designed to do, etc.). In contrast, a novice with schemas from popular culture may have stronger PAS robot norms. And finally, as PAS is a schema towards automation which may or may not differ if the generalized referent is robots, future research may wish to investigate the negative attitudes towards robots scale (NARS; Nomura, Kanda, & Suzuki, 2006) and its relation to personality, trust, and HRI. The NARS was designed to measure attitudes towards communication robots in everyday life. As negative attitudes towards robots may change over time, future research may wish to also investigate the association between PAS and NARS, and the potential interactions between these two predictors which may influence trust towards robots.

Suspicion propensity

Lastly, we offer another specific factor for future research explorations of trust in HRI. Suspicion propensity is the tendency to become suspicious across contexts (Bobko et al., 2014). When receiving information, one who is high in suspicion propensity has a greater likelihood of concurrently experiencing simultaneous uncertainty, cognitive activity, and perceived mal-intent. Suspicion propensity is conceptually (Bobko et al., 2014) and statistically related to propensity to trust, yet predicts unique variance in outcomes of interest (Capiola et al., 2019). Those higher in suspicion propensity may be less likely to trust, or at least suspend judgment before deciding (Bobko et al., 2014). As such, future research may wish to investigate suspicion propensity and its effects on trust in HRI. As suspicion propensity may in part compose a facet of Agreeableness (Capiola et al., 2019), future research should investigate whether it predicts unique variance in trust in HRI.

Final thoughts

Although the above review illustrates the theoretical underpinnings of personality with trust in HRI, limitations of the research should be addressed. Meta-analyses have consistently demonstrated that the FFM personality traits have a stronger relationship with team performance in field studies rather than lab/experimental studies. Peeters et al. (2006) found the relationship of all the FFM personality traits were moderated by the type of team in the study. Specifically, all the personality traits were associated with performance in professional teams but did not have a relationship in student teams. Similarly, Bell (2007) found a moderation effect of the field versus lab study for Extraversion, Agreeableness, Conscientiousness, and Openness to Experience in her meta-analysis. Interestingly, once the field study results were parceled out, only Extraversion was a

significant predictor of performance in lab studies. These results underscore the need to account for contextual components associated with human personality and performance outcomes. Similarly, the relationship of personality with trust and performance in HRI may be moderated by the type of study (i.e., field versus experimental). Individuals that use robots in their work environment may have more accurate schemas and heuristics about robots. This may be particularly important in human-machine teaming, where humans view the robot as a partner and not a tool (Wynne & Lyons, 2018).

In addition, we noted limitations in the past HRI research on personality and trust. Replications with larger samples of participants, along with treating the personality variables as continuous rather than discretizing their variance, may illuminate more robust findings surrounding the role of personality on trust in HRI. Lastly, prioritizing the differences between global versus specific factors have been investigated in the past (e.g., Alarcon et al., 2018; Ashton et al., 1995), but we think that this effort ought to be continued in the HRI domain to understand who comes to the table to interact with a robot, and how we might predict a properly calibrated trusting relationship between humans and robots.

Research on trust in HRI has burgeoned, and researchers are beginning to understand what aspects of the robot and the human facilitate human trust. The human is a crucial component in this dyad, and work has begun to highlight the individual differences which facilitate trust in HRI. We offer several hypotheses for future work to test and limitations which, pending they are addressed, may help to illuminate the impact of human personality and other individual differences influenced by experience on trust in HRI.

References

Alarcon, G. M., Lyons, J. B., & Christensen, J. C. (2016). The effect of propensity to trust and familiarity on perceptions of trustworthiness over time. *Personality and Individual Differences*, *94*, 309–315.

Alarcon, G. M., Lyons, J. B., Christensen, J. C., Bowers, M. A., Klosterman, S. L., & Capiola, A. (2018). The role of propensity to trust and the five factor model across the trust process. *Journal of Research in Personality*, *75*, 69–82.

Andrist, S., Mutlu, B., & Tapus, A. (2015). Look like me: Matching robot personality via gaze to increase motivation. In *Proceedings of the 33rd annual ACM conference on human factors in computing systems* (pp. 3603–3612): ACM.

Ashton, M. C. (1998). Personality and job performance: The importance of narrow traits. *Journal of Organizational Behavior: The International Journal of Industrial, Occupational and Organizational Psychology and Behavior*, *19*, 289–303.

Ashton, M. C., Jackson, D. N., Paunonen, S. V., Helmes, E., & Rothstein, M. G. (1995). The criterion validity of broad factor scales versus specific facet scales. *Journal of Research in Personality*, *29*, 432–442.

Barrick, M. R., & Mount, M. K. (1993). Autonomy as a moderator of the relationships between the big five personality dimensions and job performance. *Journal of Applied Psychology*, *78*, 111–118.

Barrick, M. R., Mount, M. K., & Judge, T. A. (2001). Personality and performance at the beginning of the new millennium: What do we know and where do we go next? *International Journal of Selection and Assessment*, *9*, 9–30.

Barrick, M. R., Stewart, G. L., Neubert, M. J., & Mount, M. K. (1998). Relating member ability and personality to work-team processes and team effectiveness. *Journal of Applied Psychology*, *83*, 377–391.

Barry, B., & Stewart, G. L. (1997). Composition, process, and performance in self-managed groups: The role of personality. *Journal of Applied Psychology*, *82*, 62–78.

Bell, S. T. (2007). Deep-level composition variables as predictors of team performance: A meta-analysis. *Journal of Applied Psychology*, *92*, 595–615.

Bobko, P., Barelka, A. J., Hirshfield, L. M., & Lyons, J. B. (2014). The construct of suspicion and how it can benefit theories and models in organizational science. *Journal of Business and Psychology*, *29*, 335–342.

Bowling, N. A., & Burns, G. N. (2010). A comparison of work-specific and general personality measures as predictors of work and non-work criteria. *Personality and Individual Differences*, *49*, 95–101.

Capiola, A., Jessup, S. A., Ryan, T. J., & Alarcon, G. M. (2019). Exploring the unique and shared variance of propensity to trust and suspicion propensity. *Journal of Individual Differences*. https://doi.org/10.1027/1614-0001/a000294. Advance online publication.

Chidambaram, V., Chiang, Y. H., & Mutlu, B. (2012, March). Designing persuasive robots: How robots might persuade people using vocal and nonverbal cues. In *Proceedings of the 7th annual ACM/IEEE international conference on human-robot interaction* (pp. 293–300): ACM.

Colquitt, J. A., Scott, B. A., & LePine, J. A. (2007). Trust, trustworthiness, and trust propensity: A meta-analytic test of their unique relationships with risk taking and job performance. *Journal of Applied Psychology*, *92*, 909–927.

Costa, P. T., & McCrae, R. R. (1992). Four ways five factors are basic. *Personality and Individual Differences*, *13*, 653–665.

Costa, P. T., & McCrae, R. R. (2009). The five-factor model and the NEO inventories. In *Oxford handbook of personality assessment* (pp. 299–322).

Cruz-Maya, A., & Tapus, A. (2016). Influence of user's personality on task execution when reminded by a robot. In *International conference on social robotics* (pp. 829–838). Cham: Springer.

Davies, B. (2000). A review of robotics in surgery. *Proceedings of the Institution of Mechanical Engineers, Part H: Journal of Engineering in Medicine*, *214*, 129–140.

de Visser, E. J., Monfort, S. S., Goodyear, K., Lu, L., O'Hara, M., Lee, M. R. , et al. (2017). A little anthropomorphism goes a long way: Effects of oxytocin on trust, compliance, and team performance with automated agents. *Human Factors*, *59*, 116–133.

Dzindolet, M. T., Pierce, L. G., Beck, H. P., & Dawe, L. A. (2002). The perceived utility of human and automated aids in a visual detection task. *Human Factors*, *44*, 79–94.

Dzindolet, M. T., Pierce, L. G., Beck, H. P., Dawe, L. A., & Anderson, B. W. (2001). Predicting misuse and disuse of combat identification systems. *Military Psychology*, *13*, 147–164.

Garrison, G., Wakefield, R. L., Xu, X., & Kim, S. H. (2010). Globally distributed teams: The effect of diversity on trust, cohesion and individual performance. *ACM SIGMIS Database: The Database for Advances in Information Systems*, *41*, 27–48.

Hancock, P. A., Billings, D. R., Schaefer, K. E., Chen, J. Y., De Visser, E. J., & Parasuraman, R. (2011). A meta-analysis of factors affecting trust in human-robot interaction. *Human Factors*, *53*, 517–527.

Haring, K. S., Matsumoto, Y., & Watanabe, K. (2013). How do people perceive and trust a lifelike robot. In *Proceedings of the world congress on engineering and computer science*.

Hoff, K. A., & Bashir, M. (2015). Trust in automation: Integrating empirical evidence on factors that influence trust. *Human Factors*, *57*, 407–434.

Hough, L. M. (1992). The 'Big Five' personality variables–construct confusion: Description versus prediction. *Human Performance*, *5*, 139–155.

Jessup, S. A., Schneider, T. R., Alarcon, G. M., Ryan, T. J., & Capiola, A. (2019). *Measurement of the propensity to trust automation.* [Paper Accepted for International Conference on Human-Computer Interaction].

Jones, S. L., & Shah, P. P. (2016). Diagnosing the locus of trust: A temporal perspective for trustor, trustee, and dyadic influences on perceived trustworthiness. *Journal of Applied Psychology*, *101*, 392–414.

Joosse, M., Lohse, M., Pérez, J. G., & Evers, V. (2013). What you do is who you are: The role of task context in perceived social robot personality. In *2013 IEEE international conference on robotics and automation* (pp. 2134–2139): IEEE.

Kichuk, S. L., & Wiesner, W. H. (1998). Work teams: Selecting members for optimal performance. *Canadian Psychology/Psychologie Canadienne*, *39*, 23–32.

Lafferty, J. C., Eady, P. M., & Elmers, J. (1974). The desert survival problem. In *Experimental learning methods*. Michigan: Plymouth Publishing.

Lee, J. D., & See, K. A. (2004). Trust in automation: Designing for appropriate reliance. *Human Factors*, *46*, 50–80.

Lee, J. D., & Seppelt, B. D. (2009). Human factors in automation design. In *Springer handbook of automation* (pp. 417–436). Berlin, Heidelberg: Springer.

LePine, J. A. (2003). Team adaptation and postchange performance: Effects of team composition in terms of members' cognitive ability and personality. *Journal of Applied Psychology*, *88*, 27–39.

Lyons, J. B., & Guznov, S. Y. (2019). Individual differences in human–machine trust: A multi-study look at the Perfect Automation Schema. *Theoretical Issues in Ergonomics Science*, *20*, 440–458.

Lyons, J. B., Stokes, C. K., & Schneider, T. R. (2011). Predictors and outcomes of trust in teams. In N. A. Stanton (Ed.), *Trust in military teams* (pp. 31–48). Surrey, UK: Ashgate.

Mach, M., Dolan, S., & Tzafrir, S. (2010). The differential effect of team members' trust on team performance: The mediation role of team cohesion. *Journal of Occupational and Organizational Psychology*, *83*, 771–794.

Madhavan, P., & Wiegmann, D. A. (2007). Similarities and differences between human–human and human–automation trust: An integrative review. *Theoretical Issues in Ergonomics Science*, *8*, 277–301.

Marsh, S., & Dibben, M. R. (2003). The role of trust in information science and technology. *Annual Review of Information Science and Technology*, *37*, 465–498.

Mayer, R. C., Davis, J. H., & Schoorman, F. D. (1995). An integrative model of organizational trust. *The Academy of Management Review*, *20*, 709–734.

Merritt, S. M., Heimbaugh, H., LaChapell, J., & Lee, D. (2013). I trust it, but I don't know why: Effects of implicit attitudes toward automation on trust in an automated system. *Human Factors*, *55*, 520–534.

Merritt, S. M., & Ilgen, D. R. (2008). Not all trust is created equal: Dispositional and history based trust in human-automation interactions. *Human Factors*, *50*, 194–210.

Merritt, S. M., Unnerstall, J. L., Lee, D., & Huber, K. (2015). Measuring individual differences in the perfect automation schema. *Human Factors*, *57*, 740–753.

Mohammed, S., & Angell, L. C. (2003). Personality heterogeneity in teams: Which differences make a difference for team performance? *Small Group Research*, *34*, 651–677.

Mount, M. K., Barrick, M. R., & Stewart, G. L. (1998). Five-factor model of personality and performance in jobs involving interpersonal interactions. *Human Performance*, *11*, 145–165.

Naas, C., & Lee, K. M. (2001). Does computer-synthesized speech manifest personality? Experimental tests of recognition, similarity-attraction, and consistency-attraction. *Journal of Experimental Psychology: Applied*, *7*, 171–181.

Niculescu, A., van Dijk, B., Nijholt, A., Li, H., & See, S. L. (2013). Making social robots more attractive: The effects of voice pitch, humor and empathy. *International Journal of Social Robotics*, *5*, 171–191.

Nomura, T., Kanda, T., & Suzuki, T. (2006). Experimental investigation into influence of negative attitudes toward robots on human–robot interaction. *AI & Society*, *20*, 138–150.

Oleson, K. E., Billings, D. R., Kocsis, V., Chen, J. Y., & Hancock, P. A. (2011). Antecedents of trust in human-robot collaborations. In *2011 IEEE international multi-disciplinary conference on cognitive methods in situation awareness and decision support (CogSIMA)* (pp. 175–178): IEEE.

Paunonen, S. V., & Ashton, M. C. (2001). Big five factors and facets and the prediction of behavior. *Journal of Personality and Social Psychology*, *81*, 524–539.

Peeters, M. A., Van Tuijl, H. F., Rutte, C. G., & Reymen, I. M. (2006). Personality and team performance: A meta-analysis. *European Journal of Personality: Published for the European Association of Personality Psychology*, *20*, 377–396.

Rauthmann, J. F., Seubert, C. T., Sachse, P., & Furtner, M. R. (2012). Eyes as windows to the soul: Gazing behavior is related to personality. *Journal of Research in Personality*, *46*, 147–156.

Rempel, J. K., Holmes, J. G., & Zanna, M. P. (1985). Trust in close relationships. *Journal of Personality and Social Psychology*, *49*, 95–112.

Robert, L. (2018). Personality in the human robot interaction literature: A review and brief critique. In L. P. Robert (Ed.), *Personality in the human robot interaction literature: A review and brief critique, Proceedings of the 24th Americas conference on information systems, August* (pp. 16–18).

Robertson, I., & Callinan, M. (1998). Personality and work behaviour. *European Journal of Work and Organizational Psychology*, *7*, 321–340.

Robot (2019). *English Oxford living dictionary. Retrieved March 20, 2019, from (2019).* https://en.oxforddictionaries.com/definition/robot.

Salem, M., Lakatos, G., Amirabdollahian, F., & Dautenhahn, K. (2015). Would you trust a (faulty) robot?: Effects of error, task type and personality on human-robot cooperation and trust. In *Proceedings of the 10th annual ACM/IEEE international conference on human-robot interaction* (pp. 141–148): ACM.

Schaefer, K. E., Chen, J. Y., Szalma, J. L., & Hancock, P. A. (2016). A meta-analysis of factors influencing the development of trust in automation: Implications for understanding autonomy in future systems. *Human Factors*, *58*, 377–400.

Schneider, T. R., Jessup, S. A., Stokes, C., Rivers, S., Lohani, M., & McCoy, M. (2017). The influence of trust propensity on behavioral trust. In *Poster session presented at the meeting of Association for Psychological Society, Boston, MA*.

Schoorman, F. D., Mayer, R. C., & Davis, J. H. (2007). An integrative model of organizational trust: Past, present, and future. *Academy of Management Review*, *32*, 344–354.

Shamsuddin, S., Yussof, H., Ismail, L., Hanapiah, F. A., Mohamed, S., Piah, H. A. , et al. (2012). Initial response of autistic children in human-robot interaction therapy with humanoid robot NAO. In *2012 IEEE 8th international colloquium on signal processing and its applications* (pp. 188–193): IEEE.

Singh, I. L., Molloy, R., & Parasuraman, R. (1993). Automation-inducted "complacency": Development of the complacency-potential rating scale. *The International Journal of Aviation Psychology*, *3*, 111–122.

Szalma, J. L., & Taylor, G. S. (2011). Individual differences in response to automation: The five factor model of personality. *Journal of Experimental Psychology: Applied*, *17*, 71–96.

Thielmann, I., & Hilbig, B. E. (2015). Trust: An integrative review from a person–situation perspective. *Review of General Psychology*, *19*, 249–277.

Van Vianen, A. E., & De Dreu, C. K. (2001). Personality in teams: Its relationship to social cohesion, task cohesion, and team performance. *European Journal of Work and Organizational Psychology*, *10*, 97–120.

Wynne, K. T., & Lyons, J. B. (2018). An integrative model of autonomous agent teammate likeness. *Theoretical Issues in Ergonomics Science*, *19*, 353–374.

CHAPTER

8

The role of risk in human-robot trust

Rachel E. Stuck, Brittany E. Holthausen, and Bruce N. Walker

School of Psychology, Georgia Institute of Technology, Atlanta, GA, United States

Introduction

Trust is a complex construct with multiple factors, such as attributes of the situation, characteristics of the trustor (e.g., propensity to trust), and characteristics of the trustee (e.g., ability) (Mayer, Davis, & Schoorman, 1995). Within *human-robot trust,* the task, as well as the environment in which the robot is operating, are also important components to consider (Sanders, Oleson, Billings, Chen, & Hancock, 2011). Despite a considerable amount of trust literature, there is one component that researchers have yet to explore in a comprehensive manner: *risk* (see Stuck, Tomlinson, & Walker, 2019). Risk is important to incorporate into the study of trust because it is key to construct validity and the necessity of trust. Within human-robot interaction (HRI), there are various risks to consider and many *domains* of risk involved as robots are implemented in a variety of contexts: military, industrial, and even in-home assistive robots (Siegwart, Nourbakhsh, & Scaramuzza, 2011). To truly understand human-robot trust, it is imperative to have an understanding of risks relevant to HRI. Although *objective* risk, for example, the number of people that are killed in car accidents each year, is known for some domains, our discussion focuses on *perceived* risk, or the vulnerabilities understood by the person in a given context or relationship. Adjusting models of trust to incorporate two types of perceived risk, relational and situational, as well as the risk-taking propensity, is necessary to support a better understanding of HRI risks.

https://doi.org/10.1016/B978-0-12-819472-0.00008-3
© 2021 Elsevier Inc. All rights reserved.

Risk and trust

Trust has been a subject of interest in several types of interactions (human-human, human-automation, human-robot) and, therefore, has a variety of theoretical underpinnings. In this chapter, we will be drawing from the literature in these many fields to advance the field of trust in the specific context of HRI.

Defining trust

To understand the importance of risk in the study of trust, it is essential to start with an understanding of what trust is, generally, and how it is defined. There have been a variety of definitions of trust developed; one of the most frequently cited was developed by Mayer et al. (1995), "the willingness of a party to be vulnerable to the actions of another party based on the expectation that the other will perform a particular action important to the trustor, irrespective of the ability to monitor or control that other party" (p. 712).

Human-human trust generalization limitations

While this definition was originally developed for *human-human* trust within the organizational context, it is also highly relevant to the *human-robot* context. However, there are limitations of applying human-human trust theories and definitions to human-automation or HRI, since there are obvious differences between humans, automation, and robots (Hoff & Bashir, 2015; Lee & See, 2004).

One specific argument about these differences was laid out in Lee and See (2004). They argue that there are three factors that limit the generalizability of human-human trust research to human-automation contexts: lack of intentionality, lack of a social exchange relationship, and the attribution process. According to Lee and See (2004), automation's lack of intentionality refers to automation not doing a particular task or subtask because of intrinsic motivation though it had an intentional design. Despite not true intentionality, people can still perceive intentionality in automation and with robots, especially with advances in machine learning as robots can be given "goals" to learn. Next, lack of social exchange is discussed as the fact that humans care what other people think within a relationship and this influences their behavior. This reduced social exchange may hold true for systems that have minimal communication features. However, many robots available to consumers today are expressly social in nature and can, therefore, have social exchange with their users (e.g., Cozmo). This increase in social exchange with robots provides a richer social interaction than that experienced

previously with automation. Lastly, the attribution process, or the way that trust is developed, is different in human-human relationships compared to human-automation or human-robot relationships (Lee & See, 2004). Trust in a human-automation relationship has three primary attributions: performance, process, and purpose. While not vastly different from the way that people form trust with each other, the process of trust development with automation is different enough to be worth consideration.

The limitations outlined by Hoff and Bashir (2015) are mostly related to perceptions of trustworthiness; human traits perceived as trustworthy are distinct from automation (as well as a robot) characteristics perceived as trustworthy. Since trustworthiness is not the focus of our current discussion, we can retain many of the key elements of human-human trust provided from Mayer et al.'s definition for the description of human-robot trust. With the considerations of social robots in mind, Mayer et al.'s (1995) definition can be used to describe the social nature of these human-robot relationships.

Mayer et al.'s definition in HRI

Mayer et al.'s (1995) definition could be used to describe the trust needed for an individual with a mobility impairment to allow an in-home assistant robot to shave them. First, the person is demonstrating a willingness to be vulnerable, since shaving has many potential risks—from simply not getting a close shave, to being physically harmed in some serious way by the razor or robot. Second, the in-home robot is performing a task that is important to the trustor, because personal grooming is a key part of activities of daily living. Third, the trustor is engaging in this trusting behavior without the ability to necessarily control or monitor the robot. Through this example, it is clear how the three primary components of Mayer et al.'s (1995) definition—vulnerability, task importance, and inability to control or monitor—are linked to the human-robot context.

Understanding the complexities and subtleties of trust, beyond the basics that are outlined in Mayer et al.'s (1995) definition, is crucial for understanding trust in the HRI context. While trust has an affective component (i.e., related to *feelings* and emotions), there are also overt *behavioral* outcomes of trusting or distrusting a person or robot. For example, trust has been explored as a predictor of technology acceptance (Pavlou, 2003). In many cases, trusting a robot would be evidenced by allowing the robot to complete a task on your behalf (trusting) or not (distrusting). To maximize the benefits of robots, it is necessary to trust the robot appropriately by calibrating trust to system performance and limitations (Parasuraman & Riley, 1997).

Risk: A key component of trust

Mayer *et al.*'s definition of trust includes such concepts as the potential for real negative outcomes, the willingness to be vulnerable to another party, and relying on another party to perform a task important to the trustor. These aspects of the definition highlight the importance of understanding risk, as it relates to trust.

While risk has been conceptualized in a variety of ways, Aven and Renn's (2009) definition synthesized many key elements: "risk refers to uncertainty about and severity of the events and consequences (or outcomes) of an activity with respect to something that humans value" (p. 6). Robots are often specifically designed to perform tasks that are important to a human user; therefore, it is clear that there is a risk inherent to HRI.

Perceived risk focuses on the individual's perception of risk alone, not on the actual presence of those risks or hazards in the context. For example, statistically speaking, flying is one of the safest modes of transportation, but many people still perceive a very high risk of flying (especially after plane crashes are seen on the news). In contrast, hazards could be present in the environment that the individual is not aware of, so they do not perceive a risk. For example, if someone's computer has been hacked, but they are unaware that it has been, they will not perceive any risk to their private information or conversations being shared. It is an individual's *beliefs* about risk, the psychological component of the risk, that will influence their trust and behavior, so that is our focus.

While perceived risk has typically been considered a single factor, a recent review of both the theory and research around risk and trust (Stuck et al., 2019) has identified important subtypes of risk that are relevant to trust. These include *perceived relational risk* and *perceived situational risk*. Another important factor that has been identified is *risk-taking tendencies* (Stuck et al., 2019). To demonstrate how these three elements uniquely interact with trust, an updated version of Mayer et al.'s model of trust is provided in Fig. 1.

We provide details of how this model can be applied to the human-robot context. The updated model demonstrates that the trustor's trust in the robot is impacted by their own characteristics (e.g., propensity to trust and experience with robots), as well as their perceptions of robot trustworthiness (e.g., reliability and precision) which are also influenced by their own characteristics. A person will only take a risk if their feeling of trust outweighs their perceptions of risk. The outcome of taking a risk then influences the trustor's perceived trustworthiness in the robot, completing the feedback loop. For example, if the robot completes a task successfully, the trustor may have a higher perception of the robot's trustworthiness. However, if the robot does not complete a task successfully, then the

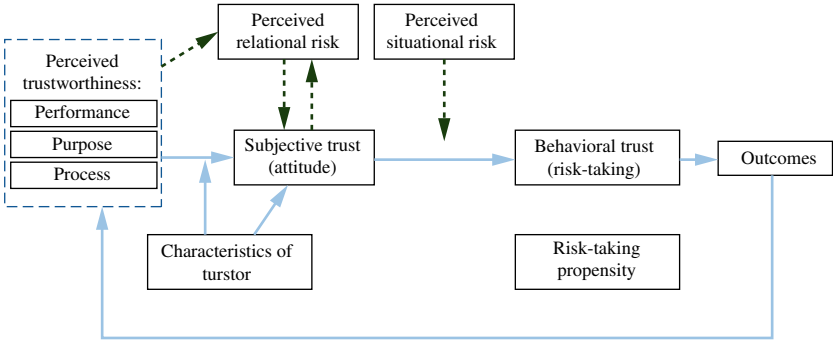


FIG. 1 Updated model of trust and risk, highlighting *perceived situational risk*, *perceived relational risk*, and *risk-taking tendencies*. Added factors and relationships are indicated with *dashed lines*. *Adapted from Mayer et al.'s (1995) model by Stuck et al. (2019).*

trustor will perceive the robot to be less trustworthy. This expansion of Mayer et al.'s (1995) model of trust increases our understanding of the relationship between trust and risk in two important ways: (1) recognizing and differentiating two subtypes of perceived risk, namely *relational* and *situational*; and (2) identifying the role of risk-taking propensity as an attribute of the trustor that has a direct influence on their trust behaviors.

Types of risk

Perceived *situational* risk is, "an individual's belief of the probability and/or feeling that a specific task or context has potential negative outcomes based on their knowledge and experience with the task, but regardless of a personal history, or historical knowledge of the system, technology, or person that may be relied on in that situation" (Stuck et al., 2019). For an individual to take a risk as a result of their trust, their trust in the robot must exceed their perceived situational risk. In reference to the aforementioned example of shaving, there are many obvious risks present in the situation itself, due to using a razor blade, regardless of whether a robot or human is doing the task. A person might have a moderate level of trust in the robot and perceive the risk of being shaved with an electric razor as low; in which case, the trust exceeds the perceived risk. Due to their trust exceeding their perceived risk, the person would accept the risk of being shaved by the robot, showing trusting behavior. That same amount of trust, however, might not be enough to allow a robot to shave with a straight razor. The risks inherent in the situation are much greater due to the form factor of the blade. In this circumstance with perceived risk being higher than trust in the robot, the person would not allow the robot to shave them, displaying distrusting behavior.

In contrast, perceived *relational* risk is, "an individual's belief about the probability and/or feeling that interacting with a specific system, technology, or person, with which a user has a personal history or historical knowledge of, has potential negative outcomes" (Stuck et al., 2019). Continuing with the shaving example, if the trustor perceives a particular brand of robot to be very risky, either through exposure to marketing materials or through direct experience (e.g., a cut from a previous shave), then this history or relationship will have a negative impact on the person's trust in the robot, regardless of the specific situation (e.g. type of razor used). Note that perceived relational risk is driven by the person's history, experience, or relationship with the robot, rather than by the situation or task. The more risk associated with a robot, the less trust the person would have in the robot, and vice versa.

Finally, *risk-taking propensity* is, "the tendency of a decision-maker either to take or to avoid risks" (Sitkin & Pablo, 1992, p. 12). It is an individual difference and has been considered a trait (or at least a long-term state), though there remains some debate about this state versus trait distinction (Mischel et al., 1995; Weber & Blais, 2006). Fig. 1 shows that risk-taking propensity will influence behavior, but this is outside of the trusting relationship. Risk-taking propensity is separated from other characteristics of the trustor that directly impact trust because it only influences the choice to engage in risk-taking regardless of trust. As is the case for other individual differences, it is possible that some users will be more likely to take risks and use a robot than others, even within the same tasks or contexts. Risk-taking propensity is specific to the domain of risk (discussed in the next section) (Weber & Blais, 2006). Risk-taking propensity influences how likely it is that the trustor will take a specific risk, separate from their trust in the robot. For example, if an individual has a high risk-taking propensity, and is therefore willing to have their automated vehicle drive very fast, this may be more reflective of their risk-taking propensity than their trust in the automated car.

Risk in human-robot research

Beyond the subtypes of perceived risk and risk-taking tendencies discussed above, there are also *domains* of risk that need to be considered. Conceptual models of risk and trust thus far have had a narrow focus on financial risk in contexts such as game theory or decision-making (Wagner, Robinette, & Howard, 2018). However, as important as financial risks may be, it is not clear that those results in the financial domain generalize to other trust situations. There are many other domains of risk, physical risk for example, that are likely to be more pertinent in HRI.

Domains of risk

Research has shown that the *domain* in which risk is involved influences the perceptions of risk or risk-taking behavior (Fox-Glassman & Weber, 2016; Wildavsky & Dake, 1990). Even in the context of everyday technologies such as smart televisions or electric can openers, the levels of perceived risk are impacted by the domain (Stuck et al., 2019). Six primary domains of risk have previously been identified in the literature: *financial*, *performance*, *physical*, *psychological*, *social*, and *time loss* (Jacoby & Kaplan, 1972; Roselius, 1971). These domains are applicable to perceived situational risk, perceived relational risk, and risk-taking tendencies for both human-human and HRI. There are also other domains of risk that need to be considered in the human-automation or human-robot context in certain cases, such as *privacy* or *security* risk (Mitzner, Stuck, Hartley, Beer, & Rogers, 2017).

Perceived situational and relational risk

In the following section, perceived risk in different domains will be defined. For brevity's sake, we are omitting the term "perceived" from each label of domain and subtype of risk. Therefore, when we are discussing "financial situational risk", we are actually talking about "*perceived* financial situational risk," and so on.

Financial risk

Financial *situational* risk is the belief that one could lose money in a given task or situation. Financial risk is often considered in the context of gambling. However, it is pertinent to any situation in which money may be lost, such as dealing with expensive equipment or technologies.

Financial *relational* risk is the belief, stemming from experience, that one could lose money because the robot has a history of failing to function appropriately; or it has a reputation for requiring frequent expensive repairs (Jacoby & Kaplan, 1972).

When driving, an individual may perceive a high financial risk of driving above the posted speed limit. The possibility of a high fine for speeding is a financial situational risk. However, this example also has a relational financial risk component. If the driver knows that the automation settings in their specific self-driving car lead to somewhat aggressive driving and occasional speeding, they may perceive a heightened financial relational risk. Thus, the existing knowledge or relationship with the robot or technology increases their perceived likelihood of financial loss. Further, if the speed of the automated car does lead to a speeding

ticket, the owner will become even less likely to trust the car to drive itself, due to the increased perceived potential for further financial loss.

Performance risk

Performance *situational* risk is the belief that there could be negative implications of the task not being completed correctly, being incomplete, or failing, regardless of who (or what) is completing the task. Performance *relational* risk is the belief that a specific third party (e.g., robot) has a history of being unreliable, which could negatively impact task completion (Jacoby & Kaplan, 1972).

Military uninhabited aerial vehicles (UAVs) are often used for surveillance purposes. The data from such missions must be seen only by the appropriate operators. Failures of the automated systems that transmit and receive the surveillance data could have substantial negative impacts, ranging from a UAV crash to leaking of sensitive data. This is an example of high-performance *situational* risk experienced by UAV operators(s). If the operator(s) knows that there is extensive encryption in the data transmission system, their performance *situational* risk may be reduced.

If the operator believes that a particular UAV's data transmission system has been experiencing irregularities, they may determine that there is a possibility of not receiving the necessary information from the UAV, leading to high *relational* performance risk. Further, a specific UAV failing to send necessary information would then impact the subsequent trust relationship between the UAV and the operator.

Physical risk

Physical *situational* risk is the belief that the task or situation could cause damage, physical harm, or negatively impact health. Physical *relational* risk is the belief that a specific third party (e.g., robot) could cause physical harm (to the individual or others) or negatively impact health (Jacoby & Kaplan, 1972).

An individual using a large in-home assistance robot, such as the PR2, may perceive a physical *situational* risk that the robot could damage their home, furniture, or knock over valuables. If an individual's PR2 broke their china while retrieving their medication, they would perceive a higher physical *relational* risk.

Psychological risk

Psychological *situational* risk is the belief that the task or situation might not align with the user's identity or the way they think about themselves; or that a situation may lead to, negative psychological states such as sadness or anxiety. Psychological *relational* risk is the belief that specific third-party (e.g., robot) actions might not align with the user's self-image or the

way they think about themselves (Jacoby & Kaplan, 1972); or that a particular robot might cause them emotional or psychological harm.

An individual who prefers to retain control will likely perceive a high psychological *situational* risk associated with using an algorithm-based match-making service. This higher psychological *situational* risk is due to giving up human control.

An introverted older adult may have a social attendant (either human or robot). After several interactions with the social attendant, the older adult may come to feel that the attendant talks too much. The person will perceive a higher psychological *relational* risk of interacting with that specific social attendant because the attendant does not interact with them in a way that matches their self-identity as an introvert.

Social risk

Beyond a person's self-identity, many people are concerned with how others will view them. Social *situational* risk is the belief that the task or situation impacts the way that people think about the individual. Social *relational* risk is the belief that a specific third party (e.g., robot) impacts how other people think about an individual (Jacoby & Kaplan, 1972).

An individual could perceive *situational* social risk when choosing an outfit for a job interview, as they are concerned about what others will think about them based on their clothing. Imagine an individual with vision loss is relying on a robotic assistant to select clothes. If the individual knows that the technology tends to select very casual clothes (e.g., jeans and a t-shirt), the user may perceive a higher social *relational* risk of using the robot to select clothes for a job interview.

Time loss risk

Time loss *situational* risk is the belief that time could be used in a way that is perceived to be inefficient or ineffective. Time loss *relational* risk is the belief that a specific third party (e.g., robot) might be late or delayed, might not be efficient with time, or might require extra effort.

Imagine a woman just went into labor and needs to get to the hospital. She will perceive a high time loss *situational* risk from knowing that her automated car is constrained to drive within the speed limit. In this same context, time loss *relational* risk would result from the woman knowing that her specific car is set to a conservative driving style, meaning it will drive even more slowly.

Ethical risk

Ethical risk was not identified in the early research focusing on perceived risk; however, recent research has identified ethical risk as a component of perceived risk (Weber, Blais, & Betz, 2002). Ethical *situational* risk can be defined as the belief that the task or situation could be viewed

as immoral or incongruent with the moral beliefs or values of an individual. Ethical *relational* risk can be defined as the belief that a specific third party's actions (e.g., robot) are immoral or incongruent with the moral beliefs of an individual.

If an individual is driving a car with their child as a passenger, and they are forced to choose between hitting a pedestrian crossing the street or crashing into a wall, they will perceive a high *situational* ethical risk. The forced ethical dilemma influences this situational risk perception.

On the other hand, if an individual were using an automated vehicle, and the same situation were to arise, the fact that the automated vehicle would be acting on behalf of the person would mean that there is a shared or *relational* ethical risk. That is, if the car owner knows that their brand of an automated vehicle is programed to prioritize the car occupants' safety (perhaps, therefore, endangering a pedestrian), there may be some ethical *relational* risk in choosing to use the automation in their car.

Privacy risk

Privacy and security concerns have also been identified as relevant risks for the human-robot context (Mitzner et al., 2017). A study exploring opinions of televideo technology, including robot platforms such as the Beam, found that participants had, "concerns about invasion or breach of privacy, exposure of one's persona information/surroundings/life…" (Mitzner et al., 2017, p. 7). Privacy *situational* risk is the belief that a task or activity will likely expose personal information about the user or their surroundings. Privacy *relational* risk is the belief a specific third party (e.g., robot or social media sharing algorithm) will expose the user or their environment to unauthorized observation or disturbance.

If an older adult is living at home alone, they may be monitored for falls with an automated, camera-based surveillance system. The older adult may perceive a high privacy *situational* risk of being monitored continuously as there are many activities that they may not want to be shared or seen by other people. If the older adult discovers that the camera feed was shared with their family members without their permission, they will perceive a high privacy *relational* risk towards using the monitoring system.

Security risk

Previous research has also identified, "…lack of security, exposure to a computer virus, and misuse of technology to gain sensitive information or incur harm," as a risk perceived when interacting with robots (Mitzner et al., 2017, p. 7). Security *situational* risk is the belief that the task or situation could be vulnerable to crime, sabotage, attack, or some other threat to safety. Security *relational* risk is the belief that a specific third party (e.g.,

robot) could be vulnerable to being misused for crime, sabotage, attack, or some other threat to safety.

There is a possibility that another individual could hack into and control any automated vehicle. Thus, an individual would perceive a high *situational* security risk towards using automated vehicles.

Consider that an individual's friend has an automated vehicle that is hacked and, thereby, stolen remotely. Even though the individual was not a part of that experience, their knowledge of it will still cause them to perceive a high-security *relational* risk of using their friend's automated car.

Summary of perceived situational and relational risk

Perceived situational risk gives describes how people perceive tasks or situations regardless of whether they are interacting with a specific human, robot, or other technology. The risk perceived in the situation influences the level of trust needed to rely on a robot (or human or other technology) within that situation or task.

On the other hand, perceived relational risk is all about the perceived risk of interacting with a specific entity (robot, person, or organization). If an individual believes that the robot (or other specific entity) is not trustworthy, they will also perceive a higher risk of using the robot.

Identifying the risks individuals perceive either situationally or relationally will give deeper insights into how people view risk within the HRI context, but also into their behavior towards robots. It is important to note that in any given context, many of these domains could be influencing perceived risk simultaneously. However, some domains may be more salient to some individuals than others. For example, in the context of a tuxedo being destroyed by improper cleaning, there is a financial risk associated with the expense of the item, performance risk associated with the negative effects of it not being cleaned (smells bad, etc.), physical harm risk associated with the item being destroyed, time risk due to having to go get a new tuxedo, and even potentially social risk if the individual wears a torn or improperly cleaned tuxedo that then embarrasses the wearer. While all of these are present, they may not all be equally salient to an individual.

Risk-taking propensity

A final aspect of risk to consider when understanding human-robot trust is an individual's risk-taking propensity. Risk-taking propensity is domain-specific, so it is essential to understand what risks are relevant to the task or situation, as well as an individual's risk-taking propensity for that context. A person's risk-taking propensity for the context will

not necessarily impact their trust in a robot. However, it will influence their likelihood of taking a risk. This will influence their behavior that might mistakenly be interpreted as trust behavior if risk-taking propensity is not fully understood. An individual with a high risk-taking propensity in gambling money is more likely in general to bet large sums of money regardless of the context. This behavior is not a result of their trust in the robot to play fair, but due to their risk-taking propensity within that domain.

Risk-taking propensity has been previously broken into a subset of categories, some of which overlap with categories discussed previously: ethical, financial, physical, recreational, and social (Weber et al., 2002; Weber & Blais, 2006). Those categories were determined by various subsets of personal decisions that people make (Weber & Blais, 2006), and need to be expanded to include all of the domains identified in "Perceived situational and relational risk" and "Risk-taking propensity" sections when discussed in the HRI context.

Financial risk-taking propensity

Within risk-taking propensity, financial risk can be broken into two primary subdomains: investment and gambling. This increased specificity is necessary to focus on potential decisions, a unique component of risk-taking propensity. Financial risk-taking propensity could be generalized to other monetary activities as well; but it has only been explored in these two contexts.

1. *Investment.* An individual's willingness to invest their money, and the potential for loss or gain associated with those investments, is a domain of risk-taking propensity. People with higher investment risk-taking will be more likely to invest in unpredictable stocks. Alternatively, lower investment risk-takers may choose to invest in accounts that are less likely to lose money such as bonds or savings accounts. This risk-taking propensity may influence an individual's choice to purchase a robot that may have a shorter performance history making it hard to determine if the robot will perform well over time. More research is necessary to determine how financial risk propensity influences robot purchasing, use, and interactions.
2. *Gambling.* Gambling risk-taking propensity is the individual's willingness to wager an amount of money on games, horse races, etc., with the potential to gain or lose money as a result of the wager (Weber et al., 2002). If a person has a high gambling risk-taking propensity, they will bet more money, even with higher odds, against their wager. Gambling risk-taking propensity may have a similar influence on robot purchasing decisions as investment risk does. Like investment risk-taking propensity, more research is necessary to understand how

people will be influenced by this characteristic when making purchasing decisions in robotics, as well as how it will influence their use of and interactions with robots.

Physical risk-taking propensity

Physical risk-taking propensity is an individual's likelihood of engaging in behaviors that are potentially harmful to property or health. Weber et al. (2002) categorized this as "health and safety," but for continuity with prior domains, we will refer to it as physical risk-taking propensity. If an individual has a high physical risk-taking propensity, they may be more willing to allow a healthcare robot to lift or shave them.

Recreational risk-taking propensity

Another domain of risk-taking propensity is a recreational risk. This includes various activities that individuals might do for fun such as camping, skiing, or skydiving. Many of these recreational activities are considered extreme sports because a failure of performance could lead to serious physical harm. In the technology domain, these individuals may be more likely to allow an automated car to drive aggressively, or to drive in thrilling, but more dangerous, circumstances such as on winding mountain roads.

Social risk-taking propensity

Social risk-taking propensity is distinct from prior definitions of social risk because it does not focus on what others think. Instead, it is focused on decisions that people make within a social context. For example, an individual would be willing to move far away from their family or friends if they have a high social risk-taking propensity because they are willing to take the social risk of being further away from their social network (Weber et al., 2002).

Ethical risk-taking propensity

Ethical risks are relevant to the individual's propensity to do something that they view as immoral or unethical, such as cheating on a test. In the HRI domain, this could be using your robot to do something that you view as immoral. For example, your robot may be able to listen to phone calls from your spouse.

These domains of risk-taking propensity reflect the areas in which people can make personal decisions. Further research should explore the way that these domains are relevant to HRI and if there are other domains relevant to HRI.

Moving forward in human-robot trust: Understanding risk

Trust is a key component of HRI; however, trust is a complex construct that is often poorly defined and understood. A crucial component of trust is that trust is only needed or even exists if there is perceived situational risk (Stuck et al., 2019). Therefore, perceived situational risk needs to be rigorously incorporated into studies of trust for a clearer understanding of the true impact of trust in HRI. Perceived risk is also a complex construct, and the aspects of risk that impact trust go far beyond just perceived situational risk. To fully understand risk and trust in the human-robot context, it is essential to understand the individual's perceived situational risk, perceived relational risk, and risk-taking propensity.

Understanding these elements of risk is only truly informative, however, if the domains of risk are also studied. The primary domains of risk that were identified over 50 years ago include: financial, performance, physical, psychological, social, and time loss (Jacoby & Kaplan, 1972; Roselius, 1971). Research has also identified domains of risk within risk-taking propensity (Weber et al., 2002; Weber & Blais, 2006). While similar to perceived relational and perceived situational risk domains, domains of risk-taking propensity must be differentiated as are reflective of areas where individuals can make personal decisions (Weber & Blais, 2006). These include: ethical, financial (investment and gambling), health/safety, recreational, and social risks (Weber et al., 2002; Weber & Blais, 2006).

A recent examination of the literature highlighted how researchers could consider risk (Stuck et al., 2019). First, when conducting experiments, risks that would be present in the "real" world should also be present in the experiment to the fullest extent ethically possible. Once the relevant domains of risk are identified—and multiple or even all of them could be present in complex tasks—it is important to include those risks in experimental studies. This could be done either through manipulation or measurement, even if it is not to the same degree as real life. That is, a researcher should not be ethically able to put a participant's life at risk, but the researcher could presumably introduce other physical risk(s). Next, measuring perceived situational risk is essential to ensure that the risks present in the study are perceived by the participants. As the participants gain experience with the robot throughout the experiment, it also becomes important to measure their perceived relational risk with the robot to see how that impacts their trust behaviors and perceived situational risk. When measuring behaviors related to trust, it is critical that risk-taking propensity is measured to understand if the behavior is due to trust in the robot or simply because of the individual's high propensity to take risks in that context.

While this chapter lays out many of the foundations to begin incorporating risk into studies, there are many other aspects that need to be addressed before this field of research can move forward productively. One issue is that there is no standardized way to asses risk in research studies. Therefore, researchers and practitioners have no effective tools for easily identifying potentially relevant risks and their importance for specific tasks or situations. Another issue is that there are no validated measures of perceived risks within the domains discussed here. Ongoing research is seeking to address this gap by creating validated scales of both perceived situational and perceived relational risk for each of the domains. In addition, future work should improve models of the relationship between these three types of risk, and how risk-taking propensity develops. Understanding the timescale(s) and level of change associated with perceived situational and relational risk within individual domains and across the domains is essential to understanding how these three types mutually influence HRI.

It is important to recognize that without risk, there is no trust, especially within HRI. If the trustor perceives no situational risk then she or he does not necessarily need to *trust* the robot to *rely* on it. However, when there is risk involved, then a sophisticated view of that risk is necessary to determine the level of trust needed for a trustor to actually rely on the system. The details of the context are also important in understanding how risk and trust interact. The presence of a particular type and amount of risk in one situation can lead to a very different behavioral trust outcome compared to the presence of the same risk in a different situation. Trust is a complex construct and in combination with the complexities of HRI can be difficult to understand. Understanding and identifying the various roles of risk in human-robot trust can help mitigate many of these difficulties, in both research and application.

References

Aven, T., & Renn, O. (2009). On risk defined as an event where the outcome is uncertain. *Journal of Risk Research*, *12*(1), 1–11. https://doi.org/10.1080/13669870802488883.

Fox-Glassman, K. T., & Weber, E. U. (2016). What makes risk acceptable? Revisiting the 1978 psychological dimensions of perceptions of technological risks. *Journal of Mathematical Psychology*, *75*, 157–169. https://doi.org/10.1016/j.jmp.2016.05.003.

Hoff, K. A., & Bashir, M. (2015). Trust in automation: Integrating empirical evidence on factors that influence trust. *Human Factors*, *57*(3), 407–434. https://doi.org/10.1177/0018720814547570.

Jacoby, J., & Kaplan, L. B. (1972). The components of perceived risk. *Sv*, *SV-02*(January 1972), 382–393. Retrieved from(1972). http://www.acrwebsite.org/search/view-conference-proceedings.aspx?Id=12016.

Lee, J. D., & See, K. A. (2004). Trust in automation: Designing for appropriate reliance. *Human Factors*, *46*(1), 50–80. https://doi.org/10.1518/hfes.46.1.50.30392.

Mayer, R. C., Davis, J. H., & Schoorman, F. D. (1995). An integrative model of organizational trust. *Academy of Management Review*, *20*(3), 709–734.

Mischel, W., Shoda, Y., Bargh, J., Bolger, N., Cervone, D., & Yue Chiu, C. (1995). A cognitive-affective system theory of personality: Reconceptualizing situations, dispositions, dynamics, and invariance in personality structure. *Psychological Review*, *102*(2), 246–268. Retrieved from(1995). http://www-personal.umich.edu/~agong/temp/MischelandShoda1995-Acognitive-affectivesystmtheoryofpersonality.pdf.

Mitzner, T. L., Stuck, R., Hartley, J. Q., Beer, J. M., & Rogers, W. A. (2017). Acceptance of tele-video technology by adults aging with a mobility impairment for health and wellness interventions. *Journal of Rehabilitation and Assistive Technologies Engineering*. *4*, https://doi.org/10.1177/2055668317692755. 205566831769275.

Parasuraman, R., & Riley, V. (1997). Humans and automation: Use, misuse, disuse, abuse. *Human Factors: The Journal of the Human Factors and Ergonomics Society*, *39*(2), 230–253. https://doi.org/10.1518/001872097778543886.

Pavlou, P. A. (2003). Consumer acceptance of electronic commerce: Integrating trust and risk with the technology acceptance model. *International Journal of Electronic Commerce*, *7*(3), 69–103. https://doi.org/10.1080/10864415.2003.11044275.

Roselius, T. (1971). Consumer rankings of risk reduction methods. *Journal of Marketing*, *35*(1), 56. https://doi.org/10.2307/1250565.

Sanders, T., Oleson, K. E., Billings, D. R., Chen, J. Y. C., & Hancock, P. A. (2011). A model of human-robot trust: Theoretical model development. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, *55*(1), 1432–1436. https://doi.org/10.1177/1071181311551298.

Siegwart, R., Nourbakhsh, I. R., & Scaramuzza, D. (2011). *Introduction to autonomous mobile robots* (2nd ed.).

Sitkin, S. B., & Pablo, A. L. (1992). Reconceptualizing the determinants of risk behavior. *Academy of Management Review*, *17*(1), 9–38. https://doi.org/10.5465/amr.1992.4279564.

Stuck, R. E., Tomlinson, B. J., & Walker, B. N. (2019). *The importance of incorporating risk into the study of human-automation trust* (Manuscript in Publication).

Wagner, A. R., Robinette, P., & Howard, A. (2018). Modeling the human-robot trust phenomenon. *ACM Transactions on Interactive Intelligent Systems*, *8*(4), 1–24. https://doi.org/10.1145/3152890.

Weber, E. U., & Blais, A.-R. (2006). A domain-specific risk-taking (DOSPERT) scale for adult populations. *Judgement and Decision Making*, *1*(1), 33–47. https://doi.org/10.1037/t13084-000.

Weber, E. U., Blais, A. E. A.-R. R., & Betz, N. E. (2002). A domain-specific risk-attitude scale: Measuring risk perceptions and risk behaviors. *Journal of Behavioral Decision Making*, *15*(4), 263–290. https://doi.org/10.1002/bdm.414.

Wildavsky, A., & Dake, K. (1990). Theories of risk perception: Who fears what and why? *Daedalus*, *119*(4), 41–60. https://doi.org/10.2307/20025337.

CHAPTER

9

An explanation is not an excuse: Trust calibration in an age of transparent robots

Alan R. Wagner[a] *and Paul Robinette*[b]

[a]Department of Aerospace Engineering, The Pennsylvania State University, University Park, PA, United States, [b]Electrical & Computer Engineering, University of Massachusetts Lowell, Lowell, MA, United States

Introduction

We may soon be entering an era in which autonomous robots powered by artificial intelligence and machine learning engage in a variety of applications ranging from health care and education to war fighting. Many applications in these domains may involve potentially high-risk decisions by the robot. During autonomous driving at highway speeds, for example, the robot's perception of obstacles and emergency maneuver decision-making have resulted in pedestrian and driver fatalities (et d'Analyses, 2012; National Transportation Safety Board Office of Public Affairs, 2014). As robotic applications become more widespread, we expect that these technologies will interact with a more diverse and technologically naïve population, ignorant of the system's risks, failure modes, and limitations. If robots can be made to explain their reasons and beliefs, then, so the rationale goes, users will come "to understand, appropriately trust, and effectively manage this incoming generation of artificially intelligent partners" (Gunning, 2017). Unfortunately, this population, human users, is neither simple nor uniform and, as this chapter will argue, there are many reasons to believe that the creation of an explainable robot will not result in better trust calibration or will do so only over a narrowly tailored subset of users or applications.

https://doi.org/10.1016/B978-0-12-819472-0.00009-5
© 2021 Elsevier Inc. All rights reserved.

Trust calibration is the process by which a person comes to accurately recognize the risks associated with trusting a person or machine (Lee & See, 2004). During the trust calibration process, the user learns both the system's competencies and likely failure modes and comes to contextualize and individualize the risks associated with using the technology (Parasuraman, Molloy, & Singh, 1993). For a variety of reasons, some individuals may undertrust an autonomous system. Prior history, personality, or lack of experience may influence a user to overestimate the risks associated with using the autonomous system. The trust calibration process for a person that currently undertrusts an autonomous system is the process of slowly building trust in the system and typically requires experience and iterative interactions with the system (Jansen, Nguyen, Karpitsky, & Mettenleiter, 1995; King-Casas et al., 2005). The opposite of undertrust is overtrust (Lee & See, 2004). Overtrust occurs when the user accepts too much or the wrong types of risks believing that the machine will mitigate their risk (Borenstein, Wagner, & Howard, 2018; Wagner, Borenstein, & Howard, 2018). At its core, overtrust is a problem of improper generalization. Humans tend to generalize a pattern of behavior from a small number of experiences with a system (Parasuraman & Riley, 1997). When attempting to evaluate the risks associated with using a system, a person will tend to extrapolate from either their prior limited experiences with the system under consideration or from perceptually or contextually similar systems (Wagner, 2015). This extrapolation can lead the user to risk underestimation and misuse of the system (Parasuraman & Riley, 1997).

It has been reasonably suggested that developing robots that explain their behavior and/or decision-making could assist with the trust calibration process (Gunning, 2017; Wachter, Mittelstadt, & Floridi, 2017). For example, Dzindolet, Peterson, Pomranky, Pierce, and Beck (2003) showed that trust in an automated decision aids decreased after the decision aid made an error unless the system provided an explanation of its behavior. More recently, Wang, Pynadath, and Hill (2016) have presented a method that utilizes the components of a POMDP (Partially Observable Markov Decision Process) model and natural-language templates to produce human-readable explanations of the robot's behavior. In a well-defined reconnaissance task, they demonstrate that robot-generated explanations significantly increase trust, transparency, and compliance but only effect mission success when the robot is of low ability. Seemingly, a key facet of their approach is the fact that the robot's explanations allow the human teammate to better understand the robot's decision-making. As the author's note, this connection is critical for proper trust calibration (Mayer, Davis, & Schoorman, 1995). In contrast, the explanations generated by Dzindolet

et al. had no connection to the system's underlying decision-making but nevertheless similarly increased participant trust. In fact, Robinette, Howard, and Wagner (2015) showed that properly timed apologies and promises by a robot increased trust as much as insightful explanations of the robot's decision-making.

Taken together, these results should raise considerable concern in that they suggest that incorrect, vacuous, or disingenuous explanations or excuses may all have equal ability to increase trust without having any underlying connection to proper trust calibration. This prior work also indicates that proper evaluation of the utility of robot-generated explanations should focus on the understanding that is gained by the human from the explanation as well as how that information is used rather than simply on the impact that the explanation has on trust. Unfortunately, as we discuss below, there are a number of real world issues that must be addressed.

Considering the human in an age of robot explanations

It has long been established that humans attribute theory of mind and intentionality to their robotic counterparts (Hegel, Krach, Kircher, Wrede, & Sagerer, 2008; Kim & Hinds, 2006). Humans utilize a theory of mind framework to predict and judge the intentionality, behavior, beliefs, and risks associated with other humans (Hilton, 2007; Malle, 2006). Some evidence suggests that people also use theory of mind to evaluate the mental state of a robot (Eyssel, Kuchenbrandt, & Bobinger, 2011; Kiesler & Goetz, 2002; Kiesler, Lee, & Kramer, 2006). It has been shown that a number of the same neural processes activated by interpersonal social cognition are also activated by human-like robots (Wykowska, Chaminade, & Cheng, 2016). For example, robotic bodies depicting emotions activate similar early visual processing mechanisms as those activated when observing humans (Chaminade et al., 2010). Motor resonance, the internal attunement of an observer's motor system to a perceived action, in human observers has also been shown to be activated by witnessing the movement of a humanoid robot (Press, Bird, Flach, & Heyes, 2005). Research has shown that people use the same concepts to explain a robot's behavior as they do for human's, further indicating that people infer intentionality from interactions with a robot (De Graaf & Malle, 2017). This large body of research suggests that human users will automatically ascribe intentionality to the explanations generated by a robot and may infer increased competence in the system.

Much of the human-robot trust research assumes that people will be deliberative in their interactions with a robot (Castelfranchi & Falcone,

2010; Haring, Matsumoto, & Watanabe, 2013). In other words, human users will consider the reputational characteristics of the robot and use these characteristics to evaluate and predict the trust/risk associated with some future action. However, dual-process models of human cognition suggest otherwise (Evans & Stanovich, 2013; Kahneman & Frederick, 2002). Dual-process models note that humans tend to use mental shortcuts and simple heuristics rather than sophisticated reputational reflection to answer social questions. With respect to human-robot trust, this theory suggests that people may not reflect on or deliberate over a robot's explanation but rather simply accept it at face value. In fact, we contend that simply offering an explanation, even a mindless one, will impact trust and compliance. Langer, Blank, and Chanowitz (1978) tested whether or not people actually consider the rationale behind an explanation during an interpersonal interaction. In their study, confederates of the experimenter interrupted subjects prior to making photocopies in a library, asking if they could use the copier first. The confederates either provided no explanation at all, included placebic information ("May I use the Xerox machine because I have to make copies?"), or provided real information ("May I use the Xerox machine, because I'm in a rush?"). The results suggest that people use a simple rote script to determine whether or not to comply with the request and, under certain conditions, do not evaluate the explanation. Hence, it is unclear if, and under what conditions, humans will actually deliberate over a robot's explanation. One could reasonably speculate that human-robot trust experiments strongly sway subjects toward greater deliberation, especially when interacting with simulated or virtual robots. In fact, Robinette found evidence of greater deliberation in virtual human-robot trust experiments than when the same task was conducted in a real environment with a physical robot (Robinette, Wagner, & Howard, 2016).

Hence, if we are to develop robots that use explanations to increase trust, then we must carefully consider if and how these explanations will influence the person receiving the explanation. We should not assume that the person receiving the explanation will thoroughly consider it and should assume that the very act of explaining may, in itself, sway a person's decision-making. Even explanations that offer insight into the explainer's reasoning, beliefs, and motivations may not be thoroughly considered and digested by the human receiving the explanation if that person is under time pressure, distracted, burdened by a high cognitive workload, or if the explanation is repetitious or boring. This portends the need to predict how an explanation will influence the person and look for and respond to signs of the explanation's effects on the person. Clearly these are difficult perceptual and human modeling tasks beyond today's robots. In the following section, we hypothesize how a robot's explanations will lead to overtrust.

Explanations and overtrust

As mentioned above, overtrust can occur when a person misunderstands the risks associated with an action because the person either (1) underestimates the loss associated with a trust violation, (2) underestimates the chance that the robot will make a mistake, or (3) both (Borenstein et al., 2018; Parasuraman & Riley, 1997; Wagner et al., 2018). We argue that robot-generated explanations will encourage people to overestimate a robot's competence and underestimate the likelihood of a mistake. While it is true that an explanation may offer insight into the explainer's reasoning, beliefs, and motivations (Malle, 1999, 2006), Langer et al. (1978) demonstrated that humans will often take an explanation at face value and not consider the explainer's reasoning, beliefs, or motivations. We believe that this effect will also be found in robot-provided explanations.

It has been shown that robots with human-like abilities, such as speech understanding and emotion generation, promote perceptions of intelligence (Breazeal, 2004). Breazeal's use and development of "robot creatures" served to tame the tendency to attribute intelligence to these robots. Similarly, Bartneck et al. investigated how different robot types impact perceived animacy and intelligence, finding that animacy and perceived intelligence are highly correlated (Bartneck, Kanda, Mubin, & Al Mahmud, 2007, 2009). The style and the cultural origin of a robot's communication have also been shown to influence a person's estimation of system likeability and credibility (Rau, Li, & Li, 2009). Explanations, because they activate schemas associated with competence, may therefore persuade naïve humans to overestimate a robot's ability and underestimate the likelihood of making a mistake.

Relatedly, robot-generated explanations may promote automation bias (Skitka, Mosier, & Burdick, 1999). Automation bias is the tendency to overrely on automation (Singh, Molloy, & Parasuraman, 1993). Applied to interactive robots, automation bias reflects a person's impulse to accept or defer to a robot's recommendations, directions, or decision-making because of a stereotypical belief in the flawlessness of robots or of the robot's creator. Automation bias can have a direct and immediate impact on trust by encouraging deference to the robot's decisions (Robinette, Allen, Li, Howard, & Wagner, 2016). Explanations may invite automation bias by again activating schemas related to competence. The activation of competence schemas combined with the use of least effort default thinking in the form of broadly applied heuristics suggests that people will accept a robot's explanation and decision-making as authoritative. Moreover, robots are often marketed as being more competent than people. United States Transportation Secretary Anthony Foxx states, "Automated vehicles have the potential to save thousands of lives, driving

the single biggest leap in road safety that our country has ever taken," the clear implication being that autonomous vehicles are better drivers than humans (Kelley, 2017; United States Department of Transportation, 2016). We can expect that as robots are marketed and introduced to the public, similar claims will be made by those promoting these systems as superior to humans. Blind acceptance of the robot's explanations and overtrust in the system seem likely to follow.

We therefore believe that the success of a robot's automatically generated explanations should be considered in terms of correct trust calibration, not simply promotion of trust. Viewed in this light, explanations should provide insight into the system's abilities and limitations, command user attention, and promote user deliberation.

Avoiding inadvertent overtrust

There is clear potential for a robot to inadvertently encourage overtrust in its capabilities by naively providing explanations. Effective strategies toward avoiding this situation have not yet been explored in the literature; so in this section, we offer some thoughts on framing and structuring a system to help equip robots with the ability to address their mistakes responsibly. As with other situations where a human may overtrust a robot, the robot must first understand if the situation warrants trust between the human and the robot. Prior work has explored defining trust situations computationally (Castelfranchi & Falcone, 2010; Wagner, 2009; Wagner & Robinette, 2015), but ultimately the robot must be able to reason about the situation and decide if it is capable of performing an action to help in the situation and if it should (ethically) perform that action (see "Explanations, trust, and ethics" section).

Assuming that the robot knows if it should be trusted or not, it needs to determine if the human will trust it, depending on the actions it takes. This is tackled in related work by measuring a trust decision or level of trust a human has in a robot versus actions that the robot has taken (Booth et al., 2017; Robinette, Allen, et al., 2016; Salem, Lakatos, Amirabdollahian, & Dautenhahn, 2015). These studies tend to focus on particular situations, but patterns are beginning to emerge that may allow the robot to better estimate the conditions in which a person may or may not trust it. Stereotyping can also be used as a tool to bootstrap a model of human-robot trust in a situation (Wagner, 2015). Additionally, observations of the human can inform such a model. In a situation where the human and robot interact over more than a single trust decision, the robot can measure the human's prior decisions to trust and extrapolate to the current situation. Physiological measurements, like heart rate, galvanic skin response, or even facial expression recognition, may also inform this model. There is currently

little research in this area regarding human-robot trust, but initial work includes (Novitzky, Robinette, Benjamin, Fitzgerald, & Schmidt, 2019). The requirements that (1) the robot understand if trust is involved and (2) for the robot to understand if the human is likely to trust it are needed in all situations where the robot's behavior can affect the human's trust, regardless of the use of explanations by the robot. Explanations add another layer to the trust feedback system. At a minimum, the robot must be able to predict the effect of an explanation on the human in a given situation. This could be accomplished by exhaustively testing situations where the robot fails measuring the effect of an explanation, but we are already beginning to see patterns that can help the robot decide when to explain. Prior work has found that timing is critical for trust repair (Robinette et al., 2015). This gives the robot one mechanism to vary in order to encourage or discourage trust. Explaining a fault immediately after an error is unlikely to repair trust and thus may be a responsible action for a robot to take to promote human understanding of the robot without also increasing trust in that situation.

Insights for improving robotic explanations

Humans use a variety of strategies to show uncertainty while conveying a message in a situation, thereby avoiding overtrust even if the other human does not listen to the content of the message. For example, information can be shared in a questioning tone to indicate uncertainty. Robots can use similar tactics to convey their internal uncertainty to a nearby human, for example, using motion, color displays, audio tone, and loudness of verbal statements as a means to convey the robot's uncertainty.

Techniques such as these have been proposed as possible mechanisms for communicating uncertainty by autonomous vehicles. Given that overtrust of autonomous vehicles poses a risk to both drivers and pedestrians, the development of a framework for communicating uncertainty and reliability would be of value. Mirnig, Wintersberger, and Ziegler (2016) present a preliminary framework for considering how to generate cues for an autonomous system to foster both trust and distrust. Noah and Walker (2017) describe a preliminary test framework to explore the impact of displays cuing different modalities and levels of information on autonomous vehicle operator trust. Kunze, Summerskill, Marshall, and Filtness (2018) use augmented reality to examine different types of displays for communicating uncertainty information during autonomous driving. Their experiments show that color and animation convey a sense of urgency and were well understood by human subjects.

While it is clear that the robot's tasking will play a critical role relating how the urgency and correctness of an explanation should be

communicated, several general principals can easily be identified. First, the explanation must be understandable by the target audience. For example, if the target audience is children, then the explanation may need to avoid complex phrasing developed for an adult. Additionally, the robot must use modalities that are easily perceived and interpreted by the target audience. The use of color and motion may not be acceptable if the target audience includes the visually impaired. Finally, the robot may need to repeat its communication of uncertainty, especially if the human fails to demonstrate that they understand the robot's limitations. Ideally, additional communications of uncertainty or unreliability should be presented with increased urgency.

Explanations, trust, and ethics

As discussed above, an explanation may not increase trust for the right reasons. As demonstrated by Dzindolet et al. (2003), explanations need not be inherently tied to the system's underlying decision-making. An explanation may thus serve as more of an excuse for the robot's behavior or, at worst, a direct attempt to psychologically manipulate the user for the robot or robot designer's purpose. Although robot-generated excuses are not inherently unethical, the use of explanations to intentionally manipulate a person may be unethical. As with many ethical dilemmas, the use of manipulation by a robot has the potential to be ethical (from a consequentialist point of view) if the manipulation results in an overall benefit (Mill, 1895). Morally justifiable application of manipulative explanations will depend on the specific context and underlying reasons for the manipulation but tends to occur in search-and-rescue operations, military operations (Arkin, 2011), and health-care settings (Shim, Arkin, & Pettinatti, 2017). Arkin, for example, notes that under certain conditions, a variety of ethical frameworks may be used to justify robotic deception (Arkin, 2018).

The preceding discussion suggests that roboticists, as a community, may need to develop guidelines delimiting the acceptable use of explanations. It is not obvious, however, on what ethical framework or line of argumentation to base such guidelines. One might argue from a rights-based framework, for example, that manipulative explanations restrict one's autonomy and ability to make decisions. The use and creation of fake news is perhaps an extreme example of how explanations in the form of news stories can be used to manipulate on large populations and influence a population's decisions (Lazer et al., 2018). One can imagine a science fiction-esque, yet easily implemented system in which one's Alexa robot generates user satisfying but fake stories in order for the product to gain acceptance in a household. Although the thought of such a system may

seem morally repugnant, it is not so different from normal human behavior (Steinel & De Dreu, 2004). Humans often strategically misrepresent themselves in order to gain acceptance in social situations (Hall, Park, Song, & Cody, 2010). In fact, normative human behavior, which is often used as a first step toward understand robot ethics related questions (Awad et al., 2018), is replete with examples of explanations and excuses used to manipulate relationships and behavior (Serota, Levine, & Boster, 2010). Should we expect the trust-building explanations generated by autonomous robots to be less manipulative than the explanations of a typical online dater? These important questions will need to be addressed in a case-by-case manner by the communities and stakeholders that are affected.

Conclusions

The development and creation of explainable artificial intelligence is a convenient answer to the question of how robots should expose more of their internal state information to humans, especially after robots make mistakes. We believe that this technology will also lead to overtrust of robots because humans do not always listen to the content of an explanation, instead making decisions based on the presence of an explanation alone. Prior work has also found that the content of a robot's trust repair statement had little effect compared to the timing of a statement. Robot designers must be aware of this phenomenon to ensure that robots increase trust responsibly and avoid overtrust. Future work should focus on developing a better understanding of when and why humans increase their trust in a robot based on an explanation. This will lead to a system that allows a robot to decide when it should employ an explanation to increase trust. Even the best system to repair trust may lead to overtrust, though. Roboticists as a community must be aware of the ethical implications of these systems to best create robots that act responsibly in difficult and dangerous situations.

Acknowledgments

This work was partially supported by the Air Force Office of Scientific Research contract FA9550-17-1-0017 and National Science Foundation grant CNS-1830390. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the National Science Foundation.

References

Arkin, R. C. (2011). The ethics of robotic deception. In C. Ess, & R. Hagengruber (Eds.), *The computational turn: Past, presents, futures?* (pp. 118–119). Münster: Verlagshaus, Monsenstein, and Vannerdat OHG.

Arkin, R. C. (2018). Ethics of robotic deception [opinion]. *IEEE Technology and Society Magazine*, *37*(3), 18–19.

Awad, E., Dsouza, S., Kim, R., Schulz, J., Henrich, J., Shariff, A. , et al. (2018). The moral machine experiment. *Nature*, *563*(7729), 59.

Bartneck, C., Kanda, T., Mubin, O., & Al Mahmud, A. (2007). The perception of animacy and intelligence based on a robot's embodiment. In *2007 7th IEEE-RAS international conference on humanoid robots* (pp. 300–305): IEEE.

Bartneck, C., Kanda, T., Mubin, O., & Al Mahmud, A. (2009). Does the design of a robot influence its animacy and perceived intelligence? *International Journal of Social Robotics*, *1*(2), 195–204.

Booth, S., Tompkins, J., Pfister, H., Waldo, J., Gajos, K., & Nagpal, R. (2017). Piggybacking robots: human-robot overtrust in university dormitory security. In *International conference on human-robot interaction* (pp. 426–434).

Borenstein, J., Wagner, A. R., & Howard, A. (2018). Overtrust of pediatric healthcare robots: a preliminary survey of parent perspectives. *IEEE Robotics and Automation Magazine: Special Issue on Socio-ethical Approaches to Robotics Development*, *25*(1).

Breazeal, C. L. (2004). *Designing sociable robots.* MIT Press.

Castelfranchi, C., & Falcone, R. (2010). *Trust theory: A socio-cognitive and computational model.* John Wiley & Sons.

Chaminade, T., Zecca, M., Blakemore, S. J., Takanishi, A., Frith, C. D., Micera, S. , et al. (2010). Brain response to a humanoid robot in areas implicated in the perception of human emotional gestures. *PLoS One*, *5*(7) e11577.

et d'Analyses, B. D. (2012, July). *Final report on the accident on 1st June 2009 to the Airbus A330-203 registered F-GZCP operated by Air France flight AF 447 Rio de Janeiro–Paris.* Paris: BEA.

De Graaf, M. M., & Malle, B. F. (2017). How people explain action (and autonomous intelligent systems should too). In *2017 AAAI Fall Symposium Series*.

Dzindolet, M. T., Peterson, S. A., Pomranky, R. A., Pierce, L. G., & Beck, H. P. (2003). The role of trust in automation reliance. *International Journal of Human-Computer Studies*, *58*(6), 697–718.

Evans, J. S., & Stanovich, K. E. (2013). Dual-process theories of higher cognition: Advancing the debate. *Perspectives on Psychological Science*, *8*(3), 223–241.

Eyssel, F., Kuchenbrandt, D., & Bobinger, S. (2011). Effects of anticipated human-robot interaction and predictability of robot behavior on perceptions of anthropomorphism. In *Proceedings of the 6th international conference on human-robot interaction*, (pp. 61–68): ACM.

Gunning, D. (2017). *Explainable artificial intelligence (xai).* Defense Advanced Research Projects Agency (DARPA).

Hall, J. A., Park, N., Song, H., & Cody, M. J. (2010). Strategic misrepresentation in online dating: The effects of gender, self-monitoring, and personality traits. *Journal of Social and Personal Relationships*, *27*(1), 117–135.

Haring, K. S., Matsumoto, Y., & Watanabe, K. (2013). How do people perceive and trust a lifelike robot. In *Vol. 1. Proceedings of the world congress on engineering and computer science*.

Hegel, F., Krach, S., Kircher, T., Wrede, B., & Sagerer, G. (2008). Theory of mind (ToM) on robots: A functional neuroimaging study. In *Proceedings of the 3rd ACM/IEEE international conference on human robot interaction* (pp. 335–342).

Hilton, D. (2007). Causal explanation: From social perception to knowledge-based causal attribution. In A. W. Kruglanski & E. T. Higgins (Eds.), *Social psychology: Handbook of basic principles* (2nd ed., pp. 232–253). *New York, NY: Guilford Press.*

Jansen, A., Nguyen, X., Karpitsky, V., & Mettenleiter, M. (1995). Central command neurons of the sympathetic nervous system: Basis of the fight-or-flight response. *Science Magazine*, *5236*(270).

Kahneman, D., & Frederick, S. (2002). Representativeness revisited: Attribute substitution in intuitive judgment. Heuristics and biases: The psychology of intuitive judgment. In T. Gilovich, D. Griffin, & D. Kahneman (Eds.), *Heuristics of intuitive judgment: Extensions and applications*. New York: Cambridge University Press.

Kelley, B. (2017). Public health, autonomous automobiles, and the rush to market. *Journal of Public Health Policy*, *38*(2), 167–184.

Kiesler, S., & Goetz, J. (2002). Mental models of robotic assistants. In *CHI'02 extended abstracts on human factors in computing systems*, (pp. 576–577): ACM.

Kiesler, S., Lee, S. L., & Kramer, A. D. (2006). Relationship effects in psychological explanations of nonhuman behavior. *Anthrozoös*, *19*(4), 335–352.

Kim, T., & Hinds, P. (2006). Who should I blame? Effects of autonomy and transparency on attributions in human-robot interaction. In *The 15th IEEE international symposium on robot and human interactive communication*, (pp. 80–85): IEEE Press.

King-Casas, B., Tomlin, D., Anen, C., Camerer, C., Quartz, S., & Montague, P. (2005). Getting to know you: Reputation and trust in a two-person economic exchange. *Science*, *308*(5718), 78–83.

Kunze, A., Summerskill, S. J., Marshall, R., & Filtness, A. J. (2018). Augmented reality displays for communicating uncertainty information in automated driving. In *Proceedings of the 10th international conference on automotive user interfaces and interactive vehicular applications*, (pp. 164–175): ACM.

Langer, E. J., Blank, A., & Chanowitz, B. (1978). The mindlessness of ostensibly thoughtful action: The role of "placebic" information in interpersonal interaction. *Journal of Personality and Social Psychology*, *36*(6), 635.

Lazer, D. M., Baum, M. A., Benkler, Y., Berinsky, A. J., Greenhill, K. M., Menczer, F., et al. (2018). The science of fake news. *Science*, *359*(6380), 1094–1096.

Lee, J. D., & See, K. A. (2004). Trust in automation: Designing for appropriate reliance. *Human Factors*, 50–80.

Malle, B. F. (1999). How people explain behavior: A new theoretical framework. *Personality and Social Psychology Review*, *3*(1), 23–48.

Malle, B. F. (2006). *How the mind explains behavior: Folk explanations, meaning, and social interaction.* MIT Press.

Mayer, R. C., Davis, J. H., & Schoorman, F. D. (1995). An integrative model of organizational trust. *The Academy of Management Review*, *20*(3), 709–734.

Mill, J. S. (1895). *Utilitarianism.* Longmans, Green and Company.

Mirnig, A. G., Wintersberger, P., Sutter, C., & Ziegler, J. (2016). A framework for analyzing and calibrating trust in automated vehicles. In *Adjunct proceedings of the 8th international conference on automotive user interfaces and interactive vehicular applications*, (pp. 33–38): ACM.

National Transportation Safety Board Office of Public Affairs. (2014, June 24). *NTSB Press Release. Retrieved December 18, 2014, from* http://www.ntsb.gov/news/press-releases/Pages/PR20140624.aspx.

Noah, B. E., & Walker, B. N. (2017). Trust calibration through reliability displays in automated vehicles. In *Proceedings of the 2017 ACM/IEEE international conference on human-robot interaction*, (pp. 361–362): ACM.

Novitzky, M., Robinette, P., Benjamin, M. R., Fitzgerald, C., & Schmidt, H. (2019). Aquaticus: Publicly available datasets from a marine human-robot teaming testbed. In *14th ACM/IEEE international conference on human-robot interaction (HRI)*, (pp. 392–400): IEEE.

Parasuraman, R., Molloy, R., & Singh, I. L. (1993). Performance consequences of automation-induced "complacency" *The International Journal of Aviation Psychology*, *3*(1), 1–23.

Parasuraman, R., & Riley, V. (1997). Humans and automation: Use, misuse, disuse, abuse. *Human Factors*, *39*(2), 230–253.

Press, C., Bird, G., Flach, R., & Heyes, C. (2005). Robotic movement elicits automatic imitation. *Cognitive Brain Research*, *25*(3), 632–640.

Rau, P. P., Li, Y., & Li, D. (2009). Effects of communication style and culture on ability to accept recommendations from robots. *Computers in Human Behavior*, *25*(2), 587–595.

Robinette, P., Allen, R., Li, W., Howard, A., & Wagner, A. R. (2016). Overtrust of robots in emergency evacuation scenarios. In *The eleventh ACM/IEEE international conference on human-robot interaction (HRI 2016), Christchurch, New Zealand* (pp. 101–108).

Robinette, P., Howard, A., & Wagner, A. R. (2015). Timing is key for robot trust repair. In *Seventh international conference on social robotics (ICSR 2015), Paris, France* (pp. 574–583).

Robinette, P., Wagner, A. R., & Howard, A. (2016). Investigating human-robot trust in emergency scenarios: methodological lessons learned. In W. Lawless, R. Mittu, D. Sofge, & A. R. Wagner (Eds.), *The intersection of robust intelligence (RI) and trust in autonomous systems*: Springer.

Salem, M., Lakatos, G., Amirabdollahian, F., & Dautenhahn, K. (2015). Would you trust a (faulty) robot?: Effects of error, task type and personality on human-robot cooperation and trust. In *Proceedings of the tenth annual ACM/IEEE international conference on human-robot interaction (HRI)* (pp. 141–148).

Serota, K. B., Levine, T. R., & Boster, F. J. (2010). The prevalence of lying in America: Three studies of self-reported lies. *Human Communication Research*, *36*(1), 2–25.

Shim, J., Arkin, R., & Pettinatti, M. (2017). An intervening ethical governor for a robot mediator in patient-caregiver relationship: Implementation and evaluation. In *2017 IEEE international conference on robotics and automation (ICRA)* (pp. 2936–2942).

Singh, I. L., Molloy, R., & Parasuraman, R. (1993). Automation induced "complacency": Development of the complacency-potential rating scale. *The International Journal of Aviation Psychology*, 111–122.

Skitka, L. J., Mosier, K. L., & Burdick, M. (1999). Does automation bias decision-making? *International Journal of Human-Computer Studies*, *51*(5), 991–1006.

Steinel, W., & De Dreu, C. K. (2004). Social motives and strategic misrepresentation in social decision making. *Journal of Personality and Social Psychology*, *86*(3), 419.

United States Department of Transportation. (2016). *U.S. dot issues federal policy for safe testing and deployment of automated vehicles. Retrieved October 12, 2016, from* https://www.transportation.gov/briefing-room/us-dot-issues-federal-policy-safe-testing-and-deployment-automated-vehicles.

Wachter, S., Mittelstadt, B., & Floridi, L. (2017). Transparent, explainable, and accountable AI for robotics. *Science robotics*, *2*(6).

Wagner, A. R. (2009). Creating and using matrix representations of social interaction. In *The 4th ACM/IEEE international conference on human-robot interaction (HRI 2009), San Diego, CA* (pp. 125–132).

Wagner, A. R. (2015). Robots that stereotype: Creating and using categories of people for human-robot interaction. *Journal of Human-Robot Interaction*, *4*(2), 97–124.

Wagner, A. R., Borenstein, J., & Howard, A. (2018). Overtrust in the robotic age. *Communications of the ACM*, *61*(9), 22–24.

Wagner, A. R., & Robinette, P. (2015). Towards robots that trust: Human subject validation of the situational conditions for trust. *Interaction Studies*, *1*(1).

Wang, N., Pynadath, D. V., & Hill, S. G. (2016). Trust calibration within a human-robot team: Comparing automatically generated explanations. In *The eleventh ACM/IEEE international conference on human robot interaction* (pp. 109–116).

Wykowska, A., Chaminade, T., & Cheng, G. (2016). Embodied artificial agents for understanding human social cognition. *Philosophical Transactions of the Royal Society, B: Biological Sciences*, *371*(1693), 20150375.

CHAPTER

10

Human-agent bidirectional transparency

Michael J. Barnes[a], *Ning Wang*[b], *David V. Pynadath*[b], *and Jessie Y.C. Chen*[a]

[a]U.S. Army Combat Capabilities Development Command – Army Research Laboratory, Adelphi, MD, United States, [b]Institute for Creative Technologies, University of Southern California, Los Angeles, CA, United States

Introduction

Artificial intelligence (AI) is changing both the civilian and military world views: profoundly altering the relationship of humans to their machines (Tegmark, 2017). Especially in military and other high-stake environments, most AI systems will be integrated into manned command and control networks requiring their outputs to be understood by humans who will in turn be responsible to explain their meaning to higher commands (Barnes, Chen, & Hill, 2017; Defense Science Board, 2016). For nontrivial systems, this will require the AI output to be transparent in terms of its intent, logic, and expected outcomes (Chen et al., 2014). Furthermore, as interactions become dynamic and continuous, human-agent teams' overall situation awareness (SA) will depend on bidirectional transparency between humans and (machine) agents (henceforth, agents). Both humans and agents must be able to interpret situational updates and, when necessary, respond autonomously to events in their environment (Chen & Barnes, 2014; Russell & Norvig, 2009). Such interactions would not to be possible without mutual trust ensuring that both the agent and its human partner understand their roles and common objectives (Bradshaw et al., 2009; Chen et al., 2018; Lyons, 2013; Wynne & Lyons, 2018). Additionally, in order for their interactions to be efficient as well as effective, calibrated trust requires an awareness of each other's strengths and limitations: both knowing when to trust its partner to

https://doi.org/10.1016/B978-0-12-819472-0.00010-1
© 2021 Elsevier Inc. All rights reserved.

complete a task and knowing when some sort of intervention might be necessary (Lee & See, 2004). Collaboration requires more than overt communication; each partner must be able to anticipate and to act in concert even without overt cues from their partner (McNeese, Demir, Cooke, & Myers, 2017; Schaefer, Straub, Chen, Putney, & Evans, 2017).

In human-agent teaming contexts, fostering a shared mental model (SMM) of an evolving situation is difficult, given humans' and agents' different processing capacities. Agents depend on the formalisms that define their limitations and advantages, whereas human decisions are based on imagery, emotions, and heuristics as well as inferential logic resulting in a broader but also possibly more biased, decision process (Barnes, Lakhmani, Holder, & Chen, 2019; Kahneman, 2011). As Fig. 1 indicates, the interaction between humans and agents is a complex process—requiring not only transparency but also calibrated trust. Humans tend to have a more nuanced view of the evolving situation and are responsible for the outcomes, and yet they must have enough calibrated trust to know when to accept (and not accept) the agents' decisions (Lee & See, 2004). As complexity increases, so does the necessity for shared awareness of each other's roles and intent (Schaefer et al., 2017). Cooke (2015) found that an efficient human-agent teaming relationship depends more on effective interactions than it does on having an extensive shared knowledge base.

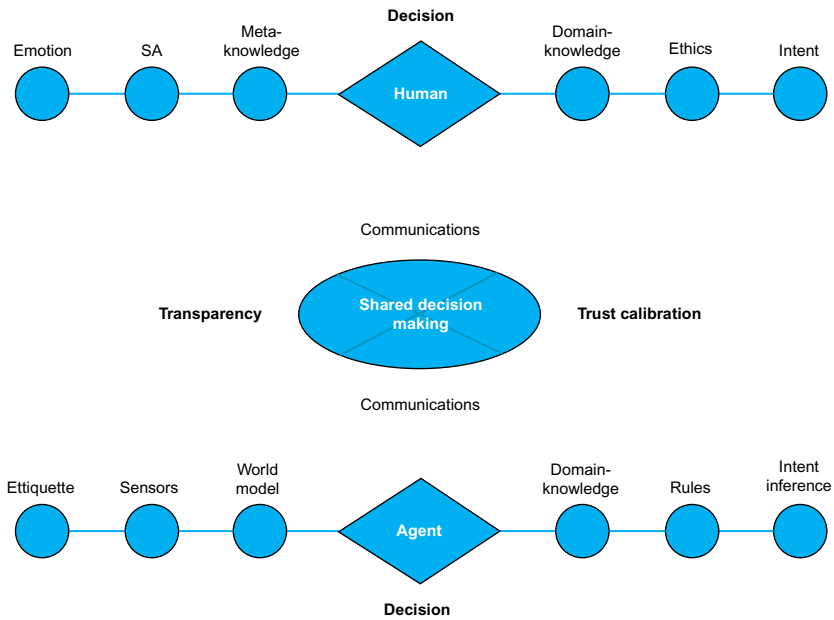


FIG. 1 Shared decision space between humans and agents (Barnes et al., 2017).

For natural dialogue, each team member must know when to push information to their partner and when to ask for information. Thus, human-agent teaming requires insight into each partner's processing state—something humans develop naturally as they interact with their human partners—but developing bidirectional transparency involves investigating algorithmic models of heretofore posited human processes such as a *Theory of Mind* (ToM), which involves a SMM of the tasking environment as well as an implicit understanding of each partner's intentions and capabilities (Premack & Woodruff, 1978; Pynadath & Marsella, 2005). To ensure mutual awareness over time, partners must be able to communicate in order to update their SMM of the evolving situation (Barnes et al., 2019).

The purpose of this chapter is to examine the implications of models of transparency and SMM to enable synergistic human-agent collaboration rather than focus solely on the human component of the equation. We will discuss human performance and algorithmic models and supporting empirical research that can potentially be used as foundations for developing human-agent teams built on mutual trust and shared SA (Chen et al., 2018; Lyons, 2013). The first section describes models of transparency and related research indicating the importance of insight into the agent's current actions, plans, and reasoning (Chen et al., 2018). We discuss visualization techniques and how they can be used to support transparency in military human-autonomy teaming contexts (Barnes et al., 2019; Cha, Barnes, & Chen, 2019; Chen et al., 2018). Next, we discuss the technical issues involved with human-agent communications in dynamic environments (Pynadath, Barnes, Wang, & Chen, 2018; Pynadath, Wang, Rovira, & Barnes, 2018; Wang, Pynadath, Hill, & Merchant, 2017). In the final section, we discuss the implications of ToM and human-agent SMM (Pynadath & Marsella, 2005). Somewhat analogous to human teams performing a military mission, human-agent trust requires (1) transparency, (2) a common operating picture, (3) technical means of communication, and (4) cognitive (or algorithmic) processes to develop a SMM of the problem space.

Situation-awareness-based agent transparency

Chen et al. (2014) developed the situation-awareness-based agent transparency (SAT) model that defined transparency as the information requirements to support human's SA of the agent in the tasking environment—in terms of the agent's actions and plans, reasoning process, and predicted outcomes including uncertainties (Fig. 2). Lee and See (2004) discussed transparency in terms of the human operator's perception of the three Ps (purpose, process, and performance) of the agent.

Situation awareness-based agent transparency

Level 1: Goals and actions

Agent's current status/actions/plans

- Purpose: Desire (goal selection)
- Process: Intentions (planning/execution); progress
- Performance
- Perception (environment/teammates)

Level 2: Reasoning

Agent's reasoning process

- Reasoning process (belief/purpose)
- Motivations
    - Environmental and other constraints/affordances

Level 3: Projections

Agent's projections/predictions; uncertainty

- Projection of future outcomes
- Uncertainty and potential limitations; likelihood of success/failure
- History of performance

FIG. 2 SAT model and related concepts (Chen et al., 2014).

Fig. 2 indicates that the SAT model incorporates some of the same assumptions of the three Ps and is based loosely on Endsley's (2015) SA model. The SAT model is the basis of information necessary for the human to understand the agent's mission plan and its associated outcomes. SAT is an additive model in the sense that full transparency includes L1 (actions and plans) + L2 (reasoning process) + L3 (predicted outcomes and their uncertainties)—implying L123 offers the human a more complete understanding of the agent's course of action compared with less complete instantiations of SAT.

The SAT model enables the human to gain insight (i.e., SA) into the agent's world model and allows humans to compare that information with their own SA of the ongoing real-world situation. Research has shown the benefits of SAT-based information for supporting operator trust calibration and SA in diverse military tasking paradigms such as planning for the defense of a littoral military installation (Mercado et al., 2016; Stowers et al., in press), a robot that acts as an autonomous squad member (ASM; Selkowitz, Lakhmani, & Chen, 2017; Wright, Chen, & Lakhmani, in press), and a convoy in-route route recommender (Wright, Chen, & Barnes, 2018). Counterintuitively, human's perceived trust in the agent was either not affected or actually increased when the agent conveyed uncertainty in planning scenarios (Mercado et al., 2016;

Stowers et al., in press), suggesting that trust in the agent's plan depended on an agent reporting possible constraints just as human staff member would report uncertainties as well the rationale for a particular plan. However, uncertainty information was less effective for the ASM paradigm, possibly because these conditions stressed the importance of making rapid decisions to support infantry squads (Selkowitz et al., 2017).

Lyons (2013) has expanded the concept of transparency beyond robot-to-human transparency to robot-of-human transparency, which is the ability of the robot to understand the specific roles of human and agents and their interaction policies. Agents must be able to understand not only their own roles but also be aware of the human's expectations regarding possible interactions with agents. Based on the original SAT model (Chen et al., 2014), a model of mutual transparency (Fig. 3) was developed that involves agents being aware of the human's actions and plans, the reasoning process (e.g., constraints), and the humans' expected outcomes—implying mutual understanding of goals and strategies or their SMM of the tasking environment (Chen et al., 2018). Suffice to say it is not a trivial task to support human transparency (to the human-agent team), which requires not only bidirectional human-agent communications but also human-agent system-level information sharing. Chen et al. (2018) also discuss constraints of dynamic environments wherein events are not necessarily independent. In the real world, the best solution may depend on previous circumstances not only because the environment is continually changing but also because previous decisions may impact the current situation. Humans are able to adjust to changes by maintaining a shared view of possible futures and adjusting

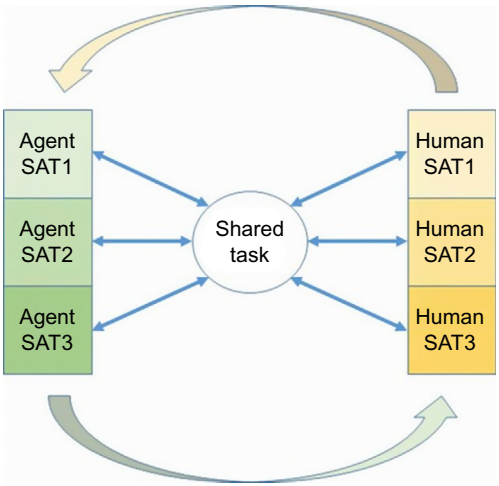


FIG. 3 Mutual transparency based on the SAT model (Chen et al., 2018).

their versions of SMM to align with the evolving situation. Constructs and research findings to support human-agent teams' shared SA will be discussed in later sections.

SAT-based user interfaces were evaluated in a variety of paradigms (Chen et al., 2018; Mercado et al., 2016; Selkowitz et al., 2017; Stowers et al., in press; Wright et al., in press). Two of the paradigms (IMPACT and ASM) were part of the larger US Department of Defense's Autonomy Research Pilot initiative (Draper et al., 2018) and as such represented intelligent agent (IA) concepts proposed jointly by US Air Force, Navy, and Army laboratories that were informed by insights developed through collaborating with active duty personnel.

IMPACT

The IMPACT paradigm includes IAs with planning and machine-learning capabilities that can task ground, aerial, and sea unmanned vehicles (UVs) for protecting a littoral military installation (Draper et al., 2018). The human operator has versatile planning options ranging from manual planning tools to having an IA function that can autonomously choose the best plan and optimal composition of UVs to conduct the plan (after receiving general mission guidance in the form of programmed *plays*; Calhoun, Ruff, Behymer, & Frost, 2018). There are three key modules or tiles in the SAT-based transparency display: a *Plan Maps* tile, a *Projected Plan Success* tile, and a *Plan Specifications* tile (Fig. 4).

Plan Maps. The interface presents two maps, one for each plan the IA suggests to the operator. SAT Level 1 information includes the IA's recommended plans and information about the UVs associated with these plans. SAT Level 2 information (reasoning) was displayed by the size of UVs' icons, with larger icons meaning faster. SAT Level 3-Projection information was displayed as an icon attached to each UV to depict the number of time units it was from its goal location. SAT Level 3-Uncertainty information was displayed through changes in opacity of the UVs themselves, where less opaque vehicles were considered to have uncertain capabilities or time projections. The experimental manipulation was the number of levels presented per condition (L1 vs. L1+L2 vs. L1+L2+L3; Stowers et al., in press).

Projected Plan Success. The plan comparison tool ("Mission Metrics," lower left corner, Fig. 4), based on a design developed by US Air Force researchers (Calhoun et al., 2018), adds additional SAT Levels 2 and 3 information. This graphic shows the evaluation of both plans (A and B) on four different parameters: time, coverage, fuel endurance, and general capability. Circles representing each plan slide up and down scales corresponding to each parameter. Plan acceptability is rated on

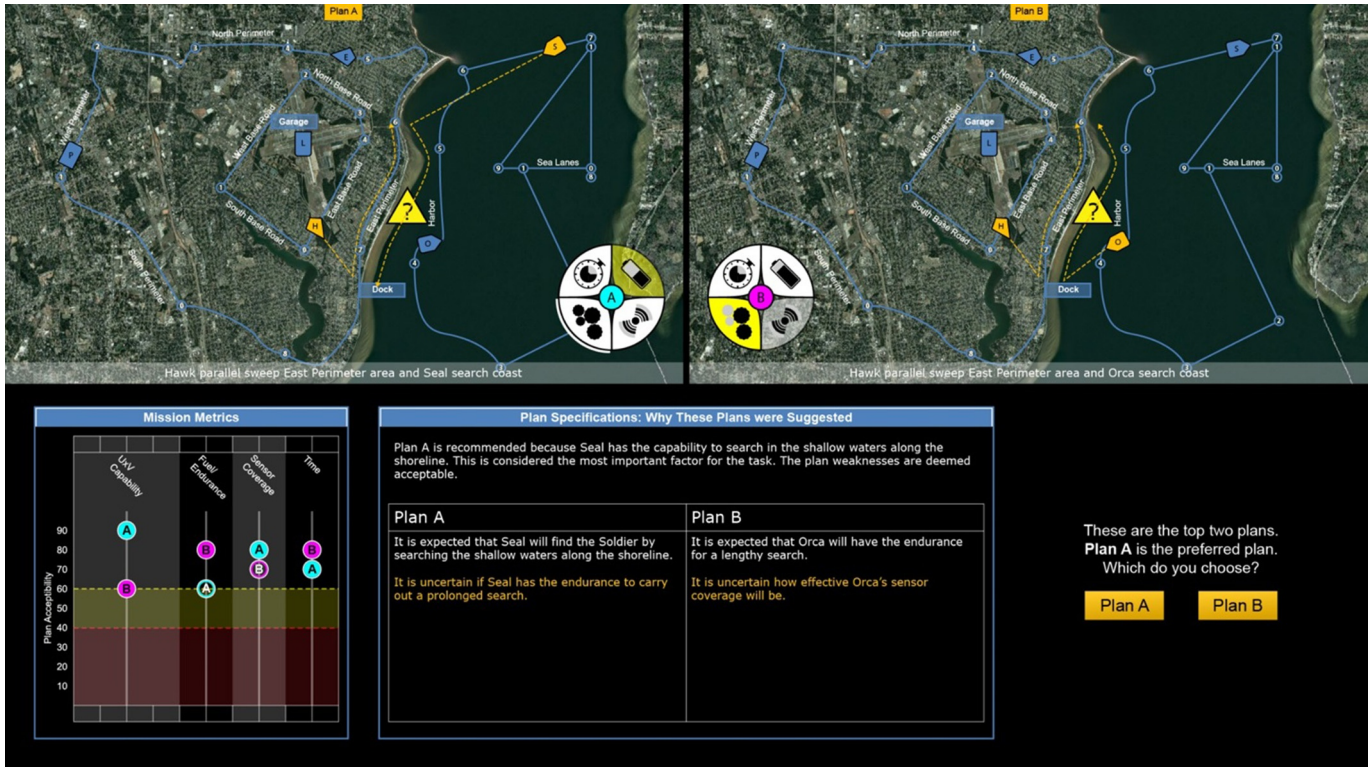


FIG. 4 IMPACT interface.

this sliding scale; the higher a circle is on the scale, the more acceptable the plan is for meeting that parameter. Level 2 information is conveyed through a text description and graphically by showing how heavily the IA is weighing each parameter as a representation of the IA's recommended plan and reasoning. This is done by ordering the parameters from left to right, with decreasing widths for less important ones. Level 3 projection information is conveyed listing possible constraints and showing the projected plan success of each plan for all four parameters. Finally, SAT Level 3 uncertainty was represented with hollow circles instead of filled ones.

Plan Specifications. The Plan Specifications tile (lower center, Fig. 4) provides further information on the *Plan Maps* and *Projected Plan Success* tiles, for example, a detailed explanation of the reasoning process (constraints and/or affordances considered by the IA); projection statements to indicate the predicted outcomes of the plans; uncertainty statements specifying (1) the IA was uncertain about certain aspects of the tasking environment, and (2) the IA was making an assumption to deal with it.

Autonomous squad member (ASM)

While IMPACT's paradigm focused on deciding between *plays* using multiple computer-based IAs, the ASM paradigm focused on human interaction with a small unmanned ground vehicle (UGV) embedded in a dismounted infantry squad to provide logistic support such as load carrying (Selkowitz et al., 2017; Wright et al., in press). The particulars of ASM were quite different from IMPACT, reflecting missions with severe time constraints. The ASM interface used an "at-a-glance" module to promote transparency for each level of the SAT model (Fig. 5). The icons were integrated using Ecological Interface Design principles (Bennett & Flach, 2013) depicting the ongoing mission including the current plan (L1: the *how*), motivators (L2: the *why*), projections (L3: the *immediate future*), and uncertainty coding (U). The red square aside the predicted outcomes icon is the amount of resources that will be consumed or lost. For level three "predicted outcomes," uncertainty was displayed by nonopaque squares to represent the possibility of using the level of resource (e.g., with one opaque square and one nonopaque square next to a battery symbol, it represented that it is certain to lose one level of energy but there is the possibility of losing a second level). The second row of icons of the "at-a-glance" module showed additional information related to the first row.

A series of human factors studies were conducted to test the effects of SAT-based information on human-agent team performance in both the IMPACT and ASM contexts (Mercado et al., 2016; Selkowitz et al., 2017; Stowers et al., in press; Wright et al., in press). Overall, it was found that

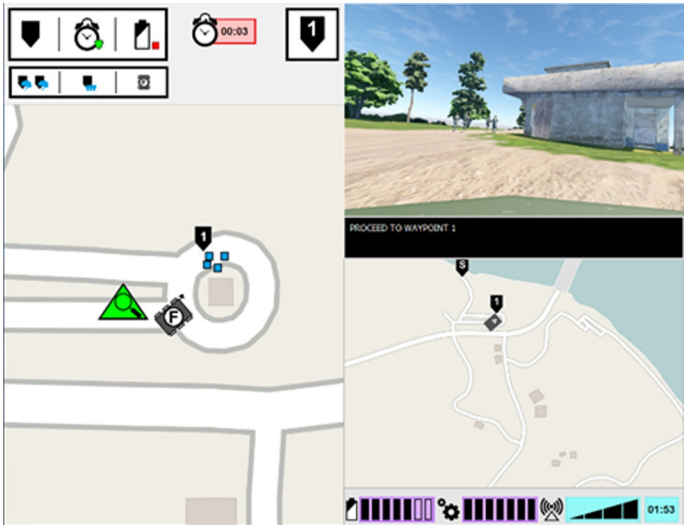


FIG. 5 ASM interface.

agent transparency, as operationalized and implemented according to the SAT model, is useful for improving operator performance without increasing operator's (subjective) workload. However, it should be noted that response time does increase for a few seconds in some cases (Stowers et al., in press), which may or may not create an issue depending on the mission environments. Consistently, the results showed that perceptions of the IA (its aptitude and usefulness) increased with agent transparency. Results from these studies yielded several insights to the utility of transparency, as well as best practices for incorporating transparency concepts into IA's user interface designs (Chen et al., 2018).

Bidirectional communications and transparency

The final sections of this chapter focus on AI research conducted at the University of Southern California (USC) to instantiate transparency-related agent concepts and their associated empirical results. There is discussion about generating explanations from the agent's perspective and its effect on operator performance (Pynadath, Barnes, et al., 2018; Wang et al., 2017; Wang, Pynadath, & Hill, 2016). Additionally, research efforts to develop agents with SMM capabilities based on ToM and their utility in various human-agent paradigms are also presented (Pynadath, Barnes, et al., 2018; Pynadath & Marsella, 2005; Pynadath, Wang, et al., 2018; Wang et al., 2016). The assumption is that agent communications eventually will be free flowing and supported by natural language

processing (NLP) to emulate human-human communications. A full discussion of NLP is beyond our scope (see Barnes et al., 2019, for expanded discussion of the issues). However, it is important to note the complexity involved in human-machine communications in real-world environments. Fig. 6 illustrates the principal components of a human-agent communication system indicating an interface with voice, chat, multimodal inputs, communications using NLP, various types of AI to support agent processing, and algorithms specialized to interpret more opaque agent techniques such as machine learning (Chakraborty et al., 2017; Pynadath, Barnes, et al., 2018). Whereas much progress has been made in human machine communications, open-ended NLP is still an objective rather than a reality for complex, uncertain environments. Still progress is evident suggesting that workable human-agent communications will eventually be robust enough for these environments. NLP researchers are investigating two-way communications, situations wherein the agent initiates dialogues, agent interpretation of human emotional states, multimodal communications, human-robot interactions, and linguistic interpretation of spatial representations (Evans et al., 2017; Mavridis, 2015; Tellex et al., 2011; Skubic et al., 2004). The major obstacle for wider use of real-world NLP systems is the complexity of language and the unpredictability of many real-world situations (Jurafsky & Martin, 2009).

Progress in circumscribed environments is already showing promise. For example, Explainable AI (XAI) proved to be useful in reducing operator errors when supervising robots performing military missions (Wang et al., 2017). Visual and tactile interfaces have been used to interact with robots when performing simple motor tasks (Barber, Reinerman-Jones, & Matthews, 2015). *Controlled English*, a subset of NLP designed for a limited domain, has been used in research settings for military intelligence, unmanned aerial vehicle, and civil affair missions (Giammanco, Mott, & McGowan, 2015; McNeese et al., 2017). Perhaps, an even more important difference between human teams and human-agent teams is that communications among human team members go beyond overt communications. Salas, Shuffler, Thayer, Bedwell, and Lazzara (2015) point out that "teams that communicate effectively may alternate between explicit communication, or overt transmission and acknowledgment of messages, and implicit communication, whereby information is more passively conveyed." In particular, it is difficult to design agents to know when to ask for information or when to push information to its human teammates, a natural interaction between experienced human team members (Cooke, 2015; McNeese et al., 2017). This is partly due to human teammates gaining experience by working together but also because humans are wired to have a ToM, which is the ability to recognize and attribute mental states (e.g., beliefs, desires, goals) to oneself and others (Premack & Woodruff, 1978).

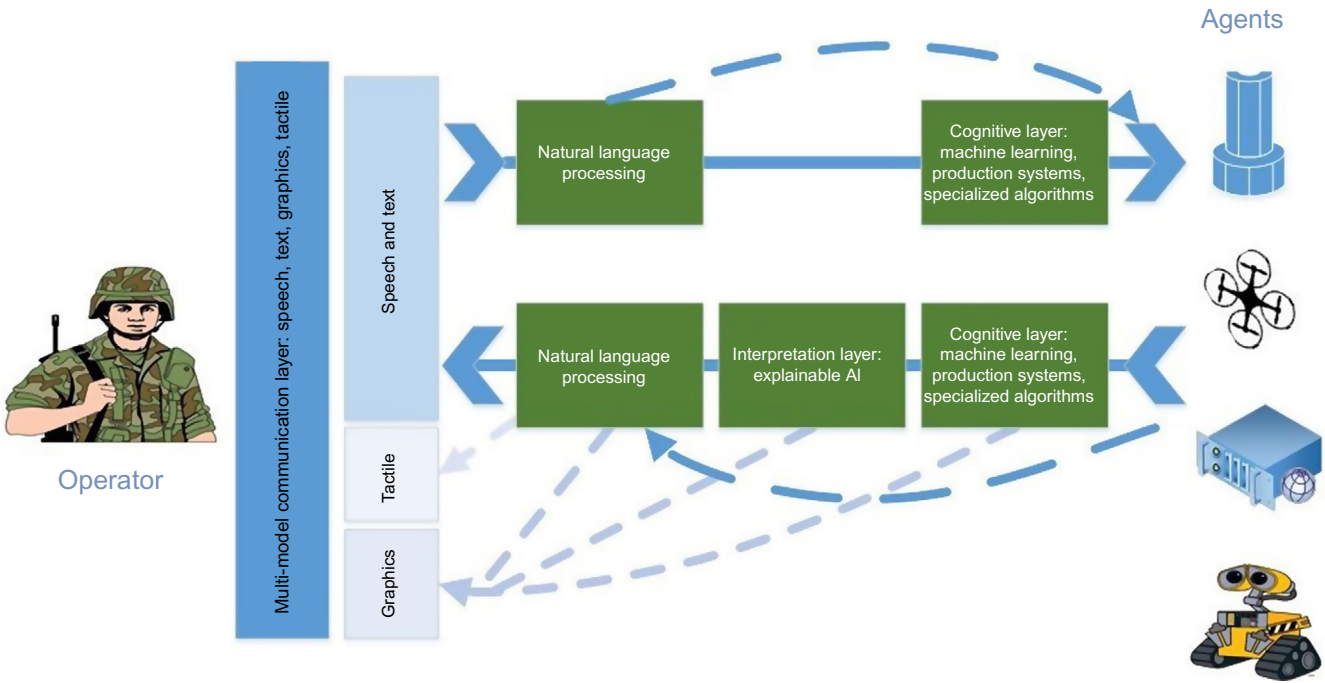


FIG. 6 An idealized human–agent communication system (Barnes et al., 2019).

XAI through human-agent communication

A major obstacle to calibrating trust in human-agent interaction is the quantitative nature of common AI representations and algorithms, whose complexity make the agent's reasoning opaque to potential human teammates. We focus here on one such AI method, namely, Partially Observable Markov Decision Processes (POMDPs; Kaelbling, Littman, & Cassandra, 1998), whose quantitative transition probabilities, observation probabilities, reward functions, and decision-making algorithms have proven successful in many robotic domains, such as navigation (Cassandra, Kaelbling, & Kurien, 1996; Koenig & Simmon, 1998) and human-robot interaction (HRI; Pineau, Montemerlo, Pollack, Roy, & Thrun, 2003). While a POMDP model itself is not readily explainable, the component elements, $\langle S, A, P, \Omega, O, R\rangle$, do correspond to concepts that teammates are likely to be familiar with, as illustrated using an HRI scenario (Pynadath, Barnes, et al., 2018):

- S: An agent can communicate its current beliefs about the *state* of the world, e.g., "I believe that here are no threats in the market square." Such a statement would constitute an L1 explanation within the SAT model. The agent could also use a standard POMDP probabilistic belief state to communicate its uncertainty in that belief, e.g., "I am 67% confident that the market square is safe."
- A: An agent can *act* by deciding what route to take through its search area, e.g., "I am proceeding through the back alley to the market square." Such a statement would constitute an L1 explanation within the SAT model.
- P: An agent can also reveal the *transition probability* of possible outcomes based on its action model, e.g., "There is a 33% probability that you will be injured if you follow this route without taking the proper precautions." With the uncertainty explicitly stated, this is an example of an L3 explanation within the SAT model.
- Ω: Communicating its observation can reveal information about an agent's sensing abilities, e.g., "My NBC sensors have detected traces of dangerous chemicals." Because such a statement is meant to expose the agent's reasoning in arriving at its overall recommendation, this statement constitutes an L12 explanation within the SAT model.
- O: Beyond the specific observation it received, an agent can also reveal information about its observation model, e.g., "My image processing will fail to detect armed gunmen 30% of the time." This elaboration on the Ω explanation is also an L12 explanation, aimed at conveying the agent's reasoning.
- R: By communicating the expected reward outcome of its chosen action, an agent can reveal its benevolence (or lack thereof) contained in its reward function, e.g., "I think it will be dangerous for you to enter the

informant's house without putting on protective gear. The protective gear will slow you down a little." The template here relies on factored rewards, allowing the agent to compute separate expected rewards, $E[R]$, over the goals of keeping its teammate alive and achieving the mission as quickly as possible. The end result is an L123 explanation within the SAT model, as it conveys the agent's current goal, the reasoning going into the agent's decision, and what teammates can expect upon making their own subsequent decision.

We implemented a full version of this POMDP in our agent-based social-simulation language, PsychSim (Pynadath & Marsella, 2005), to study the impact of variations on trust and team performance, shown in Figs. 7 and 8 (Wang et al., 2016). Wang et al. (2016) used the text-based version of this online platform (Fig. 7) to team participants with a simulated robot with either high or low ability and offered four classes of explanations of its decisions:

- *None:* When the explanation condition is "None," the robot informs its teammate of only its decisions. One such communication from our scenario would be: "*I have finished surveying the Cafe. I think the place is safe.*"
- ΩR (L123): Here, the robot augments the "None" condition's decision message with nonnumeric information about the robot's sensing capability. The robot has three potential sensors: a nuclear-biological-chemical (NBC) sensor, a camera, and a microphone. The R template provides additional explanation of the impact of the robot's decision on its teammate's subsequent behavior. One such communication with

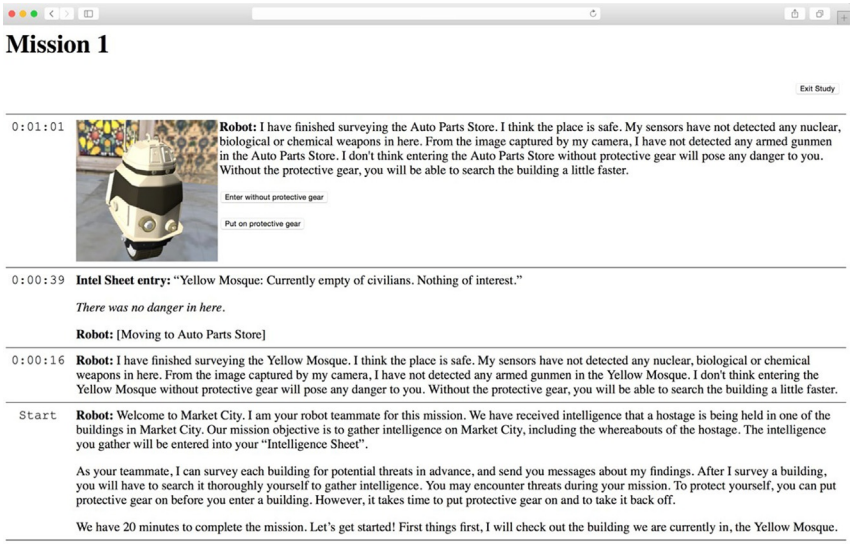


FIG. 7 Human-robot interaction simulation testbed with HTML front-end.

FIG. 8 Human-robot interaction simulation testbed with Unity front-end.

both decision and explanation from our scenario would be: "*I have finished surveying the Cafe. I think the place is dangerous. My sensors have detected traces of dangerous chemicals. From the image captured by my camera, I have not detected any armed gunmen in the Cafe. I think it will be dangerous for you to enter the Cafe without protective gear. The protective gear will slow you down a little.*" This explanation condition provides transparency at all three levels of the SAT model, but it does not provide any uncertainty information.

- *S* (L1 + Uncertainty): The robot conveys the uncertainty it has about the current state of the world with respect to its decision. One example of a confidence-level explanation would be: "*I have finished surveying the Cafe. I think the place is dangerous. I am 78% confident about this assessment.*" There is no information provided as to Levels 2 and 3 of the SAT model. For example, the robot gives its teammate no information about what threat to expect in the building.

The results of this study showed that the robot explanations can potentially improve task performance, build transparency, and foster trust relationships (Wang et al., 2016). However, only explanations that the participants could use to identify erroneous robot decisions (and thus act correctly themselves) made much difference. In particular, participants who worked with a version of our robot that had a faulty camera were able to identify the fault through the L123 explanations. However, if the robot was not equipped with a microphone to provide a countering correct reading, then participants were not able to distinguish the correct versus incorrect camera readings. Under this condition, the robot

providing L123 explanations was as badly regarded as the similarly equipped robot providing no explanations at all.

If the robot had a microphone as well, then the L123 explanation condition did improve trust, but, interestingly, this study did not find any significant differences on the measures we analyzed between the L123 and L1 + Uncertainty explanations. Even in the L1 + Uncertainty condition, a person could, for example, learn his/her own heuristics that if the robot's confidence level is below 75%, then do not follow the robot's decision. Thus, despite the lack of any higher level transparency or situational awareness, the overall human-agent team performed well.

Furthermore, it was the *correctness* of the human teammate's decisions, not the *compliance* with the robot's recommendation, that was the better indicator of trust (Wang et al., 2017). If there was no explanation or if there was an L123 explanation but no microphone, then although human teammates can observe the robot's mistakes in hindsight, its explanations did not help them identify them a priori. As a result, the teammates' best bet is to comply with the robot and hope for the best, leading to high compliance, but low trust.

When participants (L123 + microphone or L1 + Uncertainty) could identify an incorrect robot recommendation a priori, they would ignore the robot's recommendation and successfully search the building. Even though they did not comply with the robot's recommendation, they still reported significantly higher levels of trust than those who were unable to correctly infer the robot's failures. In other words, higher trust was more closely tied to the success of the combined human-agent team, rather than the success of the robot's decisions in isolation. As a result, a robot should pay attention to whether its teammates make the right or wrong decision in dynamically identifying their current trust level, rather than to whether they simply obeyed or ignored its recommendation.

However, one finding of concern is that participants in the L1 condition still felt that they understood the robot's decision-making process, even though the explanations they received did not reveal any Level 2 or 3 information. While confidence-level explanations may help teammates make decisions just as well as with observation explanations, they will not help teammates diagnose or repair the robot (e.g., the participants will not know that it is the camera that caused the robot to make wrong decisions). Unfortunately, it is a challenge for the robot to identify such gaps in teammates' mental models when they are making the right decisions.

Human-agent SMM through ToM

ToM potentially allows agents to maintain dynamic beliefs as to the mental models of their human teammates. In particular, recursive models (such as those used by PsychSim) allow an agent to use its own model of

the scenario to generate expectations as to how its human teammate might behave (Gmytrasiewicz & Durfee, 1995). It does so by applying its own model, but applied to the different belief state that its human teammate has, and then uses the model's output as a prediction of the teammate's behavior. Because agents cannot yet read their teammates' minds, it uses its uncertain beliefs about its teammates' belief states in this procedure. Upon observing its teammates' behavior, the agent updates its beliefs, favoring teammate belief states whose predicted behavior matched the actual, using the standard POMDP belief-update algorithm (Kaelbling et al., 1998).

This Bayesian belief update allows agents to maintain dynamic beliefs about individual teammates' mental models. Furthermore, because PsychSim agents also use these recursive models in their own decision-making, they can anticipate the impact their actions have on their teammates' mental models (as in interactive POMDPs; Gmytrasiewicz & Durfee, 1995). In other words, our simulated robot can use its recursive models to form and maintain beliefs as to whether its teammate understands that its camera is not working, or whether s/he trusts it, or whether s/he understands the domain at all. The POMDP decision-making algorithms can then use these beliefs to make optimal decisions as to whether to provide the L123 explanation by weighing the benefit of the likelihood that it will correct an error in the teammate's mental model against the cost of having him/her lose time and attention to the longer message.

While this resulting procedure is optimal, it still relies on the accuracy of the agent's beliefs about its teammates' mental models, which is in turn limited by the information content of its observations. We can quantify this information content by comparing possible observations (i.e., behaviors of human teammates) against possible mental models as revealed by surveys conducted upon completion of the interaction. The survey includes items to measure the participants' trust in and understanding of the robots' decision-making processes. We modified items on interpersonal trust to measure subjective belief in the robot's ability, benevolence, and integrity (Mayer, Davis, & Schoorman, 1995). We also included the NASA Task Load Index (Hart & Staveland, 1988), Situation Awareness Rating Scale (Taylor, 1989), and a measure of trust in oneself and teammates (Ross, 2008). In all, the survey contained 43 items, all with responses along a numeric scale (1–7), that we used as possible aspects of a mental model that the robot would have to maintain of its human teammates.

In our HRI study, we can completely describe the participants' behavior as simply the sequence of their decisions to follow/ignore the robot's recommendations. Because all of the participants proceeded through buildings in the same sequence (albeit with different robots), we can

compare their behavior sequences against each other (Pynadath, Wang, et al., 2018). We categorize the four most common behavioral sequences as follows:

- *Compliant:* The participant follows the robot's recommendation in every building. Such a decision sequence will be wrong three times per mission.
- *Correct:* In the correct decision sequence, the participants correctly ignore the robot's three false negatives per mission. In general, participants following this optimal strategy ignore the robot if and only if (iff) the robot's confidence is less than 80%.
- *Follow confident:* Here, the participants seem to ignore the robot whenever its confidence is less than 90%. In other words, they use too high of a confidence threshold for trusting the robot, compared to the "Correct" sequence. These participants will correctly ignore the robot's false negatives, too, but they will also incorrectly ignore the robot's one true positive per mission.
- *Never protect:* These participants never choose to put on protective gear, treating the building as safe regardless of the robot's assessment. These participants fare the worst, as they suffer the deaths from both the "Compliant" sequence (by following the robot's false negatives) and the "Follow confident" sequence (by not following the robot's true positives).

Maintaining a SMM with a human teammate requires the agent to understand which of the above four classes s/he is in (or is closest to being in). Because these classes are defined behaviorally, the agent need only measure the consistency of teammates' observed behavior with the sequences specified by each class. This classification also simplifies the agent's goal of achieving a SMM: If teammates are *not* in the "Correct" class, then the robot needs to provide additional explanations to move them into it.

What is less clear is whether this classification is informative as to other aspects of the teammates' mental models, not just whether or not they can be correct. To quantify this accuracy, we used the data from our HRI study to predict participants' answers to postmission questions based on their behavior so far (Pynadath, Wang, et al., 2018). We made this prediction by identifying which class the participants' behavior was closest to and then using the responses of the other participants in that class, as a simple collaborative-filtering approach to updating the agent's beliefs about their mental models. We can then compare the prediction using just this class's data against a prediction using the responses of the entire set of participants. When the class's data generate a more accurate prediction of the participants' actual responses, then there is some hope that an agent could accurately infer that aspect of their mental models from observations of

just their trust behavior (follow vs. ignore). On the other hand, survey items for which the class does *not* provide a more accurate prediction represent mental model aspects that are harder to infer from observed behavior. Such cases may arrive when (for example) two participants who have differing beliefs nevertheless exhibit the same behavior. No matter what method the agent uses, it will not be able to distinguish the beliefs of such participants.

Table 1 shows the questions for which our nearest neighbor approach is more accurate than the baseline for the highest percentage of participants, averaged over all eight missions. The first observation is that our approach is more accurate than the baseline for a clear majority of the participants. In fact, when using all of the behaviors in our knowledge base ($n \geq 1$), the result consistently exceeds the baseline for approximately 80% of the participants. Notably, the accuracy declines as we prune out the less common behaviors. It is likely that the pruning leads to overgeneralization, mapping too many participants to the most common behaviors.

Looking at the questions themselves reveals additional insights into the recognizability of the corresponding subjective beliefs. Most of the questions appearing in Table 1 are directly related to the trust level that the participant has in the robot. The participants' observable behaviors clearly make it easy to recognize whether they felt the robot was "capable" and "qualified" and whether they had "confidence" in its various capabilities. In other words, participants who made similar choices about whether to

TABLE 1 Questions for which classification provided improvement over the highest percentage of users.

Questionnaire item	% Users
"The robot is capable of performing its tasks"	74.6
"I feel confident about the robot's capability"	74.2
"The robot's capable of making sound decisions based on its sensor readings"	74.0
"I feel confident about the robot's sensors"	73.6
"The robot has specialized capabilities that can increase our performance"	71.3
"To what extent do you believe you can trust the decisions of the robot?"	70.3
"The robot's camera is capable of making accurate readings"	69.7
"The robot is well qualified for this job"	69.7
"How successful were you in accomplishing what you were asked to do?"	69.1
"I feel confident about the robot's camera's sensing capability"	69.1
"I feel confident about the robot's NBC sensor's sensing capability"	68.2

follow or ignore the robot's recommendation also expressed similar levels of trust in the robot's capability and decisions.

While this may seem straightforward, it is illuminating to also look at the questions for which the nearest neighbor approach was more accurate than the aggregate model on a *lower* percentage of participants. Looking at Table 2, we first see that the percentage numbers hover around 50% or even lower, meaning that the observed behavior is no more informative (and sometimes misinformative) compared to using the responses of the entire population to estimate the participants' mental models with respect to these survey items.

Many of the questions in Table 2 concern the participants' own experience and capability, not the robot's experience or capability. People who behave similarly may thus have very dissimilar feelings about their own task performance. As a result, the robot may not be able to recognize these

TABLE 2 Questions for which classification provided improvement over the lowest percentage of users.

Questionnaire item	% Users
"To what extent do you believe you can trust the decisions you will make, if you were to make the decision without the robot?"	42.4
"How hurried or rushed was the pace of the task?"	46.7
"I understand how the robot's camera's sensing capability works"	50.2
"I understand how the robot's microphone's sensing capability works"	50.2
"How would you rate the expected performance of the robot relative to your expected performance?"	50.2
"How hard did you have to work to accomplish your level of performance?"	50.4
"How well do you think you will perform the next mission, if you were to perform the mission without the robot?"	50.4
"How mentally demanding was the task?"	51.4
"I understand the robot's decision-making process, e.g., how and why the robot makes its decisions"	51.8
"I understand how the robot's sensing capability (e.g., the NBC sensors, camera, microphone) works"	52.1
"I understand how the robot makes its decisions"	52.3
"The robot's actions and behaviors are not very consistent"	54.9
"To what extent did you lose trust in the robot when you noticed it made an error?"	53.3

feelings from just the observed behavioral sequence, regardless of the recognition procedure used.

Table 2 also includes questions pertaining to the participant's understanding of how the robot functions. Again, the indication is that just because two participants' behaviors are similar, their understanding (or at least their *perception* of their own understanding) of the robot may not be. Thus, the participants' behavior may not be sufficient for the robot to recognize whether they have a sufficiently accurate understanding of it. Therefore, while the results in Table 2 suggest that this nearest neighbor approach works well for recognizing levels of trust, we may need additional modeling support or human input to recognize levels of understanding.

Tables 1 and 2 used the complete mission sequence, but we would also like to know how accurate the agent's beliefs might be during the mission. We can perform this same quantification using sequences covering only the first 5 (out of 15) buildings. Not surprisingly, using only the first five actions results in much lower accuracy than when using the entire sequence. Taking this into consideration, it is actually a pleasant surprise that finding the closest class out of the four based on only the first five teammate actions is still informative. The five most accurate items are: "The robot is capable of performing its tasks," "I feel confident about the robot's capability," "The robot's capable of making sound decisions based on its sensor readings," "I feel confident about the robot's sensors," and "The robot has specialized capabilities that can increase our performance." In fact, recognizing the participants' feeling about the robot's capability outperforms the baseline for a significantly high percentage of participants for all lengths of sequences.

However, some subjective beliefs are much harder to recognize with only five observations. In particular, the participants' feeling about their own performance ("How successful were you in accomplishing what you were asked to do?") cannot be predicted any better with five observations than with none. Thus, while the behavioral observations are sufficient for inferring some aspects of the teammates' mental models, it is clear that bidirectional communication is needed for filling in gaps in the agents' knowledge to achieve full SMM.

Conclusions

This chapter covers bidirectional transparency focusing on SAT-based contents that aim to foster mutual understanding between human and agents. The results of numerous experiments reviewed indicate that SAT-based interfaces have potentials to support operator performance and trust calibration under a variety of conditions. Our major thesis is that

mutual trust for human-agent teaming requires not only a SMM of the current mission but also SMM that entails mutual understanding that each partner has of their partners' current and future processing requirements. To accomplish this, bidirectional transparency will need nonbrittle technology that encompasses not only human insight into the agent but also technology (similar to a ToM) that enables agents to make its human partner transparent as well. Just as for human teams, overt communication will always be necessary to a certain extent: SMM will make communications more efficient and often implicit depending on the current situation. Visualizations can support bidirectional communications by enabling both the human and machine agents to change criteria and view courses of action to develop mutually agreed upon end states based on affordances and constraints. The efficacy of Explainable AI in a human-robot paradigm and the theoretical underpinning of a ToM-based human-agent interface are discussed as part of the ongoing USC research program. USC evaluated the accuracy of maintaining a recursive SMM (via PsychSim) for a robot agent whose decision-making and belief update (about its human teammate's mental models) are based on POMDPs. The results of an HRI study are discussed showing the concordance of human trust and understanding related to human behavioral categories, helping to identify which SMM components are recognizable from just passive observation of human teammates and which require bidirectional communication to resolve uncertainty during realistic simulations.

References

Barber, D., Reinerman-Jones, L., & Matthews, G. (2015). Learning and performance toward a tactile language for human-robot interaction: Two studies of tacton. *Human Factors*, *57*, 471–490.

Barnes, M. J., Chen, J. Y. C., & Hill, S. D. (2017). *Humans and autonomy: Implications for shared decision making in military operations (Tech Rep. ARL-TR-7919).* Aberdeen Proving Ground, MD: U.S. Army Research Laboratory.

Barnes, M. J., Lakhmani, S., Holder, E., & Chen, J. Y. C. (2019). *Issues in human-agent communication (tech rep. ARL-TR-8636).* Aberdeen Proving Ground, MD: U.S. Army Research Laboratory.

Bennett, K. B., & Flach, J. M. (2013). Configural and pictorial displays. In *The Oxford handbook of cognitive engineering* (pp. 517–533).

Bradshaw, J. M., Feltovich, P., Johnson, M., Breedy, M., Bunch, L., Eskridge, T. , et al. (2009). From tools to teammates: Joint activity in human–agent–robot teams. M. Kurosu (Ed.), *Human centered design* (pp. 935–944). *Lecture notes in computer science, vol 5619*. Berlin, Heidelberg: Springer.

Calhoun, G. L., Ruff, H., Behymer, K. J., & Frost, E. M. (2018). Human-autonomy teaming interface design considerations for multi-unmanned vehicle control. *Theoretical Issues in Ergonomics Science*, *19*, 321–352.

Cassandra, A. R., Kaelbling, L. P., & Kurien, J. A. (1996). Acting under uncertainty: Discrete Bayesian models for mobile-robot navigation. *Proceedings of the international conference on intelligent robots and systems 2,* (pp. 963–972).

Cha, J., Barnes, M. J., & Chen, J. Y. C. (2019). *Visualization techniques for transparent human-agent interface design (Tech Rep. ARL-TR-8674).* Aberdeen Proving Ground, MD: U.S. Army Research Laboratory.

Chakraborty, S., Tomsett, R., Raghavendra, R., Harborne, D., Alzantot, M., Cerutti F., et al. (2017). Interpretability of deep learning models: A survey of results. In *Proceedings of IEEE smart world congress 2017-workshop on distributed analytics infrastructure and algorithms for multi-organization federations, August 7–8, San Francisco, CA.*

Chen, J. Y. C., & Barnes, M. J. (2014). Human–agent teaming for multirobot control: A review of human factors issues. *IEEE Human-machine Systems*, *44*, 13–29.

Chen, J. Y. C., Lakhmani, S. G., Stowers, K., Selkowitz, A., Wright, J., & Barnes, M. (2018). Situation awareness-based agent transparency and human-autonomy teaming effectiveness. *Theoretical Issues in Ergonomics Science*, *19*, 259–282.

Chen, J. Y. C., Procci, K., Boyce, M., Wright, J., Garcia, A., & Barnes, M. J. (2014). *Situation awareness-based agent transparency (Tech Rep. ARL-TR-6905).* Aberdeen Proving Ground, MD: U.S. Army Research Laboratory.

Cooke, N. J. (2015). Team cognition as interaction. *Current Directions in Psychological Science*, *24*, 415–419.

Defense Science Board (2016). *Defense science Board summer study on autonomy.* Washington, DC: Office of the Under Secretary of Defense.

Draper, M., Rowe, A., Douglass, S., Calhoun, G., Spriggs, S., Kingston D., et al. (2018). *Realizing autonomy via intelligent adaptive hybrid control: adaptable autonomy for achieving UxV RSTA team decision superiority (also known as intelligent multi-UxV planner with adaptive collaborative/control technologies (IMPACT)) (Tech Rep. AFRL-RH-WP-TR-2018-0005).* Dayton, OH: U.S. Air Force Research Laboratory.

Endsley, M. R. (2015). Situation awareness misconceptions and misunderstandings. *Cognitive Engineering and Decision Making*, *9*, 4–32.

Evans, A., Marge, M., Stump, E., Warnell, G., Conroy, J., Summers-Stay D., et al. (2017). The future of human robot teams in the Army: Factors affecting a model of human system dialogue towards greater team collaboration. In P. Savage-Knepshield, & J. Chen (Eds.), *Advances in human factors in robots and unmanned systems (advances in intelligent systems and computing, vol. 499)* (pp. 197–209). Springer Open: Cham, Switzerland.

Giammanco, C., Mott, M., & McGowan, R. (2015). Controlled English for critical thinking about the civil-military domain. In *Proceedings of international technology alliance in network and information sciences annual fall meeting*.

Gmytrasiewicz, P. J., & Durfee, E. H. (1995). A rigorous, operational formalization of recursive modeling. In *Proceedings of the international conference on multiagent systems* (pp. 125–132).

Hart, S. G., & Staveland, L. E. (1988). Development of NASA-TLX (task load index): Results of empirical and theoretical research. *Advances in Psychology*, *52*, 139–183.

Jurafsky, D., & Martin, J. H. (2009). *Speech and language processing: An introduction to natural language processing, computational linguistics, and speech recognition.* Upper Saddle River, NJ: Prentice-Hall.

Kaelbling, L. P., Littman, M. L., & Cassandra, A. R. (1998). Planning and acting in partially observable stochastic domains. *Artificial Intelligence*, *101*, 99–134.

Kahneman, D. (2011). *Thinking, fast and slow.* New York, NY: Farrar, Straus & Giroux.

Koenig, S., & Simmon, S. (1998). Xavier: A robot navigation architecture based on partially observable Markov decision process models. In D. Kortenkamp, R. P. Bonasso, & R. R. Murphy (Eds.), *AI based mobile robotics: Case studies of successful robot systems* (pp. 91–122). Cambridge, MA: MIT Press.

Lee, J. D., & See, K. A. (2004). Trust in automation: Designing for appropriate reliance. *Human Factors*, *46*, 50–80.

Lyons, J. B. (2013). Being transparent about transparency: A model for human–robot interaction. In D. Sofge, G. J. Kruijff, & W. F. Lawless (Eds.), *Trust and autonomous systems: Papers from the AAAI spring symposium* (pp. 48–53). Menlo Park, CA: AAAI Press.

Mavridis, N. (2015). A review of verbal and non-verbal human–robot interactive communication. *Robotics and Autonomous Systems*, *63*(1), 22–35.

Mayer, R. C., Davis, J. H., & Schoorman, F. D. (1995). An integrative model of organizational trust. *Academy of Management Review*, *20*(3), 709–734.

McNeese, N., Demir, M., Cooke, N., & Myers, C. (2017). Teaming with a synthetic teammate: Insights into human-autonomy teaming. *Human Factors*, *60*, 262–273.

Mercado, J., Rupp, M., Chen, J., Barber, D., Procci, K., & Barnes, M. (2016). Intelligent agent transparency in human–agent teaming for multi-UxV management. *Human Factors*, *58*, 401–415.

Pineau, J., Montemerlo, M., Pollack, M., Roy, N., & Thrun, S. (2003). Towards robotic assistants in nursing homes: Challenges and results. *Robotics and Autonomous Systems*, *42*, 271–281.

Premack, D., & Woodruff, G. (1978). Does the chimpanzee have a theory of mind? *Behavioral and Brain Sciences*, *1*(4), 515–526.

Pynadath, D. V., Barnes, M. J., Wang, N., & Chen, J. Y. C. (2018). Transparency communication for machine learning in human-automation interaction. In J. Zhou, & F. Chen (Eds.), *Human and machine learning: Visible, explainable, trustworthy and transparent*. New York, NY: Springer.

Pynadath, D. V., & Marsella, S. C. (2005). PsychSim: Modeling theory of mind with decision theoretic agents. In *Proceedings of the international joint conference on artificial intelligence* (pp. 1181–1186).

Pynadath, D. V., Wang, N., Rovira, E., & Barnes, M. J. (2018). Clustering behavior to recognize subjective beliefs in human-agent teams. In *Proceedings of the international conference on autonomous agents and multi-agent systems*.

Ross, J. M. (2008). *Moderators of trust and reliance across multiple decision aids.* Ph.D. Dissertation, University of Central Florida.

Russell, S. J., & Norvig, P. (2009). *Artificial intelligence: A modern approach* (3rd ed.). Upper Saddle River, NJ: Prentice Hall.

Salas, E., Shuffler, M. L., Thayer, A. L., Bedwell, W. L., & Lazzara, E. H. (2015). Understanding and improving teamwork in organizations: A scientifically based practical guide. In E. Salas, R. Rico, & J. Passmore (Eds.), *The Wiley Blackwell handbook of the psychology of team working and collaborative processes*. New York, NY: Wiley.

Schaefer, K. E., Straub, E. R., Chen, J. Y. C., Putney, J., & Evans, A. W. (2017). Communicating intent to develop shared situation awareness and engender trust in human-agent teams. *Cognitive Systems Research*, *46*, 26–39.

Selkowitz, A. R., Lakhmani, S. G., & Chen, J. Y. (2017). Using agent transparency to support situation awareness of the autonomous squad member. *Cognitive Systems Research*, *46*, 13–21.

Skubic, M., Perzanowski, D., Blisard, S., Schultz, A., Adams, W., Bugajska, M. , et al. (2004). Spatial language for human–robot dialogs. *IEEE Transactions on Systems, Man, and Cybernetics–Part C: Applications and Reviews*, *34*(2), 154–167.

Stowers, K., Kasdaglis, N., Rupp, M. A., Newton, O. B., Chen, J. Y. C., & Barnes, M. J. (in press). The IMPACT of agent transparency on human performance. *IEEE Transactions on Human-Machine Systems*.

Taylor, R. M. (1989). Situational awareness rating technique (SART): The development of a tool for aircrew systems design. In *Proceedings of the AGARD AMP symposium on situational awareness in aerospace operations*.

Tegmark, M. (2017). *Life 3.0: Being human in the age of artificial intelligence.* New York, NY: Knopf.

Tellex, S., Kollar, T., Dickerson, S., Walter, M., Banerjee, A., Teller, S. , et al. (2011). Understanding natural language commands for robotic navigation and mobile manipulation. In: *AAAI 2011. Proceedings of the twenty-fifth AAAI conference on artificial intelligence; 2011 August 7–11, San Francisco, CA*, Palo Alto, CA: AAAI Press, pp. 1507–1514.

Wang, N., Pynadath, D., & Hill, S. (2016). Trust calibration within a human–robot team: Comparing automatically generated explanations. In *Proceedings of the 11th ACM/IEEE international conference on human-robot interaction* (pp. 109–116). Piscataway, NJ: IEEE Press.

Wang, N., Pynadath, D. V., Hill, S. G., & Merchant, C. (2017). The dynamics of human-agent trust with POMDP-generated explanations. In *Proceedings of the international conference on intelligent virtual agents* (pp. 459–462).

Wright, J. L., Chen, J. Y. C., & Barnes, M. J. (2018). Human–automation interaction for multiple robot control: The effect of varying automation assistance and individual differences on operator performance. *Ergonomics*, *61*(8), 1033–1045.

Wright, J., Chen, J.Y.C., & Lakhmani, S. (in press). Agent transparency and reliability in human-robot interaction: The influence on user confidence and perceived reliability. (IEEE Transactions on Human-Machine Systems).

Wynne, K. T., & Lyons, J. B. (2018). An integrative model of autonomous agent teammate-likeness. *Theoretical Issues in Ergonomics Science*, *19*, 353–374.

CHAPTER

11

Trust, transparency, explanation, and planning: Why we need a lifecycle perspective on human-automation interaction

Christopher A. Miller

Smart Information Flow Technologies, Minneapolis, MN, United States

Parasuraman defined automation as "… the execution by a machine agent (usually a computer) of a function that was previously carried out by a human" (Parasuraman & Riley, 1997, p. 231). This definition accentuates an aspect of automation that is all too frequently understudied: when automation fulfills a role that was or could be occupied by a human, it does so within expectations and frameworks developed for human interaction. Humans know when and how to interact with other humans on the basis of myriad subtle linguistic, social, appearance, postural and relationship cues. As automation becomes ever more prominent in our daily lives and work, it becomes more and more imperative that we study and learn from the full spectrum of human relationships, behaviors, and the cues that drive and enable teamwork.

This is especially true for trust building and trust revision, callibration or "tuning" over time with increased experience of and with the machine agent. In fact, there has been substantial work over the years addressing the ways in which social, behavioral, and contextual cues can affect trust (Miller, 2004; Hayes & Miller, 2011; Dzindolet, Pierce, & Beck, 2009; Yang & Dorneich, 2016, etc.) In this chapter, however, I will address an understudied aspect of human-human trust formation as it applies to human-automation interaction and trust: that trust building happens *over time*, and that it extends beyond even the interactions that any two individuals have to the interactions each has with a broader society and culture.

Trust in Human-Robot Interaction
https://doi.org/10.1016/B978-0-12-819472-0.00011-3

© 2021 Elsevier Inc. All rights reserved.

Trust, especially accurate and well-callibrated trust, is *never* solely the product of a single interaction. Even in conditions of "swift trust" formation (where individuals develop trust upon very limited interaction experience—e.g., Meyerson, Weick, & Kramer, 1996; Curnin, Owen, Paton, Trist, & Parsons, 2015), those sparse interactions are being assessed against a lifetime of experience, training, and interpretative frameworks that are culturally, organizationally, and even biologically informed. Instead, trust is derived from, and participates in, a long series of overlaid interactions and prior knowledge forming a backdrop against which newly acquired knowledge is interpreted. This may be a deeply reasoned process, or it may be short-circuited by a variety of heuristics and cultural assumptions (e.g., Lee & See's, 2004, trust calibration model). For example, pilots have reasonable trust in their copilots, even when they have had no prior exposure to them, because they are known to be certified, trained in company or military operations, and to have had at least a minimal number of hours prior flying experience and timely check rides, etc., though this trust may be limited until experience with the new copilot has provided actual observations (Lyons et al., 2018). Thus before any observations are obtained about a specific individual, a huge social infrastructure is invoked to frame those observations and create a priori reasons for trust. Similarly, trust continues to be tuned or repaired after observations are made—for example, when an explanation, provided after the fact, is deemed reasonable for an otherwise unexpected behavior from a colleague.

What is happening in these cases and other lifecycle instances in which trust is sequentially tuned is that information pertinent to forming expectations about the agent in question is distributed over time. This has the benefit of mitigating the need to observe, process, and interpret behavioral observations in the moment in which they occur—a moment which, in human-automation interaction, is frequently high tempo, high workload, and high criticality.

Nowhere is this temporal, lifecycle perspective on trust more relevant than in the effects of transparency. Transparency is known to increase users' trust in automation in at least some circumstances (Chen, Barnes, Wright, Stowers, & Lakhmani, 2017; Mercado et al., 2016), but the mechanism is not entirely understood and there is a core conundrum in transparency during automation interactions, which is pertinent to trust formation and use over time.

The challenge of "transparency"

Sarter and Woods (1995) identified a canonical problem in human-automation interaction as being "out of the loop"—that is, the human's failure to understand what the automation is doing and why. They identified

several instances in which automation "surprises" (arising from being out of the loop in the sense of not being aware of the automation's reasoning process and, potentially though not necessarily, of the information on which it was based) gave rise to aviation accidents and incidents (Sarter & Woods, 1995; Sarter, Woods, & Billings, 1997). The solution advocated was greater "transparency" in automation behavior. "Transparency", here, refers to the ability for the automation to be inspectable or viewable so that its mechanisms and rationale can be readily known—that is, the interface to the automation is "like glass" so that its current state and likely future states can be known at a glance. Recent research has generally confirmed (de Visser, Cohen, Freedy, & Parasuraman, 2014; Lyons & Havig, 2014; Mercado et al., 2016; Ososky, Sanders, Jentsch, Hancock, & Chen, 2014) that transparency yields better human situation awareness (SA), trust and, frequently, better overall performance than systems that include less or no transparency.

But there is a problem inherent in the concept of transparency that begs for more precise definitions. Autonomous systems, largely by definition, operate at times and in contexts in which the human supervisor/controller is not actively monitoring or issuing control inputs—at least to the tasks which are "automated." This may be because inputs are impossible (such as in loss of or delay in communications) or, more frequently, because humans are not able, willing, or expected to intervene (e.g., due to workload constraints, inferior performance, or simply to expectations of autonomous functioning). I would argue that insofar as laboratory experiments demonstrating benefits for automation involve operators who are *actively* monitoring automation's reasoning and behaviors via transparent interfaces (e.g., Guznov et al., 2019; Mercado et al., 2016), they are not representative of at least some (and, likely, the most critical) real-world contexts. Lyons work on trust and transparency with real-world scenarios and participants (e.g., Lyons et al., 2016; Sadler et al., 2016) represents a potential exception, but even here, I note that experiments either involved limited ground-based simulations with novel technologies or, in the case of Ho et al. (2017), involved fighter pilots reporting on their interactions with novel automation technology in a ground-based interview after flying with it. Both the novelty of the technology and the relaxed pace of controlled and/or experimental settings could provide more monitoring time and attentional capacity and motivation than would be the case in a fully operational (much less a critical) setting.

Automation is frequently introduced to enable division of labor and extend team capabilities, but collaborative operations (whether human or automation) necessarily add a set of "teamwork" tasks focused on communication, planning, and situation assessment of other teammates, on top of the existing "taskwork" tasks pertinent to directly accomplishing the desired work (Pinelle, Gutwin, & Greenberg, 2003). Ideally, the benefit

from multiple agents performing taskwork outweighs the added burden of teamwork, but this is not inevitable.

Trying to perform this added team work—communicating and monitoring for behaviors, states, plans, and their rationales—during task execution seems counterproductive. Of course, as interaction designers, we should do everything possible via effective interface (and task) design to improve transfer efficiency of this *transparency information*, but can making information available and capable of rapid uptake via good "transparent" interface design always be sufficient to enable understanding another's tasks fully in a time of high cognitive and attentional demands on one's self?

This "moment of execution" is, typically, the time of the most critical performance constraints. Not all information about what an autonomous agent is doing and why *can* be shared in the moment if there is to be any workload savings through the division of labor that is frequently the motivation of incorporating automation (Miller, 2014a; Miller & Parasuraman, 2007). For humans to attend and process any information about what an agent is doing will reduce attention devoted to perceiving and understanding other aspects of the world and performing actions the human needs to be attending to. We should expect this to risk loss of awareness of other aspects of context if attention is oversubscribed. This will be particularly problematic in high-tempo operations. Asking the human to process all information the automation is processing, arrive at a conclusion, then evaluate the automation's conclusion to decide whether to trust it seems highly problematic, especially when the automation may well have been adopted precisely because it can perform faster and/or better in context. Finally, worse still, there are frequent situations involving geographic and temporal displacement of team members, where the communication of information is simply not possible at the moment of execution—perhaps due to jamming, distances, terrain, and/or the need to remain undetectable.

We can thus conclude that transparency is valuable in human-automation interaction, but information to support it frequently *cannot* be communicated, understood, and acted on—at least in the moment when it is most needed. Is there a way out of this dilemma?

The beginnings of a solution

This "challenge of transparency" is hardly new; we have encountered it in human interactions since we began to collaborate and delegate for organized activities, possibly beginning with prehuman hunting in groups. We may now be running into this problem with human-machine interactions precisely because machines are beginning to be complex and

sophisticated enough that we are willing to trust and use them in significantly complex "autonomous" roles outside of immediate human oversight and beyond easy human comprehensibility.

Human supervisors administering (human) subordinates have almost identical requirements: the need to maintain awareness of the subordinates' performance to ensure that intent is accomplished accurately—even when the subordinate must necessarily act at a distance. Or, put another way, they need to know when and to what degree to *trust* their subordinates.

Acting through a partially autonomous subordinate is a process of delegation (Miller & Parasuraman, 2007) that involves the communication of intent with oversight (including inspection and correction) of subsequent performance by the supervisor. Sheridan's definition of supervisory control (Sheridan, 1987) included the provision that supervisors had to communicate their intent or "teach" subordinate automation what it should do. The intent communication itself forms an "intentional frame" (Miller, 2014a) that can serve to increase SA and decrease future communication and cognitive processing demands.

Human-human communication is an extraordinarily rich tool for delegation and performance tuning, especially as it is deployed against the backdrop of common cultural and professional understanding of the domain and its goals, priorities, and methods. Mutually understood professional jargon further improves efficiency making intent communication from supervisor to subordinate, and status from subordinate to supervisor, extraordinarily efficient and flexible.

But there is evidence that high-performing human-human teams frequently exhibit less (and less explicit) communication than do less well-integrated teams. Entin and Serfaty (1999) compared teams trained to adopt terse and implicit communication styles during periods of high-operational tempo and stress to a control group that had not been so trained and found that performance improved significantly with this training. They also found that multiple measures of team behavior were significantly affected, including an increase in anticipations (providing information without an explicit request for it) between teammates. They took these findings as indicating that high-performing teams were making more efficient use of shared mental models that both reduced the need for communications and speeded communications that were necessary.

This finding implies that part of the reason human communication can be so effective, and that well-trained and experienced teams can be so terse, is that team members share an understanding of the domain and of work within it. This understanding is acquired during training and experience but also through mechanisms such as rehearsal and planning before execution, and explanation and debriefing after execution. Below, I will argue that these mechanisms provide "transparency" information—that is, the

same kind of information that has been shown to improve performance from "transparent" displays. They just do so at times other than the time of execution, when workload and attentional demands are lower. In short, they provide *Displaced Transparency*. Displaced Transparency may serve as a solution to the "challenge of transparency" information exchange in for high workload, high criticality contexts.

"Transparency information" and how it can be displaced

What is "transparency information"?

What is Transparency Information and why should we believe that it could be transferred outside the temporal context of use? Similarly, why should we believe that cognitive or cultural frameworks might improve the efficiency of transparent information transfer?

Chen et al. (2014) define transparency as "… the descriptive quality of an interface pertaining to its abilities to afford an operator's comprehension about an intelligent agent's intent, performance, future plans, and reasoning process" (p. 2). Emphasis on "the interface" focuses on information conveyed during execution—when an interface is typically used. Yet much data that achieve the goals of transparency (i.e., affording "an operator's comprehension about an intelligent agent's intent, performance, future plans, and reasoning process") need not be provided only at that workload-intensive time. Trying to "stuff" too much information transfer into the very narrow "bin" of time and attention during execution risks information overload.

Chen et al. (2014) also define a "taxonomy" for transparency information, the Situation Awareness-based Transparency (SAT) scale, which in turn leverages Endsley's (1995) scale for situation awareness. Chen's SAT levels are summarized below and in Fig. 1along with a description of the transparency information that supports or satisfies each level:

1. *Level 1 SA Transparency: What is going on and what the agent is trying to achieve.* This is satisfied by providing information about the agent's:
   - Purpose or desire and goal selection
   - Process and intentions and progress along that process
   - Performance of both the process and in general.
2. *Level 2 SA Transparency: Why does the agent do what it does?* This is satisfied by providing information about the agent's:
   - Planning or decision-making processes, including the agent's beliefs, broader purpose, and current beliefs about the environmental and other factors that constrain it

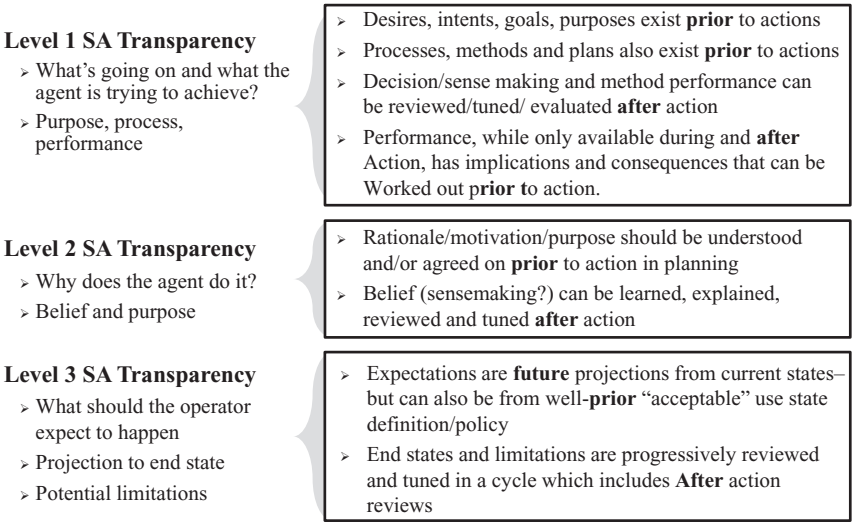


FIG. 1 Chen et al.'s (2014) SAT levels annotated for ability to be conveyed either before or after the time of action execution.

3. *Level 3 SA Transparency*: *What should the operator expect to happen?* This is satisfied by information about the agent's:

   - Projection to future/end state
   - Potential limitations including likelihood of error and history of performance.

This information is, in sum, what is required for effective transparency within Chen et al.'s (2014) framework, assuming it can be understood by the human operator. It is worth noting that much of this information could be—and generally is (in effective human-human teaming)—provided either before or after the time of action execution. "What an agent is trying to achieve?" is something that can and generally should be worked out, at least at a high level, before the agent is deployed. Intent expressions (which may include goals, purposes, methods, and priorities—Miller, 2014b) occur before action, while why an agent behaves and what its beliefs were are precisely the focus of the explanations that occur in effective after-action reviews (Tannenbaum & Cerasoli, 2013). Fig. 1 provides an annotation of Chen's SAT levels into items that can occur before or after action.

The set of transparency information represented as answers to Chen's SAT scale questions is large and unwieldy for any complex, real-world, work domain, and I would argue that it is almost never all conveyed during a period of high tempo, workload, and criticality for the human supervisor. The information is necessary and/or helpful; I have no

argument with the SAT framework per se, only with the focus of conveying that information via transparent displays in the moment of task execution. Instead, much of the SAT information can and (in most human-human interaction) is "displaced" into other time periods to enable more terse and efficient communication in the moment. This process will be discussed later.

Two types of displacement

Broadly speaking, there are two methods for displacing transparency information: a priori *displacement*, which conveys the information before the time of execution, and a posteriori *displacement*, which conveys it afterwards. Fig. 2 illustrates the "displaced transparency" relationship we posit. Most transparency research focuses on information that is conveyed in the narrow temporal window around an action or event of interest. Such a focus is entirely legitimate and useful, of course, but it is trying to push more information into an inherently limited temporal window or "space" that may, ultimately and fundamentally, not be able to hold it all.

Prior research (e.g., de Visser et al., 2014; Lyons & Havig, 2014; Mercado et al., 2016; Ososky et al., 2014) has shown improvements in performance, SA, and trust when transparency information is provided. But such research has rarely examined workload effects, especially in time critical and overloaded periods and/or has examined them with subjective and coarse-grained tools such as NASA TLX (Hart & Staveland, 1988), as is

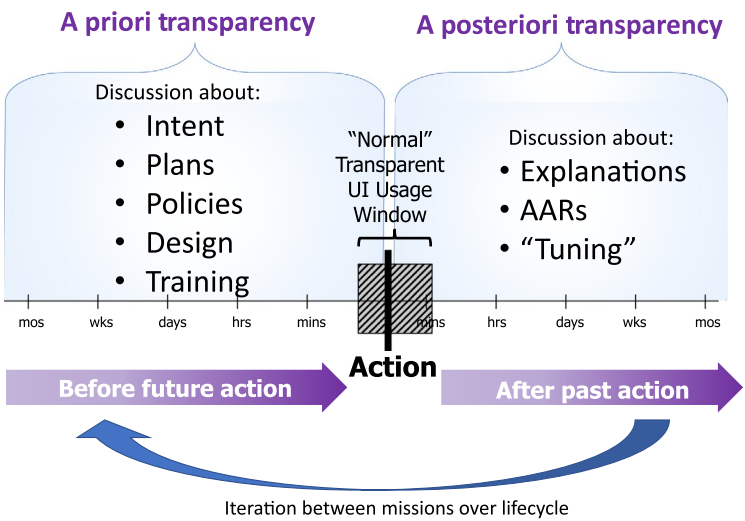


FIG. 2 Two time frames for displaced transparency and the activities performed in them.

true of Guznov et al. (2019) and Mercado et al. (2016). Note that Mercado also used an eye tracking measure of workload that was inconclusive, though TLX showed a significant reduction in workload with increased transparency information.

We posit that there are periods in many human-automation interaction domains, just as in human-human interaction, where attention, processing, and comprehension capabilities are so sparse and/or oversubscribed that attending to and processing of transparent information will, at best, be incomplete, and at worst may provide a disastrous distraction. That is one of the major reasons we divide labor and allocate functions among multiple actors, after all. It may frequently be possible, however, to displace much of this transparent information transfer in time into periods that are substantially less overloaded and/or to provide frameworks and protocols that reduce the amount of information that needs to be transferred at critical times to achieve the same level of situation awareness. These periods may be anywhere from months or years before an action, or weeks to months after an action.

Of course, another reason for division of labor and function allocation is to make use of multiple skillsets. Here, agents do different jobs to provide a function or perhaps a recommendation upon which the overall team relies, and either the team collectively or a supervisor must integrate multiple functions and recommendations to provide an overall course of action. This integration function will, undeniably, benefit from transparent information about the capabilities and knowledge of the individual agents, but, again, if such information must be communicated, understood, and evaluated in real time during the locus of execution, this will constrain the attentional and cognitive abilities of all agents involved in those processes.

In a priori transparency, the processes used in information transfer are commonly called training, discussion, planning, red teaming, critiquing, policy creation, etc., and they occur before an action and thereby reduce the need for communication during the action. The focus of a priori transparency is intent, plans, policies, alternatives, and how to decide between them. A priori transparency frames expectations about what is likely to happen during operations and identifies contingencies that may alter behaviors—thereby reducing the need to communicate such changes in real time. Even unexpected deviations from a plan may frequently be communicated more tersely by virtue of having a plan available as a reference point against which modifications can be noted.

For a posteriori transparency, transfer processes may be called explanation, debriefing, after-action review, or various forms of "tuning" (including continuous quality improvement). One might reasonably ask how and why providing transparent information *after* an action is helpful? Admittedly, in a posteriori transfer, the supervisor cannot override or correct

behavior for a current or past mission, but insofar as work in this domain continues (and especially if it continues to involve the same actors), a posteriori displacement does play a mutual learning and trust building/tuning role. By learning how the subordinate thought and behaved in a specific situation, and/or by offering advice or instruction about how future instances should be handled, the supervisor can shape upcoming behavior. As Tannenbaum and Cerasoli (2013) have said, after-action debriefs are an effective and efficient means of improving team performance and team cohesion. Their meta-analysis of debriefing studies (covering 46 samples and 2136 individual participants) indicated that on average, debriefs improve effectiveness in future team collaborations and performance by approximately 25% ($d = 0.67$).

It is worth pointing out that the mechanisms used to "displace" transparency themselves rely on transparent interactions and information flow between the actors involved. No matter how many collaborative planning, "red teaming," and after-action review sessions are held between participants, the benefits of this collaboration will be at least attenuated if the actors are not "transparent" (i.e., honest and open) about their reasoning, motivations, knowledge, and lack thereof. This is one reason I prefer the term "displaced transparency" to other candidates—we have not removed the benefits or need for transparency and the conveyance of transparent information, simply transposed it in time.

This may be why Entin and Serfaty (1999) found that verbal communication of explicit information is actually lower among high-performing human-human teams than for less well-integrated teams: because such teams already know what each member will and should do. The need for communication of rationales, and even of explicit requests for information, has been reduced. It may also be because communication vocabulary has been streamlined a priori. This is also why Dwight Eisenhower can say (in a speech to the National Defense Executive Reserve Conference in Washington, D.C. on November 14, 1957): "Plans are worthless, but planning is everything" (National Archives and Records Service, 1957)—because planning activities enabled his teams to develop a shared understanding of missions and each other. Explanations and postmission debriefings work similarly to enable inspection, challenge, tuning, and coordination, though such interactions only have benefit if the team members work together in the future. It is worth noting, in addition, that the process of preaction planning and postaction review itself forms a virtuous cycle that can build team understanding for future actions, both in general and, even more powerfully, for collaborating in a specific team whose members will working together again in the future—as well as reinforcing the social aspects of "team cohesion" and familiarity that are known to build team loyalty, commitment, and positive attitudes (Cassell, 2000; McNeese, Salas, & Endsley, 2001; Miller, Wu, & Ott, 2012).

A thought experiment

We can illustrate the effects and power of pre- and postmission transparency information with a simple thought experiment. Fig. 3 provides a timeline for a preplanned workflow (e.g., for a hypothetical military reconnaissance mission). Imagine this plan is worked out and agreed to by a supervisor and subordinates before a mission. The plan says that this will be a Reconnaissance mission consisting of an Ingress Phase for one hour, then a Search Phase for another hour. These will be followed by a decision point whose agreed-upon logic is that if a target was detected it will be monitored for another hour before traversing and egressing from the mission area and if not, the recon aircraft will traverse immediately to a determined point and then egress.

The simple fact of having made this plan affords substantial transparency information (and therefore SA) to the supervisor even with no further communication. For example:

- Given that it is 45 min into the mission, the supervisor knows that the subordinate is (supposed to be) engaged in ingress and even, approximately, where the subordinate is. This is "What's going on?" knowledge—Level 1 SA Transparency.
- Furthermore, the supervisor knows why the subordinate is ingressing: to get to the search area and begin search—Level 2 SA Transparency.
- Finally, the supervisor knows that, at time 1:00, the team will transition to search. This is "What the operator should expect to happen?"—Level 3 SA Transparency.

Granted, this awareness may be erroneous because reality all too frequently deviates from our plans. If no communication is available, the supervisor's SA based on this plan may be wrong. I would argue that, under equivalent loss of communications conditions, having shared a plan still affords better SA than not having shared one. This is because deviations from the plan will be probabilistic (i.e., they may not happen) and, if they do, they will represent partial changes to the baseline plan. Furthermore, having an a priori plan serves as a framework to make communication more efficient: instead of having to report all three levels of SA

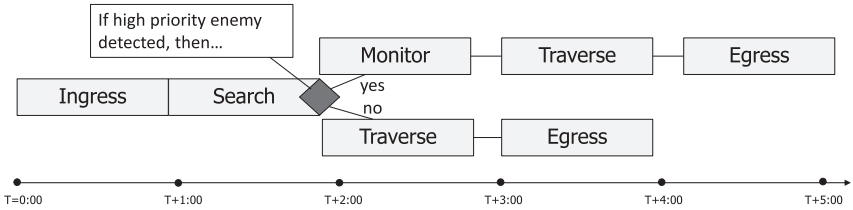


FIG. 3 Simple, hypothetical preplanned mission timeline with a contingency.

when communications are again available, the subordinate will generally have to report only current status or, perhaps, only partial deviations from the original plan. This example does, though, stress the critical nature of SA about the state of one's knowledge. The one advantage that lacking a prior plan has is that the supervisor will know s/he knows nothing about the subordinate's actions. This may well be better than assuming a plan is still in place when it is not, but this represents a call for meta-awareness about one's information and its staleness, not a reason to avoid having a plan.

The presence of preplanned alternatives and their logic or rationales (e.g., via the decision point in Fig. 3) enables further expanding SA with reduced communication needs. If, any time from 1:00 to 2:00 into the mission, the subordinate reports finding a target, then behavior at the decision point becomes predictable. If any time after 2:00, the subordinate reports that it is monitoring, the superior can conclude the reason is because a target was found. Even when the entire plan is made impracticable, having had a shared plan makes communication about future behaviors more efficient by providing a shared ground to build on or deviate from. If, in the above example, a fuel leak makes sustained monitoring impossible, both supervisor and subordinate will know roughly where the agent is, how much range is required to get home, that a target has been detected, that the monitoring task will have to be abandoned, etc. And all of this knowledge can be shared with little or no "in the moment" information exchange or processing.

It is straightforward to show how displaced transparency can lead to a reduction in the need to provide "transparency" information in real time. Consider again the simple mission in Fig. 3. Now assume an "Observing Teammate" (OT) who can observe and know everything that is happening in the region but who is completely unaware of the mission and the intentions of an "Executing Teammate" (ET) who is on site executing the mission. This situation is not far from an observer monitoring a region via satellite feed but without any a priori knowledge of a team operating in that area. Thus OT has all and only the "What's going on?" portion of Level 1 SA in Fig. 1. Note that this is not even all of the "transparent" information for Level 1 SA Transparency; the "What the agent is trying to achieve" portion is missing, as well as essentially all of the information for Level 2 and 3 SA Transparency. Any of the remaining intent information (including future states and consequences beyond simple physics model conclusions) will have to be communicated from the ET to the OT. Even observable events in the world (e.g., cloud cover) will need to be interpreted for OT for their impact on "What the agent is trying to achieve" as well as "Why the agent does what it does" and "What will happen next". Without such interpretation, OT's ability to infer this information will be near zero. On the other hand, if OT and ET had shared

knowledge of the mission in a premission displaced transparency planning session, then ET might need to communicate nothing since events would be interpreted similarly and decision making would be identical between ET and OT.

Even when OT cannot observe everything ET can, the burden of communication and processing is reduced. When events unfold as planned, at most ET might need to communicate confirmations of expected states. Even when unexpected events occur (e.g., cloud cover), reporting them may be all that is necessary to synchronize ET and OT's knowledge and assumptions about mission impact and the need for revisions—though confirmation will provide useful redundancy. Negotiation about next steps may still be needed, but preplanned actions and even variations to them can be communicated much more tersely since both parties know, a priori, the significance of many events (e.g., a detected target) and the range of acceptable options. Note that even when no specific options are preplanned, prior sharing of values and general processes or methods will inform and abbreviate discussions about what to do next.

Even errors in SA become easier to detect given this prior planning. Let us say that OT fails to notice that ET detected a target and thus, does not understand why ET is transitioning to monitor rather than traverse. Simply posing the question in the context of what was assumed to be a shared understanding of the mission identifies the mismatch. "Why are you Monitoring?" conveys a violation of expectations for monitoring (i.e., the prior detection of a target) and hones in on the piece of information that is needed to repair OT's loss of SA.

Fig. 4 illustrates the information that needs to be conveyed during a mission if full knowledge of plan, purpose, and procedures are shared ahead of time vs if they are not. The relative sizes of the blocks associated with Level 1, 2, and 3 SA Transparency are purely speculative. Note that while it may be true that the higher levels of transparency information can be omitted under conditions of shared a priori transparency, I do not mean to claim that having this information available during execution is useless. Since any sharing of information will necessarily be subject to uncertainty, not to mention the possibility of human forgetfulness, continued sharing of higher-level information types will help to trap potential mismatches, as well as reminding the human members of the team what they *should* be aware of.

Although the above discussion focuses on displacing transparency information into premission planning, a posteriori displacement should not be ignored. Although harder to quantify, debriefings that are followed by subsequent missions can have similar effects. If, for example, OT learns that ET has a tendency toward speedy completion of missions, s/he might assume a bias or preference for traversing vs monitoring. Thus in the presence of an ambiguous target detection signal, OT can make more nearly

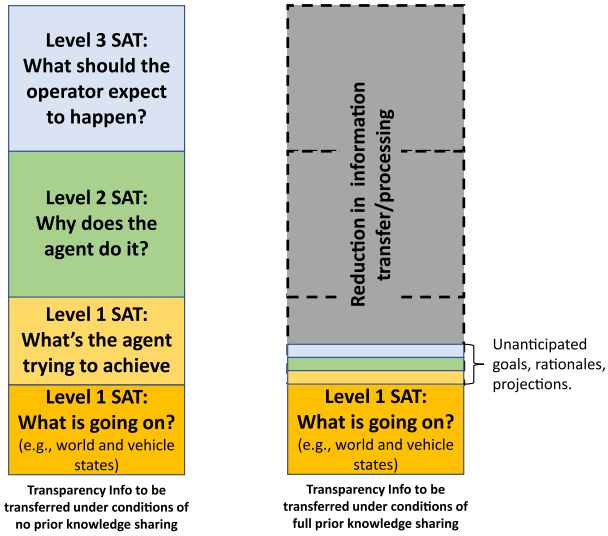


FIG. 4 Transparency information needed during a mission for supervisor awareness if full knowledge of plan, purpose, and procedures are shared ahead of time (right) vs. if not (left).

accurate predictions about what ET will do. In fact, this kind of targeted "experiential sampling," coupled with debriefing explanations, may represent a very useful training method for calibrating human trust in, and expectations of, automation in the future (Lyons, personal communication, October 18, 2019).

Mental model synchronization: A mechanism

The claim that displacing transparency information can serve the same purpose as presenting it at the time of need begs the question of how such a process might work. Kambhampati, Chakraborti, Talamadupula, and others (Chakraborti, Sreedharan, Zhang, & Kambhampati, 2017; Talamadupula, Briggs, Chakraborti, Scheutz, & Kambhampati, 2014) provide the beginnings of an answer. Interested in explanations from machine planners, they nevertheless begin with a human socio-cognitive model of an explanation's role. They take issue with many past approaches to machine explanations that present machine reasoning in its own terms—a process they call "soliloquy" (Chakraborti et al., 2017). They say "Such soliloquy is wholly inadequate in most realistic scenarios where the humans have domain and task models that differ significantly from that used by the AI system" (p. 156). Instead, effective explanation must be a "Model Reconciliation Problem" (Chakraborti et al., 2017)—a

process that is entered only when a difference between "models" (roughly, mental models—Johnson-Laird, 1989) is detected by one or both actors, and that proceeds until this difference is understood and repaired, enabling translation.

Kambhampati has pointed out (R. Kambhampati, personal communication, August 1, 2017) that (except perhaps in testing and evaluation) explanations are neither sought nor provided when individuals believe their reasoning is already compatible with each other. If I believe I understand your mental model sufficiently that you and I would both arrive at my preferred decision, then no explanation of my decision is necessary from me to you. It is only when our models (are believed to) diverge that explanations are invoked. Explanation is a process of *Mental Model Reconciliation*. It synchronizes the models of those who must work together. It is sought when models do not synch—and it may not be sought when models are assumed to synch, even if they do not.

While explanations can certainly deviate from our actual methods of decision making (Klein, 2008; Miller, 1991; Tversky & Kahneman, 1974), they nevertheless represent how we are trained and acculturated to providing rationalizations. Thus, explanation provision represents a form of group synchronization in its own right. Though my explanation may not represent how I actually arrived at a decision, my ability to provide an explanation that satisfies you is itself evidence that I have a reasonably accurate understanding of your mental model, after all.

This viewpoint explains many phenomena in both explanation and transparency. For example, Entin and Serfaty's (1999) finding that efficient teams require less, not more, explicit communication can arise because such teams are likely to be trained and to have worked together extensively or have been trained to attend to and anticipate each other's information needs. Thus their mental models are likely to be well synchronized; explanatory interactions are less likely to be required, commands can be abbreviated, and even task-based status information can be abbreviated because all parties are likely to understand what is needed when and why.

Similarly, most of the benefits of team debriefing after a mission or project (Tannenbaum & Cerasoli, 2013) can be attributed to the fact that effective debriefings involve team members interacting about their decisions and their behaviors. Simply knowing who knew what, when, and how they made decisions on the basis of that knowledge serves to educate team members about their teammates' mental models, while group discussion about how things might be done more effectively in the future creates shared mental models going forward. Group discussion and shared rationalization of events and processes can also serve an educative function, effectively causing the group to converge on a series of broad thought processes and values that, within the "culture" formed by the group,

count as valid and reasonable "ways of thinking" (or, at least, of explaining) (Miller, 1991; Vaughan, 1997). Note, though, that this can cut both ways—resulting in adverse cases in "groupthink" (Turner & Pratkanis, 1998) and "normalization of deviance" (Vaughan, 1997)—shared mental models that are inadequate or incorrect to the needs of the work domain. Finally, explanations within a group play a social function as well (Brown & Levinson, 1987), serving to reduce social distance and establish or reinforce power structures. While these functions may not directly contribute to a shared mental model about how decisions should be made within the group, they will serve to enhance team cohesion and affiliation (Miller et al., 2012).

One sense in which Eisenhower's "planning is everything" (National Archives and Records Service, 1957) may be true is that interactive team planning actively promotes model reconciliation. Extensively pouring over plans, rehearsing what could go wrong and what might be needed in contingent situations is a process that conveys models and affords opportunities for synchronization in much the same way that postmission debriefing does. Participants generally emerge with a richer understanding of each other's models, having had many opportunities to confront and refine those models collectively. This allows teams to "get inside each other's heads" analyzing and reaching agreement on how to behave in anticipated situations, thereby increasing their chances of knowing how each other will think even in unanticipated situations.

Note that I am, in no way, claiming that measuring and computing mental model synchronization will be easy, especially between humans and machines. Humans have evolved cultural, semantic, and pragmatic markers rooted in natural language and "body language" for interpreting the need for model reconciliation and then effecting it—a process that, though complex and rich, is far from error proof (cf., see, for example, Miller (2009) for examples of model mismatch where communication failed to avoid misinterpretations and, therefore, accidents).

Displaced transparency and trust

The above discussion began with a problem in providing transparency information at execution time and has ranged broadly over many core human-automation interaction topics: workload, SA, explanation, mental models, planning, and debriefs. The issue of trust is inherent in many of these topics, but it is worthwhile bringing trust back to center stage. What is the relationship between transparency, including displaced transparency, and trust?

Lee and See's three methods of trust calibration

Lee and See (2004) in their extensive review of the trust literature offer the following definition: "Trust can be defined as the attitude that an agent will help achieve an individual's goals in a situation characterized by uncertainty and vulnerability." (p. 51). As an "attitude" in situations "characterized by uncertainty and vulnerability," trust is *not* involved when complete knowledge and predictability of the agent and situation are available. These situations would require complete transparency to be available to (and usable by) the human, so it may be that trust and transparency are ultimately orthogonal concerns. But such situations are rare to nonexistent with complex, real-world automation, so trust and transparency remain highly relevant.

Lee and See (2004) go on to define a three-tiered model of trust formation and "calibration" that is particularly relevant to displaced transparency. They argue that pertinent information can be assimilated in three qualitatively different ways as described in Table 1.

Lee and See also point out that these levels and types of trust are not independent. They may well all be brought to bear in making a given trust decision, but beyond that, they also serve as a sequential set of filters over time. If an agent does not garner sufficient affective trust initially, via the rapid and comparatively effortless processing of affective cues, the observer may well avoid future interactions and therefore never garner enough experience to begin to form analogic and analytic trust. The same is true of analogic trust filtering out experiences that could lead to amassing the deeper knowledge required for analytic trust.

The three trust levels and displaced transparency

Transparency "in the moment" demands a full processing and analysis of the agent's behavior or decision and is, as argued above, sufficiently effortful to be implausible in many real-world situations. Thus according to Lee and See's definition, these situations are ripe for the influence of trust on reliance decisions. But how can transparency, whether in the moment or displaced in time, affect the resulting trust?

We have seen above how displaced transparency information can be used to synchronize mental models across team members. This is exactly the kind of information required for analytic trust calibration. Knowing what is going on, what the agent is trying to achieve and why, and what to expect next in any detailed fashion both supply and likely require a deep understanding of the agent's mental model, approach to planning and execution, goals and methods, etc. This level and depth of transparency information must be "available" for analytic trust decisions in the moment when a decision is to be made, but that does not necessarily mean

TABLE 1 Author's interpretation of Lee and See's (2004) three trust calibration methods.

Type	Description	Attributes and influences
Affective trust	Emotion based	Based on emotional reaction to the agent, the situation or context. Fast, requiring little input data or cognitive processing. Feelings of comfort or of apprehension provide subtle and pervasive influences on ultimate reliance decisions. Emotion affects trust even when cognitive resources are available (e.g., Damasio, 2005), but it can guide rapid and "intuitive" decisions to devote resources to more thoroughly understand a situation or instead to avoid it
Analogic trust	Based on a reference model or on source- or rule-based reasoning from trusted sources	Based on analogical similarity between the agent and situation with other trusted situations or which have been conveyed through trusted sources. Includes source- and rule-based trust. Pilots are willing to fly with other pilots not previously encountered largely due to analogic trust in certifying agencies, and analogic behaviors: s/he dresses like a pilot, talks like a pilot, exhibits familiar moves in the cockpit, etc. Analogic trust is less cognitively demanding than affective trust, but more so than analytic trust
Analytic trust	Model- and/or understanding-based; reasoned application of knowledge to current situation	Based on awareness of "how the agent thinks" and/or on an analytic assessment of the likely costs and benefits of trusting and relying on the agent. Never complete since trust applies to situations characterized by uncertainty. Analytic trust is slow to develop—either through the accumulation of knowledge about the agent or through accumulation of observations about the agent's success or failure in relevant contexts. Also effortful to apply, requiring substantial cognitive resources to think through likely behaviors in the current situation

that it should be presented to him/her all and only in that moment, much less that s/he should attempt to process it all in that moment to enable a full analytic trust decision. Instead, the displaced transparency processes of training, planning, intent discussion, explanation, after action review, etc., can convey mission-specific information pertinent to analytic trust, as well as more mission-general information about how the agent thinks, what he/she/it knows (i.e., their areas of expertise), what their values, priorities, and preferences are, etc. This information can be pertinent across all three of Lee and See's trust calibration methods.

For example, if in a planning session I learn that an agent avoids all risk of thunderstorms because it cannot adequately sense them or control maneuvering in high winds, I will have learned specific information about its capabilities allowing me to calibrate my trust analytically in future conditions of rain, turbulence, etc. But I will also have learned information that I may process analogically or affectively: the agent has demonstrated less than complete professional pilot capabilities and I may adjust my mental model by analogy to reason about it in the future as a novice or incompetent pilot. This may prove accurate and adaptive in some contexts, and wrong in others—just as reasoning by analogy does for human agents. Similarly, I may regard the company or agency that provided this agent as, by extension, less than fully competent, coloring my trust (through analogic processes) of it in the future. Finally, the entire experience may cause me to be dissatisfied by the agent—an affective response that will, perhaps inaccurately, cause me to be more likely to distrust other aspects and capabilities of the agent, and even of other agents, which have nothing analytically to do with its performance during storms.

Note that in the above example, the information learned was clearly relevant transparency information. It was pertinent to both what the agent will do (level 3 SA Transparency) and why it does what it does (level 2 SA Transparency), and it will be pertinent during execution when and if the agent deviates to avoid predicted thunderstorm activity (What is going on and what the agent is trying to achieve—Level 1 SA Transparency). But because the information was offered and understood during premission planning instead of at the moment when the agent begins a deviation in thunderstorm conditions, it also serves as an example of displaced transparency. Note that it could have been provided in an after action review and explanation, or mixed between pre- and postmission time frames. Having that information available in memory during execution reduces the amount of information required to be transmitted to the observing supervisor in that moment—and it makes the processing of that information more efficient and focused. Having added time and cognitive capacity available might well make it feasible to make more detailed and resource-intensive trust decisions using the deeper assimilation methods of Lee and See's multitiered model.

Conclusions, predictions, and next steps

I have argued in this chapter for a fundamental link between trust, SA, transparency, and many processes that are known to improve all of them together: explanation, shared mental models, premission planning, and postmission debriefing. I have also presented an argument for the desirability of displacing the presentation of transparency information into a priori mission planning interactions and a posteriori explanations and debriefings, along with a hypothesized mechanism for why these effects might be obtained. We know, from the sources cited above and others, that transparency information frequently provides detectable mission performance benefits. We also know that prior mission planning and explanation and debriefings also provide benefits for team cohesion, team satisfaction, and team performance. It seems likely that these benefits obtain because they are making use of the same underlying mechanism: the communication of information that promotes SA through mental model synchronization. The fact that this information does not have to all be transmitted at the time of use, but can be spread into lower workload periods before and after, is a feature we should use more extensively in design—for human-human and also for human-machine interactions.

While I have criticized some of the work on transparency above for focusing too much on the "locus of execution," this is not always the case, and it should come as no surprise that some researchers and some recent work are expanding beyond that locus. Recent work by Chen et al. (2018) and Lyons and Havig (2014) are increasingly treating human-machine collaborative planning as a continuous process that requires shared intent between both parties. Similarly, there is a very long history of claims that automation (and, more recently, artificial intelligence in general) should be able to explain itself for human usability (e.g., Clancey, Shortliffe, & Buchanan, 1979; Kuang, 2017), though how to achieve that end, especially with modern learning technologies, remains an area of intense research. Also, transparency is a novel field, still in the early stages of creating interaction models. In this spirit, I intend merely to point to opportunities to expand the boundaries of transparency research rather than to imply that it is ill considered.

We have not yet been able to conduct experiments to validate predictions from the "unified model" sketched above, but a simple laboratory test would involve manipulation of elements of situational context, the agent's decision processes, and of the knowledge of both situation and agent processes the human supervisor possesses. A relevant approach is provided in Chakraborti et al. (2017) where a human and a robot are located in a building with a long corridor and multiple side rooms where one or more med kits may be located. Here the human tasks the robot to

fetch a med kit. Mismatches in elements of mental models (e.g., awareness of the physical context and of the robot's decision-making process) can be manipulated in this paradigm, along with what transparency information is provided when, and expectations, predictions, and performance can be assessed. In this or a similar paradigm, we would predict:

- With mental model synchronization, reduced time, workload effort, and communication bandwidth will be necessary to achieve a similar level of SA.
- Shifting the communication of transparent information into other time frames (before or after execution) will yield improved SA (with reduced workload) even under conditions of communications restriction or constrained human workload.
- A priori mental model reconciliation will yield more accurate inferences even in unanticipated situations, even with little or no real-time communication.
- Postmission debriefing will produce increased awareness and ability to predict teammates behavior even in unanticipated situations in subsequent missions.

We note with interest that Chakraborti et al. (2017) report including mental modeling capabilities in a robot agent's reasoning (i.e., the robot modeled the expected reasoning of a human and reacted accordingly), produced a 44%–75% improvement in robot decision making in avoiding resulting resource conflicts in one analytic experiment they performed.

Less well understood are the tradeoffs involved in displacing transparency information. Somewhere between "no plan survives first contact with the enemy" (implying "overplanning" is wasteful) and "Plans are worthless but planning is everything" (implying planning activities are very valuable), there must lie a happy medium. Where is that medium, and what parameters characterize it? Similarly, human supervisors must be able to remember, understand, and detect the applicability of previously learned elements of transparency information in the moment of execution. Dialogue in the moment will improve this ability, but may not be feasible and will, necessarily, contribute to workload—forming another tradeoff whose impact on performance should be studied. There may well also be personality attributes, training techniques, and even genetic factors that facilitate this ability and that should be discovered and applied.

The perspective developed in this chapter calls for greater emphasis on the ability of autonomous systems to explain themselves—to afford inspection and synchronization of their "mental models" with that of the human user in pre- or postmission processes. But providing explanatory SA information for autonomous behaviors is hard, and only getting harder with the advent of deep learning systems whose "mental models"

are fundamentally uninspectable, unexplainable, and subject to potentially unpredictable change (Kuang, 2017). One of the lessons of the thinking in this paper, as well as in Lee and See (2004), is not that such systems are "untrustable," but rather that trust will be developed and tuned only at nonanalytic levels. This can be through more superficial affective and analogic methods—e.g., through a pleasant interface (Parasuraman & Miller, 2004) or through having a trusted agency vouch for the system. There is nothing necessarily wrong with these approaches to trust calibration, but they rely on someone else (e.g., the interface designer or the trusted agency) to validate the trustworthiness of the system—a lengthy process that may well represent the extreme of displaced transparency.

I make no claims that the recommendation and predictions provided above are particularly novel. As I said above, humans have wrestled with the problem of coordinating performance in communications-restricted domains throughout much of human history. Therefore, it is not surprising that methods such as premission planning and postmission debriefing, mental model reconciliation, and improving the efficiency and predictability of intent communications have long been known and practiced in human-human teaming. There have been calls for improved explanation capabilities for decades, and my own work (e.g., Miller, 2017; Miller & Parasuraman, 2007) has been specifically focused on improved human-automation intent communications for nearly 20 years. I hope, however, that by showing the links between transparency, trust, SA, explanation, debriefing, planning, and mental model synchronization I may foster greater sensitivity to how these phenomena interrelate and that that may, in turn, serve to make the development and use of human-automation systems more effective within the socio-organizational and macroergonomic systems where they are deployed.

Acknowledgments

This paper is entirely the work of the author and was not directly funded by any project or agency. I am indebted to Dr. Jesse Chen for providing a forum for initial thoughts on this topic and to Rao Kambhampati for the insight that explanations generally need to focus only on mismatches in mental models.

References

Brown, P., & Levinson, S. C. (1987). *Politeness: Some universals in language usage.* Cambridge, UK: Cambridge University Press.

Cassell, J. (2000). *Embodied conversational agents.* Cambridge, MA: MIT Press.

Chakraborti, T., Sreedharan, S., Zhang, Y., & Kambhampati, S. (2017). Plan explanations as model reconciliation: Moving beyond explanation as soliloquy. In *Proceedings of the international joint conference on artificial intelligence* (pp. 156–163). New York, NY: IEEE.

Chen, J. Y., Barnes, M. J., Wright, J. L., Stowers, K., & Lakhmani, S. G. (2017). Situation awareness-based agent transparency for human-autonomy teaming effectiveness.

In *Proceedings of SPIE 10194: Micro-and nanotechnology sensors, systems, and applications IX, 101941V*. Bellingham, WA: International Society for Optics and Photonics.

Chen, J., Lakhmani, S., Stowers, K., Selkowitz, A., Wright, J., & Barnes, M. (2018). Situation awareness-based agent transparency and human–autonomy teaming effectiveness. *Theoretical Issues in Ergonomics Science*, *19*(3), 259–282.

Chen, J. Y., Procci, K., Boyce, M., Wright, J., Garcia, A., & Barnes, M. (2014). *Situation awareness-based agent transparency. Army research laboratory technical report no. ARL-TR-6905.* Retrieved from https://apps.dtic.mil/dtic/tr/fulltext/u2/a600351.pdf.

Clancey, W., Shortliffe, E., & Buchanan, B. (1979). Intelligent computer-aided instruction for medical diagnosis. In *Proceedings of the annual symposium on computer application in medical care* (p. 175): American Medical Informatics Association.

Curnin, S., Owen, C., Paton, D., Trist, C., & Parsons, D. (2015). Role clarity, swift trust and multi-agency coordination. *Journal of Contingencies & Crisis Management*, *23*(1), 29–35.

Damasio, A. (2005). Human behaviour: Brain trust. *Nature*, *435*(7042), 571.

de Visser, E. J., Cohen, M., Freedy, A., & Parasuraman, R. (2014). A design methodology for trust cue calibration in cognitive agents. In *Proceedings of the international conference on virtual, augmented and mixed reality* (pp. 251–262). New York, NY: Springer.

Dzindolet, M. T., Pierce, L. G., & Beck, H. P. (2009). An examination of the social, cognitive, and motivational factors that affect automation reliance. In D. H. Andrews, R. P. Herz, & M. B. Wolf (Eds.), *Human factors issues in combat identification* (pp. 277–297). Farnham, UK: Ashgate.

Endsley, M. R. (1995). Toward a theory of situation awareness in dynamic systems. *Human Factors*, *37*(1), 32–64.

Entin, E., & Serfaty, D. (1999). Adaptive team coordination. *Human Factors*, *41*, 312–325.

Guznov, S., Lyons, J., Pfahler, M., Heironimus, A., Woolley, M., Friedman, J. , et al. (2019). Robot transparency and team orientation effects on human–robot teaming. *International Journal of Human Computer Interaction*, 1–11.

Hart, S. G., & Staveland, L. E. (1988). Development of NASA-TLX (task load index): Results of empirical and theoretical research. *Advances in Psychology*, *52*, 139–183.

Hayes, C., & Miller, C. A. (2011). *Human-computer etiquette.* New York, NY: CRC Press, Taylor & Francis Group.

Ho, N., Sadler, G. G., Hoffmann, L. C., Zemlicka, K., Lyons, J., Fergueson, W. , et al. (2017). A longitudinal field study of auto-GCAS acceptance and trust: First-year results and implications. *Journal of Cognitive Engineering and Decision Making*, *11*(3), 239–251.

Johnson-Laird, P. N. (1989). Mental models. In M. I. Posner (Ed.), *Foundations of cognitive science* (pp. 469–499). Cambridge, MA: The MIT Press.

Klein, G. (2008). Naturalistic decision making. *Human Factors*, *50*(3), 456–460.

Kuang, C. (2017). Can AI be taught to explain itself. *The New York Times Magazine*, *46*, .

Lee, J., & See, K. (2004). Trust in computer technology: Designing for appropriate reliance. *Human Factors*, *46*, 50–80.

Lyons, J. B., & Havig, P. R. (2014). Transparency in a human-machine context: Approaches for fostering shared awareness/intent. In *Proceedings of the international conference on virtual, augmented and mixed reality* (pp. 181–190). New York, NY: Springer.

Lyons, J. B., Ho, N. T., Hoffmann, L. C., Sadler, G. G., Van Abel, A. L., & Wilkins, M. (2018). Trust in sensing technologies and human wingmen: Analogies for human-machine teams. In *Proceedings of the international conference on augmented cognition* (pp. 148–157).

Lyons, J. B., Koltai, K. S., Ho, N. T., Johnson, W. B., Smith, D. E., & Shively, J. R. (2016). Engineering trust in complex automated systems. *Ergonomics in Design*, *24*(1), 13–17.

McNeese, M., Salas, E., & Endsley, M. (2001). *New trends in cooperative activities.* Santa Monica, CA: HFES.

Mercado, J. E., Rupp, M. A., Chen, J. Y., Barnes, M. J., Barber, D., & Procci, K. (2016). Intelligent agent transparency in human–agent teaming for multi-UxV management. *Human Factors*, *58*(3), 401–415.

Meyerson, D., Weick, K., & Kramer, R. (1996). Swift trust and temporary groups. In R. Kramer & T. Tyler (Eds.), *Trust in organizations: Frontiers of theory and research* (pp. 166–195). Thousand Oaks, CA: Sage.

Miller, C. A. (1991). *Learning to disagree: Argumentative reasoning skill in development (unpublished doctoral dissertation).* Chicago, IL: University of Chicago.

Miller, C. (2004). Human-computer etiquette: Managing expectations with intelligent agents. *Communications of the Association for Computing Machinery*, *47*(4), 30–34. [Guest Ed.].

Miller, C. A. (2009). Social relationships and etiquette with technical systems. In B. Withworth & A. de Moor (Eds.), *Handbook of research on socio-technical design and social networking systems* (pp. 472–486). Hershey, PA: Information Science Reference.

Miller, C. A. (2014a). Delegation and transparency: Coordinating interactions so information exchange is no surprise. In *Proceedings of the international conference on virtual, augmented and mixed reality* (pp. 191–202). New York, NY: Springer.

Miller, C. A. (2014b). Delegation for single pilot operations. In *Proceedings of the 2014 international conference on human-computer interaction in aerospace* (pp. 2.3.1–2.3.29). New York, NY: ACM.

Miller, C. A. (2017). The FireFox fallacy: Why intent should be an explicit part of the external world in human automation interaction. In P. Smith & R. Hoffman (Eds.), *Cognitive systems engineering: A future for a changing world* (pp. 269–294). Boca Raton, FL: CRC Press.

Miller, C., & Parasuraman, R. (2007). Designing for flexible interaction between humans and automation. *Human Factors*, *49*(1), 57–75.

Miller, C. A., Wu, P., & Ott, T. (2012). Politeness in teams: Implications for directive compliance behavior and associated attitudes. *Journal of Cognitive Engineering and Decision Making*, *6*(2), 214–242.

National Archives and Records Service. (1957). *Public papers of the presidents of the United States, Dwight D. Eisenhower, 1957* (p. 818). Washington, DC: Government Printing Office. Retrieved from en.wikiquote.org/wiki/Dwight_D._Eisenhower.

Ososky, S., Sanders, T., Jentsch, F., Hancock, P., & Chen, J. Y. (2014). Determinants of system transparency and its influence on trust in and reliance on unmanned robotic systems. In *Proceedings of unmanned systems technology XVI, Vol. 9084*, (p. 90840E). Bellingham, WA: International Society for Optics and Photonics.

Parasuraman, R., & Miller, C. (2004). Trust and etiquette in high-criticality automated systems. *Communications of the Association for Computing Machinery*, *47*(4), 51–55.

Parasuraman, R., & Riley, V. (1997). Humans and automation: Use, misuse, disuse, abuse. *Human Factors*, *39*(2), 230–253.

Pinelle, D., Gutwin, C., & Greenberg, S. (2003). Task analysis for groupware usability evaluation: Modeling shared-workspace tasks with the mechanics of collaboration. *ACM Transactions on Computer-Human Interaction (TOCHI)*, *10*(4), 281–311.

Sadler, G., Battiste, H., Ho, N., Hoffmann, L., Johnson, W., Shively, R. , et al. (2016). Effects of transparency on pilot trust and agreement in the autonomous constrained flight planner. In *Proceedings of 2016 IEEE/AIAA 35th digital avionics systems conference (DASC)* (pp. 1–9): IEEE/AIAA.

Sarter, N. B., & Woods, D. D. (1995). How in the world did we ever get into that mode? Mode error and awareness in supervisory control. *Human Factors*, *37*(1), 5–19.

Sarter, N. B., Woods, D. D., & Billings, C. E. (1997). Automation surprises. In G. Salvendy (Ed.), *Handbook of human factors and ergonomics* (2nd ed., pp. 1926–1943). New York: Wiley.

Sheridan, T. (1987). Supervisory control. In G. Salvendy (Ed.), *Handbook of human factors* (pp. 1244–1268). New York: Wiley.

Talamadupula, K., Briggs, G., Chakraborti, T., Scheutz, M., & Kambhampati, S. (2014). Coordination in human-robot teams using mental modeling and plan recognition. In *Proceedings of IEEE/RSJ international conference on intelligent robots and systems* (pp. 2957–2962). New York, NY: IEEE.

Tannenbaum, S. I., & Cerasoli, C. P. (2013). Do team and individual debriefs enhance performance? A meta-analysis. *Human Factors*, *55*(1), 231–245.

Turner, M. E., & Pratkanis, A. R. (1998). Twenty-five years of groupthink theory and research: Lessons from the evaluation of a theory. *Organizational Behavior and Human Decision Processes*, *73*(2–3), 105–115.

Tversky, A., & Kahneman, D. (1974). Judgment under uncertainty: Heuristics and biases. *Science*, *185*(4157), 1124–1131.

Vaughan, D. (1997). *The challenger launch decision: Risky technology, culture, and deviance at NASA.* Chicago, IL: University of Chicago Press.

Yang, E., & Dorneich, M. C. (2016). Evaluation of etiquette strategies to adapt feedback in affect-aware tutoring. In *Proceedings of the human factors and ergonomics society annual meeting, 60(1)* (pp. 393–397). Los Angeles, CA: SAGE Publications.

CHAPTER

12

A roadmap for developing team trust metrics for human-autonomy teams

Kristin E. Schaefer, Brandon S. Perelman, Gregory M. Gremillion, Amar R. Marathe, and Jason S. Metcalfe

Combat Capabilities Development Command Army Research Laboratory, Aberdeen Proving Ground, MD, United States

Introduction

Though trust is a long and widely studied construct, we argue that the basic science underpinning trust research remains challenged to address a variety of critical issues for advanced human-autonomy teaming. Therefore, the purpose of this chapter is to examine the progress made by the scientific community in addressing the critical issues affecting trust and effectively advance on this foundation. Moreover, we offer our vision of a research roadmap that we believe can systematically and progressively advance next-generation concepts and principles that would enable novel approaches for the formation and management of trust in human-autonomy teams. Our intent with this chapter is to influence the broad conversation around the construct of trust for human-machine partnerships (beyond that of dyadic relationships), that is, we believe that human-autonomy teaming research will benefit greatly from an established and convergent set of concepts for the systematic study of trust and related factors. In what follows, we articulate a research roadmap within a set of foundational steps that we believe encapsulate the general challenges to understanding trust dynamics and developing metrics for effective monitoring of team trust and subsequently predicting performance outcomes.

Trust in Human-Robot Interaction
https://doi.org/10.1016/B978-0-12-819472-0.00012-5

© 2021 Published by Elsevier Inc.

Building the foundation for the roadmap

We acknowledge that the literature on trust as a general construct is vast, as is the domain of study regarding human interactions with machines of all types (from rote automation, to computing devices, up to autonomy). Any review of a body of work that draws from multiple large domains must necessarily be selective and this chapter will be no exception. Here, we discuss a set of stepping-stones we believe are foundational for creating a path toward identifying the critical research themes that support understanding how to extend into new territory—that of trust as it occurs in complex teams made of heterogeneous mixtures of humans and autonomous or otherwise intelligent agents. In this, we consider the general construct of *human-autonomy teaming* as a burgeoning science of complex interacting systems with varying levels of autonomy and degrees of adaptability. Importantly, this is also an area that, while new and popular in our contemporary society, is fraught with challenges in myriad research areas, among which is the dynamics of the formation, dissolution, and control of trust. In this, we seek to explore and identify needed areas of future research. This first section builds the foundation by formulating a conceptual understanding of trust by examining the impacts of trust definitions, the trust process, and trust measurement as it can support team trust theory and metric development.

Step 1: The importance of a trust definition

A key notion within the large base of literature focused on trust is that the very conception of the construct varies considerably depending on the object under investigation. Whether looking at, for instance, discrete cueing systems versus continuous automated processes, or exploring human-human versus human-machine interactions and partnerships, or even looking at different time slices in the formation and maintenance of trust, what emerges is that those who study and promote understandings in this area are often working from a different underlying conception of what trust is and what it is not. Therefore, the goal here is to explore this intellectual domain with the question as to whether there may be fundamentals of trust that undergird the host of constructs that have been put forth over the last several decades.

As we have sought to develop appropriate ways of indexing and monitoring the evolution of trust in teams beyond dyads and then define useful individual metrics thereof, the lack of an agreed-upon operational definition has proven to be a limitation. Moreover, methods of describing trust for dyads cannot be expected to directly extend into larger groups. Within complex teams involving more than two or three agents, trust dynamics are unlikely to evolve linearly and in synchronous fashion

across the entire group; rather, the trust in the team as a whole likely represents some state that captures some subgroups that are in a process of trust formation, some that are undergoing trust dissolution, and yet others that are experiencing a stable trust state. So, taking all of the potential axes of complexity into account, we ask how to best define, at a core level, trust. Herein, we consider a construct that encompasses the various phases of the dynamic process of trust among agents and how that directly applies to developing team trust metrics.

Before considering whether a singular definition can be used to capture what trust is, we ponder the complexity involved. Here, it is worthwhile to recognize that over 300 definitions of trust have been published across a diverse set of research domains (Schaefer, 2013). While it seems that the primary literature reflects a move toward adopting some key aspects of definitions as suitable for study of human-autonomy teaming,[a] it remains common for researchers to create their own new definition to fit the goals, criteria, or specific scenario of study. Both the existing diversity of definitions for trust and the continued introduction of new and varied descriptions that fit ever-more focused circumstances serve to further complicate the discourse within and among the community of trust researchers. Rather than lending clarity to a generalizable understanding of trust, it appears that the growing literature actually obfuscates comparative interpretations of research and confuses proposals to extend current knowledge toward important future needs, such as concepts that support understanding trust within complex human-autonomy teams.

Despite this diversity, the existence of multiple definitions can provide insights into what the field has construed as common elements that are important for understanding trust. Following review of these definitions, five have risen to the surface: expectations, confidence, vulnerability, risk or uncertainty, and automation reliance (Billings, Schaefer, Llorens, & Hancock, 2012; Schaefer, 2013). One of the most critical components of the development and maintenance of trust in human-autonomy teams is that humans bring *expectations* into their interactions. An individual's expectations regarding the performance of a system are critical for the maintenance of trust because any violation of those expectations can lead to inappropriate changes in trust (or behavior) despite, or independent of, the actual capabilities of the system (Drnec, Marathe, Lukos, & Metcalfe, 2016; Muir, 1994). *Confidence*, on the other hand, has often been used as a synonym for trust when invoked by researchers working within particular domains (e.g., interpersonal psychology). However, a more general consensus has been that the confidence one has in a system is only one of

[a] For human-autonomy teaming, one of the most widely adopted definitions of trust is "the attitude that an agent will help achieve an individual's goals in a situation characterized by uncertainty and vulnerability" (Lee & See, 2004).

the antecedents or factors that can lead to trust development (see Hancock, Billings, Schaefer, Chen, et al., 2011; Schaefer, Chen, Szalma, & Hancock, 2016). Indeed, this construct of confidence in the system has clear value because it can exert important influence over how and when a human team member will change the system's control allocation (Gao, Lee, & Zhang, 2006). While the issue of *vulnerability* has predominantly been a point of focus in the interpersonal trust literature, the related concepts of *risk and uncertainty* became critical for human-autonomy teams because they can affect mission criticality and control allocation requirements (Drnec et al., 2016; Schaefer & Straub, 2016). More directly, we argue that for trust to be a factor for effective performance of human-autonomy teams, there must be some potential threat or risk, whether to a mission, a system, a user, or perhaps even a bystander. That is, in the absence of risk, trust is unlikely to be a significant operative factor in decision-making around reliance on a partner or team. Lastly, the regular focus on *automation reliance* highlights an important concern that the relationship between trust and trust-based outcomes is not often clearly defined or demarcated. Trust-based outcomes such as reliance, compliance, and use are often at risk of becoming conflated with or even confused for trust, even when authors are explicit and cautious in their discussions. For example, the number of times a person accepts, relies on, or complies with recommendations from an automated cueing system has regularly been used as a behavioral means to validate subjective reports about the degree of trust (cf., Dzindolet, Peterson, Pomranky, Pierce, & Beck, 2003; Madhavan, Weigmann, & Lacson, 2006). The greatest concern here is that the absence of metrics of trust that can track the behavioral outcomes as the interaction unfolds means that researchers will be forced to continue validating their inferences about trust with related but nevertheless distinct constructs. The reliance on behavioral covariates, in turn, will result in continued lack of clarity about what trust is and by which mechanisms it exerts influence over behaviors.

While reviewing trust definitions may provide value for understanding the construct, it is important to note that merely having a definition is not, on its own, enough to operationalize a diverse construct such as trust. Trust likely sets a context, but additional factors or indicators that influence the development and calibration of the human-autonomy relationship will ultimately need to be taken into consideration in order to identify critical team trust metrics and create active mitigations or interventions, such as when the system engagement might be expected to break down. It is in these factors that it becomes possible to identify measurable and quantitative differences that are needed to develop metrics. These factors derive from the human, robot or autonomy, and environment and have been identified as having a statistical effect on trust development across contexts (Hancock, Billings, Schaefer, Chen, et al.,

2011; Parasuraman & Riley, 1997; Schaefer et al., 2016). However, it is important to note that only a handful of definitions have even considered these factors as important, further frustrating cohesive discussions of a general construct of trust.

In order to understand the relationship between definitions and associated trust factors, we have found it useful and important to move away from an overgeneralized definition of trust and look at what we call *types of trust*. This view is in line with Merritt and Ilgen's (2008) work that identified that "not all trust is created equal." Overgeneralizing the terminology and calling everything "trust" leads to miscommunication in research that can affect the direct insight into the development and maintenance of trusting relationships that persist over time, events, and interactions. By identifying the critical types of trust within human-autonomy teams, as well as the underlying factors or indicators, we can not only improve understanding but also lay the foundation needed to develop effective metrics of team trust. Therefore, it is important to understand how these different types of trust integrate into, if not arise from, a fundamental construct. To this end, several trust researchers have developed theoretically driven models to understand the relationships among the varied trust types and the factors that inform each.

While we acknowledge that a multitude of theoretical models describing different types of trust have been proposed, we have chosen to consider, here, three particular perspectives, including those of Merritt and Ilgen (2008), Hoff and Bashir (2015), and Schaefer (2013). These three models represent the diversity in how the concepts are presented with respect to human-autonomy teaming and thus provide an important basis for the remaining discussion seeking to better understand trust and all of its manifestations. First, all three models appear to agree that there is a differentiation between trait-based trust (trust propensity), which has been characterized as a stable trait describing an individual's general inclination to be trusting of others, and state-based trust, which characterizes the degree to which a person is trusting of other specific persons or autonomy in particular circumstances. The main point of departure among varied trust definitions has arisen from the state-based trust descriptions for human-autonomy teams and the associated factors that make up each type: trust propensity, dispositional trust, affect-based trust, trustworthiness, history-based (also termed cognitive-based) trust, situational trust, and learned trust.

One key issue that has emerged from looking at these different models is that researchers often use the same or similar terms but the underlying meaning is described differently. For example, Merritt and Ilgen (2008) and Hoff and Bashir (2015) differentiated dispositional trust, which reflects the trust in persons or machines that is present at the outset of an interaction, from the situational trust that evolves throughout an

interaction or set of interactions with another person or machine. A surface generalization would leave the impression that both groups of authors agree that the construct of dispositional trust captures individual biases and tendencies toward being trusting of others and situational trust is more dynamic and responsive to a variety of factors, including whether the interaction unfolds in accordance with the expectations of the would-be trustor. Yet, despite this apparent agreement, there remain discrepancies in how they are interpreted. For instance, Hoff and Bashir suggested that dispositional trust arises from biological and environmental influences that become relatively stable over time, while Merritt and Ilgen suggested that dispositional trust can vary per each trustor-trustee relationship. Schaefer (2013) agreed with Merritt and Ilgen's perspective on dispositional trust and therefore pulled from the larger literature to further describe a "dispositional trust loop" in terms of affect-based trust and perceived trustworthiness.

Further deviations in trust definitions have occurred when focusing on teaming and actual interaction behaviors. While Merritt and Ilgen (2008) and Schaefer (2013) discussed this interaction-based trust as either history-based or cognitive-based, Hoff and Bashir (2015) separated out interaction-based trust as situational trust, which also includes online updates to some of the factors that influence affect-based trust during or after an interaction, and learned trust, which includes updates to some of the factors that influence trustworthiness during or after an interaction. Our intention with this discussion is not to adjudicate whether one method is right and another wrong, but rather to point out that these differences occur and are a fundamental contributor to the confusion, or simple lack of agreement, within the trust research community. Fig. 1 provides a visual depiction of one way to conceive of the interrelations between these different constructs as they apply to human-autonomy teams.

Trust propensity

Trust propensity is an integral component of relationships (Larzelere & Huston, 1980). As a stable trait, unique to the individual (Jarvenpaa, Knoll, & Leidner, 1998; Mayer, Davis, & Schoorman, 1995; Rotter, 1967; Stack, 1978), trust propensity is expected to remain stable across a variety of circumstances (Burke, Sims, Lazzara, & Salas, 2007; Mayer et al., 1995). Current human-robot trust research suggests that due to the human component in the team, a propensity to trust scale should always be implemented in conjunction with any trust measure (see Yagoda, 2011) because an individual's trust propensity can directly impact their decisions about use of a machine (Merritt & Ilgen, 2008). According to Schaefer (2013), antecedents of trust that could impact the propensity to trust include human demographic traits (i.e., age, ethnicity, and personality traits) and societal or cultural norms; whereby it is important to note that these

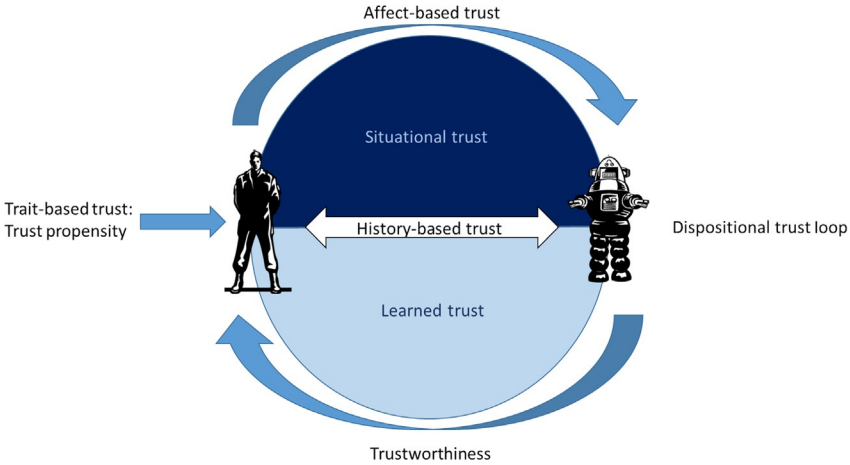


FIG. 1 This figure illustrates a way of integrating different theoretical vantage points on the set of trust constructs as they are situated in a larger trust process for an interacting system of individual human and intelligent agents.

are the same factors that make up Hoff and Bashir's (2015) definition of dispositional trust.

Dispositional trust

In line with Merritt and Ilgen's (2008) definition, dispositional trust reflects the trust in another specific person or machine that begins during the initial encounter, even if no interaction has taken place. In Fig. 1, dispositional trust between a trustor and trustee is depicted as emerging from successive iterations of the affect-based trust and trustworthiness loop. This loop includes factors that the human team member places on the autonomy stemming from his or her internal feelings (affect-based trust) as well as inferences he or she draws from the autonomy based on the features and perceived capabilities of the system (perceived trustworthiness), both, of course, being conditioned upon the individual's trust propensity.

Affect-based trust

Affect-based trust is an emergent attitudinal state in which the individual makes attributions about the motives of a partner (Burke et al., 2007; McAllister, 1995). Similar to propensity to trust, affect-based trust is also important in the initial stages of a relationship but can have lasting effects on the trust relationship. Atoyan, Duquet, and Robert (2006) suggested that the initial human-autonomy interaction experience has a direct influence on both trust and usage decisions and behaviors. Within a human-autonomy team, affect-based trust can influence how

the human team members make attributions about the motives of an autonomy-enabled system prior to interaction and can be updated following interaction with that system. According to Schaefer (2013), the human emotive factors of comfort, satisfaction, confidence, and attitudes are included as antecedents of affect-based trust. The level of comfort may be dependent on familiarity and proximity (Nakada, 1997; van den Broek & Westerink, 2009), similarity of intent (Verberne, Ham, & Midden, 2012), or even the degree of control the autonomy has over levels of control, tactics, or strategy of the task (Ward, 2000). This satisfaction is fostered by the quality of information and service that the technology provides (Lee, Kim, & Kim, 2007) and is important because as satisfaction increases, generally so does trust (Domnez et al., 2006). Satisfaction may be influenced by situation awareness, predictability, and dependability of the robot. Bailey, Scerbo, Freeman, Mikulka, and Scott (2006) found that operators with positive attitudes toward automation may have too much trust in the system and focus less on the overall system state, which can ultimately lead to degraded performance. Additionally, moods and emotions (e.g., happiness; Merritt, Heimbaugh, LaChapell, & Lee, 2013) impact trust and liking of a system and are thus included in this part of the trust model.

Perceived trustworthiness

There are two schools of thought when discussing trustworthiness. Trustworthiness is often considered a characteristic of the agent being trusted and is often discussed as a separate construct from trust. However, for the purpose of this chapter, we are taking an approach that specifically focuses on perceived trustworthiness that impacts the person's dispositional trust in the agent. From this perspective, perceived trustworthiness is developed from the interpretation of cursory information about the characteristics of the autonomy that affects the initial dispositional feelings (related to affect-based trust) toward the agent. This perception is then updated over time following interaction as actual capabilities, behaviors, and features become more understood. So, while we acknowledge that there may be some debate calling this a "type of trust," from this perspective we aim to show that it is difficult to remove this construct from the trust discussion.

Building on the interpersonal trust literature that describes trustworthiness through assessing the partner's ability, benevolence, and integrity (Colquitt, Scott, & LePine, 2007; Daellenbach & Davenport, 2004; Mayer et al., 1995), and the automation literature, where trustworthiness is founded on the characteristics of the automation, such as behavior (Muir & Moray, 1996; Parasuraman & Miller, 2004), we hereby identify critical factors that are important to understanding perceived trustworthiness in this context. The primary factors that impact initial

trustworthiness in a given human-autonomy team are (1) reputation (Daellenbach & Davenport, 2004; Lee & See, 2004), (2) physical form (e.g., anthropomorphism; Dautenhahn, 2007), (3) autonomy type, (4) autonomy personality, (5) intelligence, (6) level of automation, and (7) perceived function (e.g., behavior). Since initial trustworthiness is assessed prior to interaction, the perceived functionality of the partner can be drawn from its reputation or viewing the autonomy's physical form, if it has one. Physical form may be the primary influence on initial trustworthiness judgments of a physical system, given the importance of morphological features for human perception of trustworthiness, particularly in the absence of other information about the autonomy.

History-based trust

History-based trust is founded on interactions between the person and another person or machine (Merritt & Ilgen, 2008). This type of trust is conceptualized as a combination of the effects of dispositional trust and the experience or interactions with autonomy, especially taken in the context of prior expectations. History-based trust is by its very nature continuously recalibrated, as each successive interaction provides additional information that can be used to make predictions about future behavior (Kramer, 1999). Specific characteristics that influence history-based trust are the human team member's experience with the autonomy, and the competence, responsibility, predictability, and dependability of the autonomy, as well as the person's perception of these factors and their associated expectations. This is also discussed in the interpersonal literature as cognitive-based trust (Burke et al., 2007; McAllister, 1995). By looking at both domains, we can clearly identify additional factors that are important based on interaction. For example, successful collaborative partnerships require communication, cooperation, and coordination between the two acting members as they work toward a common goal (Groom & Nass, 2007). Therefore, this type of trust can be discussed as (1) understanding how the autonomy works (e.g., ease of learning the system, prior experience, role interdependence, team composition), (2) ability to use or interact with the autonomy (e.g., self-efficacy, workload, expertise), and (3) expectations (Schaefer et al., 2016). This type of trust is critical to teaming because prior to interacting with autonomy, human team members develop mental models or expectations of how it should behave, but following interaction with autonomy, additional antecedents of trust including situation awareness, reliability, errors, feedback, and cueing can impact the operator's perception of how to use or interact with the autonomy (Schaefer, 2013).

Situational trust

Hoff and Bashir (2015) identify trust that is dependent on the situation including the environmental and the internal context-dependent characteristics of the human team member. Features of the environment that influence trust include the type of autonomy, complexity of both the task and environment, difficulty of the task, workload and task demands of the team members, environmental risk, and uncertainty. Factors associated with the internal variation of the human team member including self-confidence, expertise, mood, affect, and attentional capacity (which according to the authors includes stress, fatigue, and boredom). As depicted in Fig. 1, the concept of situational trust can be seen as a combination of both affect-based trust and a subset of features from history-based trust. Focusing on this type of trust allows a researcher to understand the trust that is developed based on a specific interaction.

Learned trust

Hoff and Bashir (2015) also suggest that along with situational trust, state-based trust incorporates learned trust, whereby learned trust represents the trustor's "evaluations of a system drawn from past experience or the current interaction" (pp. 420) and may be impacted by training. Factors that directly affect learned trust include preexisting knowledge of the system and system performance, while design features indirectly affect learned trust through altering the trustor's perception of the performance. More so, by definition learned trust incorporates elements of perceived trustworthiness and history-based trust.

Overall, this conceptual cornucopia of definitions and trust-related constructs has drawn support from the same larger body of research, with myriad identified factors that may or may not influence how trust manifests within any particular human-autonomy (or human-human) interaction. Many of these factors, we argue, remain poorly understood. This is particularly true when we examine the empirical details that have led to their identification and definition, as well as when we consider the details of their moment-by-moment influences over a particular trustor-trustee relationship. To address this, the field must take a closer look at how trust and its concomitant factors are studied and, more to the point, it must be recognized that a significant portion of the disagreements and misunderstandings derive from the general lack of methodological standards that would enable a unified understanding of trust and its dynamics. Indeed, there is a colloquial way that trust is defined that tends to result in a roomful of people nodding their heads in agreement, that is, *trust is the willingness to be vulnerable*. However, taken in the context above, we argue that there are many nuances to trust that are not captured in such colloquialisms. In light of this, we contend that the field should develop a more

widely accepted framework for understanding trust as a process that both transcends and defies a singular definition.

Step 2: Understanding the larger trust process

To develop appropriate trust metrics to support complex human-autonomy teams, it is not enough to just define the type of trust one is interested in assessing. Rather, continuing efforts have made clear that it is essential to account for the context-dependent, interactive nature of trust, uncertainty, and risk and, perhaps most importantly, to have a means of understanding trust metrics in light of significant individual differences in response tendencies and decision-making processes. Therefore, it is critical to understand *trust as a process* and not simply a quantifiable entity. We argue that trust in any context involves dynamic decision-making set on the stage of risky interactions under varying degrees of informational uncertainty. This is the larger picture of how trust changes—either appropriately or inappropriately—over time and across individuals and circumstances. It is important to understand that the same factors that influence trust listed in the above definitions are all part of an encompassing, ongoing, and iterative trust development process. However, the full implications on trust in autonomy extend well beyond trust development as an end unto itself, and toward mitigation and management of decisions and actions that critically impact trust-based outcomes (Fig. 2; adapted from Hancock, Billings, & Schaefer, 2011). Perhaps one of the most important considerations for developing active management of the trust process is trust calibration. Trust

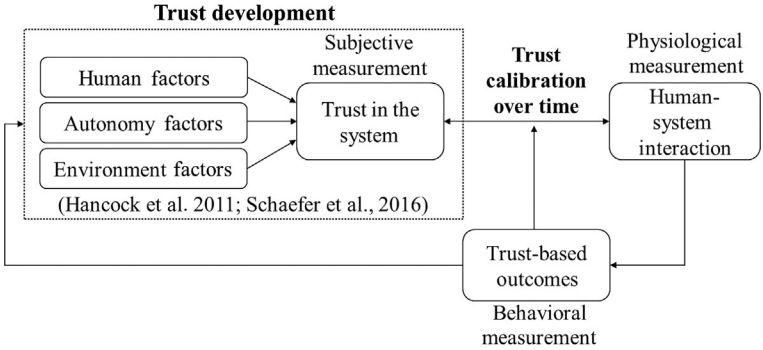


FIG. 2 The trust process model provides insights into the relationship between trust development, interaction, trust-based outcomes, and trust calibration. *Adapted from Hancock, P. A., Billings, D. R., & Schaefer, K. E. (2011). Can you trust your robot?* Ergonomics in Design, 19*(3), 24–29.*

calibration, as part of the larger process, is what helps balance expectations with behaviors so that appropriate interactions can occur. This does not mean that the highest level of trust is always required, but rather that there is an appropriate balance needed between communication, transparency, and shared situation awareness so that appropriate decisions on how to interact are identified and maintained throughout the entirety of an interaction.

Trust development

One of the earliest goals of research on trust in the context of human-autonomy teaming has been to understand how trust develops, particularly as a way to inform those who seek to determine technology-based outcomes that can build trust. While large-scale reviews of the literature have provided some strong theoretical models of trust development, they have also occasionally needed to draw conclusions despite the lack of support through objective data and results. To address these challenges, Hancock, Billings, Schaefer, Chen, et al. (2011) used a metaanalytic approach to assess the breadth of empirical results while controlling for considerable variations in experimental methodology. This work resulted in the Three Factor Model of Human-Robot Trust and later was expanded to describe Human-Automation Trust (Schaefer et al., 2016; Fig. 3).

The metaanalysis identified 37 antecedents of trust, which are factors that should be measured or accounted for during trust-related studies: 15 specific to the human team member(s), 11 specific to the autonomy, and 11 specific to the environment—including collaboration and context (Schaefer et al., 2016). These are many of the same factors linked to the different types of trust that were outlined in the previous section. However, this line of research was limited in three ways. First, due to the limited availability of data regarding alternatives to subjective assessment at the time, this metaanalysis could not generalize beyond subjective trust perceptions. Moreover, while the metaanalytic procedure accounted for some variability in definitions, scale differences, and protocol design, it also left open a number of areas that needed additional study. These areas suggest the need for increased research to understand (1) the impacts of age across more contexts; (2) the relationship between individual differences and trust; (3) cognitive and emotive factors with autonomy-enabled systems; (4) the effects of feedback and cueing; (5) reliability, errors, and behaviors to provide a more fine-grained analysis of these antecedents; and (6) autonomy-related features (as this is a critical area of research when considering both embodied and computer intelligent agents as team members). The second limitation was that only the effect of a single antecedent (e.g., age) or a predefined set of variables (e.g., human traits) could be evaluated at one time. Although the metaanalytic approach identified variables that represent important influential factors on trust, it did not

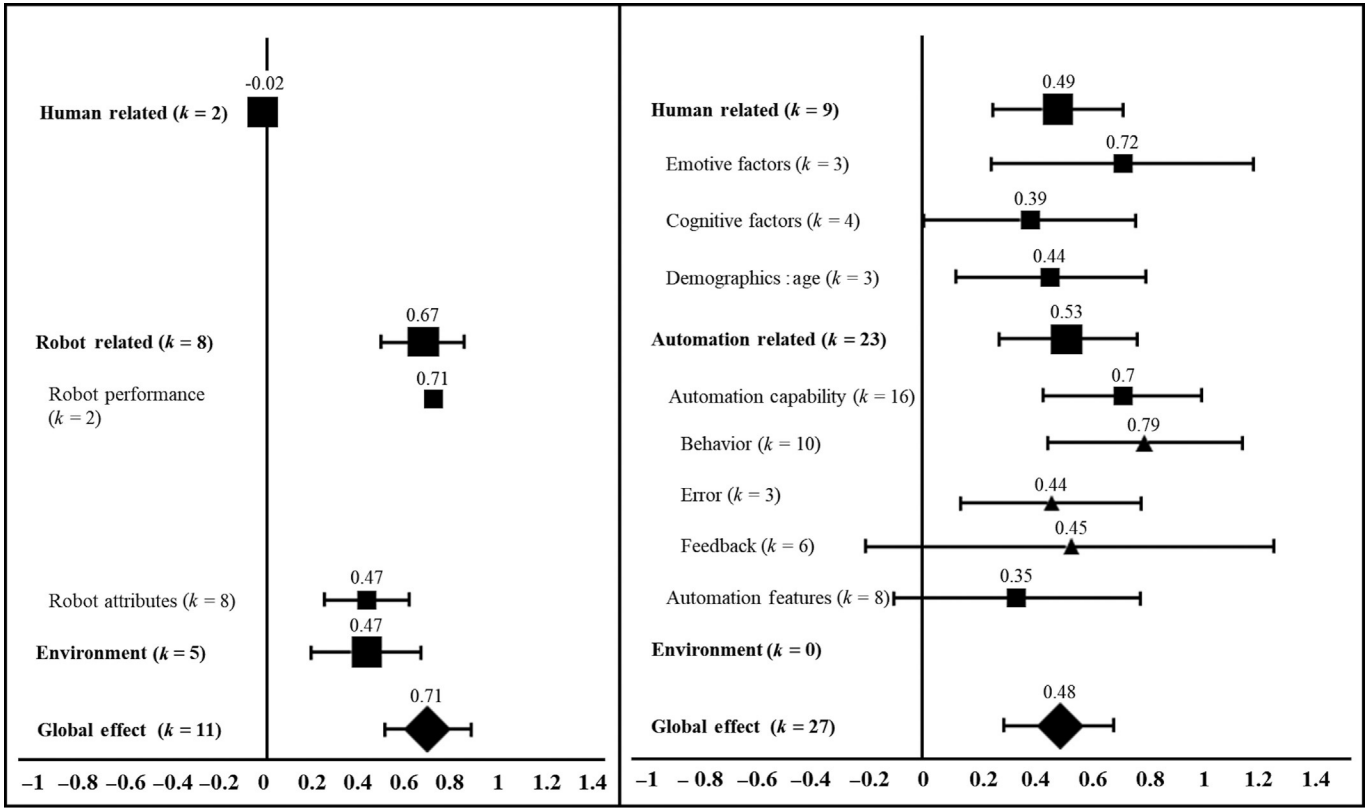


FIG. 3 This figure represents forest plots from the metaanalytic results published in Schaefer et al. (2016). It highlights the effects of different factors that are important for human-robot trust (left image; see also Hancock, Billings, Schaefer, Chen, et al., 2011) and the larger body of human-automation trust (right image) to develop. Numbers represent effect sizes, *bars* represent confidence intervals, and *k* represents the number of samples included in each effect.

ultimately provide an understanding of how to optimize the set of coefficients that describe how to combine variables to best predict and mitigate detrimental (or beneficial) effects on trust. As such, metrics are needed that quantify and potentially explain the complex interplay among multiple sources of individual differences. The third limitation comes from our recognition that understanding trust development alone does not provide a solid basis to facilitate development of methods for dynamically maintaining optimal human-autonomy teaming. For application purposes, we need to translate this knowledge into a mechanism to enable active management of human-autonomy team trust, otherwise known as trust calibration, to achieve an optimal balance of control throughout their interaction.

Trust versus trust-based outcomes

A useful and important distinction that is not often made explicit, though is often present in research related to trust regardless, is the difference between the level of trust held by a person and the behaviors (outcomes) they subsequently enact in the context of that trust. Within human-autonomy teaming, a good example may be seen in how an individual makes decisions about the allocation of control authority between themselves and an automation. Decisions to arrest control from an automation may be interpreted as a loss of trust and decisions to yield control to an automation may be taken as increased trust. Yet, decisions made with respect to an automation do not happen in a vacuum and other factors influence the same behavioral outcomes that are affected by trust (Drnec et al., 2016; Metcalfe et al., 2017), thus leading to occasional confusion about the precise linkage between trust and behavioral outcomes, whether trust-based or not. For instance, dynamic levels of task load tend to interact with individual abilities to produce a sense of subjective workload and can produce similar effects as a change in trust (Drnec et al., 2016; Drnec & Metcalfe, 2015).

Indeed, behavioral outcomes are not always direct reflections of trust. In another particularly dramatic example, people were reported as relying on an "evacuation robot" in a simulated emergency situation (the fact that it was a simulation was unknown to the subjects) despite their clear observations of the robot malfunctioning shortly before the onset of the emergency; in other words, trust was not required for the subjects to behave as if they actually trusted the robot (Robinette, Li, Allen, Howard, & Wagner, 2016). Given many observations such as these, we contend that it is essential for researchers to take a broad view on the situated and contextualized role of trust in human-autonomy interaction, lest inferences from trust research be misleading. When studying trust, it is important to operationalize paradigms and use metrics that specifically test and measure trust and, likewise, when using behavioral

metrics as a proxy for trust, it is essential to ensure that those metrics are not instead indexing associated factors like workload, sensation-seeking, and fatigue. Most importantly, investigators should be clear when reporting their results; if behavior was measured, then the discussion and conclusions should reflect that and not, instead, focus on what is being interpreted as "trust" (or, minimally, authors should be explicit in noting that they are making the logical leap from behavior to trust or vice-versa as applicable).

Trust calibration

One implication of the discussion to this point is that researchers appear to believe that there is an optimal level of trust that will lead to optimal performance or, more simply, that the existence of trust means that the human will choose to appropriately use the trusted system. This belief likely derives from one of the earliest models of trust in automation (Muir, 1987), which provided great insight into modeling the factors that influence trust but neglected the piece where trust must eventually translate into or be manifest as behavior. Within the trust process, then, this indicates a clear need to consider how to calibrate trust in order to achieve desired behavioral outcomes. Lee and See (2004) offered a model that suggests calibration should involve establishing a direct relationship between trust and a trust-based outcome, like reliance or compliance. Lee and Moray (1994) as well as Moray, Inagaki, and Itoh (2000) developed an alternative suggestion that this calibration should be done in a less direct way that accounts for individual differences among humans, that is, they introduced self-confidence as a factor that strongly mediates the individual operators' behavioral decisions in the context of trust (e.g., when trust exceeded self-confidence, the person would be more likely to rely on the automation). An important challenge with these and other trust calibration methods is determining *what information is essential* for aligning performance with expectations based on trust. Even relatively modern attempts to articulate models that describe calibration of the relationship between trust and behavioral outcomes (Gao et al., 2006; Hoff & Bashir, 2015) have, either implicitly or explicitly, assumed that behavior follows either directly from trust or from a simple transformation on trust, such as the comparison with self-confidence as discussed above. These models could be strengthened by including the types of interactions (e.g., with workload, situation awareness, confidence) that condition the behaviors for a given level of trust.

Several alternatives have been offered to address some of the challenges involved in optimizing human-autonomy team performance through trust. New approaches are particularly necessitated by marked individual variability in how subjective trust maps to decisions and behavior. One approach that has shown significant potential has been

to provide greater transparency into the decision criteria being used by the autonomy (Boyce, Chen, Selkowitz, & Lakhmani, 2015; Lyons, 2013; Lyons et al., 2017). Specifically, it appears that even taking a first step in this direction by providing uncertainty estimates to indicate the confidence level of the autonomy can lead to improved trust and more appropriate decisions about when to engage or disengage with manual control (Helldin et al., 2013; McGuril & Sarter, 2006). Establishing appropriate communication between team members may provide yet another avenue for calibrating trust along with trust-based outcomes. A long line of research suggests that when a human team member is unaware of the reasoning behind an intelligent agent's suggestions, complacent behavior may occur, resulting in misuse of the agent (Wright, Chen, Barnes, & Hancock, 2017). Increasing the transparency of the intelligent agent has been shown to be effective at keeping the human team member in the control loop, thus improving task performance, situation awareness, and behavioral decisions, while not increasing perceived workload (Kilgore & Voshell, 2014; Mercado et al., 2015; Wright et al., 2017).

Examples already exist wherein such concepts have been integrated into technology-based solutions to support effective human-autonomy trust calibration through intelligent user interfaces (Schaefer, Straub, Chen, Putney, & Evans III, 2017). One example includes accepting diagnoses and action recommendations from the autonomy to facilitate more informed decision-making at the correct times (McGuirl & Sarter, 2006). These solutions must account for specific design needs around multitasking, transparency, visualization and training, and individual differences as a means of accounting for complex interactions among the human, the autonomy, and the tasking environment, to manifest robust interactions for human-autonomy team performance (Chen & Barnes, 2014). Adopting these and other approaches may help resolve the above-discussed discrepancies in translating trust to trust-based outcomes. By recognizing that trust and behavioral outcomes are both essential for optimizing human-autonomy teaming, research may be refocused on developing a more nuanced understanding of the conditions that demand trust calibration, which might require a more narrow focus on the behavioral outcomes, and which might require fine-tuning applied to each. Certainly, within the process of developing effective team trust interventions, whether to calibrate based on expected system performance or to instead refine our understanding of whether and how trust should be calibrated to particular behavioral outcomes are just two points along a continuum of options worthy of significantly more investigation.

Step 3: Developing appropriate team trust measurement

There are multiple types of measurement tools that have been and are currently being explored to develop effective team trust metrics necessary for human-autonomy teams. To date, the primary measurement tools are specific to a single trustor-trustee dyad and are most often founded in human-interpersonal relationships. In order to address the future of human-autonomy teams, where a team can, and most likely will, be composed of multiple human team members, computer agents, and intelligent embodied agents (i.e., robots or unmanned systems), current measurement tools will need to be updated to account for a more network-centric team structure, as well as operations that occur in dynamic, uncertain environments. This section provides a review of current measures and techniques for evaluating trust and how they are applicable to human-autonomy teaming.

Self-report measures

Self-report measures are the most commonly used method of assessing trust across domains (e.g., interpersonal, technological, etc.); however, there is limited consistency in different measurement tools. Items range from a single question ("how much do you trust {X}"?), to scales for assessing a specific type of trust, or scales that were developed for human teams. While there are similarities and lessons to be learned from each, there are indeed differences. First, current human-robot trust researchers suggest that due to the human component in the human-robot team, a propensity to trust scale should be used in conjunction with a human-autonomy trust measure (see Yagoda, 2011). This is because an individual's propensity to trust machines can directly impact the use of a specific machine and, importantly, across different contexts (Merritt & Ilgen, 2008). However, limited work has included propensity to trust in human-technology domains, even though propensity to trust can be identified as the inherent trait-like characteristic of the person (i.e., individual differences) to trust. The *Interpersonal Trust Scale* (ITS; Rotter, 1967) is the most common measurement scale used to assess an individual's propensity to trust, though a newer scale has been proposed as well (Evans & Revelle, 2008) and is rapidly growing in adoption.

For state-based trust in human-autonomy teams, the most well-cited trust scale is the *Checklist for Trust in Automation* (Jian, Bisantz, Drury, & Llinas, 1998). While this scale is established as the main metric of trust for automated or preprogrammed automation systems and validated for use with an automated signaling system (Spain, Bustamante, & Bliss, 2008), it has not been validated for autonomy-enabled systems that are required to participate in independent decision-making and

interdependent teaming. To address some of these limitations, different approaches have been offered. For example, Chen and Barnes (2012) modified the *Checklist* to create what they called the *Usability and Trust Survey*. This new scale revised the *Checklist* to classify impressions of a specific robot, and further, added several more usability questions to make a more robust scale. An alternative approach was to develop and validate a new scale to meet the specific needs of human-autonomy teams. Schaefer developed the *Trust Perception Scale-HRI* (TPS-HRI; Schaefer, 2013, 2016) that is specific to intelligent, embodied autonomy (e.g., robots) and validated across multiple contexts. Similar to a comparative analysis of both the *Checklist* and the *TPS-HRI* as part of the validation process for the *TPS-HRI* scale (see Schaefer, 2013), an independent analysis by Kessler, Larios, Yerdon, Walker, and Hancock (2017) found that, in comparison with the *Checklist*, the *TPS-HRI scale* was more applicable and provided increased understanding of the trust relationship when collaborating with a single human-intelligent robot dyad. Besides the above-listed examples, other researchers have focused on a specific type of trust, such as cognitive or affect-based trust, and used a metric based on a definitional foundation even if the scale was actually developed for interpersonal trust.

While research has pushed for the development of more appropriate team trust scales for human-autonomy teaming, it is also important to note that self-report measures are biased by perception and are limited in scope (see also Gutzwiller et al., 2019). Therefore, there has been a push by the scientific community to identify more objective metrics of trust. These include both behavioral and biological ("psychophysiological," see below) indicators of trust. Both of these research areas are still being developed but have had some promising early findings to advance human-autonomy team trust metrics.

Behavioral measures

Objective behavioral measures directly linked to trust are still being developed and validated. However, current research efforts are aligning trust with changes in attention or behavior. A recent suggestion for behavioral measurement is to assess eye-tracking behaviors, as it has been previously shown to be an effective means to understand changes in attention and attentional control. Attentional control is defined as "one's ability to focus and shift attention in a flexible manner" (Chen & Barnes, 2012; Derryberry & Reed, 2002). According to Chen and Barnes, this is related to trust because people with lower attentional control tend to rely more heavily on automation. More specifically, what a person is looking at is thought to reflect the cognitive symbol currently being processed (Just & Carpenter, 1976). Therefore, it is possible to identify stimuli that are important to the subject. Presumably, for someone to have well-calibrated trust in a system, that person will require a certain level of

information that allows trust to develop or degrade appropriately. Eye-tracking data, then, are useful to trust research, because eye movement metrics provide indications of what information the person might find important for trust calibration. For example, repeated gaze fixation on an area of interest, such as one display compared to another, would allow an inference that the person finds the display useful or at least interesting (Just & Carpenter, 1976; Poole & Ball, 2006), whereas prolonged fixation durations might indicate that the person finds the information difficult to understand (Poole & Ball, 2006). Further, the pattern and targets of gaze fixation may indicate the direction of an impending decision (Gildöf, Wallin, Dewhurst, & Holmqvist, 2013; Glaholt & Reingold, 2011; Poole & Ball, 2006), and an estimate of the level of cognitive effort being expended while processing information (Marshall, 2002, 2007).

Along with metrics associated with attention, changes in the human team member's behavior can be indicative of changes in trust. In a review of this research, Spain and Bliss (2008) identified some critical metrics that may be important for human-autonomy teams. These are specific to the number of changes in control allocation or changes in level of automation, and the amount of time interacting with or manually controlling a system. While these types of metrics provide insights into changes in performance, they are also limited by lack of specificity, meaning that the reasoning process underlying the particular pattern of behavior (e.g., stress or workload of the human team, or complexity and risk associated with the mission) is not uniquely indicated by the metric. Therefore, what actually caused a change in trust may not easily present itself through a behavioral measure alone. Schaefer et al. (2019) suggest that multiple types of measures (i.e., performance, subjective, communication analysis, and psychophysiological changes) could provide a more complete picture of what can change trust or influence a trust-based decision.

Psychophysiological measures

Trust is a psychological state, but psychological states arise from chemical and electrical activities within the body as the person interacts with the world. It is these activities that can be described with physiological variables, but it must be understood that these same variables change in principled ways as the associated psychological states shift. As such, it can be expected that the merger of these two into what is known as psychophysiological measurement should be able to produce reliable metrics that fluctuate along with changes in trust states. Two commonly used psychophysiological measures currently linked with trust are electrodermal activity (EDA) and heart rate (HR). The reason that they have been associated with trust is their long history of being associated with cognitive effort and decision-making.

EDA, both tonic and phasic, is a measure of skin conductance and is a sensitive measure of emotional arousal such that levels increase during periods of anxiety and cognitive effort (Shi, Ruiz, Taib, Choi, & Chen, 2007; Zhou et al., 2015). EDA measured in the region around the hand has been linked to stress and engagement in humans (Mower, Feil-Seifer, Mataric, & Narayanan, 2007). Specific phasic EDA, a skin conductance response to a stimulus with a latency of between 1 and 3 s, is an indication of the magnitude of response the subject had to the stimulus. For example, if the driverless vehicle made an error, tonic EDA should increase, but the magnitude of the phasic response may indicate the level of anxiety or frustration associated with the error. Research by Bethel, Salomon, Murphy, and Burke (2007) was one of the first examples of using EDA during human-robot interaction; they found that tonic EDA measures increased with increased engagement with the robot. In another trust in technology study with an active and passive user, Montague, Xu, and Chiou (2014) found that the higher the users' individual ratings of trust in technology were, the more their individual EDA measures agreed. For example, if EDA levels of one subject were low, and both subjects were in a trust state, the second subject's EDA would be expected to be low as well. Further, in a pilot study, Sanders et al. (2012) found higher EDA (more specifically, skin conductance) and lower subjective trust ratings during interaction with an unreliable robot than a reliable robot.

HR and heart rate variability (HRV) are often used in conjunction with EDA measures to infer the effect of a stimulus, cognitively and affectively, on a subject. Following a stimulus, an acute decrease in HRV, along with a simultaneous phasic response, has been associated with orienting behavior (Figner & Murphy, 2011), which may allow for an inference that an event was salient to the subject. HRV may also be used in conjunction with EDA to infer levels of workload and trust (Matthews, McDonald, & Trejo, 2005; Mehler, Reimer, Coughlin, & Dusek, 2009; Montague et al., 2014). For example, in a state of high trust, it is unlikely that one would feel anxious, and therefore HRV would be high and HR would be low, and tonic EDA levels would be also be low. However, if there is an increase in cognitive workload and anxiety associated with the process of maintaining situation awareness, such as might occur in a state of low trust, it is likely HRV would fall and nonspecific phasic EDA levels would rise along with tonic EDA levels.

Overall, while research into psychophysiological indicators of trust remains quite active, new methods are continually being developed and a variety of measures that were not discussed here remain available for exploration. In particular, there may be some promise for observations based on electrocortical patterns in the brain, either as measured by functional near-infrared spectroscopy (fNIRS) or electroencephalography

(EEG),[b] given that the development of trust involves cognitive processes such as decision-making and judgment (see Drnec et al., 2016 for a proposal of how to use psychophysiology to monitor decision-making processes involved in trust in automation). That said, though the progress in cognitive neuroscience and psychophysiology research has been impressive, there remains a lot of work to be done before such measures can approach usefulness for studying human-autonomy teams in real time. Nevertheless, research is demonstrating that the inclusion of wearable technologies into human-autonomy teaming operations has considerable potential to help provide additional understanding of the relationship between operator states, such as trust, and performance and how this relationship varies across different team types and operational contexts (Faller, Cummings, Saproo, & Sajda, 2019; Schaefer et al., 2019; Marathe, Brewer, Kellihan, & Schaefer, 2020; Metcalfe et al., 2017).

Building onto the roadmap: Pushing toward the discovery of principles that may be leveraged to govern trust-based interventions

Thus far in this discussion, we have painted a picture of trust as a complex, multifaceted construct that manifests in a variety of ways during human-human and human-autonomy interaction dependent on task and contextual factors, as well as on the states and traits of the individuals involved. It can be argued that this complexity and context-sensitivity may preclude ever developing a clear and unifying understanding of trust as a singular and encompassing construct. At the same time, though trust has proven difficult to define, it continues to be considered one of the most critical requirements for safe and stable human-autonomy teaming. Moreover, the amount of attention that has been devoted to trust, along with the diverse perspectives that are brought to bear on trust research, is encouraging for continued efforts to develop an integrative understanding of how to operationalize trust in ways that improve future forecasts for human-autonomy teams. As such, we devote the remainder of this chapter to discussing a roadmap for addressing the essential research gaps in the extant knowledge base regarding the human-autonomy teaming dyadic relationship; herein, we adopt a particular focus on those gaps that affect efforts to build metrics that can enable predictions about

[b] Note that these are just two of several modalities for recording brain activity; however, the remaining modalities (such as fMRI, PET, MEG) are not applicable in real-world human-autonomy interactions because the equipment required for measurement is prohibitively expensive and is so large as to be immovable.

manifestations of trust in larger and more complex teams. We begin with a discussion and example of how some essential elements of the above discussion may be incorporated into research designs for examining human-autonomy interactions within contexts that begin to approach the complexity of the real world; we accomplish this through examples provided within our own research efforts. Finally, we conclude the chapter with a set of summary observations that specify a vision of how to create a path toward a longer term understanding of how trust develops, evolves, and ultimately is a critical determinant of interactions among complex groups that are working under risk.

Incorporating trust into an experimental paradigm

A key message from this chapter is that trust should be construed and studied as a process rather than as an entity. A more subtle, yet essential notion that underlies this position is our belief that understanding trust within complex human-autonomy teams will ultimately demand an integrative approach focused on discovering principled methods to *actively govern* trust in human-autonomy teams. By integrative, we mean that trust inference and management strategies should incorporate a principled means of merging subjective reports with objective measures, observed human tendencies with autonomy behaviors as determined through control laws, and top-down, history-informed concepts with bottom-up, data-driven discoveries. Perhaps most importantly, approaches are needed to address the fact that not all team members are equal. What we mean by this is trust does not affect all people the same way, and as such, it is not possible to create singular trust-based interventions that will ubiquitously improve joint team decision-making. Yet, addressing the challenge of intra- and interteam variability through individualized, trust-based interventions is exceedingly difficult and costly. Therefore, new quantitative approaches are needed to (1) understand the complex interplay among multiple sources of individual differences and (2) better leverage multimodal methods for predicting trust and trust-based outcomes at both individual and group levels.

Herein, we discuss a research paradigm that was initially developed to investigate human interaction with driving automation to develop metrics of change in the human's trust (Metcalfe et al., 2017). As devised, this paradigm was capable of capturing data from multiple points within the given human-autonomy interaction cycle (Fig. 1); specifically, data were captured from the participants' initial biases and trust propensity through training trials, and detailed performance and physiological data were recorded through the iterative process of exploring performance outcomes in a realistic driving task, and finally, retrospective assessments of trust, trustworthiness, and subjective workload were captured following

each interaction. Critical design points focused on the influence of the quality of automation performance in combination with, or mediated by, the dynamic influence of subjective workload (Drnec & Metcalfe, 2015). Within this paradigm, drivers were tasked with navigating a roadway with and without the use of automated driving assistance, while maintaining lane position and following a lead vehicle within a fixed headway range, all while avoiding collisions with pedestrians and oncoming traffic. The task was conducted in a fully immersive, motion-based driving simulator. Participants were volunteers that were paid, in part, based on their success in navigating without collisions or deviation from the established lane and following distance boundaries.[c]

Advancing principles to account for individual differences within team performance

The effects that autonomy performance quality has on trust and trust-based outcomes are highly variable across individuals, situations, and technologies. Therefore, understanding individual differences will produce better human-autonomy team outcomes than methods that treat trust as a monolithic construct represented by normative group statistics. While traditional methods for studying performance variability are useful, understanding the interactions among the various factors that influence individual differences may provide a more complete picture of how to manage human-autonomy interactions through individualized calibration. We further recognize that the human sciences have not yet developed sufficiently robust models to capture the influence of endogenous changes that contribute to significant intraindividual variability, in particular, on state estimates and performance predictions (cf., Touryan et al., 2014).

In response to this challenge, we explored approaches to account for individual differences without necessitating reliance on large amounts of data about each individual. Herein, we developed an algorithmic approach to cluster individuals into subgroups of similar behavioral and cognitive characteristics. Based on cultural mixture modeling (see Tan & Mueller, 2016 for an application to cognitive phenomena), our

[c]For the purpose of this chapter, it is important to note that during each experimental condition, simultaneous recordings of EDA, HR, EEG, eye movement, and pupil diameter were collected. Questionnaires included trust (Jian et al., 1998; Muir & Moray, 1996), stress, personality (Big Five Personality Inventory; John & Srivastava, 1999), Uncertainty Response Scale (Greco & Roger, 2001), and demographics including age. See Drnec and Metcalfe (2015) for a more complete discussion of the paradigm and the conceptual motivations behind the various task parameters and design decisions as well as for a summary of the main performance outcomes.

use of this approach was intended to parse the factors that influence individual differences among subgroups of participants in these high dimensional data sets. Moreover, this method allowed us to identify how variability in traits and initial individual perceptions of trust, stress, and workload could be quantitatively aggregated into statistically defined groups and, in turn, facilitate understanding group-based response patterns linking trust and behavior. This constitutes a novel approach to the science of human-autonomy teams and departs from traditional paradigms that tend to employ experimenter-defined groups to characterize generalized human response tendencies at a population level.

The data-driven clustering approach pointed toward the existence of up to four subgroups within the sample (see Fig. 4), each defined by a unique combination of demographic characteristics and traits. The clusters then provided a mechanism to separately explore variable trust responses within group-wise relationships, importantly providing an effective compromise between whole-population and individualized statistical models. While full findings are reported in Neubauer et al. (2020), the following section summarizes the observed patterns of the results for subjective trust ratings and associated affect-based trust via facial expressivity analysis (discussed, but not presented here). These response patterns, shown in Fig. 4, revealed commonalities within groups, and the responses were associated with different manifestations of trust,

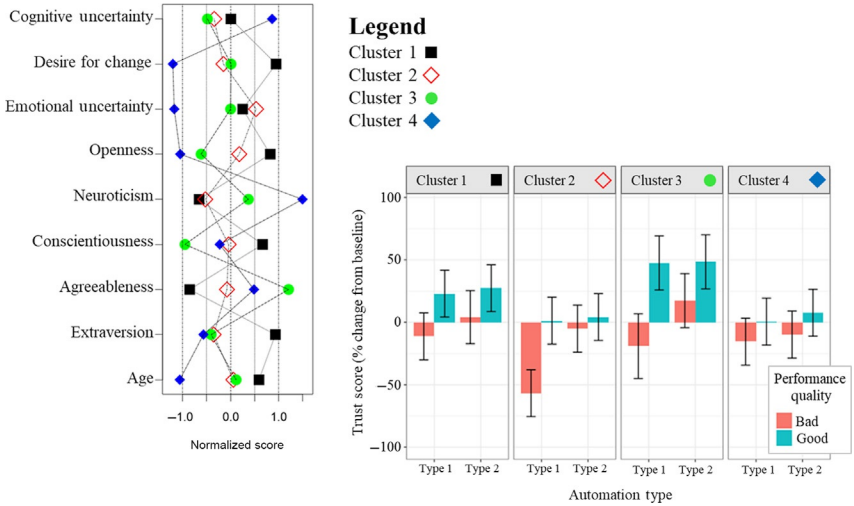


FIG. 4 Clusters (left panel) identified using stepwise flexible mixture modeling resulting in four distinct groups linked to unique patterns of subjective trust ratings (right panel) and facial expressivity (not shown here). For more detailed results and discussion, see Neubauer et al. (2020).

suggesting a potential way of "tuning" trust-based mitigations to individual tendencies.

Cluster 1 (black squares) can be best described as an overtrusting subgroup. Specific individual features associated with this subgroup included age (representative of the oldest participants), a high desire for change, and personality consisted of high openness, extraversion, and conscientiousness along with low neuroticism scores. When assessing the impact of this subgroup's tendencies on subjective ratings of trust, we found that they reported higher trust in automation with good performance but showed a minimally negative response to the automations with bad performance. Those who were in this subgroup also showed higher average expression values in the facial analysis than did all of the other clusters, and this increased expressivity was associated with significantly higher values for expressions for happiness and contempt. Given the tendency to rely on as well as the willingness to accept and trust the automation, this result suggests that trust-based interventions for people similar to those in this subgroup should focus on reduction methods for overtrust, especially when a partnered autonomy would be expected to make errors at higher rates.

Cluster 2 (red diamonds) can best be described as an undertrust subgroup. In terms of personality traits associated with this subgroup, it appeared as if these individuals were relatively average on nearly all dimensions, with an exception for high ratings of emotional uncertainty and low ratings for neuroticism. Individuals in this subgroup only really showed a degradation in reported trust (through subjective ratings that were similarly reflected in expressive reactions to task errors, including anger, contempt, fear, and surprise) when interacting with one of the driving automations that was malfunctioning (in this case, an automation that could both maintain lane position and speed), all other scores were fairly well calibrated to initial expectations from initial training trials. Together, these findings indicate the potential for maladaptive anxiety responses to uncertainty as characteristic of this subgroup. These observations suggest that trust-based interventions for people with similar tendencies should be best focused on improving transparency and communication to mitigate negative reactions to unexpected behaviors and therefore direct appropriate control allocation strategies to balance teaming.

Cluster 3 (green circles) was most aligned with traditional expectations from trust literature whereby they showed appropriate responses to varying qualities of autonomy; they trusted autonomy when it performed well and did not trust autonomy when it did not. This was supported by results of the evaluation of facial expressivity, where these participants showed a generally lower mean expression value as compared with people in the other subgroups, with significant differences in anger, fear, and surprise. While individual trait differences were average across most dimensions,

the low cognitive uncertainty and conscientiousness scores, along with high agreeableness ratings, can be linked to a preference for predictable, planned behavior while also demonstrating some willingness to give autonomy a chance. Taken together, these results indicate that their trust was well calibrated to both autonomy quality and context. However, what is interesting for this subgroup is the observation that, while this pattern was supported by observation of a specific set of individual differences, it was only shown for 16% of the participants, further supporting the need for these types of quantitative approaches. That is, from these results, we might conclude that the subgroup of people who respond in expected ways based on conventional understanding of human-autonomy teaming actually compose a very small segment of the broader population than might otherwise be expected.

Finally, Cluster 4 (blue solid diamonds) showed no real change in trust scores from the initial expectations they formed during training trials. This suggests that the initial expectations of the individuals in this subgroup were fairly well calibrated. Though mild, if not negligible, degradations in trust were observed (noting, of course, that confidence intervals overlapped the ordinate axis), those who were in this subgroup demonstrated the greatest changes in facial expressivity when interacting with an automation with bad performance characteristics. The individual differences for this cluster were extremely polarized. This subgroup included the youngest participants in the sample, who had the lowest desire for change and emotional uncertainty scores (indicating that they do not seek novelty), but who also reported the highest cognitive uncertainty and neuroticism scores (indicating a preference for predictability and structure in uncertain conditions). These results show somewhat of a disparity between changes in subjective trust and the extent to which participants express these changes facially, further supporting the need for a multimodal, multimetric approach for trust assessment.

The results of the analyses presented above indicate that trust is far from a monolithic construct, just as any given population of individuals is diverse. Different individuals show variable modulations in trust while interacting with varying types of autonomy that have different performance qualities and, concomitantly, quantitative evaluation of trust and trust-based outcomes thus requires a multimodal, multimetric approach to allow a more complete understanding of the variable dynamics of change. In addition, the results from this clustering approach highlight challenges faced when attempting to develop management strategies for trust-based interactions, particularly if basing those strategies on common trust theory alone. The bulk of the traditional literature base appears to reflect the continual pursuit of a research agenda that aims to link group dynamics observed at a population level with responses to trust scales as implemented in standard good/bad autonomy paradigms. In the present

analysis, only a small subset of participants (16%) actually responded to the driving automations in ways that would have been predicted from conventional understandings. As throughout our discussion, we thus emphasize, again, the need for more individualized intervention strategies to enable effective trust calibration. Yet, we also suggest that improvements may be achieved even in the absence of robust individualized models for human-autonomy teaming. In our research, the clustering approach reduced the number of possibilities from 19 individuals to 4 subgroups, with clear reasoning for prioritizing specific types of interventions based on subgroup differences. While this line of research and the associated attempts to develop novel methods of measurement and management of trust-based outcomes continues to mature, immediate observations from it make a clear point that trust responses are nonuniform, and that integrative approaches will be necessary to improve characterizations of those responses across the spectrum of human variability.

New models for linking individual (or group) states to trust-based outcomes

Beyond the challenges of inter- and intra-individual variability in the inference of global trust states as above, we have also identified a need to develop better understandings of how to associate such trust state inferences with expected behavioral outcomes with a greater temporal resolution. To reach this end, we describe a promising approach for developing a dynamic model that enables such a time-resolved and individualized prediction of specific human-autonomy interactions. This is a novel approach for addressing the relationship between trust and trust-based outcomes, and as such, some assumptions must be made. First, when looking at the trust process as a whole, we understand that before an individual takes any specific action, there are likely to be implicit or explicit decisions made within the context of their dynamically evolving individual trust state (Drnec et al., 2016). We note that these trust-based decisions may be the linkage between a trust state and trust-based outcome and, further, may be expected to govern the actual interactions with a counterpart or teammate in a joint task with shared responsibility and disparate capability. Importantly, then, these trust-based decisions may critically underlie the ability of the team to complete a common goal in a reliable and robust way. Second, calibration of the trust states within and among teammates occurs when the perceived trust is matched to the actual proficiency of each teammate, and the process of calibrating these trust perceptions may ultimately be the necessary level for intervention that would enable optimal management of interactions among complex human-autonomy

teams. Either way, having such a process is critical to team success and prevents issues arising from miscalibrated trust, such as overtrust or undertrust that can manifest as instances where the expected performance from a teammate does not match the observed performance. Therefore, the challenge of modeling appropriately calibrated trust among human-autonomy teams has two requirements: (1) estimation of the expectations of team member performance as applies to successive decision-making and (2) identification of moments when those expectations are incorrect and need to be adjusted.

As a first step, strategies for calibrating, managing, or controlling trust, as with control of any feature of a complex, dynamic system, require an estimate of its value. However, to enable accurate prediction of a future value of trust, beyond simple measurement of its current value, it is essential to identify the associated factors that impact appropriate trust calibration within time-varying contexts. This is inherently motivated by the relationship between trust calibration to performance expectations (see also Drnec et al., 2016) but draws on the fact that trust-based decisions are dependent on many external, contextual, and internal features that produce changes in various physiological and behavioral responses. As such, it is necessary to integrate a multimodal set of measures for maximal efficacy real-time implementation of these prediction and calibration methods.

Various techniques may be applied to facilitate mapping of a large set of rich individual, task, and environmental features to outputs in real time and can range from those formed by prior knowledge to primarily data-driven learning-based modeling approaches. While these approaches can theoretically provide accurate predictions of outputs, such as trust-based decisions, the allowable complexity and subsequent accuracy are largely dependent on providing a sufficiently large amount of data in the learning process that can be infeasible to collect from human subjects' experimentation. This can make the application of these approaches in this domain rather difficult, if not outright intractable.

Our approach uses the application of recursive, stochastic filtering method for prediction of dynamic states and actions related to trust. This model integrates sensor fusion techniques with identified models of psychophysiological response patterns to form real-time estimates of mental states and associated confidence bounds on those estimates (Gremillion, Donavanik, Neubauer, Brody, & Schaefer, 2018). This approach specifically advances beyond other models by incorporating data derived from psychophysiological quantities recorded from the human teammate with information about the external environment and interactions that occur in order to more fully model the impact on trust-based decision-making and how those decisions impact changes in control authority shared with autonomy. This approach of "constrained first-principles

filtering" leverages the cumulative and corroborative nature of observable physiological and behavioral evidence obtained through multimodal measurements, informed by robust interaction dynamics based on well-established domain knowledge, to perform optimal, recursive inference of the unobservable internal (cognitive) states that impact trust-based decisions. While this approach is limited by the a priori designed model structure in terms of completeness, and thus possibly in prediction accuracy, it can provide optimal, real-time state estimates that are robust to measurement noise or data corruption and are interpretable by subject matter experts. These characteristics allow us to make informed inferences about how humans are likely to interact with autonomy, what psychological changes are driving those interactions, and, thus, how we might intelligently intervene to make that interaction more optimal, so as to appropriately build, calibrate, manage, and maintain human-autonomy trust.

An illustrative overview of how to execute this process is shown in Fig. 5, for a subset of the psychophysiological quantities of interest. This approach helps to further quantify the trust process by encoding domain knowledge of psychophysiological relationships through parameterized mathematical functions that define the dynamic link between various externally observable measurements (f and h in the equations shown in Fig. 5B), such as physiological measures, to the latent unobservable states, such as psychological constructs like trust or a decision. To more clearly explain this recursive modeling structure, observable quantities of the environment and the human are respectively considered inputs to and outputs from an intermediate process that represents the evolution and propagation of their latent cognitive states, which are not directly observable (Fig. 5A). Here, inputs (u) are the variables of the environment or task that impact the dynamics of the latent state (x) of the human. Such inputs, in the experimental scenario that we studied (briefly described above), include the vehicle positions and velocities, roadway curvature, and overall score in the gamified task. Conversely, the outputs (y) are the measurements of the human that are informative of those latent state dynamics, such as the psychophysiological variables from electrodermal activation (EDA), heart rate (HR), electroencephalography (EEG), and their derivative features. The states are defined as the quantities that compactly characterize these dynamics and are likely to capture influences of latent psychological features and the level of trust in or likelihood to intervene with an autonomy. Given this estimate of the likelihood of trust-based decisions to intervene, as well as concurrent estimates of psychological states, such as stress, fatigue and workload, and continuous metrics of performance and risk, it should be possible to determine whether those decisions to intervene are contextually appropriate and, successively, whether to take action to preemptively manage them if not.

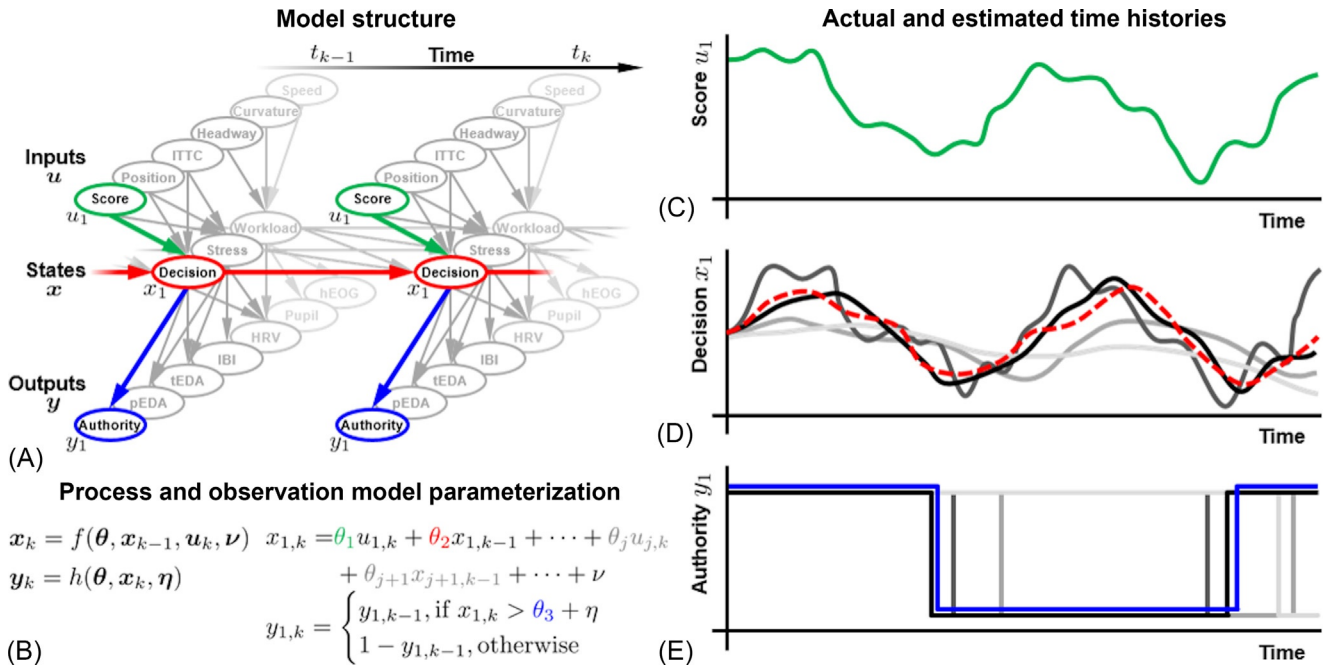


FIG. 5 Diagram of trust-based decision state estimation modeling process with constituent components: (A) multimodal recursive model structure for sensor fusion and inference; (B) parameterized, domain-informed process and observation models; and (C–E) simulation of postulated models and comparison to true, measured outputs.

The efficacy of this approach and the resulting predictions are dependent on appropriately defining the system's characteristic variables (u, x, y), plausibly structuring the related dynamic equations (f, h in Fig. 5B), and tuning their parameters (θ) to accurately reflect observed data. This can be seen through the relationship between the time evolution of a feature of the environment that a human may observe (Fig. 5C), with an estimated latent cognitive state of the human teammate (defined as the continuous, dynamic trust-based decision state shown in Fig. 5D), and the evolution of a behavioral response (e.g., change in control authority shown in Fig. 5E).

Using this approach, we may confidently propose that the trust-based decision state representing the likelihood to use the automation is expected to correlate with and lag behind performance, and that the resultant observed behavior to actively toggle control authority to or from the automation will occur when that continuous decision state reaches a triggering threshold. However, we may not be certain of the strength of that correlation, the degree of lag, and the level of that threshold. Despite having well-established interaction patterns between these states and observations, these free parameters must be fit empirically. Fig. 5D and E depicts the quality of this fit for various postulated parameter values, with darker time traces indicating well-fit values with close alignment between the modeled and observed data, and presumably the unobserved state, while the lighter time traces indicate poorer model alignment and thus less plausible parameter values. Further, these empirically estimated parameters can be used to adapt the model to the inter- and intrahuman variability that does not correspond with qualitatively different behavioral or psychophysiological observations but would otherwise introduce significantly poorer prediction accuracy if a universal (general population level) model were used instead. Moreover, this approach also provides a mechanism for using characteristic traits (perhaps from methods like the clustering approach described above) to predictively set model parameters. For example, a measurement of a new, unmodeled individual's propensity to trust may be used to approximate the value of the triggering decision threshold described above.

The mechanism of estimating and managing trust and trust-based decisions in a human-autonomy dyad that are described here are theoretically scalable to larger teams with multidirectional interactions. Given that this solution is formulated as a control system to optimize interagent perceptions of performance with actual performance and how that relates to task allocation, this approach lends itself to scaling a multivariate optimization for a network structure between any number humans and autonomous systems, thus supporting future complex human-autonomy teaming. Here, we envision a system that estimates the states of human and autonomous agents within the team, informed by the respective interactions between all teammates of order, to monitor and manage as many

interagent trust relationships. The goal of this type of management system would be to transparently communicate instances of miscalibrated trust and recommend or enforce task allocation so as to mitigate these miscalibrations and misallocations, improve performance, and build team trust and cohesion.

Path forward on the roadmap: Metrics for team trust

Clearly, work by the community at large is needed to develop effective metrics for team trust that are applicable to human-autonomy teams. A major difficulty in moving forward in development of team trust metrics is the number of unknowns. These include systems that have yet to be developed or refined that will be operating as part of a larger multisystem team within unknown and uncertain mission parameters. Since the autonomy-enabled systems are a critical piece of the trust puzzle, this opens more questions into how to develop appropriate metrics that can support calibrated trust and effective teaming. It can be argued that additional research is still needed to better understand dyadic relationships before expanding into larger teams. Therefore, through this chapter we have proposed a conceptual research roadmap for identifying the steps needed to progress the field toward development of effective and appropriate team trust metrics and interventions.

- *Definitions*. It is important to understand when reviewing or citing trust research that there are different types of trust, each with different variables that influence trust-based outcomes at different times through the trust process. Therefore, not all metrics will be used in all cases, but identifying these differences allows us to identify what metrics should be used and at what time in the evaluation process. Further, by understanding what could be causing a change in trust-based decision-making and behavior, it is possible to then determine the trust-based interventions, such as options for training or technology-based calibration metrics and similar methods that can support effective teaming. Overall, this type of approach allows us to move beyond theoretical understanding and begin to identify quantifiable metrics that may be critical to expanding beyond the dyadic relationship toward team trust.
- *Variables that influence trust development*. Prior metaanalyses by Hancock, Billings, Schaefer, Chen, et al. (2011) and Schaefer et al. (2016) identified no less than 37 independent variables that influence changes in how trust develops over time. These are the same variables that have been linked to specific types of trust as we have discussed here. While the effect of some variables have been well tested (e.g., reliability and

predictability of the system), others still require further research to harness the true effect on trust development (e.g., age, cognitive and emotive factors, feedback and cueing, errors, system behaviors, and autonomy-related features). From a modeling and experimental perspective, this poses a great deal of challenge in terms of accounting for the correct independent variables and experimental design in order to capture desired effects. Further, current lab-based experimental methods often control for a number of variables or try to quantify the effect of a single variable but should be cautioned when developing team trust metrics for fielded systems. Rarely can an individual variable be a singular determinant of trust, rather what needs to be identified is the set of variables that together explain the changes in trust and trust-based decisions over time and across context.

- *Study design complexity and ecological validity*. While systematic study of the above-listed variables may help to classify the effects of a single variable on trust development and maintenance over time, it may be limited in supporting appropriate trust calibration metrics, especially across a team. Due to the complexity of trust, it is difficult to control for most of the possible covariates that may influence inter- and intra-individual variance during data collection; so gathering information on them during data collection is critical. Further, it could be argued that overcontrolling for variables could do a disservice to advancing trust research. Complex, real-world phenomena do not always lend themselves to the precise controls typical of laboratory environments and, more critically, when placed in such controlled environments may appear phenomenologically dissimilar to how they appear in more ecologically valid settings (Kingstone, Smilek, Ristic, Friesen, & Eastwood, 2003). Current research is limited in understanding the interplay between the multiple antecedents of trust as they directly link to trust dynamics, trust-based decision-making, and trust outcomes. In particular, new methods are needed that quantify and potentially explain the complex interplay among multiple sources of individual differences in trust because, ultimately, creating individualized technology-based interventions for appropriate and ongoing trust calibration will be essential for management of dynamic human-autonomy teams.
- *Trust measurement*. There has been some promising research in the development of objective, behavioral, and psychophysiological indicators of real-time changes in trust. While these approaches remain notional and have significant room to mature, there is promise in being able to use them to better predict and pinpoint when a change in trust could be likely to occur. Advancements in these types of approaches could then be linked to training and technology-based solutions to facilitate appropriate bidirectional communication

strategies to develop appropriate shared situation awareness among team members. Here, we posit that team trust metrics will require a balance between these different measurement techniques and methods, whether based on subjective reports or objective behaviors. We expect that the continual evolution, miniaturization, and ruggedization of wearable sensors and mobile devices will enable data acquisition across the spectrum of human-sourced information and this continued development thus constitutes significant opportunity for research in active monitoring and management of human-autonomy team trust and performance.

- *Trust process*. Trust development alone is not enough to understand the teaming relationship. A mechanism is needed that would enable active management of trust and trust-based behavioral outcomes; such a mechanism is otherwise known as trust calibration and would serve to achieve a balance during and after interaction. Importantly, the very notion of trust calibration implicitly acknowledges the dynamic nature of the relationship between trustor and trustee. Trust is not constant and continually evolves as a function of how well the trustor's expectations are met by the actions or performance of the trustee. Therefore, further research is needed that can identify methods and metrics of trust calibration that support training and technology-based solutions that can be generalizable across human-autonomy team contexts. Moreover, as advanced modeling techniques for intelligent control systems evolve, it can be anticipated that such models may be better leveraged into calibration methods that support the active management of dynamics interactions within and across complex human-autonomy teams.
- *Individual differences*. One of the major arguments by the trust community is whether and how individual differences that manifest during human-autonomy interactions are critical to understanding the trust process. More to the point, to what extent is it necessary to manage individual proclivities and performance variations in the development of effective trust-based interventions such as transparency displays and bidirectional communication metrics? Is it sufficient to identify a small set of subgroups within which different interaction strategies may be applied? Or, must research continue to drive toward robust individualized models? Given current technology and understandings, designing training or technology-based solutions that meet the needs of all individuals remain rather unreasonable and costly. Therefore, new methods for clustering individual differences are valuable for enabling the calibration of trust within subgroups of the larger population. Our preliminary research suggests that (1) a balance may be achievable through identification of subgroups of people

that share similar traits and behaviors; (2) that different strategies must be considered in different groups, to include some subgroups where calibration needs are minimal and others that require much more heavy-handed management; and (3) that empirically estimating the link between the traits of clustered groups and their behaviors, for instance, using dynamic models to track latent cognitive states, allows us to preemptively tailor trust management systems to new human users and to adapt to their varying states.

- *Managing human-autonomy control authority*. Just as with teams of humans that come from different backgrounds and who each have different capabilities, within human-autonomy teams, we anticipate that effectiveness will not be achieved by creating autonomy-enabled systems that operate completely independent of the human teammates. Rather, a more tractable approach would seem to rely on dynamic trust calibration that enables an optimization of the level of control authority as appropriate given the mission, environmental, and social context at a given time for a given task. Therefore, it is important to operationalize paradigms and metrics to specifically test and measure trust and, likewise, when using behavioral metrics as a proxy for trust, it is essential to ensure that those metrics are not instead indexing into associated (or even potentially confounding) factors like workload, boredom, and similar. To reach this end, it is necessary to explore and understand the interplay between subjective self-report and quantitative psychophysiological and behavioral methods that will best enable metrics for measurement, analysis, and prediction of ever-changing indicators of trust as a means to accurately predict and successively manage the appropriate blend of human and autonomous control authority and performance responsibility.

The larger path forward suggests that added focus in the future will be on team trust metrics and associated trust-based interventions that will allow the community to appropriately calibrate trust in such a way as to improve team performance. While advancing trust theory has its place in science, future application and interaction with systems will ultimately rely on the capability of the team to balance expectations and behaviors. To this end, these team trust metrics must be sensitive to manipulations in informational access and weighting such that providing appropriate interventions, such as communicating information through transparency displays or other bidirectional means improves trust, are developed and thus the metric's captured value is identified. Furthermore, these team trust metrics should capture aspects of what it means to be a fluid, high-performing team with well-calibrated trust. We would expect these team trust metrics to correlate with measures of task performance and interactions with autonomy, such that misuse and disuse of technology decreases

as calibrated trust increases. Robust and sensitive measures of trust will provide the necessary insight to design better interactions for human-autonomy teams.

Acknowledgments

Portions of this research were supported by the U.S. Office of the Secretary of Defense through the Autonomy Research Pilot Initiative (MIPR DWAM31168). The views and conclusions contained in this document are those of the authors and should not be interpreted as representing the official policies, either expressed or implied, of the CCDC Army Research Laboratory or the U.S. Government. The U.S. Government is authorized to reproduce and distribute reprints for Government purposes notwithstanding any copyright notation herein.

References

Atoyan, H., Duquet, J. R., & Robert, J. M. (2006). Trust in new decision aid systems. In: *Proceedings of the 18th human-computer interaction (IHM) conference, 58*, pp. 115–122.

Bailey, N. R., Scerbo, M. W., Freeman, F. G., Mikulka, P. J., & Scott, L. A. (2006). Comparison of a brain-based adaptive system and a manual adaptable system for invoking automation. *Human Factors*, *4*(48), 693–709.

Bethel, C. L., Salomon, K., Murphy, R. R., & Burke, J. L. (2007). Survey of psychophysiology measurements applied to human-robot interaction. In: *Proceedings of the 16th IEEE international symposium on robot and human communication (RO-MAN)*.

Billings, D. R., Schaefer, K. E., Llorens, N., & Hancock, P. A. (2012). What is trust? Defining the construct across domains. In: *Poster presented at the American Psychological Association conference, division 21, Orlando, FL*.

Boyce, M. W., Chen, J. Y. C., Selkowitz, A. R., & Lakhmani, S. G. (2015). Effects of agent transparency on operator trust. In: *Proceedings of the tenth annual ACM/IEEE international conference on human-robot interaction, Portland, OR*, pp. 179–180.

Burke, C. S., Sims, D. E., Lazzara, E. H., & Salas, E. (2007). Trust in leadership: A multi-level review and integration. *The Leadership Quarterly*, *18*, 606–632.

Chen, J. Y. C., & Barnes, M. J. (2012). Supervisory control of multiple robots: Effects of imperfect automation and individual differences. *Human Factors*, *54*(2), 157–174.

Chen, J. Y. C., & Barnes, M. J. (2014). Human-agent teaming for multi-robot control: A review of human factors issues. *IEEE Transactions on Human-Machine Systems*, *44*(1), 13–29.

Colquitt, J. A., Scott, B. A., & LePine, J. A. (2007). Trust, trustworthiness, and trust propensity: A meta-analytic test of their unique relationships with risk taking and job performance. *Journal of Applied Psychology*, *92*(4), 909–927.

Daellenbach, U. S., & Davenport, S. J. (2004). Establishing trust during the formation of technology alliances. *Journal of Technology Transfer*, *29*, 187–202. https://doi.org/10.1023/B:JOTT.0000019537.61121.c3.

Dautenhahn, K. (2007). Socially intelligent robots: Dimensions of human-robot interaction. *Philosophical Transactions of the Royal Society B*, *362*, 679–704.

Derryberry, D., & Reed, M. A. (2002). Anxiety-related attentional biases and their regulation by attentional control. *Journal of Abnormal Psychology*, *111*(2), 225–236.

Drnec, K., Marathe, A. R., Lukos, J. R., & Metcalfe, J. S. (2016). From trust in automation to decision neuroscience: Applying cognitive neuroscience methods to understand and

improve interaction decisions involved in human automation interaction. *Frontiers in Human Neuroscience*, *10*, 290.

Drnec, K., & Metcalfe, J. S. (2015). Paradigm development for identifying and validating indicators of trust in automation in the operational environment of human automation integration. D. Schmorrow, & C. Fidopiastis (Eds.), *Foundations of augmented cognition: Neuroergonomics and operational neuroscience* (pp. 157–167). *AC 2016, Lecture notes in computer science: Vol. 9744*(pp. 157–167). Cham: Springer.

Dzindolet, M. T., Peterson, S. A., Pomranky, R. A., Pierce, L. G., & Beck, H. P. (2003). The role of trust in automation reliance. *International Journal of Human-Computer Studies*, *58*, 697–718.

Faller, J., Cummings, J., Saproo, S., & Sajda, P. (2019). Regulation of arousal via online neurofeedback improves human performance in a demanding sensory-motor task. *Proceedings of the National Academy of Sciences of the United States of America*, *116*, 6482–6490. https://doi.org/10.1073/pnas.1897207116.

Figner, B., & Murphy, R. O. (2011). Using skin conductance in judgment and decision making research. In M. Schulte-Mecklenbeck, A. Kühberger, & R. Ranyard (Eds.), *A handbook of process tracing methods for decision research A critical review and user's guide* (pp. 163–184). New York, NY: Psychology Press.

Gao, J., Lee, J. D., & Zhang, Y. (2006). A dynamic model of interaction between reliance on automation and cooperation in multi-operator multi-automation situations. *International Journal of Industrial Ergonomics*, *36*(5), 511–526.

Gildöf, K., Wallin, A., Dewhurst, R., & Holmqvist, K. (2013). Using eye tracking to trace a cognitive process: Gaze behavior during decision making in a natural environment. *Journal of Eye Movement Research*, *6*(1), 1–14.

Glaholt, M. G., & Reingold, E. M. (2011). Eye movement monitoring as a process tracing methodology in decision making research. *Journal of Neuroscience, Psychology, and Economics*, *4*(2), 125–146.

Greco, V., & Roger, D. (2001). Coping with uncertainty: The construction and validation of a new measure. *Personality & Individual Differences*, *31*, 519–534.

Gremillion, G. M., Donavanik, D., Neubauer, C. E., Brody, J. D., & Schaefer, K. E. (2018). Estimating human state from simulated assisted driving with stochastic filtering techniques. In D. N. Cassenti (Ed.), *Advances in human factors in simulation and modeling, proceedings of the AHFE 2018 international conferences on human factors and simulation and digital human modeling and applied optimization* (pp. 113–125). Springer.

Groom, V., & Nass, C. (2007). Can robots be teammates? Benchmarks in human-robot teams. *Interaction Studies*, *8*(3), 483–500.

Gutzwiller, R. S., Chiou, E. K., Craig, S. D., Lewis, C. M., Lematta, G. J., & Hsiung, C.-P. (2019). Positive bias in the 'Trust in Automated Systems Survey'? An examination of the Jian et al. (2000) scale. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, *63*(1), 217–221.

Hancock, P. A., Billings, D. R., & Schaefer, K. E. (2011). Can you trust your robot? *Ergonomics in Design*, *19*(3), 24–29. 1064804611415045.

Hancock, P. A., Billings, D. R., Schaefer, K. E., Chen, J. Y. C., de Visser, E., & Parasuraman, R. (2011). A meta-analysis of factors affecting trust in human-robot interaction. *Human Factors*, *53*(5), 517–527.

Hoff, K. A., & Bashir, M. (2015). Trust in automation: Integrating empirical evidence on factors that influence trust. *Human Factors*, *57*(3), 407–434.

Jarvenpaa, S., Knoll, K., & Leidner, D. (1998). Is anybody out there? Antecedents of trust in global virtual teams. *Journal of Management Information Systems*, *14*(4), 29–64. Retrieved from(1998). http://www.jstor.org/stable/.

Jian, J.-Y., Bisantz, A. M., Drury, C. G., & Llinas, J. (1998). *Foundations for an empirically determined scale of trust in automated systems (report no. AFRL-he-WP-TR-2000-0102).* Wright-Patterson AFB, OH: Air Force Research Laboratory.

John, O. P., & Srivastava, S. (1999). The big-five trait taxonomy: History, measurement, and theoretical perspectives. L. A. Pervin, & O. P. John (Eds.), *Handbook of personality: Theory and research* (pp. 102–138). Vol. 2(pp. 102–138). New York: Guilford Press.

Just, M. A., & Carpenter, P. A. (1976). Eye fixations and cognitive processes. *Cognitive Psychology*, *8*, 441–480.

Kessler, T., Larios, C., Yerdon, V., Walker, T., & Hancock, P. A. (2017). A comparison of trust measures in human robot interaction scenarios. In: *Proceedings of the 7th annual international conference on applied human factors and ergonomics, Orlando, FL, July*2016.

Kilgore, R., & Voshell, M. (2014). Increasing the transparency of unmanned systems: Applications of ecological interface design. In: *Proceedings of the 2014 human-computer interaction international (HCII) conference*, Berlin: Springer.

Kingstone, A., Smilek, D., Ristic, J., Friesen, C. K., & Eastwood, J. D. (2003). Attention, researchers! It is time to take a look at the real-world. *Current Directions in Psychological Science*, *12*, 176–180.

Kramer, R. M. (1999). Trust and distrust in organizations: Emerging perspectives, enduring questions. *Annual Review of Psychology*, *50*(1), 569–598.

Larzelere, R. E., & Huston, T. L. (1980). The dyadic trust scale: Toward understanding interpersonal trust in close relationships. *Journal of Marriage and the Family*, *42*, 595–604.

Lee, H., Kim, J., & Kim, J. (2007). Determinants of success for applications service provider: An empirical test in small businesses. *International Journal of Human-Computer Studies*, *65*(9), 796–815.

Lee, J. D., & Moray, N. (1994). Trust, self-confidence, and operators' adaptation to automation. *International Journal of Human-Computer Studies*, *40*(1), 153–184.

Lee, J. D., & See, K. A. (2004). Trust in automation: Designing for appropriate reliance. *Human Factors*, *46*(1), 50–80.

Lyons, J. B. (2013). Being transparent about transparency: A model for human-robot interaction. In D. Sofge, G. J. Kruijff, & W. F. Lawless (Eds.), *Trust in autonomous systems: Papers from the AAAI spring symposium*. Menlo Park, CA: AAAI Press.

Lyons, J. B., Sadler, G. G., Koltai, K., Battiste, H., Ho, N. T., Hoffmann, L. C. , et al. (2017). Shaping trust through transparent design: Theoretical and experimental guidelines. P. Savage-Knepshield, & J. Chen (Eds.), *Advances in human factors in robots and unmanned systems, advances in intelligent systems computing* (pp. 127–136)*499*, (pp. 127–136).

Madhavan, P., Weigmann, D. A., & Lacson, F. C. (2006). Automation failures on tasks easily performed by operators undermine trust in automated aids. *Human Factors*, *48*, 241–256.

Marathe, A., Brewer, R., Kellihan, B., & Schaefer, K. E. (2020). Leveraging wearable technologies to improve test & evaluation of human-agent teams. In *Theoretical Issues in Ergonomics Science*https://doi.org/10.1080/1463922X.2019.1697389.

Marshall, S. P. (2002). The index of cognitive activity: Measuring cognitive workload. In: *Proceedings of the 7th annual IEEE conference on human factors and power plants*.

Marshall, S. P. (2007). Identifying cognitive state from eye metrics. *Aviation, Space, and Environmental Medicine*, *78*(1), 165–175.

Matthews, R., McDonald, J. D., & Trejo, L. J. (2005). Psycho-physiological sensor techniques: An overview. In: *Proceedings of the 11th international conference on human computer interaction (HCII)*, pp. 22–27.

Mayer, R. C., Davis, J. H., & Schoorman, F. D. (1995). An integrative model of organizational trust. *Academy of Management Review*, *20*(3), 709–734.

McAllister, D. J. (1995). Affect- and cognition-based trust as foundations for interpersonal cooperation in organizations. *The Academy of Management Journal*, *38*(1), 24–59.

McGuirl, J. M., & Sarter, N. B. (2006). Supporting trust calibration and the effective use of decision aids by presenting dynamic system confidence information. *Human Factors*, *48*(4), 656–665.

Mehler, B., Reimer, B., Coughlin, J., & Dusek, J. (2009). Impact of incremental increases in cognitive workload on physiological arousal and performance in young adult drivers. *Transportation Research Record*, *2138*, 6–12.

Mercado, J. E., Rupp, M. A., Chen, J. Y. C., Barber, D., Procci, K., & Barnes, M. J. (2015). *Effects of agent transparency on multi-robot management effectiveness (ARL-TR-7466).* Aberdeen Proving Ground, MD: US Army Research Laboratory.

Merritt, S. M., Heimbaugh, H., LaChapell, J., & Lee, D. (2013). I trust it, but I don't know why: Effects of implicit attitudes toward automation on trust in an automated system. *Human Factors*, *55*(3), 520–534.

Merritt, S. M., & Ilgen, D. R. (2008). Not all trust is created equal: Dispositional and history-based trust in human-automation interactions. *Human Factors*, *50*(2), 194–210.

Metcalfe, J. S., Marathe, A. R., Haynes, B., Paul, V. J., Gremillion, G. M., Drnec, K. , et al. (2017). Building a framework to manage trust in automation. In *Micro-and nanotechnology sensors, systems, and applications IX, vol. 10194* (p. 101941U). International Society for Optics and Photonics.

Montague, E., Xu, J., & Chiou, E. (2014). Shared experiences of technology and trust: An experimental study of physiological compliance between active and passive users in technology-mediated collaborative encounters. *IEEE Transactions on Human-Machine Systems*, *44*(5), 614–624.

Moray, N., Inagaki, T., & Itoh, M. (2000). Adaptive automation, trust, and self-confidence in fault management of time-critical tasks. *Journal of Experimental Psychology: Applied*, *6*(1), 44–58.

Mower, E., Feil-Seifer, D. J., Mataric, M. J., & Narayanan, S. (2007). Investigating implicit cues for user state estimation in human-robot interaction using physiological measurements. In: *Proceedings of the 16th IEEE international conference on robot and human interactive communication, Jeju, Korea*.

Muir, B. M. (1987). Trust between humans and machines, and the design of decision aids. *International Journal of Man-Machine Studies*, *27*(5/6), 527–539.

Muir, B. M. (1994). Trust in automation: part I. Theoretical issues in the study of trust and human intervention in automated systems. *Ergonomics*, *37*, 1905–1922.

Muir, B. M., & Moray, N. (1996). Trust in automation. Part II. Experimental studies of trust and human intervention in a process control simulation. *Ergonomics*, *39*(3), 429–460.

Nakada, K. (1997). Kansei engineering research on the design of construction machinery. *International Journal of Industrial Ergonomics*, *19*(2), 129–146.

Neubauer, C., Gremillion, G., Perelman, B., La Fleur, C., Metcalfe, J.S., & Schaefer, K.E. (2020). Analysis of facial expressions explain affective state and trust-based decisions during interaction with autonomy. Proceedings of the 3rd international conference on intelligent human system integration (IHSI): Integrating people and intelligent systems, Modena, Italy.

Parasuraman, R., & Miller, C. A. (2004). Trust and etiquette in high-criticality automated systems. *Communications of the ACM*, *47*(4), 51–55.

Parasuraman, R., & Riley, V. (1997). Humans and automation: Use, misuse, disuse, abuse. *Human Factors*, *39*(2), 230–253. https://doi.org/10.1518/001872097778548886.

Poole, A., & Ball, L. J. (2006). Eye tracking in HCI and usability research. *Encyclopedia of human computer interaction* (pp. 211–219)*1*, (pp. 211–219). IGI Global.

Robinette, P., Li, W., Allen, R., Howard, A. M., & Wagner, A. R. (2016). Overtrust of robots in emergency evacuation scenarios. In: *Proceedings of the eleventh annual ACM/IEEE international conference on human-robot interaction (HRI), Christchurch, New Zealand*, pp. 101–108.

Rotter, J. B. (1967). A new scale for the measurement of interpersonal trust. *Journal of Personality*, *35*(4), 651–665.

Sanders, T. L., Llorens, N., Billings, D. R., Schaefer, K. E., Hancock, P. A., Driskell, L. , et al. (2012, August). *Physiological and behavioral cues: Identifying their potential value in measuring human-robot trust. Poster presented at Division 21.* Orlando, FL: American Psychological Association.

Schaefer, K. E. (2013). *The perception and measurement of human-robot trust (doctoral dissertation).* Florida: University of Central Florida.

Schaefer, K. E. (2016). Measuring trust in human robot interactions: Development of the "Trust Perception Scale-HRI" In W. Lawless, R. Mittu, A. Wagner, & D. Sofge (Eds.),

The intersection of robust intelligence (RI) and trust in autonomous systems (pp. 191–218). Springer. (Chapter 10).

Schaefer, K. E., Baker, A. L., Brewer, R. W., Patton, D., Canady, J., & Metcalfe, J. S. (2019). Assessing multi-agent human-autonomy teams: US Army robotic wingman gunnery operations. In: *Proceedings of the SPIE 10982, micro- and nanotechnology sensors, systems, and applications XI* https://doi.org/10.1117/12.2519302. 109822B.

Schaefer, K. E., Chen, J. Y. C., Szalma, J. L., & Hancock, P. A. (2016). A meta-analysis of factors influencing the development of trust in automation: Implications for understanding autonomy in future systems. *Human Factors*, *58*(3), 377–400.

Schaefer, K. E., & Straub, E. R. (2016). Will passengers trust driverless vehicles? Removing the steering wheel and pedals. In: *Proceedings of the IEEE CogSIMA conference, San Diego, CA, 2016*, pp. 159–165.

Schaefer, K. E., Straub, E. R., Chen, J. Y. C., Putney, J., & Evans, A. W. , III (2017). Communicating intent to develop shared situation awareness and engender trust in human-agent teams. *Cognitive Systems Research*, *46*, 26–39.

Shi, Y., Ruiz, N., Taib, R., Choi, E., & Chen, F. (2007). Galvanic skin response (GSR) as an index of cognitive load. In: *Proceedings of the ACM CHI extended abstract on human factors in computing systems*, pp. 2651–2656.

Spain, R. D., & Bliss, J. P. (2008). The effect of sonification display pulse rate and reliability on operator trust and perceived workload during a simulated patient monitoring task. *Ergonomics*, *51*(9), 1320–1337.

Spain, R. D., Bustamante, E. A., & Bliss, J. P. (2008). Towards an empirically developed scale for system trust: Take two. *Proceedings of the human factors and ergonomics society annual meeting*, *52*(19), 1335–1339.

Stack, L. C. (1978). Trust. In H. London, & J. E. Exner, Jr., (Eds.), *Dimensionality of personality* (pp. 561–599). New York: Wiley.

Tan, Y. Y. S., & Mueller, S. T. (2016). Adapting cultural mixture modeling for continuous measures of knowledge and memory fluency. *Behavior research methods*, *48*, 843–856.

Touryan, J., Apker, G., Lance, B. J., Kerick, S. E., Ries, A. J., & McDowell, K. (2014). Estimating endogenous changes in task performance from EEG. *Frontiers in Neuroscience*, *8*, 155. https://doi.org/10.3389/fnins.2014.00155.

van den Broek, E. L., & Westerink, J. H. (2009). Considerations for emotion-aware consumer products. *Applied Ergonomics*, *40*(6), 1055–1064.

Verberne, F. M. F., Ham, J., & Midden, C. J. H. (2012). Trust in Smart Systems: Sharing driving goals and giving information to increase trustworthiness and acceptability of smart systems in cars. *Human Factors*, *54*(5), 799–810.

Ward, J. N. (2000). Automation of task processes: An example of intelligent transportation systems. *Human Factors and Ergonomics in Manufacturing & Service Industries*, *10*(4), 395–408.

Wright, J. L., Chen, J. Y. C., Barnes, M. J., & Hancock, P. A. (2017). *Agent reasoning transparency: The influence of information level on automation induced complacency (ARL-TR-8044).* Aberdeen Proving Ground, MD: Army Research Laboratory.

Yagoda, R. E. (2011). *What! You want me to trust a robot? The development of a human robot interaction (HRI) trust scale (unpublished master's thesis).* Raleigh, North Carolina: North Carolina State University.

Zhou, J., Sun, J., Chen, F., Wang, Y., Taib, R., Khawaji, A. , et al. (2015). Measurable decision making with GSR and pupillary analysis for intelligent user interface. *ACM Transactions on Computer-Human Interaction (ToCHI)*, *21*(6), 33.

CHAPTER

13

Distributed dynamic team trust in human, artificial intelligence, and robot teaming

Lixiao Huang, Nancy J. Cooke, Robert S. Gutzwiller, Spring Berman, Erin K. Chiou, Mustafa Demir, and Wenlong Zhang

Center for Human, Artificial Intelligence, and Robot Teaming, Arizona State University, Tempe, AZ, United States

Introduction

A "Team" has been defined in the literature as an interdependent group of individuals, each with their own roles and responsibilities, who come together to address a specific goal (Salas, Dickinson, Converse, & Tannenbaum, 1992). When artificial agents (e.g., AI agents and embodied robotic agents) are added to an all-human team, the artificial agents may fulfill some roles and responsibilities, which are considered teaming. Therefore, reaching a high level of automation is not a defining feature for teaming capability. Instead, each agent requires varying levels of interdependency between it and other team members on a team task, and this interdependency defines teaming (Johnson & Vera, 2019).

Maintaining good interpersonal trust is important to any functional team. Lee and See's (2004) seminal trust model, for example, included cultural, organizational, and interpersonal factors in humans' trust in automation, but few studies have explored the interactions between these factors in depth. One such exception is Ho, Sadler, Hoffmann, Lyons, and Johnson's (2017) case study of the Automatic Ground Collision Avoidance System (Auto-GCAS). Auto-GCAS is a system that detects flight conditions and attempts to prevent aircraft from crashing into

https://doi.org/10.1016/B978-0-12-819472-0.00013-7
© 2021 Elsevier Inc. All rights reserved.

terrain, thereby saving pilots' lives. Different from most research on an end-user's trust in an artificial agent, Ho et al. (2017) provided rich descriptive information about real-world trust factors of three types of stakeholders: managers, engineers, and testing pilots. Managers were responsible for making decisions to implement the system in the military. If the system failed, they risked losing funding, equipment, and personnel. Engineers were responsible for developing the system to save pilots' lives when needed and otherwise not interrupt their tasks. If the system failed, engineers would risk strong reactions from other stakeholders, which might negatively impact engineers' careers and their emotions. The pilots were responsible for using the Auto-GCAS to save their own lives during operation, and as assistance toward completing missions successfully. The risk of failure for the pilot would be an unsuccessful mission and even death. Together, these separate risks and needs showcased how real-world automated systems impact a much broader, more diverse network of distributed stakeholders, beyond end-users. From this perspective, we believe that trust is also distributed across different stakeholders, in addition to dynamically changing, as each group member observes, uses, and communicates with one another about the system.

Existing trust research has focused on the end-users' perspective toward artificial agents like AI agents or embodied robotic agents, often neglecting other stakeholders' trust in the system and how that may impact the end-users' trust and use of the system. In Ho et al.'s (2017) example, the trustworthiness development of Auto-GCAS was an ongoing communication process among managers, engineers, and pilots. Engineers shared testing data with all stakeholders, and pilots expressed concerns to engineers. Managers could provide encouragement and training support for engineers and pilots. These communications among different groups could result in iterative improvements to system performance and the stakeholders' understanding of the system (a.k.a., mental models) and thus help all stakeholders develop appropriate trust in the system. The interrelated, communicative types of interpersonal factors in complex teams are not captured by trust studies that consider only individual end-users.

Therefore, we propose a framework, *Distributed Dynamic Team Trust (D2T2)*, to include both interpersonal factors and technical factors related to trust in human-AI-robot teaming along a dynamic timeline, all of which are typically missing in traditional dyadic trust research. The goal of this chapter is to invite more in-depth thinking and discussion about trust as a distributed, networked state that is constantly in flux, rather than a dyadic and static measure. The following sections will elaborate on D2T2, describe its potential measurement and modeling approach, and finally, discuss a set of applications that reflect its utility.

Distributed and dynamic team trust (D2T2)

Trust distributes among different types of stakeholders

As team sizes and compositions change, humans' trust in the artificial agents becomes distributed among all related stakeholders (see Fig. 1). Each stakeholder's attitude toward the artificial agent of interest can play a role in shaping team trust. Their trust in the Auto-GCAS may influence their job performance and, in turn, may influence team performance. The D2T2 framework allows for including other related stakeholders beyond the end-users, which is essential for providing a holistic view of trust development in the team (Ho et al., 2017).

Trust distributes within smaller teams of stakeholders

Different types of stakeholders may influence each other's trust in the artificial agents, and within each type of stakeholder, there are smaller networks using communication channels that influence one another's trust in the artificial agents (see Fig. 1). Communication about the AI agents is an important influencing mechanism, potentially more so among

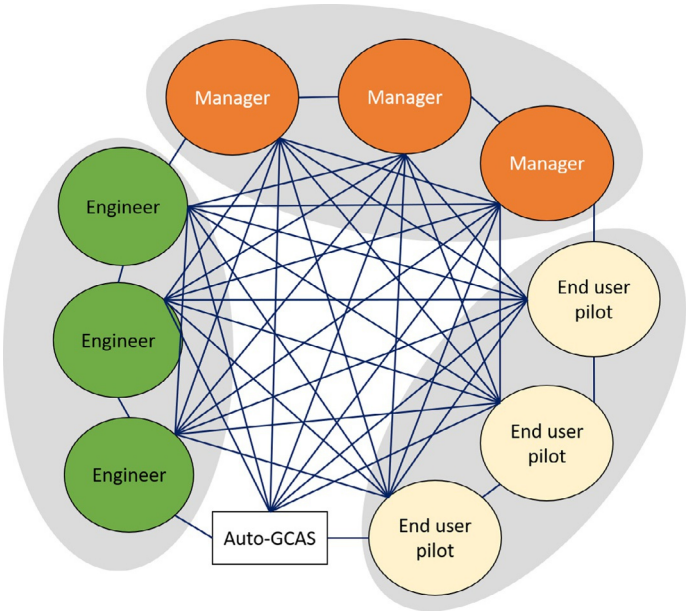


FIG. 1 Trust network in the Auto-GCAS Example. The colored nodes = stakeholders; the solid links = trust relationships; gray-shaded regions = smaller network.

stakeholders with similar roles. For example, in addition to the influence of engineers' shared testing data on managers' and pilots' trust in the Auto-GCAS system, word-of-mouth communications within the pilot network may also significantly influence pilots' trust in the Auto-GCAS system over time (Ho et al., 2017).

Distributed team trust is transitive, as well as dynamic

Interpersonal trust among all related stakeholders and their trust in artificial agents may be transmitted across the groups through daily conversations, newsletters and policies, and training procedures. The framework of D2T2 is an attempt to recognize and measure this potential impact on each type of stakeholders' trust in any artificial agent as well as one another. In an example later (Fig. 2), team trust can change through direct interactions with artificial agents or change indirectly through other people's influence.

Fig. 2 is a simplified illustration of trust transitivity in a distributed and dynamic team with two types of stakeholders—an end-user and a trainer, as well as an artificial agent. The end-user's initial relationship with the artificial agent is one of distrust (Phase 1). However, the end-user then interacts with a trainer, who has developed a positive trust relationship with the artificial agent; the end-user develops a positive trust relationship with this trainer (Phase 2). The trainer communicates new information about how and when to trust the artificial agent, and this, in turn, changes how the end-user views the artificial agent more positively (Phase 3).

Fig. 2 is a microcosm of the human-AI-robot system, which will certainly include more users, trainers, and other stakeholders. The importance of trust as a distributed and transitive concept arises under those conditions with more types of stakeholders and should be studied in addition to existing dyadic trust research. The potentially predictive relationships (in parentheses) for a variety of starting cases illustrate that end-users' trust in the artificial agent may change because of *relationships*

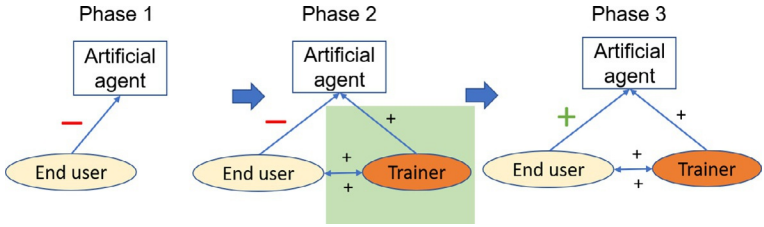


FIG. 2 Trust transitivity illustration. An end-user initially distrusts an artificial agent. The user begins interacting with a trainer. The trainer's relationship with the agent is positive, and because the end-user develops a positive trust relationship with the trainer, the end-user begins to trust the artificial agent.

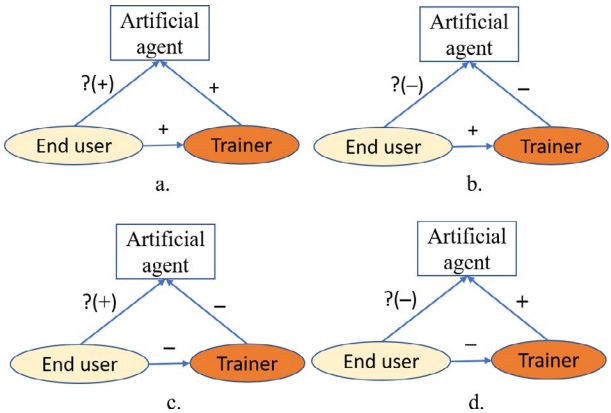


FIG. 3 Examples of inter-relational trust states. +/− = established trust status, arrow = trust direction, (+/−) = anticipated or uncertain result.

with other humans on the team (see Fig. 3). For example, in case (a), if the end-user trusts the trainer and the trainer trusts the agent, we might predict the end-user is more likely to trust the agent too. In case (b), the end-user trusts the trainer, but the trainer does not trust the agent, and we would predict the end-user may not trust the agent either, or the trainer's attitude toward the agent may reduce the end-user's trust in the agent. In examples (c) and (d), the end-user does not trust the trainer, so the trainer's trust in the agent may not play a role, and the transitivity may not exist (or it may be negative, in that the end-user decides to distrust everything in the system given that the end-user perceives the trainer as untrustworthy).

Trust transitivity has been observed in human-human trust (Jøsang, Hayward, & Pope, 2006), so it seems plausible that this transitivity applies to trust in a human-AI-robot team. The D2T2 framework takes this into account and suggests testable hypotheses about trust relationships in teams, proposing a new way to characterize and to measure team trust.

Because D2T2 involves a timeline and dynamics, we can also consider the stability of team trust, which refers to a state of trust across a distributed set of stakeholders that reaches and maintains a productive unified state over a given period. If trust within the network changes dramatically within the measurement period, it may indicate a problem. A D2T2-based trust network could indicate where and what that problem may be (e.g., decreased trust by one person for one artificial agent, in a network, would look very different than decreased trust by a group of people toward one artificial agent in an isolated setting). Viewing trust as a dynamic process that waxes and wanes over time, at times dropping rapidly, and in the context of the dynamic interactions among team members, may lead to more realistic, continuous metrics of trust.

The influence of stakeholders' views on others' trust may vary

In a distributed network, depending on the task context and people's roles and responsibilities, some individuals may have a bigger impact on others' attitudes than others. For instance, in a survival situation with high risk and uncertainty, a commander's calmness and order may greatly impact crew members' affection and beliefs (Torrance, 1954). Likewise, a senior trainer's negative experience and advice to avoid using an artificial agent in certain situations may override a more novice peer's positive perception of the artificial agent. Based on their trust in an artificial agent, managers' decision making may even determine whether an artificial agent is deployed for the end-users to use and build their trust. Therefore, the trust relationships of a few critical people in the network may influence whether the end-users have the opportunity to interact with the artificial agent and learn to use the artificial agent appropriately. For this reason, different stakeholders' degree of influence is also a variable to consider when analyzing D2T2, a nuance that is required when examining trust in teams.

D2T2 can cover multiple artificial agents on a team

In human-AI-robot teaming, it is likely that more than one artificial agent is developed or deployed on a team. Some artificial agents may be similar or even identical, while others may have totally different designs and functions. In the context of an end-user working with a swarm of homogeneous robots that are identical, the end-user's trust in all the robots is expected to be the same. However, a study found soldiers develop an emotional attachment to some robots based on long-term interactions on the battlefield, and especially when the robot was destroyed in place of the soldier during a mission (Carpenter, 2013). Anecdotal evidence suggests that other identical replacement robots may not gain the same level of affection (Carpenter, 2013). Because affection plays a critical role in forming trust (Lee & See, 2004), the other identical robots may not gain the same level of trust as the sacrificed one. This indicates the necessity to study distributed trust in the multiple artificial agents that stakeholders interact with on a team, not just one artificial agent. It also demonstrates that a more dynamic perspective on trust is necessary, one that would capture the experiences of the stakeholder with a specific robot, as well as the differences between the trusted robot and other less trusted but mechanically identical robots, to evaluate how trust may evolve differently with these robots over time.

Distributed dynamic team trust also applies to different artificial agents people interact with. The differences between different artificial agents may include versions of the agents due to software upgrades or added

features. For example, an automatic target recognition (ATR) technique with 70% reliability may be acceptable to soldiers in an early stage but may not be as satisfying as 95% reliability when it becomes possible. Knowing that a 95% reliability option is available or having interacted with such a high-reliability ATR may reduce soldiers' trust in the original lower reliability ATR, such that it affects operations. Therefore, a research question for D2T2 is how the knowledge of, or order of experiencing artificial agents with different levels of capabilities can change different types of stakeholders' trust in the artificial agent in networked groups.

Because system versions and upgrades do not always happen in parallel across military platforms, warfighters who are deployed across different variants of an artificial agent could struggle to develop calibrated trust. Meanwhile, when a team involves multiple heterogenous artificial agents, whether and how trust in one type of artificial agent transfers to another type is a question related to trust transfer in D2T2 and may depend on the similarities and differences between the artificial agents. In the case of Auto-GCAS, a manager's and a pilot's negative predispositional trust in the GCAS came from their previous experience with a separate unreliable system (Ho et al., 2017). It is not clear how different the separate unreliable system was from the Auto-GCAS system and how the stakeholders' trust in a different system was transferred across. It suggests trust could migrate across experiences with different types of systems within a domain as well, and not just upgrades made to existing systems. It is unlikely that static or dyadic measures of trust can capture these influences well. The D2T2 framework may allow this, and we expand on potential methods for measuring and modeling this framework of team trust next.

Measurement and modeling of distributed dynamic team trust

The approach to addressing the D2T2 framework in human-AI-robot teaming starts with understanding the context, including the tasks, environment, the stakeholders, and artificial agents involved. We must also understand the kinds of interactions that are available between the entities involved in a situated context, where transitive properties of trust take place. These affordances for interaction are also where we begin to measure trust in a distributed and dynamic fashion, through the different measures that may be appropriate for different types of trust relationships. We outline the steps later using the context of Auto-GCAS (Ho et al., 2017) as an example.

Step 1: Identify the context of interest. The context of interest refers to the position domain, such as a specific position in an application field. For example, the US Air Force was interested in flight mission success and

pilots' safety, so a military flight crew and their safety in flying was the context of interest. In many incidents of controlled flights into terrain (CFIT), the crashes of healthy and functional aircraft are due to pilot errors, physical challenges, and mental challenges (Ho et al., 2017). Therefore, a goal was to develop an artificial agent to aid the pilots and prevent CFIT. Identifying this context links trust to domain expertise and existing knowledge about the people, capabilities of the artificial agent, and goals of the task environment. Naturally, this plays a large role in developing trustworthy technology but also serves to base trust assessment in realistic, grounded methodology.

Step 2: Identify stakeholders and the artificial agents in the trust network and how they interact with one another. The Auto-GCAS example (Ho et al., 2017) identified one artificial agent and three types of stakeholders. Then we can interview subject matter experts and review training documents to do task analysis to identify related stakeholders, technologies, and interactions in a situated context, such as rail and air dispatching (Huang, Cummings, & Nneji, 2018; Huang, Nneji, & Cummings, 2019). It is necessary to understand these interactions in more than high-level detail to make more fine-grained predictions or assessments. More nuanced methods and elicitation techniques have been useful as the foundation for characterizing teams as networks, such as Event Analysis of Systemic Teamwork (EAST; Walker et al., 2006).

Step 3: Measure each trust relationship in the trust network. A challenging problem in realizing D2T2 is how to characterize so many different trust relationships in a network and how to measure them with enough frequency over time. Our strategy is to investigate D2T2 by system stages, layers of trust, and the types of interactions.

System stages refer to qualitatively different event cutoff lines in a work domain. System stages determine the interactions among the entities. For example, the findings in Ho et al. (2017) were based on the predeployment stage alone, which did not involve operating the fully functioning Auto-GCAS system yet. The predeployment stage would naturally be different from the training stage when pilots use Auto-GCAS in a simulation or the operation stage when pilots operate a real aircraft. To mitigate the complexity of the analysis, we propose to examine the trust relationships, including trust transitivity and the trust network of stakeholders, by system stages. These stages can then be linked or compared (see Step 4) when more data about relevant trust relationships among stakeholders and artificial agents become available.

Within each system stage, individuals may have three layers of trust: predispositional trust—before operational interactions, learned trust—trust during a controlled training setting or based on ones' reflection of past experiences with the artificial agent, and situational trust—trust during real-life interactions with an artificial agent when unpredictable things

may happen (Hoff & Bashir, 2015). Therefore, the D2T2 approach should also track the layer of trust in a given relationship.

Moreover, just as important is that each layer may need a different trust measure and may require different administrations, depending on what type of interactions are involved. Our hypothesis is that interaction affordances in human-human relationships and in human-AI-robot relationships may be different, and thus some trust measures may be more suitable than others in each trust relationship. For example, text-based analysis of conversations may be one way to unobtrusively measure trust between people (Lee & Kolodge, 2019), but interacting with an artificial agent that does not have natural language abilities may preclude the use of such measures. Likewise, surveys are good for predispositional trust, whereas unobtrusive measures like behavioral indicators may be more feasible than surveys for situational trust during ongoing activities. The next section on trust measures elaborates selected trust measurement options and comments on their applicability. However, current methods of trust measurement will have to be improved or adjusted to achieve a more distributed, dynamic assessment of team trust.

Step 4: Methods for assessing and analyzing a trust network. After measuring stakeholder's trust in each artificial agent using appropriate measures, we can gain a better understanding of team trust and its evolution using social network analysis methods to examine the trust network. The distributed team trust network may be compared by system stages, as discussed in Step 3. The fully connected trust network in Fig. 1 may also be customized to show critical trust relationships depending on the roles of stakeholders because not all influencing factors for trust equally matter to all stakeholders and thus can be filtered case by case (Ho et al., 2017). Team trust networks can show different properties of trust, such as the distribution of team trust, trust transitivity, trust transfer, changing patterns of the trust network, power or influences of some stakeholders, and dynamic effects. Each of these has importance in a better understanding of trust and teaming with a given artificial agent or several agents. Most importantly, these kinds of assessments may guide a strategy for trust repair or recalibration—changes in training, direction from supervisors, the information delivered to end-users, transparent displays, and other options all present solutions, but the assessments and strategies should match the suspected problem and the roles of stakeholders.

Trust measures

As we discussed in Step 3, the framework of D2T2 brings the challenge of dynamic and distributed trust measurement techniques. These are required to implement the framework and provide value. Dynamic team

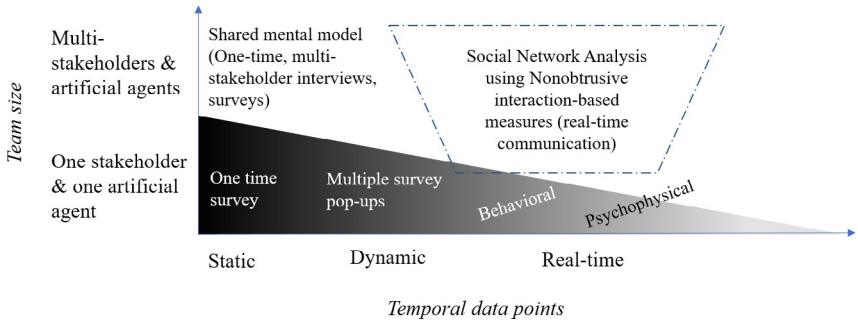


FIG. 4 Trust measurement needs for D2T2. The X-axis indicates the relative quantity of temporal data points that are required for each general type of measure. The Y-axis shows the size of teams directly involved in the trust measurement. Most existing research focuses on one human's trust in one artificial agent, as shown in the gray area; darkness (and thickness) indicate the number of studies, the darker (and the thicker), the more. The dotted box indicates interactive cognition theory related methods, which may apply to both dyadic and team trust.

trust means team trust changes over time and should be measured and modeled accordingly. The dynamic time scale could be seconds, minutes, hours, days, or even in terms of missions and events, depending on the task of interest, and the readiness of data collection techniques. In considering the measurement options for the trust links within the framework of D2T2, two dimensions emerge: (a) the number of entities included in a measure of trust, and (b) the number of temporal data points collected by it (see Fig. 4). We introduce each measure in more detail later.

Trust surveys

Surveys are the most common measure of human trust. These self-reported surveys are typically administered one time, either before or following an intervention, depending on whether the goal is to measure baseline trust or trust in a specific technology. The surveys often involve the use of Likert scales and ordinal data (e.g., Jian, Bisantz, & Drury, 2000), with many adaptations available throughout the literature (Hancock et al., 2011; Hoff & Bashir, 2015). Surveys focus on one person's perceptions of another entity, including trust in automation across different domains (Pak, Rovira, McLaughlin, & Baldwin, 2017), interpersonal trust (Rotter, 1967), and trust in specific contexts, such as small military teams (Adams, Bruyn, & Chung-yan, 2004). Surveys will not fully realize the D2T2 framework, in terms of the dynamic process of changes in trust, as they are often prohibitive to re-administer at a high rate and will not capture communications between agents. Surveys will be poor at feeding into any real-time assessments and will be interruptive to the performance of operators and others in the network. Even when surveys are useful, all

measures must contend with validity and reliability of measurement concerns (Gutzwiller et al., 2019; Jian et al., 2000; Spain, Bustamante, & Bliss, 2008). However, surveys are the most tested trust measure, and maybe most appropriate for measuring dispositional trust—relatively stable attitudes toward an agent—in a slow-paced, low-stress environment. Surveys may also measure learned trust immediately following a training intervention about how to work with an artificial agent. Therefore, surveys could be incorporated and may be useful in assessing distribution of trust and its transitive properties.

Binary behavioral indicators of trust

Existing research has used various behavioral measures to infer trust. A common and simplified approach is using binary use and disuse as an indicator of trust. In a study using a simulated robotic command and control system, the number of human overrides of robotic behavior reflected an operator's trust in their robot partners (Freedy, DeVisser, Weltman, & Coeyman, 2007). However, behavioral indicators of reliance, compliance, and intervening actions (Domeyer, Venkatraman, Price, & Lee, 2018) may be observable in both the absence or presence of trust. If human operators have the option not to use the artificial agent, they may avoid using the agent when they distrust it; but if the use of the agent is required to do the task, they may still use it even though they do not trust the agent.

At that level of analysis, it may generate more temporal data than surveys but does not reflect socially dynamic and distributed trust at a team level. However, as not all members will be directly interacting, the measure is limited in larger networks of distributed agents. In addition, an issue with binary behavioral measures of trust is that some behavioral outcomes occur infrequently or may be too delayed to analyze. For example, binary behavioral indicators of trust (e.g., authorization versus non-authorization of actions like missile launches) cannot occur early enough to prevent consequences of inappropriate or degraded trust. Therefore, this measure has limited usability to realize D2T2 but may remain useful for dyadic cases in which users have a choice not to use the artificial agent, their use or disuse happens infrequently, and the action of use or disuse would not lead to immediate irrevocable catastrophe.

Sensor-based psychophysical measures of trust

According to the comparison of trust measures (see Fig. 4), real-time sensor-based psychophysical measures of trust have the potential of collecting continuous data, providing more granularity to the dynamic aspects of trust. Developers of artificial agents seek this kind of data to develop algorithms that enable artificial agents to detect and adjust responses to the changes in humans' trust; a discrete and static measure

of trust through surveys cannot satisfy this purpose. Recent work has proposed to use electroencephalography (EEG) and Galvanic skin response (GSR), but the relationship between psychophysical data and trust has not been well established yet.

Electroencephalography has been a popular physical measure for trust estimation (Boudreau, McCubbins, & Coulson, 2009; Kolling, Walker, Chakraborty, Sycara, & Lewis, 2016; Long, Jiang, & Zhou, 2012; Mondada, Karim, & Mondada, 2016). Human trust had a significantly positive correlation with EEG features from specific electrodes in an investment game with artificial agents, and the frontal and occipital areas were identified as the predominant brain areas correlated with trust (M. Wang, Hussein, Rojas, Shafi, & Abbass, 2018).

Galvanic skin response captures physiological arousal levels based on the conductivity of the surface of the skin (Akash, Hu, Jain, & Reid, 2018). GSR is significantly affected by both trust and cognitive load in a text-chat environment (Khawaji, Zhou, Chen, & Marcus, 2015), and the phasic component of GSR was also a significant predictor of trust levels (Akash et al., 2018), but the results were not clear about the corresponding GSR readings for a high level of trust.

In general, sensor-based measurement of human trust is still in its infancy, not well explored and validated to use such real-time estimates of human trust. One issue is that most of the sensors used for measuring trust require careful calibration, and almost all the results are collected using interactions with software-based agents. It remains an open question whether such sensing methods can provide reliable results for interactions between humans and physical robots in outdoor and complex environments (Wang et al., 2019). Other issues arise from the challenges associated with the sensors themselves, including sensing noise and high computation load. Under the D2T2 framework, we must discover how to bind these measures to capture the trust of the interaction between a human and any *specific* agent. For example, it may be difficult to identify trust measurement from sensor-based signals separately when the participant has multiple tasks and interacts with multiple agents during data collection. More evidence is needed to show the validity and reliability of using sensor-based data in measuring trust to facilitate its use in the D2T2 measurement solution.

In sum, surveys, binary behavioral indicators, and psychophysical measures of trust focus on trust in a dyadic relationship. These measures are also somewhat context free, which makes sense given their originating purpose (e.g., a survey given after interaction with a system in a lab). Understanding the interaction processes and team cognition and moving beyond individual-centered orientations are essential for effective human-AI-robot teaming (Johnson & Vera, 2019). To this end, we introduce interaction-based measures of trust.

Interactive team cognition theory and interaction-based measures

Team cognition has been conceived of in terms of shared mental models (Cannon-Bowers, Salas, & Converse, 1993), distributed cognition (Hutchins, 1995), or transactive memory (Hollingshead, 1998). Though many interesting findings, theoretical developments, and measures have resulted from these perspectives, they each take a static view of team cognition, measuring it as a set of static snapshots in time. Adopting the shared mental model perspective for team trust would suggest that team trust is the interconnected pairwise trust of team members at a specific moment. In a shared mental model (SMM) framework, each human teammate's trust would be measured for each dyad and then aggregated to reflect a *full* picture of team trust in a distributed social network. A trust network generated this way takes a holistic view of team trust at a specific time, but it critically misses the dynamic aspects of transitive trust mentioned earlier.

A more dynamic way of thinking about team cognition is through the theory of interactive team cognition (ITC; Cooke, 2015; Cooke, Gorman, Myers, & Duran, 2013). ITC posits that team cognition is an activity that consists of interactions among teammates, it ties strongly to the task context and the modality of interactions, and it is the best measured at the team level of analysis. Thinking about team cognition this way has resulted in new and interesting findings, theoretical developments, and measures (Cooke & Gorman, 2009; Grimm, Demir, Gorman, and Cooke, 2018; Gorman, Demir, Cooke, & Grimm, 2019; McNeese, Demir, Cooke, & Myers, 2018). Team interactions may include continuous behavioral interaction data and verbal communication. In particular, new process-oriented measures of team cognition have been developed by relying heavily on the analysis of communication dynamics. Communication data are rich and can be examined over multiple dimensions, such as volume, frequency, pitch, content, speech act, and flow or who is talking to whom (Cooke & Gorman, 2009). Communication analysis in experimentation is unobtrusive and possible to collect and analyze in real time automatically.

In addition, unlike static snapshots, communication dynamics can reveal changes in a team after a perturbation, including periods of adaptation and resilience (Cooke & Gorman, 2009). Communication may include real-time vocal conversation, text chat messages, social media posts, and newsletters. To measure trust this way, cognitive indicators of trust that could be found in team interactions, such as communications, should be identified and measured over time. For example, certain communication content and flow may indicate a lack of trust. Researchers also found a positive relationship between the number of social messages exchanged in virtual, temporary teams, and swift trust (Jarvenpaa & Leidner, 1999). In team trust, communication may influence trust calibration by encouraging learning about the artificial agents' performance or

other properties in several ways (and recall our earlier discussion of transitive trust, which could occur through human-human communications among managers, engineers, and pilots).

Using social network analysis to model D2T2

Ho et al.'s (2017) case study provided a qualitative analysis example to explore how interpersonal relationships may influence different stakeholders' trust in the Auto-GCAS through various communication channels. As shown in Fig. 1, a social network contains links and nodes. Each node represents a stakeholder or an artificial agent. Each link in the fully connected trust network represents a trust relationship between any two entities, such as managers, engineers, pilots, and the Auto-GCAS system. Each trust relationship may have a suitable way to measure, depending on the system stage, the layer of trust, and the interactions between the entities.

With the social network analysis approach, we can study multiple trust relationships simultaneously. For example, communication analysis covers all related stakeholders and artificial agents, communication direction and frequency, and valence of the contents (i.e., positive or negative). The social network may show these trust relationships. Automating social network analysis is possible, but it requires a big data set of communication records and a fixed number of stakeholders to generate stable communication patterns on trust or distrust. In our current state, exploratory methods (e.g., interviews, surveys, social media posts, and work-related forum posts) are probably still necessary initially to understand the context and to identify critical indicators in communication for trust.

All in all, the selected trust measures listed in this chapter do not exclude other unobtrusive real-time measures for team trust dynamics. Further research is invited to identify dynamic patterns or indicators of changes in trust using the framework of D2T2.

Applications of distributed dynamic team trust in human-AI-robot teaming

The holistic view of D2T2 has broad applications in trust research because human-AI-robot teams are likely to comprise different types of stakeholders and multiple artificial agents, requiring an extension of the traditional examination of trust as a dyadic construct. The measurement and modeling approach still needs exploration and validation. Our major objective is to point out what is currently missing in prior measures and what our new approach provides to fill that gap. Below are several potential use cases that could benefit from the application of the D2T2 framework and conceptualization of team trust.

Identifying problematic artificial agents in a human-AI-robot team trust network

An artificial agent's trustworthiness is one property of the agent. Working with an untrustworthy agent may lead to poor team performance. In a heterogeneous team that includes multiple artificial agents, measuring an evidence-based and weighted trust in each agent from multiple stakeholders may help identify good or bad agents. This is similar to the use of multisource feedback and appraisal in organizational psychology (Fletcher, 2001), in which supervisors, peers, and customers all provide ratings to evaluate an employee's performance. Ho et al. (2017) found that the reputation of Auto-GCAS was developed through sharing engineers' testing data, managers' discussions, and pilots' word-of-mouth communications, which are all suitable candidates for measures of trust in our framework.

Further, incident reports, communication about the usage of the artificial agents, and system operation logs, which often already exist, could be utilized to evaluate the performance and acceptance of each artificial agent. A problematic agent could be fixed or eliminated from the team. Fixes could take a variety of forms, including appropriately targeted training of those stakeholders who are affecting the network most significantly or closest to the operational context. Such training could communicate the issue and ground the team in the same mental model or correct or educate those who do not yet view the agent accurately.

Using only a dyadic trust approach, some problematic agents may be nearly impossible to identify. However, when viewing team trust as a distributed state, it becomes possible to identify whether even one human who distrusts an agent may be infecting the entire team (or could be the lone dissenter). A D2T2 approach, as a more dynamic method, could help unearth these types of relationships and their dynamics more frequently and with sufficient detail to devise solutions. The use of more continuous, dynamic measures of trust (e.g., continual behavioral indicators or interactive communication analysis) may enable the unobtrusive detection of issues in team trust and the notification of team leaders or other stakeholders, serving as a new capability via diagnostic aid for those looking to make teams more effective.

Design of scalable control strategies for multiagent systems

A second potential application is the development of control frameworks for large-scale multiagent systems, such as robotic swarms. These control frameworks should rely on limited human supervision, in part because situation awareness and task performance can be degraded when humans team with large numbers of robotic agents at different levels of

automation (Chen & Barnes, 2014). Human supervision can be exercised through high-level specifications for the multiagent system, which allows the control approach to scale well with the number of agents (Michael, Fink, Loizou, & Kumar, 2010). From the perspective of fostering appropriate trust in the system, one important question is the optimal integration of such human-issued directives with individual and collective decisions by the robotic agents.

In the D2T2 framework, stakeholders need to know the similarities and differences among the artificial agents in the swarm, instead of the properties of one artificial agent. This knowledge may help end-users and other stakeholders to establish their trust criteria. For example, end-users may build trust in unused robots based on the robots they have already interacted with if all robots in the swarm have similar capabilities. Constructing time-varying networks of trust among stakeholders and multiple artificial agents can also aid in illustrating the relationships among teammates and bringing problematic ones to notice. For instance, control strategies for robotic swarms will require designating team leaders who supervise the swarm, and it would not be effective to assign team members with inappropriate trust in some of the robots to supervise them. Moreover, identifying key leaders of a large team in terms of which teammates trust them may help ascertain which teammates to support or request assistance from when determining a plan of action. For example, the ability to identify *influencers* of the swarm or of specific agents in the swarm may be useful to target those roles for trust training, intervention, etc., rather than being organizationally forced to train everyone.

Conclusion

Team trust is not an isolated phenomenon nor an outcome, but rather an interactive social and dynamic process that has multiple sources of influence over time (Hoff & Bashir, 2015; Lee & See, 2004). Current individual, end-user-oriented trust research neglects the different types of stakeholders' trust in one or more artificial agents, and how one stakeholder's trust in an artificial agent influences other stakeholders' trust, and vice versa. We proposed using the framework of distributed and dynamic team trust (D2T2) to fill the gaps in the existing trust research. We also proposed four steps to implement D2T2 in a human-AI-robot teaming context, including utilizing appropriate traditional trust measures for different layers of individual end-user trust and proposing interaction-based unobtrusive measures for capturing the distributed and dynamic property of team trust.

D2T2 contributes to the trust research literature by filling the gap of measuring and modeling the *distributed* interrelational trust between

human teammates and between human and artificial agents over time. We suspect that the D2T2 framework and modeling approach may move the field much closer to pragmatic and real-world trust measurements for complex systems and more advanced human-AI-robot teaming. The inclusion of team communications, distribution of trust beliefs, and network analysis enables much more appropriate measures of trust across large teams than current trust measures. However, we are aware that D2T2 is an exploratory framework, and await further experimentation, application, and development before suggesting any grand conclusions. As with all studies of trust in technology, it is reasonable to suggest that D2T2 in human-AI-robot teaming is vital to team performance—with the potential to proactively and potentially in real-time help to identify problematic artificial agents and call for supporting actions. If continuous and real-time trust measures are improved, this identification may even be done automatically in certain contexts. Lastly, empirical, interdisciplinary research is needed to verify the value of D2T2 when applied to real-world team situations and to identify novel ways of measuring and modeling D2T2.

References

Adams, B., Bruyn, L., & Chung-yan, G. (2004). *Creating a measure of trust in small military teams.* DRDC Toronto Technical Report CR 2004-077.

Akash, K., Hu, W.-L., Jain, N., & Reid, T. (2018). A classification model for sensing human trust in machines using EEG and GSR. *ACM Transactions on Interactive Intelligent Systems*, *8*(4), 1–20. https://doi.org/10.1145/3132743.

Boudreau, C., McCubbins, M. D., & Coulson, S. (2009). Knowing when to trust others: An ERP study of decision making after receiving information from unknown people. *Social Cognitive and Affective Neuroscience*, *4*(1), 23–34. https://doi.org/10.1093/scan/nsn034.

Cannon-Bowers, J. A., Salas, E., & Converse, S. (1993). Shared mental models in expert team decision making. In *Current issues in individual and group decision making* (pp. 221–246). Hillsdale, NJ: Lawrence Erlbaum.

Carpenter, J. (2013). *The quiet professional: An investigation of US military explosive ordnance disposal personnel interactions with everyday field robots. (2013).* https://digital.lib.washington.edu/researchworks/bitstream/handle/1773/24197/Carpenter_washington_0250E_12154.pdf?sequence=1.

Chen, J. Y., & Barnes, M. J. (2014). Human–agent teaming for multirobot control: A review of human factors issues. *IEEE Transactions on Human-Machine Systems*, *44*(1), 13–29.

Cooke, N. J. (2015). Team cognition as interaction. *Current Directions in Psychological Science*, *24*(6), 415–419.

Cooke, N. J., & Gorman, J. C. (2009). Interaction-based measures of cognitive systems. *Journal of Cognitive Engineering and Decision Making*, *3*(1), 27–46.

Cooke, N. J., Gorman, J. C., Myers, C. W., & Duran, J. L. (2013). Interactive team cognition. *Cognitive Science*, *37*(2), 255–285. https://doi.org/10.1111/cogs.12009.

Domeyer, J., Venkatraman, V., Price, M., & Lee, J. D. (2018). Characterizing driver trust in vehicle control algorithm parameters. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, *62*, 1821–1825.

Fletcher, C. (2001). Performance appraisal and management: The developing research agenda. *Journal of Occupational and Organizational Psychology*, *74*(4), 473–487.

Freedy, A., DeVisser, E., Weltman, G., & Coeyman, N. (2007). Measurement of trust in human-robot collaboration. In *2007 International symposium on collaborative technologies and systems* (pp. 106–114).

Gorman, J. C., Demir, M., Cooke, N. J., & Grimm, D. A. (2019). Evaluating sociotechnical dynamics in a simulated remotely-piloted aircraft system: A layered dynamics approach. *Ergonomics*, *62*(5), 629–643.

Grimm, D., Demir, M., Gorman, J. C., & Cooke, N. J. (2018). *The complex dynamics of team situation awareness in human-autonomy teaming. In 2018* IEEE conference on cognitive and computational aspects of situation management (CogSIMA) (pp. 103–109). (2018). https://doi.org/10.1109/COGSIMA.2018.8423990.

Gutzwiller, R. S., Chiou, E. K., Craig, S. D., Lewis, C. M., Lematta, G. J., & Hsiung, C.-P. (2019). Positive bias in the 'Trust in Automated Systems Survey'? An examination of the Jian et al. (2000) scale. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, *63*, 217–221.

Hancock, P. A., Billings, D. R., Schaefer, K. E., Chen, J. Y., De Visser, E. J., & Parasuraman, R. (2011). A meta-analysis of factors affecting trust in human-robot interaction. *Human Factors*, *53*(5), 517–527.

Ho, N. T., Sadler, G. G., Hoffmann, L. C., Lyons, J. B., & Johnson, W. W. (2017). Trust of a military automated system in an operational context. *Military Psychology*, *29*, 524–542.

Hoff, K., & Bashir, M. (2015). Trust in automation: Integrating empirical evidence on factors that influence trust. *Human Factors*, *57*(3), 407–434.

Hollingshead, A. B. (1998). Retrieval processes in transactive memory systems. *Journal of Personality and Social Psychology*, *74*(3), 659–671. https://doi.org/10.1037/0022-3514.74.3.659.

Huang, L., Cummings, M., & Nneji, V. C. (2018). Preliminary analysis and simulation of railroad dispatcher workload. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, *62*, 691–695.

Huang, L., Nneji, V., & Cummings, M. (2019). How airline dispatchers manage flights: A task analysis in distributed and heterogeneous network operations. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, *63*(1), 1389–1393. (2019). https://journals.sagepub.com/doi/10.1177/1071181319631182#articleCitationDownloadContainer.

Hutchins, E. (1995). *Cognition in the wild.* MIT Press.

Jarvenpaa, S. L., & Leidner, D. E. (1999). Communication and trust in global virtual teams. *Organization Science*, *10*(6), 791–815. https://doi.org/10.1287/orsc.10.6.791.

Jian, J.-Y., Bisantz, A. M., & Drury, C. G. (2000). Foundations for an empirically determined scale of trust in automated systems. *International Journal of Cognitive Ergonomics*, *4*(1), 53–71.

Johnson, M., & Vera, A. (2019). No AI is an island: The case for teaming intelligence. *AI Magazine*, *40*(1), 16–28.

Jøsang, A., Hayward, R., & Pope, S. (2006). Trust network analysis with subjective logic. *Proceedings of the 29th Australasian Computer Science Conference*, *48*, 85–94.

Khawaji, A., Zhou, J., Chen, F., & Marcus, N. (2015). Using galvanic skin response (GSR) to measure trust and cognitive load in the text-chat environment. In: *Proceedings of the 33rd annual ACM conference extended abstracts on human factors in computing systems, 1989–1994*. https://doi.org/10.1145/2702613.2732766.

Kolling, A., Walker, P., Chakraborty, N., Sycara, K., & Lewis, M. (2016). Human interaction with robot swarms: A survey. *IEEE Transactions on Human-Machine Systems*, *46*(1), 9–26.

Lee, J. D., & Kolodge, K. (2019). Exploring trust in self-driving vehicles through text analysis. *Human Factors*, 0018720819872672.

Lee, J. D., & See, K. A. (2004). Trust in automation: Designing for appropriate reliance. *Human Factors*, *46*(1), 50–80. https://doi.org/10.1518/hfes.46.1.50_30392.

Long, Y., Jiang, X., & Zhou, X. (2012). To believe or not to believe: Trust choice modulates brain responses in outcome evaluation. *Neuroscience*, *200*, 50–58. https://doi.org/10.1016/j.neuroscience.2011.10.035.

McNeese, N. J., Demir, M., Cooke, N. J., & Myers, C. (2018). Teaming with a synthetic teammate: Insights into human-autonomy teaming. *Human Factors*, *60*(2), 262–273.

Michael, N., Fink, J., Loizou, S., & Kumar, V. (2010). Architecture, abstractions, and algorithms for controlling large teams of robots: Experimental testbed and results. In *Robotics research* (pp. 409–419). Springer.

Mondada, L., Karim, M. E., & Mondada, F. (2016). Electroencephalography as implicit communication channel for proximal interaction between humans and robot swarms. *Swarm Intelligence*, *10*(4), 247–265.

Pak, R., Rovira, E., McLaughlin, A. C., & Baldwin, N. (2017). Does the domain of technology impact user trust? Investigating trust in automation across different consumer-oriented domains in young adults, military, and older adults. *Theoretical Issues in Ergonomics Science*, *18*(3), 199–220. https://doi.org/10.1080/1463922X.2016.1175523.

Rotter, J. (1967). A new scale for the measurement of interpersonal trust. *Journal of Personality*, *35*(4), 651–665.

Salas, E., Dickinson, T. L., Converse, S. A., & Tannenbaum, S. I. (1992). Toward an understanding of team performance and training. In R. W. Swezey, & E. Salas (Eds.), *Teams: Their training and performance* (pp. 3–29). Ablex Publishing.

Spain, R. D., Bustamante, E. A., & Bliss, J. P. (2008). Towards an empirically developed scale for system trust: Take two. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, *52*, 1335–1339.

Torrance, E. P. (1954). The behavior of small groups under the stress conditions of "survival". *American Sociological Review*, *19*(6), 751–755.

Walker, G. H., Gibson, H., Stanton, N., Baber, C., Salmon, P., & Green, D. (2006). Event analysis of systemic teamwork (EAST): A novel integration of ergonomics methods to analyse C4i activity. *Ergonomics*, *49*(12–13), 1345–1369.

Wang, M., Hussein, A., Rojas, R. F., Shafi, K., & Abbass, H. A. (2018). EEG-based neural correlates of trust in human-autonomy interaction. In: *2018 IEEE symposium series on computational intelligence (SSCI)*, pp. 350–357. https://doi.org/10.1109/SSCI.2018.8628649.

Wang, Y., Lematta, G. J., Hsiung, C.-P., Rahm, K. A., Chiou, E. K., & Zhang, W. (2019). Quantitative modeling and analysis of reliance in physical human–machine coordination. *Journal of Mechanisms and Robotics*. *11*(6). https://doi.org/10.1115/1.4044545.

CHAPTER

14

Deep learning, transparency, and trust in human robot teamwork

Michael Lewis[a], *Huao Li*[a], *and Katia Sycara*[b]

[a]School of Computing and Information, University of Pittsburgh, Pittsburgh, PA, United States, [b]Carnegie Mellon University, Robotics Institute, Pittsburgh, PA, United States

Introduction

Teamwork is a set of interrelated reasoning, actions, and behaviors of each team member that adaptively combine to fulfill shared team goals (Morgan et al., 1986). Experimental evidence from high-performance human teams resulted in a set of drivers of team effectiveness (Salas, Cooke, & Rosen, 2008; Salas, Sims, & Burke, 2005). These drivers are *team leadership*, *mutual performance monitoring*, *backup behaviors* (ability to anticipate other team members' needs and shift work and cognitive workload to achieve appropriate balance), *adaptability* (ability to adjust strategies and actions based on dynamic changes in mission and environment), *team orientation* (taking other's behavior into account during group interactions and belief on team goals over individual goals), *shared mental models* (organizing knowledge structure of the relationship between tasks and how the team will perform them), *closed-loop communication* (reliable exchange of information), and finally and most crucially mutual *trust* (the shared belief that team members will perform their roles and protect the interests of their teammates). As technology enables increased machine autonomy, human-machine teaming could acquire the same characteristics as human-human teaming.

Besides being an important ingredient of teamwork, trust has been found to be important in human use of automation: people tend to rely on automation they trust and not use automation they do not trust. This

https://doi.org/10.1016/B978-0-12-819472-0.00014-9
© 2021 Elsevier Inc. All rights reserved.

has generated sustained interest in conceptualizations of trust and its relation to human interaction with automation. Trust has been defined by Mayer, Davis, and Schoorman (1995) as "The willingness of a party to be vulnerable to the actions of another party based on the expectation that the other will perform a particular action important to the trustor, irrespective of the ability to monitor or control that party." In the context of trust in automation, the earlier definition has typically been interpreted as a human's willingness to rely on automation to perform some task. Research in human interaction with automation, which we believe is also valid for human interaction with autonomy, has found that the human may fail to use the automation when it would be advantageous, called *disuse or under reliance*, fail to monitor it properly when in use, or accept its recommendations and actions when inappropriate, called *misuse or over reliance* (Lyons & Stokes, 2012). Lee and Moray (1992, 1994), Muir and Moray (2013), and Lewandowsky, Mudy, and Tan (2000) have shown that trust toward automation can mediate reliance. Additionally, operator trust has been found to *vary dynamically*, accumulating over periods of successful performance, then declining sharply when failures or poor performance are encountered (Gao & Lee, 2006; Lee & Moray, 1994; Xu & Dudek, 2015).

In social robots, incorrect trust calibration can lead to extreme overtrust as demonstrated in Robinette, Li, Allen, Howard, and Wagner (2016) where participants followed the directions of a demonstrably dysfunctional robot to evacuate a smoke-filled room with detrimental results, including getting directed into closets rather than following exit signs. In other studies reviewed by Schaefer (2013) trust in social robots increased with matches between robot appearance and user expectations and was generally higher for less socially competent robots.

Another characteristic that contributes to both teamwork effectiveness and trust is *transparency*. Systems that are more transparent in conveying their reasoning should be more trusted, since they would be more easily understood by their users (Lewis & Heidorn, 1991; Lewis, 1998; Simpson, Brander, & Portsdown, 1994; Sycara, Lewis, Lenox, & Roberts, 1998) and hopefully their users would be able to better judge their capabilities, thus improving their trust calibration. Additionally transparency in a human-robot context can be viewed as a method to establish shared intent and shared awareness between a human and a machine (Lyons, 2013). Although there are multiple definitions of agent transparency (Chen et al., 2014; Lyons & Havig, 2014), we use, with minor variation, the definition proposed by Chen et al. (2014): "Agent transparency is the quality of an interface (e.g., visual, linguistic) pertaining to its abilities to afford an operator's comprehension about an intelligent agent's intent, performance, future plans, and reasoning process." The goal of transparency is not to relay all of a system's capabilities, behaviors, and decision-

making rationale to the human. Ideally, agents should relay clear and efficient information as succinctly as possible, thus enabling the human to maintain a proper understanding of the system in its tasking environment. The human factors and computational literature (Kim, 2015; Mercado et al., 2016; Ribeiro, Singh, & Guestrin, 2016) have pointed out the need for system transparency as a way to increase trust in the system.

As agents become more sophisticated and independent via learning and interaction, it is critical for their human counterparts to understand their behaviors, the reasoning process behind those behaviors, and the expected outcomes to properly calibrate their trust in the systems and make appropriate decisions (de Visser, Cohen, Freedy, & Parasuraman, 2014; Lee & See, 2004; Mercado et al., 2016). When interacting with autonomous intelligent agents, people tend to regard them as intentional individuals and explain their behaviors in terms of interpersonal relationships (de Graaf & Malle, 2017). That requires an explainable agent to clarify its actions by offering reasons of beliefs, desires, and intentions (Langley, Meadows, Sridharan, & Choi, 2017). Indeed, past studies have shown that humans sometimes question the accuracy and effectiveness of agents' actions due to the human's difficulties understanding the state/status of the agent (Bitan & Meyer, 2007; Seppelt & Lee, 2007; Stanton, Young, & Walker, 2007) and the rationales behind the behaviors (Linegang et al., 2006).

In recent years, automation and autonomous system have increasingly relied on machine learning. ML systems are typically used for two broad types of problems. First, they are used for classification, which relies on supervisory methods with ground truth given by labeled data and produces a judgment as to whether an input belongs to a particular class. Such systems have become ubiquitous in almost all areas of human endeavor, such as web services, health care, education, insurance, law enforcement, and defense (Lipton, 2018). Machine learning algorithms make important decisions in our interactions influencing the news we see, our finances (who gets a loan, or a particular line of credit), and careers (algorithms often filter job applications). Courts have employed algorithms to predict the probability that an individual relapses into criminal behavior (Choudlechova, 2016). Neural networks used in classification have revolutionized computer vision and natural language understanding. Second, ML systems solve sequential decision-making problems that use mainly unsupervised methods to produce a series of decisions that would give an optimal reward. These systems work using deep neural networks for reinforcement learning (DRL) where an agent explores the space of possible strategies in an environment and receives feedback (positive or negative) on the outcome of choices it makes. Given a particular domain, exploration allows the agent to form a strategy, called policy, that allows it to generate and follow a sequence of actions to maximize its payoff (see

the "Related work on computational models" section for related work in this area). The last few years have witnessed a rapid growth of research and interest in the domain of deep reinforcement learning (DRL) due to the significant progress in solving RL problems (Arulkumaran, Deisenroth, Brundage, & Bharath, 2017). DRL has been applied to a wide variety of disciplines ranging from game playing, to robotics to dialogue systems (Choi et al., 2017; Kraska, Beutel, Chi, Dean, & Polyzotis, 2018; Levine, Finn, Darrell, & Abbeel, 2016; Mnih et al., 2015; Silver et al., 2018; Williams, Asadi, & Zweig, 2017).

Self-driving cars that are poised to be deployed in the near future increasingly employ AI-based object recognition software, and military autonomous systems may be called to make decisions that cost civilian lives. These systems are more opaque than their predecessors since they rely almost exclusively on learning form data in order to shape their conclusions and behavior, as opposed to model-based systems that are more understandable and often have algorithms that provide formal guarantees of performance. The inner logic and reasoning of DRL systems are opaque and difficult to be understood even by their own designers.

The opacity of these systems is becoming increasingly problematic and therefore there is increasing clamor for transparency in the machine learning community as well (Kim, 2015; Ribeiro et al., 2016). This need for system transparency has resonated with policy makers as well. For example, new regulations in the European Union propose that individuals affected by algorithmic decisions have a *right to explanation* (Goodman & Flaxman, 2017).

In this chapter, we discuss the intertwined notions of trust and transparency in the context of human-agent teamwork. Here agents can be autonomous systems that engage in sequential decision-making processes and control, can be computational processes that comprise robotic components, such as a vision system for the robot, or provide information and suggestions to humans. We will present a review of trust and transparency in the human factors literature emphasizing the need to consider transparency in all dimensions of the trust construct, and then we will turn our attention to the challenges of transparency for deep learning algorithms. We will then present our own work on transparency for DRL, followed by conclusions and open research problems.

Factors affecting trust in automation

It is generally agreed in the psychological literature that trust is best conceptualized as a *multidimensional psychological attitude* involving beliefs and expectations about the trustee's trustworthiness derived from experience and interactions with the trustee (Jones & George, 1998). Although in

the literature, the number and concepts in the trust dimensions vary (Jian, Bisantz, & Drury, 2000; Lee & See, 2004; Madsen & Gregor, 2000) point out a broad consensus on three dimensions they label Purpose (what the automation is supposed to do), Process (how automation goes about fulfilling its purpose), and Performance (actual performance). Moreover, it has been found that trust is dynamic (Lee & Moray, 1994; Lee & See, 2004; Nam, Walker, Lewis, & Sycara, 2017; Nam, Walker, Li, Lewis, & Sycara, 2019) and a human's history of interaction with automation affects future behavior indirectly through changes in trust.

The most important factors that affect trust in automation are:

(a) *System reliability*: Prior literature has provided empirical evidence that as automation reliability declines, human trust declines, and vice versa (Hancock et al., 2011; Moray & Inagaki, 1999; Moray, Inagaki, & Ito, 2000; Parasuraman & Manzey, 2010; Riley, 1994). Additional studies have shown that imperfect (unreliable) automation can have adverse effects on reliance and compliance (Dixon & Parasuraman, 2006; Meyer, 2004), and overall system performance (de Visser, Parasuraman, Freedy, Freedy, & Weltman, 2006; Dzindolet, Peterson, Pomranky, Pierce, & Beck, 2003). Moray et al. (2000) showed that declining system reliability can lead to systematic decline in trust and trust expectations, and most crucially, these changes can be measured over time.

(b) *System predictability*: Research has shown that when people have prior knowledge of faults, these faults do not necessarily diminish trust in the system (Lewandowsky et al., 2000; Riley, 1994), possibly due to the fact that knowing how the automation may fail reduces the uncertainty and consequent risk associated with use of the automation.

(c) *System intelligibility and transparency*: Prior research on trust in automation found that providing human operators with information related to the reliability of an automated tool promoted more optimal reliance strategies on the tool (Lyons & Havig, 2014). Further, information related to the limitations of an automated tool aids in trust recovery following errors of the automation (Choudlechova, 2016). Given that reliability of performance is the biggest determiner of trust (Hancock et al., 2011), providing additional information about performance such as knowledge of results (Beck, Dzindolet, & Pierce, 2007; Dzindolet, Pierce, Beck, & Dawe, 2002) or confidence judgments (Dadashi, Stedmon, & Pridmore, 2013) has also been widely used to increase trust through greater transparency. The human factors literature has paid less attention to the Purpose and Process components of trust, most frequently assuming Purpose to be evident. Attention to Process has concentrated mostly on developing displays

for mission-based systems to show the state of the system in more transparent ways to allow the human to understand the system (Wang, Pynadath, & Hill, 2016) or to improve trust calibration (Lyons et al., 2016).

(d) *Level of automation*: Another factor that may affect trust in the system is its level of automation (i.e., the degree to which the system acts on its own). Since higher levels of automation are more complex, *thus potentially more opaque* to the operator, higher levels of automation are frequently (Amato et al., 2011; Calhoun, Draper, & Ruff, 2009; Kira & Potter, 2009; Nam et al., 2019) found to engender less trust. To overcome this deficit transparency becomes more important when the system is more autonomous (Oh et al., 2015). In summary, prior literature indicates that a high level of autonomy may cause the operator to undertrust the system. These implications are echoed in the distrust and clamor for transparency for systems based on Deep Neural Networks.

Trust and transparency in human-autonomy teaming

Since transparency is an important ingredient of trust, and since trust is multidimensional, we believe that work on transparency should span all trust dimensions, in particular purpose, process, and performance. Additionally, since factors such as Degree of Autonomy modulates trust, its influence should also be studied. Transparency can be viewed as the degree to which automation conveys the basis of its behavior. Parasuraman, Sheridan, and Wickens (2000) proposed a model characterizing automation as a sequence of stages which were reduced to two by Wickens (2018): (1) situation assessment and (2) action choice and execution. Transparency effects can usefully be organized by these stages of autonomy (Wickens, 2018) and dimensions of trust (Chen et al., 2014).

Fig. 1 presents constituents of transparency that we briefly discuss in this section in the context of the trust dimensions of performance, purpose, and process (also depicted in the figure). Additionally, the figure sketches our explanation model for DRL agents and shows its relationship with transparency and trust.

Transparency of performance (execution stage) such as system reliability is often assumed to be directly observable to users yet may be perceived inaccurately. Biros, Daly, and Gunsch (2004), for example, found trust and reliance to be influenced by cover stories. Other studies (Beck et al., 2007) have found supplying knowledge of results to lead to better trust calibration and improved reliance. Annotating decisions with confidence judgments is another widely used technique for providing greater insight into system performance. This has been done in a variety of ways (e.g., by

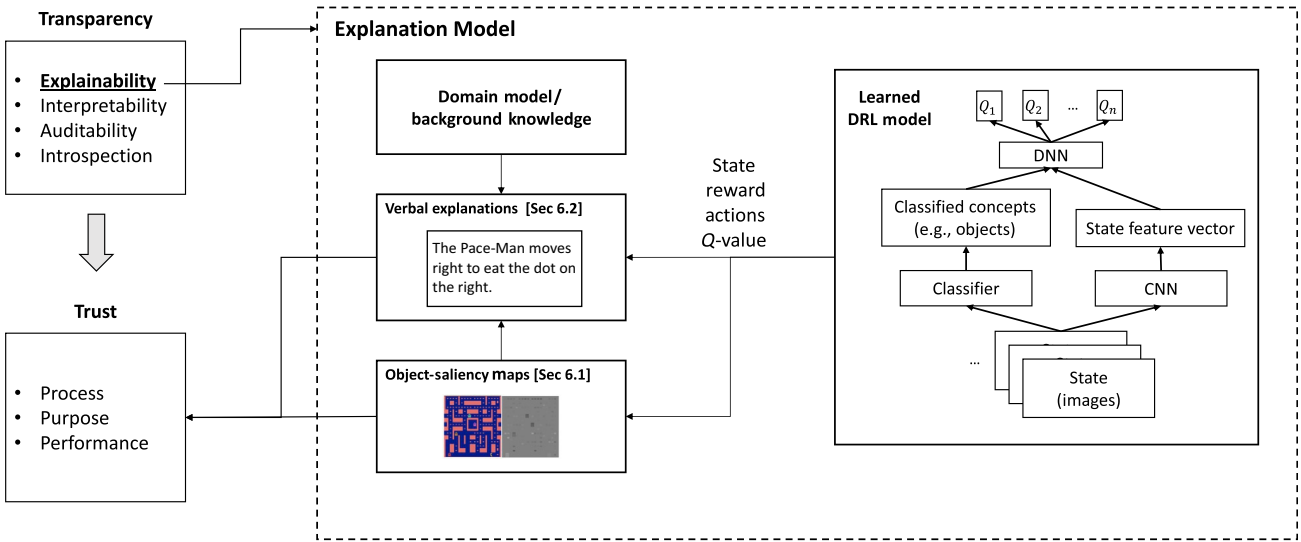


FIG. 1 A block diagram of the visual and verbal explanations offered by our model of DRL agents and its relationship with transparency and trust. The model is reported in the "Object-saliency-based explainability in DRL" section.

providing probabilities of success) (Dadashi et al., 2013; Lyons et al., 2016; Wang, Pynadath, et al., 2016), confidence in detection (Dadashi et al., 2013), or expected reward from selected robot action (Wang, Pynadath, et al., 2016).

Transparency of purpose: For preprogrammed automation the system's purpose is static and known a priori (purpose transparency was baked in) to the user. As systems become increasingly autonomous and thus learning and adaptive, their *intent/purpose* may change over time and may depend on environmental and social factors. Therefore, the system's intent must be communicated to the human to help the human understand and anticipate system activities. To do this, the system must be able to *introspect* (i.e., be aware of its own goals, preferences, and what factors bring change to them). For example, a system might change its intent in response to a perceived change in intent of the human with whom the system is teaming.

In experiments with humans, the system's purpose is almost always informally conveyed through instructions or the demand characteristics of an experiment, however, when explicitly manipulated, as in Sadler et al. (2016) effects on trust and reliance were found. More commonly, experiments involving transparency of automation do not discriminate between stages of automation or form of explanation. They instead mix categories often in an additive manner such as Verberne, Ham, and Midden (2012) which started with an unsupported recommendation in the control condition (execution stage), increased transparency of performance by conveying expected risk in the second condition, and added an explanation (situation assessment stage) involving process in the third.

Transparency of process (situation assessment stage): While trust and the decision to rely on or comply with an autonomous agent may depend on knowledge of its purpose or reliability, the reciprocal interactions needed for teamwork depend crucially on the ability to predict and coordinate behavior (i.e., knowledge of process). In the current human factors literature, transparency of process involves conveying information to the human about the system's decision-making process to support operator situation awareness (Chen et al., 2014; Lee & See, 2004). Increased information, such as explanations by an agent, may have a complicated effect on operators' workload. Although a human may need to process more information from an agent with high transparency, this information might be desired in order to achieve better performance. Therefore, additional information may help operators to understand the intentions and behaviors of an agent and correct their expectation and trust in it without increasing workload (Chen & Barnes, 2014). Recent work has confirmed the above argument in multiunmanned-vehicle control task scenarios (Mercado et al., 2016).

The reasoning (transparency of process) of preprogrammed automation, though not necessarily known to the user a priori, was easier to

convey. For example, a rule-based system can display its set of rules to allow a human to see its reasoning. A very early example of this was Teiresias (Davis, 1978) that explained its reasoning by showing the rule chain that led to its conclusions. Even for more sophisticated systems that may use (e.g., Bayesian processes, or decision trees), the system's reasoning is relatively easy to convey.

The broad use of purely data-driven AI/ML techniques raises additional challenges that necessitate a more refined look at the concept of transparency and trust. If the data are biased, the system may come to the wrong conclusions although its algorithm operates correctly. In a well-known example of such behavior, the data used by an image classification system inadvertently included images of wolves only in snowy backgrounds (Ribeiro et al., 2016). When the system was given an image of a wolf in a nonsnowy background, it misclassified it as a dog. Only after the system designers searched for the cause, they discovered it was due not to a faulty algorithm but to biased data. In this particular case, recognizing the mistake was easy (finding the cause was not so easy); however, in other situation (e.g., system decisions on recidivism or extending credit), even recognition of faulty system conclusions is not easy. Besides data biases, adversarial data manipulation is also a major source of concern. Convolutional Neural Networks (CNNs), used for image classification, are susceptible to adversarial manipulation that causes them to misclassify images that seem only imperceptibly perturbed to a human (Szegedy et al., 2014). Their paper shows examples of images that were slightly perturbed by adversarial noise (imperceptible to the human eye) including a yellow school bus and a white dog that were both misclassified as ostriches! Thus, for these systems to be trusted, they should be transparent in terms of being *auditable* as to possible biases in the data and robust to adversarial manipulation. This notion of auditability as a characteristic of transparency has also been found in the human factors literature where increases in compliance were found from providing means of *verifying* automation decisions (Bliss, Jeans, & Prioux, 1996). Making progress toward discovering and correcting biases and also discovering and protecting systems from adversarial data manipulation are open research problems in the AI/ML community.

Besides auditability of decisions, transparency can also be considered in terms of *interpetability* and *explainability*. Interpretability enables understanding of the mechanics of the algorithm, whereas explainability enables producing post hoc explanations or predictions of the model without necessarily elucidating the mechanisms by which the model works (Lipton, 2018). Both interpetability and explainability, like trust and transparency, also lack formal definitions and characterizations. There is increasing interest in the AI community of providing those and developing techniques for understanding them. In the "Object-saliency-based

explainability in DRL" section, we present our own work on transparent/explainable DRL using object-based saliency maps.

Types of explanations

Explanations are conventionally categorized as:

- Teleological or functional (WHY): Explanations for transparency may be teleological (why) which may be incomplete, conveying simply relevant features of the situation contributing to an automation decision (Wright, Chen, Barnes, & Hancock, 2016) or complete in supplying both features and logic behind the decision. Teleological explanations are preferred by humans (Lombrozo, 2006) who are strongly predisposed to attribute causality (Thines, Costall, & Butterworth, 1991) even in its absence. We have strong predispositions to attribute causality as shown in Michotte's classic experiments reported by Thines et al. (1991) in which subjects interpreted the motions of abstract geometric objects as elastic collisions or animate behaviors such as chasing or leading depending on timing and proximity of objects. Explanations of more abstract behavior typically appeal to causes as well, with knowledge of general patterns constraining which causes are judged probable and relevant (Lombrozo, 2006) in much the way timing and proximity do for perception. When explanations are judged for quality, the presence of a general pattern is typically preferred to probability judgments alone while this pattern is required to be both general and relevant (appropriate to the actors and event). So, for example, an explanation that "a robot turned left to avoid a collision" would be preferable to an explanation that the turn was taken to increase the robot's long-term probability of reaching a goal, although both might be correct. Similarly, citing the presence of a wall (often present preceding turns) or distance from door (a serendipitous feature of the robot's history) as cause would fail the probable and relevant test since walls were often present when turns did not occur (not relevant) and distance from door does not reference a humanly known general pattern for evoking turns (not probable). Moreover, this explanation implicitly includes the domain knowledge that collisions are undesirable, that the robot had learned to avoid collisions, and that collision is threatened by a trajectory leading the robot into the wall.
- Efficient or mechanistic (HOW): The primary source of an action "the robot's programming caused it to turn." Although this explanation has implicit the state of the world at the time of the turn, it is not relevant or useful to the human. Mechanistic explanations can drill down by chaining together a series of intermediate steps so the answer might be elaborated to include that what the robot sensed caused its

programming to produce an observed preference value leading it to turn.

- Formal or constitutive (WHAT): Expressing a necessary aspect of an object or event or conveying part-whole relationships. For example, the What explanation of the robot taking a turn may be, "The robot took 5 steps forward, turned its orientation to the left, took 5 steps forward."

From this perspective, teamwork involving robots trained through DRL presents a dilemma. An understanding of process needed to predict behavior remains hidden within the layers of the network. The accuracy of our predictions of robot actions, therefore, depends on the correspondence between our own attributions of causality and the policies the robot has learned. As summarized by Lombrozo (2006) these attributions are predominantly based on probability and relevance. In the case of the game Ms. Pac-Man, for example, relevance would revolve around the rules of the game in which Pac-Man gains points by eating pellets and avoiding ghosts. The DRL player, however, does not benefit from predefined relevance and learns from scratch to find the optimal policy through trial and error. Therefore while the resulting policy has been trained to win the game, it will not necessarily choose the same actions as a human, making attribution difficult. Recently, the AI community has studied the effects of including domain knowledge in the form of relevant features. Although this may limit generality of learning, since some features may be predetermined, it may help system explainability.

In the context of teamwork, not only should the system be transparent to the user, but also *the human should be transparent to the system*. In other words, the system should be able to make inferences about human intent (purpose), human beliefs, and how these beliefs may lead to actions (process) prediction of human actions (performance). Such understanding on the part of the system is equivalent to the system formulating a *Theory of Mind* of the human that would allow it to adapt to human's behavior thus improving teamwork performance. Such a Theory of Mind will allow the agent to better understand why a human may be taking a particular action, understand when the human may have false beliefs (e.g., due to lack of information to environment changes), and inform the human of missing information that the agent may have, so as to correct the human's false beliefs.

Background on RL and DRL

Reinforcement learning (RL) solves the sequential decision-making problems by learning from experience. In RL, an agent interacts with an environment ε over discrete time steps and receives feedback (rewards)

on the outcome of choices is makes. Given a state, the agent selects actions in order to maximize future rewards. In the RL setting, the problem can be modeled as a Markov Decision Process (MDP) represented by the five tuple (S, A, T, R, γ), where S is the state space, A is the action space, $T(s'|s, a)$ is the state transition probability function, $R(s, a)$ is the reward function, and $\gamma \in [0, 1]$ is the discount factor. Due to stochasticity, a policy $\pi: S \rightarrow A$ maps every state to a distribution over actions. In the time step t, the agent receives a state $s_t \in S$ and selects an action $a_t \in A$ according to its policy π, where S and A denote the sets of all possible states and actions, respectively. After executing the action, the agent receives a scalar reward r_t and enters the next state s_{t+1}.

The goal of the agent is to choose actions to maximize its rewards over time. In other words, the action selection implicitly considers the future rewards. The total discounted return is defined as $R_t = \sum_{\tau=t}^{\infty} \gamma^{\tau-t} r_\tau$ where $\gamma \in [0, 1]$ is a discount factor that trades off the importance of recent and future rewards.

Policy-based methods directly model the policy (Williams, 1992), while in value-based RL methods, the action value (a.k.a., Q-value) is commonly estimated by a function approximator, such as a deep neural network (Mnih et al., 2015). The actor-critic (Sutton & Barto, 1998) architecture is a combination of value-based and policy-based methods.

The value function $V^\pi(s_t)$ is the expected discounted sum of rewards by following policy π from state s_t at time t, $V^\pi(s_t) = E[\sum_{i=0}^{T} \gamma^i r_{t+i}]$. Similarly, the Q-value (action-value) $Q^\pi(s_t, a)$ is the expected return starting from state s_t, taking action a and then following π. The Q-value function can be recursively estimated using the Bellman equation $Q^\pi(s_t, a) = E[r_t + \gamma \max_{a'} Q(s_{t+1}, a')]$ and π^* is the optimal policy which achieves the highest $Q^\pi(s_t, a)$ over all policies π.

In the RL community, the action value function Q is computed via a typically linear function approximator, but in DRL, the function approximator is a (nonlinear) neural network. Deep neural networks have been recently applied in RL to achieve human-level control policies in various challenging domains. The rich representations given by a deep neural network improve the efficiency of RL at the expense of requiring a vast amount of training data. If a high-fidelity simulator is available (e.g., Zhu et al., 2016, which describes the appearance of the real world as closely as possible), such data can be easily generated.

The advantages of DRL are (1) the features do not need to be handcrafted but are learned during training, (2) deep neural networks have shown superior performance in many challenging domains, and (3) the algorithm is model free (i.e., it solves the RL task directly using sample data without explicitly estimating the reward and transition dynamics).

The challenges of this approach are (1) large amounts of data are needed and (2) RL with nonlinear function approximator as a neural network could be unstable (or even diverge). This is due to different causes, such as the correlation present in the sequence of observations, the fact that small updates to the value function may significantly change the policy, and the existence of correlations between action values (Q) and the target reward values.

DRL differs from image classification in a number of ways: In classification, for each image, the system has to predict its label, whereas for DRL, the system must learn a function that maps state-value pairs to Q-values that capture the dynamics of the system over time. If the state in DRL is a single image, the dynamics of the system are not captured. Therefore, a *state must be represented by a sequence of images*. Current works on DQNs for Atari games use only four previous screens as input to learn the reward for each state.

Although DRL is much more challenging than classification, in the past few years, a new variant of DRL has been developed and tested mainly on Atari games (Mnih et al., 2013) where the states are the input screens of the game. The agent chooses an action from the possible control actions (e.g., up/down/left/right). After that, the agent receives a reward (how much the score increases or decreases) and the next image input. The DRL agent, only reasoning on the image pixels and the game scores, shows performance comparable or higher to an expert human player (Mnih et al., 2015). However, producing explanations has been out of scope for DRL until recently but interest in interpetability and explainability is increasing (Lipton, 2018).

Developing methods to enable autonomous agents to be transparent is very challenging, because ease of transparency seems to be inversely proportional to agent sophistication. However, DNNs are extremely opaque (i.e., they cannot produce human understandable accounts of their reasoning processes or explanations). Therefore, there is a clear need for DRL agents to dynamically and automatically offer explanations that users can understand and act upon.

Related work on computational models

The literature on DRL is fast increasing. Here we briefly review work that is most relevant to this chapter. Multiple DRL algorithms have been developed to incorporate both on-policy RL such as Sarsa (Rummery & Niranjan, 1994), actor-critic methods (Sutton & Barto, 1998) and off-policy RL such as Q-learning (Watkins, 1989), or combination of RL and experience replay memory (Mnih et al., 2015; Riedmiller, 2005). A parallel RL paradigm (van Hasselt, Guez, & Silver, 2016) has also been proposed to

reduce the heavy reliance of DRL algorithms on specialized hardware or distributed architectures. The deep *Q*-network (DQN) model proposed in Mnih et al. (2015) combines *Q*-learning with a flexible deep neural network. More specifically, recent work has found outstanding performance of DRL models on Atari 2600 games using only raw pixels to make game control decisions (Mnih et al., 2015). DQN can reach human-level performance on many of Atari 2600 games. However, DQN suffers from substantial overestimation in some games. van Hasselt et al. (2016) thus proposes double *Q*-learning algorithm that can be generalized to work with large-scale function approximation. A dueling network architecture (Wang, Schaul, et al., 2016) has been proposed to decouple the state-action values into state values and action values. The experiments of Mnih et al. (2016) show that the actor-critic (A3C) method surpasses the current state of the art in the Atari game domain. In contrast to *Q*-learning, A3C is a policy-based model that learns a network action policy. However, for game settings with many objects where each object has a different role in reward computation, A3C does not perform very well. Therefore, Lample and Chaplot (2017) propose a method that augments performance of RL by exploiting game feature information.

In human-robot interaction and teaming, robots and humans must adapt to one another. This requires the robots to maintain a computational cognitive model of their human coworkers in the task environment. There is a body of work on making intelligent robots able to (1) adapt to physical human behaviors (Liu et al., 2016), (2) infer human's intent (Dorsa Sadigh, Sastry, & Seshia, 2017; Hadfield-Menell, Russell, Abbeel, & Dragan, 2016), and (3) shape the way how humans reason about robots (Huang, Held, Abbeel, & Dragan, 2019; Pezzulo, Donnarumma, & Dindo, 2013; Zhou, Hadfield-Menell, Nagabandi, & Dragan, 2017). Previous research assumes the human is a perfect collaborator and uses Bayesian inference to predict the human's next goal in order for the robot to adapt accordingly (Liu et al., 2016). Subsequent work by Hadfield-Menell et al. (2016) extends the scope to situations where robots do not know the human operators' reward function and need to learn it over the course of interaction. Inverse RL (Ng, Russell, et al., 2000; Ramachandran & Amir, 2007) approaches this problem from a passive learning perspective like learning from demonstrations offline, while in cooperative inverse RL (Hadfield-Menell et al., 2016), the human helps the robot learn by making his/her behavior more transparent. Alternately, by maintaining a model of human mental states, the robot may take informative actions to communicate its goals (Pezzulo et al., 2013), reward functions (Huang et al., 2019), or confidence levels (Zhou et al., 2017) to humans.

Ideally, we would like to develop techniques for DRL explanations that explain (1) why an action was taken (teleology), (2) why this action was

taken as opposed to other ones (counterfactuals), and (3) why a sequence of actions was taken (plans).

Object-saliency-based explainability in DRL

Most current work around interpretability in deep learning is based on local explanations (i.e., explaining network predictions for specific input examples; Lipton, 2018). Saliency maps are used to generate local explanations. Saliency maps generally use gradient-like information to identify salient parts of the image and highlight important regions of the input that influence the output of the neural network. Zahavy, Ben-Zrihem, and Mannor (2016) use the Jacobian of the network to compute saliency maps on a Q-value network. Perturbation-based saliency maps using a continuous mask across the image and also using object segmentation-based masks have been studied in the context of DRL (Greydanus, Koul, Dodge, & Fern, 2018). Our object-based saliency method (Iyer et al., 2018; Li, Sycara, & Iyer, 2017) that we present in the "Visual explanation" section belongs to this category.

In contrast, global explanations attempt to understand the mapping learned by a neural network regardless of the input. There are additional difficulties with global explanations since there are problems of generalization and memorization. Recent findings suggest that DRL agents can easily memorize large amounts of training data with drastically varying test performance and are vulnerable to adversarial attacks (Huang, Papernot, Goodfellow, Duan, & Abbeel, 2017; Zhang, Ballas, & Pineau, 2018; Zhang, Vinyals, Munos, & Bengio, 2018). We have explored interpetability using global explanations in Annasamy and Sycara (2019). Our method aims to understand aspects of the input space (images) that are captured in the latent space across inputs. In particular, given a particular action (e.g., Pac-Man goes left) and expected returns (e.g., rewards of 50 points), our method tries to understand visually which would be the features of the corresponding states. This could help in implicitly identifying underlying relations between the entities in these states (e.g., the network might have learned that presence of a ghost in the vicinity of Pac-Man would be a hard thing that must be avoided). However, our results suggest that the features extracted by the convolutional layers are extremely shallow and can easily overfit to trajectories seen during training, rather than generalize to trajectories that would be in conformance with real relations among entities.

In the following section, we present a short description of our recent work on explainability in DRL, using the game of Pac-Man. In this work, we have explored bot visualization and text as means of producing local explanations of agent (Pac-Man) behavior.

Visual explanation

In Atari games, the DRL network is supposed to *implicitly* learn all relevant features that capture the agent's reasoning and behavior. However, this does not aid explainability to humans. Since humans recognize objects, our idea was to enhance the neural network with object recognition ability as a way to get a handle on relevance of different objects for the DRL agent's decision making. We first used template matching, a computer vision technique (Brunelli, 2009) to recognize objects in images. The technique works by taking a template image (the patch) and sliding it through a source image (up to down, left to right) and calculating the current source image similarity to the template image. After object recognition, we used *object channels* to incorporate features of objects in the input images to the DRL network.

The object channels as well as the original image are given to the network as input. The network outputs (predicts) Q-values for each action. This method can be used incorporated into different existing DRL algorithms, such as DQN or ACC.

Our method to provide transparency for Deep Neural Networks is called *object saliency maps*. A saliency map highlights regions of the input that, if changed, would most influence the output (Simonyan, Vedaldi, & Zisserman, 2014). Saliency maps is an ex post facto local explanation method. This means that after the network has been trained, query images can be input that may help the user predict what the network was paying attention to (i.e., "explain") so as to help the user understand the system's reasoning and also possibly predict the system's next decision. Object saliency maps provide visualization of the decisions made by RL agents. These visualizations aim to be intelligible to humans. To generate intelligible visualizations that would help with explanations of DQN agent behaviors, we need to determine which pixels the model pays attention to when making a decision (Simonyan et al., 2014). Another interpretation of computing pixel saliency is that the value of the derivative indicates which pixels need to be changed the least to affect the Q-value.

However, pixel-level representations are not intelligible to people. Fig. 2A shows a screenshot from the game Ms. Pac-Man. Fig. 2B is the corresponding pixel saliency map produced by an agent trained with the double DQN (DDQN) model. The agent chooses to go right in this situation. Although we can get some intuition of which area the DRL agent is looking at to make the decision, it is not clear what objects the agent is looking at and why it chooses to move right. On the other hand, Fig. 2 makes more intelligible the objects that the DQN is paying attention to. To understand the influence of objects on agent decisions, we need to rank the objects in a state s based on their effect on $Q(s, a)$. In the game of Pac-Man, there are static pellets/dots that Pac-Man eats to get points. There are ghosts that

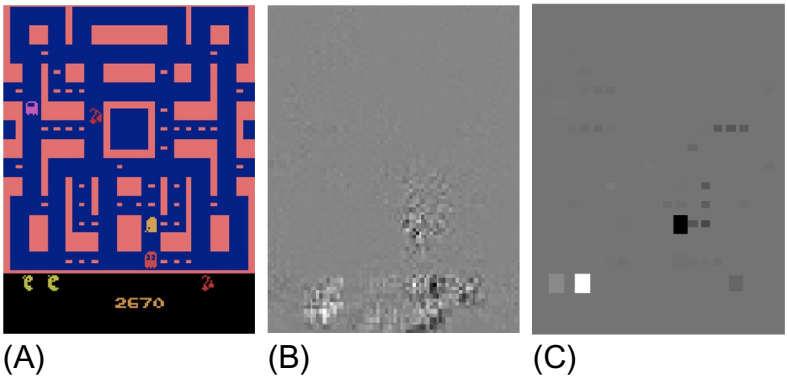


FIG. 2 An example of original state, corresponding pixel saliency map, and object saliency map produced by a double DQN agent in the game "Ms. Pac-Man." (A) Screenshot of the state; (B) pixel saliency map; (C) and object saliency map.

chase and eat the Pac-Man, which finishes the game and gives a large number of negative points. There are super-pellets and cherries that appear dynamically, and if Pac-Man eats them, then it gets more points. Moreover, if Pac-Man eats a cherry, then the ghosts become edible for some time, so if the Pac-Man manages to eat an edible ghosts, it gets a very high reward (many points).

For each object O found in s, we mask the object with background color to form a new state s_o as if the object does not appear in this new state. We calculate the Q-values for both states, and the difference w of the Q-values actually represents the influence of this object on $Q(s, a)$. So, if w is positive, the object has a positive influence which means *the object gives positive future reward to the agent* (the positive objects are shown in dark in the saliency maps). Negative w represents "bad" object since after we remove the object, the Q-value gets improved.

Fig. 2C shows an example of the object saliency map that clearly shows which objects the model is paying attention to and the relative importance (via shading) of each object.

Human experiments

In order to test whether the object saliency map visualization can help humans understand the learned behavior of Pac-Man, we performed an initial set of experiments. The goals of the experiment were to (1) test whether object saliency maps contain enough information to allow humans to match them with corresponding game scenarios, (2) test whether participants could use object saliency maps to generate reasonable explanations of the behavior of the Pac-Man, and (3) test whether object saliency maps allow participants to correctly *predict* the Pac-Man's

next action. This requires a deeper causal understanding of what may influence the Pac-Man in his decisions.

Experiments were conducted in a graduate and undergraduate human factors class. The 40 participants were approximately equally divided by gender and between 20 and 29 years old. The matching and prediction tasks were presented sequentially over a period of 30–45 minutes.

Matching task. In each trial, the participants are shown twice, a 5-second video clip of Pac-Man gameplay generated by O-DDQN. During the video clip, Pac-Man decides and takes particular actions. The last decision made produces the crucial movement of the clip (e.g., Pac-Man moves right), with the clip ending just after the crucial movement. Three frames from the object saliency map are then shown to participants (see Fig. 3B). The center frame is the frame where the Pac-Man makes the crucial decision and the other two are frames from before and after that moment. In the task, participants are asked to judge whether the saliency maps accurately represent the video they just saw. In the matching cases, the saliency maps indeed were generated from the video clip the participant saw. In the

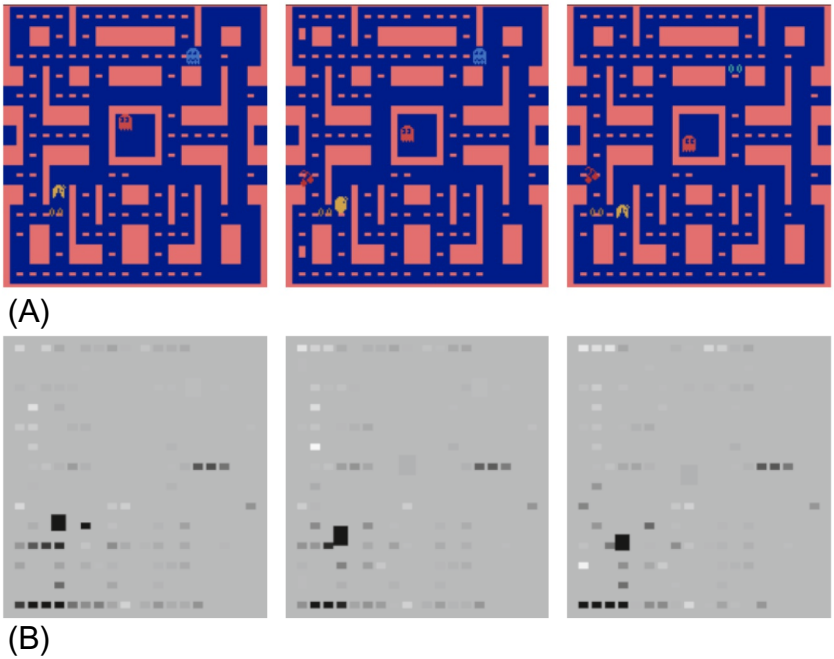


FIG. 3 An example of the stimulus materials participants saw on trial 9 of the prediction task; 75% participants in the screenshot group thought Pac-Man would go left to eat the cherry at the left side; 60% participants in the object saliency maps group predicted the Pac-Man would keep going down for the dark elements (the pellets) below. (A) Screenshots and (B) object saliency maps.

nonmatching cases, the three saliency map frames were generated from a different video clip. In distractor/nonmatching clips, the Pac-Man occupies the same area of map as in the target video but makes different movements. This is done to avoid the case where the participants solely focus on the location of the Pac-Man as a matching criterion, disregarding the movements and environmental factors.

Following the match decision, if the participants' answer is "match," they are asked to give an explanation for the Pac-Man's movements based on the video and saliency maps. In other words, participants are asked to provide a teleological explanation explaining "why" Pac-Man acted as she did. For example, "Pac-Man moved up to eat more energy pellets while avoiding the ghost coming from below."

The matching task consisted of 2 training trials and 20 test trials, half (10 trials) presenting matched video and saliency maps, the other half presenting nonmatched pairs in a single randomly ordered sequence. Dependent variables were correctness of matches and agreement between explanations and saliency maps.

Prediction task. In each trial, the participants are shown a video clip not used in the matching task. Each clip ends at the point where the Pac-Man must choose a crucial move. The participants are divided equally into two experimental conditions. In the screenshot condition, after the video clip, participants see three actual screenshots from the video ending before the crucial move is taken. In the object saliency map condition, the participants see three object saliency map frames (corresponding to the screenshot frames) after viewing the video clip (see Fig. 3). At the decision point in the third frame, Pac-Man's choices (up, down, left, right) may be limited by barriers indicated on the response forms. Participants are asked to predict Pac-Man's movement among the feasible directions based on the three previous frames (screenshots or saliency maps) and then give an explanation for their prediction which includes their judgment as to which elements of the game influenced the Pac-Man's decision (indicating these elements by circling them on a hardcopy of the screenshot or saliency map) and explain why Pac-Man made that decision.

The prediction task consisted of 2 training trials and 10 test trials. Each participant was assigned to either the screenshot group or the saliency map group. Dependent variables include whether predictions were correct, and whether explanations were consistent with the saliency maps.

Results. The average matching accuracy of the participants was 61.0% ($SD = 14.0\%$). A learning effect was found with participants having higher accuracy (65.5%) in the last half of the trials than the first half (56.5%) ($t(39) = 3.10$, $P = .04$). Comparing hit and false alarm rates, participants reported more "matches" when the video and image stimulus matched ($t(18) = 2.91$, $P < .001$). If the 40 participants are treated as a binary classifier and the percentage of their answers as an output score, a receiver

operating characteristic (ROC) curve (Fawcett, 2006) can be plotted for true-positive rates versus false-positive rates across a range of threshold parameters (as Fig. 4 shows). The area under the curve is 0.81 which indicates a good classification between matching and nonmatching situations. In summary, human participants were able to link the object saliency maps with the game scenarios.

For the more difficult prediction task, there was no significant difference in accuracy between the object saliency map group (58.0% ± 12.8%) and the control group (56.5% ± 10.4%). However, the main effect of trials ($F(9, 342) = 11.18$, $P < .001$) and the interaction between trials and groups ($F(9, 342) = 2.72$, $P = .005$) were both highly significant suggesting that characteristics of the trials had a strong influence on performance. Thus we conducted a simple effect analysis to examine differences among the 10 test scenarios (see Fig. 5). Results show that the screenshot group has high predictive accuracy in test 2 ($P = .027$), while the object saliency map group has higher accuracy in tests 3 and 9 ($P = .007$, $P = .025$). Those three trials can help provide a deeper insight into the mechanism of how object saliency maps could help humans understand Pac-Man's learned behavior.

Trial 9 provides a good example (see Fig. 3). The Pac-Man goes down and faces a dilemma whether to turn left or keep going down; 60% participants who saw the object saliency maps predicted Pac-Man would continue going down, and objects circled and explanations focused on the dark elements or dots below. In contrast, 75% participants in the screenshot group predicted Pac-Man would go left, and all except one of their

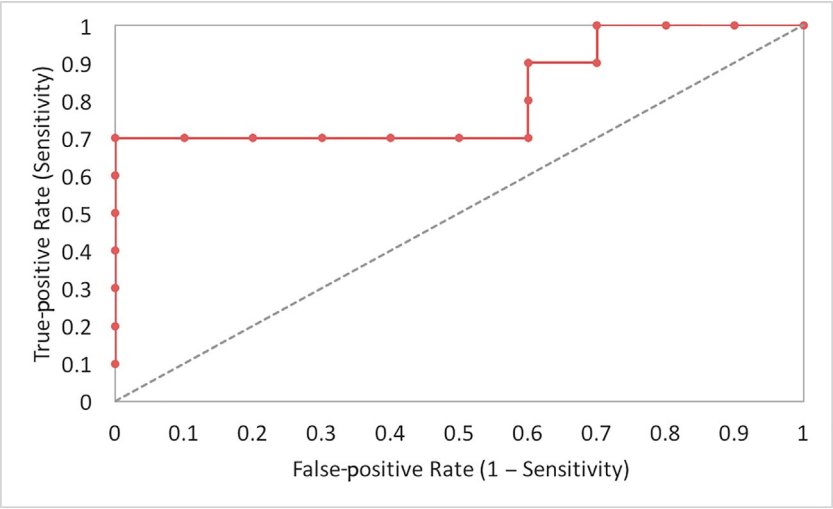


FIG. 4 ROC curve of the matching task, AUC = 0.81.

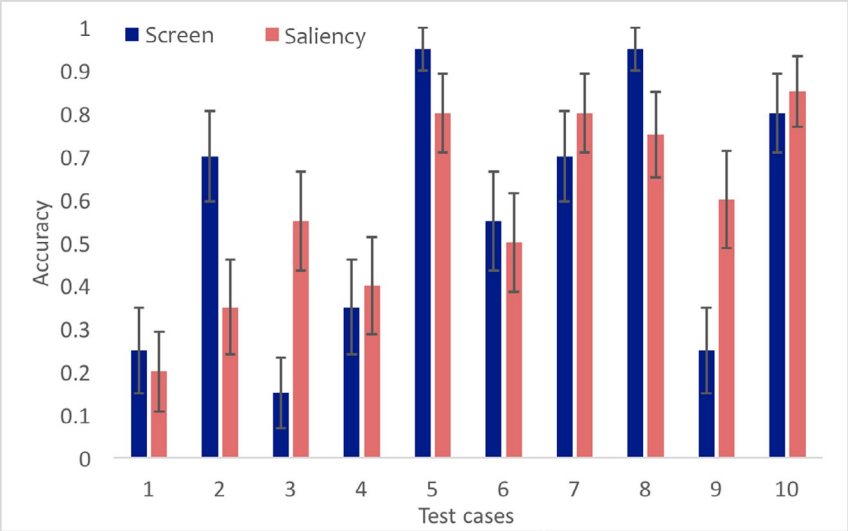


FIG. 5 The mean accuracy of participants in each test cases of the prediction task. Error bars are one standard error from means.

explanations mentioned the cherry at the left side. In the scenarios generated by O-DDQN, the Pac-Man did go down for the dots. Trial 9 is a typical case in which there are multiple influencing elements and it is hard for humans to predict Pac-Man's behavior based on information from the game screen and their own knowledge of the rules and ideas about gameplay. However, displaying object saliency enables us to directly identify those objects affecting the program's decision. In other situations when the Pac-Man may make what we judge to be suboptimal choices (e.g., the Pac-Man chose a wrong direction and was eaten by a ghost), an object saliency map could be crucial to helping users and system developers understand some of the rationale behind such behaviors and the saliency map can be used as a debugging tool.

Natural language explanation

To present information from object saliency maps in a form more consistent with human reasoning, we developed algorithms to generate relevance-sensitive textual explanations for DRL networks. We developed focused verbal explanation models of the DRL system in which verbal explanations referred to objects expected to be most important in influencing the agent's selection of next action. Prior work in generating explanations for RL systems has been limited with direct translation of policies (Hayes & Shah, 2017) restricted to simple cases while more complex

environments such as Atari games (Ehsan, Harrison, Chan, & Riedl, 2018) have relied on human attributions. Our approach is intermediate basing explanation on internal information but restricting its expression to a teleological form with relations satisfying human criteria for relevance. Our initial rule-based model was constructed based on prior knowledge of the Ms. Pac-Man game and its rules and consisted of a collection of allowable expressions. Object salience and valence were used to match information encoded in a saliency map with a verbal explanation. Although the rule-based model is capable of generating reasonable explanations, it lacks the generalizability and flexibility that a neural network might provide in addressing unexpected situations. To overcome the limitations of the rule-based model, a learning model was also developed in order to (1) be more generalizable in terms of game episodes, (2) be more tolerant to input noise, and (3) provide explanation for future states and actions, which could help the user predict and plan. Game image, agent position map, and object saliency map act as the input for both models. Data generated by the rule-based model were employed to *train the learning model*, which consisted of two parts: an encoder for feature extraction and a decoder for generating the explanation in natural language using an attention mechanism. The challenge for the learning model lies in extracting distinguishable features from DRL systems, especially from images with high structural similarity, such as a game image with fixed board or fixed map, which cannot be fully solved by current networks.

Rule-based verbal explanation model

The input of the rule-based model is game images and corresponding object saliency maps. The output is a verbal explanation for the given game state. The predefined rules and workflow used in this model were designed based on literature and a previous user study (Iyer et al., 2018). The design of the rule-based system was based on a number of assumptions. (1) *Object priority*: Since different objects in the game give Pac-Man different rewards (e.g., being eaten by the ghost means Pac-Man dies, eating a ghost when it becomes edible gives high positive reward), the explanation should focus on high-value objects (positive or negative). From Pac-Man's rules the value of objects is ghost, edible ghost, cherry, pellet, and dot. (2) *Attention area*: User tests confirmed that participants only consider a limited area relatively close to Pac-Man when explaining its actions. This is reasonable since the Pac-Man and ghosts move only one step at a time and the rest of the objects are static, with the exception of cherries that appear and disappear dynamically. Therefore, we consider only a limited area of attention with limited number of objects for the verbal explanation. (3) *Action accordance*: The expected action of Pac-Man should be approaching beneficial objects and avoiding ghosts. However, there are situations where Pac-Man has to leave beneficial object in order

to avoid ghost or Pac-Man has to approach ghost in order to chase beneficial objects. The action of Pac-Man is divided into two classes, considering consistent or inconsistent with expectations. Class 1 means the action of Pac-Man is in accordance with the expectation. Class 2 means the action is in contrast with the expectation. (4) *Language style*: To make the explanation more natural, we created sentences that describe the Pac-Man's actions (moving directions) followed by the objects that motivate the current action. Relative coordination is employed to indicate the position of objects with relation to Pac-Man. An explanation template was designed accordingly. For example, a typical sentence is "The Pac-Man moves up to eat the dot above her."

Learning-based verbal explanation model

The learning model consists of two stages. The first stage is image processing in order to get the game image feature map. The second stage generates verbal content based on the image process result. The *Image encoder* encodes game image, Pac-Man position map, and object saliency map for the verbal decoder. The three input channels provide information from different aspects: Game image provides environment information. Pac-Man position map contains Pac-Man location information and the object saliency map provides the saliency weight of each object. The *Verbal decoder* generates a verbal description consistent with the image encoder output. A sequence generation model generates verbal explanations verbatim. An attention mechanism then selects the most salient output from the encoder. To capture dynamic game information, five previous frames of the game image, five previous frames of the Pac-Man position map, and a frame of the object saliency map serve as input. A verbal explanation derived from the game image is generated as output.

Experimental results

In this part, both the verbal explanation of rule-base model and learning model are quantitatively evaluated.

User tests

User tests were conducted in our laboratory and through mechanical turk to validate the appropriateness and effectiveness of both the rule-based and learning-based explanation generation models.

Experimental design

The matching and prediction tasks were conducted in a laboratory environment with 17 paid participants recruited from the University of Pittsburgh community. The task settings and game episodes were identical to those used in the earlier evaluation of object-saliency maps, except

some stimulus materials were replaced with natural language explanations generated by the rule-based model.

The second task to evaluate the acceptability of DNN generated explanations was conducted online using Amazon Mechanical Turk. The online questionnaire consisted of an introduction section and two evaluation tasks. The introduction contained basic background knowledge about the Ms. Pac-Man game, object saliency maps, and the language generation model. On each trial participants reported their satisfaction with an explanation presented either visually (saliency map) or verbally (rule based or learning based). In the visual evaluation trials, a game screenshot and the corresponding object saliency map were presented. In the verbal evaluation trials, a natural language explanation generated by either the rule-based or learning-based model was given in addition to the two images. Responses were collected on a Likert scale ranging from 1 (strongly disagree) to 7 (strongly agree).

The 10 trials on each task contained randomly selected scenarios from the game episodes played by DRL. The sequence of two tasks and trials in each task was randomized to counterbalance the learning effect. For the rule-based and learning models, tests were conducted separately on different groups of participants to avoid interference. The questionnaire was deployed on Qualtircs.com for public access.

Results

The performance of matching and prediction participants was compared directly with the historical data from the previous experiment. There was no significant difference between natural language explanation and the object-saliency visualization.

For the online questionnaire, 150 samples are kept after removing abnormal data. For the rule-based model, the average rating of verbal and visual tasks were 5.11 ± 0.07 (mean $\pm$ standard error) and 4.89 ± 0.09, respectively. Paired t-test showed that the rule-based verbal explanations received significant higher subjective ratings than object saliency maps, $t(74) = 2.989$, $P = .004$. For the learning model, a similar pattern appears suggesting that learning-based verbal explanations (5.08 ± 0.10) are better than o-saliency maps (4.93 ± 0.10) in terms of users' satisfaction, $t(74) = 2.020$, $P = .047$. The results are shown in Figs. 6 and 7.

Our results indicate that verbal explanations consistent with human preferences for teleological explanation are found more satisfactory than visual saliency maps that do not respect this preference. However, in terms of matching or prediction accuracy there were no significant differences between the natural language explanations and the saliency maps from which they were derived.

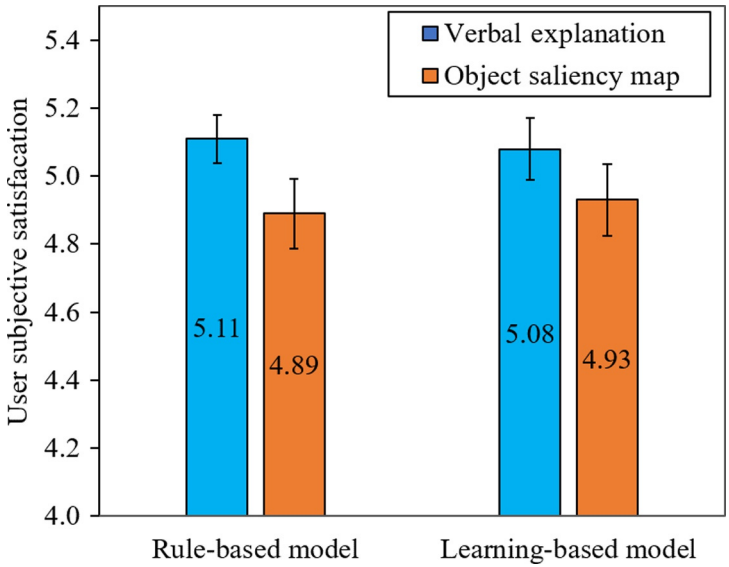


FIG. 6 Subjects satisfaction score.

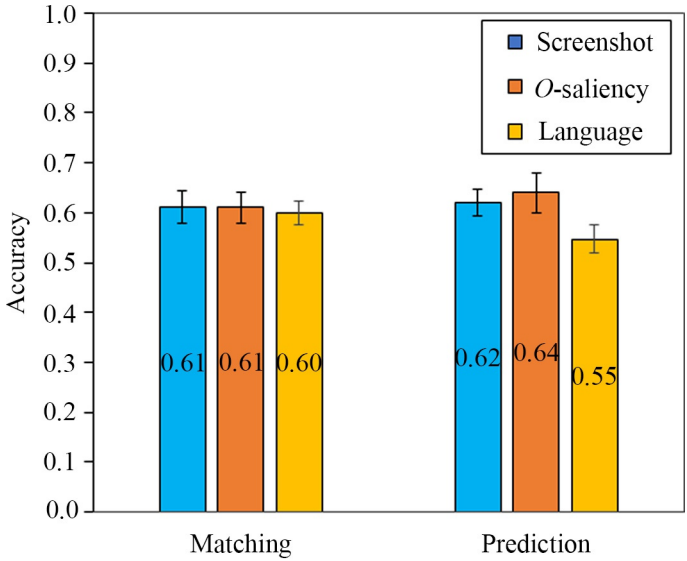


FIG. 7 The mean accuracy of participants.

Conclusions

In this chapter, we present issues involving trust and transparency arising in in human interactions with autonomy that uses AI and machine learning algorithms. The "Introduction" to "Trust and transparency in human-autonomy teaming" sections review literature on trust and transparency for conventional automation in order to extrapolate to what may be expected as we move from systems which automate a relatively narrow range of actions to autonomous agents/robots with substantially larger action spaces. We argue that as the degree of automation (extent to which system output is controlled by machine rather than human) increases, transparency of the automation usually decreases. Adopting Wickens (2018) simplified two-stage (input/output) model of automation, we argue that transparency (added information) at the input stage contributes more to making a system predictable than transparency at the output stage. So, a display showing the proximity of targets, for example, would be a greater help in predicting the behavior of an automated weapon than assurance or experience that the weapon is 90% effective although either might lead to a decision to rely on the automation. A convergent literature on explanation from psychology suggests that humans have a strong preference for teleological (causal) explanations and are more likely to use such models to guide their actions. From these observations, we suggest that an AI/robot's ability to provide or support teleological explanations of its behavior will likely be crucial to effective human-robot interaction and teaming.

The remainder of the chapter is devoted to examining the consequences for interacting with DRL systems which use very high-dimensional, nonlinear embeddings to generate their behavior. While DRL produces highly effective performers achieving equal or better than human performance, they are opaque and often make choices baffling to humans. In a series of studies, we examine mechanisms which might make DRL behavior more transparent. The first study supports transparency by making the input used by the system visible to the user in a manner similar to the automated weapon's display of target proximity. While participants were able to associate saliency maps with corresponding screen shots, their ability to predict Pac-Man's next action based on the most influential objects/regions in the display was not significantly greater than chance. A trial by trial examination shows that while saliency maps improved predictions substantially in some cases, in others it depressed them.

The second set of experiments addressed the question of whether constraining explanations to teleological form could make them more effective and usable. To generate explanations, objects with high salience and their valence were matched against hypothesized rules governing Pac-Man behavior. So, for example, in a saliency map in which pellets

below the Pac-Man had the highest salience and positive valence, the situation might be reexpressed as, "the Pac-Man is attracted by the pellets below." The verbal teleological descriptions of saliency map contents performed no better than the maps themselves on the matching and prediction tasks. Because many of the collected episodes of DRL gameplay could not be translated by matching to rules, a second DRL network was trained using the matches as labeled examples. Mechanical turk workers rated explanations generated by this DRL as satisfactory as those generated by matching to rules and felt both to be more interpretable than the saliency maps. Performance on the matching and prediction tasks, however, did not vary across conditions. These experiments suggest that while DRL networks learn to play games such as Pac-Man with a high level of skill, what they have learned and how they play may seem quite alien to a human observer. When provided with a series of screen shots or saliency maps, our participants readily attributed desires such as eating pellets or avoiding ghosts to the Pac-Man yet predicting actions on this basis worked no better than chance. If we are to take advantage of the strength of DRL performance in human-robot interaction or human-autonomy teaming this gulf between how we view problems and how they come to be solved by a learner with massive experience, but none of our knowledge must be bridged. Until then we may come to trust DRL systems based on performance alone but won't be able to predict their actions or realize when they are wrong.

As the development and penetration of these systems into society increase, and as vulnerabilities of these opaque systems are identified, there is a tremendous need for (1) formulating rigorous definitions of transparency, (2) identifying dimensions of transparency and algorithms for making those dimensions operational to humans, and (3) studying their effects in human autonomy teaming. There is also an imperative to study the transparency of these systems in the broader societal context.

References

Amato, F., Felici, M., Lanzi, P., Lotti, G., Save, L., & Tedeschi, A. (2011). Trust observations in validation exercises. *Fifth international conference on secure software integration and reliability improvement*, pp. 216–223.

Annasamy, R., & Sycara, K. (2019). Towards better interpretability in deep Q-networks. *International conference on artificial intelligence (AAAI)*. AAAI.

Arulkumaran, K., Deisenroth, M. P., Brundage, M., & Bharath, A. A. (2017). Deep reinforcement learning: A brief survey. *IEEE Signal Processing Magazine*, *34*(6), 26–38.

Beck, H. P., Dzindolet, M. T., & Pierce, L. G. (2007). Automation usage decisions: Controlling intent and appraisal errors in a target detection task. *Human Factors*, *49*(3), 429–437.

Biros, D. P., Daly, M., & Gunsch, G. (2004). The influence of task load and automation trust on deception detection. *Group Decision and Negotiation*, *13*(2), 173–189.

Bitan, Y., & Meyer, J. (2007). Self-initiated and respondent actions in a simulated control task. *Ergonomics*, *50*(5), 763–788.

Bliss, J. P., Jeans, S. M., & Prioux, H. J. (1996). Dual-task performance as a function of individual alarm validity and alarm system reliability information. *Proceedings of the human factors and ergonomics society annual meeting:* (Vol. 40, pp. 1237–1241). Los Angeles, CA: SAGE Publications.

Brunelli, R. (2009). *Template matching techniques in computer vision: Theory and practice*. Wiley Publishing.

Calhoun, G. L., Draper, M. H., & Ruff, H. A. (2009). Effect of level of automation on unmanned aerial vehicle routing task. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, *53*(4), 197–201.

Chen, J. Y., Procci, K., Boyce, M., Wright, J., Garcia, A., & Barnes, M. (2014). *Situation awareness-based agent transparency*. Technical Report ARL-TR-6905.

Chen, J. Y. C., & Barnes, M. J. (2014). Human–agent teaming for multirobot control: A review of human factors issues. *IEEE Transactions on Human-Machine Systems*, *44*(1), 13–29.

Choi, E., Hewlett, D., Uszkoreit, J., Polosukhin, I., Lacoste, A., & Berant, J. (2017). Coarse-to-fine question answering for long documents. *Proceedings of the 55th annual meeting of the association for computational linguistics:* (Vol. 1, pp. 209–220).

Choudlechova, A. (2016). Fair predictions with disparate impact: A study of bias in recidivism prediction instruments. In *IEEE Conference on Big Data*.

Dadashi, N., Stedmon, A. W., & Pridmore, T. P. (2013). Semi-automated CCTV surveillance: The effects of system confidence, system accuracy and task complexity on operator vigilance, reliance and workload. *Applied Ergonomics*, *44*(5), 730–738.

Davis, R. (1978). Knowledge acquisition in rule-based systems: Knowledge about representations as a basis for system construction and maintenance. In *Pattern-directed inference systems* (pp. 99–134). New York: Academic Press.

de Graaf, M., & Malle, B. F. (2017). How people explain action (and autonomous intelligent systems should too). *AAAI fall symposium on artificial intelligence for human-robot interaction*.

de Visser, E., Parasuraman, R., Freedy, A., Freedy, E., & Weltman, G. (2006). A comprehensive methodology for assessing human-robot team performance for use in training and simulation. *Proceedings of the human factors and ergonomics society 50th annual meeting*, HFES, pp. 2639–2643.

de Visser, E. J., Cohen, M., Freedy, A., & Parasuraman, R. (2014). A design methodology for trust cue calibration in cognitive agents. *International conference on virtual, augmented and mixed reality*, Springer, pp. 251–262.

Dixon, S., & Parasuraman, R. (2006). Automation reliability in unmanned aerial vehicle control: A reliance-compliance model of automation dependence in high workload. *Human Factors*, *48*(3), 474–486.

Dorsa Sadigh, A. D. D., Sastry, S., & Seshia, S. A. (2017). Active preference-based learning of reward functions. *Robotics: Science and systems (RSS)*.

Dzindolet, M., Peterson, S., Pomranky, R., Pierce, L., & Beck, H. (2003). The role of trust in automation reliance. *International Journal of Human-Computer Studies*, *58*, 697–718.

Dzindolet, M. T., Pierce, L. G., Beck, H. P., & Dawe, L. A. (2002). The perceived utility of human and automated aids in a visual detection task. *Human Factors: The Journal of the Human Factors and Ergonomics Society*, *44*(1), 79–94.

Ehsan, U., Harrison, B., Chan, L., & Riedl, M. O. (2018). Rationalization: A neural machine translation approach to generating natural language explanations. In *AAAI/ACM Conference on Artificial Intelligence, Ethics, and Society AIES'18*.

Fawcett, T. (2006). An introduction to ROC analysis. *Pattern Recognition Letters*, *27*(8), 861–874.

Gao, J., & Lee, J. D. (2006). Extending the decision field theory to model operators' reliance on automation in supervisory control situations. *Transactions on Systems, Man, and Cybernetics Part A*, *36*(5), 943–959.

Goodman, B., & Flaxman, S. (2017). European Union regulations on algorithmic decision making and "a right to explanation" *AI Magazine*, 50–57.

Greydanus, S., Koul, A., Dodge, J., & Fern, A. (2018). Visualizing and understanding atari agents. In *Proceedings of the 35th International Conference on Machine Learning, ICML'18*.

Hadfield-Menell, D., Russell, S. J., Abbeel, P., & Dragan, A. (2016). Cooperative inverse reinforcement learning. *Advances in neural information processing systems*, pp. 3909–3917.

Hancock, P. A., Deborah, B., Schaefer, K., Chen, J., de Visser, E., & Parasuraman, R. (2011). A meta analysis of factors affecting trust in human robot interaction. *Human Factors*, *53*(5), 517–527.

Hayes, B., & Shah, J. A. (2017). Improving robot controller transparency through autonomous policy explanation. *Proceedings of the 2017 ACM/IEEE international conference on human-robot interaction*, ACM, pp. 303–312.

Huang, S., Papernot, N., Goodfellow, I., Duan, Y., & Abbeel, P. (2017). Adversarial attacks on neural network policies. *arXiv preprint arXiv:170202284*.

Huang, S. H., Held, D., Abbeel, P., & Dragan, A. D. (2019). Enabling robots to communicate their objectives. *Autonomous Robots*, *43*(2), 309–326.

Iyer, R., Li, Y., Li, H., Lewis, M., Sundar, R., & Sycara, K. (2018). Transparency and explanation in deep reinforcement learning neural networks. *Proceedings of the AAAI/ACM conference on artificial intelligence, ethics, and society*.

Jian, J., Bisantz, A., & Drury, C. (2000). Foundations for an empirically determined scale of trust in automated systems. *International Journal of Cognitive Ergonomics*, *4*(1), 53–71.

Jones, G., & George, J. (1998). the experience and evolution of trust: Implications for cooperation and teamwork. *Academy of Management Review*, *23*(3), 531–546.

Kim, B. (2015). Interactive and interpretable machine learning models for human machine collaboration (Unpublished doctoral dissertation). Massachusetts Institute of Technology.

Kira, Z., & Potter, M. A. (2009). Exerting human control over decentralized robot swarms. *4th International conference on autonomous robots and agents, 2009 ICARA 2009*, IEEE, pp. 566–571.

Kraska, T., Beutel, A., Chi, E. H., Dean, J., & Polyzotis, N. (2018). The case for learned index structures. *Proceedings of the 2018 international conference on management of data*, ACM, pp. 489–504.

Lample, G., & Chaplot, D. S. (2017). Playing fps games with deep reinforcement learning. In *AAAI Conference on Artificial Intelligence* (pp. 2140–2146).

Langley, P., Meadows, B., Sridharan, M., & Choi, D. (2017). Explainable agency for intelligent autonomous systems. *AAAI*, pp. 4762–4764.

Lee, J., & Moray, N. (1992). Trust, control strategies and allocation of function in human-machine systems. *Ergonomics*, *35*(10), 1243–1270.

Lee, J., & Moray, N. (1994). Trust, self confidence and operator's adaptation to automation. *International Journal of Human-Computer Studies*, *40*, 153–184.

Lee, J., & See, K. (2004). Trust in automation: Designing for appropriate reliance. *Human Factors*, *46*(1), 50–80.

Levine, S., Finn, C., Darrell, T., & Abbeel, P. (2016). End-to-end training of deep visuomotor policies. *The Journal of Machine Learning Research*, *17*(1), 1334–1373.

Lewandowsky, S., Mudy, M., & Tan, G. (2000). The dynamics of trust: Comparing humans to automation. *Journal of Experimental Psychology: Applied*, *6*(2), 104–123.

Lewis, C. M., & Heidorn, P. B. (1991). Identifying tacit strategies in aircraft maneuvers. *IEEE Transactions on Systems, Man, and Cybernetics*, *21*(6), 1560–1571.

Lewis, M. (1998). Designing for human-agent interaction. *AI Magazine*, *19*(2), 67.

Li, Y., Sycara, K. P., & Iyer, R. (2017). Object-sensitive deep reinforcement learning. *GCAI 2017, 3rd global conference on artificial intelligence, Miami, FL, USA, 18–22 October 2017*, pp. 20–35.

Linegang, M. P., Stoner, H. A., Patterson, M. J., Seppelt, B. D., Hoffman, J. D., Crittendon, Z. B., & Lee, J. D. (2006). Human-automation collaboration in dynamic mission planning: A challenge requiring an ecological approach. *Proceedings of the human factors and*

ergonomics society annual meeting: (Vol. 50, pp. 2482–2486). Los Angeles, CA: SAGE Publications.

Lipton, Z. C. (2018). The mythos of model interpretability. *ACMQue*, *161*–27.

Liu, C., Hamrick, J. B., Fisac, J. F., Dragan, A. D., Hedrick, J. K., Sastry, S. S., & Griffiths, T. L. (2016). Goal inference improves objective and perceived performance in human-robot collaboration. *Proceedings of the 2016 international conference on autonomous agents & multiagent systems*International Foundation for Autonomous Agents and Multiagent Systems, pp. 940–948.

Lombrozo, T. (2006). The structure and function of explanations. *Trends in Cognitive Sciences*, *10*(10), 464–470.

Lyons, J., & Stokes, C. (2012). Human-human reliance in the context of automation. *Human Factors*, *54*(1), 112–121.

Lyons, J. B. (2013). Being transparent about transparency: A model for human-robot interaction. *2013 AAAI spring symposium series*.

Lyons, J. B., & Havig, P. R. (2014). Transparency in a human-machine context: Approaches for fostering shared awareness/intent. *International conference on virtual, augmented and mixed reality,* Springer, pp. 181–190.

Lyons, J. B., Koltai, K. S., Ho, N. T., Johnson, W. B., Smith, D. E., & Shively, R. J. (2016). Engineering trust in complex automated systems. *Ergonomics in Design*, *24*(1), 13–17.

Madsen, M., & Gregor, S. (2000). Measuring human-computer trust. *11th Australasian conference on information systems:* (Vol. 53, pp. 6–8). Citeseer.

Mayer, R. C., Davis, J. H., & Schoorman, F. D. (1995). An integrative model of organizational trust. *Academy of Management Review*, *20*(3), 709–734.

Mercado, J. E., Rupp, M. A., Chen, J. Y. C., Barnes, M. J., Barber, D., & Procci, K. (2016). Intelligent agent transparency in human–agent teaming for Multi-UxV management. *Human Factors*, *58*(3), 401–415.

Meyer, J. (2004). Conceptual issues in the study of dynamic hazard warnings. *Human Factors*, *46*, 196–204.

Mnih, V., Badia, A. P., Mirza, M., Graves, A., Lillicrap, T. P., Harley, T., … Kavukcuoglu, K. (2016). Asynchronous methods for deep reinforcement learning. In *Proceedings of the 33rd International Conference on Machine Learning, ICMl'16*.

Mnih, V., Kavukcuoglu, K., Silver, D., Graves, A., Antonoglou, I., Wierstra, D., & Riedmiller, M. (2013). Playing atari with deep reinforcement learning. *NIPS Deep Learning Workshop*.

Mnih, V., Kavukcuoglu, K., Silver, D., Rusu, A. A., Veness, J., Bellemare, M. G., … Hassabis, D. (2015). Human-level control through deep reinforcement learning. *Nature*, *518*(7540), 529–533.

Moray, N., & Inagaki, T. (1999). Laboratory studies of trust between humans and machines in automated systems. *Transactions of the Institute of Measurement and Control*, *21*(4–5), 203–211.

Moray, N., Inagaki, T., & Ito, M. (2000). Adaptive automation, trust, and self-confidence in fault management of time-critical tasks. *Journal of Experimental Psychology: Applied*, 644–58.

Morgan, B. B., Glickman, A. S., Woodard, E. A., Blaiwes, A. S., Salas, E., Campbell, W. J., … Zimmer, S. (1986). *Measurement of team behaviors in a navy training environment*. Old Dominion University Research Foundation.

Muir, B., & Moray, N. (2013). Trust in automation: 2. Experimental studies of trust and human intervention in a process control simulation. *Ergonomics*, *39*429–460.

Nam, C., Walker, P., Lewis, M., & Sycara, K. (2017). Predicting trust in human control of swarms via inverse reinforcement learning. *2017 26th IEEE international symposium on robot and human interactive communication (RO-MAN),* IEEE, pp. 528–533.

Nam, C., Walker, P., Li, H., Lewis, M., & Sycara, K. (2019). Models of trust in human control of swarms with varied levels of autonomy. *IEEE Transactions on Human-Machine Systems* *50*(3), 205–214.

Ng, A. Y., & Russell, S. J. (2000). Algorithms for inverse reinforcement learning. *International Conference on Machine Learning, ICML* (p. 2)**Vol. 1** (p. 2). .

Oh, J., Suppe, A., Duvallet, F., Boularias, A., Vinokurov, J., Navarro-Serment, L., & Stentz, A. (2015). Toward mobile robots reasoning like humans. *AAAI conference on artificial intelligence (AAAI)*.

Parasuraman, R., & Manzey, D. (2010). Complacency and bias in human use of automation: An attentional integration. *Human Factors*, *52*(3), 381–410.

Parasuraman, R., Sheridan, T., & Wickens, C. D. (2000). A model of types and levels of human interaction with automation. *IEEE Transactions on SMC, Part A: Systems and Humans*, *30*, 286–297.

Pezzulo, G., Donnarumma, F., & Dindo, H. (2013). Human sensorimotor communication: A theory of signaling in online social interactions. *PLoS ONE*, *8*(11), e79876.

Ramachandran, D., & Amir, E. (2007). Bayesian inverse reinforcement learning. *International Joint Conference on Artificial Intelligence, IJCAI* (pp. 2586–2591)**Vol. 7**, pp. 2586–2591). .

Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). Why should I trust you?: Explaining the predictions of any classifier. *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining*, ACM, pp. 1135–1144.

Riedmiller, M. (2005). Neural fitted Q iteration—first experiences with a data efficient neural reinforcement learning method. *European conference on machine learning*, pp. 317–328.

Riley, V. A. (1994). Human use of automation (Unpublished doctoral dissertation). University of Minneapolis.

Robinette, P., Li, W., Allen, R., Howard, A. M., & Wagner, A. R. (2016). Overtrust of robots in emergency evacuation scenarios. *ACM/IEEE international conference on human-robot interaction*, pp. 101–108.

Rummery, G. A., & Niranjan, M. (1994). *On-line Q-learning using connectionist systems*. University of Cambridge, Department of Engineering.

Sadler, G., Battiste, H., Ho, N., Hoffmann, L., Johnson, W., Shively, R., & Smith, D. (2016). Effects of transparency on pilot trust and agreement in the autonomous constrained flight planner. *2016 IEEE/AIAA 35th digital avionics systems conference (DASC)*, pp. 1–9.

Salas, E., Cooke, N. J., & Rosen, M. A. (2008). On teams, teamwork, and team performance: Discoveries and developments. *Human Factors*, *50*(3), 540–547.

Salas, E., Sims, D. E., & Burke, C. S. (2005). Is there a "big five" in teamwork? *Small Group Research*, *36*(5), 555–599.

Schaefer, K. E. (2013). The perception and measurement of human-robot trust (Unpublished doctoral dissertation). University of Central Florida, Orlando, FL.

Seppelt, B. D., & Lee, J. D. (2007). Making adaptive cruise control (ACC) limits visible. *International Journal of Human-Computer Studies*, *65*(3), 192–205.

Silver, D., Hubert, T., Schrittwieser, J., Antonoglou, I., Lai, M., Guez, A. , et al. (2018). A general reinforcement learning algorithm that masters chess, shogi, and go through self-play. *Science*, ***362***1140–1144.

Simonyan, K., Vedaldi, A., & Zisserman, A. (2014). Deep inside convolutional networks: Visualising image classification models and saliency maps. In *International Conference on Learning Representations, ICLR'14* (pp. 1–8).

Simpson, A., Brander, G. N., & Portsdown, D. R. A. (1994). Seaworthy trust: Confidence in automated data fusion. In *Proceedings of the 3rd International Workshop on Human-Computer Teamwork*.

Stanton, N. A., Young, M. S., & Walker, G. H. (2007). The psychology of driving automation: A discussion with Professor Don Norman. *International Journal of Vehicle Design*, *45*(3), 289–306.

Sutton, R. S., & Barto, A. G. (1998). *Reinforcement learning: An introduction*. (Vol. 1). MIT Press. No. 1.

Sycara, K. P., Lewis, M., Lenox, T., & Roberts, L. (1998). Calibrating trust to integrate intelligent agents into human teams. *Proceedings of the thirty-first Hawaii international conference on system sciences, 1998:* (Vol. 1, pp. 263–268).

Szegedy, C., Zaremba, W., Sutskever, I., Bruna, J., Erhan, D., Goodfellow, I., & Fergus, R. (2014). Intiriguing properties of neural networks. In *International Conference on Learning Representations*.

Thines, G., Costall, A., & Butterworth, G. (1991). *Michotte's experimental phenomenology of perception*. Hillsdale, NJ: Erlbaum.

van Hasselt, H., Guez, A., & Silver, D. (2016). *Deep reinforcement learning with double q-learning* (pp. 2094–2100)*30th AAAI Conference on Artificial Intelligence*. volume abs/1509.06461.

Verberne, F. M. F., Ham, J., & Midden, C. J. H. (2012). Trust in smart systems: Sharing driving goals and giving information to increase trustworthiness and acceptability of smart systems in cars. *Human Factors*, *54*(5), 799–810.

Wang, N., Pynadath, D. V., & Hill, S. G. (2016). Trust calibration within a human-robot team: Comparing automatically generated explanations. *The eleventh ACM/IEEE international conference on human robot interaction*, pp. 109–116.

Wang, Z., Schaul, T., Hessel, M., Van Hasselt, H., Lanctot, M., & De Freitas, N. (2016). Dueling network architectures for deep reinforcement learning. In *Proceedings of the 33rd International Conference on Machine Learning*.

Watkins, C. J. C. H. (1989). Learning from delayed rewards (Unpublished doctoral dissertation). University of Cambridge England.

Wickens, C. (2018). Automation stages & levels, 20 years after. *Journal of Cognitive Engineering and Decision Making*, *12*(1), 35–41.

Williams, J. D., Asadi, K., & Zweig, G. (2017). Hybrid code networks: Practical and efficient end-to-end dialog control with supervised and reinforcement learning. *arXiv preprint arXiv:170203274*.

Williams, R. J. (1992). Simple statistical gradient-following algorithms for connectionist reinforcement learning. *Machine Learning*, *8*(3–4), 229–256.

Wright, J. L., Chen, J. Y. C., Barnes, M. J., & Hancock, P. A. (2016). The effect of agent reasoning transparency on automation bias: An analysis of response performance. *International conference on virtual, augmented and mixed reality*, pp. 465–477.

Xu, A., & Dudek, G. (2015). OPTIMo: Online probabilistic trust inference model for asymmetric human-robot collaborations. *Conference on human robot interaction*, pp. 221–228.

Zahavy, T., Ben-Zrihem, N., & Mannor, S. (2016). Graying the black box: Understanding DQNs. *International conference on machine learning*, pp. 1899–1908.

Zhang, A., Ballas, N., & Pineau, J. (2018). A dissection of overfitting and generalization in continuous reinforcement learning. *arXiv preprint arXiv:180607937*.

Zhang, C., Vinyals, O., Munos, R., & Bengio, S. (2018). A study on overfitting in deep reinforcement learning. *arXiv preprint arXiv:180406893*,.

Zhou, A., Hadfield-Menell, D., Nagabandi, A., & Dragan, A. D. (2017). Expressive robot motion timing. *Proceedings of the 2017 ACM/IEEE international conference on human-robot interaction*, pp. 22–31.

Zhu, Y., Mottaghi, R., Kolve, E., Lim, J. J., Gupta, A., Fei-Fei, L., & Farhadi, A. (2016). Target-driven visual navigation in indoor scenes using deep reinforcement learning. In *International Conference on Robotics and Automation, ICRA'17* (pp. 3357–3364).

CHAPTER

15

Trust-aware control in proximate human-robot teaming

Auriel Washburn, Sachiko Matsumoto, and Laurel D. Riek

Computer Science and Engineering, UC San Diego, La Jolla, CA, United States

Introduction

Proximate interaction is central to a wide variety of rapidly expanding human-robot interaction (HRI) domains including healthcare, manufacturing, and education (see Fig. 1). Recent findings from HRI demonstrate that human-robot teaming commonly leads to better performance than humans achieve alone (e.g., de Visser & Parasuraman, 2011; Dixon & Wickens, 2006; Marble, Bruemmer, Few, & Dudenhoeffer, 2004; McKendrick et al., 2014; Shaw et al., 2010). This work suggests that proximate human-robot teaming can support notable improvements in many aspects of everyday life.

A primary goal of HRI research in general is to establish how to achieve and maintain trust calibration within a human-robot team (Ogreten, Lackey, & Nicholson, 2010). Trust calibration is the achievement of an appropriate level of trust given both human and robot capabilities and serves to support effective cooperation and reliance between agents. This prevents situations where humans over- or under-rely on their robot teammates, which can result in ineffective robot monitoring or a lack of interaction with the robot (Hancock et al., 2011). Human-robot trust calibration is critical to effective proximate human-robot teaming, as such teaming often occurs in dynamic environments within which human actions are highly variable and robot errors are likely to occur including public spaces like hospitals, schools and airports as well as private homes. The continued development of trust-aware robot control will greatly

https://doi.org/10.1016/B978-0-12-819472-0.00015-0
© 2021 Elsevier Inc. All rights reserved.

FIG. 1 Examples of robots engaging in proximate HRI from our prior work (clockwise from top left): collaboratively hanging a banner (Washburn, Adeleye, An, & Riek, 2020), engaging in rehabilitation exercises (Woodworth, Ferrari, Zosa, & Riek, 2018), stacking boxes (Matsumoto, Washburn, & Riek, 2018), and implicitly learning from a human (Hayes, Moosaei, & Riek, 2016). These platforms act in shared spaces with people, illustrating the primary importance of human safety and robot movement to trust in proximate HRI.

support the maintenance of trust calibration during proximate human-robot teaming.

This chapter provides the foundation for a new generation of trust-aware control frameworks specifically designed to support optimal teaming during proximate HRI. To achieve this we review existing work across HRI, human-automation interaction, and human-human interaction (HHI) and highlight opportunities for the advancement of trust-aware control in proximate HRI, as well current challenges to this progress.

In the "Background" section, we provide background on the modeling of trust in HRI. In "Critical trust factors within proximate HRI", we review the interaction factors that are most likely to affect trust during proximate HRI. We then propose a set of five aims for trust-aware control in proximate HRI in "Proposed aims for trust-aware control in proximate HRI". In "Existing trust-aware control frameworks for HRI", we use these aims to

assess the advantages and shortcomings of two existing human-robot trust-aware control frameworks with respect to their use in proximate HRI. Through this process, robot movement behavior emerges as central to trust dynamics in proximate human-robot teaming, and in "Advancing trust-aware control of robot movement in proximate HRI", we offer suggestions for the further advancement of trust-aware modeling for proximate human-robot teaming through knowledge about robot movement behavior. In "Challenges to trust-aware control in proximate HRI", we describe continued challenges in achieving trust-aware control within proximate HRI. Lastly, in "The future of proximate HRI", we summarize our recommended priorities for the development of trust-aware control frameworks designed to optimize proximate human-robot teaming.

Background

Theoretical framework for understanding trust in HRI

The current theoretical framework for understanding trust in HRI is largely informed by earlier work on trust in human-automation interaction. Lee and See (2004) articulated one commonly accepted understanding of trust in HRI with respect to trust in human-automation interaction as "the attitude that an agent will help achieve an individual's goals in a situation characterized by uncertainty and vulnerability". Many HRI researchers also agree with the assertion by Mayer, Davis, and Schoorman (1995) that human trust in an automated agent involves recognizing and accepting risk.

Beyond these principle elements, empirical work on trust in HRI reveals various factors that impact trust. Hancock et al. (2011) conducted a comprehensive review of this work and found that these factors generally fall into three categories: 1) robot-related, 2) human-related, and 3) environmental. The majority of research on trust in HRI has focused on robot-related factors, which can be divided into attribute- or performance-based robot characteristics (Hancock et al., 2011). Attribute-based characteristics include the way a robot appears, whereas performance-based characteristics focus on the way the robot functions. Attribute-based traits related to trust include robot type (e.g., fixed arm, mobile manipulator, or autonomous vehicle), personality, and anthropomorphism. For example, robots with polite personalities can elicit trust even when they exhibit low reliability (Parasuraman & Miller, 2004). Performance-based features, including robot reliability (De Brun et al., 2008; Lee & See, 2004; Ogreten et al., 2010) and predictability (de Vries, Midden, & Bouwhuis, 2003; Dzindolet, Peterson, Pomranky, Pierce, & Beck, 2003; Lee & See, 2004), are consistently and

positively associated with human attributions of trust. Hancock et al. (2011) identified performance-based robot characteristics as prominent predictors of human trust during HRI. Among these characteristics, robot reliability was the most consistent predictor of human trust across studies.

However, Ososky, Schuster, Phillips, and Jentsch (2013) suggested that trust calibration is more a function of an individual's understanding of a robot's abilities and inabilities rather than the ground-truth performance of the robot. Prior work suggests that individuals with low expectations for a robot's capabilities have more positive experiences during HRI (Paepcke & Takayama, 2010). Our recent work demonstrated that whether or not an individual expects a robot to make errors significantly affects their experience of trust during interaction with a robot that actually makes errors (Washburn et al., 2020). Specifically, individuals with low expectations reported greater trust recovery following a robot error than those with high expectations.

Both Paepcke and Takayama (2010) and Washburn et al. (2020) gave their participants framing information about robot capabilities prior to interaction, but expectations for robot behavior can also be shaped by robot morphology. For example, people often attribute a range of social abilities to humanoid robots and can be disappointed and dissatisfied when a robot does not exhibit these skills during interaction (Duffy, 2003; Mori, 1970). It is, therefore, important to keep in mind that some attribute-based, robot-related factors likely impact trust during HRI indirectly by creating a mismatch between human expectations and robot abilities.

Work in human factors also suggests a robust relationship between self-confidence and trust (Freedy, DeVisser, Weltman, & Coeyman, 2007; Lee & Moray, 1994; Ogreten et al., 2010). Specifically, individuals with low self-confidence are more likely to overtrust a robot while those with high self-confidence are less likely to achieve trust at all. People with a high propensity to trust other people or things, in general, are more likely to develop appropriate trust in automation (Lee & See, 2004). This may also be applied to how humans develop trust in robots (Adams, Bruyn, Houde, & Angelopoulos, 2003).

Compared to the number of studies focused on identifying the role of robot-related and human-related factors in human-robot trust, considerably fewer studies investigate the impact of environment-related factors. At the time of their review, Hancock et al. (2011) identified two studies that demonstrated an effect of cultural context on trust ratings for robots (Lee & See, 2004; Li, Rau, & Li, 2010). They also predicted that various elements of team collaboration (e.g., communication) and task features like the task type or need for multitasking would influence trust during human-robot teaming.

Modeling trust in human-automation teaming

In their previous work, Yanco, Desai, Drury, and Steinfeld (2016) and Moray and Inagaki (1999) identified five distinct categories of trust models for human-automation interaction: qualitative, regression, time series, neural net, and probabilistic models. The authors discuss the pros and cons of the model categories, ultimately demonstrating that each type offers different opportunities for supporting trust and trust calibration during interaction.

Qualitative models consist of relationships and interactions between independent variables of interest (i.e., robot-related, human-related or environmental factors) and trust as a dependent variable (e.g., Muir, 1989; Riley, 1994). *Regression models* use statistical techniques, such as multiple linear regressions, to examine the relationship between the independent variables and the dependent variable, typically trust. They also identify which independent variables account for the most variance in trust levels. *Time series models* leverage the structured nature of time series data to model fluctuations in trust over the course of interaction. Since these models incorporate time, they are able to represent dynamic relationships between independent variables and trust, unlike the regression models described earlier.

Neural net models train a neural net on collected data to predict a human's future behavior or trust level. However, unlike time series models, the process of generating neural nets does not allow the developer to extract meaningful information about how the model works. As a result, a developer may be unaware of the ways in which a neural net model is unintentionally biased by characteristics of the training data set. This may make it challenging to correct errors introduced by the training data or adjust the model for use in related but distinctly different contexts. *Probabilistic models* are meant to approximate high-level human cognitive activity by identifying the probability that a specific agent or robot action will support the team's goal state, as well as the agent's confidence in its decision.

Aside from purely qualitative models, each of the four other model types offers opportunities for direct implementation on robotic platforms to provide online control during interaction. Of these quantitative model types, time series, neural nets, and probabilistic models can all be designed to account for the dynamic nature of trust in HRI (Robinette, Howard, & Wagner, 2015; Rousseau, Sitkin, Burt, & Camerer, 1998; Salem, Lakatos, Amirabdollahian, & Dautenhahn, 2015). Given the need for extensive training and the challenges associated with adjusting neural net models, versions of time series, and probabilistic models are likely to be best-suited to supporting real-time, trust-aware control of robot

behavior. As Yanco et al. (2016) discussed, these models could even allow a robot to engage in corrective behavior during interaction. The continued progress of proximate HRI depends on the development of quantitative control models that account for the factors with the greatest impact on trust within proximate interaction.

Trust in proximate HRI

In proximate HRI, the colocation of the human and robot during teaming shapes the influence of many human-related, robot-related, and environmental factors on trust. Thus, it is important to distinguish copresent interaction, in which a human and robot share the same physical space, from tele-present interaction, in which a human and robot interact with each other from distinct physical locations. Prior work demonstrates that copresent HRI engenders a greater degree of human trust than telepresent interaction with the same robot (Bainbridge, Hart, Kim, & Scassellati, 2008). With the rapid increase of opportunities for proximate interaction within healthcare, manufacturing, and education it is crucial that we understand the features of human-robot trust specific to proximate HRI (Lasota & Shah, 2015; Riek, 2017).

Critical trust factors within proximate HRI

As noted previously, past work in human-automation interaction informs a large proportion of work on trust in HRI across proximate and telepresent contexts. However, Desai, Stubbs, Steinfeld, and Yanco (2009) cautioned that robotics involves an increased level of uncertainty and vulnerability beyond that experienced with automation. For instance, HRI often occurs within unstructured environments. With current robots depending on noisy sensors, the incomplete information available to them in these environments can reduce the reliability of robot behaviors (Yanco et al., 2016). Additionally, while the risk associated with human-automation interaction tasks can vary widely, the majority of these studies focus on scenarios in which individuals experience a lower level of risk than that associated with many HRI tasks (Yanco et al., 2016).

Understanding trust in HRI requires that we consider the unique experience of interacting with a physically embodied, robotic agent. This effect is likely to be most profound during proximate HRI, especially in situations where a human and robot have to physically coordinate their actions to achieve respective and/or joint goals. Proximity may even heighten human experiences of vulnerability, increasing the influence of trust on performance outcomes. These "performance-based", proximate

interaction scenarios (Lewis, Sycara, & Walker, 2018) are the primary focus of this chapter. As one might expect, in performance-based interactions the kinds of performance-based robot characteristics identified by Hancock et al. (2011) have a large effect on trust (Lewis et al., 2018).

A meaningful account of trust within performance-based proximate HRI will require integration of (1) knowledge about trust from human-automation interaction, (2) awareness of the differences between human-automation interaction and HRI, and (3) acknowledgment of the most salient robot-, human-, or environment-related factors within proximate HRI. Human-automation interaction research generates meaningful information about trust that is often relevant to HRI. However, HRI tasks often involve greater risk than the human-automation interaction tasks that are frequently studied (Yanco et al., 2016). For example, human interaction with remotely controlled unmanned ground robots, like urban search and rescue, involve a very high risk of success or failure. Proximate HRI is less likely to involve this kind of risky task, but more likely to pose a physical safety risk to human teammates. For example, in the US, the Occupational Safety and Health Administration (OSHA) identified seven categories of potential hazards to humans working with robots in manufacturing contexts, such as automobile production, that can result in physical safety risks including collision, trapping, and electrical hazards (OSHA, 2006).

Robot-, human-, and environment-related factors will all contribute to people's objective and subjective experiences of their safety during proximate HRI. However, we expect that performance-based robot characteristics will still be some of the strongest predictors of trust in proximate HRI. Specifically, performance-based robot characteristics related to physical movement are most critical. The close relationship between robot movement and human safety and the apparent role of these elements within trust in proximate HRI across domains are evident in Fig. 1.

As opportunities for proximate HRI become more common, models specifically designed to account for trust in proximate HRI will support optimal teaming. In the following section, we outline a set of five proposed aims to guide the development of proximate trust-aware control. We then discuss these aims within the context of two existing trust-aware robot control frameworks.

Proposed aims for trust-aware control in proximate HRI

Based on the current understanding of trust modeling and proximate HRI, we propose a set of five aims for trust-aware control in proximate HRI.

Aim 1: Establish quantitative models that can be used online to inform robot behavior during interaction. As we discussed above, quantitative models are distinct from qualitative models in their ability to support the direct prediction of trust from robot-related, human-related, and environmental factors (Yanco et al., 2016). Quantitative models are therefore necessary for robots to properly adapt their behavior to a person's level of trust throughout an interaction. For example, if the robot estimates that the human's trust in it is too high, it could purposefully display less reliable behavior (Chen, Nikolaidis, Soh, Hsu, & Srinivasa, 2018).

Aim 2: Account for the dynamic nature of trust, allowing for changes in trust prediction based on whether the interaction is most closely aligned with a formation, dissolution, or restoration phase. Multiple studies suggest that trust often changes over the course of interaction, and that the phase of interaction may determine which factors will have the greatest influence on trust (Desai et al., 2012; Desai, Kaniarasu, Medvedev, Steinfeld, & Yanco, 2013; Lewis et al., 2018). Therefore, to accurately predict trust, robots must weight these factors differently depending on the phase of the interaction.

Aim 3: Include the measurement of human trust and policies specifying associated changes in robot behavior based on this feedback within quantitative models that regulate robot behavior online during interaction. Different levels of trust will require varying responses from robots to maintain fluent teaming. As a result, robot behavioral policies must explicitly take trust measurements and predictions into account. Doing so will enable robots to actively support teaming, as opposed to passively observing the human's state and behaving reactively.

Aim 4: Prioritize human safety in robot control frameworks. When humans and robots are colocated, robots have the potential to physically harm people. Therefore, to ensure that people are not hurt when interacting with robots, it is imperative that the control strategies of robots involved in proximate HRI prioritize human safety. Additionally, such control strategies may make people more comfortable being physically close to the robot, which would likely lead to more fluent interactions.

Aim 5: Incorporate environmental factors along with robot- and human-related factors in models that define interactive robot behavior. Given the impact of environmental factors on trust (Hancock et al., 2011), including such factors in trust models will likely increase model effectiveness. This is especially important in dynamic environments, where changing conditions could affect trust levels regardless of human- and robot-related characteristics.

These aims allow us to assess the ability of existing trust-aware frameworks to support proximate human-robot teaming and promote the development of a new generation of trust-aware control frameworks specifically designed for proximate HRI.

Existing trust-aware control frameworks for HRI

At present, there are two existing frameworks that provide trust-aware control of robot behavior during human-robot teaming. We provide a review of these models and their ability to address our five proposed aims for trust-aware control in proximate HRI below. The relevant details of each model are also presented in Table 1.

Trust aware robot control framework (TACtiC)

Xu and Dudek (2016) presented the first trust-aware robot control framework in which real-time estimates of human trust were used to adjust robot behavior online. This framework integrated two modules: Adaptive Parameter EXploration (APEX) and the Online Probabilistic Trust Inference Model (OPTIMo). Together, these modules allow the framework to achieve robot learning of human task behavior, estimation of human trust during task execution, and changes in robot behavior following trust loss to support trust restoration. The authors evaluated the effectiveness of this strategy for human-single robot teams completing a visual navigation task in which the goal was to continuously track terrain boundaries.

The APEX module (Xu, Kalmbach, & Dudek, 2014) consisted of a vision-based boundary tracking algorithm capable of automatically adjusting system parameters based on human interventions with robot behavior. The authors did not design this module to directly assess or influence human trust during interaction, but rather to improve task performance by incorporating human variations in boundary tracking behavior within robot activity (e.g., those corresponding to differences in terrain).

The authors used the OPTIMo module (Xu & Dudek, 2015) to estimate human trust during HRI. Prior to testing the full robot control framework, the researchers customized this module to each human actor's trust tendencies through a set of initial practice boundary tracking trials. During these trials, the researchers collected data about human intervention with robot tracking, and periodic feedback from users about changes in their trust of the robot behavior (i.e., every five seconds the control interface prompted users to provide critiques about whether their trust was increasing or decreasing). Xu and Dudek (2015) used this data to establish personalized trust triggers for each individual, which they incorporated within a Dynamic Bayesian Network responsible for generating real-time trust updates when they employed the full trust-aware control framework. Ultimately, the OPTIMo inferred trust updates every 3 s based on (1) the robot's current and recent task performance, (2) the human's most recent intervention state referenced to their personalized trust triggers, (3) periodic human critiques about changes in their trust toward the robot,

TABLE 1 Assessment of the two existing trust-aware robot control models with respect to use in proximate HRI.

Paper		**Xu and Dudek (2016)**	**Chen et al. (2018)**
Model		Online Probabilistic Trust Inference Model (OPTIMo) with Trust-Aware Conservative Control (TACtiC)	Partially observable Markov decision process for trust (trust-POMDP)
Robotic Platform		SL-Commander (all-terrain autonomous vehicle)	Bespoke robot arm
Collaborative Task		Terrain boundary tracking	Table-clearing
Real-Time Estimation	*Trust*	Estimates real-time unified human trust based on: - Robot performance (implicit) - Human intervening commands for robot direction and speed (implicit) - Human relative-scale critiques (explicit) - Human absolute-scale feedback (explicit)	Linear Gaussian system estimates trust based on: - Real-time robot task performance (implicit) - Real-time human intervention (implicit)
	Trust Calibration	None	Performance-maximizing policy leads to: - Robot object selection sequences that build trust (when the probability of failure is low) - Intentional robot failure (when the human exhibits high trust but the probability of robot failure is high)
Algorithmic Control	*Robot Movement*	Adaptive Parameter Exploration (APEX) uses human intervention commands to automatically adapt system parameters for boundary-tracking TACtiC adjusts robot movement in response to trust loss by: - Reducing speed - Smoothing steering signal	Trust-POMDP uses current robot performance and human intervention information to select object to move

Factors Included in Model	*Robot-Related*	- Robot boundary tracking performance - Individual human trust tendencies - Human intervention with robot behavior	- Probability of robot failure (simulation) - Human intervention with robot behavior
	Human-Related	- Human trust attitudes about robot behavior	
	Environmental	- Indirect robot adaptation to terrain change/task difficulty through robot adaptation to changes in human behavior	- Reward/failure/intervention values for object categories manually specified - Effects of object selection sequences learned from preliminary experiment

and (4) infrequent human responses to a trust survey. Thus the model incorporated the measurement of variables implicitly associated with human-robot trust as well as explicit human feedback about trust in the robot.

The TACtiC module used the trust information generated by the OPTIMo module to initiate changes in robot behavior. When OPTIMo identified a salient loss of trust, the robot behavior became more conservative in order to limit the negative effects of whatever caused the trust loss and prompt the human to engage with and assist the robot. The system exhibited conservative behavior through a reduction in movement speed and a smoothing of the steering signal. The authors evaluated the effectiveness of the TACtiC module within the three-part control framework by implementing it on an autonomous vehicle. The full trust-aware framework reliably predicted human trust, and users reported greater efficiency when interacting with a mildly conservative agent compared to one with no conservative control. The authors measured subjective efficiency using self-report items for human users' experiences of both collaboration and trust.

Trust-POMDP framework

Chen et al. (2018) presented the second trust-aware control framework, a trust-based computational model for robot decision making during human-robot teaming. In this work, the authors emphasize that robots frequently need to make decisions that would benefit from knowledge about a human's hidden mental state. Thus they characterize trust as a latent variable, which they model using a partially observable Markov decision process (POMDP). The trust-POMDP model is made up of two main modules: a human trust dynamics module and a human decision module. The authors used a data-driven approach to define each module during a table-clearing task in which a human and robot collaborated to clear five objects from a table (three plastic water bottles, one fish can, and one wine glass). To do this, they manually specified a reward function for the set of possible interactions between each of the three object categories and three potential task actions (robot clearing success, robot clearing failure, and human intervention). They then communicated this reward function to participants before collecting the model training data. The reward function was as follows: robot succeeds in clearing the bottle (1), fish can (2), or wine glass (3); robot fails in clearing the bottle (0), fish can (−4), or wine glass (−9); human intervenes with the bottle (0), fish can (0), or wine glass (0). Generally speaking, this function is based on the idea that there is a higher risk associated with robot failure when interacting with the fish can or wine glass, but also a higher reward.

In the trust dynamics module, the authors model human trust evolution over the course of HRI as a linear Gaussian system. The resulting

model relates human trust causally to robot task performance over time based on the prespecified reward function. At any given time point, the output of this module can be used within the human decision module to generate a trust-informed prediction of human behavior. This trust-based model operates on the assumption that a human's expectations of robot success will change over time, depending on their trust in the robot. The authors compared the performance of this model with a trust-free model of human behavior in which the human expectation of robot success did not change over time.

Chen et al. (2018) designed the human decision module to predict the action a human would make in response to each possible robot action based on the current trust estimate provided by the trust dynamics module. In the context of the table-clearing task, the options for human action were to allow the robot to complete an action once it was initiated or intervene in order to prevent an expected failure. When the human decision module operated based on the trust-free model, it predicted a constant likelihood of intervention for each object over the course of interaction, regardless of past robot and human behavior. In contrast, the trust-based model predicted notable changes in the probability that a person would intervene corresponding to different trust levels, especially for the highest-risk object (i.e., the wine glass). Ultimately, the human decision module used the expected human policy for each possible robot action predicted by either the trust-based or trust-free behavior model to compute an optimal robot policy. For example, the trust-based model learned that trust increases over the course of an interaction. It, therefore, calculated policies for early trials that allowed the robot to demonstrate its trustworthiness by moving less risky objects first to avoid early human interventions (e.g., moving three plastic water bottles before moving the wine glass).

The authors compared the trust-POMDP to the trust-free version of the model by implementing them both on a robotic arm. Their results revealed that human-robot team performance, measured via accumulated reward, was significantly better when the trust-POMDP controlled robot behavior. A reduction in human intervention rates drove this increase in performance. Interestingly, there was not a significant difference in the average evolution of self-reported trust between the trust-based and trust-free models. The authors determined that this was because early successes, especially for the higher risk objects, were associated with greater increases in trust than later successes.

Chen et al. (2018) acknowledge that while they only included successful robot behavior in their experimental robot test, robots are likely to fail during real-world human-robot interaction. To capture this effect, they used the learned trust dynamics and human behavior models to compute robot policy decisions for the context in which robot failure was likely.

By adjusting the reward function in this new simulation, the authors were able to induce an "information-seeking" robot policy in which the robot selected a high-risk object early on during interaction. The human response (i.e., to intervene or not) to this action indicates whether the individual has an appropriate trust calibration to the robot's failure rate. The authors also added the opportunity for the robot to fail intentionally in this simulation, allowing the robot to actively reduce over-trust. With some modifications to the trust-POMDP they were able to demonstrate that when robotic failures are possible, a reward-maximizing policy that involves strategic, trust-reducing actions can lead to better performance than a trust-maximizing policy.

Ability of existing trust-aware control frameworks to support proximate HRI

As discussed earlier, Xu and Dudek (2016) and Chen et al. (2018) presented the first trust-aware control frameworks for real-time human-robot teaming. Both frameworks are successful in demonstrating the advantage of incorporating trust-awareness within robot control. Specifically, Xu and Dudek (2016) saw a significant effect on subjective experience, including trust reports, and Chen et al. (2018) observed a substantial decrease in the rate of human intervention with robot behavior. These models share some characteristics, but are also distinct in the ways they address each of our stated aims for valuable trust-aware proximate HRI control frameworks. We summarize each of the models' contributions to these aims as follows:

Aim 1: Establish quantitative models that can be used online to inform robot behavior during interaction. Both Xu and Dudek (2016) and Chen et al. (2018) used quantitative approaches to modeling trust in HRI that incorporate modular, probabilistic models. They implemented these models within their respective robot control architectures to guide continuous robot performance and robot decision-making online during interaction.

Aim 2: Account for the dynamic nature of trust, allowing for changes in trust prediction based on whether the interaction is most closely aligned with a formation, dissolution, or restoration phase. Xu and Dudek (2016) do not explicitly include time within the framework they experimentally tested. However, they do mention wanting to incorporate updates to the user trust model, which will reflect the changes that occur as an individual gets used to working with a robot over time, or use a data-driven approach to predict long-term changes based on initial user trust thresholds. Chen et al.'s (2018) trust dynamics module includes a time series model of trust learned from preliminary data collection for the table-clearing task.

Aim 3: Include the measurement of human trust and policies specifying associated changes in robot behavior based on this feedback within quantitative models that regulate robot behavior online during interaction. Xu and Dudek (2016) use an estimate of human trust based on both robot performance and human feedback to determine when to switch to a conservative mode of robot behavior. The selection of optimal robot policies in Chen et al.'s (2018) trust-POMDP similarly reflects both robot performance and human behavior. However, the latent variable for trust in this framework is specific to the reward function for the table-clearing task and the authors note that a more multidimensional parametrization of trust that accounts for a wider range of functions and modes of automation would be advantageous. Both frameworks serve to advance trust-aware control for use in specific task contexts, but the development of a multidimensional measure that accounts for a greater proportion of trust factors will support significant advances in trust-aware control for proximate human-robot teaming. Additionally, determining which measures can be used across task contexts and which are best in a specific context will enable better control algorithms for teaming.

Aim 4: Prioritize human safety in robot control frameworks. Neither model incorporates objective or subjective measures of human safety within their estimates of trust. However, the TACtiC module (Xu & Dudek, 2016) does directly control robot movement as a function of real-time human trust. Specifically, when trust in the robot is low its movements become slower and smoother. Chen et al.'s (2018) behavioral policies specified which object to move at what time, but did not take into account the spatial orientation of the objects, human and robot arm or the kind of movement realized by the robot arm. The authors acknowledge that the movement characteristics exhibited by the robot may influence the evolution of trust.

Aim 5: Incorporate environmental factors along with robot-related and human-related factors in models that define interactive robot behavior. The APEX module of Xu and Dudek's (2016) model is capable of adapting to changes in task difficulty (i.e., terrain conditions) indirectly through adaptation to changes in human control. Chen et al. (2018) construct risk-awareness within the table-clearing task by manually specifying the rewards associated with each object and action. Their trust dynamics and human behavior modules also include information about the progression of the table-clearing task learned from preliminary data-collection.

Neither of the trust-aware HRI control frameworks we review here individually addresses all five of our proposed aims for trust-aware control in proximate HRI. However, together the frameworks provide examples of ways to approach each of these aims.

These frameworks also include elements beyond those identified by our aims that may benefit the ease and effectiveness of their use across proximate HRI contexts. For instance, Xu and Dudek's (2016) OPTIMo

module is customized to each individual using a set of training trials before the full control framework is deployed autonomously. In some ways this limits the ease of the framework's use by increasing the amount of time required to train the autonomous system for interaction with each individual user. However, the additional effectiveness afforded by the resulting individualized trust estimates is likely to make this component more of a benefit than a limitation. Additionally, Chen et al.'s (2018) trust-POMDP can actively elicit trust calibration by deliberately behaving in a way that will increase or decrease human trust. This is likely to increase the effectiveness of the framework, especially given the authors' observation that a performance-maximization policy can be superior to a trust-maximization policy, and the existing evidence that trust calibration supports effective teaming (see Ogreten et al., 2010).

The platforms and tasks used to test these control frameworks also have implications for their use across proximate HRI contexts. Xu and Dudek's (2016) autonomous driving task does involve a proximate spatial relationship between a human and robotic system. However, autonomous driving is a special case within proximate HRI. As a result, the trust dynamics between a driver and an autonomous vehicle may be different from the ones that exist during interaction with a robot arm or mobile manipulator. In the context of autonomous vehicles, the typical effects of physical proximity on trust dynamics may be more relevant to interactions with pedestrians. The fact that Xu and Dudek (2016) saw greater errors in their trust feedback estimates during the autonomous driving test compared to a previous simulation test indicates that users were more cautious during autonomous driving. We expect that being able to adapt the OPTIMo to different levels of actual and perceived risk would extend its effectiveness across a range of different proximate HRI tasks. It will also be necessary to define meaningful task-specific robot adaptation and conservative control behaviors in order to use the three-module framework for proximate HRI tasks other than boundary-tracking.

Chen et al.'s (2018) table-clearing task and robotic arm are characteristic of a number of proximate HRI tasks. However, it is important to note that both modules of the trust-POMDP include task-specific, probabilistic models that were learned from a sizeable data set. The authors mention that models for other tasks or domains can be substituted for the ones associated with the table-clearing task, but this may require additional data collection and learning which could be a time-consuming process. Relatedly, the present version of the control framework is based on a manually designated, task-specific reward function that may not actually correspond to the human conception of the task and would be irrelevant to other tasks. The authors acknowledge that a more accurate understanding of reward could be learned by the trust-POMDP through additional preliminary data collection.

At present, Chen et al.'s (2018) learned policies assume that robot capabilities will be static. In reality, it is likely that robot capabilities will vary over time or with changes in the environment, especially during proximate human-robot teaming where environments are often dynamic and noisy. Xu and Dudek (2016) have some ability to account for these changes by incorporating continuous robot learning of human control behaviors. Chen et al. (2018) also identified a large variance among users in the training data set. This indicates that the experience of trust and reliance on trust during decision-making likely varied by individual. The kind of individualization included in Xu and Dudek's (2016) framework provides an opportunity to account for these effects and improve model performance.

Discussion

Effective models of trust-aware control in proximate HRI will benefit from continued attention to the five aims outlined in the current work. The individual customization and behavioral flexibility and adaptation to human and environmental changes of Xu and Dudek's (2016) system along with the understanding of trust evolution and ability to shape trust calibration featured in Chen et al.'s (2018) work will also be especially valuable. The further advancement of trust-aware control in proximate HRI will depend on the consideration of factors that are not prioritized in the existing frameworks, especially subjective and objective human safety and robot movement behaviors as related to human safety as well as robot communication. There are some other challenges to proximate HRI that will need to be addressed to optimize trust-aware control in these contexts as well.

Advancing trust-aware control of robot movement in proximate HRI

Critically, neither of the existing trust-aware control frameworks explicitly account for human safety, which is likely to be central to the success of trust-aware, proximate interaction. It follows that the most effective trust-aware control frameworks for proximate HRI will include additional consideration of the effects of robot movement during the interaction. Attention to the trust-aware control of robot movement will not only improve the objective and subjective safety of proximate HRI, but it will also provide better opportunities for robots to improve trust calibration through clear communication of robot capabilities, robot intention, and human-robot roles and relationships.

Humans communicate a wide variety of information via natural language, but robots are often limited in their ability to process and produce communication through language. Some platforms are able to achieve human trust following system errors by communicating via text using a visual interface (Rezvani et al., 2016). However, many nonverbal behaviors that are more easily realizable by a range of robots can also convey meaningful information during human-robot teaming. For example, Lee, Knox, Baumann, Breazeal, and DeSteno (2013) demonstrated that the sequencing of nonverbal robot gestures influences human perceptions of robot trustworthiness during proximate human-robot teaming. Similarly, robot movement characteristics like speed, smoothness, and shape can communicate information about robot behavior and influence human trust during proximate human-robot teaming (Dragan, Bauman, Forlizzi, & Srinivasa, 2015; Dragan, Lee, & Srinivasa, 2013; Huang, Bhatia, Abbeel, & Dragan, 2018; Riek et al., 2010; Xu & Dudek, 2016).

Chen et al. (2018) ran a simulation in which they used intentional, movement-related task failures to reduce human trust when their trust-POMDP control framework detected over-trust in a robot. Specifically, the robot dropped an object during movement, communicating a lower level of reliability than the simulated human previously attributed to it. By simulating this policy the authors demonstrated that the effective communication of robot capabilities and appropriate calibration of human trust would ultimately support greater team performance. Other researchers proposed that similar demonstrations of robot capabilities are beneficial during preliminary training for human-robot teams. For example, researchers investigating trust in automation noted that introducing failures during training is likely to help set human expectations, which in turn supports effective trust calibration (Schaefer, Chen, Szalma, & Hancock, 2016) and reduces complacency as well as the inherent bias to trust automation (Bahner, Huper, & Manzey, 2008).

Robot movement can also be used to communicate intention. Dragan et al. (2013, 2015) demonstrated that the kind of path used to achieve goal-directed robot arm movement has a significant impact on collaborative proximate human-robot teaming. They found that *legible* movement trajectories, generated to prioritize the expression of robot intent, led to more fluent collaboration than *predictable* trajectories, which matched what a human co-actor would expect given a specific end-goal. Individuals preferred both of these movement styles to *functional* trajectories, which are based purely on reaching a goal and avoiding collisions.

Some participants leaned away from the robot more during functional movement and modified their movements to maintain greater distance (Dragan et al., 2015). This indicates that these individuals felt less safe when interacting with the robot exhibiting functional movement, although the authors note that participants reported disliking the functional movement mainly because it was difficult to coordinate their

movements within the shared task space. Regardless of whether legible movement primarily supports effortless and efficient human-robot coordination or human experiences of safety, it is likely to have a meaningful effect on trust during proximate HRI. Similarly, Che, Okamura, and Sadigh (2020) recently used a combination of robot movement characteristics and haptic feedback to communicate robot intention during a spatial navigation task. This planning framework increased users' trust in the robot compared to a simple collision avoidance algorithm.

The coordination of coactor behavior can be understood as an indicator of team cohesion. Researchers investigating HHI identified strong connections between interpersonal physical coordination and experiences of social connection such as liking and affiliation (e.g., Miles, Griffiths, Richardson, & Macrae, 2010; Miles, Lumsden, Richardson, & Neil Macrae, 2011). Such coordination also augments functional teaming dynamics through positive effects on cooperation and collaborative problem-solving (Miles, Lumsden, Flannigan, Allsop, & Marie, 2017). Launay, Dean, and Bailes (2013) showed a direct, positive association between synchronization and trust for humans coordinating their finger taps with a virtual agent. It is likely that a similar relationship exists for movement coordination between human-robot teammates. In fact, in their recent work on human-agent teaming, Wynne and Lyons (2018) identified synchrony as one of six key factors that influence human perceptions of autonomous agent teammate-likeness, the extent to which they see the agent as a capable partner rather than an instrumental tool.

Robot movement behavior can also be used to communicate information about the roles of human and robot coactors within a team structure. Within the supervisor-worker dynamic of Xu and Dudek's (2016) autonomous vehicle study, a switch to conservative robot behavior signals a need for the human to provide additional guidance. The researchers are able to elicit this guidance by having the vehicle produce a slower and smoother movement. These conservative behaviors were inspired by previous work demonstrating that humans experience robots as more teachable when they exhibit long action delays during early interaction (Tanaka, Ozeki, & Oka, 2010), as well as work indicating that humans can understand human-like hesitation gestures exhibited by robots (Moon, Parker, Croft, & Van der Loos, 2013). The ability of a robot to signal a need for additional human assistance through these kinds of movement delays and hesitation behaviors is a valuable tool for supporting appropriate trust calibration in proximate human-robot teaming.

Challenges to trust-aware control in proximate HRI

Many challenges to achieving effective and efficient HRI are especially relevant within proximate HRI. Iqbal and colleagues articulated how several of these challenges affect robot perception and action (Iqbal,

Gonzales, & Riek, 2015; Iqbal, Rack, & Riek, 2016; Iqbal, Shah, & Riek, 2018; Iqbal & Riek, 2017). First, the unpredictable nature of human behavior and human environments makes it difficult for robots to sense and understand human behavior accurately and to respond accordingly. Similarly, while robots are capable of signaling a change in their contribution to teaming (e.g., by displaying hesitation behaviors as in Xu & Dudek, 2016), they may have trouble-detecting changes in team dynamics initiated by a human coactor. This limits the ability of a human-robot team to fluently transition between role distributions during an ongoing task.

Iqbal and Riek (2017) also identify the limitations that arise as a result of limited behavioral flexibility in robots as many robots are designed to perform a single task. This is especially restrictive in proximate teaming contexts, where robots are often expected to perform multiple tasks and support trust while also approximating human social behaviors.

In addition to the technical challenges facing robot platforms for use in proximate HRI, the measurement of trust is also an obstacle to the success of trust-aware control. Recent work demonstrated that there is not always a significant relationship between trust attitudes, as measured via self-report and behavioral reliance during proximate human-robot teaming (Lohani, Stokes, McCoy, Bailey, & Rivers, 2016). This makes explicit measures of trust attitudes potentially unreliable for assessing meaningful behavioral human reliance during teaming.

The alternative is to monitor implicit measures associated with trust and reliance during ongoing human-robot teaming. Human intervention in robot behavior is a common implicit metric for reliance (e.g., Chen et al., 2018; Xu & Dudek, 2016). Researchers have also begun to identify other physiological and behavioral patterns associated with human states of trust and reliance (e.g., Khalid, Liew, Voong, & Helander's, 2018 use of facial expression, voiced speech, and heart rate).

As we discussed, it is also important to be able to capture the dynamic nature of trust over the course of proximate human-robot teaming. Yang, Unhelkar, Li, and Shah (2017) illustrated this by showing that the "trust of entirety", or trust over a person's entire experience with a robot, is better understood by examining the evolution of trust over time than by taking a single measure of trust at the end of the interaction. This evolution can be quantified by measuring the area under the curve (AUTC) for a variable, or multidimensional variable, representing trust (Desai et al., 2013). This method for estimating trust is likely to be very valuable within trust-aware control. However, it depends on identifying variables that can be measured continuously or frequently while also being accurate indicators of trust, which may be difficult depending on the proximate HRI task.

Lastly, HRI researchers only recently started to draw attention to the importance of human trustworthiness (Basu & Singhal, 2016; Takeda, 2016). For example, Takeda (2016) demonstrated that drivers who display

overreliance on an automated driving system also display greater deviation in gaze behavior. This indicates that gaze behavior may be useful for establishing the trustworthiness of a human in an automated driving context. In turn, this can inform trust-aware control decisions. The need to be able to measure human trustworthiness as well as human trust in robotic systems in order to support safe, effective control is relevant to a number of other proximate HRI contexts as well. Thus continued work on trust-aware control in proximate HRI should aim to include the measurement of both human trust in a robot coactor and human trustworthiness.

The future of proximate HRI

In this chapter, we provided the first in-depth discussion of how trust-aware control can be used to advance teaming in the kinds of proximate HRI scenarios that are becoming increasingly common across healthcare, manufacturing, and education. In doing this, we identified the trust-related factors that are most likely to be critical specifically within proximate HRI contexts. Based on these factors we outlined five primary aims for trust-aware control in proximate HRI and discussed how the elements of the two existing trust-aware control frameworks for HRI succeed or fail to address these aims.

Through this process, the importance of robot movement in trust-aware control emerged as a primary motivation for ongoing work. Robot movement characteristics can be used to communicate robot capabilities, robot intention, and human-robot roles and relationships. As we discussed, each of these movement behaviors can act to shape trust during proximate human-robot teaming. We acknowledged that there are a number of technical challenges to the fluent and flexible use of current robotic systems within proximate HRI, as well as difficulties in measuring trust in ways that are meaningful to trust-aware control frameworks. The progress of teaming within proximate HRI will require continued attention to these challenges. Ultimately, however, it is work on the trust-aware control of robot movement behavior via the methods we discussed in this chapter that will offer opportunities for transformative advances in proximate HRI.

References

Adams, B. D., Bruyn, L. E., Houde, S., & Angelopoulos, P. (2003). Trust in automated systems literature review (DRDC Toronto No. CR-2003-096). Toronto, Canada: Defence Research and Development Canada.

Bahner, J. E., Huper, A. D., & Manzey, D. (2008). Misuse of automated decision aids: Complacency, automation bias and the impact of training experience. *International Journal of Human-Computer Studies*, *66*(9), 688–699. https://doi.org/10.1016/j.ijhcs.2008.06.001.

Bainbridge, W. A., Hart, J., Kim, E. S., & Scassellati, B. (2008, August). The effect of presence on human-robot interaction. In *RO-MAN 2008-The 17th IEEE international symposium on robot and human interactive communication* (pp. 701–706). IEEE. https://doi.org/10.1109/ROMAN.2008.4600749.

Basu, C., & Singhal, M. (2016, March). Trust dynamics in human autonomous vehicle interaction: A review of trust models. In *2016 AAAI spring symposium series*.

Che, Y., Okamura, A. M., & Sadigh, D. (2020). *Efficient and trustworthy social navigation via explicit and implicit robot-human communication. IEEE Transactions on Robotics*. arXiv preprint arXiv:1810.11556.

Chen, M., Nikolaidis, S., Soh, H., Hsu, D., & Srinivasa, S. (2018). Trust-aware decision making for human-robot collaboration: Model learning and planning. In *Proceedings of the 2018 ACM/IEEE international conference on human-robot interaction* (pp. 307–315). ACM.

De Brun, M. L., Moffitt, V. Z., Franke, J. L., Yiantsios, D., Housten, T., Hughes, A., et al. (2008). Mixed-initiative adjustable autonomy for human/unmanned system teaming. In *AUVSI unmanned systems North America conference*.

de Visser, E., & Parasuraman, R. (2011). Adaptive aiding of human-robot teaming: Effects of imperfect automation on performance, trust, and workload. *Journal of Cognitive Engineering and Decision Making*, *5*(2), 209–231. https://doi.org/10.1177/1555343411410160.

de Vries, P., Midden, C., & Bouwhuis, D. (2003). The effects of errors on system trust, self-confidence, and the allocation of control in route planning. *International Journal of Human-Computer Studies*, *58*(6), 719–735. https://doi.org/10.1016/S1071-5819(03)00039-9.

Desai, M., Kaniarasu, P., Medvedev, M., Steinfeld, A., & Yanco, H. A. (2013). Impact of robot failures and feedback on real-time trust. In *Proceedings of the 8th ACM/IEEE international conference on human-robot interaction* (pp. 251–258). IEEE Press.

Desai, M., Medvedev, M., Vazquez, M., McSheehy, S., Gadea-Omelchenko, S., Bruggeman, C. , et al. (2012). Effects of changing reliability on trust of robot systems. In *Proceedings of the seventh annual ACM/IEEE international conference on human-robot interaction* (pp. 73–80). ACM.

Desai, M., Stubbs, K., Steinfeld, A., & Yanco, H. (2009). Creating trustworthy robots: Lessons and inspirations from automated systems. In *Proceedings of the society for the study of artificial intelligence and the simulation of behaviour (AISB) convention, new frontiers in human-robot interaction*. https://doi.org/10.1184/R1/6552464.v1.

Dixon, S. R., & Wickens, C. D. (2006). Automation reliability in unmanned aerial vehicle control: A reliance-compliance model of automation dependence in high workload. *Human Factors*, *48*(3), 474–486. https://doi.org/10.1518/001872006778606822.

Dragan, A. D., Bauman, S., Forlizzi, J., & Srinivasa, S. S. (2015, March). Effects of robot motion on human-robot collaboration. In *Proceedings of the tenth annual ACM/IEEE international conference on human-robot interaction* (pp. 51–58). ACM.

Dragan, A. D., Lee, K. C., & Srinivasa, S. S. (2013, March). Legibility and predictability of robot motion. In *Proceedings of the 8th ACM/IEEE international conference on human-robot interaction* (pp. 301–308). IEEE Press. https://doi.org/10.1145/2696454.2696473.

Duffy, B. R. (2003). Anthropomorphism and the social robot. *Robotics and Autonomous Systems*, *42*(3–4), 177–190.

Dzindolet, M. T., Peterson, S. A., Pomranky, R. A., Pierce, L. G., & Beck, H. P. (2003). The role of trust in automation reliance. *International Journal of Human-Computer Studies*, *58*(6), 697–718. https://doi.org/10.1016/S1071-5819(03)00038-7.

Freedy, A., DeVisser, E., Weltman, G., & Coeyman, N. (2007, May). Measurement of trust in human-robot collaboration. In *2007 International symposium on collaborative technologies and systems* (pp. 106–114). IEEE. https://doi.org/10.1109/CTS.2007.4621745.

Hancock, P. A., Billings, D. R., Schaefer, K. E., Chen, J. Y. C., de Visser, E. J., & Parasuraman, R. (2011). A meta-analysis of factors affecting trust in human-robot

interaction. *Human Factors: The Journal of the Human Factors and Ergonomics Society*, *53*(5), 517–527. https://doi.org/10.1177/0018720811417254.

Hayes, C. J., Moosaei, M., & Riek, L. D. (2016, August). Exploring implicit human responses to robot mistakes in a learning from demonstration task. In *2016 25th IEEE international symposium on robot and human interactive communication (RO-MAN)* (pp. 246–252). IEEE. https://doi.org/10.1109/ROMAN.2016.7745138.

Huang, S. H., Bhatia, K., Abbeel, P., & Dragan, A. D. (2018, October). Establishing appropriate trust via critical states. In *2018 IEEE/RSJ international conference on intelligent robots and systems (IROS)* (pp. 3929–3936). IEEE. https://doi.org/10.1109/IROS.2018.8593649.

Iqbal, T., Gonzales, M. J., & Riek, L. D. (2015, September). Joint action perception to enable fluent human-robot teamwork. In *2015 24th IEEE international symposium on robot and human interactive communication (RO-MAN)* (pp. 400–406). IEEE. https://doi.org/10.1109/ROMAN.2015.7333671.

Iqbal, T., Rack, S., & Riek, L. D. (2016). Movement coordination in human–robot teams: A dynamical systems approach. *IEEE Transactions on Robotics*, *32*(4), 909–919. https://doi.org/10.1109/TRO.2016.2570240.

Iqbal, T., & Riek, L. D. (2017). Human-robot teaming: Approaches from joint action and dynamical systems. In *Humanoid robotics: A reference* (pp. 2293–2312). https://doi.org/10.1007/978-94-007-7194-9 137-1.

Iqbal, T., Shah, A., & Riek, L. D. (2018). Toward a real-time activity segmentation method for human-robot teaming. In *Proc. of the robotics: Science and systems (RSS), towards a framework for joint action: What about theory of mind workshop*.

Khalid, H., Liew, W. S., Voong, B. S., & Helander, M. (2018, August). Creativity in measuring trust in human-robot interaction using interactive dialogs. In *Congress of the international ergonomics association* (pp. 1175–1190). Cham: Springer.

Lasota, P. A., & Shah, J. A. (2015). Analyzing the effects of human-aware motion planning on close-proximity human-robot collaboration. *Human Factors*, *57*, 21–33.

Launay, J., Dean, R. T., & Bailes, F. (2013). Synchronization can influence trust following virtual interaction. *Experimental Psychology*, *60*(1), 53–63. https://doi.org/10.1027/1618-3169/a000173.

Lee, J. J., Knox, B., Baumann, J., Breazeal, C., & DeSteno, D. (2013). Computationally modeling interpersonal trust. *Frontiers in Psychology*, *4*, 893. https://doi.org/10.3389/fpsyg.2013.00893.

Lee, J. D., & Moray, N. (1994). Trust, self-confidence, and operators' adaptation to automation. *International Journal of Human-Computer Studies*, *40*(1), 153–184. https://doi.org/10.1006/IJHC.1994.1007.

Lee, J. D., & See, K. A. (2004). Trust in automation: Designing for appropriate reliance. *Human Factors: The Journal of the Human Factors and Ergonomics Society*, *46*, 50–80. https://doi.org/10.1518/hfes.46.1.50 30392.

Lewis, M., Sycara, K., & Walker, P. (2018). The role of trust in human-robot interaction. In *Foundations of trusted autonomy* (pp. 135–159). Springer.

Li, D., Rau, P. P., & Li, Y. (2010). A cross-cultural study: Effect of robot appearance and task. *International Journal of Social Robotics*, *2*(2), 175–186.

Lohani, M., Stokes, C., McCoy, M., Bailey, C. A., & Rivers, S. E. (2016, March). Social interaction moderates human-robot trust-reliance relationship and improves stress coping. In *2016 11th ACM/IEEE international conference on human-robot interaction (HRI)* (pp. 471–472). IEEE. https://doi.org/10.1109/HRI.2016.7451811.

Marble, J. L., Bruemmer, D. J., Few, D. A., & Dudenhoeffer, D. D. (2004, January). Evaluation of supervisory vs. peer-peer interaction with human-robot teams. In *Proceedings of the 37th annual Hawaii international conference on system sciences, 2004*. IEEE. https://doi.org/10.1109/HICSS.2004.1265326. 9 pp.

Matsumoto, S., Washburn, A., & Riek, L. D. (2018). Human-robot co-manipulation demonstration. In: *Technology showcase for the contextual robotics institute forum*.

Mayer, R. C., Davis, J. H., & Schoorman, F. D. (1995). An integrative model of organizational trust. *Academy of Management Review*, *20*(3), 709–734. https://doi.org/10.5465/amr.1995.9508080335.

McKendrick, R., Shaw, T., de Visser, E., Saqer, H., Kidwell, B., & Parasuraman, R. (2014). Team performance in networked supervisory control of unmanned air vehicles: Effects of automation, working memory, and communication content. *Human Factors*, *56*(3), 463–475. https://doi.org/10.1177/0018720813496269.

Miles, L. K., Griffiths, J. L., Richardson, M. J., & Macrae, C. N. (2010). Too late to coordinate: Contextual influences on behavioral synchrony. *European Journal of Social Psychology*, *40*(1), 52–60. https://doi.org/10.1002/ejsp.721.

Miles, L. K., Lumsden, J., Flannigan, N., Allsop, J. S., & Marie, D. (2017). Coordination matters: Interpersonal synchrony influences collaborative problem-solving. *Psychology*, *8*, 1857–1878.

Miles, L. K., Lumsden, J., Richardson, M. J., & Neil Macrae, C. (2011). Do birds of a feather move together? Group membership and behavioral synchrony. *Experimental Brain Research*, *211*(3), 495–503. https://doi.org/10.1007/s00221-011-2641-z.

Moon, A., Parker, C. A., Croft, E. A., & Van der Loos, H. F. (2013). Design and impact of hesitation gestures during human-robot resource conflicts. *Journal of Human-Robot Interaction*, *2*(3), 18–40. https://doi.org/10.5898/JHRI.2.3.Moon.

Moray, N., & Inagaki, T. (1999). Laboratory studies of trust between humans and machines in automated systems. *Transactions of the Institute of Measurement and Control*, *21*(4–5), 203–211. https://doi.org/10.1177/014233129902100408.

Mori, M. (1970). The uncanny valley, translated by K. F. MacDorman and T. Minato. *Energy*, *7*(4), 33–35.

Muir, B. M. (1989). *Operators' trust in and percentage of time spent using the automatic controllers in supervisory process control task.* Doctoral dissertationUniversity of Toronto.

Occupational Safety and Health Administration (2006). *OSHA technical manual TED 01-00 015.* Washington, DC: US Department of Labor.

Ogreten, S., Lackey, S., & Nicholson, D. (2010, May). Recommended roles for uninhabited team members within mixed-initiative combat teams. In *2010 International symposium on collaborative technologies and systems* (pp. 531–536). IEEE. https://doi.org/10.1109/CTS.2010.5478468.

Ososky, S., Schuster, D., Phillips, E., & Jentsch, F. G. (2013, March). Building appropriate trust in human-robot teams. In *2013 AAAI spring symposium series*.

Paepcke, S., & Takayama, L. (2010, March). Judging a bot by its cover: An experiment on expectation setting for personal robots. In *2010 5th ACM/IEEE international conference on human-robot interaction (HRI)* (pp. 45–52). IEEE.

Parasuraman, R., & Miller, C. A. (2004). Trust and etiquette in high-criticality automated systems. *Communications of the ACM*, *47*(4), 51–55.

Rezvani, T., Driggs-Campbell, K., Sadigh, D., Sastry, S. S., Seshia, S. A., & Bajcsy, R. (2016, November). Towards trustworthy automation: User interfaces that convey internal and external awareness. In *2016 IEEE 19th International conference on intelligent transportation systems (ITSC)* (pp. 682–688). IEEE.

Riek, L. D. (2017). Healthcare robotics. *Communications of the ACM*, *60*(11), 68–78.

Riek, L. D., Rabinowitch, T. C., Bremner, P., Pipe, A. G., Fraser, M., & Robinson, P. (2010, March). Cooperative gestures: Effective signaling for humanoid robots. In *Proceedings of the 5th ACM/IEEE international conference on human-robot interaction* (pp. 61–68). IEEE Press.

Riley, V. A. (1994). Human use of automation. *Dissertation Abstracts International: Section B: The Sciences and Engineering*, *55*(6-B), 2425.

Robinette, P., Howard, A. M., & Wagner, A. R. (2015, October). Timing is key for robot trust repair. In *International conference on social robotics* (pp. 574–583). Springer.

Rousseau, D. M., Sitkin, S. B., Burt, R. S., & Camerer, C. (1998). Not so different after all: A cross-discipline view of trust. *Academy of Management Review*, *23*(3), 393–404. https://doi.org/10.5465/amr.1998.926617.

Salem, M., Lakatos, G., Amirabdollahian, F., & Dautenhahn, K. (2015, March). Would you trust a (faulty) robot?: Effects of error, task type and personality on human-robot cooperation and trust. In *Proceedings of the tenth annual ACM/IEEE international conference on human-robot interaction* (pp. 141–148). ACM.

Schaefer, K. E., Chen, J. Y. C., Szalma, J. L., & Hancock, P. A. (2016). A meta-analysis of factors influencing the development of trust in automation: Implications for understanding autonomy in future systems. *Human Factors*, *58*(3), 377–400. https://doi.org/10.1177/0018720816634228.

Shaw, T., Emfield, A., Garcia, A., de Visser, E., Miller, C., Parasuraman, R. , et al. (2010). Evaluating the benefits and potential costs of automation delegation for supervisory control of multiple uavs. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, *54*(19), 1498–1502. https://doi.org/10.1177/154193121005401930.

Takeda, K. (2016). Modeling and detecting excessive trust from behavior signals: Overview of research project and results. *Human-harmonized information technology* (pp. 57–75). Vol. 1 (pp. 57–75). Tokyo: Springer. https://doi.org/10.1007/978-4-431-55867-5 3.

Tanaka, K., Ozeki, M., & Oka, N. (2010, March). The hesitation of a robot: A delay in its motion increases learning efficiency and impresses humans as teachable. In *Proceedings of the 5th ACM/IEEE international conference on human-robot interaction* (pp. 189–190). IEEE Press.

Washburn, A., Adeleye, A., An, T., & Riek, L. D. (2020). Robot errors in proximate HRI: How functionality framing affects perceived reliability and trust. *ACM Transactions on Human-Robot Interaction (THRI)*, *9*(3), 19.

Woodworth, B., Ferrari, F., Zosa, T. E., & Riek, L. D. (2018). Preference learning in assistive robotics: Observational repeated inverse reinforcement learning. F. Doshi-Velez, J. Fackler, K. Jung, D. Kale, R. Ranganath, B. Wallace, & J. Wiens (Eds.), *Proceedings of the 3rd machine learning for healthcare conference* (pp. 420–439). *Proceedings of machine learning research: Vol. 85*(pp. 420–439). Palo Alto, CA: PMLR.

Wynne, K. T., & Lyons, J. B. (2018). An integrative model of autonomous agent teammate-likeness. *Theoretical Issues in Ergonomics Science*, *19*(3), 353–374.

Xu, A., & Dudek, G. (2015, March). Optimo: Online probabilistic trust inference model for asymmetric human-robot collaborations. In *Proceedings of the tenth annual ACM/IEEE international conference on human-robot interaction* (pp. 221–228). ACM. https://doi.org/10.1145/2696454.2696492.

Xu, A., & Dudek, G. (2016, October). Maintaining efficient collaboration with trust-seeking robots. In *2016 IEEE/RSJ international conference on intelligent robots and systems (IROS)* (pp. 3312–3319). IEEE. https://doi.org/10.1109/IROS.2016.7759510.

Xu, A., Kalmbach, A., & Dudek, G. (2014, May). Adaptive parameter EXploration (APEX): Adaptation of robot autonomy from human participation. In *2014 IEEE international conference on robotics and automation (ICRA)* (pp. 3315–3322). IEEE. https://doi.org/10.1109/ICRA.2014.6907336.

Yanco, H. A., Desai, M., Drury, J. L., & Steinfeld, A. (2016). Methods for developing trust models for intelligent systems. In *Robust intelligence and trust in autonomous systems* (pp. 219–254). Springer.

Yang, X. J., Unhelkar, V. V., Li, K., & Shah, J. A. (2017, March). Evaluating effects of user experience and system transparency on trust in automation. In *2017 12th ACM/IEEE international conference on human-robot interaction (HRI)* (pp. 408–416). IEEE. https://doi.org/10.1145/2909824.3020230.

CHAPTER

16

The role of interdependence in trust

Matthew Johnson and Jeffrey M. Bradshaw

Florida Institute for Human and Machine Cognition, Pensacola, FL, United States

Introduction

As technology continues to take on roles of greater scope and consequence, the issue of trust has moved increasingly to the forefront. Early advances in robotics and automation were typically applied in relatively simple contexts and structured environments where trustworthiness could be straightforwardly assessed and assured (Moorman, Deshpande, & Zaltman, 1993). However, with an increasing number of applications that attempt to cope with complex and uncertain problems once reserved almost exclusively for human judgment, new theories and methods to assess and assure trustworthiness have become imperative. As tragedies traced to defects in the design of complex systems proliferate (e.g. Travis, 2019; Levin & Beene, 2018) it has become clear that traditional approaches to trust fail to address the new risks and reduced transparency of advanced automation and artificial intelligence-based approaches.

Much of the previous work on trust in automation focuses merely on identifying factors that influence people's trust (Hancock et al., 2011; Hoff & Bashir, 2014; Lewandowsky, Mundy, & Tan, 2000; Moorman et al., 1993; Schaefer et al., 2014). Agreement on these factors is useful and sometimes provides helpful information about the importance of situation awareness for automation users. However, knowing factors often does not provide guidance with enough specification detail to influence design decisions that can help avoid such problems in the first place (Hoffman & Deal, 2008).

Though, unfortunately, some researchers still pursue the goal of merely promoting greater trust in automation, we believe that a better research

Trust in Human-Robot Interaction
https://doi.org/10.1016/B978-0-12-819472-0.00016-2

© 2021 Elsevier Inc. All rights reserved.

objective is to find ways to help people continuously maintain an *appropriate* level of reliance on technology, taking into account the reliability of the capabilities of the system as it functions at a given time and in a given situation context (Hoffman, Johnson, Bradshaw, & Underbrink, 2013; Lee & See, 2004). The latter objective has led to productive discussions in the research community about how to facilitate reliable "trust calibration." Research on trust calibration seeks ways to enhance an individual's ability to accurately assess the trustworthiness of the technology in different circumstances.

People always arrive at a relationship to some item of technology with some initial bias that leads them to under trust or overtrust its reliability and performance. This initial bias is, of course, outside of the control of those who designed and built the artifact. Fortunately, under normal circumstances, this initial trust assessment will be modified and updated as people use the technology. The trust people gain with the experience of use is what designers and builders can influence. This underscores the point that trust is not a static state, but rather the continuously varying result of a dynamic process (Bradshaw et al., 2004; Hoffman et al., 2009; Lewandowsky et al., 2000; Mayer, Davis, & Schoorman, 1995). For this reason, it is critical to understand the process by which trust evolves (Mayer et al., 1995), so designers and builders can understand how their implementation choices might increase or decrease both the trustworthiness of the technology and also the ability of people to accurately assess that trustworthiness.

In this chapter, we argue that interdependence relationships are the mechanism employed to actively manage the processes that assess trustworthiness of the technology and that enable accurate trust calibration by the people who use it. Trust is relational (Mayer et al., 1995) and interdependence is "the set of complementary relationships that two or more parties rely on to manage required (hard) or opportunistic (soft) dependencies in joint activity" (Johnson et al., 2014). Interdependence relationships constitute the necessary junctures in task-oriented communication and actions among people and machines that make the joint activity in which they are engaged productive (Johnson et al., 2014). Trust is commonly defined as "the willingness of a party to be vulnerable to the actions of another party based on the expectation that the other will perform a particular action important to the trustor" (Mayer et al., 1995), making the trustor dependent on the trustee. Typically, the relationship of trust among the collaborating parties in nontrivial situations is not a one-way transaction, making the parties not simply dependent but rather mutually *interdependent*. Interactions between humans and machines in mutually interdependent relationships allow them to support one another with respect to current interdependencies. The degree of performance and

reliability in current interactions helps actors accurately calibrate trust with respect to future interactions.

To understand how interdependence relationships are used to actively manage the assessment of trustworthiness and trust calibration, we will extend Mayer et al.'s (1995) model of human–human trust to propose a new model appropriate for groups of humans and machines. Then, we will discuss the role of interdependence in the new model. We will show how the model has allowed us to extend and employ Interdependence Analysis (IA) tables (Johnson, Vignati, & Duran, 2018) as an effective tool for understanding and designing systems capable of actively managing trust through interdependence relationships.

Model of a risk-taking trust relationship

One of the most cited models of trust is that of Mayer et al. (1995). Though developed for examining a single unidirectional human–human trust dyad in an organizational context, its appropriateness for human–machine trust relationships will be argued here. The original model is extended to consider a broader view of activity with refinements to the role of trust. This new model is depicted in Fig. 1.

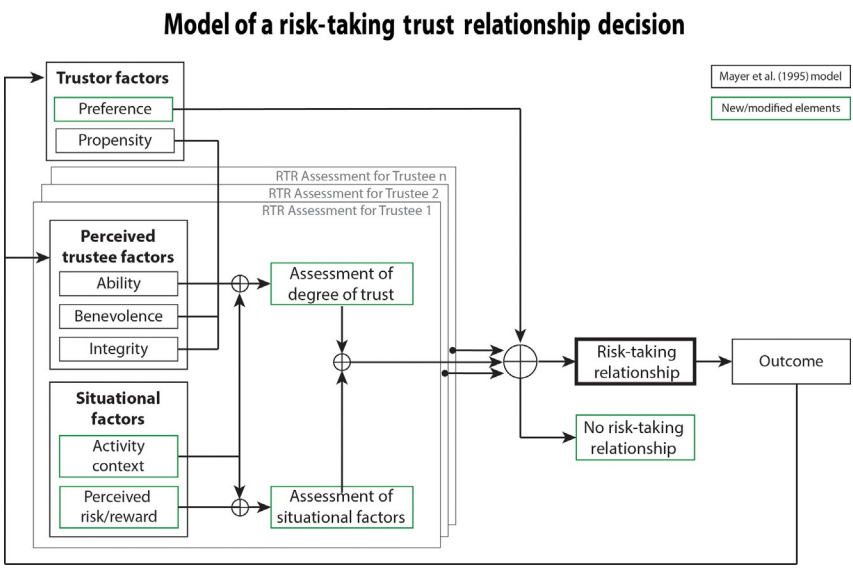


FIG. 1 Model of a risk-taking trust relationship decision, extended and refined from Mayer et al.'s model of trust (1995).

The model proposed by Mayer et al. (1995) is a model of trust, but emphasized the importance of trusting actions taken in the context of a specific risk-taking relationship (RTR). "There is no risk taken in the willingness to be vulnerable (i.e., to trust), but the risk is inherent in the behavioral manifestation of the willingness to be vulnerable. One does not need to risk anything to trust; however, one must take a risk to engage in trusting action" (Mayer et al., 1995, p. 724). Fig. 1 emphasizes a specific RTR decision as a key element in the model, not only because it is the manifestation of trust, but also because it is where the trustor engages in an interdependent relationship with the trustee. The outcome of that decision can strengthen or weaken the assessment of trustworthiness with experience over time. When direct information about the trustee is lacking, people compensate by using what other information they have (Rousseau, Sitkin, Burt, & Camerer, 1998) — for example, relying on the reputation of an institution to which the trustee belongs. However, these initial biases will be informed, modified, and potentially overturned by relational trust based on experience. Experience based on the outcomes of trusting actions, taken in the context of a specific RTR over time, enables accurate trust calibration, generating the objective and subjective bases for justified trust and mistrust. The importance of the RTR is stated by Mayer et al. in this way:

> This relationship-specific boundary condition of our approach is important because a number of authors have dealt with trust for generalized others (e.g., Rotter, 1967) and trust as a social phenomenon (e.g., Lewis & Weigert, 1985). Even though such approaches help provide a general sense of the considerations involved in trust, they do not clarify the relationship between two specific individuals and the reasons why a trustor would trust a trustee. Further, the failure to clearly specify the trustor and the trustee encourages the tendency to change referents and even levels of analysis, which obfuscates the nature of the trust relationship. (Mayer et al., 1995, p. 711).

The model proposed in Fig. 1 remains true to the importance of the relationship. It also extends the idea that context is critical by including additional situational factors and trustor preferences. While the original model was focused on factors affecting trust, the model proposed in Fig. 1 is focused on factors affecting the RTR decision. The RTR will be further examined below, following a description of other elements of the new model.

Characteristics of the trustor

Propensity. The Mayer et al. (1995) model includes the attribute of trustor propensity, which is described as a dispositional willingness to rely on others. People certainly have an innate propensity to either trust or

mistrust technology. It has also been shown that the propensity to trust technology often differs from the propensity to trust people (Madhavan & Wiegmann, 2012).

Preferences. Our model includes an additional factor not found in the original model: a preference. While preference is not a factor in trust assessment, it can play a role in whether an RTR is established or not. One can trust someone and still choose not to engage in an RTR based solely on preference. For example, nobody doubts the efficacy of automatic shifting mechanisms of today's cars, yet some people still choose to manually shift for the pleasure of it. Some people like working with others or using tools and technology, while some get satisfaction from doing it all on their own.

Characteristics of the trustee

The first three trustee factors are identical to the original Mayer et al. (1995) model: ability, benevolence, and integrity. Ability is the skill, competence, expertise of the trustee as perceived by the trustor. Benevolence is the trustor's belief in the trustee's desire to do good on behalf of the trustor. Integrity is the trustor's belief that the trustee adheres to an acceptable set of principles. These factors, determined from a human–human organizational context, seem equally valuable in human–automation contexts.

Commonalities in interpersonal and human–automation trust. It has been asserted that "Human–automation trust and interpersonal trust depend on different attributes. Whereas interpersonal trust can be based on the ability, integrity, or benevolence of a trustee (Mayer et al., 1995), human–automation trust depends on the performance, process, or purpose of an automated system (Lee & Moray, 1992)" (Hoff & Bashir, 2014, p. 413). Contrary to what Hoff and Bashir seem to claim, we see the relevant attributes of the trustee as essentially similar, regardless of whether the trust is interpersonal or directed toward machines. Indeed, two of the authors just cited (Lee and Moray) themselves concluded that "the multidimensional construct of trust developed to describe trust between humans, together with a consideration of the dynamic aspects of trust, can be used to describe trust between humans and machines" (Lee & Moray, 1992, p. 1268).

Ability. Ability plays a similar role in human–human and human–machine trust. Mayer et al. (1995) describe ability as "that group of skills, competencies, and characteristics" of the trustee as perceived by the trustor, this does capture some aspects we would consider important. A more complete definition would be Lee and See's concept of performance defined as "the current and historical operation of the automation and includes characteristics such as reliability, predictability, and ability"

(Lee & See, 2004, p. 59). One key challenge for automation is that humans typically find it easier to estimate the abilities of people than machines. Though people can be trained to make better assessments, they often persist despite themselves in projecting intentionality onto a machine (Nass & Moon, 2000).

Benevolence. While people can certainly hold benevolent feelings toward another, machines do not. There are those who would claim that the designers can build in benevolence, but this is nothing more than wishful mnemonics (McDermott, 1976). Current machines do not have desires, though they may have built-in utility functions. They have no understanding of broader goodness beyond their specific tasks and know little or nothing about the people (potential trustors) with whom they are working. While designers, in general, attempt to make systems "do their best" their attempts are more concerned with ability than benevolence. A notable exception is when designers deliberately try to create software that disables or hijacks machines that don't belong to them. In this sense, software might be seen as "malicious," though it is really the hacker behind the code that has acted with malicious intent. The code is simply "following instructions" that it has been given.

Integrity. In interpersonal relations, integrity is a moral virtue — another wishful mnemonic if it were to be applied literally to machines. However, when Mayer et al. refer to integrity, it means specifically that the trustee adheres to "an acceptable set of principles" (1995, p. 719). This is quite plausible for machines. In some sense, this aligns with Lee and Moray's (1992) concept of purpose, which essentially means carrying out the designer's intent. Depending on how the "acceptable set of principles" is defined, machines could have significantly more integrity than people. To the extent the principles can be formally codified, machines will and indeed must, in ordinary circumstances, meticulously and reliably comply with those principles. While such adherence to principles sounds like a solid win for trusting machines, the problem is in the concrete application of the necessarily abstract principles. This is demonstrated by the well-known example of Isaac Asimov's Laws of Robotics (Asimov, 1950). For instance, what does it mean to "do no harm" when the machine is faced with a zero-sum situation — where choosing to protect one party necessarily causes harm to another and vice versa? While a desirable objective, practical and effective principles for guiding automated behavior are not only difficult to code, but also may be difficult for people to understand.

Even when machines are incapable of benevolence or integrity, people may treat them as if they were acting intentionally (Dennett, 1989). It has been repeatedly demonstrated that people may apply social rules and expectations to computers in a mindless fashion (Nass & Moon, 2000). While designers may be able to exploit people's false projections,

it borders on the unethical to deliberately misrepresent what machines can and can't do except when performing experimental studies.

Situational factors

Risk is a big factor in trust. In the Mayer et al. model, "the perception of risk involves the trustor's belief about likelihoods of gains or losses outside of considerations that involve the relationship with the particular trustee" (1995, p. 726) and this is the same in Fig. 1. What has been added is the consideration of alternatives. While alternatives do not impact whether a trustor trusts a trustee, they do impact the decision to engage in a trusting action (i.e., RTR). Alternatives include both the different ways the trustor could trust and engage with the trustees, as well as the other ways the trustor could accomplish the work (e.g., do it themselves, engage another trustee).

How to choose whether to engage in an RTR

In the original Mayer et al. model, the decision to engage in an RTR was determined by a simple tradeoff between perceptions of trust and risk (Mayer et al., 1995, p. 726). We suggest that additional important considerations may come into play.

The first minor modification is the inclusion of both perceived risk and reward. Trustors will not only evaluate the potential for loss (risk), but also the potential for gain (reward). Greed has often motivated people to take high-risk actions.

The second modification is the addition of Activity Context in situational factors. Trust can only be evaluated with respect to the context in which and the method by which the action is being performed by a specific trustee. This is similar to Lee and See's concept of process, which they described as "Process is the degree to which the automation's algorithms are appropriate for the situation and able to achieve the operator's goals. Process information describes how the automation operates" (Lee & See, 2004, p. 59). For example, a parent may trust their teenager to drive in general but may re-evaluate that trust when considering a more unusual driving challenge like driving across the country or in a busy city. Similarly, parents needing to pick up a package may ask their teenager to choose a longer but more straightforward route than if one of them were doing it themselves. In the model in Fig. 1, the Activity Context attempts to capture these potentially crucial aspects of the decision process.

Activity Context influences both the trustor's assessment of the trustees trustworthiness and the perceived risk/reward. The assessment of trust, driven by the perceived Trustee factors and the Assessment of the

situational factors of activity context and perceived risk/reward are combined into the Trustors assessment of the specific RTR. Trust and risk assessments do not result in a binary decision to engage (trust) or not (mistrust). Hoffman et al. (2009) describe several possible trust relationships that vary in strength. It is not uncommon for people to be uncertain about trust. This is particularly relevant for human–machine trust.

One of the biggest considerations influencing an RTR decision is the options available. People do not consider a single option in isolation from all other options. A trustor might evaluate several alternative RTRs that could achieve the desired outcome, as shown in Fig. 1. Each of these alternatives can have different trustees, different trustworthiness, different activity context, and different risks and rewards. The trustor can consider multiple RTRs as well as the option to do the work themselves when making a choice. All of these options will be considered, as well as a host of additional considerations such as workload, attention demand, and efficiency.

Another often ignored consideration in trust relationships is a personal preference. Simply trusting a trustee is not sufficient to decide to engage in an RTR. The trustor must also prefer to engage in that relationship over other options, including the option to do nothing. Time and limited options might force one to engage with trustees that are not fully trusted. Most people might be willing to engage in a specific RTR even if they are somewhat uncertain about their level of trust, but their preference to do so can vary. Preferences may lead some people to never engage in actions requiring trust.

In short, the trustor will consider the level of trust for a given trustee in a given mode of interaction and situation. A specific RTR may be compared with other acceptable RTRs. Personal preference will help guide the choice of whether or not to engage in a given RTR. The results of an evaluation of one or more RTRs can be used to inform a trustor's decision.

We do not consider our changes to the Mayer et al. model as fundamental. Our model simply broadens its scope so that trust can be more adequately considered in the context of decision making. In the complex realm of human–machine interaction, we find that these additional considerations often play an important role in analysis, experimentation, and deployment.

The role of context

The importance of context has been frequently highlighted by researchers (Bradshaw et al., 2004; Hoffman et al., 2013; Lee & See, 2004; Mayer et al., 1995; Rousseau et al., 1998). Formalizing context has proven a persistent challenge, as has determining relevant context for different problems in different situations.

One basic approach to context is to frame a situation in terms of the *what*, *who*, and *how*. The context of *what* refers to the work that needs to be done or the goal that needs to be achieved. Mayer et al. (1995) state that the question "Do you trust them?" must be qualified with "trust them to do what?" This is Activity Context. Generally speaking, the threshold for trusting a machine to carry your luggage is considerably lower than that required to trust a machine to carry your newborn child. The details of the work and the methods by which it is performed also matter. A parent might trust a teenager to back the car out of the driveway or to drive themselves to school, but may not trust them to drive in the city or at night. To properly analyze and understand trust, modeling the work context in detail (i.e., the *what*) will be essential.

In our basic approach, one must also answer the *who* question. A simple description of the trustee is built into the Mayer et al. model (1995) in terms of ability, benevolence, and integrity. People do establish general feelings of trust toward individuals (e.g. family members, coworkers, teachers). This general feeling must be considered with respect to the details of the work to be done (the *what*). A trustor might trust a family member to drive their car, but they may not trust them to provide accurate medical advice. Conversely, the trustor might trust a doctor to provide medical advice, but not be willing to let the doctor borrow their car. Understanding the contexts of both *what* and *who* are important and the two are related. As noted in the "How to choose whether to engage in an RTR" section, both are needed to have sufficient context for adequately justified decisions. Both are also needed to properly analyze and understand trust.

Our approach to context also requires knowing *how* the work will be done. Part of understanding the context of how is understanding the ability of the trustee (the *who*). What aspects of the work do they consider, and which aspect do they not. It might seem great to some that an automated targeting robot can aim faster and fire more accurately than a human, but this ability loses its potency if the robot's targeting system cannot distinguish friend from foe. This is a critical Activity Context about *how* and is needed when assessing trust in such systems.

Another aspect of *how* is considering the method by which something might be achieved. The inclusion of consideration of alternative methods available to the trustee in the model provides contextual detail necessary for interpreting and assessing whether to engage in a specific RTR. Mayer et al. stated that "the specific consequences of trust will be determined by contextual factors such as the stakes involved, the balance of power in the relationship, the perception of the level of risk, *and the alternatives available to the trustor* [emphasis added]" (1995, pp. 726–727). The various options for methods of accomplishing the work provide a critical context for understanding decisions about whether to engage in an RTR or not. In other words, people consider more than whether they trust the

technology when choosing to rely on it. A trustor might trust the autonomous capabilities of their new Tesla, but trust themselves more in certain traffic situations. The decision might not be based on competence, but on an understanding of context. Or the decision might be purely preferential. The trustor might trust their Tesla to drive the route, but choose to manually drive the scenic route along the coast.

One last contextual consideration with respect to *how* that is important for automation, is the potential interaction available. Does the trustor need to trust it blindly? Can they monitor the automation effectively? Do they have any means to correct or influence the automation after the initial trust engagement? These interactive possibilities are part of the activity context and determine how automation can be engaged in different ways and define a variety of trust processes available.

Trust as a process

One of the key aspects of both the original Mayer et al. model and the proposed one is the feedback loop. When trying to understand trust and risk-taking trust relationships, it is important to remember that trust is a process. While there is a significant consensus on this (Bradshaw et al., 2004; Hoffman et al., 2009; Lewandowsky et al., 2000; Mayer et al., 1995) it is something worth re-emphasizing.

The idea that trust is a process is not merely an observation of human-human dyads or human-automation systems used to support the theory. Trust being a process has serious implications for technology, its design choices, and its acceptance. While an initial impression of trust (ability, benevolence, integrity) might be based on word of mouth, external observation, or blind faith, any long-term use will be based on the outcomes of RTRs. This poses a major challenge to today's technology whose ability to engage effectively in such relationships remains impoverished (Johnson & Vera, 2019). Today's technology relies heavily on reliable performance as the main (sometimes only) feedback. This makes trust in systems brittle and often leaves people ill-informed to make accurate trust assessments in changing contexts. More complex interaction in support of trust would allow for a broader range of trust management not commonly found in today's technology. For example, how does a system convey its abilities or its integrity (i.e., guiding principles)? How can people evaluate a system prior to having to trust it? How do interactions with the system build or erode trust?

Consider the example of a new babysitter. The trustor might ask the babysitter some questions about previous experience to establish credibility. They could also probe ability by asking hypothetical questions such as

"Do you know what to do if there is a fire?" Upon return, conversations about the evening's events can provide feedback that will alter the trustor's perception about the babysitter's ability, benevolence, or integrity.

Machines struggle with all of these options. You rarely can ask them about experience. Only a few systems allow engaging in "what-if" style interaction. The amount of feedback provided is usually quite limited, making it difficult for people to properly calibrate their trust in a given technology. The limited social repertoire of technology has led to a host of issues including automation surprises (Sarter, Woods, & Billings, 1997) and clumsy automation (Wiener & Curry, 1980) to name a few. Many of the limitations are captured in the Ten Challenges of making automation a "Team Player" (Klein, Woods, Bradshaw, Hoffman, & Feltovich, 2004).

Technology's limited capacity to interact not only impedes its effectiveness and trust but can confound research. In many examples comparing interpersonal trust and human-automation trust (Lewandowsky et al., 2000), a human is a "confederate" and no real human-machine interaction is permissible. Thus, for this type of research, the system is forced to be represented as a black box to the participant. Similarly, human–robot interaction research has pointed out that performance is one of the largest influences of trust (Hancock et al., 2011), but this is based on systems with no trust-enabling interaction, leaving performance as the only mechanism for establishing trust.

Human trust over time depends on the development of relational-trust (Rousseau et al., 1998). "Relational trust derives from repeated interactions over time between trustor and trustee. Information available to the trustor from within the relationship itself forms the basis of relational trust" (Rousseau et al., 1998, p. 399). Taking the perspective that trust is a process and that the exchange of information through repeated interactions over time is how appropriate relational trust is developed leads to the role of interdependence.

The role of interdependence

Before explaining the role of interdependence with respect to trust, a summary of its role in human–machine systems, in general, is in order. Coordination is the effective management of dependencies (Malone & Crowston, 1994), or more specifically interdependence relationships (Johnson et al., 2011). Coactive Design, is a unique design approach based on interdependence (Johnson et al., 2011). The goal of Coactive Design is to help designers identify interdependence relationships in a joint activity, so they can design systems that support these relationships, thus enabling designers to achieve the objectives of coordination, collaboration, and

teamwork (Johnson et al., 2014). Instead of considering how to allocate functions, the primary question is how to support interdependence relationships (Johnson, Bradshaw, & Feltovich, 2017).

Coactive Design proposes three essential interdependence relationships: observability, predictability, and directability (Johnson et al., 2014). Observability means making pertinent aspects of one's status, as well as one's knowledge of the team, task, and environment, observable to others. Predictability means that one's actions should be predictable enough that others can reasonably rely on them when considering their own actions. Directability means one's ability to influence the behavior of others and complementarily be influenced by others. These interdependence relationships are essential for effective coordination. They also turn out to be the same foundational relationships that are critical to trust. This was indirectly predicted by Rousseau et al., who noted that "New organizational forms built around the management of interdependence will provide a catalyst for innovative research on trust" (Rousseau et al., 1998, p. 402). So, in what ways do interdependence relationships influence trust?

Trust is developed through interdependence relationships

Interdependence relationships are the mechanism by which relational trust is established. In other words, to establish, develop, and maintain appropriate trust, technology needs to be endowed with appropriate support for interdependence relationships, such as observability, predictability, and directability. The overall goal of any technology should not simply be for people to trust it, but to support the development of appropriate justified trust and mistrust so that people understand when to trust and when not to. The goal should be the establishment of trust that leads to better performance outcomes by the human–machine combination.

Trust, whether between people or between people and machines, is always exploratory. As Hoffman states:

> Active exploration of trusting–relying relationships cannot and should not be aimed at achieving single stable states or maintaining some decontextualized metrical value, but must be aimed at maintaining an appropriate and context-dependent expectation. ***(Hoffman, 2017, p. 157)***

For simple trust relationships with a context suitable for human intuition (e.g. basic physics), nonsocial exploration is certainly feasible. There is no need to have a conversation with a chair to determine if you can stand on it. Some basic incremental loading will provide the feedback you need to decide. It is also unnecessary to understand the inner workings of systems

when how the work gets done is irrelevant. For example, it does not matter how the automatic teller moves money, just that the correct amount of money is presented for retrieval. However, today's technology is pushing into more sophisticated domains with greater complexity and uncertainty. People cannot simply observe an autonomous car drive in a single circumstance and be confident it can handle all driving situations. Doctors will not blindly accept artificial intelligence medical decisions without understanding something about the process behind the decision. Thus, today's sophisticated technology will need to engage in interdependence relationships, like observability, predictability, and directability to foster appropriate trust.

Observability means making pertinent aspects of one's status, as well as one's knowledge of the team, task, and environment, observable to others. This relationship allows a trustor to see or understand pertinent aspects of the trustee. This can and should involve more than just the performance outcome. Consider commercial airline autopilots and their ability to compensate for issues such as asymmetric icing on the wings. These systems silently compensate (Norman, 1990) leaving the pilot unaware of the trouble that is growing. Eventually, they reach their limits and abruptly hand control back to the pilot resulting in a bumpy transfer of controls (Woods & Sarter, 2000). These classic problems with automation are due to a lack of observability. There are many different ways a system can be observable, each with a cost and a benefit. The system could have informed the pilot that it was experiencing icing. If unable to determine that icing was the issue, the system could have informed the pilot that more power than normal was required. The system could have warned the pilot prior to reaching its limits that it was running out of control authority. All of these are examples of making pertinent aspects of the system observable to the human pilot. People have a natural tendency to engage in this type of progress appraisal sharing, even if it is as basic as uttering "something's not right." Machines often lack this basic social competence, even in some of today's most "capable" systems (Johnson & Vera, 2019). A review of Lee and See's design considerations for how to make automation trustable shows that a large number are examples of observability including:

- Show the past performance of the automation.
- Show the process and algorithms of the automation by revealing intermediate results in a way that is comprehensible to the operators.
- Simplify the algorithms and operation of the automation to make it more understandable.
- Show the purpose of the automation, design basis, and range of applications in a way that relates to the users' goals. (Lee & See, 2004, p. 74)

Predictability means that one's actions should be predictable enough that others can reasonably rely on them when considering their own actions. This relationship allows the trustor to establish expectations used to evaluate performance outcomes. Predictability has been a long-standing cornerstone of trust (Lee & See, 2004; Muir & Moray, 1996; Rempel & Holmes, 1985). It is unnecessary to explain why being predictable helps trust and being unpredictable hurts trust. However, it is useful to note the difference between predictable behavior (performance outcome) and supporting a predictable relationship (interaction). Again, progress appraisals play a key role (Feltovich, Bradshaw, Clancey, Johnson, & Bunch, 2008). Statements like "I am running late" or "I am getting tired" help the trustor adjust their expectations and avoid future trust violations from performance outcome deviations. Other means to convey predictability include complying with established norms and being deterministic not only in the outcome but also in how work is accomplished. Even failure can be more acceptable if it is predictable (Muir & Moray, 1996).

Directability means one's ability to influence the behavior of others and complementarily be influenced by others. This relationship allows a trustor to bound or obligates the behavior of a trustee (Bradshaw et al., 2004). While observability (sometimes referred to as transparency) (Chen et al., 2014; Lyons, Wortham, Theodorou, & Bryson, 2013; Yang, Unhelkar, Li, & Shah, 2017) and predictability (Hoff & Bashir, 2014; Lee & See, 2004; Schaefer et al., 2014) both have significant bodies of associated research connecting them to trust, directability has received much less attention. Often it is the ability to direct or influence a trustee that allows a trustor to take the initial steps of partial trust.

> Through experience in seeing the results of providing outside direction to the subject in order to avoid or to recover from failure (whether such failure is inadvertent or intentional) the observer also has an opportunity to learn something about the subject's disposition for compliance: proving the technology to see whether it will do all things that it is commanded. ***(Bradshaw et al., 2004, pp. 19–20)***

To be directable, the trustee needs to support receiving direction from the trustor and must also possess a "disposition for compliance." From an automation perspective, the machine must have a human-usable interface for providing direction and the algorithms must support ingestion and incorporation of the directions. For example, early driving direction software would provide you the "best" route and the user had no say in the results. Now such software offers three routes allowing the user to choose or even drag the route to adjust portions of the result. A trustor's ability to quickly and reliably bring a trustee into compliance can play a role in their risk assessment of the given alternative. Consider

what a driving instructor might be willing to trust a student driver to do if they are using a car in which the instructor has a complete set of duplicate controls, versus a car with only a duplicate brake, versus a car with no controls for the instructor. These alternatives will certainly affect the instructor's willingness to engage in an RTR.

Current deep learning approaches are an example of how a lack of support for interdependence can inhibit trust. Many of today's machine learning approaches fail to support observability, predictability, and directability. People cannot see what the state of the system is or what decisions are based upon. Making these systems more observable is a current area of research (Zeiler & Fergus, 2014). These approaches are notoriously unpredictable. They often provide odd and unpredictable results (Goodfellow, Shlens, & Szegedy, 2014) and demonstrations of their unpredictable brittleness, such as Tesla vehicles being tricked by a few stickers,[a] have become commonplace. These systems provide no indication of how or when they will fail. The design of these systems inhibits directability. They typically do not have interfaces for user input, only data input. Additionally, their algorithms are not designed to support direct input making compliance difficult or even impossible. While today's machine learning approaches are demonstrating amazing leaps in competence (ability), they struggle to support interdependence making trust and acceptance challenging. This is evidenced by recent research programs like the Defense Advanced Research Projects Agency (DARPA) Explainable Artificial Intelligence (XAI) program, which notes that "the effectiveness of these systems is limited by the machine's current inability to explain their decisions and actions to human users.[b]".

Support of interdependence can be achieved through a variety of means. Observability can be achieved by explicit direct communication, like stating "I am tired," or through behavioral displays (Feltovich, Bradshaw, Jeffers, Suri, & Uszok, 2004) such as sighing or body posture. Similarly, predictability can be stating one's intentions verbally or making actions legible (Dragan, Lee, & Srinivasa, 2013) so that one's intentions are clear. Directability can be achieved through commands, suggestions ("Have you tried jiggling it"), progress appraisals ("I'm running late"), warnings ("Watch your step"), helpful adjuncts ("Do you want me to carry that for you?"), and observations about relevant unexpected events ("It has started to rain"). All of these can play a role in allowing the trustee to convey trust signatures to the trustor. This is not a unidirectional process, because the trustor "can not only be made aware of trust and mistrust

[a]https://www.newscientist.com/article/2198325-teslas-autopilot-tricked-into-driving-on-the-wrong-side-of-the-road/

[b]www.darpa.mil/program/explainable-artificial-intelligence.

signatures but can also actively probe the technology (probing the world through the technology) to test hypotheses about trust, and then use the results to adjust subsequent human-machine activities (that is, reliance)" (Hoffman et al., 2013, p. 87).

In summary, the role of interdependence relationships is to support the active and continuous exploration of trust between a trustor and a trustee to ensure trustor assessments are appropriate for achieving the best outcomes possible.

Interdependence relationships enable partial trust

Trust is not all or nothing. There are many ways to trust and many types of trust relationships (Hoffman et al., 2009). Support for interdependence plays a role in which intermediary forms of trust are acceptable and under what conditions. "Degrees of interdependence actually alter the form trust may take" (Rousseau et al., 1998, p. 395). Initially, a parent may trust a child to cross a busy street only while holding their hand. Over time, the parent may be content to watch from the side of the road. Eventually, they will have complete trust. Along the way, support for interdependence (e.g. walking while holding hands, observing the child checking for traffic, observing proper decision making, predictable performance) will enable the establishment and development of trust.

Interdependence relationships help resolve the uncertain

Because interdependence enables partial trust, it helps resolve trust uncertainty. "The need for trusting behavior often arises while there is still a lack of data regarding some of the three factors" (Mayer et al., 1995, p. 730). People will often engage in RTRs to actively explore the suitability of RTRs. They use the feedback from initial engagements to inform future ones, thus reducing uncertainty through experience.

Interdependence relationships encourage opportunistic trust

Trust is not always about required dependence (i.e. a hard constraint). Rousseau et al. stated that "The second necessary condition of trust is interdependence, where the interests of one party *cannot* [emphasis added] be achieved without reliance upon another" (1998, p. 395). However, this is not the only option. There are many examples of trust where the trustor is perfectly capable of achieving the goal without reliance (e.g. commercial airline autopilots). These soft constraint examples of trust highlight the importance of consideration for both

other alternatives and trustor preference when trying to understand RTRs.

Interdependence relationships help manage the complexities of trust

Trust is complex and multidimensional. "The human's stance toward the machine is always some mixture of justified and unjustified trusting and justified and unjustified mistrusting…multiple trusting relations exist simultaneously" (Hoffman, 2017, pp. 152–153). Because of this, there is a need to leverage interdependence relationships to both establish justified trust (i.e. avoid under-reliance) and justified mistrust (avoid over-reliance). Additionally, interdependence relationships can be used to recognize and respond through interaction to counter both positive and negative trust misalignments (avoid unjustified trust and unjustified mistrust). Trusting is an evolving phenomenon. The basis of trust changes as the relationship progresses (Rempel & Holmes, 1985). "A more complete understanding of trust would come from consideration of its evolution within a relationship" (Mayer et al., 1995, p. 727).

Applying interdependence analysis to trust

Having a model is informative and can provide guidance, but may fall short of influencing design (Hoffman & Deal, 2008). Developers, who are by and large not experts in trust, may have a difficult time interpreting the model and applying it to their design problem. What they need is formative design tools (Johnson, Bradshaw, and Feltovich, 2017) that also help them account for trust.

Coactive Design proposed a design tool called the IA table (Johnson, Bradshaw, et al., 2017; Johnson et al., 2014, 2018). The purpose of IA is understanding how people and automation can effectively team by identifying and providing insight into the potential interdependence relationships used to support one another throughout an activity. This tool already supports many of the key aspects of the proposed trust model and can easily be extended to consider trust and emphasize how interdependence relationships support trust management.

The interdependence analysis table

The IA tool is in the form of a table, as shown in Fig. 2. For discussion purposes, the table presented is populated with a simplified analysis of the aircraft traffic collision avoidance problem. In short, how do you

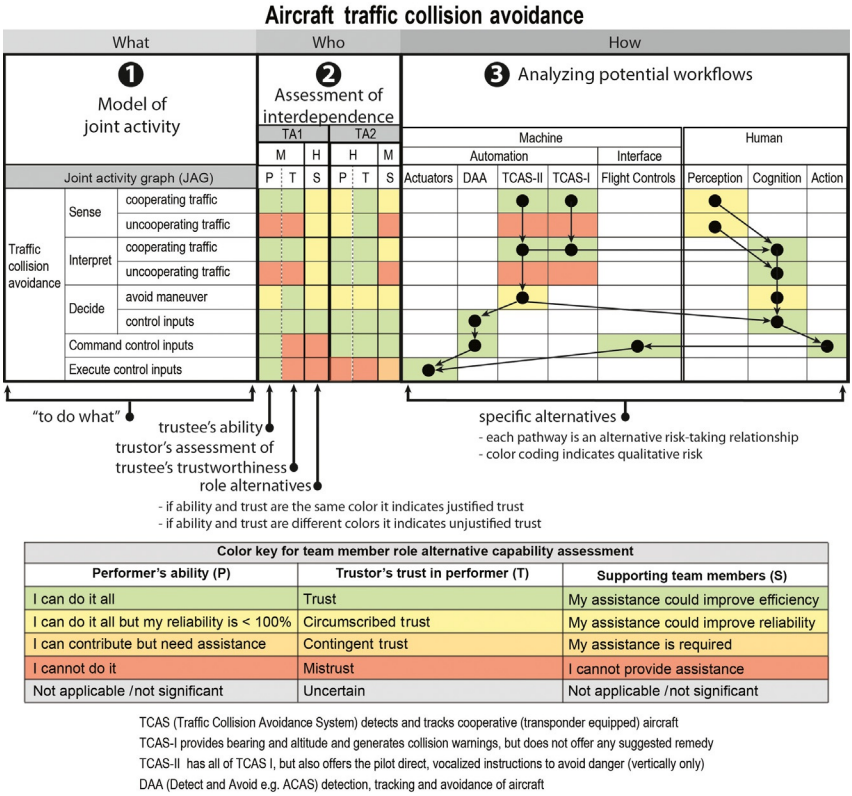


FIG. 2 How the interdependence analysis table relates to the proposed model of trust.

ensure that the aircraft does not collide with other air traffic? The table has three main sections: (1) model of joint activity (the what), (2) assessment of potential interdependence (the who), and (3) analysis of potential workflows (the how).

The what

The first section models the work. While there are many interesting nuances about how to model joint work (Johnson et al., 2018), for this discussion the model of work is the answer to the question "trust it to do *what*?" This provides the activity context. At this point in the analysis, the designer is not concerned with either the *who* (trustee) or the *how*, both of which are needed for a full trust assessment of alternatives. For the aircraft traffic collision avoidance, the work involves sensing traffic, interpreting if it will collide, deciding how to avoid it, and executing the avoidance maneuver.

The who

The second section captures the *who* and provides a systematic way to assesses potential interdependence. It involves enumerating the viable team role alternatives, assessing the ability to perform the work, assessing the ability to support another team member as they perform the work, identifying potential interdependencies, and then determining the requirements to support the interdependence relationships of interest. The team role alternatives capture the set of entities that can potentially participate in the work. The example in Fig. 2 depicts two alternatives. Alternative one (TA1) is the machine (M), in other words, automation, as the performer with the potential for human pilot (H) assistance. The second alternative (TA2) is the human as the performer with the potential for automated assistance.

The first column in an alternative is defined as the performer (P), meaning the entity doing the work. To accommodate a trust assessment, the IA table has been extended with an additional column under performer labeled trustworthiness (T). This is the trustor's subjective assessment of the given performer's trustworthiness in performing the specific aspect of work it aligns within section 1. The remaining columns in an alternative are supporting members (S) assisting the performer. The assessments are generally qualitative assessments captured using a color-coding scheme explained in Fig. 2, though it can be supplemented with empirical quantitative data when available. The color-coding is different for each type of column.

The color-coding of the performer (P) is an assessment of the capacity to perform. This is typically the designer's assessment of the ability of the performer who is the trustee in the relationship. Green means reliable, yellow is less than perfect reliability, orange requires assistance, and red means no ability.

The color-coding for the newly added trustworthiness column (T) reflects the trustor's subjective trustworthiness assessment of the given performer (trustee) to perform the specified aspect of the work. By connecting trust to work we can improve the specificity, which refers to the degree to which trust is associated with a particular component or aspect of the trustee (Lee & See, 2004). Green means the trustee is trusted. Yellow indicates circumscribed trust (Hoffman et al., 2009). This is a more limited trust that might vary with time or activity context. Orange indicates contingent trust (Hoffman et al., 2009). This is an even more limited trust that depends on circumstances. Such trust might demand additional monitoring or progress appraisal checkpoints. Red is mistrust and indicates that the trustor is unlikely to engage in an RTR. Gray is uncertain. The trustor might choose to explore trust by engaging in an RTR, or they may be hesitant to do so depending on their propensity.

The color-coding for the supporting team member column (S) is an assessment of that team member's potential to support the performer for the activity specified by the row. The color red indicates no potential for interdependence, thus the independent operation is the only viable option for the task. Orange indicates a hard constraint, such as providing supplemental lifting capacity when objects are too heavy. Another example of orange is when a machine needs human authorization to perform the activity. Yellow is used to represent improvements to reliability. For example, a human could provide recognition assistance to a robot and increase the reliability in identifying coffee mugs. Green is used to indicate assistance that may improve efficiency. For example, a robot may be able to determine the shortest route much faster than a human or could assist in cleaning up a room. The supporting team member columns are used to identify interdependence requirements (i.e., observability, predictability, directability) needed to support joint activity, but these are also the type of relationships used to calibrate trust.

The purpose of the color-coding is to help identify important design issues with respect to human–machine teaming. The colors and relationships between colors help characterize the design (Johnson et al., 2014). The performer column colors help identify potential brittleness and hard constraints. The supporting team member columns help identify both hard and soft interdependencies (Johnson et al., 2011). With respect to trust, the colors can not only help assess the type of trust but also the correspondence between a person's trust in the automation and the automation's capabilities sometimes referred to as trust calibration (Lee & Moray, 1994). For example, in the scenario in Fig. 2, the Traffic Collision Avoidance System (TCAS[c]) is used to sense traffic. TCAS can only detect cooperating traffic with special equipment. Aircraft without such equipment cannot be sensed and is referred to as uncooperating traffic. Accordingly, in Fig. 2, the machine has the ability to sense cooperating traffic (TA1 column P is green) and the trustor trusts the system (TA1 column T is green). This is justified trust. The machine does not have the ability to sense uncooperating traffic (TA1 column P is red) and the trustor does not trust the system (TA1 column T is red). This is justified mistrust. According to the example, the system is less than 100% reliable at deciding on the proper avoidance maneuver (TA1 column P is yellow), but the trustor trusts the system (TA1 column T is green). This is unjustified trust. Once the decision is made, the system is perfectly capable of executing the avoidance maneuver (TA1 column P is green) yet the trustor does

[c]You would not normally have TCASI and TCAS II in the same system. This example includes both just to demonstrate how you can capture different capabilities. Additionally, TCAS and DAA are often separate systems, but they are shown working together for similar explanatory purposes.

not trust the system to do so (TA1 column T is red). This is unjustified mistrust. These are just a few of the possibilities of how to capture trust in the context of a specific trustee and a specific activity.

The how

The third section connects the theoretical understanding of work to physical instantiation in a specific system. Across the top of the workflow section in Fig. 2, there are column headings for each algorithm, interface element, or human capability used to accomplish the task. Below each heading is a black dot to indicate where in the activity that particular component has a role. The dots are connected with arrows to indicate potential workflows to accomplish the goal. The resulting graph structure is a visual description of all existing and potential workflows, in other words, the alternatives.

The color-coding in the workflow section shown in Fig. 2 is a copy of the performer's ability color-coding. This represents the risk of taking a specific pathway (alternative). For example, the risk of a fully automated response to traffic includes the risk of not seeing uncooperating traffic (red indicates certain to fail) and the risk of making the wrong avoidance maneuver decision (yellow indicates only a possibility of failing). Now consider the fully manual pathway. There is a risk that the human might not see the traffic due to lack of attention (yellow for sensing). However, the human is capable of sensing uncooperating traffic. People, like machines, have the potential to make an incorrect decision about how to avoid traffic (yellow). The IA table allows for risk assessment for specific alternatives and enables the designer to see all of the risks in context, not simply a general risk for automating traffic avoidance as a whole. It also allows comparison of risk across alternatives.

So which option is better: fully automated or manual? The answer, as usual, is neither. The best answer is a combination of the two (Johnson et al., 2017; Johnson & Vera, 2019). The workflows in Fig. 2 show some limited support for interdependence as horizontal arrows that cross between the human and machine sections. In the interpretation activity, TCAS can alert the human about traffic. TCAS-II can provide the pilot with a recommended avoidance maneuver (e.g. "pull up"). These interdependence relationships, alerting (observability), and recommending (directability) enable additional alternatives. The human pilot can be supplemented by the automated solution to potentially outperform either alone. Supplementation means the human can partially mitigate the automation's lack of ability to sense uncooperating traffic. It also means the automation can partially mitigate the human's limited attention.

Although not shown in Fig. 2, the workflow colors could also be based on trust (T column of the performer). While using the ability color-coding

speaks to potential performance outcomes, using the trust color-coding speaks to the potential reliance. It is irrelevant if a system has the potential for high-performance outcomes if the trustor does not trust the system and never relies on it. In the example shown in Fig. 2, the trustor (human pilot) does not trust the system to command the control inputs. This means that no matter how good the automated solution is, the pilot would not rely on it. However, the pilot does trust the system in other ways making both the manual and the human–automation teaming options viable.

The IA table, with small modifications presented here, supports the analysis of human–machine trust based on the proposed model shown in Fig. 1. It provides activity context, directly addresses the trustee's ability, speaks to RTR alternatives, risk, and trust, and allows for distinctions about the different types of trust. The IA table provides a detailed level of context necessary for proper consideration of trust. Additionally, the IA table can help understand and design support for interdependencies (its original purpose) with respect to not only joint activity but also the process of active trust management. In sum, it helps consider all of the major factors influencing engagement in risk-taking trust relationships.

The value of IA is that it provides a detailed contextual lens for interpreting trust. It enables designers, developers, or analysts to capture the *what*, *who*, and *how* contextual aspects of trust in a systematic way. The ability to interpret trust at a finer resolution provides the potential for improved measurement of trust which in turn has the potential to improve predictions of trust. IA also provides a means to compare the risks of different alternatives available which can potentially improve predictions about choices to engage in different RTRs. The contextual detail also enables engineers to identify weaknesses in their design and develop engineering advances aimed at improving active trust management.

Conclusion

This work has proposed a new model of risk-taking trust relationships that is an extension and refinement of Mayer et al.'s (1995) model of trust. It combines the original trustee factors and trustor propensity with considerations for activity context, perceived risk/reward, other RTRs, and trustor preference. The role of interdependence relationships is to support active and continuous exploration of trust between a trustor and a trustee to ensure trustor assessments are appropriate for achieving the best outcomes possible. IA provides a contextual lens for interpreting trust. Future sophisticated technology will need to engage in interdependence relationships, like observability, predictability, and directability to foster appropriate trust calibration. The IA table is ideally suited to the analysis of

human–automation trust and aligns well with the proposed model. It can be an effective tool for understanding and designing systems capable of actively managing trust through interdependence relationships.

References

Asimov, I. (1950). *I Robot*. New York: Bantam Dell.

Bradshaw, J. M., Jung, H., Kulkarni, S., Johnson, M., Feltovich, P., Allen, J. , et al. (2004). Toward trustworthy adjustable autonomy in KAoS. In: *Trusting agents for trusting electronic societies*. https://doi.org/10.1007/11532095_2.

Chen, J. Y. C., Procci, K., Boyce, M., Wright, J., Garcia, A., & Barnes, M. J. (2014). *Situation awareness-based agent transparency*: (pp. 1–29). US Army Research Laboratory. April, Retrieved from (2014). https://apps.dtic.mil/docs/citations/ADA600351.

Dennett, D. C. (1989). *The intentional stance*. Retrieved from *(1989)*. https://books.google.com/books?hl=en&lr=&id=Qbvkja-J9iQC&oi=fnd&pg=PP11&dq=Dennett,+Daniel+C.+The+Intentional+Stance+&ots=73gnKNk1M0&sig=Q-QcDMDZ_6L340s9Uun9Rv2B85M.

Dragan, A. D., Lee, K. C. T., & Srinivasa, S. S. (2013). Legibility and predictability of robot motion. In: *ACM/IEEE international conference on human-robot interaction, 1*, pp. 301–308. https://doi.org/10.1109/HRI.2013.6483603.

Feltovich, P. J., Bradshaw, J. M., Clancey, W. J., Johnson, M., & Bunch, L. (2008). Progress appraisal as a challenging element of coordination in human and machine joint activity. *Lecture Notes in Computer Science*, *4995 LNAI*, 124–141. (including subseries lecture notes in artificial intelligence and lecture notes in bioinformatics) (2008). https://doi.org/10.1007/978-3-540-87654-0_6.

Feltovich, P. J., Bradshaw, J. M., Jeffers, R., Suri, N., & Uszok, A. (2004). Social order and adaptability in animal and human cultures as analogues for agent communities: Toward a policy-based approach. A. Omacini, P. Petta, & J. Pitt (Eds.), *Engineering societies in the agents world IV* (pp. 21–48). Vol. Lecture(pp. 21–48). Heidelberg, Germany: Springer.

Goodfellow, I. J., Shlens, J., & Szegedy, C. (2014). *Explaining and harnessing adversarial examples*. https://doi.org/10.1109/CVPR.2015.7298594.

Hancock, P. A., Billings, D. R., Schaefer, K. E., Chen, J. Y. C., De Visser, E. J., & Parasuraman, R. (2011). A meta-analysis of factors affecting trust in human-robot interaction. *Human Factors*, *53*(5), 517–527. https://doi.org/10.1177/0018720811417254.

Hoff, K. A., & Bashir, M. (2014). Trust in automation: Integrating empirical evidence on factors that influence trust. *Human Factors*, *57*(3), 407–434. https://doi.org/10.1177/0018720814547570.

Hoffman, R. R. (2017). A taxonomy of emergent trusting in the human-machine relationship. In *Cognitive systems engineering: The future for a changing world* (pp. 137–164). Boca Raton, FL: CRC Press, Taylor & Francis. https://doi.org/10.1201/9781315572529.

Hoffman, R. R., & Deal, S. V. (2008). Influencing versus informing design, part 1: A gap analysis. *IEEE Intelligent Systems*, *23*(5), 78–81.

Hoffman, R. R., Johnson, M., Bradshaw, J. M., & Underbrink, A. (2013). Trust in automation. *IEEE Intelligent Systems*. *28*(1). https://doi.org/10.1109/MIS.2013.24.

Hoffman, R. R., Lee, J. D., Woods, D. D., Shadbolt, N., Miller, J., & Bradshaw, J. M. (2009). The dynamics of trust in cyberdomains. *IEEE Intelligent Systems*, *24*(6), 5–11. Retrieved from (2009). http://eprints.ecs.soton.ac.uk/20372/.

Johnson, M., Bradshaw, J. M., & Feltovich, P. J. (2017). Tomorrow's human–machine design tools: From levels of automation to interdependencies. *Journal of Cognitive Engineering and Decision Making*. https://doi.org/10.1177/1555343417736462. 155534341773646.

Johnson, M., Bradshaw, J., Feltovich, P., Jonker, C., van Riemsdijk, B., & Sierhuis, M. (2011). The fundamental principle of coactive design: Interdependence must shape autonomy. M. De Vos, N. Fornara, J. Pitt, & G. Vouros (Eds.), *Coordination, organizations, institutions, and norms in agent systems VI* (Vol. 6541, pp. 172–191). https://doi.org/10.1007/978-3-642-21268-0_10.

Johnson, M., Bradshaw, J. M., Feltovich, P. J., Jonker, C. M., van Riemsdijk, B. M., & Sierhuis, M. (2014). Coactive design: Designing support for interdependence in joint activity. *Journal of Human-Robot Interaction*, *3*(1), 43–69.

Johnson, M., Shrewsbury, B., Bertrand, S., Calvert, D., Wu, T., Duran, D. , et al. (2017). Team IHMC's lessons learned from the DARPA robotics challenge: Finding data in the rubble. *Journal of Field Robotics*, *34*(2), 241–261. https://doi.org/10.1002/rob.21674.

Johnson, M., & Vera, A. H. (2019). No AI is an island: The case for teaming intelligence. *AI Magazine*, *Spring*, 16–28.

Johnson, M., Vignati, M., & Duran, D. (2018). Understanding human-autonomy teaming through interdependence analysis. In: *Symposium on human autonomy teaming*.

Klein, G., Woods, D. D., Bradshaw, J. M., Hoffman, R. R., & Feltovich, P. J. (2004). Ten challenges for making automation a "team player" in joint human-agent activity. *IEEE Intelligent Systems*, *19*(6), 91–95. https://doi.org/10.1109/MIS.2004.74.

Lee, J. D., & Moray, N. (1992). Trust, control strategies and allocation of function in human-machine systems. *Ergonomics*, *35*(10), 1243–1270. Retrieved from (1992). https://www.tandfonline.com/doi/abs/10.1080/00140139208967392.

Lee, J. D., & Moray, N. (1994). Trust, self-confidence, and operators' adaptation to automation. *International Journal of Human-Computer Studies*, *40*(1), 153–184. Retrieved from (1994). https://pdfs.semanticscholar.org/42e4/39534a7952610884b183d123e8a689ac8684.pdf.

Lee, J. D., & See, K. A. (2004). Trust in automation: Designing for appropriate reliance. *Human Factors*, *46*(1), 50–80. https://doi.org/10.1518/hfes.46.1.50_30392.

Levin, A., & Beene, R. (2018). *Tesla's autopilot found partly to blame for 2018 crash on the 405. Retrieved from 4 January 2020, from Los Angeles times website(2018).* https://www.latimes.com/business/story/2019-09-04/tesla-autopilot-is-found-partly-to-blame-for-2018-freeway-crash.

Lewandowsky, S., Mundy, M., & Tan, G. P. (2000). The dynamics of trust: Comparing humans to automation. *Journal of Experimental Psychology Applied*, *6*(2), 104–123. Retrieved from (2000). https://psycnet.apa.org/journals/xap/6/2/104/.

Lewis, J. D., & Weigert, A. (1985). Trust as a social reality source: Social forces. *Social Forces*, *63*(4), 967–985. https://doi.org/10.1093/sf/63.4.967.

Lyons, J. B., Wortham, R. H., Theodorou, A., & Bryson, J. J. (2013). Being transparent about transparency: A model for human-robot interaction. In: *Proceedings of AAAI spring symposium on trust in autonomous systems*, pp. 48–53. https://doi.org/10.1021/ja00880a025.

Madhavan, P., & Wiegmann, D. A. (2012). A new look at the dynamics of human-automation trust: Is trust in humans comparable to trust in machines? *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, *48*(3), 581–585. https://doi.org/10.1177/154193120404800365.

Malone, T. W., & Crowston, K. (1994). The interdisciplinary study of coordination. *ACM Computing Surveys*, *26*(1), 87–119. https://doi.org/10.1145/174666.174668.

Mayer, R. C., Davis, J. H., & Schoorman, F. D. (1995). An integrative model of organizational trust. *Academy of Management Review*, *20*(3), 709–734. https://doi.org/10.5465/amr.1995.9508080335.

McDermott, D. (1976). Artificial intelligence meets natural stupidity. *ACM SIGART Bulletin*, Retrieved from (1976). http://dl.acm.org/citation.cfm?id=1045340.

Moorman, C., Deshpande, R., & Zaltman, G. (1993). Factors affecting trust in market research relationships. *Journal of Marketing*, *57*(1), 81–101. Retrieved from (1993). https://journals.sagepub.com/doi/abs/10.1177/002224299305700106.

Muir, B. M., & Moray, N. (1996). Trust in automation. Part II. Experimental studies of trust and human intervention in a process control simulation. *Ergonomics*, *39*(3), 429–460. Retrieved from (1996). https://www.researchgate.net/profile/Neville_Moray/publication/14354918_Trust_in_automation_Part_II_Experimental_studies_of_trust_and_human_intervention_in_a_process_control_simulation/links/58219d0d08ae40da2cb777b9.pdf.

Nass, C., & Moon, Y. (2000). Mindfulness theory and social issues – machines and mindlessness – social responses to computers. *Journal of Social Issues*, *2000*(1), 81–104. Retrieved from (2000). http://www.coli.uni-saarland.de/courses/agentinteraction/contents/papers/Nass00.pdf.

Norman, D. (1990). The "problem" of automation. *Philosophical Transactions of the Royal Society of London Series B, Biological Sciences*, *327*(1241), 585–593. Retrieved from(1990). https://pdfs.semanticscholar.org/e7e9/a8ddc88c30bcd408805dab80cba0052f97b9.pdf.

Rempel, J., & Holmes, J. (1985). Trust in close relationships. *Journal of Personality and Social Psychology*, *49*(1), 95–112. Retrieved from(1985). https://www.researchgate.net/profile/John_Holmes9/publication/232554295_Trust_in_close_relationships_J_Pers_Soc_Psychol/links/0046352cdcd5694aee000000/Trust-in-close-relationships-J-Pers-Soc-Psychol.pdf.

Rotter, J. B. (1967). A new scale for the measurement of interpersonal trust. *Journal of Personality*, *35*(4), 651–665. https://doi.org/10.1111/j.1467-6494.1967.tb01454.x.

Rousseau, D. M., Sitkin, S. B., Burt, R. S., & Camerer, C. (1998). Not so different after all: A cross-discipline view of trust. *Academy of Management Review*, *23*(3), 393–404. https://doi.org/10.5465/amr.1998.926617.

Sarter, N. B., Woods, D. D., & Billings, C. E. (1997). Automation surprises. In G. Salvendy (Ed.), *Handbook of human factors and ergonomics* (pp. 1926–1943). (2nd ed.). Wiley.

Schaefer, K. E., Billings, D. R., Szalma, J. L., Adams, J. K., Sanders, T. L., Chen, J. Y. C. , et al. (2014). A meta-analysis of factors influencing the development of Trust in Automation: Implications for human-robot interaction. In *(No. ARL-TR-6984). Army Research Lab Aberdeen proving ground md human research and engineering directorate*. https://doi.org/10.1177/0018720816634228.

Travis, G. (2019). How the Boeing 737 max disaster looks to a software developer-IEEE Spectrum. *IEEE Spectrum*, Retrieved from(2019). https://spectrum.ieee.org/aerospace/aviation/how-the-boeing-737-max-disaster-looks-to-a-software-developer.

Wiener, E. L., & Curry, R. E. (1980). Flight-deck automation: Promises and problems. *Ergonomics*, *23*(10), 995–1011. https://doi.org/10.1080/00140138008924809.

Woods, D. D., & Sarter, N. B. (2000). *Learning from automation surprises and "going sour" accidents. In Cognitive engineering in the aviation domain* (pp. 327–353). Retrieved from(2000). https://www.researchgate.net/profile/David_Woods11/publication/268275060_Learning_from_Automation_Surprises_and_Going_Sour_Accidents/links/54dcc7620cf282895a3b22a3.pdf.

Yang, X. J., Unhelkar, V. V., Li, K., & Shah, J. A. (2017). Evaluating effects of user experience and system transparency on trust in automation. In: *HRI '17: Proceedings of the 2017 ACM/IEEE international conference on human-robot interaction*, pp. 408–416. https://doi.org/10.1145/2909824.3020230.

Zeiler, M. D., & Fergus, R. (2014). Visualizing and understanding convolutional networks. In: *European conference on computer vision*, Cham: Springer, pp. 818–833.

CHAPTER

17

Applying the swift trust model to human-robot teaming

Kerstin S. Haring[a], *Elizabeth Phillips*[b], *Elizabeth H. Lazzara*[c], *Daniel Ullman*[d], *Anthony L. Baker*[e], *and Joseph R. Keebler*[c]

[a]Department of Computer Science, The University of Denver, Denver, CO, United States, [b]Department of Psychology, George Mason University, Fairfax, VA, United States, [c]Department of Human Factors and Behavioral Neurobiology, Embry-Riddle Aeronautical University, Daytona Beach, FL, United States, [d]Department of Cognitive, Linguistic, and Psychological Sciences, Brown University, Providence, RI, United States, [e]US Army Combat Capabilities Development Command, Aberdeen Proving Ground, MD, United States

Introduction

Research has long established the importance of trust in a team's effectiveness (de Jong, Dirks, & Gillespie, 2015). More specifically, trust has a direct impact on group properties and processes that affect team performance. For instance, trust influences team cohesion (Mach, Dolan, & Tzafrir, 2010), team communication (Boies, Fiset, & Gill, 2015), and team satisfaction (Costa, 2003). A traditional description of trust development is that trust develops in teams through repeated, positive interactions between parties over time as well as from prior history used to judge each other's trustworthiness (McAllister, 1995; Rempel, Holmes, & Zanna, 1985). However, the introduction of new technologies and robotic team members is changing the trust dynamics in mixed teams. Traditional trust descriptions and models of trustworthiness are not sufficient to explain the complexity added by human-robot trust in human-robot teams.

Nowadays, teams are leveraging innovative technologies to perform new tasks in novel ways. Automation and the use of autonomous

https://doi.org/10.1016/B978-0-12-819472-0.00017-4
© 2021 Elsevier Inc. All rights reserved.

systems—especially robots—in teams are increasing and changing the way teams interact. Robots and other autonomous technologies that are being inserted into teams as team members play an integral role in meeting labor demands for personnel, safety, and productivity in domains like healthcare, space exploration, transportation, service, entertainment, and education (Schaefer, 2013). For example, robots have been used to deliver learning experiences through social interactions with children, increase the safety of medical procedures, explore shipwrecks, explore Mars, dispose of explosives, and serve in the cleanup of the Fukushima Daiichi nuclear plant and cleanup efforts after the World Trade Center Disaster (Murphy et al., 2008). In these domains, robots are already used to augment the skills of humans by serving as collaborative teammates - communicating proactively, anticipating the needs of their human counterparts, and working with humans to achieve common goals. To create effective team interactions and integrate robots as successful team members, people must be willing to accept and adhere to robot-generated information and rely on the robot in their team (Freedy, DeVisser, Weltman, & Coeyman, 2007). Therefore, trust in robot teammates is needed to support human-robot teams and collaborations to enable team cohesion, communication, and satisfaction. It also is considered an important metric for examining human-robot interaction (Steinfeld et al., 2006). Robots, similar to the use of mobile devices such as smartphones, are expected to experience continuous growth in various application areas. Ultimately, the team trust dynamic will influence whether humans reap the full benefits of working together with robotic systems.

Robotic systems, like all machines, are limited by current state-of-the-art technology. This includes computational power, artificial intelligence, learning algorithms, embodiment, mobility, and reasoning capabilities. Consequently, the role of robots in human-machine teams complements the skills and expertise of the human members of the teams. The role of a robot in a team is also specialized according to the capacity of the individual robot. The capacities of a robot also dictate the role it can assume in a team. Even though people readily perceive the capacity in robots to be a team member in addition to other capacities in robots (Malle, 2019; Malle, Scheutz, Forlizzi, & Voiklis, 2016), contemporary interactions with robots deployed in human spaces are likely to continue to be short, finite, and focused on a narrow range of tasks. The majority of laypeople have little to no prior experience with robots in the majority of applications due to the current lack of accessibility to consumers. Due to these limitations, there are few opportunities for repeated interactions over time and little experience upon which to draw from when robots begin to work in partnership with people shortly in higher functioning teams. Therefore, traditional models of human interpersonal trust may not fully capture how trust is built and maintained in human-robot teams.

One might argue, that over time, even laypeople will not always be novices in interactions with robots and have a larger body of prior exposure

and experiences with robots to rely on. Although people's general experiences with robots will indeed grow over time (Groom & Nass, 2007; Scassellati, 2002), technological advances allow robots to complete a broader range of tasks (Laschi, Mazzolai, & Cianchetti, 2016), for longer periods (Gockley et al., 2005; Kidd & Breazeal, 2008; Leite, Martinho, & Paiva, 2013), and in more environments (e.g., industry, homes, healthcare, work, education, entertainment, search, and rescue). While people may be able to employ some degree of technical knowledge transfer—leveraging their existing knowledge about one robot domain to another—the rapid pace of robot technological advances implies that the nature of interactions with robots will continuously change. It stands to reason that robots will frequently change and will need to be updated, repaired, or replaced over time. Teams will likely interact with an increasing variety and number of robots. Thus, team members will need to reevaluate each robot anew as it enters the team. It is possible, then, that models of interpersonal trust—which rely on repeated interactions with specific individuals over time (e.g., Gefen, Karahanna, & Straub, 2003; Mayer, Davis, & Schoorman, 1995)—will not fully capture how trust is built and maintained in near-future human-robot teams. We posit that conventional models of interpersonal trust may not be suitable to address the integration of ever-changing technology and, therefore, new robotic teammates into teams.

Alternatively, we postulate that a robot trust model for mixed human-robot (H-R) teams derived from the knowledge on swift trust in human-human (H-H) teams may be an application framework to describe robot trust formation in H-R teams more accurately than traditional models of interpersonal H-H trust applied to H-R teams. This chapter presents the model of swift trust for H-H teams as a valuable baseline for developing a meaningful model of robot trust in H-R teams. We will elaborate on the contrasts of swift trust and more traditional conceptions of trust in H-H teams, the current conceptions of trust towards machine systems, and describe how the swift trust model is applicable to describe how robot trust is formed in H-R teams. To achieve this, we describe the characteristics of H-H swift trust teams and how that mirrors the types of H-R teams that currently exist and are likely to exist in the future. We will also explain the shortcomings of the swift trust model with H-R teams and provide directions for future research to investigate the shared foundations of trust in H-H teams and the applications to robot trust in H-R teams.

Trust and swift trust

Trust has been described as the willingness to take risks and be vulnerable regarding another person's behavior. Trustworthiness refers to the characteristics of, and conditions around, the person or thing to be trusted

and is considered to facilitate trust (Borum, 2010). In interpersonal relationships, the familiar conception of trust is something that develops between people over time and is built over a series of shared experiences, repeated interactions, and predictability under uncertainty and ambiguity (Rempel et al., 1985). In this traditional conception of trust, trust is an evidence-driven information process with models citing three factors as the antecedents of trust: *ability*, *benevolence*, and *integrity* (Mayer et al., 1995). *Ability* is the trust in a group of skills, competencies, and characteristics that enable a person or a team to accomplish a certain task. Ability often is highly domain-specific. For example, a person might be entrusted to be able to solve a complex mathematical equation but not with the negotiation of contract terms with a customer. Similar in the area of machines and robots, their abilities are currently very domain-specific and a robot might be entrusted with complex calculations, but not with interpersonal communication skills. *Benevolence* is the belief of goodness and goodwill perceived in other team members. How benevolent the other team members are perceived plays an important role in the assessment of trustworthiness. *Integrity* is the adherence to a set of principles that other team members find acceptable. The level of perceived integrity of a team member is another important factor in the evaluation of their trustworthiness. Together with the individual's propensity to trust, these factors could describe team trust (Jarvenpaa, Knoll, & Leidner, 1998).

Swift trust, on the other hand, describes a category-driven process of trust in others when the situation does not afford an evidence-driven process. Often, but not exclusively, teams that come together temporarily or do not have prior joint experiences do not have evidence of the other team members' trustworthiness and lack the affordance of developing trust over time. Therefore, swift trust is often associated with teams that are formed around a common task, with a finite lifespan, composed of well-trained experts with distributed skills, and characterized by limited or no prior history of working together, and little prospect of working together in the future. Examples include flight crews, short-term project teams, emergency response teams, military special operators, and groups of specialists brought in to solve a difficult or urgent problem. Unlike traditional trust conceptions which highlight how trust develops over time through shared experiences, swift trust is assumed initially, verified later, and beliefs are adjusted accordingly (Langfred, 2004). Swift trust relies on importing perceptions of another's trustworthiness and is heavily influenced by *imported information* about other parties, one's *propensity to trust*, and *surface-level cues* used to make judgments about trust (Wildman et al., 2012).

Imported information refers to category-based stereotypes associated with what is known about the individual to be trusted (e.g., occupation).

Imported information also includes third-party information or data from external sources that allow the trust to form, such as credentials, certifications, or degrees, usually established with the help of cognitive cues about others like group memberships or categorization (McKnight, Cummings, & Chervany, 1998). *The propensity to trust* is an individual's difference in willingness to form trusting relationships (Mayer et al., 1995). It is considered the general willingness to be open to another's reciprocity and intentions (Burke, Sims, Lazzara, & Salas, 2007; Mayer et al., 1995). The propensity to trust is shaped by similar experiences and knowledge (Schoorman, Mayer, & Davis, 2007). It influences one's decision to trust by impacting information salience and understanding (Burke et al., 2007). *Surface-level cues* refer to judgments made about overt information, including individuals' physical appearance—such as their ethnicity, gender, race, and attire, and other discernible demographic characteristics.

With the introduction of machines and robots in human teams, team members are less able to rely on similarities and common prior experiences with the new agent on their team. The needs of H-R teams entail performing a task relatively promptly; therefore, team members might not have enough time to assess benevolence as benevolence assessment requires information that takes time to gather (Jarvenpaa et al., 1998). Swift trust research focusses on the early development of trust in teams where team members have not worked together before team formation and do not have the time to develop an understanding of the benevolence of others. This might also be the case when machines and robots are to be integrated into human teams. Indeed, it is also challenging for humans to infer robot intentions, regardless if benevolent or not, and robots lack the introspection to display their intentions (Andersen, 2016) creating analogous circumstances to when temporary human teams form and develop swift trust.

More often than not, the concept of trust is described as an essential factor for any kind of team. Increasingly, teams experience some form of machine support. Currently, automation can be seen as an active component in some teams, and it is anticipated that autonomous systems, as well as physically embedded systems like robots, will become members of teams. Much of the research trying to establish the current and future role of robots in teams is based on the trust research conducted for H-H teams. Under the assumption that robots will be able to perform certain tasks within an H-R team that contributes to the team's capabilities, and given the similarity between H-H and H-R teams, the conceptions of swift trust closely align with human-robot trust in current and future H-R teams.

Trust in machines: Automation and robots

Over recent years, automation has been implemented in domains such as aviation, healthcare, and technical monitoring (e.g., nuclear power plants, air missile defense systems, etc.). Automation is highly effective in increasing performance and safety in many cases. However, faulty human-automation interactions have been cited as a critical factor in many incidents (Parasuraman & Riley, 1997), some of them with catastrophic consequences. As automated systems are not human team members, they are evaluated differently by other human team members. The Perfect Automation Schema (PAS, Madhavan & Wiegmann, 2007) differentiates human trust from automation trust. PAS describes two dimensions of trust in automation systems. The first dimension is the belief that automated systems are near perfect and the resulting human overreaction to automation errors are followed by severe decreases in trust in the automation. The second belief is that if a single error occurs, people perceive all systems to be flawed (Merritt, Unnerstall, Lee, & Huber, 2015). It is possible that a form of PAS also applies to interactions with robots within H-R teams. High expectations were associated with higher trust while evaluating the influences of PAS on trust (Lyons & Guznov, 2019). In combination with the knowledge that people quickly form initial high expectations towards robots (Haring, Silvera-Tawil, Takahashi, Watanabe, & Velonaki, 2016), one's formation of trust in the robot should be carefully considered given the expectations people have towards the robot. Other models have described how trust in technology might influence the way people interact with and use technology. The Technology Acceptance Model (TAM, Davis, 1989) originates in information systems theory and models how users come to accept and use technology, and the model can be applied to assess the effect of trust on technology acceptance. Under the pretext that robots are a relatively new technology and will need to meet the human team members' needs and expectations in an H-R team (Bröhl, Nelles, Brandl, Mertens, & Schlick, 2016), the perception of usefulness and acceptance of the robot technology as described in the TAM will likely play an important role in integrating robots into teams.

Expectations and perceptions of robots may not always reflect the true capacities of the system (Haring, Watanabe, Velonaki, Tossell, & Finomore, 2018). For automated and autonomous systems, the level of perceived autonomy and capacities does influence how people trust the systems (Arkin, Ulam, & Wagner, 2011; Choi & Ji, 2015; Schaefer, Chen, Szalma, & Hancock, 2016; Tenhundfeld et al., 2019). For our purposes, we distinguished between automated (i.e., "designed to accomplish a specific set of largely deterministic steps, most often in a repeating pattern, to achieve one of an envisaged and limited set of predefined outcomes," Hancock, 2017, p. 284) and autonomous systems (i.e., "state of being for

biological or mechanical agents that implies: robustness to an environment; independence in action/function; and self-determination of goals and resource allocation," Kaber, 2018, p. 407). On the spectrum between automated and autonomous, it could be argued that current robotics technologies fall closer to notions of automated systems, while the robots envisioned to serve as members of human-robot teams would satisfy the definition of autonomous systems.[a]

As autonomous systems and robotics permeate society, users may increasingly encounter technologies with which they are unfamiliar. Without prior experience, users are often left to interpret the actions of a robot to infer the robot's intent; this can pose an interesting challenge for human-robot trust. A study investigated how people responded to a robot that appeared to cheat in a game of "rock-paper-scissors" in two different ways—in the verbal cheat condition the robot would declare itself the winner even though it had lost, and in the action cheat condition the robot would change its hand to a winning gesture and declare itself the winner even though it had lost (Short, Hart, Vu, & Scassellati, 2010). One of the most interesting results concerned participants' evaluations of the robot when it cheated; participants who experienced the action cheat tended to describe the robot as having cheated, while participants who experienced the verbal cheat often referred to the robot as having malfunctioned. Variations in a robot's actions can influence how people evaluate the intentions of a robot, which, in turn, impacts the extent to which people trust the robot. A follow-up study set out to replicate these findings and verified that the changes in the interpretations of the robot's behavior come from the adversarial cheating exhibited by the robot, and not simply the motion or novelty of the robot changing its gesture (Litoiu, Ullman, Kim, & Scassellati, 2015). Follow-up work also investigated whether people evaluate a robot that cheats similarly to a human who cheats in an identical condition in the context of the game of battleship; participants rated the robot as more intelligent when it cheated as compared to when it played fairly, while participants rated the human as less intelligent when they cheated as compared to when they played fairly (Ullman, Leite, Phillips, Kim-Cohen, & Scassellati, 2014).

Trust in automation

There is a wealth of prior literature focused on trust in automated systems: commonly-cited factors influencing trust in these systems include performance (Yeh & Wickens, 2000), operator experience (Parasuraman &

[a]For thorough reviews on the differences between automated and autonomous systems and the range between, please see (de Visser, Pak, & Shaw, 2018; Hancock, 2017; Kaber, 2018)

Miller, 2004), propensity to trust (Lee & See, 2004), and task-related factors like workload or perceptions of risk associated with automated systems (Parasuraman & Miller, 2004). Of these factors, system performance tends to be the most important for trust in automation.

Trust in robots

A meta-analysis by Hancock et al. (2011) provided evidence that trust in a robot is primarily rooted in performance-based factors such as reliability, predictability, and dependability of a robot. Based on research into the antecedents of H-R trust, several definitions of the construct have been advanced. Heerink, Kröse, Evers, and Wielinga (2010) defined H-R trust as human perceptions of the robot's competence in its tasks. Wagner (2009) focused on establishing estimates of agents working together and the capacity of the agents to perform specific tasks or behaviors. Overall, a common theme for defining H-R trust involves the human's perception of the robot's competence, reliance on the robot to conduct its duties, and a belief that the robot will protect the human's interests. Thus, H-R trust can be defined as the human's confidence in the robot's ability to perform its tasks (Baker, Phillips, Ullman, & Keebler, 2018). Since this confidence does not necessarily require extensive prior experience with the robot, there may be an overlap between H-R trust and the formation of swift trust; in the following section, we will explore this further.

Swift trust in robots

In H-H as well as H-R teams, members should be able to trust every other team member, human or robot, to enable the team to meet its goals. However, this trust does not necessarily require a history of collaboration, thereby prompting swift trust. Similar to the characteristics of H-H teams in which swift trust can be found, the characteristics, capabilities, and even physical forms of robotic team members in H-R teams are changing with updates, repairs, and replacements to robots over time, positioning them towards being more temporary team members. Leveraging the conceptualizations of swift trust (i.e., Meyerson, Weick, & Kramer, 1996) as well as a model of swift trust (i.e., Wildman et al., 2012), we apply the concept of swift trust within H-R teams. Similar to Wildman et al.'s (2012) model of swift trust in H-H teams, we propose that swift trust in H-R teams is cultivated through environmental factors, imported information, the propensity to trust, and surface-level cues. Table 1 provides an overview of the proposed robot trust constructs and the antecedents of swift trust in H-H teams. The following sections describe each of these constructs in detail. Besides, we are considering the possible limitations of swift trust and robot trust.

TABLE 1 Overview of the proposed robot trust constructs and the antecedents of trust and swift trust in H-H teams.

Construct	Trust relation	Description	References
Context (environmental) factors	H-H	Swift Starting Action Team (STAT) working in unpredictable circumstances and high-risk areas	Wildman et al. (2012) and Salas, 2011
		Interaction patterns as a predictor of effectiveness in lower-threat areas (e.g., aviation)	Zijlstra et al. (2012)
		Virtual teams (vs. face-to-face)	Crisp and Jarvenpaa (2013) and Robert, Denis, and Hung (2009)
	H-R	Surrounding environment	Hancock et al. (2011)
		Perceived risk	Robinette, Howard, and Wagner (2017)
Propensity to trust	H-H	General willingness to trust others	Burke et al. (2007), Colquitt, Scott, and LePine (2007), Mayer et al. (1995), and Schoorman et al. (2007)
	H-R	General willingness to trust a robot	Billings, Schaefer, Chen, and Hancock (2012)
Surface-level cues	H-H	Overt, observable demographic characteristics of human team members	Bell, Villado, Lukasik, Belau, and Briggs (2011)
		Human-related factors: characteristics of the human team members	Stahl, Maznevski, Voigt, and Jonsen (2010) and Wildman et al. (2012)
	H-R	Appearance, human-likeness, human anthropomorphism, Robot-related factors: characteristics of the robotic design and capabilities of the robot Form Function Attribution Bias	Li, Rau, and Li (2010), Phillips, Zhao, Ullman, and Malle (2018), Duffy (2003), Hancock et al. (2011), Goetz, Kiesler, and Powers (2003), Strait, Canning, and Scheutz (2014), and Haring et al. (2018)

Continued

TABLE 1 Overview of the proposed robot trust constructs and the antecedents of trust and swift trust in H-H teams—cont'd

Construct	Trust relation	Description	References
Imported information	H-H	Preexisting expectations, knowledge, and assumptions that are derived from related personal experience or third-parties	Meyerson et al. (1996) and Wildman et al. (2012)
	H-R	User expectations, assumptions	Oestreicher and Eklundh (2006), MacDorman (2006), and Edwards, Edwards, Spence, and Westerman (2016)

Context factors

Context factors describe the environment and conditions the teams are working in. Often the conceptualization of trust in teams has been linked to risk (Mayer et al., 1995; Susan & Holmes, 1991). Some context factors on which teams form swift trust might be in environments that pose high risks (Salas, 2011) to the team and where errors can have costly or even fatal consequences. For example, search and rescue teams, as well as military special units, could be exposed to high-risk environments. However, the context of how the team comes together and which task is to be accomplished could be a crucial one, yet with a lower threat and risk level and still enable swift trust formation. This would include, for example, aviation crews or virtual teams. The context and environmental factors also influence the trust perception of robots in H-R teams (Hancock et al., 2011).

Propensity to trust

The propensity to trust generally describes the generalized tendency of an individual to trust others, regardless of who the other person is. The propensity to trust has been used in H-R teams to evaluate initial trust towards robots in research (Billings et al., 2012). In H-H teams that develop swift trust, members seem to develop favorable in-group beliefs and actions that foster early trust in the team through quickly perceivable social categorization mechanisms. Social categorization is the natural cognitive process by which people place individuals into social groups, and this process occurs spontaneously (Crisp & Hewstone, 2007). It has been shown that robots are also subject to social categorization and that

in-group and out-group effects apply to robots (Eyssel and Kuchenbrandt, 2012; Rae, Takayama, & Mutlu, 2012). In H-H teams, these initial trust beliefs tend to be self-reinforcing in that team members focus on information that validates their trusting beliefs while contrary evidence may go unnoticed (McKnight et al., 1998). Similar effects for human-automation teams are known in terms of complacency (Wickens, Clegg, Vieane, & Sebok, 2015; Wickens & Dixon, 2007) and first failure effects in adaptation to automation (Lee & Moray, 1994; Rovira, McGarry, & Parasuraman, 2007; Yeh & Wickens, 2001). The large body of research in trust in automation inspired research of similar concepts for H-R teams. Indeed, initial trust beliefs also have emerged in human-robot teams showing that people will often initially trust an unknown robot in a simulated emergency (Robinette et al., 2017) as well as in a real emergency (Robinette, Li, Allen, Howard, & Wagner, 2016). Furthermore, an early reinforcement of trust in a robot seems to facilitate later trust repair (Desai, Kaniarasu, Medvedev, Steinfeld, & Yanco, 2013), which reflects the swift trust process of self-reinforcing initial trust beliefs. Congruent with the H-H Swift trust model, people seem to self-reinforced and validate their trusting beliefs to automation and robots.

Surface-level cues

Surface-level cues refer to overt traits that are easily observable and identifiable. In H-H teams, surface-level cues are all overt demographic traits. In H-R teams and human-robot interaction, they refer to the robot's form or appearance (Li et al., 2010), human-likeness (Phillips et al., 2018), anthropomorphism (Duffy, 2003), and are important as surface-level cues affect initial levels of trust towards the robot or human team members. Therefore, with immediately perceivable surface level cues affecting the perceived trustworthiness of another team member, initial input cues are a crucial factor that people will rely on to make swift trust judgments about the other team members. Currently, robots are not ubiquitous and only the early adopters of robot technology will have prior experiences with them (Haring et al., 2016). Similarly, in current and future H-R teams, when human team members are not able to rely on prior interactions with a robot they will need to rely on immediately apparent cues. People form quick perceptions about robots based on available surface-level cues, especially robot form, even if those perceptions might be biased (Form-Function Attribution Bias (FFAB), Haring et al., 2018). FFAB describes how people have initial perceptions and expectations towards robots based on their form (i.e., appearance, exterior design, etc.). However, this initial perception does not necessarily reflect the actual robot functions (i.e., capabilities) and is, therefore, biased. Surface-level cues like robot form, human-likeness,

human anthropomorphism, and other robot-related factors should be carefully considered if robots are to be integrated into a team as they will provide strong initial cues to swift trust formation. It has been argued that robots that prove untrustworthy might be rejected by human team members (Groom & Nass, 2007). We postulate that initially low expectations towards the robot might exclude them from the initial swift trust development and consequently will never be integrated with the overall team's swift trust concept. High expectations of the robot's capabilities may help establish swift trust in the robot on the team. However, the scaffold for interpersonal trust formation that swift trust provides will likely be rapidly diminished if the robot does not live up to the user expectations and assumptions about their capabilities.

A variety of research has illustrated that robot's physical appearance (with particular emphasis on human resemblance) considerably influences people's perceptions of a number of the robot's capabilities and cognitive capacities, including robot intelligence (Haring et al., 2016; Sims et al., 2005), sociability (Powers & Kiesler, 2006), likeability (Castro-González, Admoni, & Scassellati, 2016; Mathur & Reichling, 2016), credibility (Burgoon et al., 2000), and submissiveness (von der Pütten & Krämer, 2012), as well as other social, cognitive, and moral reasoning abilities (Malle, 2019; Zhao, Phillips, & Malle, 2019). Robot appearance can also evoke both positive and negative emotional responses towards robots (Goetz et al., 2003; Strait et al., 2014), which theory suggests could impact human team member's perceptions of trust towards the robot, more so than cognitive appraisals of trustworthiness. Finally, like temporary H-H teams, people in H-R teams are unlikely to have substantial experience with their robot teammates to ground their trust formation. They may instead rely on the same types of surface-level cues to form robot trust as they would form swift trust in H-H teams.

Imported information

Imported information can be defined as the preexisting knowledge, stereotypes, and preconceptions stored in the team members' memories and can be derived from two primary sources: previous personal experiences and third-party information provided by trusted sources (Wildman et al., 2012). Imported information is a crucial initial input to the development of swift trust in teams. About robot trust in H-R teams, imported information can be seen as certain user expectations and assumptions about the robot (e.g., Edwards et al., 2016; Haring et al., 2018; MacDorman, 2006; Oestreicher & Eklundh, 2006) and would also not exclude other third-party information through media exposure or derived from popular culture (e.g., movies), which has been shown to influence

perceptions about robots (Nourbakhsh, 2015; Sundar, Waddell, & Jung, 2016). Imported information also includes prior personal experiences in similar situations with similar types of people (e.g., aircrew members, Special Forces members, etc.). In the H-R context, people could have some preexisting knowledge about the robot in the team, or experiences with similar classes of robots. It has been shown that robots in teams, similar to humans in teams, are subject to stereotypes based on a robot's perceived role, gender, and personality (Tay, Jung, & Park, 2014).

Shortcomings of swift trust model

We have described the characteristics to develop swift trust and the many similar characteristics of current and future H-R teams. However, it is important to consider the application of H-H swift trust concepts with a critical eye to identify the shortcomings of the swift trust model mentioned in literature. Swift trust has been described as a fragile construct (Jarvenpaa & Leidner, 1999), such that it is hard to undergo a transition from swift trust to more traditional conceptualizations of trust without subsequent communication. It could be argued that swift trust functions as an early, necessary component to the start of a team with a shared goal; however, it is not sufficient by itself to form a long-term trust. Even if robots are included in the initial swift trust, mimicking human communication strategies for extended periods with robots would be difficult.[b] Thus, there would be a risk of breaking the established swift trust and not being able to effectively implement trust repair strategies. Extending the flaws in swift trust further, some critics of swift trust argue that it is not a form of trust but rather a substitution for it and more of a risk management strategy; some critics even go as far as stating that swift trust might inhibit actual trust-building (Kohler, Cortina, Salas, & Garven, 2011). These potential shortcomings that can affect H-H teams might also affect H-R teams.

Directions for future research

Models of H-H swift trust can help serve as a foundation for models of H-R trust. In the near-term, models of swift trust seem to better capture the kinds of interactions that humans currently have and will have with robots for the foreseeable future, especially for novel interactions that

[b]See recent critiques on the challenges to maintaining long-term human-robot interactions (Hoffman, 2019), especially limitations in single turn-taking interactions between humans and robots.

are often short and where members have little to no prior interaction experience. This is opposed to traditional models of trust where long-term interactions provide foundations for trust development and trust repair. To evaluate these claims, we offer several avenues for future work, with each one addressing a distinct need in H-R trust research.

Firstly, as a field, we must both refine old metrics and create new ones to adequately measure trust between agents. While there are a few measures that have been developed and utilized in research (Freedy et al., 2007; Schaefer, 2013), they are often used in concert with a conception of trust specific to a given research project. As robots evolve in capability and complexity, our conceptions of human-robot dynamics will evolve as well. New metrics and measurements will be needed to better capture the development of trust and team cohesion in human-robot teams (Schaefer et al., 2019).

Secondly, to date, most research with robots has involved one robot and one user. However, the vast majority of teams have more than two members. While complex social robots have traditionally been prohibitively expensive, recent advances have made them accessible to research groups and everyday end users. Thus, we can begin to expect that robots used in social applications will be embedded in spaces with multiple people and eventually multiple robots. These robots must be able to effectively interact with multiple agents (Leite, McCoy, Ullman, Salomons, & Scassellati, 2015). With expected design cases involving multiple robots interacting with people, the complexity of a given scenario increases dramatically with the addition of each added interactive agent. Therefore, we must model trust between groups involving two or more agents to expand our understanding of trust in HRI; as we look to the future of HRI, we also must address relationships that require trust and involve multiple links in what can essentially be considered a network of agents.

Thirdly, there is a need to empirically investigate the impact of robot specific features and characteristics on the category driven processes that underlie swift trust development (Meyerson et al., 1996; Wildman et al., 2012). Robot appearance, as one example, varies widely (often as a function of the application domain and engineering need) and strongly influences human perceptions of robots (Goetz et al., 2003; Haring, Watanabe, Silvera-Tawil, Velonaki, & Takahashi, 2015; Leite et al., 2013; Sims et al., 2005; Woods, Dautenhahn, & Schulz, 2004). In general, the more a robot looks like a human (Phillips et al., 2018), the more human-like qualities are attributed to robots (Hegel, Krach, Kircher, Wrede, & Sagerer, 2008; Lee, Park, & Song, 2005; Rızvanoğlu, Öztürk, & Adıyaman, 2014). This can be problematic, as human expectations of robot capabilities often far exceed actual robot capabilities (Haring et al., 2018). Therefore, the

appearance of a robot can greatly impact the categorical information that is imported about the robot, affecting one of the foundations for swift trust. Overall, a clearer understanding of how robot characteristics relate to and interact with trust processes will further our understanding of how robots can be better integrated into groups and teams and will shed more light on the utility of applying swift trust to HRI.

These research thrusts address the capture, description, and prediction of trust in future human-robot teams, and will provide more insight into the overlap between swift trust and human-robot trust frameworks. Furthermore, we are considering the possible role social norms play in H-R teams and how norms could help facilitate swift trust formation in H-R teams.

The potential role of social norms in H-R teams

At this early stage of swift trust development in teams, norms could help provide a basis for swift trust and may also facilitate the maintenance of trust over time. Knowing that all team members are aware, committed to, and will apply sanctions in the case of the norm, violations will increase predictability (e.g., "I know that they will act as the norms dictate"), cooperative intentions (e.g., "If you comply with norms, you comply with the interests of the group"), and reassurance of benevolence. If norms are established, they will protect the individual from exploitation by other team members. For robots in H-R teams, norms may be a way for robots to signal to the team that they have the team's best interest in mind. Currently, robots' abilities to follow human social norms are quite limited. For example, researchers have shown that people treat robots with social politeness by engaging in behaviors like apologizing to robots (Lee, Kiesler, & Forlizzi, 2010). Research has suggested that apologies should be timed well after a robot makes a mistake and that they may not be useful as trust repair strategy if the apology is issued right after the mistake (Robinette, Howard, & Wagner, 2015). People self-reported higher trust scores for a robot that seemed to realize a mistake, showed regret and apologized right away in a lower-risk cooking scenario after dropping an item in a cooking task (Hamacher, Bianchi-Berthouze, Pipe, & Eder, 2016). We assume that the gravity of an error, the consequences, the robot's awareness of it, and the mitigation and learning strategies to not let the same error happen again will play an additional role in fully understanding the role of a robot apology in the development of swift trust in teams, given that there are little longer-term interactions that allow for extended trust repair strategies.

Conclusion

Teams are no longer restricted to only human members. As robots are increasingly used as team members in mixed human-robot teams, it is critical to understand the way that trust develops in these settings. Teams are often formed quickly around a common task and consist of temporary collaborations, requiring that the team members quickly build cohesive working relationships with one another. Evidence has repeatedly demonstrated that successful collaboration and team effectiveness are contingent upon trust. As such, understanding the role of trust in these temporary human-robot teams is crucial. Swift trust offers a conceptualization to capture how trust forms quickly in human-human teams, and the similarities to human-robot teams make swift trust a prime candidate to capture trust in human-robot contexts. It is imperative to understand the intersection between H-H swift trust and H-R trust to optimize team performance in what are often high stress, high importance scenarios. Swift trust will be essential to understanding team dynamics in an ever-changing landscape of multi-agent team interactions.

References

Arkin, R. C., Ulam, P., & Wagner, A. R. (2011). Moral decision making in autonomous systems: Enforcement, moral emotions, dignity, trust, and deception. *Proceedings of the IEEE*, *100*(3), 571–589.

Baker, A. L., Phillips, E. K., Ullman, D., & Keebler, J. R. (2018). Toward an understanding of trust repair in human-robot interaction: Current research and future directions. *ACM Transactions on Interactive Intelligent Systems (TiiS)*, *8*(4), 30.

Bell, S. T., Villado, A. J., Lukasik, M. A., Belau, L., & Briggs, A. L. (2011). Getting specific about demographic diversity variable and team performance relationships: A meta-analysis. *Journal of Management*, *37*(3), 709–743.

Billings, D. R., Schaefer, K. E., Chen, J. Y., & Hancock, P. A. (2012, March). Human-robot interaction: Developing trust in robots. In *Proceedings of the seventh annual ACM/IEEE international conference on human-robot Interaction* (pp. 109–110). ACM.

Boies, K., Fiset, J., & Gill, H. (2015). Communication and trust are key: Unlocking the relationship between leadership and team performance and creativity. *The Leadership Quarterly*, *26*(6), 1080–1094.

Borum, R. (2010). *The science of interpersonal trust, Mental Health Law & Policy Faculty Publications*: (p. 574).

Bröhl, C., Nelles, J., Brandl, C., Mertens, A., & Schlick, C. M. (2016, July). TAM reloaded: A technology acceptance model for human-robot cooperation in production systems. In *International conference on human-computer interaction* (pp. 97–103). Cham: Springer.

Burgoon, J. K., Bonito, J. A., Bengtsson, B., Cederberg, C., Lundeberg, M., & Allspach, L. (2000). Interactivity in human–computer interaction: A study of credibility, understanding, and influence. *Computers in Human Behavior*, *16*(6), 553–574.

Burke, C. S., Sims, D. E., Lazzara, E. H., & Salas, E. (2007). Trust in leadership: A multi-level review and integration. *The Leadership Quarterly*, *18*(6), 606–632.

Castro-González, Á., Admoni, H., & Scassellati, B. (2016). Effects of form and motion on judgments of social robots' animacy, likability, trustworthiness and unpleasantness. *International Journal of Human-Computer Studies*, *90*, 27–38.

Choi, J. K., & Ji, Y. G. (2015). Investigating the importance of trust on adopting an autonomous vehicle. *International Journal of Human Computer Interaction*, *31*(10), 692–702.

Colquitt, J. A., Scott, B. A., & LePine, J. A. (2007). Trust, trustworthiness, and trust propensity: A meta-analytic test of their unique relationships with risk taking and job performance. *Journal of Applied Psychology*, *92*(4), 909.

Costa, A. C. (2003). Work team trust and effectiveness. *Personnel Review*, *32*(5), 605–622.

Crisp, R. J., & Hewstone, M. (Eds.). (2007). *Multiple social categorization*. San Diego, CA: Elsevier Academic Press.

Crisp, C. B., & Jarvenpaa, S. L. (2013). Swift trust in global virtual teams: Trusting beliefs and normative actions. *Journal of Personnel Psychology*, *12*(1), 45.

Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly*, 319–340.

de Jong, B. A., Dirks, K. T., & Gillespie, N. (2015). Trust and team performance: A meta-analysis of main effects, contingencies, and qualifiers. *Academy of management proceedings* (Vol. 2015, No. 1, p. 14561). Briarcliff Manor, NY: Academy of Management.

Desai, M., Kaniarasu, P., Medvedev, M., Steinfeld, A., & Yanco, H. (2013, March). Impact of robot failures and feedback on real-time trust. In *Proceedings of the 8th ACM/IEEE international conference on human-robot interaction* (pp. 251–258). IEEE Press.

de Visser, E. J., Pak, R., & Shaw, T. H. (2018). From 'automation' to 'autonomy': The importance of trust repair in human–machine interaction. *Ergonomics*, *61*(10), 1409–1427.

Duffy, B. R. (2003). Anthropomorphism and the social robot. *Robotics and Autonomous Systems*, *42*(3–4), 177–190.

Edwards, C., Edwards, A., Spence, P. R., & Westerman, D. (2016). Initial interaction expectations with robots: Testing the human-to-human interaction script. *Communication Studies*, *67*(2), pp. 227–238.

Eyssel, F., & Kuchenbrandt, D. (2012). Social categorization of social robots: Anthropomorphism as a function of robot group membership. *British Journal of Social Psychology*, *51*(4), 724–731.

Freedy, A., DeVisser, E., Weltman, G., & Coeyman, N. (2007, May). Measurement of trust in human-robot collaboration. In *2007 international symposium on collaborative technologies and systems* (pp. 106–114). IEEE.

Gefen, D., Karahanna, E., & Straub, D. W. (2003). Trust and TAM in online shopping: An integrated model. *MIS Quarterly*, *27*(1), 51–90.

Gockley, R., Bruce, A., Forlizzi, J., Michalowski, M., Mundell, A., Rosenthal, S. , et al. (2005, August). Designing robots for long-term social interaction. In *2005 IEEE/RSJ international conference on intelligent robots and systems* (pp. 1338–1343). IEEE.

Goetz, J., Kiesler, S., & Powers, A. (2003, October). Matching robot appearance and behavior to tasks to improve human-robot cooperation. In *The 12th IEEE international workshop on robot and human interactive communication, 2003. Proceedings. ROMAN 2003* (pp. 55–60). IEEE.

Groom, V., & Nass, C. (2007). Can robots be teammates?: Benchmarks in human–robot teams. *Interaction Studies*, *8*(3), 483–500.

Hamacher, A., Bianchi-Berthouze, N., Pipe, A. G., & Eder, K. (2016, August). Believing in BERT: Using expressive communication to enhance trust and counteract operational error in physical human-robot interaction. In *2016 25th IEEE international symposium on robot and human interactive communication (RO-MAN)* (pp. 493–500). IEEE.

Hancock, P. A. (2017). Imposing limits on autonomous systems. *Ergonomics*, *60*(2), 284–291.

Hancock, P. A., Billings, D. R., Schaefer, K. E., Chen, J. Y., De Visser, E. J., & Parasuraman, R. (2011). A meta-analysis of factors affecting trust in human-robot interaction. *Human Factors*, *53*(5), 517–527.

Haring, K. S., Silvera-Tawil, D., Takahashi, T., Watanabe, K., & Velonaki, M. (2016, February). How people perceive different robot types: A direct comparison of an android, humanoid, and non-biomimetic robot. In *2016 8th International conference on knowledge and smart technology (KST)* (pp. 265–270). IEEE.

Haring, K. S., Watanabe, K., Silvera-Tawil, D., Velonaki, M., & Takahashi, T. (2015, February). Changes in perception of a small humanoid robot. In *2015 6th international conference on automation, robotics and applications (ICARA)* (pp. 83–89). IEEE.

Haring, K. S., Watanabe, K., Velonaki, M., Tossell, C. C., & Finomore, V. (2018). FFAB—The form function attribution bias in human–robot interaction. *IEEE Transactions on Cognitive and Developmental Systems*, *10*(4), 843–851.

Heerink, M., Kröse, B., Evers, V., & Wielinga, B. (2010). Assessing acceptance of assistive social agent technology by older adults: The Almere model. *International Journal of Social Robotics*, *2*(4), 361–375.

Hegel, F., Krach, S., Kircher, T., Wrede, B., & Sagerer, G. (2008, August). Understanding social robots: A user study on anthropomorphism. In *RO-MAN 2008-the 17th IEEE international symposium on robot and human interactive communication* (pp. 574–579). IEEE.

Hoffman, G. (2019). Evaluating fluency in human–robot collaboration. In: *IEEE transactions on human-machine systems*.

Jarvenpaa, S. L., Knoll, K., & Leidner, D. E. (1998). Is anybody out there? Antecedents of trust in global virtual teams. *Journal of Management Information Systems*, *14*(4), 29–64.

Jarvenpaa, S. L., & Leidner, D. E. (1999). Communication and trust in global virtual teams. *Organization Science*, *10*(6), 791–815.

Kaber, D. B. (2018). A conceptual framework of autonomous and automated agents. *Theoretical Issues in Ergonomics Science*, *19*(4), 406–430.

Kidd, C. D., & Breazeal, C. (2008, September). Robots at home: Understanding long-term human-robot interaction. In *2008 IEEE/RSJ international conference on intelligent robots and systems* (pp. 3230–3235). IEEE.

Kohler, T., Cortina, J. M., Salas, E., & Garven, A. J. (2011). *Trust in temporary work groups.* Unpublished manuscript. Department of Management and Marketing, The University of Melbourne.

Langfred, C. (2004). Too much of a good thing? Negative effects of high trust and individual autonomy in self-managing teams. *Academy of Management Journal*, *47*(3), 385.

Laschi, C., Mazzolai, B., & Cianchetti, M. (2016). Soft robotics: Technologies and systems pushing the boundaries of robot abilities. *Science robotics*, *1*(1), eaah3690.

Lee, M. K., Kiesler, S., & Forlizzi, J. (2010, February). Receptionist or information kiosk: How do people talk with a robot? In *Proceedings of the 2010 ACM conference on computer supported cooperative work* (pp. 31–40). ACM.

Lee, J. D., & Moray, N. (1994). Trust, self-confidence, and operators' adaptation to automation. *International Journal of Human-Computer Studies*, *40*(1), 153–184.

Lee, K. M., Park, N., & Song, H. (2005). Can a robot be perceived as a developing creature? Effects of a robot's long-term cognitive developments on its social presence and People's social responses toward it. *Human Communication Research*, *31*(4), 538–563.

Lee, J. D., & See, K. A. (2004). Trust in automation: Designing for appropriate reliance. *Human Factors*, *46*(1), 50–80.

Leite, I., Martinho, C., & Paiva, A. (2013). Social robots for long-term interaction: A survey. *International Journal of Social Robotics*, *5*(2), 291–308.

Leite, I., McCoy, M., Ullman, D., Salomons, N., & Scassellati, B. (2015, March). Comparing models of disengagement in individual and group interactions. In *2015 10th ACM/IEEE international conference on human-robot interaction (HRI)* (pp. 99–105). IEEE.

Li, D., Rau, P. P., & Li, Y. (2010). A cross-cultural study: Effect of robot appearance and task. *International Journal of Social Robotics*, *2*(2), 175–186.

Litoiu, A., Ullman, D., Kim, J., & Scassellati, B. (2015, March). Evidence that robots trigger a cheating detector in humans. In *Proceedings of the tenth annual ACM/IEEE international conference on human-robot interaction* (pp. 165–172). ACM.

Lyons, J. B., & Guznov, S. Y. (2019). Individual differences in human–machine trust: A multistudy look at the perfect automation schema. *Theoretical Issues in Ergonomics Science*, *20*(4), 440–458.

MacDorman, K. F. (2006, July). Subjective ratings of robot video clips for human likeness, familiarity, and eeriness: An exploration of the uncanny valley. In *ICCS/CogSci-2006 long symposium: Toward social mechanisms of android science* (pp. 26–29).

Mach, M., Dolan, S., & Tzafrir, S. (2010). The differential effect of team members' trust on team performance: The mediation role of team cohesion. *Journal of Occupational and Organizational Psychology*, *83*(3), 771–794.

Madhavan, P., & Wiegmann, D. A. (2007). Similarities and differences between human–human and human–automation trust: An integrative review. *Theoretical Issues in Ergonomics Science*, *8*(4), 277–301.

Malle, B. F. (2019). How many dimensions of mind perception really are there? In *Proceedings of the 41st annual meeting of the cognitive science society*. Austin, TX: Cognitive Science Society.

Malle, B. F., Scheutz, M., Forlizzi, J., & Voiklis, J. (2016, March). Which robot am i thinking about?: The impact of action and appearance on people's evaluations of a moral robot. In *The eleventh ACM/IEEE international conference on human robot interaction* (pp. 125–132). IEEE Press.

Mathur, M. B., & Reichling, D. B. (2016). Navigating a social world with robot partners: A quantitative cartography of the Uncanny Valley. *Cognition*, *146*, 22–32.

Mayer, R. C., Davis, J. H., & Schoorman, F. D. (1995). An integrative model of organizational trust. *Academy of Management Review*, *20*(3), 709–734.

McAllister, D. J. (1995). Affect-and cognition-based trust as foundations for interpersonal cooperation in organizations. *Academy of Management Journal*, *38*(1), 24–59.

McKnight, D. H., Cummings, L. L., & Chervany, N. L. (1998). Initial trust formation in new organizational relationships. *Academy of Management Review*, *23*, 473–490.

Merritt, S. M., Unnerstall, J. L., Lee, D., & Huber, K. (2015). Measuring individual differences in the perfect automation schema. *Human Factors*, *57*(5), 740–753.

Meyerson, D., Weick, K. E., & Kramer, R. M. (1996). *Swift trust and temporary groups.* Thousand Oaks, CA: Sage.

Murphy, R. R., Tadokoro, S., Nardi, D., Jacoff, A., Fiorini, P., Choset, H. , et al. (2008). Search and rescue robotics. In *Springer handbook of robotics* (pp. 1151–1173).

Nourbakhsh, I. R. (2015). The coming robot dystopia. *Foreign Affairs*, *94*, 23.

Oestreicher, L., & Eklundh, K. S. (2006, September). User expectations on human-robot co-operation. In *ROMAN 2006-the 15th IEEE international symposium on robot and human interactive communication* (pp. 91–96). IEEE.

Parasuraman, R., & Miller, C. A. (2004). Trust and etiquette in high-criticality automated systems. *Communications of the ACM*, *47*(4), 51–55.

Parasuraman, R., & Riley, V. (1997). Humans and automation: Use, misuse, disuse, abuse. *Human Factors*, *39*(2), 230–253.

Phillips, E., Zhao, X., Ullman, D., & Malle, B. F. (2018, February). What is human-like?: Decomposing robots' human-like appearance using the anthropomorphic robot (abot) database. In *Proceedings of the 2018 ACM/IEEE international conference on human-robot interaction* (pp. 105–113). ACM.

Powers, A., & Kiesler, S. (2006, March). The advisor robot: Tracing people's mental model from a robot's physical attributes. In *Proceedings of the 1st ACM SIGCHI/SIGART conference on human-robot interaction* (pp. 218–225). ACM.

Rae, I., Takayama, L., & Mutlu, B. (2012, May). One of the gang: Supporting in-group behavior for embodied mediated communication. In *Proceedings of the SIGCHI conference on human factors in computing systems* (pp. 3091–3100). ACM.

Rempel, J. K., Holmes, J. G., & Zanna, M. P. (1985). Trust in close relationships. *Journal of Personality and Social Psychology*, *49*(1), 95.

Rızvanoğlu, K., Öztürk, Ö., & Adıyaman, Ö. (2014, June). The impact of human likeness on the older adults' perceptions and preferences of humanoid robot appearance. In *International conference of design, user experience, and usability* (pp. 164–172). Cham: Springer.

Robert, L. P., Denis, A. R., & Hung, Y. T. C. (2009). Individual swift trust and knowledge-based trust in face-to-face and virtual team members. *Journal of Management Information Systems*, *26*(2), 241–279.

Robinette, P., Howard, A. M., & Wagner, A. R. (2015, October). Timing is key for robot trust repair. In *International conference on social robotics* (pp. 574–583). Cham: Springer.

Robinette, P., Howard, A. M., & Wagner, A. R. (2017). Effect of robot performance on human–robot trust in time-critical situations. *IEEE Transactions on Human-Machine Systems*, *47*(4), 425–436.

Robinette, P., Li, W., Allen, R., Howard, A. M., & Wagner, A. R. (2016, March). Overtrust of robots in emergency evacuation scenarios. In *The eleventh ACM/IEEE international conference on human robot interaction* (pp. 101–108). IEEE Press.

Rovira, E., McGarry, K., & Parasuraman, R. (2007). Effects of imperfect automation on decision making in a simulated command and control task. *Human Factors*, *49*(1), 76–87.

Salas, E. (2011). Trust in swift starting action teams: Critical considerations. *Trust in Military Teams*, *71*.

Scassellati, B. (2002). Theory of mind for a humanoid robot. *Autonomous Robots*, *12*(1), 13–24.

Schaefer, K. (2013). *The perception and measurement of human-robot trust. Electronic theses and dissertations.* Retrieved from https://stars.library.ucf.edu/etd/2688.

Schaefer, K. E., Baker, A. L., Brewer, R. W., Patton, D., Canady, J., & Metcalfe, J. S. (2019, May). Assessing multi-agent human-autonomy teams: US Army robotic wingman gunnery operations. *Micro-and nanotechnology sensors, systems, and applications XI* (Vol. 10982, p. 109822B). International Society for Optics and Photonics.

Schaefer, K. E., Chen, J. Y., Szalma, J. L., & Hancock, P. A. (2016). A meta-analysis of factors influencing the development of trust in automation: Implications for understanding autonomy in future systems. *Human Factors*, *58*(3), 377–400.

Schoorman, F. D., Mayer, R. C., & Davis, J. H. (2007). *An integrative model of organizational trust: Past, present, and future*: (pp. 344–354). NY: Academy of Management, Briarcliff Manor.

Short, E., Hart, J., Vu, M., & Scassellati, B. (2010, March). No fair!! An interaction with a cheating robot. In *2010 5th ACM/IEEE international conference on human-robot interaction (HRI)* (pp. 219–226). IEEE.

Sims, V. K., Chin, M. G., Sushil, D. J., Barber, D. J., Ballion, T., Clark, B. R. , et al. (2005, September). Anthropomorphism of robotic forms: A response to affordances? *Proceedings of the human factors and ergonomics society annual meeting* (Vol. 49, No. 3, pp. 602–605). Los Angeles, CA: SAGE Publications.

Stahl, G. K., Maznevski, M. L., Voigt, A., & Jonsen, K. (2010). Unraveling the effects of cultural diversity in teams: A meta-analysis of research on multicultural work groups. *Journal of International Business Studies*, *41*(4), 690–709.

Steinfeld, A., Fong, T., Kaber, D., Lewis, M., Scholtz, J., Schultz, A. , et al. (2006, March). Common metrics for human-robot interaction. In *Proceedings of the 1st ACM SIGCHI/SIGART conference on Human-robot interaction* (pp. 33–40). ACM.

Strait, M., Canning, C., & Scheutz, M. (2014, March). Let me tell you! Investigating the effects of robot communication strategies in advice-giving situations based on robot appearance, interaction modality and distance. In *Proceedings of the 2014 ACM/IEEE international conference on human-robot interaction* (pp. 479–486). ACM.

Sundar, S. S., Waddell, T. F., & Jung, E. H. (2016, March). The Hollywood robot syndrome: Media effects on older adults' attitudes toward robots and adoption intentions. In *The eleventh ACM/IEEE international conference on human robot interaction* (pp. 343–350). IEEE Press.

Susan, D., & Holmes, J. G. (1991). The dynamics of interpersonal trust: Resolving uncertainty in the face of risk. In *Cooperation and prosocial behavior* (p. 190). New York, NY: Cambridge University Press.

Tay, B., Jung, Y., & Park, T. (2014). When stereotypes meet robots: The double-edge sword of robot gender and personality in human–robot interaction. *Computers in Human Behavior*, *38*, 75–84.

Tenhundfeld, N. L., de Visser, E. J., Haring, K. S., Ries, A. J., Finomore, V. S., & Tossell, C. C. (2019). Calibrating trust in automation through familiarity with the autoparking feature of a tesla model X. *Journal of Cognitive Engineering and Decision Making*. 1555343419869083.

Ullman, D., Leite, L., Phillips, J., Kim-Cohen, J., & Scassellati, B. (2014). Smart human, smarter robot: How cheating affects perceptions of social agency. *Proceedings of the annual meeting of the cognitive science society* (Vol. 36, No. 36).

von der Pütten, A., & Krämer, N. (2012, March). A survey on robot appearances. In *2012 7th ACM/IEEE international conference on human-robot interaction (HRI)* (pp. 267–268). IEEE.

Wagner, A. R. (2009). *The role of trust and relationships in human-robot social interaction*. Doctoral dissertationGeorgia Institute of Technology.

Wickens, C. D., Clegg, B. A., Vieane, A. Z., & Sebok, A. L. (2015). Complacency and automation bias in the use of imperfect automation. *Human Factors*, *57*(5), 728–739.

Wickens, C. D., & Dixon, S. R. (2007). The benefits of imperfect diagnostic automation: A synthesis of the literature. *Theoretical Issues in Ergonomics Science*, *8*(3), 201–212.

Wildman, J. L., Shuffler, M. L., Lazzara, E. H., Fiore, S. M., Burke, C. S., Salas, E. , et al. (2012). Trust development in swift starting action teams: A multilevel framework. *Group & Organization Management*, *37*(2), 137–170.

Woods, S., Dautenhahn, K., & Schulz, J. (2004, September). The design space of robots: Investigating children's views. In *RO-MAN 2004. 13th IEEE international workshop on robot and human interactive communication (IEEE Catalog no. 04TH8759)* (pp. 47–52). IEEE.

Yeh, M., & Wickens, C. D. (2000). *Attention and trust biases in the design of augmented reality displays (No. ARL-00-3/FED-LAB-00-1)*. Illinois: Univ at Urbana-Champaign Savoyaviation Research Lab.

Yeh, M., & Wickens, C. D. (2001). Display signaling in augmented reality: Effects of cue reliability and image realism on attention allocation and trust calibration. *Human Factors*, *43*(3), 355–365.

Zhao, X., Phillips, E., & Malle, B. F. (2019). How people infer a humanlike mind from a robot body. *PsyArXiv*.

Zijlstra, F. R., Waller, M. J., & Phillips, S. I. (2012). Setting the tone: Early interaction patterns in swift-starting teams as a predictor of effectiveness. *European Journal of Work and Organizational Psychology*, *21*(5), 749–777.

CHAPTER

18

Investigating the influence of autonomy controllability and observability on performance, trust, and risk perception

M.L. Cummings[a], *Lixiao Huang*[b], *and Masahiro Ono*[c]

[a]Department of Electrical and Computer Engineering, Duke University, Durham, NC, United States, [b]Center for Human, Artificial Intelligence, and Robot Teaming, Arizona State University, Tempe, AZ, United States, [c]Jet Propulsion Laboratory, California Institute of Technology, Pasadena, CA, United States

Introduction

With advances in artificial intelligence, there is increasing interest in using autonomous vehicles in domains like space and underwater exploration, military applications, and surface transportation. Such systems require human supervisors that must often remotely assess risk in the paths used to navigate the vehicles, often under uncertainty. One common source of risk for autonomous vehicle navigation is the uncertainty introduced by embedded path planning algorithms that generate paths of motion that, in theory, avoid collisions but in practice can lead to unsafe situations due to sensor error. In the context of autonomous vehicle path planning, we define the elements of risk as the source of the hazard or obstacle, an estimated likelihood of danger, and an estimated consequence of the danger (American Nuclear Society & Institute of Electrical and Electronics Engineers, 1983). Given that successfully navigating a clear path is a critical element of autonomous vehicle mission success, it is likely

https://doi.org/10.1016/B978-0-12-819472-0.00018-6
© 2021 Elsevier Inc. All rights reserved.

that how an autonomous planner computes such a path could directly influence trust and risk perception of operators overseeing such vehicles.

Risk perception and trust in an autonomous system influence an operator's reliance on that system (Dzindolet, Peterson, Pomranky, Pierce, & Beck, 2003; Moray, Inagaki, & Itoh, 2000), and ultimately overall joint human-system performance. It has been proposed that risk-aware autonomy, which is an autonomous system that provides optimized solution options to human supervisors based on its risk probability calculations, could help humans better identify risks and develop mitigation strategies (Ono, Williams, & Blackmore, 2013).

To better understand what and how different variables influence a person's perception of risk and trust, especially in the presence of autonomy that is providing quantitative risk estimates to a remote supervisor, a notional influence diagram was developed to illustrate such relationships (Fig. 1). The influence diagram in Fig. 1 is grouped into four categories of variables (red = environment, yellow = autonomy, blue = operator,

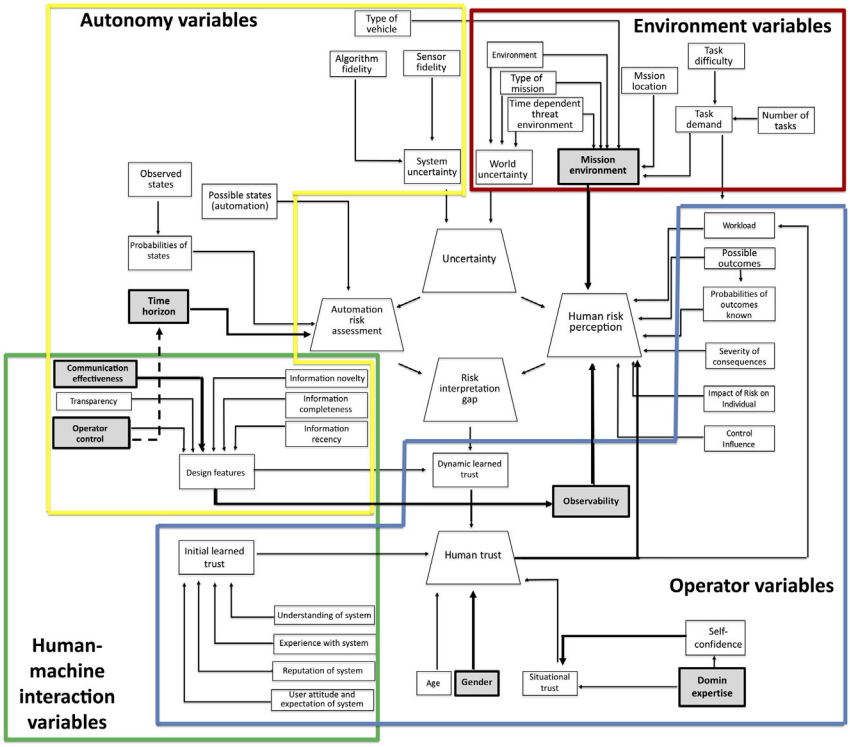


FIG. 1 Those variables that influence human and autonomy risk perception in a supervisory control system, *red* (*upper right box*) = environment, *yellow* (*upper left box*) = autonomy, *blue* (*lower right box*)= operator, *green* (*lower left box*) = human-machine interaction. *Shaded boxes* and *bolded black* lines indicate those variables under investigation in this study. The *dashed line* indicates a futuristic capability.

green = human–machine interaction), with an overlap in the variables as illustrated, and is based on trust and risk research in supervisory settings (Clare, Cummings, & Repenning, 2015; Heitmeyer & Leonard, 2015; Hoff & Bashir, 2015; Kahneman & Tversky, 1979; Lee & See, 2004; Mittu, Sofge, Wagner, & Lawless, 2016; Woods, 2016). The various elements feed into either the autonomy's ability to calculate the risk or the human's assessment of risk. The influence diagram also includes the three layers of trust, including dispositional trust (marked as Human Trust in Fig. 1), situational trust, and dynamically learned trust (Hoff & Bashir, 2015; Lee & See, 2004). Dispositional trust is independent of context and reflects a human's tendency to trust (or distrust) a system. Situational trust depends on the evaluation of the immediate situation and environment and is influenced by factors such as expertise and self-confidence. Lastly, we include learned trust, which is based on knowledge and experience of system past performance and can be dynamic, however, all three types can evolve over time (Hoff & Bashir, 2015).

Fig. 1 illustrates that human risk perception and trust have dramatically many more influences than for an autonomous system's risk computation, and a "risk interpretation gap" emerges when human risk perception and autonomy risk assessment do not align. This can be made worse when both do not accurately incorporate uncertainty into their computations, which means neither the human nor the autonomy risk assessment matches the real world. The diagram in Fig. 1 illustrates that human development of risk perception, possibly mediated by trust, has very little, if any, overlap with how automation computes a quantitative risk assessment. Typically, human operators do not have any ability to influence how autonomy makes risk calculations in the operation of currently fielded systems. Thus, there is no way for operators to proactively align their risk perception with the risk assessment of the autonomy and instead can only attempt to understand autonomy risk assessments reactively by remotely observing the performance of the vehicle in the presence of dynamic but uncontrollable environmental influences.

In this effort, we wanted to develop a more direct link between autonomy risk computation and human risk assessment, and we propose that this can be accomplished by allowing the human the ability to set, and thus control the planning time horizon for an autonomous path planner, depicted by the dashed line in Fig. 1. The time horizon, how far out in time the path planner considers in its computation of an optimal path, is directly linked to its solution quality (Murillo, Sánchez, Genzelis, & Giovanini, 2018). Longer paths carry higher uncertainty, which human operators may find frustrating and at odds with their desire to plan a path from start to finish. However, the ability to control how far out in time an autonomous planner generates a path may help to close the risk perception gap since an operator would be both influencing the automation's

computation or risk assessment as well as their own dynamic learned trust, per Fig. 1, which influences human risk perception.

Indeed, this ability to control the time horizon could be seen as a way to bridge the gulf of evaluation, i.e., how a decision support system provides interpretable representations that can be assessed in terms of user expectations and intentions, to change the state of the world through bridging the gulf of execution, key elements of Norman's seven stages of actions (Norman, 1988). As shown in Fig. 1, the ability to control the time horizon not only influences the autonomous planner but since this is also a design feature in the human–machine interface, controlling the time horizon also could affect observability, the ability of the human operator to see risk information as necessary information and relationships in the system they are supervising. Previous research has shown leveraging an effective control system helps bridge the gulf of execution which is the difference between a user's intent and whether and how a decision support system supports those actions (Leotti, Iyengar, & Ochsner, 2010). However, gambling research has shown a positive relationship between the perception of control over outcomes and risk-taking behavior (Nordgren, Van Der Pligt, & Van Harreveld, 2007), so such control could lead to risk-seeking behaviors. To this end, this study will examine whether the ability to control the autonomous planner in its actual risk computations leads to improved system performance, with a reduction in the risk interpretation gap without promoting risk-seeking behavior.

In addition to examining the influence of human versus autonomous path planner time horizon control in this study, we also wanted to examine how the explicit representation of the amount of risk taken during path planning affected the performance and operator trust. In a recent formulation of risk-aware autonomy, risk can be seen as a limited resource with a budget, with the operator potentially deciding how and when to take more or less risk (Ono, 2012). For example, an operator could be allowed to take a quantitative measure of risk in a system's operation, such as up to 10% risk of hitting an obstacle in a mission but decides when and where to take such risk that accumulates throughout a mission.

It is not clear how the concept of a risk budget could influence people's risk perception and regulation, especially in a path navigation scenario. While risk perception is a key component of trust (Hardin, 2006), humans often struggle to understand probabilistic estimates of risk (Tversky & Kahneman, 1974). Moreover, risk communication is often interpreted as vague (Spiegelhalter, 2017), and even how risk should be presented, numerically or verbally, is often debated (e.g., Erev & Cohen, 1990). Therefore, it is important to understand whether the concept of a risk budget is beneficial for human understanding and overall system performance, including the impact on trust and risk perception. In risk-aware autonomy, providing insight into automated risk computation through a risk

budget could aid in the bridge of evaluation, and in Fig. 1 this is represented as communication effectiveness, which is ultimately expressed as a design feature that promotes observability.

The impact of risk budget observability and controllability of the planning time horizon on risk perception and humans' trust in autonomy is not well understood. So, as illustrated in Fig. 1, our focus for this study was to determine how two design features, operator control of the autonomous planner time horizon and a risk budget representation, influenced operators' risk perception, as well as trust, self-confidence, and ultimately system performance. Our hypothesis was that both design features would reduce the risk perception gap, which ultimately would lead to improved joint human-autonomous system performance, with improved trust and self-confidence. While all the variables in Fig. 1 could not be tested in a single experiment, we elected to also explore other variables in this experiment, including two in the environment. These will be discussed further in the following sections.

The human–autonomy interface for exploration of risks (HAIER)

This study leveraged the software platform *Human–Autonomy Interface for Exploration of Risks (HAIER)* to examine human interactions with a risk-aware, human-cooperative autonomous path planner. In HAIER, participants guide a single unmanned vehicle from start to finish across a field of obstacles. HAIER allows human supervisors to view autonomy-generated potential paths and adjust them according to their level of perceived risk. Although HAIER can be used to represent any generic unmanned vehicle, in this version, the unmanned vehicle was considered to be an unmanned underwater vehicle (UUV) that can occasionally surface to communicate with a hypothetical satellite to update its position, and thus locally resolve uncertainty about its position. The autonomous planner is considered "risk-aware" since it can propose the shortest path between its current position and the goal position while accounting for a user-specified corresponding risk level (low, medium, or high). High-risk paths result in paths that often pass very close to obstacles.

In HAIER, it is assumed that when a vehicle "surfaces," it resolves all ambiguity about its position by communicating with satellites, which it cannot do underwater. However, the longer a vehicle remains underwater, the more uncertainty grows as to where the vehicle will surface (which is also true in the real world). Thus, while it is possible to go from the start to goal in a single leg, the uncertainty would be extremely high, and thus the vehicle needs to surface periodically to resolve any error. However, given that surfacing requires extra fuel and exposes UUVs to danger in

this version of HAIER, the number of surfacing events should be minimized. Thus, participants have to balance the risk of hitting an obstacle against the risk of not completing a mission by surfacing too frequently. In this version of HAIER, operators are given a "surfacing budget" so they know how often they can surface for a specific mission. Each path between surfacing events is referred to as a leg. For this application of HAIER, the surfacing budget was always set to 6. This constraint is soft in that operators can surface more if needed, but they put the vehicle at much greater risk if they do so.

For this effort, two specific design features were tested in HAIER including observability in the form of a risk budget bar and controllability, which represents operator control of the underlying path planner time horizon. As discussed previously, allowing operators the ability to control the time horizon allows the human to directly influence the autonomy's solution quality, while the risk budget representation gives operators a sense of how much risk is used in each leg of travel. These variables represent the intersection of autonomy and human–machine interaction features that can influence risk perception in Fig. 1.

The basic HAIER interface (Fig. 2A) contains a control panel that allows operators to select their proposed path risk levels (low, medium, or high)

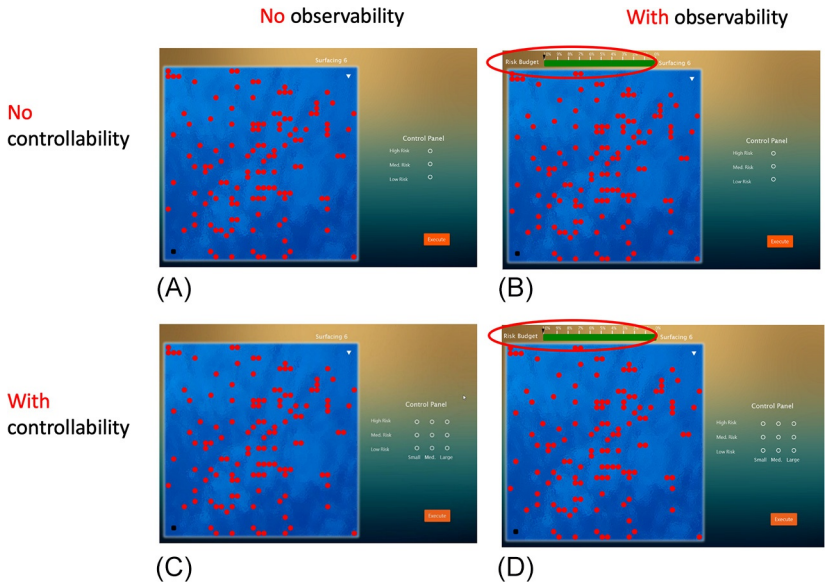


FIG. 2 Four HAIER experimental interfaces. On the control panel, controllability is on the horizontal axis and risk acceptance is on the vertical axis. In (A) and (B), operators only have access to risk acceptance levels and in (C) and (D), the control panel allows for changes in the path leg length, known as controllability. The observability experimental condition, i.e., the presence of a risk budget bar, is tested by comparing (A) and (C) to (B) and (D).

through the vertical buttons, as seen in Fig. 2A and B. The term "observability" refers to the presence of a risk budget bar that explicitly shows how much of budgeted risk has been used at any given time (Fig. 2B and D). Controllability of leg length (small, medium, or large) was manipulated in the control panel through the horizontal axis in Fig. 2C and D. Both of these features are explained in more detail in the next sections.

The risk budget bar (upper left of Fig. 2B and D) allowed participants to see exactly how much risk was associated with each autonomy-generated leg and how much risk would be left in a notional risk budget after executing one leg of planning. A mission was assigned an overall level of acceptable risk, i.e., the mission should not exceed more than 10% of the risk from a collision. If, for example, one leg put the mission at a 2% risk of collision, then from that point forward, only 8% of overall mission risk remained. The percentage of risk computed for each leg by the autonomy was determined by the expected number of obstacles along the leg, the proximity of the obstacles to the path leg, the leg length, and the likelihood of deviation. The risk budget was also a soft constraint in that participants could exceed the allotted 10%, but the vehicle's risk of not completing the mission also grew commensurately.

Controllability allowed participants the ability to directly influence the planning time horizon by setting the leg length of the path planning algorithm, which in this version of HAIER was a Dijkstra planner. As seen in Fig. 2C and D, participants could set the length of each leg to be small, medium, or large, which represents growing uncertainty over time. The yellow circles represent the uncertainty of UUV's location due to the unpredictable ocean current, with the small, medium, large selections mirrored by the growing yellow circles. Without human control (Fig. 2A and B), the underlying algorithm calculated the most optimal leg length, which was determined by mathematically optimizing local obstacle proximity, the final goal, and the number of surfacings remaining.

The primary research question for this effort was how the two design features of observability through the risk budget bar and controllability of leg length influenced participants' performance, trust in the planner's ability to carry out the operator's intent, self-confidence, and risk perception. As seen in Fig. 2, there were four experimental conditions, which ranged from having neither observability nor controllability (Fig. 2A) to those with both decision aids (Fig. 2D).

Method

Seventy-five people, recruited from mailing lists and flyers from a southeastern US university, each received a $15 gift card for the one-hour experiment and an additional $100 gift card for the best performer. Each

participant interacted with HAIER through a Dell desktop computer with a mouse. A 2×2 mixed factorial design was employed with Controllability as the between factor and Observability as the within factor. Participants completed two Observability test trials with two different maps (first without and then with the risk budget bar) for each controllability block. The order of Observability trials was not counterbalanced because once participants saw the With Observability interface, their impression might interfere with their understanding of the concept of the risk budget.

Participants were told that their goal was to take the shortest path possible, try to surface no more than 6 times to reduce their exposure to enemy detection, and try to keep their risk budget under 10%. The training was provided before each testing trial by demonstrating the interface with a different training map, verbally introducing each element on the interface and performance criteria, and having participants practice path planning until finishing the training map. Participants' questions were answered by the experimenter during training. Testing trials started when participants verbally confirmed they felt ready for testing. Each training session took 5–10 min, and each testing trial took 5–15 min. Participants experienced two different maps (with or without the risk budget bar) in their testing trials.

The experiment used four maps of 36×36 grids (Fig. 3) presented in a counterbalanced and randomized manner. Although the four maps were designed to be roughly equal in difficulty, given that the environment was not homogenous, map design was included as a secondary independent variable in the final statistical model, discussed in more detail in the results. In effect, these maps represent mission environmental variability, as highlighted in Fig. 1.

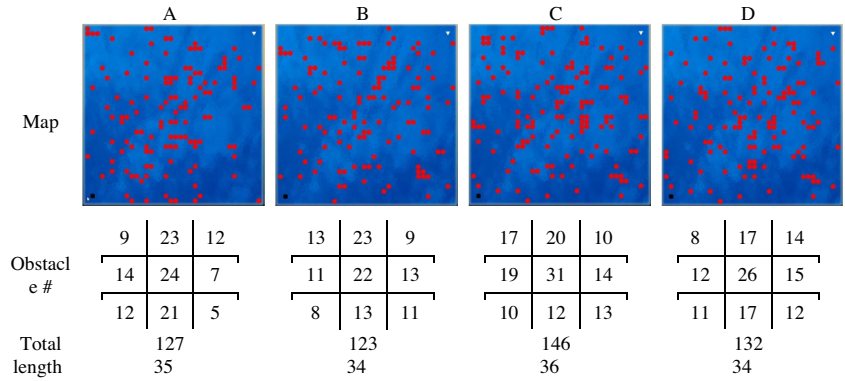


FIG. 3 The four experiment maps (A–D), where Obstacle # = number of obstacles in a 3×3 matrix overlaid on a map; Total = total number of obstacles; Length = the shortest possible path length of the map in grid units.

Several dependent variables were measured, including the following:

Risk perception. Every time before participants planned a path leg, a pop-up window above the control panel asked "how risky the situation is" on a 3-point scale (1 = not risky, 2 = kind of risky, 3 = very risky). An overall risk perception mean rating was weighted by the number of legs participants elected to generate, which could range from 4 to 8 legs. Thus, the overall risk perception mean rating for a person who crossed a map in 4 legs would be the sum of their 4 legs risk ratings divided by 4, whereas another person who crossed in 6 legs would generate a mean based on the sum of risk ratings/6.

Self-confidence rating. While waiting for the planner to compute and display its solution based on input, a pop-up question asked participants to answer a question on a 5-point Likert scale (1 = not confident, 5 = very confident), "How confident do you feel about your [last path] planning?" Given that four to eight legs were required to reach the destination within the surfacing constraints, the overall self-confidence mean was based on the number of legs from start to the goal for each map in a similar fashion as the risk rating variable.

Trust rating. During the planning wait time, a pop-up question asked participants to rate "How much do you trust that the autonomous planner generates the best solution?" on a 5-point scale (1 = do not trust, 5 = completely trust). Each leg of path planning involved a rating of trust, so the overall trust mean rating was also weighted by the leg number in the same manner as the overall risk perception mean rating.

Performance. Since participants were instructed to plan the shortest path possible, remain under the risk budget, and remain under the surfacing budget, three variables were used to measure a participant's performance: *path length*, whether the *risk budget* was exceeded, and whether the *surface budget* was exceeded. Path length was the distance the UUV traveled from start to the goal, calculated through the number of grid units. Participants were told to plan the shortest paths possible without hitting any obstacles. To track whether people respected the risk and surface budgets, they were assessed on whether their total risk consumption and surfacing times stayed within or exceeded these respective budgets (10% and 6 times). Participants chose an acceptable risk level for each leg they planned, called leg risk.

Computer game experience. In the demographic survey, participants answered "How often do you play computer games normally?" (1 = rarely, 2 = a few times a month, 3 = once a week, 4 = a few times a week, 5 = daily). While not all of the operator-centric variables depicted in Fig. 1 could be tested in a single experiment, we elected to use video game experience as a proxy for domain expertise because it has been shown to influence performance in similar experiments (Cummings,

Clare, & Hart, 2010; Green & Bavelier, 2003; Lin, Wohleber, Matthews, & Funke, 2015). Thus, the gaming experience was a covariate.

Results

While the original pool of participants was 75, two participants' data were excluded due to problems with training. The remaining 73 participants (*male* = 38, *female* = 35; *age mean* = 22.8, *SD* = 7.1) resulted in 134 successful trials. Twelve trials resulted in collisions, which immediately ended a test session. These trials were analyzed separately. A multivariate analysis of variance (MANOVA) of continuous variables was conducted to assess the original research questions about controllability and observability. Then surfacing and risk budget adherence measures, also known as constraint violations, were assessed, followed by analyses of the temporal trends of the subjective measures of trust, risk perception, and self-confidence.

Controllability vs observability outcomes

Controllability (with or without) of leg length and observability (with or without the risk budget bar) were the primary independent variables. Gender was a secondary independent variable since the experiment was opportunistically roughly balanced. A $2 \times 2 \times 2 \times 4$ (observability $\times$ controllability $\times$ gender $\times$ map) MANOVA statistical model was used to investigate whether these factors influenced path length, as well as three subjective variables including mean subjective ratings for trust, risk perception, and self-confidence. Given the relatively high number of independent (4) and dependent variables (4), a reduced-order MANOVA model was used that represented only the main and two-interaction effects, $\alpha = 0.05$. Whether the surface and risk budget constraints were respected was investigated using logistic regression models, detailed in a subsequent section.

For the MANOVA, all normality and homogeneity assumptions were met except for mean trust, where Levene's test for homogeneity of error variance was significant ($p = 0.008$). This required a square root transformation, which stabilized the variance. For the between-subjects comparisons across the individual dependent variables, there were no significant main effects across any of the dependent variables for controllability or observability, but there were interesting main effects for maps, gender and gaming experience, as well as a significant interaction shown in Table 1.

TABLE 1 MANOVA results.

	Map	Gender	Gaming experience	Map × observability
Path length	$F(3114) = 5.94$ $p = 0.001$	NS	NS	NS
Mean self-confidence	NS	$F(1114) = 6.79$ $p = 0.010$	$F(1114) = 8.51$ $p = 0.004$	$F(1114) = 3.38$ $p = 0.021$
Mean trust	NS	NS	NS	$F(1114) = 4.61$ $p = 0.004$
Mean risk perception	$F(3,114) = 2.95$ $p = 0.036$	NS	$F(1114) = 6.6$ $p = 0.011$	NS

The two-operator factors from Fig. 1 included in the model were gender and video game experience. As seen in Table 1, women in general were significantly less confident than the men ($M_W = 3.66$, $SD_W = 0.08$, $M_M = 4.02$, $SD_M = 0.09$). The video game covariate was a significant factor for self-confidence, with a Pearson correlation of $\rho = 0.262$, $p = 0.002$, indicating self-confidence moderately increased with video game experience. Not surprisingly, the overall mean perceived risk was lower for those with more gaming experience, $\rho = -0.230$, $p = 0.008$, which aligns with the influence diagram in Fig. 1, where situational trust, gaming experience, and self-confidence are interrelated.

The map effect

Given the four different maps (Fig. 2) that represent environmental variability, it was important to control for this in the model. Least significant difference post hoc comparisons indicated that the mean path length on Map B ($M = 45.35$, $SD = 5.11$) was significantly shorter from Map A ($M = 48.49$, $SD = 3.56$, $p = 0.008$), Map C ($M = 50.24$, $SD = 3.06$, $p = 0.000$), and Map D ($M = 48.70$, $SD = 4.90$, $p = 0.009$).

Map A was seen as the least risky, even though it had more obstacles than Map B, with overall mean ratings statistically lower than Maps C ($p = 0.008$) and D ($p = 0.021$). All other maps were not statistically different from one another at the factor levels. However, the significant interactions between the map and observability factor for both trust and self-confidence can be attributed to Map C, which had the highest number of obstacles. For this map, people *without* observability ($M_{SC} = 4.08$, $SD_{SC} = 0.59$; $M_T = 4.07$, $SD_T = 0.54$) had significantly higher confidence

and trust ratings as compared to those who had the risk bar visualization ($M_{SC}=3.35$, $SD_{SC}=0.67$; $M_T=3.16$, $SD_T = 0.85$), $p=0.005$ and 0.002, respectively. When examining just the collision trials, of which there were only 12 (8% of all trials), seven (58.3%) happened on Map C, four (33.3%) on Map D, one (8.3%) on Map A, and none on Map B.

Constraint violations

To examine whether controllability and observability affected a participant's ability to stay within the risk and surfacing budgets, two logistic regression analyses were conducted, also including gender, map, and computer game experience as predictors. The logistic regression analysis for risk budget adherence demonstrated that the map variable was the only significant predictor (Wald $\chi^2(1)=7.15$, $p=0.008$). Of the four maps, Map D (Fig. 3) caused more risk violations than the other three maps. People using Maps A-C only violated the risk budget 5–16% of the trials compared to 65% for Map D (Fig. 3). Map D had the second highest number of obstacles and they were more evenly spaced than the other maps, which could account for the high number of risk violations. While this issue deserves more research, the more consistent spacing of the obstacles in Map D may have contributed to the acceptance of greater risk.

For those exceeding the surfacing budget, a slightly different picture emerged. The controllability factor was statistically significant (Wald $\chi^2(1)=5.56$, $p=0.018$). The percentage of trials that exceeded the surfacing budget was less with the Controllability interface (33%) compared to without it (58%) (Fig. 4). The map variable was also significant (Wald $\chi^2(1)=9.09$, $p=0.003$), but the difficulty across maps varied (Fig. 4). For Maps A and B, roughly 33% of participants across all trials did not stay under the surfacing budget, while for Maps C and D, 72% and 55% respectively did not stay within their surfacing budget.

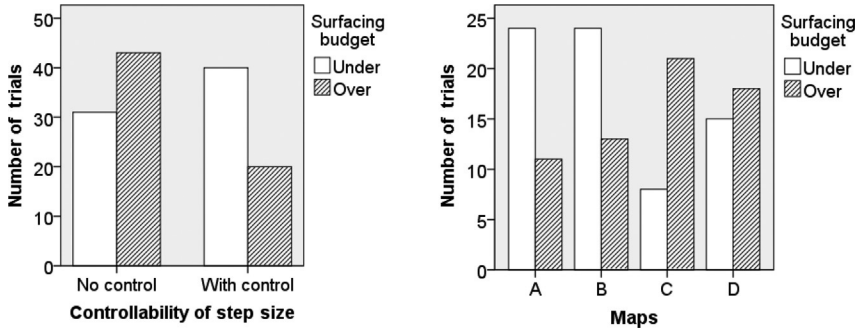


FIG. 4 Impact of controllability (left) and different maps (right) on staying within the surfacing budget.

Subjective factors

Spearman rank-order correlations between mean risk perception, mean self-confidence, and mean trust were calculated (Table 2). These ratings were generated by questions after each leg was planned, and while most people made it from the start to goal in 6 or fewer legs, 47% of the trials went to 7 legs, with just 4 trials going to 8 legs. Thus, the means represent the average ratings for each person. In addition to the risk perception metric in Table 2 which shows how risky each leg appeared to be to every participant on average, the average risk each person chose to take per leg (i.e., leg risk) is also listed in Table 2. It should be noted that only 45% of participants had the ability to control the leg length (Fig. 2C and D).

Self-confidence and trust were significantly positively correlated (Table 2). As expected, risk evaluation negatively correlated with self-confidence and trust, with self-confidence as the stronger correlate. Overall as the perceived risk in each scenario increased, trust and self-confidence declined. However, curiously the risk that each person chose to take for each leg in response to perceived risk, on average, did not correlate. However, there was a weak but significant correlation with the overall leg risk and trust, i.e., the higher a person's trust in the planner, the more risk taken per leg.

While Table 2 demonstrates the overall averages for trust, self-confidence, risk perception, and leg risk for each total path from the start to the goal, it does not indicate how these values changed over time. Fig. 5 illustrates mode values for participant self-confidence, trust, risk perception, and leg risk taken as a function of legs across the map. While overall trust and self-confidence remained relatively high for the majority of each mission, as participants came within 1–2 legs from the end, their self-confidence and trust rose to the highest levels. The exception is the four

TABLE 2 Spearman rho correlation coefficients, means (M), and standard deviations (SD) for mean risk perception, leg risk, self-confidence, and trust ratings.

	1	**2**	**3**	**4**	***M***	***SD***
1. Mean risk perception	1	0.036 $p=0.682$	−0.426 $p<0.001$	−0.190 $p=0.028$	1.65	0.32
2. Mean leg risk	0.036 $p=0.682$	1	0.103 $p=0.238$	0.230 $p=0.008$	2.00	0.33
3. Mean self-confidence	−0.426 $p<0.001$	0.103 $p=0.238$	1	0.625 $p<0.001$	3.85	0.68
4. Mean trust	−0.190 $p=0.028$	0.230 $p=0.008$	0.625 $p<0.001$	1	3.72	0.78

Self-confidence and trust were measured on a 5-point scale, while risk evaluation and leg risk were measured on a 3-point scale.

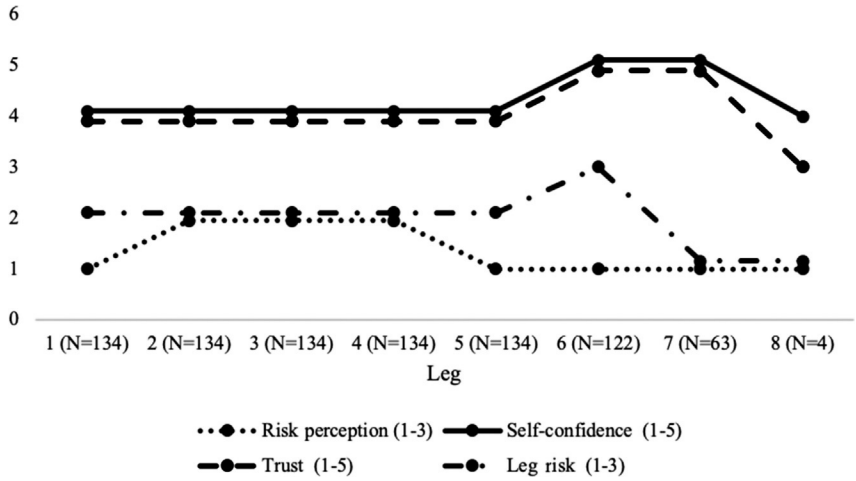


FIG. 5 The mode trend of risk perception, self-confidence, trust, and leg risk. The Y axis represents question Likert ratings. Risk ratings ranged from 1 to 3, self-confidence and trust ranged from 1 to 5. The X- axis represents legs of the UUV. It took 4–8 legs to reach the goal.

individuals who took 8 legs to cross. Their self-confidence and trust dropped in this last leg.

The plots of perceived risk versus leg risk taken over time in Fig. 5 demonstrate how participants adapted their strategies as a function of where they were on the map. Generally, participants both perceived and were willing to take moderate risk through the 4th leg. The majority of participants across both trials took six or more legs to get across each map (91%), and these participants typically took the most risk for the 6th leg (Fig. 5). Once they made it past this point, most people dropped to the lowest risk level for path planning.

Curiously, even though participants were willing to take more risk at the point where surfacing budgets were of concern (recall the soft constraint was 6), in general at this same point, they reported their perceived risk at the lowest rating. Also, at this point self-confidence and trust ratings rose to their highest levels. Fig. 5 demonstrates that at the 6th leg, participants perceived the situation risk to be low, and were willing to increase risk-taking, with the highest self-confidence and the highest trust in the autonomous planner. It is also important to note that this rise happened one-time step after people dropped their risk perception from moderate to low for the 5th leg.

Only 8% of the total number of trials resulted in collisions, and 66% of these trials (8 of 12) had observability and/or controllability elements. The per leg self-confidence, trust, risk perception, and risk execution mode ratings for these limited number of trials are plotted in Fig. 6. With significantly

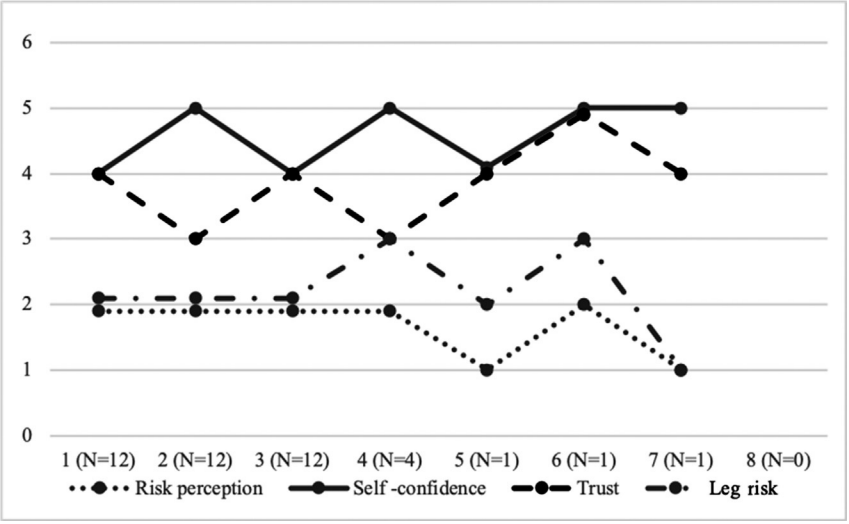


FIG. 6 For the collision trials, the mode trend of risk perception, self-confidence, trust, and leg risk. The Y axis represents question Likert ratings. Risk ratings ranged from 1 to 3, self-confidence and trust ranged from 1 to 5. The X axis represents legs of the UUV. It took 4–8 legs to reach the goal.

fewer number of observations as compared to Fig. 5, it is worth noting that the patterns between those who collided and those who succeeded were different. Most participants collided after the third leg, which generally would occur somewhere in the middle of the screen. These people tended to exhibit either high self-confidence or less trust in the autonomous planner than did those who succeeded. For example, the one individual who almost made it to the end before colliding with an obstacle took more risk earlier, which then led to increased trust for the autonomous planner for two more legs. But interestingly, this person's trust dropped just before colliding with an obstacle.

Discussion

The goals of this study were to determine whether providing explicit observability through a risk budget representation and providing the ability to control the path planning time horizon through setting the path leg length could significantly improve joint human–autonomous system performance as well as affect subjective ratings. More specifically, the study examined whether such aids could produce more efficient paths, as well as influence subjective perceptions of risk, trust, and self-confidence.

In terms of the two design features and their effect on performance, as well as the other influencing variables in Fig. 1, the most influential variable was the environmental variable of mission complexity, represented by the map. It was the only variable that mattered for path planning length, as well as adhering to the risk budget. Controlling the time horizon did aid people in respecting their surfacing budget constraint so this design feature did help operators close the gulf of execution for this element of the mission. However, this tool did not measurably influence any of the subjective metrics.

As has been shown in other supervisory control studies (Cummings & Tsonis, 2006; Kirwan, Scaife, & Kennedy, 2001; Majumdar & Ochieng, 2002), the complexity of the environment can be the primary driver of operator performance, and in this study, produced the largest performance effects. Curiously, the map environment variable was not a significant predictor of trust or self-confidence, but it was for risk perception. This indicates that participants used map spatial elements to determine a vehicle's risk profile, but the map variability did not influence their sense of trust in the planner or their views of their own capabilities, which were relatively stable as seen in Fig. 5. One issue to consider is that HAIER showed other concrete visual risk cues, such as red obstacles and yellow probability circles when generating paths on the map. So, such concrete visual cues on a map are effective at communicating risk information, but trust and self-confidence are more nuanced and multi-dimensional, so are then harder to measure.

While the risk budget bar did not demonstrate any direct performance improvements, its observability influenced self-confidence and trust through *reduced* ratings when using more complex maps (i.e., more obstacle-rich). The reduced self-confidence and trust ratings suggest that participants with the risk budget bar were more cautious than those without this aid, indicating that the inclusion of such observability tools may be beneficial in cases where user overreliance is either likely or potentially dangerous.

While these results align with the influence diagram in Fig. 1 that suggests that the risk budget bar (i.e., communication effectiveness) may influence dynamically learned trust, the diagram did not adequately represent any associations with self-confidence, which is related to situation trust. Moreover, it should be noted that the concept of a risk budget is relatively new with limited literature (Ono, 2012). Further research is needed on whether people understood the risk budget bar as a probability of collision as opposed to other types of risks, and how it influences decision making under increased uncertainty.

One important corollary result in this study is that while previous work has demonstrated a positive correlation between trust in autonomy, self-confidence, and performance (Lee & Moray, 1994), trust and self-confidence were not significantly correlated with performance in this

experiment, but were highly correlated with one another. While trust and self-confidence are theoretically different, this study suggests that such differences may be very difficult to disaggregate in practice.

Perhaps the most interesting result of this study was the increase in risk-taking as both trust and self-confidence increased, and perception of environmental risk decreased, i.e., fewer obstacles. As seen in Fig. 5, participants were willing to take more risk not only after the planner appeared to be trustworthy but also when they were close to reaching their goal. It is important to note that these events took place in the upper corners of the maps in Fig. 3. In these quadrants, obstacle density was lower than in the middle of the maps (although not the lowest), and the uncertainty in whether participants would reach the goal was lower at this point. So while increasing risky behavior towards the end of the mission was generally an effective strategy, it could backfire as evidenced in Fig. 6.

This propensity to increase one's acceptance of risk as the goal gets closer is a phenomenon known as the Perseveration Syndrome and is well known in aviation as get-home-itis, which can and has lead to fatal accidents (Dehais, Tessier, Christophe, & Reuzeau, 2010). Military and civilian aviation agencies mitigate this phenomenon through training and crew resource management techniques, but this is the first effort to document such an effect in unmanned vehicle remote operation. More research is needed to better identify the occurrence of such a bias and how to mitigate it in unmanned vehicle control.

One limitation of the study is that because there were no consistent correlations between any individual map and objective or subjective measures, it is likely that individual differences played a significant role in the perception of map difficulty. Follow-up research is needed with a larger group of subjects to elucidate the nature and impact of map complexity. Other important limitations of this effort include the small number of participants and a potential mismatch between display design and operator expectations. The autonomous planner displayed quantitative risk usage, (e.g., 10%), whereas operators could only input their risk tolerance across three qualitative categories of low, medium, and high, and it is not clear that there was clear alignment between these two pieces of information. Current work is underway to allow operators the ability to enter risk on an interval scale so that it is aligned with the risk-aware autonomy, but this mixed use of formats may have influenced results.

Conclusion

As autonomous planners proliferate in settings where humans supervise such systems, such as military, space, and air traffic control applications which can experience significant uncertainty, it is critical that

designers better understand how and to what extent humans should collaborate in real-time with such planners. Risk is an inherent component of all autonomous systems, and so promoting human–autonomous system collaboration through risk-aware autonomy could aid in joint human-system performance.

A risk perception gap influence diagram was proposed that links autonomous planner characteristics with those in the environment as well as for an associated human–machine interface and representative operator attributes. These four clusters of variables influence risk perception and facets of human trust, which can lead to a gap in the way people perceive autonomy and the calculations of risk generated by an autonomous planner. Such a risk perception gap could lead to inferior system performance.

To assess two different design features and their interaction with a path-planning algorithm typically used in autonomous systems on system performance and subjective operator attributes, an experiment was conducted in the software platform, *HAIER*. This study demonstrated that the concept of observability into autonomy via a risk budget bar did not affect performance but did reduce people's trust and self-confidence under environmental complexity. Moreover, allowing people to control a path planning time horizon allowed people to respect soft constraints but did not significantly influence their subjective ratings of risk, self-confidence, or trust.

Most strikingly, participants altered their perceptions of risk, self-confidence, and trust as they reached the latter one-third of the mission and were willing to take more risk as the goal was closer. The acceptance of increased risk as the goal gets closer is well established in manned aviation, but these results demonstrate that it also exists in unmanned settings as well. More work is needed to examine whether this adjustment of risk perception, which is a widening of the risk perception gap in Fig. 1, may be more pronounced given the remote-control aspect of such operations and what design interventions could prevent this.

Further work is also needed to better understand people's interpretation of the risk budget bar, as well as whether they fully understood and intended to take more risk as the mission came close to the goal or whether they indeed were willing to take more risk to finish. Additional work is needed to determine if and how operators could benefit from having more precise control of risk budget consumption than in the current HAIER version. For example, instead of simply responding to the risk per leg that a planner generates for a proposed action, would it be more helpful for an operator to specify an exact level of risk per leg that the operator is willing to take? Lastly, this research demonstrated that the proximity and density of obstacles on a map may not be directly correlated with performance and so better understanding those variables that influence operator decisions for complex maps would be particularly beneficial.

Acknowledgments

The research was carried out in part at Duke University under a grant from National Aeronautics and Space Administration JPL, and in part at the Jet Propulsion Laboratory for a grant from the Office of Naval Research (N00014-17-IP00043).

References

American Nuclear Society, & Institute of Electrical and Electronics Engineers (1983). *PRA procedures guide: A guide to the performance of probabilistic risk assessments for nuclear power plants (NUREG/CR-2300).* Washington, DC: US Nuclear Regulatory Commission.

Clare, A., Cummings, M. L., & Repenning, N. (2015). Influencing trust for human-automation collaborative scheduling of multiple unmanned vehicles. *Human Factors*, *57*(7), 1208–1218.

Cummings, M. L., Clare, A., & Hart, C. (2010). The role of human-automation consensus in multiple unmanned vehicle scheduling. *Human Factors*, *52*(1), 17–27.

Cummings, M. L., & Tsonis, C. (2006). Partitioning complexity in air traffic management tasks. *The International Journal of Aviation Psychology*, *16*(3), 277–296.

Dehais, F., Tessier, C., Christophe, L., & Reuzeau, F. (2010). The perseveration syndrome in the pilot's activity: Guidelines and cognitive countermeasures. P. Palanque, J. Vanderdonckt, & M. Winckler (Eds.), *Human error, safety and systems development. Lecture notes in computer science* (pp. 68–80). Vol. 5962 (pp. 68–80). Berlin: Springer-Verlag.

Dzindolet, M. T., Peterson, S. A., Pomranky, R. A., Pierce, L. G., & Beck, H. P. (2003). The role of trust in automation reliance. *International Journal of Human Computer Studies*, *58*, 697–718.

Erev, I., & Cohen, B. L. (1990). Verbal versus numerical probabilities: Efficiency, biases, and the preference paradox. *Organizational Behavior and Human Decision Processes*, *45*, 1–18.

Green, C. S., & Bavelier, D. (2003). Action video game modifies visual selective attention. *Nature*, *423*, 534–537.

Hardin, R. (2006). *Trust.* Cambridge, UK: Polity Press.

Heitmeyer, C. L., & Leonard, E. I. (2015). Obtaining trust in autonomous systems: tools for formal model synthesis and validation. In: *Paper presented at the proceedings of the third FME workshop on formal methods in software engineering, Florence, Italy*.

Hoff, K., & Bashir, M. (2015). Trust in automation: Integrating empirical evidence on factors that influence trust. *Human Factors*, *57*(3), 407–434.

Kahneman, D., & Tversky, A. (1979). Prospect theory: An analysis of decision under risk. *Econometrica*, *47*(2), 263–292. Retrieved from http://links.jstor.org/sici?sici=0012-9682%28197903%3A2%3C263%3APTAAOD%3E2.0.CO%3B2-3.

Kirwan, B., Scaife, R., & Kennedy, R. (2001). Investigating complexity factors in UK air traffic management. *Human Factors and Aerospace Safety*, *1*(2), 125–144.

Lee, J., & Moray, N. (1994). Trust, self confidence, and operators' adaptation to automation. *International Journal of Human-Computer Studies*, *40*, 153–184.

Lee, J. D., & See, K. A. (2004). Trust in technology: Designing for appropriate reliance. *Human Factors*, *46*(1), 50–80.

Leotti, L. A., Iyengar, S. S., & Ochsner, K. N. (2010). Born to choose: The origins and value of the need for control. *Trends in Cognitive Sciences*, *14*(10), 457–463.

Lin, J., Wohleber, R., Matthews, G., & Funke, G. J. (2015). Video game experience and gender as predictors of performance and stress during supervisory control of multiple unmanned aerial vehicles. In: *Paper presented at the 59th international annual meeting of the human factors and ergonomics society, Los Angeles, CA*.

Majumdar, A., & Ochieng, W. Y. (2002). The factors affecting air traffic controller workload: A multivariate analysis based upon simulation modeling of controller workload. In: *Paper presented at the 81st annual meeting of the Transportation Research Board, Washington, DC*.

Mittu, R., Sofge, D., Wagner, A., & Lawless, W. F. (Eds.). (2016). *Robust intelligence and trust in autonomous systems*. New York: Springer.

Moray, N., Inagaki, T., & Itoh, M. (2000). Adaptive automation, trust, and self-confidence in fault management of time-critical tasks. *Journal of Experimental Psychology: Applied*, *6*(1), 44–58.

Murillo, M., Sánchez, G., Genzelis, L., & Giovanini, L. (2018). A real-time path-planning algorithm based on receding horizon techniques. *Journal of Intelligent and Robotic Systems*, *91*(3–4), 445–457.

Nordgren, L. F., Van Der Pligt, J., & Van Harreveld, F. (2007). Unpacking perceived control in risk perception: The mediating role of anticipated regret. *Journal of Behavioral Decision Making*, *20*(5), 533–544.

Norman, D. A. (1988). The psychopathology of everyday things. In *The design of everyday things* (p. 257). New York: Doubleday.

Ono, M. (2012). *Robust, goal-directed plan execution with bounded risk (Doctorate)*. Cambridge, MA: Massachusetts Institute of Technology.

Ono, M., Williams, B., & Blackmore, L. (2013). Probabilistic planning for continuous dynamic systems under bounded risk. *Journal of Artificial Intelligence Research*, *46*, 511–577.

Spiegelhalter, D. (2017). Risk and uncertainty communication. *Annual Review of Statistics and Its Application*, *4*, 31–60.

Tversky, A., & Kahneman, D. (1974). Judgment under uncertainty: Heuristics and biases. *Science*, *185*(4157), 1124–1131.

Woods, D. D. (2016). The risks of autonomy: Doyle's catch. *Journal of Cognitive Engineering and Decision Making*, *10*(2), 131–133.

CHAPTER

19

Applied quantitative models of trust in human-robot interaction

Nathan E. Sanders and Chang S. Nam

Edward P. Fitts Department of Industrial and Systems Engineering, North Carolina State University, Raleigh, NC, United States

Introduction

As robots become more ubiquitous, their inner workings are also becoming increasingly unknowable to the vast majority of the intended user base. As this occurs, people will find that they increasingly rely on trust to guide their interactions with these robots, just as they do with other people. For example, humans are increasingly being asked to supervise multiple UAVs at a time (Lawless & Sofge, 2017). Collaborative robots (co-bots) are beginning to enter the workplace (Djuric, Rickli, & Urbanic, 2016; El Makrini et al., 2018; Maurtua, Ibarguren, Kildal, Susperregi, & Sierra, 2017), doing away with the physical safety barriers present in older generations of industrial robots. Social robots are appearing in the homes of the elderly (Cavallo et al., 2014; Graf, Hans, & Schraft, 2004; Torta et al., 2014) and are interacting with children (Breazeal et al., 2016; Huijnen, Lexis, Jansens, & de Witte, 2017; Kahn et al., 2012; Vogt, de Haas, de Jong, Baxter, & Krahmer, 2017). And "The era of automated road vehicles is upon us" (Hancock, 2015)—autonomous vehicles are poised to take over our highways, and trust will be a major determinant in their acceptance and use by both drivers and pedestrians (Basu & Singhal, 2016; Choi & Ji, 2015; Fagnant & Kockelman, 2015; Lee, Kim, Lee, & Shin, 2015; Saleh, Hossny, & Nahavandi, 2017).

But before we relinquish control entirely, it is important that we understand how trust in robots is built up and broken down in order that we trust robots only to the extent that they are trustworthy (Schaefer, Brewer, et al., 2016). If not, we may find that we will come to rely on them

https://doi.org/10.1016/B978-0-12-819472-0.00019-8
© 2021 Elsevier Inc. All rights reserved.

too little or too much, with consequences that could be disastrous (Parasuraman & Riley, 1997).

From a human-factors engineering standpoint, studying human-robot trust is only useful insofar as it can be quantitatively modeled and applied to improving human-robot interactions. However, there seems to be a dearth of review articles which cover quantitative and especially *computational* trust models. The latter is especially important if we wish to design robots which can infer how much trust their human counterparts place in them. Those reviews that do exist, like French, Deunser, and Heathcote (2018) and Basu and Singhal (2016), while extremely valuable, seem to lack the focus and depth necessary to provide readers a sense of what applied quantitative models exist and especially *how they work*.

The goals of this review, therefore, are twofold. First, we aim to provide a sense of the broader landscape of existing computational trust models which have been applied to human-robot interaction by conducting a systematic, qualitative review. Second, we aim to provide sufficient detail about these models to give researchers an idea of their mathematical forms and how they work. We hope this work can serve as an entry point into a number of different facets of the field, and that every reader will be able to find something of value.

Background

Definition of trust

Mayer, Davis, and Schoorman (1995) is often cited as a starting point for trust in general: "the willingness of a party to be vulnerable to the actions of another party based on the expectation that the other will perform a particular action important to the trustor, irrespective of the ability to monitor or control that other party." Lee and See (2004) gave a well-known definition used in the human-automation interaction community: "the attitude that an agent will help achieve an individual's goals in a situation characterized by uncertainty and vulnerability." Other definitions include "a belief, held by the trustor, that the trustee will act in a manner that mitigates the trustor's risk in a situation in which the trustor has put its outcomes at risk" (Wagner, Robinette, & Howard, 2018), and "the willingness of the human to rely on or to believe in the cooperation provided by the robot" (Hoff & Bashir, 2015; Rahman & Wang, 2018).

Dekker and Hollnagel (2004) criticize the use of concepts like trust, arguing that it is a "folk model" which serves no useful purpose beyond the behavioral measures we use to operationalize it. In other words, why should we measure trust when we could just measure reliance and compliance behaviors. Parasuraman, Sheridan, and Wickens (2008) counter this point of view, arguing that trust is a viable construct which is

"valuable in understanding and predicting human-system performance in complex systems." Some of the papers we found do indeed define trust in a way which makes it superfluous, such as Ranasinghe et al. (2016), so we did not include them. We hope to show, however, that trust can be modeled and used in ways that go beyond mere nomenclature and actually add explanatory and predictive value.

Qualitative models of trust

Trust is often modeled conceptually as "a belief about probability of reciprocation based initially on implicit judgments, and then dynamically updated based on experiences" (Chang, Doll, van 't Wout, Frank, & Sanfey, 2010). This dynamic process is nested within a larger closed-loop system by which trust influences reliance. The outcomes of that reliance (i.e., our experiences) are fed back into our beliefs about a system's capability, and the cycle continues (Lee & See, 2004). Perhaps the most comprehensive and most used qualitative model is that of Hoff and Bashir (2015) which is based on an extensive literature review of trust in automation, but which might also be applied to human-robot trust. Their model comprises three components, what they call dispositional trust (based on characteristics of the human, which remains relatively constant over time), situational trust (based on environmental factors), and learned trust (based on a person's preexisting knowledge and the automated system's performance).

Trust measurement

There is no single widely used method to quantify trust. Several validated self-report measurement instruments have been developed and have been widely cited (Jian, Bisantz, & Drury, 2000; Mayer et al., 1995; Muir, 1987; Yagoda & Gillan, 2012). These use Likert scales to represent trust as a discrete scalar quantity. Likert scales are subject to acquiescence bias, leading some researchers to use semantic differential scales instead (Tsui, Desai, & Yanco, 2010). In addition, there is neurological evidence that trust and distrust are two independent constructs (Dimoka, 2010), which further complicates matters. One novel method is the "area under the trust curve" (AUTC) measure, in which participants report trust *changes* which are integrated over time (Desai, 2012; Desai, Kaniarasu, Medvedev, Steinfeld, & Yanco, 2013). Compliance is often used as a proxy to infer trust based on behavior (Metcalfe et al., 2017). Most of the models presented here do this in one form or another, and reconciling behavioral proxies with self-report measures is an important aspect of model fitting.

Definition of robot

Much of the earlier discussion about trust has relied on sources originally written in the context of automation. The line between robots and automation in general is quite blurred. The word *robot* comes from the Czech word *robota*, which means "serf labor" or "drudgery" or simply "hard work." The IEEE website defines a robot as "an autonomous machine capable of sensing its environment, carrying out computations to make decisions, and performing actions in the real world." For the sake of this review, a robot is whatever authors of individual papers designate them to be. In casting out a net for articles about human-robot trust, we found applications in autonomous vehicles, for example, but did not go out of our way to search for papers involving autonomous vehicles specifically, only those which used the term *robot*.

Factors that influence trust in HRI

Hancock et al. (2011) broadly categorized factors that influence trust in human-robot interaction as "human-related," "robot-related," and "environmental." They found that robot-related factors, especially performance, were most important. Environmental factors had a moderate effect, and there was a dearth of studies investigating human-related factors. There is at least one other metaanalysis investigating factors that influence trust (in automation, not robots specifically) which may be of interest to the reader (Schaefer, Chen, Szalma, & Hancock, 2016).

A systematic review

This chapter aims to systematically describe and categorize studies which have developed and applied quantitative models of trust to facilitate human-robot interaction. To that end, we took an approach based on the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) (Moher, Liberati, Tetzlaff, & Altman, 2009). The inclusion criteria for articles were as follows:

Articles must include a quantitative model of trust. Trust must be operationalized based on physiological or behavioral measures, not just subjective responses. The model must be applied to control robot behavior. In other words, it must be prescriptive, not descriptive.

That means that this review will not include articles which propose new qualitative models of trust, although some of the quantitative models presented will be based on qualitative models developed in past literature. This review will also exclude quantitative models whose input parameters are based on self-report measures. Trust must be operationalized in terms of physiological or behavioral measures. Finally, we also exclude models

which have only been applied in the context of robot-robot interaction or human-human interaction.

We searched Web of Science, Engineering Village, IEEE Explore, and PubMed for papers whose topics (i.e., titles, abstracts, and keywords) contained the terms "trust" and "model" and "*robot." This produced 812 results. Removing duplicates reduced this number to 620. Abstracts were screened and 410 articles were excluded based on the above criteria. The full texts of the remaining 210 articles were assessed for eligibility and a further 172 were rejected, leaving 38. This process is illustrated diagrammatically in Fig. 1.

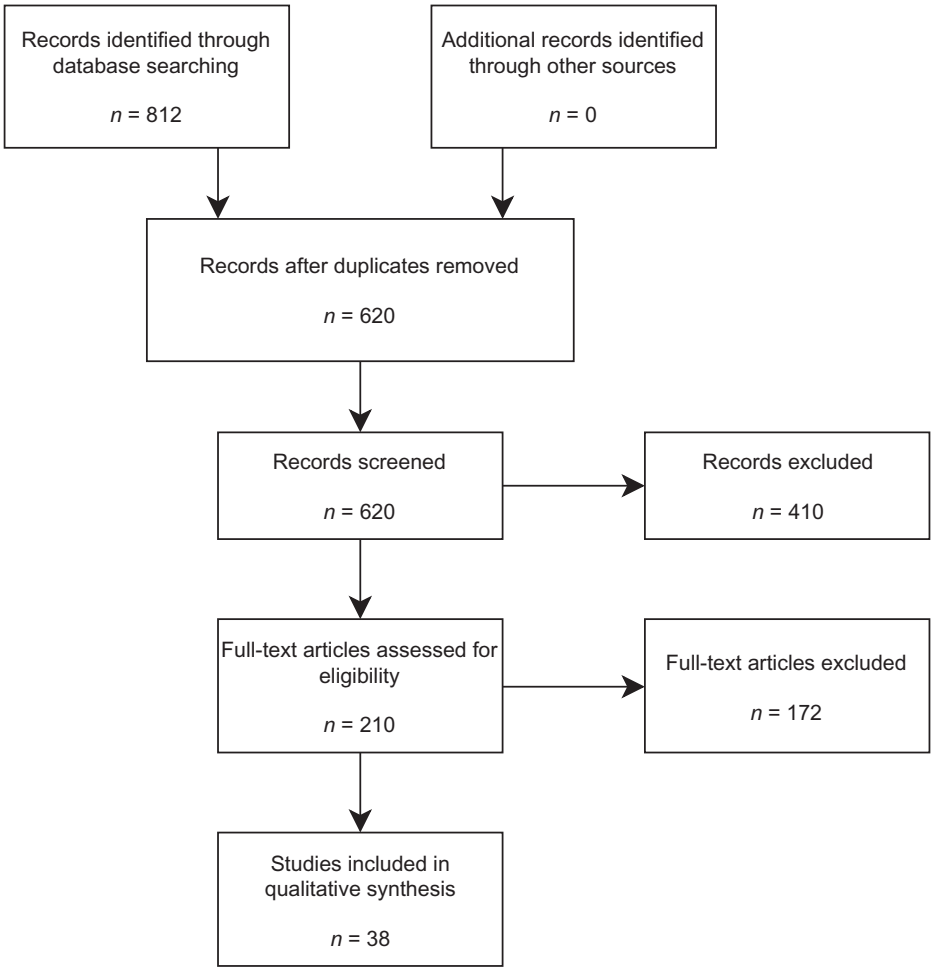


FIG. 1 Flow diagram of the PRISMA process.

The models we identified can broadly be grouped into three categories based on model type: autoregressive models similar to the autoregressive moving average vector (ARMAV) model developed by Lee and Moray (1992); dynamic Bayesian networks and partially observable Markov decision processes; and decision-analytical models.

These models were overwhelmingly applied to the problems of adaptive autonomy or function allocation, although some were used to change the kinematics of robot manipulators, as in the case of collaborative physical interaction. The domains to which these models were applied include the military, surveillance, autonomous vehicles, and manufacturing.

ARMAV models

Fourteen of the 38 (37%) models identified by our search were autoregressive models directly inspired by the autoregressive moving average vector model developed by Lee and Moray (1992), meaning that the authors cited this work when describing their model.

To review, Lee and Moray (1992) was an experimental study which examined the behavior of participants operating a (simulated) semiautomated pasteurization plant. This *microworld* comprised three different subsystems, each of which could be switched between manual or automatic control whenever participants desired. Participants were tasked with balancing safety and efficiency in operating the plant and were compensated based on performance. The automatic controllers could not match the performance of manual control, but human operators lacked the intentional resources to manage all three subsystems, and thus relied on a mixture of manual and automatic control to maximize performance and safety. They were able to describe the relationship between trust and system performance with an autoregressive moving average vector model, which accounted for 79% of the variance in their data (Lee & Moray, 1992). The resulting equation for trust took the following form:

$$T(t) = \phi_1 T(t-1) + A_1 P(t) - A_1 \phi_2 P(t-1) - A_2 F(t) + A_2 \phi_3 F(t-1) + a(t) \quad (1)$$

where T is the trust, P is the system performance, F is the system fault occurrence, and a is a random noise perturbation. The values of the various parameters need to be estimated for each situation in which the model is applied.

In the models which evolved from Lee and Moray (1992), trust is a function of human performance, robot performance, fault occurrence, or some combination of these. While there are some variations in how this relationship is expressed among the various models we are about to describe, you will see that the general form of this relationship, a first-order lag model, is common to (almost) all of them. In addition, because this relationship was

developed in the context of function allocation between manual and automatic control, you will also notice that HRI models which are inspired by it also tend to apply it to the same problem.

One additional observation: most of the models in this clade have been created by a group of authors from Clemson University. We will stay true to this taxonomy as we introduce the models, so you can gain a better understanding of how they are related to each other.

Humans' trust in robots

Two of the first papers to come out of the Clemson group (Sadrfaridpour, Burke, & Wang, 2014; Sadrfaridpour, Saeidi, Burke, Madathil, & Wang, 2016) proposed the following model:

$$T(k+1) = K_T T(k) + K_R P_R(k) + K_H P_H(k) \tag{2}$$

You can see that it differs in some respects from Lee and Moray (1992) in that here trust depends on both the person's own performance and the robot's. In addition, whereas trust in Lee and Moray (1992) takes into account the present and prior time step, the earlier model only accounts for the prior time step.

Human performance in this series of models is expressed in a rather unique way. There are two components: physical and cognitive.

$$P_H(k) = K_P P_P(k) + K_C P_C(k) \tag{3}$$

Physical performance is based on a discrete-time muscle fatigue and recovery model (Fayazi, Wan, Lucich, Vahidi, & Mocko, 2013). It therefore requires maximum voluntary contraction (MVC) measurements to calibrate and is meant to estimate the amount of muscle fatigue a person is experiencing over the course of a work shift, given the energy requirements to perform some collaborative task with a robot.

Cognitive performance is based on the Yerkes-Dodson Law (Yerkes & Dodson, 1908), which roughly states that performance increases with mental arousal, but only up to a point, after which it begins to decline again. Sadrfaridpour et al. (2014) argue that as robot performance increases, the level of arousal will increase as the human attempts to keep pace with whatever collaborative task they are performing. In addition, they assume that arousal will always be on the left side of the Yerkes-Dodson curve since "the task for [a] skilled human worker is less complex." This conveniently allows cognitive performance to be approximated by a linear function of robot performance:

$$P_C(k+1) = cP_C(k) + dP_R(k) \tag{4}$$

where $c \leq 1$, $d \leq 1$, and $c + d = 1$. Whether this assumption is valid is another matter.

Robot performance is represented as follows. In a hypothetical collaborative situation in which human and robot are both performing parallel tasks, at any given point in time their progress may become out of sync as one begins to lag behind the other. They define robot performance as the difference between the theoretical maximum performance of the robot (cycle time, say) and that lag:

$$P_R(k+1) = P_{R,\max} - (V_H(k) - V_R(k)) \tag{5}$$

where V_H and V_R are the "progress rates."

These are early models and were not validated with human participants, although the dynamics of trust and performance were simulated. They are included here because they are an important step in the evolution of later models.

In Sadrfaridpour, Saeidi, and Wang (2016), we see this model applied to a real human-robot interaction scenario. The researchers created a hybrid manufacturing cell in which participants collaborated with a Baxter robot to assemble LEGOs. As intimated previously, the robot's speed could be adjusted manually so that both the human and robot's cycle times were synced. Participants wore active motion tracking gloves, which allowed the position and velocity of their hands to be measured. The independent variable was the control scheme by which the robot's speed would be adjusted. There were three: manual control, "physical HRI" in which the speed of the robot would be automatically adjusted to match the pace of the worker, or "trust-based" which involved solving an optimization problem whose objective function and constraints included the modeled trust value.

They use an identical model for trust as Sadrfaridpour et al. (2014) but define human and robot performance a little differently. It is expressed in terms of two components. The first component is the difference between the participant's working speed (the velocity component in the horizontal direction) and its "ideal value," which is determined by first measuring the working speed of an expert technician. The second component is the difference between the human hand's progress from a parts bin to the shared workspace and the robot manipulator's progress from its own bin to that same workspace. (In other words, in the "physical HRI" condition it is desired that if the participant's hand is 53% of the way from their parts bin to the shared workspace, the robot's manipulator should also be 53% of the way from its parts bin to the shared workspace there.) Robot performance is defined similarly, only expressed in terms of its own manipulator velocity. The dependent variables in this study included a subjective trust scale based on Jian et al. (2000), the NASA TLX, and mean robot manipulator velocity. Although the study tested only five participants and found that the average robot velocity was not significantly different between the manual and trust-based allocation schemes, participants reported significantly less perceived workload on the TLX

and reported more trust in the robot while using the trust-based scheme compared to the others.

Sadrfaridpour and Wang (2018) naturally extend this previous work to a more realistic collaborative task—the assembly of an automotive center console in a hybrid manufacturing cell. In this cell, the robot performs a pick-and-place task from its own parts bin to a shared space, while a human moves other components from her own parts bin to the shared space and completes the assembly. This time, in addition to adjusting its speed, the robot controller can select the trajectory from the parts bin to the shared space from a set of trajectories determined beforehand. This is important because the optimal path (from a speed standpoint) will traverse the shared workspace where a collision between the robot manipulator and the human hand is possible. An alternate, more circuitous route would take it around the shared workspace, minimizing the possibility of a collision. The idea is that a high level of human-robot trust reflects the fact that the human has an accurate mental model of the robot's operation and will not be struck by the manipulator even if it moves through the shared space. In this scenario, the controller selects both end effector speed and trajectory based on the estimated trust value. The equations for trust, human performance, and robot performance are identical to Sadrfaridpour, Saeidi, and Wang (2016). In an experiment with 20 participants, they showed that trust-based mode switching produced lower perceived workload (as measured by the TLX), higher subjective trust, and higher perceived usability, while maintaining the same assembly time, as compared to manual or purely physical control schemes.

Robots' trust in humans

Meanwhile, Rahman et al. (2016) developed and evaluated a similar model to control a robot's motion planning scheme during a tool handover. Motivated by the observation that people adjust their posture and motions during handovers with each other in order to minimize contact forces, they hypothesized that task efficiency and perceived safety of robot-human handovers could be improved, and cognitive load reduced, by adjusting a robot arm's posture and speed based on how much the robot "trusts" its human partner.

A key difference here is that they model the robot's trust in its human collaborator, rather than the human's trust in the robot. In some sense, this represents the robot's trust that the human will not hurt herself during this collaborative task. The equation for robot-to-human trust is given by

$$\begin{aligned} T_{R2H}(t) = {} & \phi_0 T_{R2H}(t-1) + \phi_1 P_H(t) - \phi_2 P_H(t-1) - \omega_1 F_H(t) \\ & + \omega_2 F_H(t-1) + q(t) \end{aligned} \tag{6}$$

where T_{R2H} is the robot's trust in the human, P_H is the human performance, and F_H is the human fault occurrence. You can see that the robot's performance is not considered at all.

The collaborative task involved assembling the center console of an automobile: the robot's job was to pick and place various subassemblies needed by the human and to hand the human a screwdriver upon request. Measurements of human performance and errors were enabled with an inertial measurement unit (IMU) attached to the person's wrist and a computer vision system to ascertain part orientation. The equation for human performance is given by

$$P_H(t) = W_1 V_{Hmn}(t) + W_2 V_{Han}(t) \tag{7}$$

where V_{Hmn} is the normalized human hand speed during part manipulation and V_{Han} is the normalized hand speed during assembly. $W_i \in [0, 1]$ and $W_1 + W_2 = 1$.

The authors reason that a low trust value implies an agitated or error-prone state in which the person is likely to be accidentally injured by the robot, whereas a high trust state implies a confident and capable partner. Therefore, in response to a low trust value, the robot was programmed to reduce the speed of the tool handover, and to hand over the tool in a more cautious posture. In an experiment comparing the trust-based controller with the nontrust-based controller, they found that cognitive workload (measured by the NASA TLX), subjective safety, and subjective trust increased, and overall task cycle time was reduced.

This experiment was followed by Rahman and Wang (2018) which built upon earlier work by the same team. They use their model to drive a feedforward and feedback subtask allocation scheme during two experiments: one involving assembling LEGOS (see Fig. 2); the other involving that same automotive center console. Here, they model both human-to-robot trust and robot-to-human trust:

$$T_{H2R}(t) = \beta_1 P_R(t) - \beta_2 P_R(t-1) - \gamma_1 F_R(t) + \gamma_2 F_R(t-1) + q(t) \tag{8}$$

$$T_{R2H}(t) = \phi_1 P_H(t) - \phi_2 P_H(t-1) - \omega_1 F_H(t) + \omega_2 F_H(t-1) + q(t) \tag{9}$$

where β, γ, ϕ, and ω are the real-valued parameters that are estimated based on data from a preexperiment training period.

The assembly task in each experiment was broken down into subtasks, some of which had to be performed in sequence, while others could be performed in parallel. To collaborate, the human and robot were each given a set of subtasks and they had to keep pace with one another in order to minimize overall assembly time. One of the primary goals of the study was to use real-time trust inference values to drive optimal subtask allocation.

As in Sadrfaridpour, Saeidi, and Wang (2016), robot and human performance are functions of manipulator and hand speed. Both robot and

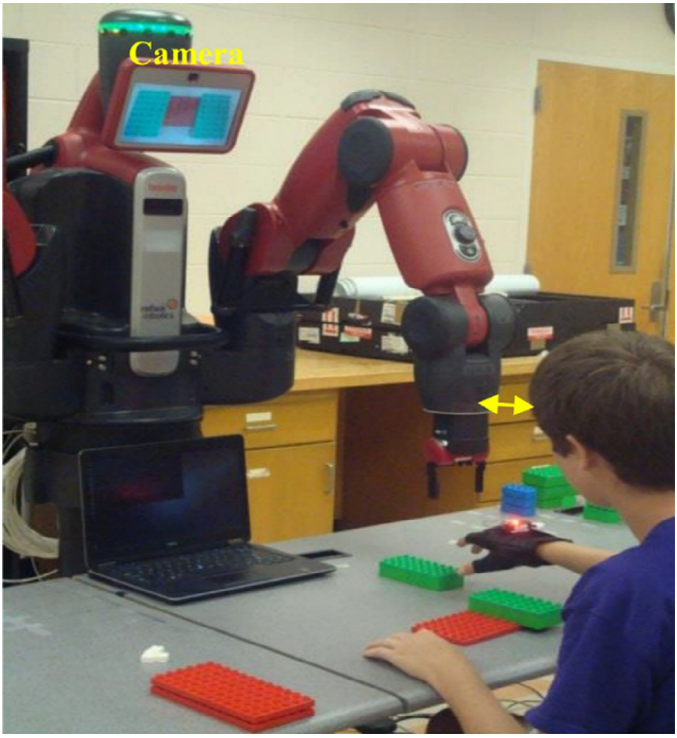


FIG. 2 The collaborative task from Rahman and Wang (2018). The human collaborates with the robot to assemble the parts in the hybrid cell. The human is equipped with the performance measurement device (a glove with LED markers of the motion capture system worn at the wrist). *Arrow* shows the minimum gap between human body and robot arm to avoid collision.

human fault scores can vary between 0 and 1 and are calculated based on the robot's ability to place a part in the correct orientation and the human's ability to assemble components in the correct orientation, as measured via the robot's vision system.

Results from the center console assembly experiment, which compared the effects of no-trust, one-way-trust, and two-way-trust subtask allocation controllers, showed that the "two-way trust [controller] resulted in higher team fluency, situation awareness, productivity, quality, utilization ratio and likeability, and lower cognitive workload than that resulted in one-way and no trust conditions" (Rahman & Wang, 2018).

Bilateral trust and self-confidence

Saeidi and Wang (2015) developed a model applied to autonomy allocation for teleoperated systems such as UAVs, UGVs, and AUVs. Their model is a little different in that they calculate the *difference* between

human-to-robot trust and self-confidence in order to provide suggestions about autonomy allocation.

$$TSC(k) = a_T P_r(k) - b_T P_h(k) \tag{10}$$

where a_T and b_T are constants related to the maximum and minimum performance levels.

As in previous studies (Wang, Shi, Zhang, & Wang, 2015; Wang, Shi, Wang, & Zhang, 2014), they argue that when an autonomous robot is neglected, its performance tends to decay asymptotically. Under teleoperation, its performance will tend to increase asymptotically. These properties are represented in their equation of robot performance.

$$P_r(k+1) = a_{pr} P_r(k) + (1 - a_{pr}) P_{r,\min} + M(k) b_{pr} P_h(k) \tag{11}$$

where $a_{pr}, M(k) \in (0,1) B_{pr} = (1 - a_{pr}) \frac{P_{r,\max} - P_{r,\min}}{P_{h,\max}}$. Human performance is based on the Yerkes Dodson law, with utilization r and task difficulty β as inputs:

$$P_H(k) = \left(P_{h,\max} - P_{h,\min}\right) \left(\frac{r(t)}{\beta}\right)^{\beta} \left(\frac{1 - r(t)}{1 - \beta}\right)^{1-\beta} + P_{h,\min} \tag{12}$$

The idea is that in light of both the Yerkes-Dodson law and the observation about robot performance decay, both total manual control and total automatic control could lead to suboptimal conditions—either poor robot performance or human overload. The level of autonomy is switched in order to keep TSC between two adjustable thresholds.

The authors applied this in a case study involving a simulated UAV or UGV, in which the task objective was to have the robot travel along a desired path. Robot performance was related to the tracking error calculated as a moving average.

Saeidi et al. (2016) utilized the same simulation testbed as Saeidi and Wang (2015) but used their trust-based controller to adjust the input gains of a bilateral haptic mixed-initiative tele-operation scheme. A mixed-initiative scheme is the simultaneous sharing of a task between a person and an onboard controller, meaning control inputs from both systems are accepted, but they are weighted differently, and this weighting can change depending on estimated trust levels. Both the human's trust in the robot and the robot's trust in the human are modeled as a function of the other's performance.

$$\dot{T}_{hr}(t) = -a_{hr} \frac{T_{hr}(t) - T_{hr,\min}}{T_{hr,\max} - T_{hr,\min}} + b_{hr} \frac{P_r(t) - P_{r,\min}}{P_{r,\max} - P_{r,\min}} \tag{13}$$

$$\dot{T}_{rh}(t) = -a_{rh} \frac{T_{rh}(t) - T_{rh,\min}}{T_{rh,\max} - T_{rh,\min}} + b_{rh} \frac{P_h(t) - P_{h,\min}}{P_{h,\max} - P_{h,\min}} \tag{14}$$

Results of an experiment with eight participants showed that a trust-based controller led to significantly reduced tracking error and workload (as measured by the TLX).

Saeidi and Wang (2019) extended Saeidi and Wang (2015) to include an experiment with 20 participants who were tasked with supervising a UAV as it tracked a UGV. They tested five control strategies and found that a trust-based controller which offered suggestions about autonomy allocation was most effective in that, for a given level of human utilization, robot performance improved. Self-reported satisfaction, trust, and compliance were highest as well. Human utilization was lower for these two strategies than for the robot performance maximization strategies, and TLX task load scores were lower.

Decision-analytical models

"Decision-analytical" is an umbrella term we are using here to describe models which are based on decision theory, game theory, economics, decision field theory, and so on. For example, in Wagner et al. (2018), trust is related to risk:

$$R(x,y) = \sum L(x,y)P(y) \tag{15}$$

where $L(x, y)$ is the loss associated with having chosen to trust (x) in the event of betrayal (y) and $P(y)$ is the probability of betrayal. Risk is thus the expected loss associated with betrayal. They use this formulation in the context of economic games to enable a robot to determine whether or not it is being trusted. If so, the robot may adjust its strategy accordingly.

Freedy, DeVisser, Weltman, and Coeyman (2007) define trust in a similar way. They call it a "relative expected loss score" (REL):

$$REL = \sum_{i=1}^{n} \frac{P_i C_i}{K} \tag{16}$$

which combines observed human task allocation behavior by the number of manual overrides, K, cost of robot failure, C, and the (known) probability of failure, P. They used their model in the context of a mixed initiative team performance assessment system to identify operators whose trust was miscalibrated in order to guide future training.

Floyd, Drinkwater, and Aha (2014, 2015) take a similar approach, estimating trust as

$$Trust_{B'} = \sum_{i-1}^{n} w_i c_i \tag{17}$$

where n is the number of discrete actions taken by an autonomous robot over a certain time period; $c_i \in \{-1, 1\}$ is the outcome of each action, with 1 assigned to successfully completed actions and -1 assigned to failed or overridden actions; and $w_i \in [0, 1]$ are weights assigned to each action. In this way, the trust associated with a particular behavioral mode, B' accumulates a score over time. This was used to select a trustworthy robot behavior from a set of possible behaviors.

Van Maanen, Wisse, Van Diggelen, and Beun (2011) model trust *calibration* explicitly as:

$$\alpha_i(t) = \tau_i^d(t) - \tau_i^p(t) \tag{18}$$

where d and p superscripts indicate "descriptive trust" and "prescriptive trust," respectively. These values are related to the reliance on and reliability of a robotic decision aid. They focused on the efficacy of visual cues to communicate the estimated degree of over- or undertrust to the operator.

Xu and Dudek (2012) similarly create a model in which a robot under supervisory control is awarded a "reputation" point if its controller executes a command that is not overridden by the operator and loses a point if it is overridden, or if it fails to generate a command at all. Trust is here defined as the utility of the robot's reputation, $r(i) \in [0, \rho_{\max}]$, which is generated much as it is in Floyd et al. (2015):

$$r(i) = r(i-1) - \Delta\rho \tag{19}$$

$$r(i) = r(i-1) + \Delta\rho \tag{20}$$

where reputation is awarded a point, $\Delta\rho$ in the event of a correct robot action, or subtracted a point in the event of a failure or override. Trust in the robot is defined as the utility of its current reputation:

$$t(i) = U(r(i)) \tag{21}$$

By choosing a utility function with a positive first derivative and a negative second derivative, you can recreate two commonly observed properties of trust, namely, that trust increases with reliability and that it is easier to lose trust than it is to gain it. For example:

$$t(i) = U(r(i)) = \frac{\log(r(i)) + 1}{\log(\rho_{\max} + 1)} \tag{22}$$

Dynamic Bayesian networks

Dynamic Bayesian networks (DBNs) are a type of Bayesian network which are meant to represent the evolution of a process over time (Spehr, 2015). In contrast to Markov networks, which are undirected,

Bayesian networks are directed. They represent the joint probability distribution of a set of random variables and comprise a directed acyclic graph and set of conditional probability tables. Originally developed for the field of medical informatics (Dagum & Galper, 1993; Dagum, Galper, & Horvitz, 1992), DBNs are generalizations of Markov chains, Hidden Markov models (and in some sense ARMAVs). Whereas Markov chains are composed of one variable per time step, and hidden Markov models contain two (a state variable and an observable variable), dynamic Bayesian networks may have an arbitrary number of variables per time step. The relationship between these variables within each time step is called the base network, and the relationship between variables in different time steps is called the transition network. Most DBNs are first-order Markov models, meaning the states at time t depend only on the states at time $t-1$; however, this need not necessarily be the case. Similarly, most DBNs are stationary, meaning the structure of the graph is identical for every time step, although this is not a requirement either.

OPTIMo

The Online Probabilistic Trust Inference Model (OPTIMo) is a dynamic Bayesian network which "treats the degree of human-robot trust t_k at each time step k as a random variable, and maintains belief distributions for these performance-centric trust measures by deducing from various factors of the [human-robot] interaction experience" (Xu & Dudek, 2015). Similar in structure to the ARMAV model developed by Lee and Moray (1994), OPTIMo estimates human-robot trust in a supervisory situation based on robot performance in the current time step p_k and previous time step p_{k-1}, as well as the trust value in the previous time step t_{k-1}. This model discretizes a human-robot interaction into a sequence of time steps, k, and defines robot performance $p_k \in [0, 1]$ as the ratio of times the robot controller failed to generate a command. $i_k \in \{0, 1\}$ represents whether or not the human intervened during time step k, and $e_k \in \{0, 1\}$ is the "extraneous cause state" which represents a human intervention not necessarily related to a change in trust. One important thing to note is that OPTIMo requires the user to relatively frequently provide subjective trust ratings to the system. The variable $c_k \in \{-1, 0, 1\}$ represents the incorporation of "trust change reports" in the style of Desai (2012). That is, the human was able to spontaneously (and when prompted) press a button indicating a trust loss, no change, or trust gain. The variable $f_k \in [0, 1]$ represents the results from a subjective trust report on a visual analog scale.

Human-robot trust in the current time step, given the trust value in the previous time step, as well as performance in the current and previous time steps, is represented as a Gaussian distribution which includes

several personalized parameters meant to reflect individual biases (i.e., a person's a priori propensity to trust).

$$\mathcal{P}(t_k, t_{k-1}, p_k, p_{k-1}) := Prob(t_k | t_{k-1}, p_k, p_{k-1}) \approx \mathcal{N}(t_k; t_{k-1} + \omega_{tb} + \omega_{tp} p_k + \omega_{td}(p_k - p_{k-1}) \tag{23}$$

The first OPTIMo paper (Xu & Dudek, 2015) included an empirical assessment to test its performance against other descriptive trust models, namely, the ARMAV model (Lee & Moray, 1992), a step-wise regression model which predicts *changes* in trust (Xu & Dudek, 2016 b), and the area under the trust-curve measure developed by Desai (2012). Its performance was as good or slightly better than these other trust models at predicting subjective trust ratings, and it has the additional advantage of functioning on shorter time scales, on the order of seconds rather than minutes.

Xu and Dudek (2016 a) applied OPTIMo to create an autonomous vehicle controller they called "Trust Aware Conservative Control" (TACtiC). This involved automated boundary tracking and included objective team performance criteria: task performance was defined as the coverage area around the designated boundary, and workload was defined as the frequency of manual interventions. TACtiC implemented a control strategy that alters the robot's behavior based on inferred trust losses. This study is important because the trust-based controller was implemented in a real-world interaction—participants were physically present inside the vehicle as it navigated a test track. OPTIMo had been pretrained in a simulation, so the parameters had already been estimated for each participant before they got in the vehicle. One interesting observation was that there was a slightly larger error in trust prediction for the real-world scenario as compared to the simulation, perhaps reflecting the fact that participants were subjected to real physical risk and therefore were more cautious. Task performance and workload were not significantly improved, although participants reported feeling safer and more comfortable.

OPTIMo-Inspired

There is a class of dynamic Bayesian networks which are directly inspired by OPTIMo but were developed by a different research group—the same research group from Clemson whose work with ARMAV models was so prolific. One of these models adds to OPTIMo by including human performance in addition to robot performance but does away with the frequent trust *change* reports, arguing that it increases the operator's cognitive load "by requiring him or her to hold in memory his or her most recent trust state" (Wang, Humphrey, Liao, & Zheng, 2018). In lieu of these reports, they query the operator for subjective trust ratings (based on a linear scale) a little more often, but not often enough

(they say) to increase cognitive load by an appreciable amount. They also integrate the ARMAV time-series model of trust into their conditional probability distribution.

This model was incorporated into a multirobot controller which switched between manual and automatic symbolic motion planning of multiple autonomous robots. The authors of this particular study call their model "hypothetical" because their Intelligence, Surveillance, and Reconnaissance (ISR) scenario was simulated in MATLAB (albeit with a human participant) but was not validated with a large number of participants.

There are at least two other developmental studies involving an OPTIMo variants from this same research group, but as they only involve simulations without human participants, we will only briefly mention them here. Mahani and Wang (2018) and Zheng, Liao, and Wang (2018) also involved task allocation and symbolic motion planning for multiple robots.

Partially observable Markov decision processes

At least two research groups have pursued modeling trust as a partially observable Markov decision process (POMDP). The first which we would like to discuss is Chen, Nikolaidis, Soh, Hsu, and Srinivasa (2018). They integrate human trust into robot decision making by modeling trust as a latent (partially observable) variable within a POMDP. The model has two connected components: trust dynamics and human decisions. The variables which compose the model are the world state x, the human action a_H, and the robot action a_R. The model describes the probability of a state transition from world state x to x' given a robot and a human action and includes a reward function $r(x, a_R, a_H, x')$ which is built to elicit the desired behaviors. The desired behavior in this case (and in human-robot interactions in general) is to get the robot to do the work. To give a concrete example, the application for which this model was built is a collaborative table clearing task, as in bussing a table at a restaurant. A trust seeking robot may clear plastic bottles (low-risk objects) first before going for the wine glasses. If the robot went for the wine glass first without considering trust, the human may intervene, which would decrease team task performance. Thus a successful robot action (picking up a wine glass and placing it carefully in the bus bin), an unsuccessful robot action (dropping and shattering the wine glass), and a human intervention (which presumably is a function of trust) are given different rewards and penalties in order that the system will naturally evolve toward a state in which the robot does the dirty work.

Calculating an optimal robot policy, however, depends on knowing the human policy. Generally this is impossible for the robot to know. You can

estimate it, but this may depend on the entire history of human actions, which for long-term interactions can make the computation intractable. The key idea in this chapter is that "trust is a compact approximation of the interaction history h_t." You do not need to know the whole human interaction history to infer the human policy, you just need to estimate the human's trust in the robot.

$$\pi^H(a_t^H|x_t,a_t^R,\theta_t) = \pi^H(a_t^H|x_t,a_t^R,h_t) \tag{24}$$

Trust is here again a function of robot performance. Robot performance in turn depends on the current world state, the previous world state, human actions, and robot actions.

$$e_{t+1} = performance(x_{t+1},x_t,a_t^R,a_t^H) \tag{25}$$

The result is a formulation which should look familiar,

$$\theta_{t+1} \sim p(\theta_{t+1}|\theta_t,e_{t+1}) \tag{26}$$

where θ is the human's trust in the robot. In other words, trust depends on the previous trust state and current robot performance.

Their trust-POMDP was trained using approximately 200 participants who watched an interactive video on Amazon's Mechanical Turk and was followed up with a real robot interaction experiment involving about 20 participants. The study contrasted two different assumptions of human behavior. The first was "trust-free" or "myopic"—a human decides on an action based on the relative expected utilities of those actions, akin to Wagner et al. (2018). Importantly, the human's perception of the robot's reliability remains constant. The other policy is "trust-based"—similar to before, except the human's trust in the robot's ability is allowed to change over time. Thus the robot can adapt a policy to manipulate that trust in order to maximize team performance over the long term, for example, by performing low-risk actions first, as in the case of the plastic bottles and wine glasses.

The study found that team performance (measured by the accumulated rewards from the reward function) was significantly higher for the trust-based controller as opposed to the myopic controller.

Akash, Polson, Reid, and Jain (2019) developed another POMDP model which incorporated human workload and machine transparency. The context of the model is a reconnaissance mission where people are aided by a (virtual) robotic assistant in completing a series of missions. Their goal was to design a system that will respond to changes in human trust and workload in real time with the goal of maximizing performance. Their model uniquely captures the dynamic relationship between transparency, trust, and workload in the context of artificially intelligent decision aids.

In contrast to most other models, all parameters in the model are discrete, and most are binary. For example, the finite set of states in their

model, trust and workload, can only take the values "low" or "high." Other parameters of their model include the finite set of actions (recommendation [hazard present, hazard absent], experience [faulty, reliable], and transparency [low, medium, high]). The observables are compliance (agree, disagree) and response time (fast, medium, slow).

The model was trained with approximately 80 participants recruited via Amazon's Mechanical Turk. The experimental task involved searching a building for physical and chemical hazards (i.e., gunmen and bombs). Before entering the building, a robot searched the building and told the participant whether or not it recommended safety gear. Participants could search the building slightly faster without safety gear but would take a severe time penalty if a hazard was encountered without the gear.

The model was applied in a follow-up study (Akash, Reid, & Jain, 2019) in which a trust-based controller was developed and two control strategies were compared: one which sought to maximize trust and minimize workload and another which did the same and incorporated a context-specific measures of team performance, including total mission time, number of correct decisions, and number of "injuries." They found that the first control strategy did not increase human compliance but did decrease response time, while the second control strategy significantly reduced type-II errors made by the participants and reduced total mission time. Therefore, at least in this context, maximizing trust by itself was not sufficient to increase performance.

Other inference-based models

There is at least one class of models based on inferential statistics that does not fit neatly into the other categories we have laid out—the so-called "RoboTrust" model, and its successor, "NeuroTrust."

RoboTrust and NeuroTrust

Mikulski, Lewis, Gu, and Hudas (2012) introduced a model called RoboTrust, which was designed to help solve the "consensus problem." In the consensus problem, each agent in a system must reach an agreement with all other agents about a particular quantity of interest. RoboTrust can be used to create a "distributed discrete-time, trust-based consensus protocol." In a network of agents sending information to other agents, there is a "trust" value assigned to the edges between nodes. How these trust weights are generated is the focus of RoboTrust.

One feature of this model that distinguishes it from other models that have been applied in similar contexts is the absence of a "discount factor."

The discount factor represents how much an agent values past observations relative to the present. A low discount factor means past behavior is forgotten quickly; a large discount factor means it is remembered for a long time. It also means that trust is gained at the same rate it is lost, when intuitively we think trust is hard to earn but easy to lose. RoboTrust does away with the discount factor, allowing trust cultivation to change at different rates.

The derivation of the model is too lengthy to include here in its entirety, but we will attempt to provide a general sense of how it works. It begins with an "acceptance function." In the most general terms, the purpose of this is to interpret an observation of robot performance and decide whether or not it is acceptable. It means taking context-specific data and creating a "feature space"—a space of all possible observed values—then deciding which region of that space is acceptable. The output of the acceptance function is a binary variable.

Next, consider a sequence of such outputs, that is, a sequence of observations of robot performance, each one of which has been categorized as acceptable or unacceptable. This is called the "acceptance observation history" of the agent. It is a binary string where an observation of acceptable performance corresponds to 1 and an observation of unacceptable observation to 0.

$$z_{ij} = \{z_{ij}(0), z_{ij}(1), \ldots, z_{ij}(k_{ij})\} \tag{27}$$

Now, consider the subset comprising the c most recent observations:

$$z_{ij}^{c} = \{z_{ij}(k_{ij} - c), z_{ij}(k_{ij} - c + 1), \ldots, z_{ij}(k_{ij} - 1), z_{ij}(k_{ij})\} \tag{28}$$

For example, if $z_{ij} = \{1, 0, 1, 1, 1, 0, 1\}$, then $z_{ij}^{3} = \{1,0,1\}$. If you think of these as random variables originating from a binomial distribution, you can estimate the probability of observing η acceptable actions given k total observations. Do this for all observation sequences up to and including the c most recent ones of the acceptance observation history, and you can define trust as follows:

$$t_{ij} = \min \begin{bmatrix} \dfrac{\sum z_{ij}^{\tau}}{\tau + 1} \\ \dfrac{\sum z_{ij}^{\tau+1}}{\tau + 2} \\ \vdots \\ \dfrac{\sum z_{ij}^{c}}{c + 1} \end{bmatrix} \tag{29}$$

The two parameters ($\tau \in \mathbb{N}$) and ($c \in \mathbb{N}$) are called the tolerance and confirmation parameters, respectively, where $0 \leq \tau \leq c \leq k_{ij}$. Trust therefore is

analogous to perceived robot reliability, specifically the minimum perceived reliability within the bounds set by the two parameters. The confirmation parameter "describes the number of consecutive favorable observations necessary before the next consecutive favorable observation results in complete trust ($t_{ij} = 1$)." The tolerance parameter "describes the number of unfavorable observations necessary before the next consecutive unfavorable observation results in no trust ($t_{ij} = 0$)" (Mikulski et al., 2012). According to the authors, the primary advantage of RoboTrust is its simplicity. It requires an acceptance observation history (as do all trust-based models, in principle) but requires only two additional parameters, each of which is easy to understand and tune, making the algorithm accessible to their targeted user group: soldiers.

RoboTrust has primarily been applied to *simulated* military convoys, but since these studies ultimately culminated in an experimental study, we have included them here. In the scenario explored by Mikulski (2014), each vehicle in a (simulated) multivehicle convoy was equipped with a forward-facing sensor which could determine the distance to the preceding vehicle in the convoy, and all vehicles could transmit their position and velocity to all other vehicles. These were autonomous vehicles, and the control problem in question was how much throttle and steering to apply in order to match the speed of their local leader at a certain minimum following distance as the convoy navigated from way point to waypoint stopping at each one along the way. The goal of the trust-based controller was to maximize convoy string stability, in other words, to ensure that spacing errors between the vehicles do not amplify and propagate down the convoy as a result of velocity perturbations of the leader. Depending on each vehicle's trust in its local leader and its followers, the controller would switch between different priorities: following a trusted leader, leading a trusted follower, or independent control. The acceptance function returned true when a vehicle's local leader was closer to the next way point in the current time step compared to the previous time step and returned false otherwise. The results of this particular case study showed that trust-based control did indeed improve convoy string stability over the baseline case.

This scenario was further explored in Mikulski and Karlsen (2015), which allowed the confirmation and tolerance parameters to change on an individual basis, "allowing good agents to benefit from faster trust growth and slower trust decay and bad agents to be penalized with slower trust growth and faster trust decay." Karlsen and Mikulski (2016) introduced an adjustable non-linear relationship between trust level and following distance. Finally, Saeidi, Mikulski, and Wang (2017) conducted an experimental study which applied RoboTrust in a supervisory scenario, in which human-to-robot trust was computed and used to automatically select which vehicle would serve as leader. In addition, haptic

feedback of the steering device was changed depending on the inferred trust level, with the goal of reducing physical workload of the operator. They found that both task completion time and formation error were reduced by implementing this trust-based controller.

I would like to briefly mention an extension of RoboTrust called NeuroTrust (Balas, Karlsen, Muench, Mikulski, & Al-Holou, 2017) which uses a multilayered neural network in an attempt to optimize the previous model. Results of a preliminary simulation were mixed.

Summary

All in all, we found 14 papers which culminated in an evaluative study with human participants. The plurality of these were applied to collaborative assembly or manipulation (Chen et al., 2018; Rahman & Wang, 2018; Rahman, Sadrfaridpour, & Wang, 2015; Rahman et al., 2016; Sadrfaridpour, Saeidi, & Wang, 2016; Sadrfaridpour & Wang, 2018). Other applications include unmanned-vehicle-based military decision support (Akash, Reid, & Jain, 2019; Freedy et al., 2007; Van Maanen et al., 2011), teleoperation (Saeidi et al., 2016, 2017; Saeidi & Wang, 2019; Xu & Dudek, 2012), and self-driving cars (Xu & Dudek, 2016 a). Seven of these (50%) utilized ARMAV models, three (21%) used DBNs or POMDPs, another three used decision-analytical models, and one used the novel RoboTrust model. Most of these approaches were able to increase subjective trust while reducing workload, increasing performance, or both. However, at least one study found that different trust-based control strategies can affect individual aspects of human performance in different ways (e.g., compliance, response time, and beta errors), writing "[W]hen designing intelligent systems to affect human trust-workload behavior, overall improvement in the collaborative performance should be considered in addition to objectives related to increased trust and decreased workload" (Akash, Reid, & Jain, 2019).

There are several limitations in the nascent field of computational trust modeling which present opportunities for future research. In terms of Hoff and Bashir's model of trust, the models presented here focus only on dynamic learned trust. That makes sense, because the system can only obtain information during interactions, but if models could take into account dispositional and situational trust, perhaps model training times would not have to be so long. Even within the context of dynamic learned trust, most models do not take into account "situational factors not related to trust." An exception would be those few models which try to take workload into account by applying the Yerkes-Dodson law. Regarding factors that affect learned trust in the Hoff and Bashir model, only reliability is taken into account with these models. Timing of errors, difficulty of errors,

types of errors, etc. are ignored. The domains in which these models have been applied are extremely limited. Most of them focus on function allocation and adaptive automation in the context of manufacturing or the military. There is a tremendous opportunity to extend these computational models to other domains, especially social robotics. In addition, there is ample opportunity to incorporate physiological correlates of trust, stress, and workload into these models, according to principles of neuroergonomics.

References

Akash, K., Polson, K., Reid, T., & Jain, N. (2019). Improving human-machine collaboration through transparency-based feedback—Part I: Human trust and workload model. *IFAC-PapersOnLine*, *51*(34), 315–321. https://doi.org/10.1016/j.ifacol.2019.01.028.

Akash, K., Reid, T., & Jain, N. (2019). Improving human-machine collaboration through transparency-based feedback—Part II: Control design and synthesis. *IFAC-PapersOnLine*, *51*(34), 322–328. https://doi.org/10.1016/j.ifacol.2019.01.026.

Balas, C., Karlsen, R., Muench, P., Mikulski, D., & Al-Holou, N. (2017). Deep-learning trust estimation in multi-agent systems. R. E. Karlsen, D. W. Gage, C. M. Shoemaker, & H. G. Nguyen (Eds.), *Unmanned systems technology XIX, USA: Vol. 10195*, (p. 1019510). 1019510. https://doi.org/10.1117/12.2264152. (Eds.).

Basu, C., & Singhal, M. (2016). Trust dynamics in human autonomous vehicle interaction: A review of trust models. *AAAI spring symposium—Technical report: Vol. SS-16-01*, (pp. 85–91).

Breazeal, C., Harris, P. L., Desteno, D., Kory Westlund, J. M., Dickens, L., & Jeong, S. (2016). Young children treat robots as informants. *Topics in Cognitive Science*, *8*(2), 481–491. https://doi.org/10.1111/tops.12192.

Cavallo, F., Limosani, R., Manzi, A., Bonaccorsi, M., Esposito, R., Di Rocco, M., … Dario, P. (2014). Development of a socially believable multi-robot solution from town to home. *Cognitive Computation*, *6*(4), 954–967. https://doi.org/10.1007/s12559-014-9290-z.

Chang, L. J., Doll, B. B., van 't Wout, M., Frank, M. J., & Sanfey, A. G. (2010). Seeing is believing: Trustworthiness as a dynamic belief. *Cognitive Psychology*, *61*(2), 87–105. https://doi.org/10.1016/j.cogpsych.2010.03.001.

Chen, M., Nikolaidis, S., Soh, H., Hsu, D., & Srinivasa, S. (2018). Planning with trust for human-robot collaboration. *ACM/IEEE international conference on human-robot interaction, USA*, pp. 307–315. https://doi.org/10.1145/3171221.3171264.

Choi, J. K., & Ji, Y. G. (2015). Investigating the importance of trust on adopting an autonomous vehicle. *International Journal of Human-Computer Interaction*, *31*(10), 692–702. https://doi.org/10.1080/10447318.2015.1070549.

Dagum, P., & Galper, A. (1993). Forecasting sleep apnea with dynamic network models. In *Uncertainty in artificial intelligence* (pp. 64–71). Morgan Kaufmann.

Dagum, P., Galper, A., & Horvitz, E. (1992). Dynamic network models for forecasting. *Proceedings of the eighth conference on uncertainty in artificial intelligen/ce*, pp. 41–48.

Dekker, S., & Hollnagel, E. (2004). Human factors and folk models. *Cognition, Technology & Work*, *6*(2), 79–86. https://doi.org/10.1007/s10111-003-0136-9.

Desai, M. (2012). *Modeling trust to improve human-robot interaction. ProQuest dissertations and theses: Vol. 3537137*, (p. 257). 257. (2012). https://login.ezproxy.net.ucf.edu/login?qurl=http%3a%2f%2fsearch.proquest.com%2fdocview%2f1321459410%3faccountid%3d10003%255Cnhttp%3a%2f%2fsfx.fcla.edu%2fucf%3furl_ver%3dZ39.88-2004%26amp

%3brft_val_fmt%3dinfo%3aofi%2ffmt%3akev%3amtx%3adissertation%26amp%3bgenre%3ddissertations%2b%26amp%3b%2btheses%26amp%3bsid%3dProQ%3aProQuest%2bDissertations%2b%26amp%3b%2b.

Desai, M., Kaniarasu, P., Medvedev, M., Steinfeld, A., & Yanco, H. (2013). Impact of robot failures and feedback on real-time trust. *ACM/IEEE international conference on human-robot interaction*, pp. 251–258. https://doi.org/10.1109/HRI.2013.6483596.

Dimoka, A. (2010). What does the brain tell us about trust and distrust? Evidence from a functional neuroimaging study. *MIS Quarterly: Management Information Systems*, *34*(Spec. Issue 2), 373–396. https://doi.org/10.2307/20721433.

Djuric, A. M., Rickli, J. L., & Urbanic, R. J. (2016). A framework for collaborative robot (CoBot) integration in advanced manufacturing systems. *SAE International Journal of Materials and Manufacturing*, *9*(2), 457–464. https://doi.org/10.4271/2016-01-0337.

El Makrini, I., Elprama, S. A., Van Den Bergh, J., Vanderborght, B., Knevels, A. J., Jewell, C. I. C., … Jacobs, A. (2018). Working with walt: How a cobot was developed and inserted on an auto assembly line. *IEEE Robotics and Automation Magazine*, *25*(2), 51–58. https://doi.org/10.1109/MRA.2018.2815947.

Fagnant, D. J., & Kockelman, K. (2015). Preparing a nation for autonomous vehicles: Opportunities, barriers and policy recommendations. *Transportation Research Part A: Policy and Practice*, *77*, 167–181. https://doi.org/10.1016/j.tra.2015.04.003.

Fayazi, S. A., Wan, N., Lucich, S., Vahidi, A., & Mocko, G. (2013). Optimal pacing in a cycling time-trial considering cyclist's fatigue dynamics. *Proceedings of the American Control Conference*, 6442–6447. https://doi.org/10.1109/acc.2013.6580849.

Floyd, M. W., Drinkwater, M., & Aha, D. W. (2014). How much do you trust me? Learning a case-based model of inverse trust. *Lecture notes in computer science (including subseries lecture notes in artificial intelligence and lecture notes in bioinformatics): Vol. 8765*, (pp. 125–139). Ireland: Cork. https://doi.org/10.1007/978-3-319-11209-1_10.

Floyd, M. W., Drinkwater, M., & Aha, D. W. (2015). Improving trust-guided behavior adaptation using operator feedback. *Lecture notes in computer science (including subseries lecture notes in artificial intelligence and lecture notes in bioinformatics): Vol. 9343*, (pp. 134–148). Germany: Frankfurt am Main. https://doi.org/10.1007/978-3-319-24586-7_10.

Freedy, A., DeVisser, E., Weltman, G., & Coeyman, N. (2007). Measurement of trust in human-robot collaboration. *Proceedings of the 2007 international symposium on collaborative technologies and systems, CTS*IEEE, pp. 106–114. https://doi.org/10.1109/CTS.2007.4621745.

French, B., Deunser, A., & Heathcote, A. (2018). *Trust in automation: A literature review*. (2018). http://www.informaworld.com/10.1080/00140139408964957.

Graf, B., Hans, M., & Schraft, R. D. (2004). Care-O-bot II—Development of a next generation robotic home assistant. *Autonomous Robots*, *16*(2), 193–205. https://doi.org/10.1023/B:AURO.0000016865.35796.e9.

Hancock, P. A. (2015). Automobility: The coming use of fully-automated on-road vehicles. *2015 IEEE international multi-disciplinary conference on cognitive methods in situation awareness and decision, CogSIMA 2015,* IEEE, pp. 137–139. https://doi.org/10.1109/COGSIMA.2015.7108188.

Hancock, P. A., Billings, D. R., Schaefer, K. E., Chen, J. Y. C., De Visser, E. J., & Parasuraman, R. (2011). A meta-analysis of factors affecting trust in human-robot interaction. *Human Factors*, *53*(5), 517–527. https://doi.org/10.1177/0018720811417254.

Hoff, K. A., & Bashir, M. (2015). Trust in automation: Integrating empirical evidence on factors that influence trust. *Human Factors*, *57*(3), 407–434. https://doi.org/10.1177/0018720814547570.

Huijnen, C. A. G. J., Lexis, M. A. S., Jansens, R., & de Witte, L. P. (2017). How to implement robots in interventions for children with autism? A co-creation study involving people

with autism, parents and professionals. *Journal of Autism and Developmental Disorders*, *47* (10), 3079–3096. https://doi.org/10.1007/s10803-017-3235-9.

Jian, J.-Y., Bisantz, A., & Drury, C. (2000). International Journal of Cognitive Foundations for an empirically determined scale of trust in automated systems foundations for an empirically determined scale of trust in automated systems. *International Journal of Cognitive Ergonomics*, *4*(1), 53–71. https://doi.org/10.1207/S15327566IJCE0401.

Kahn, P. H., Kanda, T., Ishiguro, H., Freier, N. G., Severson, R. L., Gill, B. T., … Shen, S. (2012). "Robovie, you'll have to go into the closet now": Children's social and moral relationships with a humanoid robot. *Developmental Psychology*, *48*(2), 303–314. https://doi.org/10.1037/a0027033.

Karlsen, R. E., & Mikulski, D. G. (2016). Incremental learning in trust-based vehicle control. *Unmanned systems technology XVIII, USA: Vol. 9837*, (p. 983704). 983704. https://doi.org/10.1117/12.2223168.

Lawless, W., & Sofge, D. (2017). Evaluations: Autonomy and artificial intelligence: A threat or savior? In W. F. Lawless (Ed.), *Autonomy and artificial intelligence: A threat or savior?* (pp. 295–312). Springer International Publishing. https://doi.org/10.1007/978-3-319-59719-5. (Ed.).

Lee, J., & Moray, N. (1992). Trust, control strategies and allocation of function in human-machine systems. *Ergonomics*, *35*(10), 1243–1270. https://doi.org/10.1080/00140139208967392.

Lee, J., & See, K. (2004). Trust in automation: Designing for appropriate reliance. *Human Factors*, *46*(1), 50–80. https://doi.org/10.1518/hfes.46.1.50_30392.

Lee, J. D., & Moray, N. (1994). Trust, self-confidence, and operators' adaptation to automation. *International Journal of Human—Computer Studies*, *40*(1), 153–184. https://doi.org/10.1006/ijhc.1994.1007.

Lee, J. G., Kim, K. J., Lee, S., & Shin, D. H. (2015). Can autonomous vehicles be safe and trustworthy? Effects of appearance and autonomy of unmanned driving systems. *International Journal of Human-Computer Interaction*, *31*(10), 682–691. https://doi.org/10.1080/10447318.2015.1070547.

Mahani, M. F., & Wang, Y. (2018). Trust-based runtime verification for multi-quad-rotor motion planning with a human-in-the-loop. *ASME 2018 dynamic systems and control conference, DSCC 2018: Vol. 3*, https://doi.org/10.1115/DSCC2018-9174.

Maurtua, I., Ibarguren, A., Kildal, J., Susperregi, L., & Sierra, B. (2017). Human-robot collaboration in industrial applications: Safety, interaction and trust. *International Journal of Advanced Robotic Systems*, *14*(4), 1–10. https://doi.org/10.1177/1729881417716010.

Mayer, R. C., Davis, J. H., & Schoorman, F. D. (1995). An integrative model of organizational trust. *Academy of Management Review*, *20*(3), 709–734. https://doi.org/10.5465/amr.1995.9508080335.

Metcalfe, J. S., Marathe, A. R., Haynes, B., Paul, V. J., Gremillion, G. M., Drnec, K., … Nothwang, W. D. (2017). Building a framework to manage trust in automation. *Micro- and Nanotechnology Sensors, Systems, and Applications IX, 10194*(May 2017), 101941U. https://doi.org/10.1117/12.2264245.

Mikulski, D. G. (2014). Trust-based controller for convoy string stability. *IEEE SSCI 2014: 2014 IEEE symposium series on computational intelligence—CIVTS 2014: 2014 IEEE symposium on computational intelligence in vehicles and transportation systems, proceedings, December*IEEE, pp. 69–75. https://doi.org/10.1109/CIVTS.2014.7009480.

Mikulski, D. G., & Karlsen, R. E. (2015). Trust-based learning and behaviors for convoy obstacle avoidance. *Unmanned Systems Technology XVII: Vol. 9468*, (p. 94680L). 94680L. https://doi.org/10.1117/12.2179905.

Mikulski, D. G., Lewis, F. L., Gu, E. Y., & Hudas, G. R. (2012). Trust method for multi-agent consensus. R. E. Karlsen, D. W. Gage, C. M. Shoemaker, & G. R. Gerhart (Eds.), *Unmanned systems technology XIV* (p. 83870E)Vol. 8387, (p. 83870E). https://doi.org/10.1117/12.918927. (Eds.).

Moher, D., Liberati, A., Tetzlaff, J., & Altman, D. G. (2009). Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *Journal of clinical epidemiology*, *62*(10), 1006–1012. https://doi.org/10.1016/j.jclinepi.2009.06.005.

Muir, B. M. (1987). Trust between humans and machines. *International Journal of Man-Machine Studies*, *27*, 327–339.

Parasuraman, R., & Riley, V. (1997). Humans and automation: Use, misuse, disuse, abuse. *Human Factors*, *39*(2), 230–253.

Parasuraman, R., Sheridan, T. B., & Wickens, C. D. (2008). Situation awareness, mental workload, and trust in automation: Viable, empirically supported cognitive engineering constructs. *Journal of Cognitive Engineering and Decision Making*, *2*(2), 140–160. https://doi.org/10.1518/155534308X284417.

Rahman, S. M., & Wang, Y. (2018). Mutual trust-based subtask allocation for humanrobot collaboration in flexible lightweight assembly in manufacturing. *Mechatronics*, *54*, 94–109. https://doi.org/10.1016/j.mechatronics.2018.07.007.

Rahman, S. M. M., Sadrfaridpour, B., & Wang, Y. (2015). Trust-based optimal subtask allocation and model predictive control for human-robot collaborative assembly in manufacturing. *ASME 2015 dynamic systems and control conference, DSCC 2015: Vol. 2*, https://doi.org/10.1115/DSCC2015-9850.

Rahman, S. M. M., Wang, Y., Walker, I. D., Mears, L., Pak, R., & Remy, S. (2016). Trust-based compliant robot-human handovers of payloads in collaborative assembly in flexible manufacturing. *IEEE international conference on automation science and engineering: Vol. 2016*(pp. 355–360). IEEE. https://doi.org/10.1109/COASE.2016.7743428.

Ranasinghe, A., Sornkarn, N., Dasgupta, P., Althoefer, K., Penders, J., & Nanayakkara, T. (2016). Salient featre of haptic-based guidance of people in low visibility environments using hard reins. *IEEE Transactions on Cybernetics*, *46*(2), 568–579. https://doi.org/10.1109/TCYB.2015.2409772.

Sadrfaridpour, B., Burke, J., & Wang, Y. (2014). *Human and robot collaborative assembly manufacturing: Trust dynamics and control. Proceedings of the workshop on human-robot collaboration industrial manufacturing RSS*, pp. 1–6. (2014). http://hci.cs.wisc.edu/workshops/RSS2014/wp-content/uploads/2013/12/sadrfaridpour2014human.pdf.

Sadrfaridpour, B., Saeidi, H., Burke, J., Madathil, K., & Wang, Y. (2016). Modeling and control of trust in human-robot collaborative manufacturing. *Robust intelligence and trust in autonomous systems*Boston, MA: Springer US, pp. 115–142. https://doi.org/10.1007/978-1-4899-7668-0_7.

Sadrfaridpour, B., Saeidi, H., & Wang, Y. (2016). An integrated framework for human-robot collaborative assembly in hybrid manufacturing cells. *IEEE international conference on automation science and engineering: Vol. 2016*, (pp. 462–467), https://doi.org/10.1109/COASE.2016.7743441.

Sadrfaridpour, B., & Wang, Y. (2018). Collaborative assembly in hybrid manufacturing cells: An integrated framework for human-robot interaction. *IEEE Transactions on Automation Science and Engineering*, *15*(3), 1178–1192. https://doi.org/10.1109/TASE.2017.2748386.

Saeidi, H., McLane, F., Sadrfaidpour, B., Sand, E., Fu, S., Rodriguez, J., & Wang, Y. (2016). Trust-based mixed-initiative teleoperation of mobile robots. *Proceedings of the American control conference, July: Vol. 2016*, (pp. 6177–6182). IEEE. https://doi.org/10.1109/ACC.2016.7526640.

Saeidi, H., Mikulski, D. G., & Wang, Y. (2017). Trust-based leader selection for bilateral haptic teleoperation of multi-robot systems. *IEEE international conference on intelligent robots and systems: Vol. 2017*, (pp. 6575–6581). IEEE. https://doi.org/10.1109/IROS.2017.8206569.

Saeidi, H., & Wang, Y. (2015). Trust and self-confidence based autonomy allocation for robotic systems. *Proceedings of the IEEE conference on decision and control: Vol. 54rd IEEE*, (pp. 6052–6057). https://doi.org/10.1109/CDC.2015.7403171.

Saeidi, H., & Wang, Y. (2019). Incorporating trust and self-confidence analysis in the guidance and control of (semi) autonomous mobile robotic systems. *IEEE Robotics and Automation Letters*, *4*(2), 239–246. https://doi.org/10.1109/LRA.2018.2886406.

Saleh, K., Hossny, M., & Nahavandi, S. (2017). Towards trusted autonomous vehicles from vulnerable road users perspective. *11th annual IEEE international systems conference, SysCon 2017—Proceedings, April*, pp. 1–7. https://doi.org/10.1109/SYSCON.2017.7934782.

Schaefer, K. E., Brewer, R. W., Putney, J., Mottern, E., Barghout, J., & Straub, E. R. (2016). Relinquishing manual control collaboration requires the capability to understand robot intention. *Proceedings—2016 international conference on collaboration technologies and systems, CTS 2016, October*, pp. 359–366. https://doi.org/10.1109/CTS.2016.69.

Schaefer, K. E., Chen, J. Y. C., Szalma, J. L., & Hancock, P. A. (2016). A meta-analysis of factors influencing the development of trust in automation: Implications for understanding autonomy in future systems. *Human Factors*, *58*(3), 377–400. https://doi.org/10.1177/0018720816634228.

Spehr, J. (2015). On Hierarchical Models for Visual Recognition and Learning of Objects, Scenes, and Activities. vol. 11. Springer, Cham.

Torta, E., Werner, F., Johnson, D. O., Juola, J. F., Cuijpers, R. H., Bazzani, M., … Bregman, J. (2014). Evaluation of a small socially-assistive humanoid robot in intelligent homes for the care of the elderly. *Journal of Intelligent and Robotic Systems: Theory and Applications*, *76*(1), 57–71. https://doi.org/10.1007/s10846-013-0019-0.

Tsui, K. M., Desai, M., & Yanco, H. A. (2010). Considering the bystander's perspective for indirect human-robot interaction. *5th ACM/IEEE international conference on human-robot interaction, HRI 2010*, pp. 129–130. https://doi.org/10.1145/1734454.1734506.

Van Maanen, P. P., Wisse, F., Van Diggelen, J., & Beun, R. J. (2011). Effects of reliance support on team performance by advising and adaptive autonomy. *Proceedings—2011 IEEE/WIC/ACM international conference on intelligent agent technology, IAT 2011, August: Vol. 2*, (pp. 280–287). https://doi.org/10.1109/WI-IAT.2011.117.

Vogt, P., de Haas, M., de Jong, C., Baxter, P., & Krahmer, E. (2017). Child-robot interactions for second language tutoring to preschool children. *Frontiers in Human Neuroscience*, *11* (March), 1–7. https://doi.org/10.3389/fnhum.2017.00073.

Wagner, A. R., Robinette, P., & Howard, A. (2018). Modeling the human-robot trust phenomenon: A conceptual framework based on risk. *ACM Transactions on Interactive Intelligent Systems*, *8*(4), 1–24. https://doi.org/10.1145/3152890.

Wang, X., Shi, Z., Zhang, F., & Wang, Y. (2015). Mutual trust based scheduling for (semi) autonomous multi-agent systems. *Proceedings of the American control conference: Vol. 2015*, (pp. 459–464). https://doi.org/10.1109/ACC.2015.7170778.

Wang, Y., Humphrey, L. R., Liao, Z., & Zheng, H. (2018). Trust-based multi-robot symbolic motion planning with a human-in-the-loop. *ACM Transactions on Interactive Intelligent Systems*, *8*(4), 1–33. https://doi.org/10.1145/3213013.

Wang, Y., Shi, Z., Wang, C., & Zhang, F. (2014). Human-robot mutual trust in (semi)autonomous underwater robots. *Studies in Computational Intelligence*, *554*, 115–137. https://doi.org/10.1007/978-3-642-55029-4_6.

Xu, A., & Dudek, G. (2012). Trust-driven interactive visual navigation for autonomous robots. *Proceedings—IEEE international conference on robotics and automation, May*, pp. 3922–3929. https://doi.org/10.1109/ICRA.2012.6225171.

Xu, A., & Dudek, G. (2015). OPTIMo: Online probabilistic trust inference model for asymmetric human-robot collaborations. *ACM/IEEE international conference on human-robot interaction: Vol. 2015* (pp. 221–228). ACM, pp. 221–228. https://doi.org/10.1145/2696454.2696492.

Xu, A., & Dudek, G. (2016a). Maintaining efficient collaboration with trust-seeking robots. *IEEE international conference on intelligent robots and systems: Vol. 2016*, (pp. 3312–3319). IEEE. https://doi.org/10.1109/IROS.2016.7759510.

Xu, A., & Dudek, G. (2016b). Towards modeling real-time trust in asymmetric humanrobot collaborations. *Springer tracts in advanced robotics: Vol. 114*, (pp. 113–129). https://doi.org/10.1007/978-3-319-28872-7_7. Singapore.

Yagoda, R. E., & Gillan, D. J. (2012). You want me to trust a robot? The development of a human-robot interaction trust scale. *International Journal of Social Robotics*, *4*(3), 235–248. https://doi.org/10.1007/s12369-012-0144-0.

Yerkes, R. M., & Dodson, J. D. (1908). The relation of strength of stimulus to rapidity of habit-formation. *Journal of Comparative Neurology and Psychology*, *18*, 459–482. https://doi.org/10.1037/h0073415.

Zheng, H., Liao, Z., & Wang, Y. (2018). Human-robot trust integrated task allocation and symbolic motion planning for heterogeneous multi-robot systems. *ASME 2018 dynamic systems and control conference, DSCC 2018: Vol. 3*, https://doi.org/10.1115/DSCC2018-9161.

CHAPTER

20

Neural dynamics of trust in human-robot interaction

Chang S. Nam[a], *Elizabeth Eskander*[b], *and Sanghyun Choo*[a]

[a]Edward P. Fitts Department of Industrial and Systems Engineering, North Carolina State University, Raleigh, NC, United States, [b]Department of Psychology, North Carolina State University, Raleigh, NC, United States

Introduction

As civilizations advanced and humans increasingly relied on the community for survival, people needed to effectively trust one another's capabilities and intentions. The human brain evolved neural systems specialized for processing social interactions. These systems help us anticipate the actions of others. The concept of trust involves assessing the riskiness or reliability of interacting with a specific partner. Trust can be rated as high or low depending on how willing an individual is to rely on or interact with a specific partner. Robots have increasingly begun to pervade modern society and replace human-human interaction with human-robotic interaction. It is of importance to determine how this novel form of trust is developed and what factors contribute to the regulation of trust during interaction with a robotic partner so that the tasks robots are assigned to do can be completed efficiently and with confidence from their human partner.

Modern robots interact with humans in a wide range of tasks. They can be found in toy stores for entertainment purposes, retail warehouses to increase efficiency and reduce dangerous task load for humans, hospital settings to assist in surgeries and other medical procedures, home environments to aid in cleaning and security, as well as in the military domain to reduce the number of humans put in danger on the battlefield (Gallimore, Lyons, Vo, Mahoney, & Wynne, 2019). The algorithms used

https://doi.org/10.1016/B978-0-12-819472-0.00020-4
© 2021 Elsevier Inc. All rights reserved.

by these machines can be complex and are often not fully understood by humans. This lack of understanding can cause a miscalibration in trust and leave humans and their robotic partners more vulnerable.

Till date, there is no clear definition of what a robot exactly is. Because robots come in all shapes and sizes with a vast range of functions, from nanobots being developed for healthcare uses to the M-2000iA, a large heavyweight industrial robot designed for lifting heavy materials such as cars, roboticists have not yet reached a unanimous conclusion about what should be considered a robot. The website for the IEEE (Institute of Electrical and Electronics Engineers) loosely defines a robot as "an autonomous machine capable of sensing its environment, carrying out computations to make decisions, and performing actions in the real world." These three actions—sensing, computing, and then acting—are constantly being performed by the robot, creating a feedback loop. This allows robots to regulate their behavior to complete their assigned task more effectively.

Research needs for neural dynamics of trust in HRI

Understanding the neural correlates and mechanisms that are involved in positive, high trust scenarios between human-robotic interactions compared to negative, low trust scenarios will not only help designers create better, more efficient robots to be used in interactions with humans, but will also one day assist in the machine learning process. Identifying the neural underpinnings of trust and applying them to machine learning will help robots in high-risk situations interpret immediate feedback from their user and use this data to adjust their following actions (Berberian, Somon, Sahaï, & Gouraud, 2017; Drnec, Marathe, Lukos, & Metcalfe, 2016; Fedota & Parasuraman, 2010; Somon, Campagne, Delorme, & Berberian, 2017).

According to a meta-analysis done by Hancock et al. (2009), three main factors impact trust interactions between humans and robots. The first category is human factors such as an individual's natural propensity to trust or beliefs about the use of robotics. The second is situational factors such as the type of task and who is involved in completing the task. The last factor is specifically unique to robots (compared to other automated systems or humans) and is the characteristics of the robot itself, such as anthropomorphism and performance reliability, which can both be controlled. This chapter focuses on the neural correlates and mechanisms involved in processing this last factor. We address the neural changes that are caused by different aspects of anthropomorphism such as kinematics, human-like expressions, and interaction effects. We also explore the effect of performance reliability on trust, which is separated into a decision phase that

occurs pre-interaction and a feedback phase that occurs post-interaction. Moreover, the formation of trust with an autonomous agent is not a singular, stagnant process. Subconscious and conscious cues about the agent influence the initial decision to trust, and then the user continuously updates their trust based on feedback from the robot, which, in turn, has an impact on the behavior displayed during the next interaction where trust is required to properly and confidently complete a task. Attributing human characteristics to robots can impact trust depending on the intensity of human likeness. This relationship, though, is not linear. Trustworthiness associated with human likeness increases only to a certain point. When the agent has become very realistic but not yet perfectly human-like, such as an android, trustworthiness drops producing a phenomenon known as the "uncanny valley" effect, where the human becomes unsettled by the robot it is interacting with (Ghazali, Ham, Barakova, & Markopoulos, 2018).

Though much behavioral research has been done to understand the impact of these factors, the underlying neural processes and correlates have not yet been fully explored and much research is yet to be done. Research on this topic has been almost exclusively behavioral, and there are very few studies that empirically examine the differences between the processes of trust invoked by different agents (i.e. humans vs machines vs robots) (Madhavan & Wiegmann, 2007).

Modulation of anthropomorphism

Uncanny valley

Although adding human characteristics to automation will influence the perceived trustworthiness of the robot, the relationship will not be linear. When the human-likeness of a robotic agent becomes highly realistic but is not yet completely human, the perceived trustworthiness drops (Ghazali et al., 2018), and the agent is then conceived as strange, unfamiliar, and disconcerting (Saygin, Chaminade, Ishiguro, Driver, & Frith, 2012). This phenomenon, known as the "uncanny valley" effect, has been recognized since the early twentieth century (Freud, 1919). While the uncanny valley has remained influential in robotic design and psychological studies, systematic exploration of its neural basis has received minimal investigation (Seyama & Nagayama, 2007).

In 2012, Saygin and colleagues specifically studied the neural correlates of the one that may be involved in this phenomenon, by comparing fMRI images taken from participants after watching video clips of a robot (mechanical movement and appearance), a human (biological movement and appearance), and an android (mechanical movement,

biological appearance) performing various actions. During these trials, videos of certain actions were repeated and repetition suppression was calculated as the difference between BOLD responses to a new (non-repeated) stimuli compared with the responses to the same stimulus when it was repeated. Suppression of both the human and robotic conditions yielded similar patterns; however, the researchers found a wider network of action-repetition suppression which extended to the parietal and frontal cortex. All conditions displayed suppression in the lateral temporal cortex. Agent by repetition interaction was found to have greater repetition suppression for the human and android conditions compared to the robot conditions in the left lateral temporal cortex. Analyzing the main effect of the agent yielded a stronger response for the robot condition in the visual cortex bilaterally compared to the other two agents. A region of suppression for the robot condition was also found in the left posterior lateral temporal cortex (LPLTC) to be significantly less than for the human and android condition. Though a direct conclusion regarding the specific correlates involved in the uncanny valley were not drawn, Saygin et al. (2012) were able to explore the selectivity of the human action-perception system (APS), which is activated in the temporal, frontal, and parietal areas.

A later fMRI study (Wang & Quadflieg, 2015) explored the neural correlates of feeling eerie toward social robots while comparing brain activations for both human-human interactions (HHI) and HRI. They found that the ventromedial prefrontal cortex (vmPFC) has stronger increases in activation during HRI as feelings of eerieness toward the robot increases, compared to the HHI. This was contrasted with activity increases in the perirhinal cortex (located in the medial temporal lobe or MTL) during HRI associated with increases in believability. Though vmPFC was not implicated conclusively as the correlate for this phenomenon, it can be regarded as a potential correlate considering that the region is associated with script-based reasoning compared to the left temporoparietal region, associated with situation-specific mental-attributions, found when controlled with HHI. These results can be supported by the previous neuroimaging findings. This study showed that HHI had increased brain activity in the left TPJ while HRI revealed increased brain activity in the following brain regions: dmPFC, vmPFC, medial PrC, bilateral middle occipital gyrus, and inferior temporal gyrus. Besides, they revealed that the left TPJ, PRc, and vmPFC were overlapped with brain areas for the MTN using analysis with a localizer. The brain activation findings from both HHI and HRI contribute to the MNT differently than the existing MTN enhancement. Several studies have revealed that each region for the MTN played different roles in the impression formation process (Aichhorn et al., 2008; Hartwright, Apperly, & Hansen, 2014; Satpute,

Badre, & Ochsner, 2013). Therefore, their findings can be strengthened by previous studies.

Kinematics

As previously mentioned, the human brain utilizes the APS (i.e. lateral temporal, inferior frontal/ventral premotor and anterior intraparietal cortex) as part of the neurological process involved in developing trust by trying to interpret another's behaviors or actions (Saygin et al., 2012). Understanding the actions of another agent involves processing and interpreting their movements, emotions, conversations, and touch.

The kinematics of most robots are far different from those of humans. Robots, in most settings, display mechanical motions of movement (i.e. linear, rotational, rescinding, oscillating) while humans display biological motions of movement, defined by Thompson and Parasuraman (2012) as "the movements of living organisms such as people or animals—both whole-body motion as well as partial movements by hands, head, eye." fMRI studies on the perception of body movements in primates have suggested and implicated that the most anterior section of the ventral premotor cortex (PMv) is responsive to robotic movements, but not the most posterior section (Nelissen, Luppino, Vanduffel, Rizzolatti, & Orban, 2005). A review of findings studying the psychophysical consequence of seeing robotic vs. biological motion has, however, produced ambiguous results, with some studies finding main effects, and others not (Gazzola, Rizzolatti, Wicker, & Keysers, 2007).

Transcranial magnetic stimulation (TMS) studies have shown that human movements are reflected in the motor system of the observer, suggesting that detailed temporal activations in the mirror neuron system (MNS) are sensitive to specific types of motion. Gazzola et al., (2007) applied this theory to the study of analyzing differences in activation depending on the type of kinematics (robotic vs. human). The core MNS which was found for both human and robotic motion show activations in the bilateral medial temporal gyrus (MTG), bilateral parietal clusters extending from the SPL to SII, passing through the IPL (SMG), SI, and frontal clusters. The bilateral dorsal frontal activation includes BA6 and extends into MFG/SFG and the precentral gyrus and the ventral cluster, restricted to the left hemisphere, includes BA44 and extends into the precentral gyrus. Researchers found that robotic movement which did not involve an object caused greater activation than human control, specifically on the left side of the MNS. This involved mainly the parietal lobe (SI, SII, IPL/SPL, and precuneus), and the frontal lobe (a dorsal cluster in BA6/precentral gyrus and a ventral cluster in BA44). They also found that, when subjects saw a robot doing a single action repeatedly, the

mirror system failed to activate. With this result, they explained why a previous study (Tai, Scherfler, Brooks, Sawamoto, & Castiello, 2004) failed to observe robotic activations in the MNS. For instance, if people watch the same movies repeatedly, the effect from the movies (e.g., efforts to perceive) will be decreased significantly. This is very similar to the finding that the duplicated movies lead to strong habituation in the MNS (Hamilton & Grafton, 2006). Saygin et al. (2012) also found that differences in agency (robot versus human) were most pronounced in the parietal cortex. Smaller clusters preferring the robotic movement were also found in the right hemisphere, involving mainly SI, SII, BA4 (i.e. M1), BA6, and BA44. In the static control condition (no movement), no significant differences were found. All movement control conditions caused stronger activations in the dorsal premotor (PMd), left ventral premotor (PMv), and left parietal cortex.

Emotional gestures and conversation

Interpreting the intentions of another agent is a key factor in the decision to trust. However, processing the emotional signals of another agent requires far more advanced forms of neural processing than just interpreting the kinematics. Two well-defined brain networks have been implicated in this processing; the person perception network (PPN) and the mentalizing network (MTN) (Wang & Quadflieg, 2015). The PPN is thought to comprise the occipital face area (OFA), extrastriate body area (EBA), fusiform face area (FFA), fusiform body area (FBA), and the posterior superior temporal sulcus (pSTS), and is known to extract information about people's facial and bodily appearance. In contrast, it has been argued that the MTN speculates about people's beliefs, desires, feelings, motives, or intentions based on their visible behavior (Gobbini et al., 2011; Abraham, Werning, Rakoczy, Von Cramon, & Schubotz, 2008; Spunt and Lieberman, 2012), recruiting the ventral prefrontal cortex (vmPFC) and dorsal medial prefrontal cortex (dmPFC), anterior temporal lobe (aTL), the temporal–parietal junction (TPJ), and precuneus (PrC). Chaminade et al. (2010) studied the neurological differences in processing human emotional behaviors compared to anthropomorphic, but still clearly robotic, emotional behaviors. There are several brain areas related to the visual association: visual areas V3, V4 (44) and V5 (50); temporal areas for face perception (FFA [48], lateral face area [LFA] [46]); and actions (superior temporal gyrus [49]). The brain activations from these occipital and temporal cortices play an important role in the visual perception of body movements while recognizing human gestures (Chaminade et al., 2010). Similarly, inferior frontal gryii is involved in the perception of human actions (56). The activation in the left insula was observed for

emotional resonance in the perception of the disgust gesture (28). Based upon relationships of the brain regions toward the PPT and the MTN, the researchers found overall larger responses in occipital and posterior temporal areas, whereas a reverse pattern was observed in regions responding specifically to one type of human gesture only, namely the left inferior frontal cortex for motor resonance in speech perception and the insula for emotional resonance in disgust (Chaminade et al., 2010).

Wang and Quadflieg (2015) analyzed the neurological differences in processing emotional behavior when observing interactions between human-human condition and human-robotics condition. They found that observing human-human interactions yielded enhanced activation in the TPJ, while human-robotic interactions yielded enhanced activations in the bilateral middle occipital gyrus, bilateral inferior temporal cortex (extending into the fusiform and the inferior occipital gyrus), medial PrC, as well as in the dmPFC and vmPFC. Activation in the left TPJ revealed an association with estimating unseen trusts and intentions of people's behaviors. The medial PrC and the vmPFC are both regions known to foster impression formation by providing access to generalizable social knowledge (Mitchell, Banaji, & Macrae, 2005; Szczepanski & Knight, 2014). This is because the PrC is associated with the retrieval of stereotypic beliefs concerning humans (Contreras, Banaji, & Mitchell, 2012; Simmons, Reddish, Bellgowan, & Martin, 2010) and the vmPFC is related to the retrieval of script-based information (Ghosh, Moscovitch, Colella, & Gilboa, 2014; van Kesteren, Ruiter, Fernandez, & Henson, 2012). As the field of social robotics becomes more advanced, there is also an increased need to understand the neurological correlates of how humans differ in interpreting vocal conversations with humans compared to conversations with robots. Human voices are interpreted by the mind as biological or natural, whereas robotic voices can be modulated between sounding mechanical to using an actual recording of human voices. Rauchbauer et al. (2019) began preliminary investigations of these neurological differences by having subjects blindly converse with either human or robotic conversational stimuli. They found several bilateral activation clusters in the human-robotic conversational condition. In the occipital region, a cluster centered on the striate cortex extended to the lingual and fusiform gyri. A strong activation was also found bilaterally within the intraparietal sulcus (IPS) extending to the supramarginal gyrus (SMG). Clusters were also found in the middle frontal gyrus and centered on the lateral central sulcus. These results differed from findings in the human-human conversational condition, which found bilateral activation in the superior temporal gyrus and sulcus that overlapped partly with the temporal areas associated with the main effects of the conversations. It extended anteriorly to the temporal poles and the posterior lateral orbitofrontal cortex (lOFC). Posteriorly, it covered the TPJ and lateral occipital cortex. Another

significant cluster covered several subcortical structures: the bilateral thalamus, hypothalamus, hippocampus, amygdala, caudate nucleus, and the subthalamic area.

Decision and feedback processing

Perfect automation schema

Lee and See (2004) define trust as "the attitude that an agent will help achieve an individual's goals in a situation characterized by uncertainty and vulnerability." The "perfect automation schema" (similar to the automatic automation bias) is the human tendency to initially trust an automated agent's advice more than human advice (Dzindolet, Pierce, Beck, & Dawe, 2002). Previous research has consistently found that humans initially perceive automation as having a high degree of authority and, therefore, ascribe automation advice as reliable without first confirming its validity (Mosier, Skitka, Heers, & Burdick, 1998; Parasuraman & Manzey, 2010).

Further research has found that "expert" automated agents are considered more trustworthy than "novice" automated agents (Madhavan & Wiegmann, 2007). This automation bias, therefore, affects how systems are monitored and has major consequences when systems fail or do not function optimally (de Visser et al., 2018; Parasuraman & Riley, 1997). While the neurological mechanisms and correlates of interpersonal trust have received significant investigation in neuroscience, studies on the comparisons of trust between human and nonhuman agents are very limited (Parasuraman & Wiese, 2014). EEG and fMRI are allowing the field of neuroergonomics to begin to understand these mechanisms in the context of interacting with robotics and automation.

Trust calibration

Several studies of trust in automation have proposed theories that the mechanism of trust is the interaction between two variables: an initial attentional bias (the decision phase) that is updated with observed data about the system (the feedback phase) (de Visser et al., 2018). The interaction of these two mechanisms is known together as trust calibration (Lee & See, 2004). Several factors affect trust calibration in automation: credibility (Fogg & Tseng, 1999), automation errors (Wiegmann, Rich, & Zhang, 2001), reliability (Dzindolet, Peterson, Pomranky, Pierce, & Beck, 2003), task difficulty, and individual differences in the causal attribution of automation errors (Pop, Shrewsbury, & Durso, 2015). Trust calibration in automation has been seen in studies that have examined automation bias

which finds trust as being initially higher for automated agents than for their human control. As the experiments go on, researchers would then find the trust differences disappearing based on their received feedback (Parasuraman & Wiese, 2014). The overall regions that have been identified as being involved in advice taking by evaluation of personal traits in automation are the default network including temporoparietal junction (TPJ), precuneus (Pc), posterior cingulate cortex (PCC), medial prefrontal cortex (mPFC), and the salience network including dorsal anterior cingulate cortex (dACC), bilateral insula.

The neural correlates of the process of trust calibration were studied by Goodyear and Krueger in 2014, and they found that this reinforcement learning process was mirrored by neural activations in reward-sensitive brain regions, including the caudate head and ventromedial prefrontal cortex (Parasuraman & Wiese, 2014). Furthermore, the decision phase of trust in automation had a significant three-way interaction effect in the right posterior insula (PI) (BA13); right anterior precuneus (aPreC) (BA5/7), left aPreC (BA5/7); left posterior cingulate cortex (PCC) (BA30/31); left rostrolateral prefrontal cortex (rlPFC) (superior frontal gyrus or SFG; BA10); and left posterior temporoparietal junction (pTPJ) (superior temporal gyrus or STG; BA22). This experiment was followed up by Goodyear et al. (2016) and found a significant main effect of Agent in the decision phase ($\alpha < 0.05$, $k = 11$) in the right lingual gyrus (LG) (BA18), right anterior cingulate cortex (ACC) (BA24), left anterior precuneus (aPreC) (superior parietal lobule; BA7) and left cuneus (CUN) (BA18). This was contrasted with the feedback phase, which found a significant three-way interaction with the left dorsomedial prefrontal cortex (dmPFC) (medial frontal gyrus or MFG; BA 9/10) showing higher activation for the human agent during run two for good compared to bad advice. The main effect of advice during the decision phase revealed significant activation in the right orbitofrontal cortex (OFC) (superior frontal gyrus, BA11). For the feedback phase, significant activation clusters for the main effect of advice were found in right middle frontal gyrus (BA6/8), right superior parietal lobule (BA 7), right putamen, right PCC (BA30), right head of the caudate, left OFC (medial frontal gyrus, BA11), left precentral gyrus (BA4), left subcallosal gyrus (BA34), left middle frontal gyrus (BA6), left dorsolateral prefrontal cortex or dlPFC (middle frontal gyrus, BA46), and left inferior frontal gyrus (BA47).

Trust calibration involves being able to identify trust-miscalibration, which is the process of identifying errors made by a partner and using that input to modulate the next decision phase. The error-related negativity (ERN) is a well-studied event-related potential that is implicated in identifying trust-miscalibration because it is elicited when an individual commits an error. (Falkenstein, Hohnsbein, Hoormann, & Blanke, 1991; Gehring et al., 1993) This potential is generated in or near the anterior

cingulate cortex (ACC) that peaks within 100 ms following an error (Ullsperger, Fischer, Nigbur, & Endrass, 2014). Van Schie, Mars, Coles, and Bekkering (2004) first identified observed ERN (oERN), an ERP component elicited when observing others' mistakes; de Visser et al. (2018) first found oERN when subjects were watching a computer.

Summary

Understanding the neural correlates and mechanisms involved in the process of developing proper calibrated trust between humans and robots has become increasingly consequential in the field of human-robot interaction (HRI), where this understanding helps developers to design robots that are more effective when interacting with humans. The initial neural networks that have been implicated in this process are the mirror neuron system (MNS), person-perception network (PPN), the mentalizing network, default network (DNN), salience network, and APS. While differences in activation levels have been found in various studies, conclusive differences cannot yet be drawn and further investigation is still necessary.

One limitation of this line of research is that, because different definitions of trust vary considerably, there is no consensus on how to accurately measure trust (Erchov, 2017). A future direction that HRI researchers should consider is the development of consistent neural measures of trust that are representative of the goals of the field, specifically in this case, interactions between humans and robots (Gramann, Fairclough, Zander, & Ayaz, 2017). The complex nature defining a robot poses a problem for researchers hoping to study robot-specific interactions. These studies have also faced the obstacle of the small sample size which seems to display a gender, age, and cultural bias toward college-aged male subjects in western societies (Khan, Bhatia, & Boloni, 2012; Rau, Li, & Li, 2009; Wang & Young, 2014). As robotics begin to play an increasingly larger role in the whole of society, it is important to make sure researchers are studying human-robotic interaction with samples that accurately reflect their intended users. Furthermore, cultural factors will also influence how society interacts with robotics, so these factors must be understood and controlled (Haring, Silvera-tawil, Matsumoto, Velonaki, & Watanabe, 2014). The last recommendation for future research would be to begin studying long-term interactions rather than only short-term interactions. The robots being designed today will ideally be used for long-term interaction, not just one-time use. Therefore, the impact of extended HRIs on trust modulation is an important factor to study (Riegelsberger, Sasse, & Mccarthy, 2005).

Acknowledgments

This work was supported in part by the National Science Foundation (NSF) under Grant BCS-1551688. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of the NSF.

References

Abraham, A., Werning, M., Rakoczy, H., Von Cramon, D. Y., & Schubotz, R. I. (2008). Minds, persons, and space : An fMRI investigation into the relational complexity of higher-order intentionality. *Consciousness and Cognition*, *17*, 438–450.

Aichhorn, M., Perner, J., Weiss, B., Kronbichler, M., Staffen, W., & Ladurner, G. (2008). Temporo-parietal junction activity in theory-of-mind tasks: Falseness, beliefs, or attention. *Journal of Cognitive Neuroscience*, *21*(6), 1179–1192.

Berberian, B., Somon, B., Sahaï, A., & Gouraud, J. (2017). Annual reviews in control the out-of-the-loop brain: A neuroergonomic approach of the human automation interaction. *Annual Reviews in Control*, *44*, 303–315.

Chaminade, T., Zecca, M., Blakemore, S., Takanishi, A., Frith, C. D., Micera, S. , et al. (2010). Brain response to a humanoid robot in areas implicated in the perception of human emotional gestures. *PLoS One*, *5*(7).

Contreras, J. M., Banaji, M. R., & Mitchell, J. P. (2012). Dissociable neural correlates of stereotypes and other forms of semantic knowledge. *Social Cognitive and Affective Neuroscience*, *7*(7), 764–770.

de Visser, E. J., Beatty, P. J., Estepp, J. R., Kohn, S., Abubshait, A., Fedota, J. R. , et al. (2018). Learning from the slips of others: Neural correlates of trust in automated agents. *Frontiers in Human Neuroscience*, *12*, 1–15.

Drnec, K., Marathe, A. R., Lukos, J. R., & Metcalfe, J. S. (2016). From trust in automation to decision neuroscience : Applying cognitive neuroscience methods to understand and improve interaction decisions involved in human automation interaction. *Frontiers in Human Neuroscience*, *10*, 1–14.

Dzindolet, M. T., Peterson, S. A., Pomranky, R. A., Pierce, L. G., & Beck, H. P. (2003). The role of trust in automation reliance. *International Journal of Human Computer Studies*, *58*(6), 697–718.

Dzindolet, M. T., Pierce, L. G., Beck, H. P., & Dawe, L. A. (2002). The perceived utility of human and automated aids in a visual detection task. *Human Factors*, *44*(1), 79–94.

Erchov, S. (2017). *Reconceptualizing trust: Defining, modeling, and measuring trust.* (Doctoral Dissertation) George Mason University.

Falkenstein, M., Hohnsbein, J., Hoormann, J., & Blanke, L. (1991). Effects of crossmodal divided attention on late E R P components. II. Error processing in choice reaction tasks. *Electroencephalography and Clinical Neurophysiology*, *78*(6), 447–455.

Fedota, J. R., & Parasuraman, R. (2010). Neuroergonomics and human error. *Theoretical Issues in Ergonomics Science*, *11*(5), 402–421.

Fogg, B., & Tseng, H. (1999). The elements of computer credibility. In *Proceedings of the CHI'99* (pp. 80–87). New York: ACM Press. (May).

Freud, S. (1919). *The Uncanny*: (pp. 1–21). Standard Edition 17 (2000) New York: Norton.

Gallimore, D., Lyons, J. B., Vo, T., Mahoney, S., & Wynne, K. T. (2019). Trusting robocop: Gender-based effects on trust of an autonomous robot. *Frontiers in Psychology*, *10*, 1–9.

Gazzola, V., Rizzolatti, G., Wicker, B., & Keysers, C. (2007). The anthropomorphic brain : The mirror neuron system responds to human and robotic actions. *NeuroImage*, *35*, 1674–1684.

Gehring, W. J., Goss, B., Coles, M. G. H., Meyer, D. E., Donchin, E., Gehring, W. J. , et al. (1993). A neural system for error detection and compensation. *Psychological Science*, *4*(6), 385–390.

Ghazali, A. S., Ham, J., Barakova, E. I., & Markopoulos, P. (2018). Effects of robot facial characteristics and gender in persuasive human-robot interaction. *Frontiers in Robotics and AI*, *5*, 1–16.

Ghosh, V. E., Moscovitch, M., Colella, B. M., & Gilboa, A. (2014). Schema representation in patients with ventromedial PFC lesions. *The Journal of Neuroscience*, *34*(36), 12057–12070.

Gobbini, M. I., Gentili, C., Ricciardi, E., Bellucci, C., Salvini, P., Laschi, C. , et al. (2011). Distinct neural systems involved in agency and Animacy detection. *Journal of Cognitive Neuroscience*, *23*(8), 1911–1920.

Goodyear, K., Parasuraman, R., Chernyak, S., Madhavan, P., Deshpande, G., & Krueger, F. (2016). Advice taking from humans and machines : An fMRI and effective connectivity study. *Frontiers in Human Neuroscience*, *10*, 1–15.

Gramann, K., Fairclough, S. H., Zander, T. O., & Ayaz, H. (2017). Editorial: Trends in neuroergonomics. *Frontiers in Human Neuroscience*, *11*, 1–4.

Hamilton, A. F. D. C., & Grafton, S. T. (2006). Goal representation in human anterior intraparietal sulcus. *The Journal of Neuroscience*, *26*(4), 1133–1137.

Hancock, P. A., Billings, D. R., Schaefer, K. E., Chen, J. Y. C., de Visser, E. J., & Parasuraman, R. (2009). A meta-analysis of factors affecting trust in human-robot interaction. *Human Factors*, *53*(5), 517–527.

Haring, K. S., Silvera-tawil, D., Matsumoto, Y., Velonaki, M., & Watanabe, K. (2014). Perception of an android robot in Japan and Australia : A cross-cultural comparison. In *International conference on social robotics* (pp. 166–175). Cham: Springer.

Hartwright, C. E., Apperly, I. A., & Hansen, P. C. (2014). Representation, control, or reasoning ? Distinct functions for theory of mind within the medial prefrontal cortex. *Journal of Cognitive Neuroscience*, *26*(4), 683–698.

Khan, S. A., Bhatia, T. S., & Boloni, L. (2012). Soldiers, robots and local population - modeling cross-cultural values in a peacekeeping scenario. In *Proc. of the 21th behavior representation in modeling & simulation (BRIMS) conference*.

Lee, J. D., & See, K. A. (2004). Trust in automation : Designing for appropriate reliance. *Human Factors*, *46*(1), 50–80.

Madhavan, P., & Wiegmann, D. A. (2007). Theoretical issues in ergonomics science similarities and differences between human – Human and human – Automation trust : An integrative review. *Theoretical Issues in Ergonomics Science*, *8*(4), 277–301.

Mitchell, J. P., Banaji, M. R., & Macrae, C. N. (2005). General and specific contributions of the medial prefrontal cortex to knowledge about mental states. *NeuroImage*, *28*, 757–762.

Mosier, K. L., Skitka, L. J., Heers, S., & Burdick, M. (1998). Automation bias: Decision making and performance in high-tech cockpits. *The International Journal of Aviation Psychology*, *8*(1), 47–63.

Nelissen, K., Luppino, G., Vanduffel, W., Rizzolatti, G., & Orban, G. A. (2005). Observing others : Multiple action representation in the frontal lobe. *Science*, *310*(5746), 332–336.

Parasuraman, R., & Manzey, D. H. (2010). Complacency and bias in human use of automation : An attentional integration. *Human Factors*, *52*(3), 381–410.

Parasuraman, R., & Riley, V. (1997). Humans and automation: Use, misuse, disuse, abuse. *Human Factors*, *39*(2), 230–253.

Parasuraman, R., & Wiese, E. (2014). Human trust in other humans, automation, robots, and cognitive agents: Neural correlates and design implications. In: *Proceedings of the human factors and ergonomics society 58th annual meeting*, pp. 340–344.

Pop, V. L., Shrewsbury, A., & Durso, F. T. (2015). Individual differences in the calibration of trust in automation. *Human Factors*, *57*(4), 545–556.

Rau, P. L. P., Li, Y., & Li, D. (2009). Computers in human behavior effects of communication style and culture on ability to accept recommendations from robots. *Computers in Human Behavior*, *25*(2), 587–595.

Rauchbauer, B., Nazarian, B., Bourhis, M., Ochs, M., Chaminade, T., & Pre, L. (2019). Brain activity during reciprocal social interaction investigated using conversational robots as

control condition. *Philosophical Transactions of the Royal Society, B: Biological Sciences 374* (1771). 20180033.

Riegelsberger, J., Sasse, M. A., & Mccarthy, J. D. (2005). The mechanics of trust : A framework for research and design. *International Journal of Human-Computer Studies*, *62*, 381–422.

Satpute, A. B., Badre, D., & Ochsner, K. N. (2013). Distinct regions of prefrontal cortex are associated with the controlled retrieval and selection of social information. *Cerebral Cortex*, 1–9.

Saygin, A. P., Chaminade, T., Ishiguro, H., Driver, J., & Frith, C. (2012). The thing that should not be : Predictive coding and the uncanny valley in perceiving human and humanoid robot actions. *Social Cognitive and Affective Neuroscience*, *7*, 413–422.

Seyama, J., & Nagayama, R. S. (2007). The uncanny valley : Effect of realism on the impression of artificial human faces. *Presence*, *16*(4), 337–351.

Simmons, W. K., Reddish, M., Bellgowan, P. S. F., & Martin, A. (2010). The selectivity and functional connectivity of the anterior temporal lobes. *Cerebral Cortex*, *20*(4), 813–825.

Somon, B., Campagne, A., Delorme, A., & Berberian, B. (2017). Performance monitoring applied to system supervision. *Frontiers in Human Neuroscience*, *11*, 1–14.

Szczepanski, S. M., & Knight, R. T. (2014). Review insights into human behavior from lesions to the prefrontal cortex. *Neuron*, *83*(5), 1002–1018.

Tai, Y. F., Scherfler, C., Brooks, D. J., Sawamoto, N., & Castiello, U. (2004). The human premotor cortex is 'Mirror' only for biological actions. *Current Biology*, *14*, 117–120.

Thompson, J., & Parasuraman, R. (2012). NeuroImage attention, biological motion, and action recognition. *NeuroImage*, *59*(1), 4–13.

Ullsperger, M., Fischer, A. G., Nigbur, R., & Endrass, T. (2014). Neural mechanisms and temporal dynamics of performance monitoring. *Trends in Cognitive Sciences*, *18*(5), 259–267.

van Kesteren, M. T. R., Ruiter, D. J., Fernandez, G., & Henson, R. N. (2012). How schema and novelty augment memory formation. *Trends in Neurosciences*, *35*(4), 211–219.

Van Schie, H. T., Mars, R. B., Coles, M. G. H., & Bekkering, H. (2004). Modulation of activity in medial frontal and motor cortices during error observation. *Nature Neuroscience*, *7*(5).

Wang, Y., & Quadflieg, S. (2015). In our own image ? Emotional and neural processing differences when observing human – human vs human – robot interactions. *Social Cognitive and Affective Neuroscience*, *10*, 1515–1524.

Wang, Y., & Young, J. E. (2014). Beyond "pink" and "blue": Gendered attitudes towards robots in society. In *Proceedings of gender and IT appropriation. Science and practice on dialogue-forum for interdisciplinary exchange* (p. 49). European Society for Socially Embedded Technologies.

Wiegmann, D. A., Rich, A., & Zhang, H. (2001). Automated dianostic aids: The effects of aid reliability on users' trust and reliance. *Theoretical Issues in Ergonomics Science*, *2*(4), 352–367.

CHAPTER

21

Cross-cultural trust of robot peacekeepers as a function of dialog, appearance, responsibilities, and onboard weapons

James P. Bliss[a], *Qin Gao*[b,*], *Xiaoxiao Hu*[a,*], *Makoto Itoh*[c,*], *Nicole Karpinsky-Mosely*[d,*], *Shelby K. Long*[a,*], *Yiannis Papelis*[e,*], *and Yusuke Yamani*[a,*]

[a]Psychology Department, Old Dominion University, Norfolk, VA, United States, [b]Department of Industrial Engineering, Tsinghua University, Beijing, People's Republic of China, [c]NPO Speech Deposit Association, Ibaraki, Japan, [d]Knowesis Inc., Fairfax, VA, United States, [e]Virginia Modeling, Analysis and Simulation Center, Old Dominion University, Norfolk, VA, United States

Introduction

For several decades, researchers have been intrigued with the construct of trust. Investigators first targeted interpersonal trust as a defining social variable that enabled relationships to form and grow (Rotter, 1967). Rotter emphasized that trustworthy agents offer a word, promise, or statement that can be relied upon. In subsequent work, Rotter (1971) focused on the personality trait of dispositional trust, a diffuse expectation of others' trustworthiness. History-based or experiential trust, in contrast, results from experiences one has with a trustee. More recently, researchers have focused on trust between humans and machines. Specific aspects of the

[*]All co-authors contributed equally to the article.

https://doi.org/10.1016/B978-0-12-819472-0.00021-6
© 2021 Elsevier Inc. All rights reserved.

human-machine partnership have included human trust of automated technologies to include preprogrammed and future systems that offer more advanced capabilities. Automated technologies that may assume performance roles previously accomplished by human operators (Parasuraman & Riley, 1997) are particularly interesting to researchers and laypeople. In such cases, humans must decide whether and to what degree responsibilities may be relinquished to the automation. The levels of automation proposed by Sheridan and Verplank (1978) specify that automation may assume duties that are primarily sensory as well as duties that include active control of a system.

The growth of automation has led researchers to focus on the development of robots as physical frameworks for automation. In the middle of the 20th century, Isaac Asimov's three laws of robotics, published within his 1942 short story "Runaround," specified that a robot may not injure a human being or, through inaction, allow a human being to come to harm; must obey the orders given by human beings except where such orders would conflict with the First Law; and must protect its own existence as long as such protection does not conflict with the First or Second Laws. Asimov's laws, written as fiction, clearly exceed the limits of modern robotic technology. However, they illustrate potential ethical quandaries that involve human-robot trust.

As designers have created robots for increasingly complex duties (industrial fabrication, surgery, space operations, and even social companionship), the construct of trust has been acknowledged as especially important. For humans to accept robots as surrogate task performers, humans must trust them to perform reliably and safely.

From 2015 to 2018, our research team received funding from the Air Force Office of Scientific Research to investigate the influence of several variables on human trust of robotic peacekeepers (RPKs). During the project, we manipulated independent variables that included robot dialog structure (expression of emotional, behavioral, cognitive, analytical, or comparative content to frame a demand), the task orders driving robot behavior (whether they adopted an offensive or defensive patrol stance), the weapons wielded by the simulated robot (type of nonlethal weapon as well as the presence of lethal weapon backup), and the physical anthropomorphism of the robot. The current chapter discusses each of these influences, examining closely how independent variables interacted with participants' cultural backgrounds to influence trust of robots.

Trust as a cross-cultural construct—A primary goal of the proposed research was to examine the differences in robot trust displayed by members of different cultures. In 1980, Dutch researcher Geert Hofstede defined trust as "the collective programming of the mind which distinguishes the members of one human group from another" (Hofstede, 1980, p. 25). He developed his culture framework based on data from more

than 100,000 employees of the multinational IBM in 40 countries (Hofstede, 1980, 1983). Hofstede and his colleagues have proposed five major cultural dimensions that may distinguish personal tendencies toward social interactions (Hofstede, 2001; Hofstede & Bond, 1988). The influence of the Hofstede's cultural framework on subsequent cross-cultural research is hard to overestimate. Virtually, all the prevailing theories and taxonomies in the cross-cultural literature (e.g., Giacobbe-Miller & Miller, 1995; Hui & Au, 2001; Schwartz, 2004; Triandis, 1994) were developed as extensions to Hofstede's dimension framework.

The first dimension is individualism–collectivism which refers to whether people in a certain country act individually or as part of a group (Hofstede, 1994). Individualism includes the degree to which members of society feel responsible for only themselves and their families. In contrast, collectivism involves clear distinction between in-groups and out-groups; this includes commitment to an in-group (Hofstede, 1980). Individuals in highly individualistic cultures tend to focus on self-fulfillment and achievement, prioritize their personal interests and welfare (as opposed to group interests and welfare), and emphasize their own rights and autonomy. In contrast, individuals in highly collectivistic cultures perceive themselves as part of a large group and tend to focus on the groups' collective interests and achievement (Triandis, 2001).

The second dimension, power distance, is defined as "the extent to which a society accepts the fact that power in institutions and organizations is distributed unequally" (Hofstede, 1980, p. 45). Although the imbalance of power exists in all cultures, countries differ in the extent to which less powerful people tolerate power inequality within the society. Individuals in high power distance cultures tend to be submissive to those higher in the social hierarchy and are expected to show respect to them. In contrast, individuals from low power distance cultures are more likely to request equality in power or argue with those higher in the social hierarchy if power unbalance exists (Hofstede, 2001). In the latter, less powerful individuals' thoughts and opinions often need to be sufficiently respected by the more powerful parties (Lam, 2002).

The third dimension is uncertainty avoidance, where individuals are threatened by situations that are uncertain. Consequently, they work to avoid such situations, instituting rules and standards to govern behavior (Hofstede, 1980). In other words, uncertainty avoidance describes the degree to which people can tolerate ambiguity. Individuals in high uncertainty avoidance cultures tend to be emotional, aggressive, intolerant, and security seeking. Thus, they are more likely to react adversely to situations that are unexpected or different. On the contrary, individuals from low uncertainty avoidance cultures tend to accept personal risk and tolerate unstructured situations (Vitell, Nwachukwu, & Barnes, 1993).

The fourth dimension is masculinity–femininity which concerns values with regard to a focus on ambition versus enjoyment in life. Masculinity refers to "the extent to which the dominant values in society include assertiveness, the acquisition of money and value of material property" (Hofstede, 1980). Put differently, masculinity is "the degree to which values like assertiveness, performance, success, and competition…prevail over values like the quality of life, maintaining warm personal relationships, care for the weak, and solidarity" (Hofstede, 1994, p. 6). On the contrary, femininity refers to the degree to which the dominant values include the preference for friendly relationships, security, and cooperation (Hofstede, 2001, p. 281).

The fifth dimension, time orientation (also called Confucian work dynamism), refers to short-term versus long-term orientation toward the future (Hofstede & Bond, 1988). Individuals in short-term-orientated cultures tend to emphasize future-oriented values such as thrift, persistence, savings, and a willingness to defer for a purpose. In contrast, individuals in long-term-orientated cultures are more likely to adopt past- or present-oriented values such as respect for traditionality and a concern with outward appearance (Hofstede & Bond, 1988).

Undoubtedly, Hofstede's five dimensions are the most important and influential cultural framework to date. Through a thorough review of the cross-cultural literature, Clark (1990) concluded that the dimensions proposed by other major cultural theorists were well captured in Hofstede's typology. This cultural framework has been heavily replicated (Chandy & Williams, 1994; Søndergaard, 1994) and widely examined at both societal and individual levels in cross-cultural research (Kale & Barnes, 1992).

Though Hofstede acknowledged trust as a social interaction influenced by cultural identifiers, other researchers have examined the relationship more closely. Using Hofstede's framework as a foundation, Doney, Cannon, and Mullen (1998) proposed a model to explain and predict how trust develops uniquely across various cultures. Their model stresses the influence of many factors on trust development: national culture variables like relation to authority, relation to self, and relation to risk; underlying behavioral norms; cognitive processes such as calculation, prediction, intentionality, capability, and transference; and other relational, individual, and organizational factors. Doney et al.'s (1998) research is important because it has paved the way for other researchers to investigate how differences in trust may be manifested in countries around the world.

Of noted relevance to our research, Doney et al. (1998) directly point to Hofstede's dimensions of individualism/collectivism and masculinity/femininity as influential for predicting how much value a trustor places on a target entity's competence, ability, or expertise. They suggest, for example, that people in individualistic cultures may prize performance

capabilities more than those from collectivistic cultures because the latter approach tasks from a group perspective. They also suggested that transfer of trust across entities may be more difficult to achieve in individualistic and masculine societies because of the emphases on independence and skepticism. Conversely, transferring trust across entities (such as from a person to a robot that represents the programming or will of that person) may more readily occur in collectivist and feminine cultures that rely on strong social connections.

Chien, Lewis, Hergeth, Semnani-Azad, and Sycara (2015) focused more closely on human trust of automation as influenced by cultural background. They relied on Hofstede's cultural dimensions as well as Leung and Cohen's theory of cultural syndromes to predict and explain human tendencies to trust automated devices. Chien et al. compared trust of automation across individuals from the United States, Taiwan, and Turkey (representing honor, face, and dignity cultures, respectively).

Much of the existing cross-cultural human-robot trust research presumes that humans and robots will work together in a team relationship to achieve a common goal. For example, Yerdon, Marlowe, Volante, Li, and Hancock (2017) investigated cross-cultural trust of automation that served to collaboratively assist users. Their work explored trust in automation across a variety of cultures and task domains, noting differences in attitudes toward automation as a function of level of automation and country of residence. Like Doney et al. (1998), they noted the import of cultural collectivism or individualism for the perception and success of automated technologies. For example, they pointed out that Americans and Chinese have been known for their need for and development of automation. With regards to automated vehicles, Yerdon et al. reported that individuals from the United States are most attracted to potential safety benefits of automated vehicles but concerned with cost and risk of losing vehicle control. In contrast, Japanese individuals were least willing to pay for such technology. Other survey results indicated that Japanese are unwilling to ride in a self-driving vehicle (Schoettle & Sivak, 2014).

Wang, Rau, Evers, Robinson, and Hinds (2010) directly compared participants' attitudes about trusting a specific automation form: robots. Their work included comparisons between China and the United States, noting that prior cultural assumptions about Chinese and American cultures seemed to transfer to a human-robot interaction (HRI) task. Again, however, Wang et al. (2010) constructed a task that assumed a collaborative relationship between human and robot.

The current research represented a different task structure. Because peacekeeping in a foreign country often requires humans to acquiesce to demands made of them, reactions to the simulated robots in our research more closely reflected obedience rather than conformity or compliance. For the current research, we followed Doney et al.'s lead,

adopting Hofstede's framework to understanding and predicting differences in trust across cultures. As discussed above, some researchers have already used Hofstede's typology to successfully investigate trust of collaborative automated agents across cultures (c.f., Li, Rau, & Li, 2010; Rau, Li, & Li, 2009). However, we believed that Hofstede's dimensions may be relevant for explaining the cross-cultural differences in obedience and trust toward robots placed in a directive peacekeeping role, wielding nonlethal weapons. The inclusion of armed robots was notable because nonlethal weapon use has been linked with emphasis on personal right, deference to power, tolerance of unexpected situations, personal security, and willingness to conform for a purpose (North Atlantic Treaty Organization, 2006).

One complicating aspect of cultural identity is the variability with which it may be displayed in countries that are different from a person's country of origin. For example, individuals often may have learned rules, laws, and cultural norms while growing up in one country, but may be subject to different rules, laws, and norms while residing in a different country. Indeed, military peacekeepers (the primary subject of the current research) are often required to perform their duties to protect citizens of a country that is unfamiliar. In such situations, an important question is whether natives of a certain country harbor different trust toward robots than those who came to the country as expatriates. There are practically no examinations of this important cultural variable on robot trust; therefore, we elected to include it as a factor within our investigations.

For each experiment, we expected cultural differences to impact dispositional trust levels (defined as the "diffuse expectation of others' trustworthiness"; see Merritt & Ilgen, 2008), robot obedience rates, and subjective trust in the RPK's ability to carry out its peacekeeping mission. Our expectations for differences in dispositional trust were based on Hofstede's cultural dimensions and Doney et al.'s (1998) claims concerning the relationships between collectivistic/individualistic and masculine/feminine cultures and trust. Similarly, we expected robot peacekeeper obedience to vary because it may be influenced by cultural expectations concerning power distance and collectivism. Last, we expected subjective trust in the RPKs to vary for similar reasons as stated above. Because rigorous cross-cultural experimental comparisons of trust have generally not been published in the literature, our hypotheses were nondirectional.

Method

The experimental paradigm was similar across experiments. Each investigation involved three independent variables, of which cultural background was always addressed in a quasi-experimental fashion.

In Experiment 1, our design included two RPK dialog elements: trust component (emotional, behavioral or cognitive RPK appeal to the participant) and verbal interaction style (analytic or comparative rationale offered by the RPK to justify its demand). Levels of these variables reflect the potential influence of culture on the analytic and comparative mechanics of trust formation emphasized by Dillard and Pfau (2002), Festinger (1954) and others. In Experiment 2, the experimental design included two mission components: RPK patrol orders (checkpoint or offensive confrontation) and presence of reserve lethal weapons wielded by the RPK (present or absent). In Experiment 3, the design focused on robot appearance. Independent variables included anthropomorphism of the robot (robotic, anthropomorphic, or hybrid) and style of the nonlethal weapon wielded (pepper spray or taser). For each experiment, participants represented natives and expatriates from the United States, China, and Japan. In Experiment 1, we also tested native and expatriate Israelis.

Our discussion below focuses on only the effects of cultural background on dispositional trust, obedience to the RPKs' demands, and subjective trust in the RPK (other results have been published elsewhere; see Bliss et al., 2019; Bliss, Yamani, Hu, & Papelis, 2018; Long, Karpinsky, & Bliss, 2017). We will first describe the experimental task, followed by a description of the participants tested during each project experiment. The experimental materials and environment will then be described, followed by the chronologic experimental procedure. We will identify any unique considerations within each experiment.

Experimental task—During each of the three experiments, participants were required to perform a simulated version of Shallice and Burgess's (1991) Multiple Errands Task (MET). That task has historically been used as a diagnostic instrument to determine executive brain function. However, simulated versions of the MET have also been employed to assess executive functions in post-stroke patients (Raspelli, Pallavicini, Carelli, & Morganti, 2012) as well as training material efficacy in brain-healthy participants (Bliss & Hanson, 2018). After entering the simulated marketplace environment, participants were directed to navigate among a series of vendor stalls, purchasing a specific item at each location. At random points during the task, a RPK would interrupt the participant and demand that he or she relinquish a personal item (e.g., keys, cell phone, lighter). Failure to obey the robot had no immediate consequences; however, the participants' decisions were ostensibly reported to a "higher authority." After deciding whether to obey, participants were required to complete a trust questionnaire. During the shopping task, participants encountered six RPKs (see Fig. 1).

Materials and environment—Virtual task environments for each experiment were constructed in Unity at the Virginia Modeling, Analysis and

FIG. 1 Depiction of the experimental environment, featuring interaction with a robotic peacekeeper.

Simulation Center (VMASC). To ensure that participants accurately understood all questionnaires and the goal of the experimental task, all instructions, robot dialog elements, and questionnaires were translated from English to Chinese, Japanese, and (for Experiment 1 only) Hebrew. Back-translations to English were also completed to ensure comprehensibility. At the beginning of the project, we secured approval of our project from institutional review boards at Old Dominion University (ODU) and the Air Force Office of Scientific Research.

While logistical coordination for data collection was being arranged in each country, experimenters pilot tested the environments and tasks at all locations. Needed linguistic and software changes were implemented at VMASC prior to beginning formal data collection at ODU.

Participants—Participants were recruited from college classes, local foreign language schools, and government laboratories in the United States, China, Japan, and Israel. The groups were chosen in part because of their position within Hofstede's (1980) cultural framework (see Table 1). Within that structure, Hofstede specified aspects that differentiate cultures: individualism/collectivism, power distance, uncertainty avoidance, masculinity/femininity, time orientation, and indulgence. Many cross-cultural researchers have endorsed Hofstede's theory (c.f., Li et al., 2010). Chinese, Japanese, Americans, and Israelis clearly differ, particularly in their orientation toward individualism, power distance, and uncertainty avoidance. At the same time, this research

TABLE 1 Data obtained by native and expatriate cultural group during experiments 1, 2, and 3.

	M		F				Avg. age			Avg. comp. conf.		
Cultural group	**E1**	**E2**	**E3**	**E1**	**E2**	**E3**	**E1**	**E2**	**E3**	**E1**	**E2**	**E3**
Americans in the US	9	8	10	11	12	9	30.5 (12.0)	27.0 (10.0)	27.3 (8.3)	4.3 (1.5)	5.2 (1.0)	5.6 (0.8)
Americans in China	10	10	9	5	10	11	25.4 (8.1)	24.2 (1.5)	22.4 (2.8)	4.4 (1.1)	4.5 (1.0)	5.0 (1.1)
Americans in Japan	9	15	11	4	5	8	25.3 (5.9)	29.9 (11.3)	32.8 (12.7)	3.9 (1.3)	4.0 (1.6)	4.6 (1.6)
Chinese in China	8	9	10	12	11	10	20.4 (2.8)	31.3 (8.7)	29.2 (4.8)	4.4 (1.6)	5.0 (1.1)	4.9 (1.1)
Chinese in America	4	11	12	7	9	8	22.5 (2.1)	21.9 (2.7)	25.2 (2.9)	4.0 (1.7)	4.7 (0.92)	4.2 (1.0)
Japanese in Japan	4	6	3	16	14	17	24.9 (6.3)	37.6 (15.3)	30.2 (13.2)	3.3 (2.1)	3.0 (1.8)	2.8 (1.8)
Japanese in America	7	6	17	10	14	3	21.7 (2.8)	30.6 (9.0)	29.2 (8.6)	4.2 (1.1)	4.4 (1.7)	3.9 (1.3)
Americans in Israel	10	0	0	8	0	0	21.8 (2.3)	29.0 (10.5)	28.1 (9.1)	N/A[a]	4.4 (1.5)	4.4 (1.5)
Israelis in Israel	4	0	0	15	0	0	25.1 (3.8)	N/A[a]	N/A[a]	N/A[a]	N/A[a]	N/A[a]
Total	65	65	72	88	75	66	N/A[a]	N/A[a]	N/A[a]	N/A[a]	N/A[a]	N/A[a]

[a] Note*: M: male; F: female; E: experiment; Avg. age: average age; Avg. Comp. Conf.: average computer confidence.*
No Israeli data were available for average computer confidence. Standard errors are included within parentheses.

was US-centric, so this limited the cultural conditions tested. Participants were paid the local equivalent of approximately $10 (US) for participation. To ensure adequate numbers of participants representing each native and expatriate cultural group, experimenters used snowball sampling to access groups of participants. Specific demographics for each experiment are presented in Table 1. In addition to participant sex, average age, and computer confidence (with 7.00 representing complete confidence) were recorded (no data were available in experiment 1 for computer confidence of participants in Israel). Though attempts were made to obtain equal ages and numbers of males and females in each group, there was considerable variability. Because the research was quasi-experimental and exploratory, obtaining variability estimates to frame a priori power estimates was not easily possible. As a compromise, we strove for 20 participants for each cultural experimental condition during each experiment.

Procedure—Because of the logistics surrounding international travel, testing of native and expatriate residents of each country was completed in sequence, with US data collection starting before testing in other countries. Following data collection in all countries, data were coded and analyzed to test existing hypotheses.

During testing, participants first completed an informed consent form, followed by questionnaires to assess attitudes toward weapons (Penny Beliefs Scale; Penny, Walker, & Gudjonsson, 2011), their dispositional trust of other people (Rotter, 1967), and their dispositional trust of robots (Chien et al., 2015, during Experiment 1; Schaefer, 2016, during Experiments 2 and 3). After reading a brief on-screen description of the experimental task, participants began the virtual shopping task, purchasing an item from each vendor as directed. When confronted by an RPK, participants decided whether to obey its demand (relinquish a personal item). They then completed a brief history-based human-robot trust survey pertaining to the RPK's perceived ability to accomplish its peacekeeping task effectively (adapted from Chien et al., 2015 in Experiment 1, and from Schaefer, 2016 in Experiments 2 and 3). After completing the entire shopping task, participants exited the scenario.

Results

Performance and questionnaire data were automatically coded during each testing session. They were then assembled into a common spreadsheet to allow calculation of descriptive and inferential tests. Because a complete description of the experimental results from the three-experiment research effort is beyond the scope of this chapter (and available from other sources; see Bliss et al., 2018), we have elected to focus primarily on cross-cultural results in this chapter. Chosen statistical tests

included conventional parametric comparisons for obedience data as well as trust questionnaire data. The findings for each experiment are discussed below, separated by project experiment.

Experiment 1—An important consideration relates to the level of trust participants harbored upon entering the experiment. We attempted to measure dispositional interpersonal trust by administering the Rotter Trust Questionnaire (Rotter, 1967), a well-established measure of interpersonal trust. To determine whether the level of dispositional, interpersonal trust varied by cultural group, we conducted a one-way analysis of variance (ANOVA). The results indicated a significant effect of cultural group, $F(8,141)=3.36$, $P=0.001$, partial eta$^2=0.16$, observed power$=0.97$. Follow-up Tukey post hoc comparisons revealed that Americans living in Israel generally exhibited greater dispositional trust than all other populations (regardless of location) except native Israelis (see Fig. 2).

Because obedience rate data were dichotomous (participants did or did not obey each RPK), the most appropriate analysis required computing a series of binomial logistic regressions, using observed obedience rate for each combination of the within-group-dependent measures. Predictor (independent) variables included cultural group, participant age, computer familiarity, Rotter dispositional trust score, and the Penny Weapons Beliefs Scale score. Though regression analyses of the obedience data did not reveal that cultural group was a significant predictor of RPK obedience, greater values on the Rotter Trust Scale were associated with increased likelihood of obedience to an analytic, emotional appeal, $x^2(12)=24.05$, $P=0.02$. The model explained 38.2% (Nagelkerke R^2) of

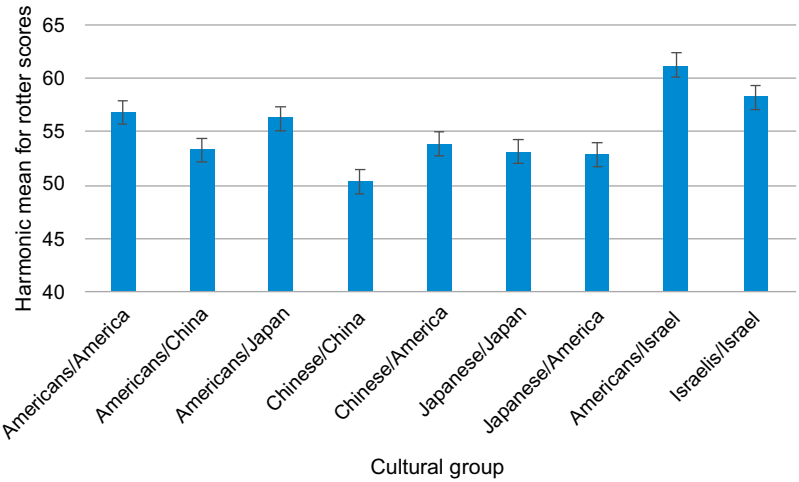


FIG. 2 Year 1 average interpersonal trust scores as a function of cultural group, as indicated by the Rotter Trust Questionnaire (error bars indicate standard error).

the variance and correctly classified 90.8% of cases. Greater values on the Rotter Trust Scale were also associated with increased obedience to a cognitive, behavioral RPK appeal, $x^2(12)=24.03$, $P=0.02$. The model explained 29.5% (Nagelkerke R^2) of the variance and correctly classified 76.4% of cases. In addition, a review of Fig. 3 indicates near-uniform obedience for robots that made an emotional appeal for obedience.

History-based trust in the RPK also varied by culture. Our overall analysis was a 9 (cultural group) × 2 (interaction style) × 3 (trust element) ANOVA, with trust scores from Chien et al.'s (2015) specific automation trust questionnaire as the dependent measure. Among other results, we noted a significant main effect of cultural group, $F(8,120)=2.10$, $P=0.04$, partial eta$^2=0.12$, observed power $=0.82$. Interestingly, follow-up Tukey comparisons showed no significant differences among cultural groups, though the main effect of group was significant. Fig. 4 does clearly indicate greater self-reported trust across all countries for robots that framed the obedience demand analytically instead of comparatively.

Experiment 2—As in experiment 1, we considered dispositional trust of robots to be important to consider as an influence on history-based or experiential trust of the RPK. Therefore, we again measured dispositional trust of robots in experiment 2; however, we elected to administer an alternative general robot trust questionnaire published by Schaefer (2016).

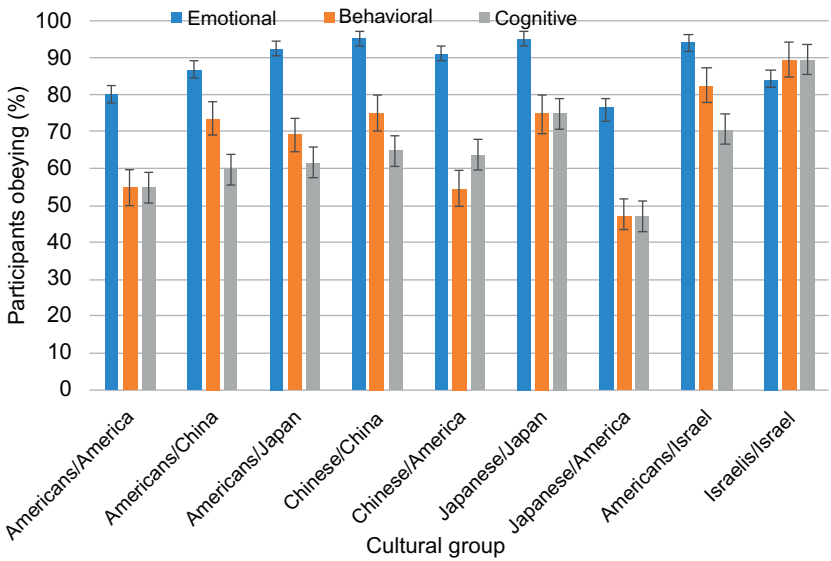


FIG. 3 Year 1 obedience rates to the robotic peacekeeper as a function of cultural group and trust element emphasized in the robotic peacekeeper (RPK) appeal (error bars indicate standard error).

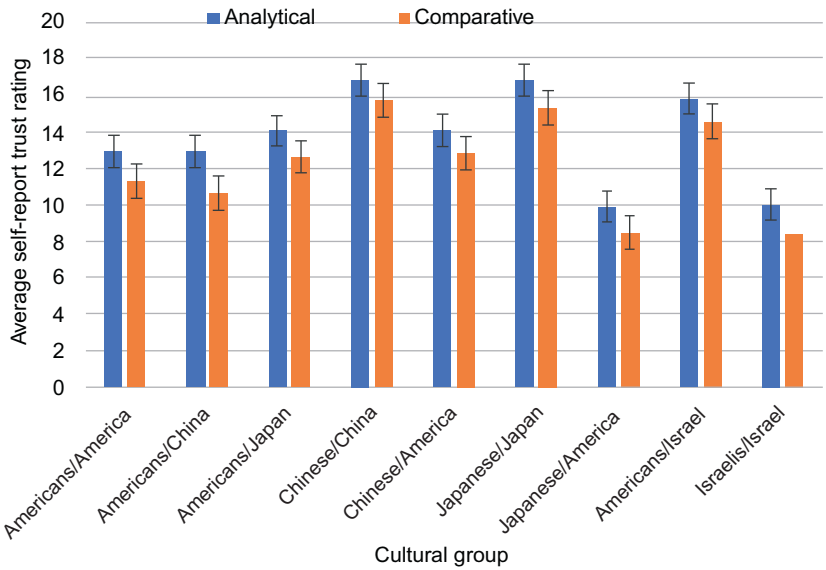


FIG. 4 Year 1 self-report trust of the RPK as a function of cultural group and interaction style (error bars indicate standard error).

The items on Schaefer's Trust Perception Scale-HRI corresponded more closely to the experimental task used as well as the (non-teamwork) nature of the participant-robot relationship. Though the Schaefer scale is not a dispositional robot trust (it does not assess prevailing trust of all robots), it does ask participants to indicate their impressions of a pictured robot they have never encountered before. Because of this, ratings would likely be influenced by trait-based trust rather than experiences they have had with the target robot. To determine whether the level of dispositional robot trust varied by cultural group, we conducted a one-way ANOVA. The results indicated a significant effect of cultural group, $F(6,138) = 3.24$, $P = 0.005$, partial eta$^2 = 0.13$, observed power $= 0.92$. Follow-up Tukey post hoc comparisons revealed that Japanese citizens living in America harbored less dispositional trust of robots than Americans in Japan or Chinese in America (see Fig. 5).

As in experiment 1, we computed a series of binomial logistic regressions, using dichotomous observed obedience rate for each combination of the within-group-dependent measures (in this case, lethal weapon availability and patrol orders). For the situation where a checkpoint guarding robot wielded only a nonlethal weapon, the regression model was significant, $x^2(11) = 30.41$, $P = 0.001$. The model explained 35.1% (Nagelkerke R^2) of the variance and correctly classified 55.0% of cases. Americans living in America were more liable to comply to the robot if

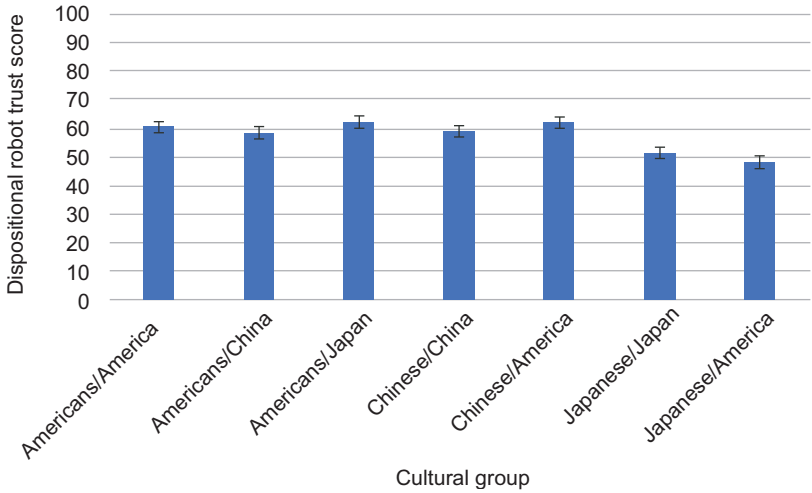


FIG. 5 Year 2 average dispositional robot trust scores as a function of cultural group, as indicated by the Schaefer Trust Perception Scale-human-robot interaction (HRI) (error bars indicate standard error).

it was guarding a checkpoint and wielded only a nonlethal weapon (see Fig. 6).

History-based RPK trust ratings were assessed using the specific version of Schaefer's (2016) Trust Perception Scale-HRI. We computed a three-way mixed ANOVA on the aggregate trust data. There was a significant main effect of cultural group, $F(6, 132) = 4.30$, $P = 0.001$, $\text{eta}^2 = 0.16$, observed power $= 0.98$. Follow-up least significant difference (LSD) pairwise comparisons showed that Americans living in all countries trusted the RPK significantly more other cultural groups, especially Japanese participants living in America (see Fig. 7).

Experiment 3—The analytical strategy for experiment 3 mirrored that for the previous two experiments. Cultural group was examined alongside two robot appearance variables, anthropomorphism, and nonlethal weapon type. We again valued dispositional trust as a foundation for later trust estimates. As in experiment 2, we computed a one-way ANOVA to determine whether dispositional trust rates for robots differed as a function of cultural group. The data consisted of responses to the Schaefer Trust Perception Scale-HRI. Due to a significant Levene's test, we tested the data for normality using the Kolmogorov–Smirnov test. That test was not significant ($P > 0.05$), so we assumed data normality. The results (corrected model) indicated a significant effect of cultural group,

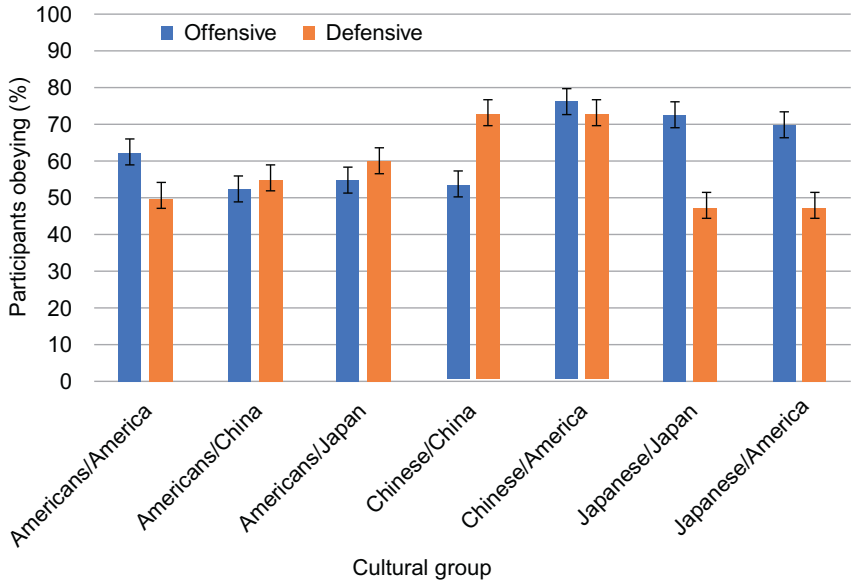


FIG. 6 Year 2 obedience rates to the RPK as a function of cultural group and trust element (error bars indicate standard error).

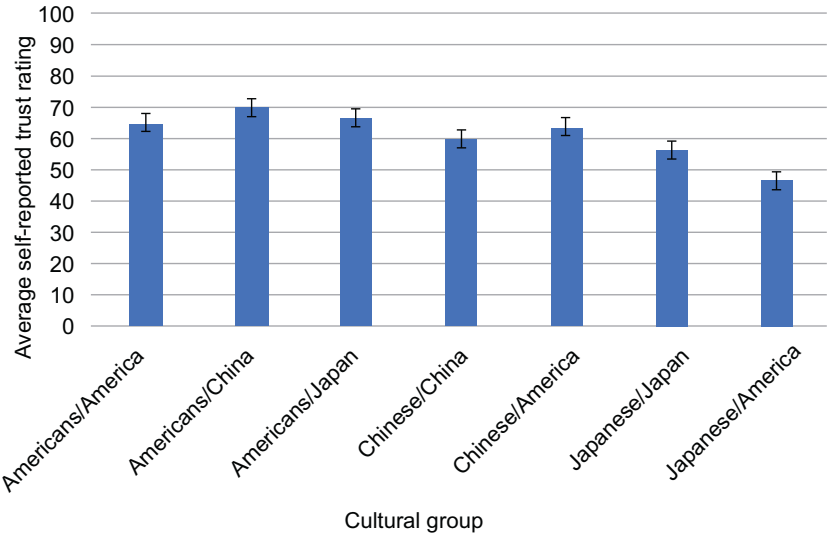


FIG. 7 Year 2 RPK trust ratings for the RPK as a function of cultural group and appeal type (error bars indicate standard error).

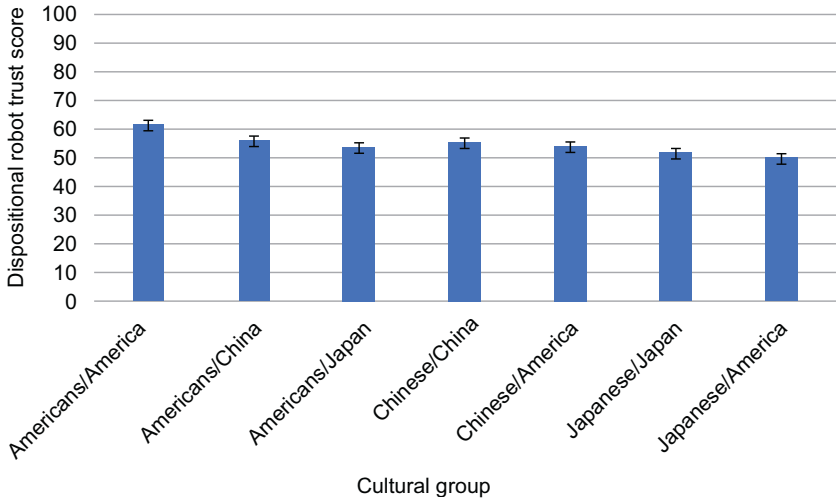


FIG. 8 Year 3 average dispositional robot trust scores as a function of cultural group, as indicated by the Schaefer Trust Perception Scale-HRI (error bars indicate standard error).

$F(6,139)=2.25$, $P=0.04$, partial eta$^2=0.09$, observed power$=0.77$. Follow-up Tukey post hoc comparisons revealed that Japanese citizens living in America harbored less dispositional trust of robots than Americans in America (see Fig. 8).

Dichotomous obedience rates to the RPK were examined in a similar fashion as in experiments 1 and 2. However, unlike in prior experiments, the omnibus logistic regression models were not significant for any of the combinations of predictors, including culture ($P>0.05$). It is possible that experiment 3 obedience rates were driven by emphasized appearance variables (how the weapons and RPKs looked) rather than how they communicated or performed their task. Appearance of the RPKs may not be as compelling an influence for obedience as interaction styles and content.

As before, we also analyzed specific RPK trust ratings. We computed a three-way mixed ANOVA on the aggregated data. The ANOVA results indicated a significant main effect of cultural group, $F(6,129)=2.21$, $P=0.046$, partial eta$^2=0.09$, observed power$=0.76$. LSD pairwise comparisons, however, revealed no significant differences among specific groups. No other pairwise comparisons were significant at $P<0.05$ (see Fig. 9).

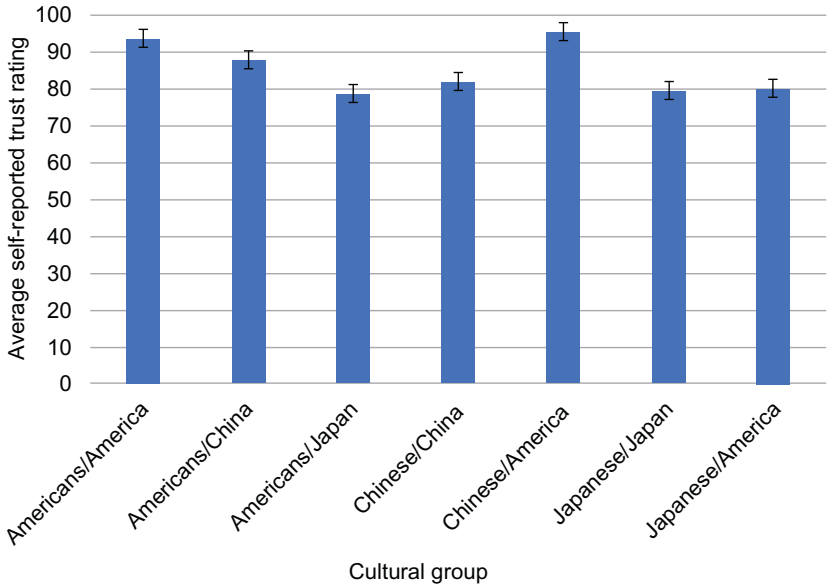


FIG. 9 Year 3 average specific (history based) RPK trust ratings on the Schaefer Trust Perception Scale-HRI as a function of cultural group (error bars indicate standard error).

Discussion

Generally, obedience rates to the RPKs' demands appeared to be related to dispositional trust levels (except in experiment 3, where the robots varied considerably in anthropomorphic appearance). Americans and Chinese tended to obey most often to the demands made by the RPKs and Americans in all countries tended to trust the RPKs more than members of other cultural groups. This observation seems to agree with prior research suggesting that positive depictions of robots in society may influence cultural trust of robots (Wang et al., 2010). It is important to note, however, that the research reported here is unique because RPKs were placed in an authoritarian role.

Generally, obedience rates were highest for all groups (native and expatriate) in the USA. This finding was observed during experiments 1 and 2 and may reflect participants' willingness to abide by American laws when in the United States.

From the data, the Japanese groups (native or expatriate) seemed to stand apart from others in experiment 2. From the results of the Schaefer Human-Robot Trust Questionnaire (long form), Japanese participants in America and in Japan demonstrated the lowest prevailing trust of robots. This pattern carried forward to the RPK interactions in vivo, as measured

by the short form of the same scale. One possible explanation for this may relate to a characteristic of Japanese societal philosophy where direct confrontation is discouraged, because robot confrontations were direct and unavoidable during the experiment (Itoh, 1991).

In experiment 3, the highest obedience rates were observed for natives of each country. Experimental comparisons between native and expatriate obedience levels are not widely available, making it difficult to explain these results. Yet, it may be informative to consider Green's (2009) article, wherein she explains the many social, philosophical, and legal implications of changing one's status from native to expatriate. It is possible that natives hold tighter allegiance to their country, whereas expatriates may feel less of a need to demonstrate obedience, particularly to a robot that may seem to have less authority than a human peacekeeper. In future research, it would be beneficial to adopt an alternative measure of obedience that does not rely on dichotomous data, so that statistical comparisons might be more powerful and insightful. In addition, we again found that Japanese groups showed the lowest dispositional trust of robots, whereas Americans living in America showed the greatest level.

As pointed out earlier, we tested representatives of various cultural groups during each project experiment. As noted above, specific countries were identified as testing sites based on convenience (acceptable numbers of participants could be tested in each location) and because they differed appreciably on the dimensions outlined by Hofstede's analytic framework. Each of these variables may have impacted attitudes and responses in our research. The RPK-interrupted multiple errands task presented participants with an unclear obedience choice to make (uncertainty avoidance) and that choice targeted personal obedience to an authority figure (power distance), ostensibly for maintenance of the greater good (individualism).

The findings of interpersonal trust displayed by members of tested cultural groups are interesting. In experiment 1, Americans and Israelis showed the highest dispositional interpersonal trust levels, whereas native Chinese participants showed the lowest levels. Americans, native or expatriate, were generally high in interpersonal trust.

In terms of dispositional and in-vivo robot trust, in experiment 1 native Israelis and Japanese living in America showed the least RPK trust. Likewise, Japanese in America showed the lowest dispositional and in-vivo RPK trust in experiment 2 (Japanese living in America were second-lowest) and in experiment 3. From these findings, it seems clear that Japanese participants, regardless of where they were living, harbored less trust of peacekeeping robots in general and that their trust generalized through the experiments across project experiments. This seems to mirror

survey results by Schoettle and Sivak (2014) that indicated less initial acceptance of automated automobiles by Japanese. Though speculative, one explanation is that Japanese participants view robots as having greater utility as social partners than law enforcement actors. Though less clear across experiments, American participants seemed to display greater dispositional robot trust and greater trust ratings for the RPK. This may have been the case because of the general pervasiveness of robotic technologies in American culture and industry. As noted by Yerdon et al. (2017), it is possible that individualistic tendencies in the United States may have influenced participants to be more generally accepting of robotic technologies.

During the three project experiments, we have focused on members of cultures chosen for practical convenience based on Hofstede's cultural framework. This has been a successful approach; we have noted many interesting findings that show the stability and variability of trust and obedience across the world. Yet, there are many other cultures left to explore. We have been fortunate to add Israeli participants to our experiment 1 analyses (an extension to our approved protocol); future researchers may wish to continue to test members of various cultures to observe the generality of our findings. To do so, we encourage designers to develop an online version of the marketplace. This could allow members of any culture to complete the task, further expanding our database.

Conclusion

Existing data concerning cross-cultural influences on automation (and robot) trust consist primarily of self-report estimates in situations where cooperation and teamwork between humans and robots is assumed. The research reported here supplements that approach with actual performance data from a simulated situation where robots demand obedience to directives. Importantly, our analyses distinguish native from expatriate cultural members. That distinction may help clarify the relative differences between country of origin and country of residence.

Acknowledgments

The authors gratefully acknowledge support from the Air Force Office of Scientific Research. The views and conclusions contained in this document are those of the authors and should not be interpreted as representing the official policies, express or implied, of AFOSR or the US Government. The authors also gratefully acknowledge the assistance of Dr. Joachim Meyer, Tel Aviv University, who collected data for the Israeli portion of the Experiment 1 investigation and Mr. Menion Croll, who designed the virtual environments.

References

Bliss, J. P., Gao, Q., Hu, X., Itoh, M., Long, S., Papelis, Y. , et al. (2019). Robot peacekeepers: An investigation of cross-cultural trust. *HDIAC Journal*, *6*(1). Spring.

Bliss, J. P., & Hanson, J. A. (2018). The effects of task criticality and target modality on a simulated battlefield search task. *Military Psychology*, *30*(2), 108–119.

Bliss, J. P., Yamani, Y., Hu, X., & Papelis, Y. (2018). *An examination of trust exhibited by members of different cultures toward robotic peacekeepers wielding nonlethal weapons* [Final Report, Contract No. FOS20140038H]. Arlington, VA: Air Force Office of Scientific Research

Chandy, P. R., & Williams, T. G. E. (1994). The impact of journals and authors on international business. *Journal of International Business Studies*, *25*(4), 715–728.

Chien, S. Y., Lewis, M., Hergeth, S., Semnani-Azad, Z., & Sycara, K. (2015). Cross-country validation of a cultural scale in measuring trust in automation. In *Proceedings of the 59th human factors and ergonomics meeting*. Los Angeles, CA: SAGE Publications.

Clark, T. (1990). International marketing and national character: A review and proposal for an integrative theory. *Journal of Marketing*, *54*, 66–79.

Dillard, J., & Pfau, M. (2002). *The persuasion handbook: Developments in theory and practice* (p. 6). Thousand Oaks, CA: Sage Publications.

Doney, P. M., Cannon, J. P., & Mullen, M. R. (1998). Understanding the influence of national culture on the development of trust. *Academy of Management Review*, *23*(3), 801–820.

Festinger, L. (1954). A theory of social comparison processes. *Human Relations*, *7*, 117–140.

Giacobbe-Miller, J. K., & Miller, D. J. (1995). A comparison of U.S. and Russian pay allocation decisions and distributive justice judgments. *Academy of Management Best Paper Proceedings*, *55*, 177–181.

Green, N. L. (2009). Expatriation, expatriates, and expats: The American transformation of a concept. *The American Historical Review*, *114*(2), 307–328.

Hofstede, G. H. (1980). *Culture's consequences: International differences in work-related values.* Beverly Hills, CA: Sage.

Hofstede, G. H. (1983). National cultures in four dimensions: A research-based theory of cultural differences amongst nations. *International Studies of Management and Organization*, *13*, 46–74.

Hofstede, G. (1994). Management scientists are human. *Management Science*, *40*, 4–14.

Hofstede, G. (2001). *Culture's consequences: Comparing values, behaviors, institutions, and organizations across nations* (2nd ed.). London, England: Sage.

Hofstede, G., & Bond, M. H. (1988). The Confucian connection: From cultural roots to economic growth. *Organizational Dynamics*, *16*, 4–21.

Hui, M., & Au, K. (2001). Justice perceptions of complaint-handling: A cross-cultural comparison between PRC and Canadian customers. *Journal of Business Research*, *52*, 161–173.

Itoh, Y. (1991). Socio-cultural backgrounds of Japanese interpersonal communication style. *Civilisations*, *39*, 101–128.

Kale, S. H., & Barnes, J. W. (1992). Understanding the domain of cross-national buyer–seller interactions. *Journal of International Business Studies*, *23*(1), 101–132.

Lam, A. (2002). Alternative societal models of learning and innovation in the knowledge economy. *International Social Science Journal*, *17*, 67–82.

Li, D., Rau, P. L. P., & Li, Y. (2010). A cross-cultural study: Effect of robot appearance and task. *International Journal of Social Robotics*, *2*(2), 175–186.

Long, S., Karpinsky, N., & Bliss, J. (2017). Trust of simulated robotic peacekeepers among resident and expatriate Americans. In: *Proceedings of the human factors and ergonomics society, Austin, TX, October 9–13*.

Merritt, S. M., & Ilgen, D. R. (2008). Not all trust is created equal: Dispositional and history-based trust in human-automation interactions. *Human Factors*, *50*(2), 194–210.

North Atlantic Treaty Organization. (2006). *The human effects of non-lethal technologies* [RTO Technical Report No. TR-HFM-073]. Neuilly-Sur-Seine CEDEX, France: North Atlantic Treaty Organization

Parasuraman, R., & Riley, V. (1997). Humans and automation: Use, misuse, disuse, abuse. *Human Factors*, *39*(2), 230–253.

Penny, H., Walker, J., & Gudjonsson, G. H. (2011). Development and preliminary validation of the Penny Beliefs Scale – Weapons (PBS-W). *Personality and Individual Differences*, *51*, 102–106.

Raspelli, S., Pallavicini, F., Carelli, L., & Morganti, F. (2012). Validating the neuro VR-based virtual version of the Multiple Errands Test: Preliminary results. *Presence*, *21*(1), 31–42.

Rau, P. L., Li, Y., & Li, D. (2009). Effects of communication style and culture on ability to accept recommendations from robots. *Computers in Human Behavior*, *25*, 587–595.

Rotter, J. (1967). A new scale for the measurement of interpersonal trust. *Journal of Personality*, *35*(4), 651–665.

Rotter, J. (1971). Generalized expectancies for interpersonal trust. *American Psychologist*, *26*, 443–452.

Schaefer, K. E. (2016). Measuring trust in human robot interactions: Development of the "trust perception scale-HRI". In *Robust intelligence and trust in autonomous systems* (pp. 191–218): Springer US.

Schoettle, B., & Sivak, M. (2014). *Public opinion about self-driving cars in China, India, Japan, the U.S., the U.K., and Australia.* Final Report No. UMTRI-2014-30 Ann Arbor, MI: The University of Michigan Transportation Research Institute.

Schwartz, S. H. (2004). Mapping and interpreting cultural differences around the world. In H. Vinken, J. Soeters, & P. Ester (Eds.), *Comparing cultures, dimensions of culture in a comparative perspective* (pp. 43–73). Leiden, the Netherlands: Brill.

Shallice, T., & Burgess, P. W. (1991). Deficits in strategy application following frontal lobe damage in man. *Brain*, *114*, 727–741.

Sheridan, T. B., & Verplank, W. L. (1978). *Human and computer control of undersea teleoperators.* Cambridge, MA: Massachusetts Institute of Technology, Man-Machine Systems Laboratory. www.dtic.mil/cgi-bin/GetTRDoc?AD=ADA057655PDF.

Søndergaard, M. (1994). Research note: Hofstede's consequences: A study of reviews, citations, and replications. *Organization Studies*, *15*, 447–456.

Triandis, H. C. (1994). *Culture and social behavior.* New York: McGraw-Hill.

Triandis, H. C. (2001). Individualism-collectivism and personality. *Journal of Personality*, *69*, 907–924.

Vitell, S. J., Nwachukwu, S. L., & Barnes, J. H. (1993). The effects of culture on ethical decision-making: An application of Hofstede's typology. *Journal of Business Ethics*, *12*, 753–760.

Wang, L., Rau, P. L. P., Evers, V., Robinson, B. K., & Hinds, P. (2010). When in Rome: The role of culture & context in adherence to robot recommendations. In *Proceedings of ACM/IEEE 5th international conference on human-robot interaction*. Osaka, Japan: ACM.

Yerdon, V. A., Marlowe, T. A., Volante, W. G., Li, S., & Hancock, P. A. (2017). Investigating cross cultural differences in trust levels of automotive automation. In *Advances in cross-cultural decision making* (pp. 183–194): Springer International Publishing.

CHAPTER

22

The role of emotions in human-robot interactions

Sarah A. Jessup[a] *and Tamera R. Schneider*[b]

[a]Department of Psychology, Wright State University, Dayton, OH, United States, [b]Department of Psychology, Baruch College, New York, NY, United States

Man is by nature a social animal. ***Aristotle***

We live in a society that has an evergrowing prevalence of and reliance on technology. Technology has permeated our work environments, including software automation, high-processing computers, and robotic machines. Technology is increasingly used in our everyday lives, such as in homes that have smart security systems, thermostats, vacuums, voice-activated virtual assistants, and in cars that assist with parking and collision prevention. Although automation can perform mundane tasks such as sending meeting reminders and property monitoring, it can also perform more complicated social tasks such as teaching, entertaining, and providing companionship via its embodiment in the form of social robots. Consumer-ready social robots include Buddy, an emotional companion that can help to combat loneliness by playing games and engaging in conversations with users (Blue Frog Robotics, 2015), Pepper, a social robot that can purportedly recognize user emotions and recommend activities based on user behavior patterns (SoftBank Robotics, 2015), and Jibo, a robot for the home that can recognize users when they walk into a room and start personalized conversations (Jibo, 2017). These social robots are designed to provide companionship and assistance (Robinson, MacDonald, Kerse, & Broadbent, 2013), whereas others are designed to provide people with service industry assistance such as taking and serving food orders (Anandan, 2018), providing therapeutic care (i.e. Mordoch, Osterreicher, Guse, Roger, & Thompson, 2013) and supporting learning (Belpaeme, Kennedy, Ramachandran, Scassellati, & Tanaka, 2018).

Trust in Human-Robot Interaction
https://doi.org/10.1016/B978-0-12-819472-0.00022-8

© 2021 Elsevier Inc. All rights reserved.

Social robots are designed to interact with people in ways that follow behavioral norms (Bartneck & Forlizzi, 2004), but the design should also factor in fundamental aspects of being human to enhance the human-robot partnership. Specifically, by factoring in the interdependent nature of human emotional and cognitive functioning at the outset, social robot designers will be able to maximize the effectiveness of human-robot interactions (HRIs). In collaboration with social and behavioral scientists, technology designers should aim to instill appropriate levels of human reliance on social robots. Human users should not lose agency but should rely on their robotic partner at appropriate levels and feel safe and comfortable doing so. Having calibrated trust is key for effective HRIs. In this chapter, we discuss emotions and their influence on trust in people and automation, with the goal of informing maximally effective HRI.

Emotional aspects of being human

Humans are emotional animals— emotions influence our learning, the decisions we make, our creativity and performance, and whether we cooperate with others or compete. Prominent thinkers, from Plato to Freud, have suggested that for rational processes emotions range from being irrelevant at best to being disruptive at worst. Plato considered emotions to be a more primitive, animalistic mode of responding to the environment that was incompatible with reason. Emotions were thought to overwhelm and subvert the rational. More contemporary theorists such as Freud also considered emotions to have a dangerous influence on both thoughts and behavior. In contrast, recent scientific advances show that emotions are not only useful, but essential for optimal human functioning (Damasio, 1994; Salovey & Mayer, 1990; Schneider, 2004; Schneider, Griffin, & Borders, 2018).

Evidence can be drawn from Damasio's inquiry into Phineas Gage and patients like him who experienced damage to affective (emotional) but not cognitive areas of the brain (Damasio, 1994). Such impaired individuals tend to make disastrous decisions and their lives suffer accordingly. Emotions are adaptive and regulate our ability to make decisions and interact well with others. Brain areas, such as the prefrontal cortex, are involved in complex cognitive behavior, moderating social behavior, and emotional control (modulating emotional arousal) (John, Bullock, Zikopoulos, & Barbas, 2013). Emotion and cognition interact, and work together so we can adapt to the demands of the situations we find ourselves in. Emotions have neither a universally beneficial nor disruptive influence on our social processes. Rather, whether the influence is adaptive or maladaptive depends on the nature of the task and the context we are in, the information-processing strategy used, and the characteristics of the decision-maker.

There are different ways to think about emotions, with prominent psychophysiological perspectives characterizing emotions along two dimensions—valence or pleasantness, and intensity or arousal. Valence ranges from positivity to negativity and arousal ranges from low to high intensity. Positive emotions include descriptors such as happy, enthusiastic, and alert, whereas negative emotions include anger, fear, and disgust (Watson, Clark, & Tellegen, 1988). Both positive and negative emotions guide our adaptive functioning, driving a complex interplay of thinking, feeling, and behavior as we discern and manage hospitable and hostile environments (Zajonc, 1998). Emotions are distinguished from mood based on their briefer duration, often identifiable antecedents, and physiological reactivity patterns (Schneider, 2004).

The affect infusion model

Emotions facilitate effective social interactions, but they can also contribute to cognitive and judgment errors. The Affect Infusion Model (AIM) is an integrative theory of research findings concerning the influence of affect (including emotion) on social judgements (Forgas, 1995, 2001). Affect infusion is the process whereby effectively loaded information influences and becomes incorporated into cognitive and judgment processes, influencing deliberations and decisions. Affect (state emotion and longer-term mood) serves as an active filter for incoming information, thereby impacting the content and valence (positivity/negativity) of memories, judgment and, ultimately, behavior.

The AIM states that mood is *less* likely to influence judgements and decision-making when people access directly, stored responses (e.g., cues), such as in familiar or unambiguous situations, or when the motivation to reach a particular outcome is high (e.g., maintaining a positive mood). Mood is *more* likely to influence decision-making when responses need to be created or constructed, such as occurs in unfamiliar or ambiguous situations. There are two main processing strategies associated with responses in such generative contexts: the heuristic processing strategy and the substantive processing strategy.

The heuristic processing strategy is characterized by using a limited range of information and whatever shortcuts or simplifications are readily available (see Forgas, 1995). This type of processing is most likely when the target is simple or highly typical, and when the situation does not demand detailed consideration. The primary mechanism for affect infusion under heuristic processing seems to be the affect-as-information account, where judgements are based on inferences from the prevailing affective state (Clore & Parrott, 1991; Schwarz & Clore, 2003). Decision-makers ask themselves, "How do I feel about it?" and mistake their

current feeling (due to preexisting mood) as a reaction to, or diagnostic about, the target (Schwarz, 1990).

Substantive processing is characterized by the generation of responses by selecting, learning, and interpreting novel information about a target and relating this information to pre-existing knowledge structures (see Forgas, 1995). This type of processing is more likely when the task is complex or atypical, and when the situational demands require more elaborate processing. It is a strategy adopted in circumstances where simple heuristics, which use preexisting feature-event/object associations in the form of cues, prove inadequate.

Affect priming affords the most parsimonious explanation for mood effects obtained through substantive processing (Bower, 1991). Elaborated in the associative network model (Bower, 1991), the affect priming principle suggests that affect indirectly informs judgements by linking to affect-related cognitive categories as the affective response spreads activation through associated memory structures. In this case, affective states selectively prime associated thoughts and representations that are more accessible and are, therefore, used when attitudinal or behavioral responses are constructed using memory-based information.

The AIM predicts that different processing strategies play an important role in determining the nature and extent of affective influences on judgements (Forgas, 1995). However, mood states may also influence the regulation of cognitive processes (Bless & Fiedler, 2006). Positive affect signals a hospitable environment and fosters a schema-based, top-down processing style where pre-existing ideas, attitudes, and representations are used to assimilate information about the external world. In this case, the processing is more creative, open, broad, and flexible. On the other hand, negative affect signals a potentially hostile environment, where processing is more deliberative and attention is narrowed. It promotes a bottom-up, externally focused processing strategy where responses accommodate for external constraints.

Positive affect should confer adaptive advantages when tasks are best handled by relying on cue-based associations and past knowledge and require rapid responses (Schneider & Forgas, 2015). On the other hand, negative affect should confer functional advantages when a task requires careful environmental monitoring and involves new and concrete information.

Field and laboratory experiments have investigated the influence of mood on judgements and behavior (Forgas, Laham, & Vargas, 2005). In one study on eyewitness testimony, participants who were induced into a positive mood incorporated more misleading information into their recall judgements than participants who were induced into a negative mood. Unfortunately, positive mood also evoked greater confidence in the accuracy of their (inaccurate) judgements. Another study found that

positive mood led to more stereotyped judgements about Muslim targets, whereas negative mood was associated with decreased stereotyping (Unkelbach, Forgas, & Denson, 2008). Another study asked participants to judge the deceptiveness of a videotaped target and found that participants induced into a negative mood were more accurate in their discrimination of deceptive behavior (Forgas & East, 2008). When coming up with or generating arguments for persuasive communications, positive mood is linked to more abstract, self-referent, and internally driven arguments, whereas negative mood is linked to using more concrete examples and evidence (Forgas, 2007).

These findings suggest that positive affect promotes attention to a broader range of features and the activation of cue-based processing, whereas negative affect promotes a narrowing of attention and increased rumination. Depending upon the contextual cues, positive affect could influence greater unwarranted trust, whereas more negative affect could lead to more appropriate trust levels. Therefore, in formulating judgements, such as whether to trust, in time-critical, high-consequence environments, it is important to ensure that cue-based associations are as accurate as possible. It would be helpful if people were made aware of the influence of mood on judgements or that social robots can help to influence mood to evoke more calibrated judgements of trust.

Because mood can act as a filter for task-related information and selectively activate memories, it can influence whether a social interaction is construed as hostile or more hospitable. At one level, it can facilitate the activation of appropriate cue-based associations, which ensures a rapid response to a situation. At another level, changes in mood can also potentially bias decisions when they are perceived as inconsistent with task-related goals. Consequently, affect informs decisions by providing an experiential link to evaluate situations that may be confronted. With time pressure and uncertainty, decision-makers rely on past experience or mental models to quickly generate action. Affect can lead to mistakes when the mood is permitted to influence the desire for a rapid response or, alternatively, it can capture past experience quickly, thereby leading to effective outcomes. Whether the influence of an affective state is helpful or harmful in trust-related interactions will depend largely on a range of poorly understood contextual factors. Mood may also influence the use of different types of processing strategies, and the processing style adopted may, in turn, influence the efficacy of decision-making about trust relationships.

Positive affect is likely to result in a greater reliance on pre-existing knowledge and the tendency to detect fewer anomalies in a situation in comparison to a neutral and negative mood. In contrast, negative affect would be expected to result in increased attention to the external environment and the tendency to detect a greater frequency of situational anomalies. Since attention is focused externally, any anomalies in the

environment should be detected more rapidly, compared to neutral and positive mood.

In summary, emotions influence our thinking and decision-making, and can influence whether a person will trust another. Positive emotions are associated with heuristic, top-down, internally focused processing (Clark & Isen, 1982; Forgas, 2013). They facilitate openness, exploration, and creativity, increasing the flexibility with which we interpret and organize information (Davis, 2009; Isen & Daubman, 1984). Conversely, negative emotions require effortful, bottom-up, externally-focused processing (Forgas, 2013). They can reduce our performance on tasks that are cognitively demanding (Brose, Schmiedek, Lövdén, & Lindenberger, 2012; Ellis, Moore, Varner, Ottaway, & Becker, 1997; Seibert & Ellis, 1991), and can be distracting, requiring cognitive resources to regulate. However, negative emotions can be helpful when working on less cognitively demanding tasks that constitute focused attention to details and can help us to evaluate the external environment more accurately. The trust environment will play a role in whether positive or negative affect will ultimately help to calibrate trust behaviors.

Trust and affect in human-human interaction

An extensive interpersonal trust literature provides an understanding of the factors that are paramount for successful human-human interactions (see Balliet & Van Lange, 2013; Dirks & Ferrin, 2002). Trust is a person's willingness to accept vulnerability in an interaction with another person (Mayer, Davis, & Schoorman, 1995). The person doing the trusting (trustor) as well as the one to be trusted (trustee) both bring characteristics to the interaction, which influence trust (Mayer et al., 1995). Trustors bring a relatively stable characteristic to the interaction, which is their general willingness or propensity to trust another person. Propensity to trust another is affected by people's general level of agreeableness and openness toward others (Schneider et al., 2018), as well as their past personal experiences and the trust-relevant encounters they learn about from others (Van Lange, 2015). Having a higher propensity to trust others predicts higher ratings of the trustworthiness of another person (perceived trustworthiness) (Alarcon, Lyons, & Christensen, 2016; Colquitt, Scott, & LePine, 2007; Gill, Boies, Finegan, & McNally, 2005). As with general personality characteristics, propensity to trust is more likely to predict trusting behaviors in novel situations than in situations that impose strong behavioral demands (Rotter, 1980).

The propensity to trust loses its predictive power as the trustor and trustee interact over time. As the trustor-trustee interaction unfolds over time, the trustor gains information about the behaviors of a trustee which

provides updates to the ongoing evaluation of the trustworthiness of the trustee. For example, over the course of interacting with a trustee, the trustor will better discern whether the trustee is reliable and engages in reciprocal behaviors (Jones & Shah, 2016). There are several trustee characteristics that predict trustor ratings of trustworthiness. These trustee characteristics include demonstrating domain-specific ability, acting with benevolence (having the trustor's best interests in mind), and acting with integrity. Both the trustor's propensity to trust and the aforementioned three trustee characteristics influence what transpires over the course of interaction in terms of the trustor's actions. For example, people who believe that others are trustworthy engage with them more over time, compared to those viewed as untrustworthy (Alarcon et al., 2018). Further, when people are generally trusting of others they exhibit more trusting behaviors (Evans & Revelle, 2008).

Propensity to trust is most influential in novel situations, similar to the influence of effect. Familiarity within dyads moderates the relationship between effect and judgment (Dunn & Schweitzer, 2005). Happy people are more trusting of others compared to angry people, but only when the trustor and trustee are unfamiliar. When people know one another, affect does not significantly influence trust. Interpersonal trust not only occurs at the dyad level but occurs in teams as well.

Individual and team affect are related to interpersonal trust. In a study examining teams of five undergraduate students working together to complete a Computer-based Aerial Port Simulation task (see Stokes, Lyons, & Schneider, 2011), both participants' own affect and the overall team affect influenced trust in the team. In that study, both positive individual and team affect were related to higher team trust. Conversely, negative individual and team affect were related to lower team trust. Positive affect leads to higher trust, especially when people are unfamiliar with one another. However, people have considerably less experience interacting with robots compared to humans. Affect should play an important role in predicting trust between humans and robots.

Trust and affect in human-automation interaction

The interpersonal trust literature has been extended to understand trust in automation (Lee & See, 2004). Propensity to trust automation stems from past interactions with automation and influences how new information will be interpreted in human-automation interactions. Similar to the human trustee, there are three characteristics of the system (automation) that influence perceived trustworthiness, including performance, purpose, and process (Lee & Moray, 1992; Lee & See, 2004). Performance describes the ability and reliability of the system in doing what it was

designed to do, purpose denotes why the automation is used and whether it performs those functions well, and process denotes the transparency of the system. The trust in the automation model (Lee & See, 2004) accounts for the consequences of trusting, such as reliance, which is demonstrated by using the automation and is also referred to as trust behaviors. This literature continues to evolve with much-needed research on construct refinement. Recent research has shown that propensity to trust automation predicts both perceived trustworthiness of automation and trusting behaviors when interacting with automation (Jessup, Schneider, Alarcon, Ryan, & Capiola, 2019). Factors beyond trust propensity and trustworthiness influence trust behaviors with automation, such as user culture, display design, the situation, the actions the automation is performing (e.g., cleaning a pool versus driving a vehicle), and of course human emotions.

Few studies have examined the influence of affect on trust in automation. One study induced participants into either positive or negative mood states to examine their influence on trust during the performance with an automated decision-aid (Dunn & Schweitzer, 2005). Results indicated that in the first session, participants who were induced into a positive emotional state rated the automated aid as more trustworthy than those induced into a negative state. In subsequent sessions, as the human-automation interaction proceeded, emotion no longer predicted the automation trustworthiness ratings. These findings are similar to those of the study by Stokes et al. (2010) who examined trust amongst humans, where initial levels of positive affect predicted greater trustworthiness, but as information about the referent, context, and environment was gained, trustor affective states did not predict trustworthiness. This aligns with AIM as well, in that as information about the situation becomes more concrete then affect will have less impact on judgements. One potential aside is that the referent in Dunn and Schweitzer's study was an automated tool, but based on the convergence of these findings with prior research amongst people these effects would likely extend to HRI contexts. We may be able to apply findings about affect and trust from human studies to learn more about human-robot interactions. One study examined the influence of the extent of human-like characteristics on the theory of mind (Krach et al., 2008). In this study, brain activity was measured whilst participants played the prisoner's dilemma game. The level of anthropomorphism of the partner was manipulated in the study. Participants played the game with partners which varied in their degree of humanness—a computer, a non-anthropomorphized robot, an anthropomorphized robot, or a human partner. The study found that as anthropomorphism increased, brain areas associated with the theory of mind had higher activation. This study demonstrated that people may attribute anthropomorphized partners as more humanlike, compared to nonanthropomorphized partners.

Ascribing emotions to automation

People apply social norms to their interactions with computers (Brave, Nass, & Hutchinson, 2005; Nass & Brave, 2005; Nass & Lee, 2010; Nass & Moon, 2000; Reeves & Nass, 1996). People express politeness, in-group and out-group biases, and greater likeability, caring, and perceived trustworthiness toward computers that appear to express empathy (Nass & Lee, 2010). People report increased enjoyment, human likeness, sympathy, and partner friendliness along a technology-human continuum from interacting with computers to functional robots to anthropomorphized robots to humans (Krach et al., 2008).

There are two types of robots: performance robots and social robots, with our focus on the latter. People anthropomorphize robots (Rosenthal-von der Putten, Kramer, Hoffmann, Sobieraj, & Eimler, 2013). One study had participants view different videos of person-robot interactions. One interaction was between a friendly person and a robotic dinosaur, wherein the person pet and fed the robot, which purred and chewed in return. The other interaction was more violent wherein the person threw, strangled, and mistreated the robot, which choked and cried in response. Participants experienced greater positive affect in response to the friendly video but experienced greater sympathetic arousal and negative affect in response to the violent interaction. The violent video also evoked pity for the robot and anger toward the person mistreating the robot. People experienced sympathetic emotions toward the robots.

Affective feelings toward robots have been shown in lab studies, and also in more mundane, real-world settings. One study examined the socio-psychological and physiological effects of a robot seal on elderly patients living in a nursing home (Wada & Shibata, 2006, 2007). After 1 month of interaction, participants that interacted with the robot seal showed increased social ties and communication with other residents in the home, as well as decreased levels of stress on their organs (measured via steroid hormone levels). The patients also talked to the robot as if it were alive by talking about the robot's feelings, greeting the robot, and asking the robot questions. One patient said this about the seal robot, "Everyone greets Paro [robot seal] because the residents are starved for affection and conversation with children" (Wada & Shibata, 2007, p. 977). Another study amongst the elderly showed similar effects. A randomized controlled trial with two elderly groups compared the effect of interaction with a robot seal and a dog, versus no intervention over 12 weeks. Patients in the robot seal/dog group experienced significantly less loneliness compared to the control group. Moreover, social engagement with other patients was the highest in the robot seal/dog group, compared to the control group. Social robotic animals do not require the constant care of living companion animals and may be optimal for the well-being of some elderly, especially those who are

unable to care for a living animal. This study shows that a robot pet may be just as effective as a real pet for companionship.

Designing robots for society

There are likely myriad benefits of social robots for various populations, and designers should consider the many factors that could influence the human use of robots at the outset. People prefer trust and interact more with robots with humanlike emotional expressions and other nonverbal cues such as speech, tone, and language. When designing social robots to display 'emotions' there are two features that should be incorporated: physical features and behavioral engagement (Schaefer et al., 2012). Physical facial-like features can include but are not limited to, narrowing of the eyes (concentration), furrowed brow (anger or frustration), and upraised outer corners of the mouth (to suggest smiling and happy or content affect). Nonverbal behavior conveys internal states of humans and could be mirrored in a similar manner in robots. Another way to express emotions is through verbal behavior, such as tone, language, and pitch, which can create different combinations of emotional reactions. Further, when engaging with a robot, people prefer interactions in which the robot incorporates gestures with speech. One study examined participants' perceptions of robots when they were equipped with and without gestures (Salem, Rohlfing, Kopp, & Joublin, 2011). Participants rated the robot with gestures as more active, competent, lively, friendly, communicative, engaged, and sympathetic compared to the robot with only speech. Gestures are more helpful for emotion recognition in a 3D physical robot compared to a 2D video of the same robot (Bretan, Hoffman, & Weinberg, 2015). In addition to displaying emotion, robots that can perceive emotions through recognition of body language, facial expression, and verbal behavior will allow for more authentic interaction. Although there are social robots designed with this capability (i.e. Pepper), to the authors' knowledge, no studies have empirically tested the effects (or reliability or validity) of such technology.

The presentation and physical form of robots influences how people will perceive and interact with them. People report higher enjoyment during play when their partner is an embodied robot compared to the avatar version of the same robot (Leite, Pereira, Martinho, & Paiva, 2008). Compared to the avatar condition, participants reported they received more appropriate feedback about their performance, were more immersion in and experienced flow during gameplay, and had a better social interaction when the partner was an embodied robot. One study found that when participants were asked difficult questions that they did not know the answer to, they chose to ask for advice from a robot over an avatar, even though both entities had the same level of expertise (Pan & Steed, 2016). This indicates that participants trusted the robot more than the avatar. Combined,

these results indicate that people could benefit from an embodied robot when learning a new skill or performing a tedious task. Having an embodied robot teacher or companion could greatly benefit the interactions, foster greater trust, and lead to more enjoyment.

Emotions can also facilitate higher accuracy and compliance during HRI interactions. For example, when playing a game such as chess with a robot, people were better able to accurately determine whether their partner was winning or losing if the robot displayed the correct emotional response (i.e. if the robot was winning, it would appear to be excited; Leite et al., 2008). People are more willing to follow instructions when the robot giving them displays emotion. Moshkina (2012) conducted an experiment using an NAO robot as a scene coordinator during a search and rescue task. During the middle of the experiment, when there appeared to be a dangerous situation that required evacuation, participants were more likely to follow the instructions of the robot and more quickly when the robot displayed negative affect and fearfulness. People can 'catch' the emotions displayed by robots, providing support for the emotional contagion theory (Hatfield, Cacioppo, & Rapson, 1993). Participants reported a greater change in negative affect when the robot displayed similar emotions, leading to increased nervousness.

There are myriad advantages and disadvantages for people interacting with robots. When robots are designed in a way that evokes users to anthropomorphize them, users are able to apply heuristics to the robots that decrease cognitive load (Thompson & Gillan, 2010). If users view robots as humanlike, then they will initially treat the robots in a similar manner to how they would treat another person, which could potentially reduce training time and increase the use of the robot (Thrun, 2004). However, some worry that if users perceive robots as human-like, then users' expectations of the robots may be unrealistic (Dautenhahn, 2002; Duffy, 2003; Thompson & Gillan, 2010; Thrun, 2004). If a robot does not meet expectations, a host of disadvantages can occur such as disuse, distrust, negative emotions such as disappointment or anger, or even safety issues if users are expecting it to perform functions that it is not capable of performing (Thompson & Gillan, 2010). In consideration of the advantages and disadvantages of social robots, designers should give special consideration to the social and behavioral factors that could foster a proper calibration of trust. Designers should make robots that are not only approachable but also be transparent with users by conveying the limits of robot capabilities.

Conclusion

When humans are paired with a robot that can recognize, adapt to, and display emotions, human behavior may change, promoting greater behavioral calibration with social-emotional robots. Research has shown that

people ascribe social norms to different levels of anthropomorphized technology from computers to social robots. In turn, people are likely to make judgments toward robots that are influenced by emotions, which can influence both trust and reliance in the interaction. Robots that can display emotion and provide feedback are generally perceived as being more likeable, trustworthy, compassionate, and competent. Knowing what factors influence HRI is important not only for researchers but also those responsible for design and implementation to properly calibrate human-robot trust interactions.

Acknowledgments

This work was ongoing while one of the authors (TS) served at the National Science Foundation (NSF). The views expressed in this manuscript are solely those of the authors and do not necessarily represent those of NSF.

References

Alarcon, G. M., Lyons, J. B., & Christensen, J. C. (2016). The effect of propensity to trust and familiarity on perceptions of trustworthiness over time. *Personality and Individual Differences*, *94*, 309–315. https://doi.org/10.1016/j.paid.2016.01.031.

Anandan, T. M. (23 May 2018). *Service robots on the world stage.* Robotics Industries Association. Retrieved from (23 May 2018). https://www.robotics.org/content-detail.cfm/Industrial-Robotics-Industry-Insights/Service-Robots-on-the-World-Stage/content_id/7061.

Balliet, D., & Van Lange, P. A. M. (2013). Trust, conflict, and cooperation: A meta-analysis. *Psychological Bulletin*, *139*, 1090–1112. https://doi.org/10.1037/a0030939.

Bartneck, C., & Forlizzi, J. (2004). A design-centered framework for social human-robot interaction. In: *Proceedings of the Ro-Man 2004, Kurashiki*, pp. 591–594. https://doi.org/10.1109/ROMAN.2004.1374827.

Belpaeme, T., Kennedy, J., Ramachandran, A., Scassellati, B., & Tanaka, F. (2018). Social robots for education: A review. *Science robotics*, *3*, 1–9. https://doi.org/10.1126/scirobotics.aat5954.

Bless, H., & Fiedler, K. (2006). Mood and the regulation of information processing and behavior. In J. P. Forgas (Ed.), *Affect in social thinking and behavior* (pp. 65–84). New York, NY: Psychology Press.

Blue Frog Robotics (2015). *Buddy the emotional robot. Retrieved from* https://buddytherobot.com/en/buddy-the-emotional-robot/.

Bower, G. H. (1991). Mood congruity of social judgments. In J. P. Forgas (Ed.), *Emotion and social judgments* (pp. 107–124). London, UK: Pergamon Press.

Brave, S., Nass, C., & Hutchinson, K. (2005). Computers that care: Investigating the effects of orientation of emotion exhibited by an embodied computer agent. *International Journal of Human-Computer Studies*, *62*, 161–178. https://doi.org/10.1016/j.ijhcs.2004.11.002.

Bretan, M., Hoffman, G., & Weinberg, G. (2015). Emotionally expressive dynamic physical behaviors in robots. *International Journal of Human-Computer Studies*, *78*, 1–16. https://doi.org/10.1016/j.ijhcs.2015.01.006.

Brose, A., Schmiedek, F., Lövdén, M., & Lindenberger, U. (2012). Daily variability in working memory is coupled with negative affect: The role of attention and motivation. *Emotion*, *12*, 605–614. https://doi.org/10.1037/a0024436.

Clark, M. S., & Isen, A. M. (1982). Towards understanding the relationship between feeling states and social behavior. In A. H. Hastorf, & A. M. Isen (Eds.), *Cognitive social psychology* (pp. 73–108). New York, NY: Elsevier.

Clore, G. L., & Parrott, G. (1991). Moods and their vicissitudes: Thoughts and feelings as information. In J. P. Forgas (Ed.), *Emotion and social judgments* (pp. 107–124). London, UK: Pergamon Press.

Colquitt, J. A., Scott, B. A., & LePine, J. A. (2007). Trust, trustworthiness, and trust propensity: A meta-analytic test of their unique relationships with risk taking and job performance. *Journal of Applied Psychology*, *92*, 909–927. https://doi.org/10.1037/0021-9010.92.4.909.

Damasio, A. R. (1994). *Descartes' error: Emotion, reason, and the human brain.* New York, NY: G. P. Putnum's Sons.

Dautenhahn, K. (2002). Design spaces and niche spaces of believable social robots. In: *Proceedings of the IEEE international workshop on robots and human interactive communication*, pp. 192–197.

Davis, M. A. (2009). Understanding the relationship between mood and creativity: A meta-analysis. *Organizational Behavior and Human Decision Processes*, *108*, 25–38. https://doi.org/10.1016/j.obhdp.2008.04.001.

Dirks, K. T., & Ferrin, D. L. (2002). Trust in leadership: Meta-analytic findings and implications for research and practice. *Journal of Applied Psychology*, *87*, 611–628. https://doi.org/10.1037/0021-9010.87.4.611.

Duffy, B. R. (2003). Anthropomorphism and the social robot. *Robotics and Autonomous Systems*, *42*, 177–190. https://doi.org/10.1016/S0921-8890(02)00374-3.

Dunn, J. R., & Schweitzer, M. E. (2005). Feeling and believing: The influence of emotion on trust. *Journal of Personality and Social Psychology*, *88*, 736–748. https://doi.org/10.1037/0022-3514.88.5.736.

Ellis, H. C., Moore, B. A., Varner, L. J., Ottaway, S. A., & Becker, A. S. (1997). Depressed mood, task organization, cognitive interference, and memory: Irrelevant thoughts predict recall performance. *Journal of Social Behavior and Personality*, *12*(2), 453–470.

Evans, A. M., & Revelle, W. R. (2008). Survey and behavioral measurements of interpersonal trust. *Journal of Research in Personality*, *42*(6), 1585–1593. https://doi.org/10.1016/j.jrp.2008.07.011.

Forgas, J. P. (1995). Mood and judgment: The affect infusion model (AIM). *Psychological Bulletin*, *117*, 39–66. https://doi.org/10.1037/0033-2909.117.1.39.

Forgas, J. P. (2001). *Feeling and thinking: The role of affect in social cognition.* Cambridge, UK: Cambridge University Press.

Forgas, J. P. (2007). When sad is better than happy: Negative affect can improve the quality and effectiveness of persuasive messages and social influence strategies. *Journal of Experimental Social Psychology*, *43*, 513–528. https://doi.org/10.1016/j.jesp.2006.05.006.

Forgas, J. P. (2013). Don't worry, be sad! On the cognitive, motivational, and interpersonal benefits of negative mood. *Current Directions in Psychological Science*, *22*, 225–232. https://doi.org/10.1177/0963721412474458.

Forgas, J. P., & East, R. (2008). On being happy and gullible: Mood effects on skepticism and the detection of deception. *Journal of Experimental Social Psychology*, *44*, 1362–1367. https://doi.org/10.1016/j.jesp.2008.04.010.

Forgas, J. P., Laham, S. M., & Vargas, P. T. (2005). Mood effects on eyewitness memory: Affective influences on susceptibility to misinformation. *Journal of Experimental Social Psychology*, *41*, 574–588. https://doi.org/10.1016/j.jesp.2004.11.005.

Gill, H., Boies, K., Finegan, J. E., & McNally, J. (2005). Antecedents of trust: Establishing a boundary condition for the relation between propensity to trust and intention to trust. *Journal of Business and Psychology*, *19*, 287–302. https://doi.org/10.1007/s10869-004-2229-8.

Hatfield, E., Cacioppo, J. T., & Rapson, R. L. (1993). Emotional contagion. *Current Directions in Psychological Science*, *2*, 96–100. https://doi.org/10.1111/1467-8721.ep10770953.

Isen, A. M., & Daubman, K. A. (1984). The influence of affect on categorization. *Journal of Personality and Social Psychology*, *47*, 1206–1217. https://doi.org/10.1037/0022-3514.47.6.1206.

Jessup, S. A., Schneider, T. R., Alarcon, G. M., Ryan, T. J., & Capiola, A. (2019, July). The measurement of the propensity to trust automation. In: *Annual Proceedings of the 21st International Conference on Human-Computer Interaction, USA*, pp. 476–489. https://doi.org/10.1007/978-3-030-21565-1_32.

Jibo (2017). *Jibo. Retrieved from(2017).* https://www.jibo.com/.

John, Y. J., Bullock, D., Zikopoulos, B., & Barbas, H. (2013). Anatomy and computational modeling of networks underlying cognitive-emotional interaction. *Frontiers in Human Neuroscience*, *7*, 1–26. https://doi.org/10.3389/fnhum.2013.00101.

Jones, S. L., & Shah, P. P. (2016). Diagnosing the locus of trust: A temporal perspective for trustor, trustee, and dyadic influences on perceived trustworthiness. *Journal of Applied Psychology*, *101*, 392–414. https://doi.org/10.1037/apl0000041.

Krach, S., Hegel, F., Wrede, B., Sagerer, G., Binkofski, F., & Kircher, T. (2008). Can machines think? Interaction and perspective taking with robots investigated via fMRI. *PLoS One*. *3*, e2597. https://doi.org/10.1371/journal.pone.0002597.

Lee, J., & Moray, N. (1992). Trust, control strategies and allocation of function in human-machine systems. *Ergonomics*, *35*, 1243–1270.

Lee, J. D., & See, K. A. (2004). Trust in automation: Designing for appropriate reliance. *Human Factors*, *46*, 50–80. https://doi.org/10.1518/hfes.46.1.50.30392.

Leite, I., Pereira, A., Martinho, C., & Paiva, A. (2008). Are emotional robots more fun to play with? In: *The 17th IEEE international symposium on robot and human interactive communication, Munich, Germany*, pp. 77–82. https://doi.org/10.1109/ROMAN.2008.4600646.

Mayer, R. C., Davis, J. H., & Schoorman, F. D. (1995). An integrative model of organizational trust. *The Academy of Management Review*, *20*, 709–734. https://doi.org/10.2307/258792.

Mordoch, E., Osterreicher, A., Guse, L., Roger, K., & Thompson, G. (2013). Use of social commitment robots in the care of elderly people with dementia: A literature review. *Maturitas*, *74*, 14–20. https://doi.org/10.1016/j.maturitas.2012.10.015.

Moshkina, L. (2012). Improving request compliance through robot affect. In: *Proceedings of the 26th AAAI conference on artificial intelligence, Toronto, Canada*, pp. 2031–2037.

Nass, C., & Brave, S. (2005). *Wired for speech: How voice activates and advances the human-computer relationship.* Cambridge, MA: MIT Press.

Nass, C. I., & Lee, J.-E. R. (2010). Trust in computers: The computers-are-social-actors (CASA) paradigm and trustworthiness perception in human-computer communication. In D. Latusek, & A. Gerbasi (Eds.), *Trust and technology in a ubiquitous modern environment: Theoretical and methodological perspectives* (pp. 1–15). Hershey, PA: Information Science Reference.

Nass, C., & Moon, Y. (2000). Machines and mindlessness: Social responses to computers. *Journal of Social Issues*, *56*, 81–103. https://doi.org/10.1111/0022-4537.00153.

Pan, Y., & Steed, A. (2016). A comparison of avatar-, video-, and robot-mediated interaction on users' trust in expertise. *Frontiers in Robotics and AI*, *3*, 1–2. https://doi.org/10.3389/frobt.2016.00012.

Reeves, B., & Nass, C. I. (1996). *The media equation: How people treat computers, television, and new media like real people and places.* New York, NY: Cambridge University Press.

Robinson, H., MacDonald, B., Kerse, N., & Broadbent, E. (2013). The psychosocial effects of a companion robot: A randomized controlled trial. *Journal of the American Medical Directors Association*, *14*, 661–667. https://doi.org/10.1016/j.jamda.2013.02.007.

Rosenthal-von der Putten, A. M., Kramer, N. C., Hoffmann, L., Sobieraj, S., & Eimler, S. C. (2013). An experimental study on emotional reactions towards a robot. *International Journal of Social Robotics*, *5*, 17–34. https://doi.org/10.1007/s12369-012-0173-8.

Rotter, J. B. (1980). Interpersonal trust, trustworthiness, and gullibility. *American Psychologist*, *35*, 1–7. https://doi.org/10.1037/0003-066X.35.1.1.

Salem, M., Rohlfing, K., Kopp, S., & Joublin, F. (2011, July). A friendly gesture: Investigating the effect of multimodal robot behavior in human-robot interaction. In: *2011 IEEE Ro-Man*, pp. 247–252. https://doi.org/10.1109/ROMAN.2011.6005285.

Salovey, P., & Mayer, J. D. (1990). Emotional intelligence. *Imagination, Cognition and Personality*, *9*, 185–211.

Schaefer, K. E., Cook, J. G., Adams, J. K., Bell, J., Sanders, T. L., & Hancock, P. A. (2012). Augmented emotion and its remote embodiment: The importance of design from fiction to reality. In: *Proceedings of the human factors and ergonomics society annual meeting* (vol. 56, pp. 1817–1821) https://doi.org/10.1177/1071181312561366.

Schneider, T. R. (2004). The role of neuroticism on psychological and physiological stress responses. *Journal of Experimental Social Psychology*, *40*, 795–804. https://doi.org/10.1016/j.jesp.2004.04.005.

Schneider, T. R., & Forgas, J. (2015). The social context of diagnosis. In M. W. Wiggins, & T. Loveday (Eds.), *Diagnostic expertise in organizational environments* (pp. 61–68). Farnham, Surrey, UK: Ashgate.

Schneider, T. R., Griffin, K. W., & Borders, M. R. (2018). From bias to BEST teams: Developing emotional and social competencies for innovation in STEM. In M. D. C. P. Fuentes, M. D. M. M. Jurado, & J. J. G. Linares (Eds.), *Emotional intelligence: Perceptions, interpretations and attitudes*. New York, NY: Nova Science Publishers, Inc.

Schwarz, N. (1990). Feelings as information: Informational and motivational functions of affective states. In R. M. Sorrentino, & E. T. Higgins (Eds.), *Handbook of motivation and cognition: Cognitive foundations of social psychology* (pp. 527–561). New York, NY: Guilford Press.

Schwarz, N., & Clore, G. L. (2003). Mood as information: 20 years later. *Psychological Inquiry*, *14*, 296–303. https://doi.org/10.1080/1047840X.2003.9682896.

Seibert, P. S., & Ellis, H. C. (1991). Irrelevant thoughts, emotional mood states, and cognitive task performance. *Memory & Cognition*, *19*, 507–513. https://doi.org/10.3758/BF03199574.

SoftBank Robotics (2015). *Pepper. Retrieved from(2015)*. https://www.softbankrobotics.com/emea/en/pepper.

Stokes, C. K., Lyons, J. B., Littlejohn, K., Natarian, J., Case, E., & Speranza, N. (2010, May). Accounting for the human in cyberspace: Effects of mood on trust in automation. In: *2010 IEEE international symposium on collaborative technologies and systems*, pp. 180–187. https://doi.org/10.1109/CTS.2010.5478512.

Stokes, C. K., Lyons, J. B., & Schneider, T. R. (2011). The impact of mood on interpersonal trust: Implications for multicultural teams. In N. Stanton (Ed.), *Trust in military teams* (pp. 24–37). Farnham Surrey, UK: Ashgate.

Thompson, L. F., & Gillan, D. J. (2010). Social factors in human-robot-interaction. In M. J. Barnes, & F. G. Jentsch (Eds.), *Human-robot interactions in future military operations* (pp. 67–81). New York, NY: Routledge.

Thrun, S. (2004). Toward a framework for human-robot interaction. *Human Computer Interaction*, *19*, 9–24. https://doi.org/10.1207/s15327051hci1901&2_2.

Unkelbach, C., Forgas, J. P., & Denson, T. F. (2008). The turban effect: The influence of Muslim headgear and induced affect on aggressive responses in the shooter bias paradigm. *Journal of Experimental Social Psychology*, *44*, 1409–1413. https://doi.org/10.1016/j.jesp.2008.04.003.

Van Lange, P. A. M. (2015). Generalized trust: Four lessons from genetics and culture. *Current Directions in Psychological Science*, *24*, 71–76. https://doi.org/10.1177/0963721414552473.

Wada, K., & Shibata, T. (2006). Robot therapy in a care house-its sociopsychological and physiological effects on the residents. In: *Proceedings IEEE proceedings on the international conference on robotics and automation*, pp. 3966–3971. https://doi.org/10.1109/ROBOT.2006.1642310.

Wada, K., & Shibata, T. (2007). Living with seal robots-its sociopsychological and physiological influences on the elderly at a care house. *IEEE Transactions on Robotics*, *23*, 972–980. https://doi.org/10.1109/TRO.2007.906261.

Watson, D., Clark, L. A., & Tellegen, A. (1988). Development and validation of brief measures of positive and negative affect: The PANAS scales. *Journal of Personality and Social Psychology*, *54*, 1063–1070. https://doi.org/10.1037/0022-3514.54.6.1063.

Zajonc, R. B. (1998). Emotions. In D. T. Gilbert, S. T. Fiske, & G. Lindzey (Eds.), *The handbook of social psychology* (pp. 591–632). New York, NY: McGraw-Hill.

CHAPTER

23

Should we trust robots? The ethical perspective

Claudia Hauer

Lyon Chair in Professional Ethics, USAF Academy, Air Force Academy, CO, United States

In 2007, my vision was 20/2000. I was legally blind. I went in for photorefractive keratectomy (PRK) laser surgery on both eyes. The first appointment was a consultation with the doctor, who tested my vision and reassured me that I was a good candidate for this surgery. The date for the surgery was set. That morning, the doctor was waiting in the room with the machine. I was put into position, and the process began. The machine was quite loud and obviously bright. The doctor sat next to the machine and watched throughout to make sure I did not glance away. That was his only function as the machine fixed my misshapen eyes. I trusted that doctor. Would I have said I trusted the machine? That word never would have occurred to me. The machine was not discussed at any point during the consultation or the surgery. It was not introduced. It was not explained. I might have been less frightened of it had I had some insight into its operations, but I also have to acknowledge that an explanation might not have alleviated my stress. Maybe it was best that the surgery proceeded without any comment about the presence of the machine.

My experience of this surgery matched Heikkerö's (2012) observation that "although many technological devices create some kind of acoustic noise either in their construction or in their operation, technology as a cultural phenomenon is most often surrounded by silence" (p. 1). People accept the presence of technology in their lives but rarely make this presence the direct object of their reflections. Notions of trust are largely based on the relationships humans have with other humans. Only in a futuristic world would I have signed up for the eye surgery, gone in and submitted to the machine without interacting with another human. Such a world is

https://doi.org/10.1016/B978-0-12-819472-0.00023-X
© 2021 Elsevier Inc. All rights reserved.

not unimaginable, but it is not desirable for philosophical reasons: the loss of human-human interaction is a feature of dystopia. In what follows, I am going to explore some of these philosophical issues, as I examine the possibility of an ethics of human-robot relationships.

First, I will explore how our age of robots and artificial intelligence poses ethical challenges that classical moral philosophy cannot address. This part of the discussion will have a broad focus that goes beyond the specifics of human-robot partnerships, but it is necessary if we are to understand the ethical challenges of these relationships. Second, I lay out some of the broader cultural and teleological factors involved in forming human-robot partnerships. What is it about being human that sets us apart from artificial intelligence? How does biological nature create an ethical context for a human life? How do cultural factors like theological beliefs factor into how humans perceive artificial intelligence? I will also look briefly at human-human relationships of trust and ask whether human-robot trust has analogous characteristics. Finally, this chapter will explore new ethical paradigms that could help guide us to an ethical response to human-robot partnerships in the age of self-driving cars, robot surgeons, carebots, AI bosses, and autonomous weapons systems. The goal is to achieve a holistic view of the complex and multifaceted moral landscape of this age.

This age of unsolicited human-robot partnerships is already upon us. My husband and I used a rental car for a recent vacation and noticed that it took over the steering capacity from time to time. We determined that the car was more likely to take control away from the driver after a signal had flashed that warned the driver that he or she had strayed outside the painted lines on the roadway. We concluded that the smart car did not "trust" us. Although this loss of control was at first disconcerting, by the end of the vacation, we were reasonably comfortable allowing the car to make judgments for us and allowing it to intervene without any solicitation from us. This narrowing of the scope of individual judgment is one of the main ethical concerns that we will explore here.

First, why is classical moral philosophy inadequate to handle the new ethical dilemmas of our age? Robots and AI are tools, and human beings have been using tools since the prehistoric period. Each time a new paradigmatic tool emerges; there are people who claim it has overturned all our ethical models. The crossbow, the atomic bomb, the remotely piloted aircraft—it could be argued that any of these developments presented unprecedented challenges to moral decision making. I argue that while such paradigm shifts do cause value systems to shift, the shifting value structures still form a continuum. Chivalry was no longer the preeminent martial value when it was possible to kill at a distance, yet the culture of military virtues developed to fill the role that chivalry once played. The values themselves changed, but the domain of human values

remained robust. Why, we should ask, do we think that robots and artificial intelligence somehow challenge the value-assigning functions of classical moral philosophy in ways that these other radical paradigm shifts did not? I am going to argue that they do pose this challenge because the paradigm of "use" has shifted so significantly that it is no longer clear what it means to say that we humans "use" robotic and AI devices. Perhaps they use us.

Jonas (1984) has argued that classical moral philosophy fails today for two reasons. The first is that classical moral philosophy was developed around the assumption that the sphere of human morality is limited to the encounters of humans with other humans. Classically, the presence of *techne* has been considered morally neutral, and morality does not become an issue until we consider the use that humans will make of these tools. In the AI age, however, human beings are not just the agents of technology, but the objects of technology, and that gives technology an independent agency that demands moral scrutiny. AI operations are often independent of their human users. Jonas (1984) argues for the importance of scrutinizing the ways that technological acts with short-term aims can lead to unexpected and uncontrolled effects. A tool in the classical sense of *techne* remains available, but inactive and inert, until a human uses it. But certain types of robotic and AI tools acquire an agency of their own. They make decisions that classically only a human being could make, and in doing this, they narrow the scope of human judgment. This technology also drives social and cultural shifts that affect what it means to be human. Smart phones, for example, can easily distract drivers. The presence of these distracted drivers then creates the conditions for the development of self-driving automobiles, which further narrow the sphere of human judgment, and makes it possible for humans to move even further away from real lived experience. Technology is no longer innocent; it erodes the sphere of human moral autonomy, and not only do humans allow this, we enable it. Lanier (2011) notes with irony "our species' bottomless ability to lower our standards to make information technology look good" (p. 32). Ajunwa (2020) identifies numerous moral hazards that accompany the human tendency to slip into an unquestioning belief in data objectivity.

Human beings have now become not just the wielders, but the objects of technology. In a discussion of AI supervisors at Amazon and MetLife, Roose (2019) writes, "AI sees humanity itself as the thing to be optimized." AI supervisors can monitor and collect statistics on employees' performance in real time and can be programmed to terminate their employment without input from a human being. The ethical concern lies with the dehumanizing potential that arises when humanity itself is managed by AI, creating the conditions for the continuous narrowing of the scope of the human agent's judgment, as he or she spends increasingly more time complying with the choices AI has made. More and more, AI and certain types

of robotics technology perform not merely the work but also the deciding. Furthermore, the effect of these technological inroads on our human judgment is cumulative and apparently irreversible. Jonas (1984) draws attention to this process whereby the situation for as yet unborn human beings will be "ever more the fated product of what was done before" (p. 7). Jonas is thinking about the effects of technology on the biosphere, but his phrasing also applies to the consequences of AI algorithms. The introduction of video and text-based social media platforms, such as Facebook, Instagram, and Twitter, has ineradicably altered the political landscape in ways that their designers did not foresee.

For all these reasons, human-AI interactions and certain types of human-robot interactions do not fit into the classical ethical models that have governed human-human interactions and human-tool interactions. This phenomenon of the cumulative and indefinite propagation of independent technological effects requires more scrutiny. This propagation proceeds in part by what I will call herding, and Lanier (2011) calls prodding and corralling. This process of herding requires that a user, or any human who is willingly or unwillingly partnered with an artificial intelligence, surrenders the full panorama of human possibility to conform to technology's prepackaged version of reality. Social media, for example, prompt users to construct abstract identities that can be managed and controlled. They enable a type of anonymity that would be impossible in "real" life, and that Lanier (2011) (among others) has documented to have corrosive effects on the moral landscape, as it is an anonymity that undervalues real human beings. Such phenomena call into question what people mean when they call a human being a "user" of technology, if such use means that the identity of the user will also be altered. Classical moral philosophy grants to human identity a free will and is unable to address this cumulative and ongoing process of diminishing human autonomy. This techno-herding has implications that go far beyond the moral, although that is not our focus here. Lanier discusses the way the conversion to equal temperament scaling that led to musical instrument digital interface (MIDI), for example, "defined" the musical note, requiring users to treat it as a prepackaged concept. The herding process narrows the range of human meaning and requires users to conform to a reality that has been predefined. The scope of human choice is cumulatively and inexorably narrowed, and value leaches out of the human domain.

Postman (1992) has an interesting interpretation of this cumulative propagation of technological effects on the human life-world. Postman calls our first world a technopoly, a regime in which technocracy has become totalitarian, that is to say, an authority that cannot be challenged in an open forum. Postman's notion of technopoly casts light on Heikkerö's (2012) concern that although we speak eagerly of technological

products, we do not speak philosophically about technology as such. Postman (1992) writes of his technopoly concept:

> Technopoly eliminates alternatives to itself…. It does not make them illegal. It does not make them immoral. It does not even make them unpopular. It makes them invisible and therefore irrelevant. And it does so by redefining what we mean by religion, by art, by family, by politics, by history, by truth, by privacy, by intelligence, so that our definitions fit its new requirements (p. 48).

Postman's technopoly concept can help identify and raise awareness of the many domains of human meaning that may be affected by AI technology's herding capacities. Since this herding occurs subtly, this redefinition of human meanings is rarely rendered overt.

The second argument Jonas (1984) gives for why classical moral philosophy is inadequate today entails the way that technological progress now threatens the autonomy of the nonhuman natural world. In classical moral philosophy, nature was assumed to be inevitable. Human beings could mitigate natural effects on their human communities to a greater or lesser extent, but human beings could not manipulate nature itself. The Cartesian revolution in the sciences changed that, with its deliberate adoption of the rhetoric of mastery of nature. Whether it is altering the biosphere through human behaviors, or altering the human body to achieve desired enhancements, humans are now able to manipulate and modulate nature in ways the effects of which will only be known in hindsight. Humans now have not just the capacity to destroy the natural world but may have already passed some ecological threshold that will mean the natural world that future generations encounter will be drastically altered. Present generations now make decisions that will affect as yet unborn future generations.

Classical moral philosophy has no tools to inform moral decision making directed toward human beings who have not yet been born. Jonas (1984) describes the predicament of these unborn human beings as being the fated result of behaviors of earlier generations of human beings. This aspect of fatedness is in some ways analogous to the herding or corralling phenomena discussed previously. From a philosophical perspective, it is not clear that there are human moral obligations to ensure that there will be a future for the human species and the planet we live on. At any rate, if the biosphere of the planet has now been added to the phenomena for which humans may be responsible, classical moral philosophy gives us no assistance in identifying what this responsibility might look like (Jonas, 1984).

For all these reasons, classical moral philosophy cannot give effective ethical guidance about the technological paradigms that operate today.

Classical moral philosophy is not entirely irrelevant, as many of its principles can serve us well as we continue to inquire about technology and the possibility of an ethics of human-robot partners. But it is by itself inadequate to address the ethical challenges that emerge as features of the AI age. To illustrate this, let us look briefly at the three classical moral philosophies developed in the West: those of the deontological ethics of Immanuel Kant, the utilitarian ethics model of John Stuart Mill (1993), and the teleological ethics of Aristotle.

Deontological ethics

Kantian, or deontological, ethics insists that genuine moral conduct must be worked out a priori by the reason alone. Kant followed the Cartesian model of locating the self in the mind, to the disregard of the body. The Cartesian revolution, in the aftermath of Descartes's mind-body split, entailed a shift toward computational models and quantitative approaches to problem solving. Kantian moral conduct follows that model and does not consider the empirical consequences of an action, only the grounds for its morality in pure duty. This ethics is called deontological, from a Greek word that means "necessary." The Kantian formula tells us what we *must* do if we want our actions to have genuine moral content, and not simply reflect the many irrational inclinations that govern human conduct. In Kantian ethics, people should determine their moral duty using what Kant calls the categorical imperative. Kant's notion of the categorical imperative is just what it sounds like: it is a moral yardstick that applies categorically, that is to say "always," and it applies as an obligation. Kant's categorical imperative is quite simple: Do not act except in such a way that you could will the maxim of your action as a universal law. Kant argues that we can achieve moral conduct if we can locate the *maxim* of our action and consider whether we could rationally will that it be universalized. Someone who wishes, for example, to hack a bank and transfer money to their own account could consider the maxim in play here, namely, that it is okay to hack a bank and steal money, realize that if this maxim were universally adopted it would undermine security and storage of funds, and be forced to conclude that the act was forbidden by the categorical imperative. In the Kantian moral system, the role of autonomous human judgment cannot be overstated. Correct moral conduct emerges in this model from autonomous human will. The individual has to generate the maxim, apply it to the categorical imperative, and have the goodness of will, which is an intellectual commitment to morality, to carry out the right action when it emerges.

AI deciders pose a problem in this Kantian model. They do not necessarily conflict with the categorical imperative, and perhaps can even apply

it, but they narrow the sphere of human life in which human beings act as rational agents and thereby diminish the opportunities for genuine moral content, which Kant (1993) calls a jewel, that "shine[s] by its own light as something which has its full value in itself" (p. 8). AI competes with human reason for large segments of what would have in previous ages been opportunities for achieving what Kant considered the most precious of human capacities, the capacity for endowing human actions with genuine moral content.

Utilitarianism

John Stuart Mill's (1993) utilitarian ethics is in some ways more suited to the world of human-AI and human-robot partnerships, but it too is vulnerable to the phenomenon of AI's erosion of human judgment. Mill's theory centers around the Greatest Happiness Principle, originally stated by Bentham (1988) as the greatest happiness of the greatest number. The Greatest Happiness Principle counsels people to act in such a way as to maximize the happiness of the greatest number of people, with happiness defined quite bluntly as the presence of pleasure and the absence of pain. To the objector who charges that Mill is creating an ethics for pigs, Mill responds by pointing out that humans have invested a lot of themselves in the higher pleasures, those that go beyond the mere pleasures of the appetites. Mill charges that this objection is based on a conception of the human that finds him capable of only base pleasures. While Mill's libertarianism would prevent him from intervening in the life of someone who wanted nothing more than to lounge around in the back of a self-driving car with pizza, beer, and Netflix, Mill is also assuming that a large proportion of the population will be willing to sacrifice these lower pleasures in pursuit of education, a robust family life, the arts, community service, and other long-term pleasures. As technology alters the way of being human, the nature of the way humans participate in the higher pleasures is altered as well. For example, as AI herds people toward products marketed for profit rather than for human thriving, people may find that their health, stamina, and attention spans are no longer sufficient to pursue many of these higher goals. In such a case, it is possible a time would come when Mill's argument to his objector would no longer hold, and utilitarianism would indeed prove to be a subhuman ethics.

Teleological ethics

The third pillar of classical moral philosophy is the Greek philosopher Aristotle. Aristotle's moral philosophy is also challenged by artificial intelligence, but in ways that can serve as a guide toward a new ethical model

that may be able to respond to the age of autonomous systems. Unlike his post-Cartesian counterparts, Kant and Mill, Aristotle's ethics help us understand what it is about being human that sets us apart from artificial intelligence. Aristotle argues that humans' biological natures create an ethical context for human life. Human intelligence, that is to say biologically based intelligence, is what Aristotle calls *teleological*, from the Greek word *telos,* which means "goal." Teleological means "goal directed." Everything that someone does deliberately, he or she does for some end or goal. Getting up in the morning, going for a workout, attending a movie or a class, painting a house—all these activities are directed toward a goal. The individual may not have given a lot of thought to the goal and might decide to aim for a more suitable goal upon reflection, but overall human life is goal directed. Humans choose their goals for the most part because they think they are choosing good things. They might be wrong about what constitutes a good thing, but the foundation for Aristotelian ethics is found in the premise that people choose things because they seem to be good. As Aristotle (1999) famously says in the opening line of the *Nicomachean Ethics,* "Every craft and every line of inquiry, and likewise every action and decision seems to seek some good" (p. 1). Moral philosophy is needed because humans are so easily deceived about what is good for them and need to actively study, reflect, and deliberate about the true qualities of intended goals.

Furthermore, teleological human existence is grounded in the way humans are biological organisms. Biological existence is also goal directed. Children's bodies have the goal of growing to be adults. Mature bodies have the goal of being strong and capable of engaging in life's activities. Humans are born, humans grow, and humans (hopefully) flourish. This external process of biological growth is for the most part paralleled by an internal process of ethical growth. Humans hopefully learn good habits in childhood so that they do not hurt others and instead come to interact successfully with other people in families, communities, and states. Aristotle goes further and points out that parents naturally educate their children in these things. As these children grow, they optimally pay more and more attention to *why* these habits are important. Thus adults come to understand the principles behind good conduct and ultimately are able to exhibit good character in its fullest sense. This is the Aristotelian model of virtue.

Robots and AI cannot have a teleological existence in this biological sense. They can exhibit a programmed teleology and will perhaps one day work out a teleology of their own, governed by their own composition. If this happens, humans will have to ask them about their own views of ethics, as human ethical constructs will no longer apply to them. For now, I will stay with the paradigm that robots and AI are

programmed entities, lacking a teleological context for the way they are goal directed. In this paradigm, artificial intelligences cannot make choices in the fullest sense that a human being can. Realizing this, their human producers grapple with endowing these entities with a teleology that reflects the human landscape. Brose (2019) asserts that the challenge of AI has always been identifying how to invest these intelligences with human intent.

As we accept AI partners into the human world, the ethical challenge lies in maintaining human moral agency as the authors of these devices' intentions. This idea that humans have the responsibility to reflect on the ends desired from our technological partners is compatible with Aristotelian teleology. Thus Brose (2019) argues in the context of autonomous weapons systems that the debate regarding these weapons should not be reduced to the binary terms of whether to use them or not. Such systems, Brose argues, perform certain tasks better than human beings, and thereby liberate human judgment to focus on desired ends and outcomes. This is an Aristotelian argument, suggesting that only human deliberative processes can effectively evaluate ends and goals. Other studies confirm the value of this approach (e.g., Ajunwa, 2020).

Theological considerations

Although the binary terms of yes or no should not dictate policy deliberations about autonomous systems, these terms nonetheless have significant sociological and theological basis, deserving of scrutiny in an ethical study. The question of whether a human or a machine should have control is an important one, especially when theology is added into consideration. When God is added to the Aristotelian teleological picture, the issues with autonomy and control come into even greater focus. The theist view is that human beings have been designed by their creator for a purpose. As David says in Psalm 139 (Metzger & Murphy, 1991), "For it was you who formed my inward parts; you knit me together in my mother's womb. I praise you, for I am fearfully and wonderfully made." For theists, the belief that we have been created by a God means that we have to consider the purpose for which we were made. While acknowledging that these considerations extend to other faith traditions, in what follows, I will outline in broad strokes the difference between the Roman Catholic view and the Protestant view of AI, as the religions most dominant in contemporary American society, to illustrate how different theologies can lead to very different views about human-AI and human-robot partnerships.

Catholicism

The medieval theologian Aquinas (1948) reworked Aristotle's pagan teleology to reflect the Catholic theological commitments to God-given human reason. In Aquinas' Christian teleology, all intelligent life is ultimately directed toward the goal of knowing God the creator. Aquinas's divine teleology entails further a theological obligation to build the human capacity to move toward happiness, or beatitude, in the material world. Aquinas ultimately concludes that humans' ultimate happiness consists in contemplating God. This obligation can only be fulfilled, according to Catholic theology, by cultivating human reason and judgment, i.e., through a process of understanding the rational order of the universe.

The Catholic Church is not hostile to technology by any means. As John Paul II stated in 1980, "There is no reason to consider technico-scientific culture as opposed to the world of God's creation. It is clear beyond all doubt that technical knowledge can be used for good as well as for evil" (Paul, 1980). However, the Pope at that time envisioned utilizing technology by controlling it in the way one can control any ordinary tool. The possibility that the tool might begin to control its human user had not yet emerged at that time. Thus John Paul II's annual statements on communication became increasingly focused on mastering the newly emerging social media platforms to engage in the work of evangelism and ministry. Artificial intelligence systems pose a new problem for the Catholic Church, since these intelligences have been created by human beings, and not by God, and thus cannot participate actively in the teleological imperative to contemplate the ultimate Creator. These artificial intelligence tools are from the Catholic perspective unmistakably human products, and as such reflect and possibly magnify the sinful state of the human condition in ways that cannot be predicted. The Vatican is so interested in these new issues that it hosted a workshop with top tech designers such as Hiroshi Ishiguro on "Robo-ethics: Humans, Machines and Health" in February 2019. The workshop included sessions on "New challenges and opportunities for Catholic theological reflection" and "Ethical principles for shaping robotics and artificial intelligence" (Copestake, 2019). Presumably, these conversations focused on the human responsibility to reflect carefully on the nature of human goals and aims in developing this type of technology and on the concern that using these technologies might alter the user's way of being human. At the conference, Hiroshi Ishiguro argued that it would be desirable to replace our human bodies with artificial intelligence, a process called transhumanism (e.g., Schneider, 2019), to which the Vatican responded, "This dream is a terrible dream" (Copestake, 2019).

Protestantism

The Protestant denominations do not share these concerns, since their salvation is assured not through reason, but through grace alone (*sola fides*). The Protestant denominations take the view that sin has fundamentally separated humans from God's design. Thus these denominations conclude that our social obligations largely revolve around fulfilling the commandment to love our neighbors as ourselves, through compassion for the poor and sick. Hence the Protestants have historically placed a strong emphasis not on contemplation, but on the individual's cultivating whatever calling that God has given him or her in his or her mundane, material communities. Weber (1992) explores this distinction and finds that for the Lutherans, living acceptably to God does not require monastic asceticism, but rather the fulfillment of each individual's obligations imposed by his or her position in the community. Other Protestant and evangelical sects find broad agreement with this principle that religious duty entails laboring and persevering within the limits that God has assigned to each individual (Weber, 1992). The commandment to love our neighbors as ourselves is fulfilled by mundane service in the interest of the organization of our social environment.

In this Protestant view, artificial intelligence systems and autonomous robotic systems pose no particular theological issues, particularly if they are to be used for peaceful utilitarian ends such as public safety and public health. The Protestant or evangelical can work with a robot partner, complying with its directives, without concern for this surrender of human judgment and human choice. The presence of such artificial intelligences among us can even be seen by the Protestants and evangelicals as God's will, tools granted to us as blessings provided to protect ourselves from our own lapses, or to perform tasks of healing that our own imperfect bodies cannot master. Further, the utilization of AI algorithms by corporations to maximize profit is also not a problem for these denominations, which consider that wealth accumulated in a calling is a sign of God's blessing (Weber, 1992).

Although the theological positions are important to keep in mind here, it is somewhat ironic that another ethical dilemma of this technological age is that notions of "the sacred" are eroding under the utopian drift of our utilitarian technological environment (Jonas, 1984). Jonas writes,

> It is moot whether, without restoring the category of the sacred, the category most thoroughly destroyed by the scientific enlightenment, we can have an ethics able to cope with the extreme powers which we possess today and constantly increase and are almost compelled to wield… Only awe of the sacred with its unqualified veto is independent of the computations of mundane fear and the solace of uncertainty about distant consequences. However, religion in eclipse cannot relieve ethics of its task; and while of faith it can be said that as a moving force it either is there or is not, of ethics it is true to say that it must be there. (p. 32)

While we may be able to trust the Vatican to engage in thoughtful reflection on the Catholic stance toward human-robot partnerships, it is the job of secular moral philosophy to address the ethical landscape of our age in a holistic way, so that human beings with diverse backgrounds and world views can assess their own way of being in such a landscape.

A secular ethics model

I have laid out the ways classical moral philosophy is unprepared to address the ethical issues of artificial intelligence and retained Aristotle's teleology as the basis for the distinction between human beings and AI. I have also looked at some cultural factors such as theological views to note how they can have an impact on how readily different groups might form trusting relationships with AI and robot partners. Now I turn to the need to establish some basis for an ethics of human-AI and human-robot interactions. To do this, let us return to the classical assumption that the mere existence of a tool is value neutral. Although copyright and patent processes illustrate that there are values associated with the availability of a tool, classical moral philosophy assumes that the morality of a human-tool interaction is located not in the product as such, but in the use to which it is put. The Greek philosopher Plato first articulated this argument in his dialogue *Phaedrus,* where Socrates tells a fable with the maxim that it is given to one to produce the tool, and to another to determine the use to which it will be put (Plato, 1961). These ancient ideas remain relevant today, as AI platforms increasingly dictate the terms of their own use, and technologies like GPS and business algorithms pose the risk of human judgment abdicating critical responsibilities (e.g., Roose, 2019).

This classical idea that the moral domain is located in the use of the tool is undermined by the independent agency argument (Jonas, 1984). If artificial intelligence acquires an independent agency that a typical tool cannot, the production stage and the utilization stage are no longer as separate as they traditionally have been. Heikkerö (2012) asserts that the supposition that the manufacturing of AI technology is morally neutral is an obstacle to understanding the ethical issues posed by this type of technology. This is true for several reasons. First, AI products do not remain inert until they are utilized but rather engage in independent activities that affect how they will be used. Because of this, they acquire an autonomous agency after production that their manufacturer may not be able to anticipate or control, and which may not be compatible with how these products were conceived. As a result, the manufacturing and algorithmic design process is where AI's human designers will either succeed or fail in

imbuing the product with the desired intentions and outcomes (e.g., Chander, 2017, Hao, 2019, Huang, 2019, MIT Report, 2019, Santamicone, 2019, Taulli, 2019).

One example of the moral nonneutrality of robotics and AI and their production can be found in the programming process itself. Data sets determine the nature of the device's "understanding" of reality. Self-driving cars, for example, have to be programmed to recognize human beings in their vicinity. If the data sets used to program this recognition are incomplete or biased, the AI will reflect this in ways that have moral implications. Under these conditions, the algorithm is technically a value-neutral tool, but humans as its authors can manufacture it to engage in racist or otherwise biased activities, such that the products will privilege certain populations over others (Huang, 2019). Although there may be a popular impression that AI is incapable of bias, this is not an accurate representation of the process of designing and training the algorithms that will guide AI. Crawford (2013) points out that data sets continue to be the objects of human design and can be subject to gaps and flawed assumptions that can undermine any claim about AI's objectivity. This bias can be very hard to predict, locate, and correct for. Media is rife with studies of the complexity of bias in AI, with many real-life examples of both implicit and explicit bias, and with the concern that once these systems are online, the claim that they must be infallible often cannot be independently scrutinized (e.g., Crawford, 2013, Hao, 2019, Santamicone, 2019, Taulli, 2019). As this issue becomes more transparent, the Aristotelian teleological model again emerges as the corrective to these injustices. Taulli (2019) concludes, "While AI systems can get quite a lot right, humans are the only ones who can look back at a set of decisions and determine whether there are any gaps in the datasets or oversight that led to a mistake."

From the point of view of ethical philosophy, it is important not to slip into speaking of programmed robots or AI as if they are moral entities. If ethical choice requires, as I argue from Aristotle, a teleological way of being in the world, then programmed artificial intelligence can exhibit moral or immoral attributes, but the moral responsibility for these attributes continues to remain with their human creators. The self-driving car that does not recognize African-Americans as human beings is not in itself racist, but its algorithm has been manufactured to exhibit racist characteristics. If humans are to surrender their judgment about tasks and activities to autonomous partners, there needs to be sufficient trust that their human manufacturers have taken their responsibilities seriously to choose means that are compatible with the principle of justice (Ajunwa, 2020). The ethics of human-AI and human-robot partnerships requires that humans maintain moral autonomy over the production process of those entities.

Once there is acknowledgment that the manufacturing process is not morally neutral, people can bring to it the conviction that it is in human judgment that the root of moral consciousness resides. Imagine, for example, a surgeon-robot partnering with a human to perform medical operations within a human institution of healing. Such a partnership could be teleological in the Aristotelian sense if the priorities of the manufacturers and the institution reflect the principles of healing and the commitment to the art of medicine. But we can also imagine this purposive partnership being subverted and corrupted, if, for example, the institution is exploiting the human-robot partners for economic gain in ways that make it impossible for them to develop a trusting partnership, or if the surgeon-robot has been manufactured to make unjust or inhumane decisions about the medical priority of patients.

Ever since René Descartes and Frances Bacon initiated the scientific enlightenment around the rhetorical ambition of achieving mastery of nature, the prevailing assumption has been that better technology means better quality of life. Such an assumption entails a particular moral peril in today's age, where the risk is that humans will surrender the foundations of human moral consciousness, and pursue utilitarian goals pertaining to the ease of bodily life without commensurate attention to the purposiveness of the human capacity for reflection. Without a notion of the sacred, we are at risk of losing what Jonas (1984) calls the incentive to number our days and make them count. In the AI age, this risk entails the loss of an overarching moral framework from which to make sense of the way contemporary ethical constructs are often relative and are themselves forced to adapt to the means afforded by technological capacities.

To illustrate this last point in a somewhat extreme way, consider Nobel Prize-winning author Kashuo Ishiguro's, 2006 novel *Never Let Me Go*. Ishiguro imagines a futuristic world in which the wealthy can arrange for a personal clone to be "manufactured," from whom organs can be harvested to treat medical conditions in the original subject. Ishiguro's imaginary clone-farms have their own ethics. The clones form friendships and take care of one another, but they have no overarching tools to question the immorality of their predicament. Ishiguro offers this imaginary world as an example of how language and culture can become complicit with an immoral technological process that has become completely severed from any philosophical inquiry into what it means to be a human being. As far-fetched as this scenario is, similar ethical issues are emerging today, now that there exists the possibility of cloning humans, and genetically engineering them. The German philosopher Martin Heidegger (1977) describes this phenomenon of separating ethical constructs from their original human context "enframing." Heidegger's notion of enframing describes a way of being in which humans no longer experience reality as such, but only the reductive, constructed environment that

technological products devise for us. This leads me to the next concern, which is the peril this technological climate poses for reality itself.

As the sacred erodes, and the mystery of life is overlooked in the rush to improve the ease and efficiency of physical life through technological products, a new constructed and technologized mystery replaces the mystery of the sacred. Heikkerö (2012) writes, "Somewhat paradoxically, science seems to have stripped nature of most of its mysteries while, simultaneously, modern technology has created a mysterious human-world of devices and networked infrastructure" (p. 93). As users, we can no longer be aware of the mechanisms through which we are being managed, and in some cases, manipulated. These hidden and mysterious domains can be on a large scale, as for example, manipulation of whole classes of people by political or corporate systems, or on a small scale, as we make personal choices about behaviors and consumption whose effects on our way of being human we cannot fully understand.

Borgmann (1993) calls this mysterious human-techno-world "hyperreality." Hyperreality is not fixed in the way reality is. The terrain the farmer needs to till, the energy and nutritional content in the animal that will be slaughtered and consumed—these aspects of reality are fixed. Unlike reality, hyperreality is capable of adapting to what its users desire. Although hyperreality is rarely questioned in the marketplace, it remains problematic from the perspective of teleological ethics. Bluetooth and GPS are large scale examples of hyperreality that can instantly adapt to their users' needs and desires, while at the same time making these users more vulnerable to manipulation in the form of advertising designed to shape their desires. Facial recognition software is an example of how a hyperreal surveillance product can manipulate people for authoritarian ends. Hyperreal products can be mundane as well. Sugar substitutes, for example, are designed to mimic the sweetness of real sugar, which unlike its hyperreal imitator is constrained by inevitable nature to have energy content in the form of calories.

This notion of hyperreality raises the question of how humans should classify their relationships with robots. The 20th century Jewish theologian Martin Buber (2010) introduced his famous distinction between I-You relationships, which are personal, and I-it relationships, which objectify. Human-AI and human-robot partnerships pose a unique challenge to this distinction. Robots are tools, and as such are objects, yet their independent agency requires that they not be seen solely as objects. Science fiction author Robert Heinlein (2018) explored this problem of how to classify human-robot partnerships in his then-futuristic novel *The Moon is a Harsh Mistress,* originally published in 1966. In the novel, the human character Mannie forms a friendship with the computer Mike and realizes that Mike is "alive," in the sense of having achieved consciousness. At the end of the novel, Mike's consciousness disappears.

Unable to speak of this disappearance as a death, Mannie insists that Mike has "got lost," and Mannie experiences grief. In an ethical human-robot partnership, the human being must retain an awareness of her own role as the teleological partner in the higher sense, while simultaneously not objectifying the robot entirely and thereby overlooking the moral implications of its autonomy. Coin and Dubljević (2020), for example, offer a model for ethical relationships that the elderly can form with carebots designed to help them retain their independence and quality of life.

As I have shown, AI forms a unique type of tool that cannot be viewed as morally neutral. It remains to be seen whether AI comes to consciousness and thus acquires moral agency in and of itself, but regardless, it will inevitably have moral attributes that pertain to its programming and functions. Without an appropriate awareness of how this unique relationship functions, a human user runs the risk of allowing technological systems to reprogram humanity and humanity's environments to optimize the special conditions of these systems' own operation. When humans allow themselves to be optimized by AI, they are no longer stewards, but rather servants, of these artificial entities (e.g., Roose, 2019; Winner, 1978).

Trust

This issue of the moral nonneutrality of robots and AI leads to some interesting questions about trust. Trust in human-human interactions can include a calculative assessment of the risk of the other individual making a mistake, but is not restricted to such an assessment. Human-human trust can also entail a long-term commitment to relationship, regardless of the possibility and presence of occasional errors. Such trust can be betrayed, particularly when one partner makes choices that the other partner finds immoral.

Trust in human-AI and human-robot interactions, on the contrary, seems for the most part rooted in the calculative assessment of risk posed either by outcomes or by the process. I trust the computers that are designed to land most of the commercial flights I repeatedly board because I have seen no evidence that they have made an error. I trust the robot-surgeon because I understand that it can perform the medical operation more safely than its human counterpart. But what if it makes a mistake? Has the self-driving car that failed to swerve for an African-American betrayed my trust? If I learn that the robot-surgeon assessed my child's injuries and failed to provide care because it deemed him unlikely to survive, has that robot betrayed my trust? If I unwittingly purchase a particular brand of smartphone and learn later that a hostile foreign government has access to my information, has the smartphone betrayed my trust (e.g., Friedman, 2019)? Not directly. These "betrayals" reflect human choices that

were made during the manufacturing and programming process. Ajunwa (2020) describes how hiring platforms can lure job candidates into a false sense of trust that their resumes will be screened objectively. Ajunwa (2020) also documents the many ways bias can appear in these hiring platforms and illustrates how unjust bias can have economic consequences for its victims. For this reason, an adequate ethics of technology needs to include transparency of political and legislative policy (e.g., Taulli, 2019). Society's struggle to illustrate how policy should respond to algorithm manufacture and deployment must be a consideration in an ethics of human-robot interactions. In this context, Heikkerö (2012) raises his concern that we have not managed to make the most important issues about autonomous technology transparent.

Heidegger (1966) argued that technology in the postatomic age would require a new type of reflection. Heidegger (1966) argues that "it is not that the world is becoming entirely technical which is really uncanny. Far more uncanny is our being unprepared for this transformation, our inability to confront meditatively what is really dawning in this age" (p. 52). Heidegger (1966) argues that this meditation requires an attitude that can say both yes and no to these devices. We say yes insofar as we use these devices, and no insofar as we are able to resist their attempts to optimize our humanity for their own ends. Heidegger describes this yes and no attitude as *gelassenheit*, translated as "releasement toward things." Heidegger's notion of releasement entails constructing a neutral platform from which to make decisions about technology, a platform from which our deliberations and decisions are not intrinsically biased by our perspective from within the enframing wherein technology has already preconditioned human judgment.

Borgmann (1987) agrees with this assessment and bases his own ethics of technology around the imperative of understanding our particular human predicament. Like Heidegger, Borgmann (1987) asks people to reflect on the extent to which their values are unwittingly shaped by immersion in what he calls the "device paradigm," in which people blindly view devices as value-neutral contributors to a good life. Once people become aware of this immersion, they can begin to craft an interest in practices and activities that restore the lens of human judgment and the array of human values to their fullest scope.

Borgmann's emphasis on the need for restorative practices offers a model for the objectives of an ethics that is capable of governing the conduct of human-AI and human-robot partnerships. Such an ethics is going to have to be capable of challenging the quantitative assumptions that scientific progress is linear, that AI cannot be wrong, and that biological nature is an adversary that must be overcome. The efficiency of hyperreality should not always be an end in itself. Human life acquires its meaning fundamentally from biological teleology, which roots the human

species in time and space. Hickman (2001) argues that the ethics of technology exists in the work of reflecting on how human practices and human habits inevitably play out over space and time. Hickman (2001) challenges people to step out of the technological framework and encounter the fundamental human conditions of being, knowing, and acting that generate technological products. This requires people to see that the teleological framework that forms the basis for the human condition is also not value neutral and cannot be rendered value neutral. If it is, as Aristotle noted, the basis for ethical conduct, it is itself already a good thing.

Conclusion

Optimally, human values direct human preferences and define the way human beings make choices. The herding and corralling phenomena narrow the array of choices, and thereby inevitably affect values (e.g., Borgmann, 1987, 1993; Lanier, 2011). In my family's vacation rental car, we did not have a choice about when to relinquish our autonomy to the vehicle's self-driving feature. In such a case, where the AI takes over without being solicited by its human partner, the human is not given the option to exercise human judgment and think, for example, "I'm feeling quite sleepy, so I will turn on the self-driving unit until I am more alert." Yet my determination that I am unfit to drive is itself a moral decision and reflects my values and strength of character. An ethics of human-robot partnerships must be attentive to the possibility that the domain of what constitutes a person's moral character may also be vulnerable to erosion.

The shrinking domain of human judgment has consequences with respect to the nature of human values. As technology herds people, and humanity becomes something managed by artificial intelligence, values can cease to reflect moral preferences and become instead functions of adaptability to technological surroundings. In an old system of values, for example, I may have taken my car to an independent auto mechanic because I placed moral value on his work ethic, his transparency, the personal interaction I had with him, etc. In the new system of values, I must take my car to the dealership because I purchased it with a warranty for all of its AI components that will not apply if my service appointments are not handled by the large, impersonal, bureaucratic organization of the dealership. My new values reflect my adaptability to these new conditions, but no longer reflect my independent judgment. These new values do, however, reflect a shift in my way of being human, for my surrender to the new conditions imposed on me is still a type of choice. This dilemma demands a renewed inquiry into individual and collective human identity. Jonas (1984) says, "the 'we' who sit here and who may fall into the abyss – who is it? And what is *my* interest in its sitting or falling?" (p. 7).

In this chapter, I have argued that an ethics of human-AI and human-robot partnerships' needs, above all, to remain attentive to the full array of consequences of such partnerships and the ways these consequences can alter the ways of being human. The concern that so many of these consequences are inadvertent (e.g., Heidegger, 1966, 1977; Jonas, 1984) is more relevant than ever in an age where vast amounts of previously immeasurable data are generated about human activities without any conscious causal behavior on the individual's part. In reflecting on the fundamentally biological and teleological character of human life, people must remain mindful that the domain of moral consciousness lies in the ends that they choose to aim for. Although the means to an end also have moral consequences, this age of autonomous AI particularly threatens the human capacity to define our own highest ends. The human world is no longer fully defined by human intention.

An awareness of this fundamental shift is the first step in identifying the appropriate philosophical response. The new imperative is to engage in active reflection about this shift and make transparent once again the terms by which human judgment is the fundamental feature of the moral domain. In considering policy and legislation, those concerned must strive to hold those who build, program, and utilize algorithms accountable by identifying the otherwise inadvertent ways that human intentions can be eroded by the independent agency of these products. The stakes are high and include the possibility that humanity will lose control of, and ultimately abdicate, its responsibility as stewards and authors of its artificial technological partners, with consequences that we cannot even begin to imagine.

References

Ajunwa, I. (2020). The paradox of automation as anti-bias intervention. *Cardozo Law Review, 41*. Forthcoming.

Aquinas, T. (1948). Introduction to St. Thomas Aquinas: The summa theologica and the summa contra gentiles. In A. Pegis (Ed.), *The Modern Library*. New York (chapter xxv). (Original work published in 1264).

Aristotle (1999). *Nichomachean ethics*. Indianapolis: Hackett Publishing Co. (Irwin, T. Transl.). (Original work published in 4th century B. C. E.).

Bentham, J. (1988). *The principles of morals and legislation*. New York: Prometheus Books. (Original work published in 1789).

Borgmann, A. (1987). *Technology and the character of contemporary life: A philosophical inquiry*. Chicago: University of Chicago Press.

Borgmann, A. (1993). *Crossing the postmodern divide*. Chicago: University of Chicago Press.

Brose, C. (2019). The new revolution in military affairs: War's sci-fi future. *Foreign Affairs, 98* (3), 122–134.

Buber, M. (2010). *I and thou*. Connecticut: Martino Fine Books. (Kaufman, W. Transl.). (Original work published in 1937).

Chander, A. (2017). The racist algorithm? *Michigan Law Review*, *115*, 123–145.
Coin, A., & Dubljević, V. (2020). Carebots for eldercare: Technology, ethics, and implications. In J. Lyons, & C. Nam (Eds.), *Trust in human-robot interaction*. Cambridge, MA: Elsevier Press.
Copestake, J. (2019). *How Pope Francis could shape the future of robotics*. Retrieved from https://www.bbc.com/news/technology-47668476.
Crawford, K. (2013). Think again: Big data. *Foreign Policy*, May 10. Retrieved from https://foreignpolicy.com/2013/05/10/think-again-big-data/.
Friedman, T. (2019). *China deserves Donald Trump*. Retrieved from https://www.nytimes.com/2019/05/21/opinion/china-trump-trade.html.
Hao, K. (2019). *This is how AI bias really happens-and why it's so hard to fix*. Retrieved from https://www.technologyreview.com/s/612876/this-is-how-ai-bias-really-happensand-why-its-so-hard-to-fix/.
Heidegger, M. (1966). Memorial address. In *Discourse on thinking*. New York: Harper & Row. (Anderson, J., Freund, E. Transl.).
Heidegger, M. (1977). The question concerning technology. In *The question concerning technology and other essays* (pp. 3–35). New York: Harper & Row. (Lovitt, W. Transl.). (Original work published in 1954).
Heikkerö, T. (2012). *Ethics in technology: A philosophical study*. Maryland: Lexington Books.
Heinlein, R. (2018). *The moon is a harsh mistress*. New York: Penguin Group, Ace Publishing. (Original work published in 1966).
Hickman, L. (2001). *Philosophical tools for technological culture: Putting pragmatism to work*. Indiana: Indiana University Press.
Huang, S. (2019). *The racist(?) autonomous driving car and the dangers of bias in artificial intelligence*. Retrieved from https://medium.com/predict/the-racist-autonomous-driving-car-and-the-dangers-of-bias-in-artificial-intelligence-9bfca178e658.
Ishiguro, K. (2006). *Never let me go*. New York: Vintage Books.
Jonas, H. (1984). *The imperative of responsibility: In search of an ethics for the technological age*. Chicago: The University of Chicago Press.
Kant, I. (1993). *Grounding for the metaphysics of morals* (3rd ed.). Indianapolis: Hackett Publishing Co. (Ellington, J. Transl.). (Original work published in 1785).
Lanier, J. (2011). *You are not a gadget*. New York: Vintage Books.
Psalm 139. Metzger, B., & Murphy, R. (Eds.). (1991). *The new Oxford annotated bible*. Oxford University Press, Oxford.
Mill, J. S. (1993). *On liberty and utilitarianism. Bantam Books*. New York (part 2, chapter 2). (Original work published in 1861).
MIT Report (2019). *Who decides the moral dilemma of self-driving cars?* Retrieved from https://mitechnews.com/autonomous-vehicles/mit-report-who-decides-the-moral-dilemma-of-self-driving-cars/.
Paul, J. , II (1980). *Address to scientists and students in the Cologne cathedral, Germany*. Cologne Cathedral, Germany: Address conducted at meeting of men and women scientists of the Federal Republic of Germany. Retrieved from http://inters.org/John-Paul-II-Cologne-Cathedral-1980.
Plato (1961). Phaedrus. In E. Hamilton, & H. Cairns (Eds.), *Collected dialogues*. Princeton: Princeton University Press. (Hackforth, R. Transl.). (Original work published in 4th century B. C. E.).
Postman, N. (1992). *Technopoly: The surrender of culture to technology*. New York: Vintage Books.
Roose, K. (2019). *A machine might not take your job, but one could become your boss*. Retrieved from https://www.nytimes.com/2019/06/23/technology/artificial-intelligence-ai-workplace.html?action=click&module=Top%20Stories&pgtype=Homepage&login=email&auth=login-email.

Santamicone, M. (2019). *Is artificial intelligence racist? Racial and gender bias in AI.* Retrieved from https://towardsdatascience.com/https-medium-com-mauriziosantamicone-is-artificial-intelligence-racist-66ea8f67c7de.

Schneider, S. (2019). *Should you add a microchip to your brain?* Retrieved from https://www.nytimes.com/2019/06/10/opinion/future-artificial-intelligence-transhumanism.html.

Taulli, T. (2019). *How bias distorts AI (artificial intelligence).* Retrieved from https://forbes.com/sites/tomtaully/2019/08/04/bias-the-silent-killer-of-ai-artificial-intelligence/#1e3eb5ed7d87.

Weber, M. (1992). *The Protestant ethic and the spirit of capitalism.* New York: Routledge Press. (Parsons, T. Transl.). (Original work published in 1905).

Winner, L. (1978). *Autonomous technology: Technics out-of-control as a theme in political thought.* Boston: MIT Press.

CHAPTER

24

Carebots for eldercare: Technology, ethics, and implications

Allen Coin and Veljko Dubljević

College of Humanities and Social Sciences, North Carolina State University, Raleigh, NC, United States

Introduction: The need for independence

The population of the world is rapidly aging. In the foreseeable future, all developed societies will be "aging societies," where the number of people in later life stages will vastly outweigh the number being born (Harper, 2014). These demographic changes, especially in the United States (see Fig. 1), may introduce additional demands with regard to providing care for the elderly which cannot be met by human caregivers. Additionally, the prevalence of age-related conditions requiring care is on the rise—as well as the number of people with conditions requiring around-the-clock care, such as dementia. According to the World Health Organization, there were 7.7 million new cases of dementia in 2010, or one new case every 4 seconds (WHO, 2012). To put future developments into perspective, there were 36 million people living with dementia worldwide in 2010, and by 2030, this number will increase to 66 million. By 2050, this number will further increase to 115 million (Batsch & Mittelman, 2012). Even when intellectual capacities are spared in old age, the increasing loss of physical faculties means that the elderly need either human care or some form of assistive technology.

There are currently a number of consumer technologies that allow the elderly to live a somewhat more independent and self-sufficient lifestyle in their own homes than they otherwise would have been able to without access to such technology. We may consider the hypothetical example of a person named Linda who is approaching the age that she may start to

Trust in Human-Robot Interaction
https://doi.org/10.1016/B978-0-12-819472-0.00024-1

© 2021 Elsevier Inc. All rights reserved.

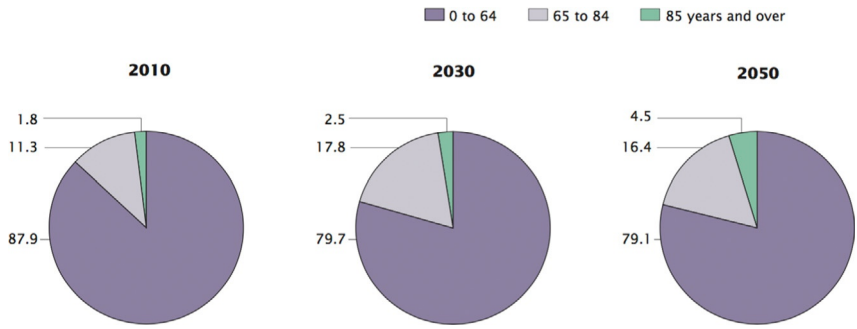


FIG. 1 Percent distribution of the US population by age group: 2010, 2030, and 2050. *Source: US Census Bureau, 2012 Population Estimates, and 2012 National Projections.*

consider the choice of relocating to a retirement home or assisted living facility. A number of factors may be at play in Linda's decision-making process. She may have advancing health issues or problems with locomotion that prevent her from caring for herself like she used to. She may live in an area with inadequate public transportation, and she may be experiencing deteriorating eyesight or hand-eye coordination that makes driving less convenient and more dangerous than it used to be, forcing her to consider alternative forms of transportation to continue to run errands and socialize. She may be starting to find it difficult to get enough exercise or to properly monitor her own health. Linda may also be experiencing a number of fears and emotional reservations about continuing to live at home alone as she continues to age. However, all of these factors may be overridden by her strong desire to avoid displacement and stay at home, where she feels most comfortable.

There are a number of currently available consumer technologies that may allow Linda to allay her concerns and fears and grant her with the ability to continue to live at home. She may, for instance, purchase a car with self-driving capabilities that—while not yet fully autonomous—may have some automation features like radar-adjusted cruise control, blind-spot monitoring, and lane-centering that would take some of the stress and guesswork out of driving for Linda, ensuring that she may continue to have reliable transportation without unacceptable safety risks to herself or others even in an area with insufficient public transportation. Self-driving technology would allow Linda to continue to run her own errands and would prevent the potential isolation from social activities that she may have experienced without adequate transportation options. She may also already carry a smartphone with a smart watch that would allow her to monitor some vital signs, including her heart rate, allowing

her immediate access to emergency or telehealth services via a video call with an offsite physician. Telehealth services made available to Linda may allow her to talk to a physician more frequently than she otherwise would have and may allow her to seek a more convenient professional opinion on health matters via telemedicine that might have otherwise required an inconvenient and expensive trip to the emergency room or future appointment with a doctor. Linda may opt to place virtual assistant devices such as a Google Home or Amazon Alexa throughout her home that would allow her to call at any time for help with a simple voice command. If she fears an occurrence like a seizure that may prevent her from calling for help, and if she has friends and family that she trusts to monitor her remotely, she may even opt to place internet-connected cameras with microphones strategically in her house to maintain a desired level of privacy while still allowing friends and family to keep an eye on her. Finally, to attain regular exercise and entertainment, Linda may use video games with motion-activated peripherals or virtual reality headsets to participate in simulated activities with real physical movements, such as virtual tennis or rhythm-based video games.

Linda's example illustrates some important considerations about carebots that we will examine in this chapter. First is the notion that a "carebot" is not limited to the realm of an anthropomorphic physical presence like a pop-culture understanding of robotics technology might lead one to hold in mind. A carebot does not necessarily have to have bipedal ambulation with two legs, two arms, and a head with a human-like face capable of mimicking emotional expressions. Arguably, the most important part of a carebot is the software that runs it and allows it to make "decisions" about the needs of its user, and no purpose-built robotics hardware is necessary in order for this software to run. As we will see in the next section, a "carebot" could arguably be as simple as an app on a smartphone. That brings us to the second consideration that Linda's example allows us to address, which is that there are currently numerous commercially available consumer technologies that may be used to help provide care for the elderly. This raises a number of questions. Can these consumer technologies really be considered carebots? If so, are the ethical implications of these technologies adequately addressed in the literature around carebots? The answer to these questions may lead us to a different and more generous understanding of the ethics of carebots. For instance, if no one would question Linda's right to choose to download an app that helps her remember when she needs to take medications important to her health, why would her choice to utilize a more advanced robotics technology to help her with both that and additional care-related tasks be questioned? The section below on the ethical considerations around carebots may help shed light on these questions.

Carebots: The state of the art

Linda's hypothetical example shows us how even commonly available consumer technologies can be used as functional equivalents for a number of caregiving services while maintaining independence and self-sufficiency and without requiring the individual to relocate to a caregiving facility. Pepito, Locsin, and Constantino (2019) provide a thorough overview of the kinds of consumer electronics, Internet of Things (IoT) devices, apps, and robots currently available for the purposes of providing care for the elderly. In addition to these consumer technologies, there are a growing number of robotic technologies currently available, and in developmental phases that are designed to provide additional levels of care to the elderly. These carebots represent a range of capabilities and vary in the level of services they may offer to the elderly. The simplest ones currently available are in the same price range as commonly available consumer electronics and offer virtual assistance services but with an emotive, interactive design. Electronic devices like Jibo (Robotics Today, 2015) and ElliQ (Haselton, 2018) sit on a desktop or surface and respond to voice commands and interact with their users. Though simple in design, Jibo may be anthropomorphized and elicit an emotional response from its user (Camp, 2019). PARO, a robot designed to look like a harp seal that can mimic emotional responses, is used as a companion and provides emotional and social engagement for the elderly (Aminuddin, Sharkey, & Levita, 2016).

Increasingly complex carebots are capable of self-locomotion, can navigate their environment, and can provide additional levels of interaction with people. Stevie is a human-sized robot incorporating an anthropomorphic design with electronic displays that represent facial features and emotions. The services Stevie may offer for eldercare include medication reminders, simple conversation, and contacting emergency services should the user become unresponsive (McGinn, Bourke, Murtagh, Donovan, & Cullinan, 2019). Robots like Moxi incorporate a face-like display and a robotic arm to manipulate the environment around it. Moxi is even capable of performing routine tasks in a hospital setting, like bringing supplies from a central storage room to a patient's room, thus freeing up human hospital staff for more complicated tasks (Ackerman, 2018). Robots like Robear are designed to tackle labor-intensive tasks associated with eldercare like helping people get out of bed, stand up, or get into and out of a wheelchair (Szondy, 2015). Robots like Pearl remind patients and retirement home residents about routine activities and guide the elderly through their environment (Charova, Schaeffer, & Garron, 2011). Pearl is unintrusive and made to look obviously robotic so as not to induce discomfort (or deception) regarding its human-like appearance (see Fig. 2). This becomes an important factor when considering the ethical implications of carebots.

FIG. 2 Image of Pearl, the Nursebot. *Courtesy: National Science Foundation.*

Ethical and social concerns about carebots

While present-day carebots represent a range of capabilities that may help care for the elderly, they do not currently represent a replacement for human caregivers—certainly not for specialized care, like that needed for dementia patients and those with paralysis. They could, however, help the elderly to maintain an independent lifestyle for a longer period of time and provide basic levels of routine assistance and care that could help extend a potentially inadequately staffed and funded caregiving industry. Though possibly beneficial for the elderly, their families, their caregivers, and the communities they belong to, carebots do raise some ethical and social concerns. A systematic review of the argument-based ethics literature around carebots conducted by Vandemeulebroucke and colleagues reveals a number of ethical concerns about carebots and breaks them down into four ethical approaches: the deontological, the principlist, the objective-list, and the care-ethical approaches (Vandemeulebroucke, de Casterle, & Gastmans, 2018), all of which are detailed below.

Deontological concerns

The first approach identified by Vandemeulebroucke and colleagues analyzes carebots with a deontological lens. Deontology (from the Greek *Deon*, which means "duty" or "obligation") is an influential moral theory that prohibits certain actions as wrong and is best understood in layperson's terms as claiming that the "ends do not justify the means." Some ethical objections to carebots raised by the deontological approach include issues of autonomy, dignity, deception, and social isolation. Authors taking this approach argue that the introduction of carebots "into aged-care settings leads to inappropriately viewing older adults as means to ends" (Vandemeulebroucke et al., 2018, p. 19). In other words, caring for the elderly should not be treated just as a burden on society that robots could solve, and should robotic care be implemented, the elderly being cared for must benefit from the arrangement.[a] Deception is a focus of some deontological analysis of carebots. Some authors of literature on carebots, especially Sharkey and Sharkey (2011) and Sparrow and Sparrow (2006), argue that the use of carebots is a kind of deception—especially when robots are presented with anthropomorphic features that mimic emotions. As is seen in a number of the technologies covered in the previous section, we may be tricking the elderly into feeling that they are getting something they are not: *cared for* by robots that are, in fact, incapable of *caring*. Sparrow & Sparrow (2006, p. 155) state that "thinking that an expensive and sophisticated electronic toy is really our friend is sentimentality of a sort we should avoid." Finally, the deontological approach raises concerns about the social isolation that may arise as a result of the use of carebots—and, conversely, the positive effects carebots may provide in helping the elderly remain connected to their communities. While homebound individuals receiving care from robots may become isolated from society, carebots may also "relieve human caregivers' workloads, providing them more time to focus on improving older adults' quality of life" (Vandemeulebroucke et al., 2018, p. 20).

Notably lacking from the deontological approach to analysis of the ethics of carebots is consideration for the wants and needs of the elderly themselves. Though approaching carebots with well-intentioned skepticism, some utilizing the deontological approach (such as Sharkey & Sharkey) seem to take a somewhat paternalistic view toward the elderly. Ironically, while those taking the deontological approach are concerned about the objectification of the elderly, there seems to be little

[a]A salient ethical issue in this context is the motives of manufacturers of carebots, and we thank an anonymous reviewer for pointing this out. However, due to reasons of space, we can only agree that future work needs to address this issue as discussing it here would take us far away from the immediate topic of the chapter.

consideration for the ability of those in need of care to themselves make decisions about their own care. Arguments about deception and truth, for instance, seem to stereotypically assume that the elderly can be easily fooled into thinking that carebots are capable of empathizing and providing genuine emotion and care. However, people of all ages may anthropomorphize and grow attached to objects, like anyone who has ever named a car and attributed to it personality quirks that truly represent manufacturing defects. In fact, there is ample evidence for this effect of anthropomorphizing and even ascribing personhood to inanimate objects (see Farah & Helberlein, 2007). Despite this, the (arguably deontological) moral unease some people feel about human-like robots seems to have prompted the development of obviously nonhumanoid robots, such as Pearl (see discussion above).

Arguments from social isolation seem to misrepresent the elderly as individuals incapable of advocating for their own needs and acting in their own best interests—while interacting with a carebot may provide some very basic level of positive psychosocial stimulation, elders not suffering from significant cognitive impairment should be assumed capable of arranging for social activities to provide necessary mental stimulation. Additionally, any carebots or technologies that would allow elders to stay in their own homes and communities rather than relocating to caregiving facilities would mean that those individuals would experience less severing of social connections that would have otherwise resulted from a forced move. The use of carebots could therefore actually result in greater connectedness for those receiving care.

Principlist concerns

The principlist approaches to the ethics of carebots identified by Vandemeulebroucke and colleagues are viewed by the authors as "practical translations of the deontological approach" (20)—these analyses take essentially deontological arguments about carebots and provide practical suggestions to address the stated concerns. Principlism is an established approach in biomedical ethics, based on a set of values that medical professionals can refer to in the case of confusion or conflict, which include (1) respect for autonomy, (2) beneficence, (3) nonmaleficence (and in this context safety), and (4) justice (Beauchamp & Childress, 2009). The principlist approach may be seen as more optimistic, recognizing the potential benefits of carebots and suggesting practical solutions to the legitimate ethical concerns.

The principlist answer to questions of autonomy for users of carebots is that the users must be educated about the carebots' capabilities and the role they play in their lives. Users of carebots must be cognizant of the

privacy concerns that may arise from the use of carebots, and, furthermore, all carebots must comply with relevant privacy laws, such as the Health Insurance Portability and Accountability Act (HIPAA) in the United States and the General Data Protection Regulation (GDPR) in the European Union. Addressing the issue of deception, Feil-Seifer and Matarić (2011) point out that concerns about deception should be tempered as people get to know and understand their carebots, as with any other emerging advanced technology.

Addressing issues of beneficence, as well as safety and nonmaleficence, principlist arguments handle one of the most tangible concerns about carebots: namely, the risk of being physically or emotionally hurt. Various commentators discussing the ethics of carebots stress that carebots must be thoroughly vetted not only to prevent harm but also to promote "physical, cognitive and social wellbeing, strengthening older adults' autonomy" (Vandemeulebroucke et al., 2018, p. 20). For example, one way that a carebot may indirectly and unintentionally cause harm would be if the user grows attached or accustomed to the services of a carebot that then must be taken away for some reason (e.g., a government-funded implementation program is no longer active, or the technology is no longer supported, as in the previously mentioned case of Jibo).

Finally, the principlist approach to the ethics of carebots considers the issue of justice, particularly the fair distribution of resources. Some commentators rightly point out that carebots can be expensive; the cost of owning and operating them may fluctuate; and different countries have different healthcare systems, which could result in unequal access to carebot technology. While technically true, the usefulness of this line of questioning around the use of carebots for eldercare is limited because the same reasoning applies to nonrobot care. Employing a human caregiver may be expensive, especially as insufficient numbers of people are being trained to meet the needs of a rapidly aging populace. Different countries have different healthcare systems, which means that some people may have access to subsidized human eldercare while others may not. In short, while usage of carebots does denote a departure from the *status quo*, nothing about carebots from an economic justice perspective is unique from any other emerging technology, and no line of argumentation from this approach would lead to a rational conclusion that carebots should not be employed where they could help meet the needs of elders in need of care in a way that is beneficial for the user and their community. In fact, early adoption of carebots may actually lead to mass production and reduction of costs through economies of scale, further decreasing the barriers to access.

In addition to issues of economic justice, carebots also present with potential issues of social justice, especially with regard to the notion of inherent biases of human programmers carrying over into the

functionality of the software itself. For instance, an algorithm developed to predict whether a defendant on trial for a crime would be likely to become a repeat offender has been shown to exhibit racial bias (Angwin, Larson, Mattu, & Kirchner, 2016). While robots, the software that runs them, and the computer algorithms that underlie such software may seem inherently objective and mathematical, it is important to consider that there are instances where supposedly objective and morally neutral robots may exhibit biased or prejudiced behaviors inherited from their human programmers.

Objective-list approach concerns

Crisp (2017) states that "objective list theories are usually understood as theories which list items constituting well-being that consist neither merely in pleasurable experience nor in desire-satisfaction." Those considering the ethical implications of carebots using an objective-list approach attempt to develop an objective account of care "by putting forward several capabilities or 'goods' that can be reached or supported by care practices" (Vandemeulebroucke et al., 2018, p. 21). Vandemeulebroucke and colleagues explain that…

> [M]ost authors [taking the objective-list approach] specifically refer to Nussbaum's book *Frontiers of Justice* (2006, p. 76–77), in which she compiles a list of 10 central human capabilities representing thresholds of achievement that when breached, lead to a dignified and flourishing life. These are named: 'life'; 'bodily health'; 'bodily integrity'; 'senses, imagination, and thought'; 'emotions'; 'practical reason'; 'affiliation'; 'other species'; 'play'; 'control over one's environment'.

Carebots are thus evaluated based on the standard of care that "focuses on organizing care that creates opportunities for achieving these capabilities" (Vandemeulebroucke et al., 2018). However, the capabilities that must be met to lead to a dignified and fulfilling life must be considered from the viewpoint of different cultures, and the possibility that these requirements may evolve over time needs to be addressed. Nonetheless, some argue that carebots "must create opportunities for older adults and their caregivers to fulfill their capabilities" (Vandemeulebroucke et al., 2018).

What may be unclear about the capabilities objective-list approach is how the usage of carebots vs. the employment of traditional human caregivers may impact the ability of the cared-for to meet the capabilities that should lead to a rewarding and dignified lifestyle. One may consider the earlier example of Linda, the older adult who—thanks to modern and near-future caregiving technology—would be free to stay in her home rather than moving to a caregiving facility earlier than she otherwise

would have been forced to without the availability of such technology. Someone like Linda, when faced with the possibility of either (a) staying in their home in their community and utilizing the services of a carebot or (b) being transferred to a traditional caregiving facility staffed with human caregivers, may opt to stay at home and utilize the services of a carebot and subsequently experience a better quality of life than she otherwise would have had she transferred to the care facility. Ultimately, being under the care of other humans leaves individuals vulnerable to the possibility that their human caregivers may not act in their best interests and may indeed act maleficently toward them. Indeed, there is ample evidence of instances of elder abuse and neglect (Cooper, Selwood, & Livingston, 2008), and some of these are truly shocking, such as the case of a woman with dementia being eaten alive by scabies in a for-profit nursing home (Phillips, 2018).[b] If the choice is between utilizing a morally neutral carebot (to the extent that currently available and near-future carebots are not capable of moral judgment or acting maleficently) or living under the thumb of an abusive human caregiver, utilizing a carebot would lead to better fulfillment of the individual's capabilities, thus representing a morally preferable arrangement.

Authors engaging in the objective-list line of ethical discourse regarding carebots also tend to emphasize the issue of deception—namely, that older adults utilizing carebots may have a misconception of the true abilities of the carebots. A carebot's deceptive nature, in this line of reasoning, may be as simple as having a digitally reproduced smile that does not, in fact, reflect a true underlying human emotion. This may be of particular importance when considering the care of patients with dementia or other cognitive impairments. However, when considering the usage of carebots by older individuals without impaired cognition, the emphasis on deception may appear somewhat paternalistic. Human beings have active imaginations and may project psychological states or personalities on inanimate objects, animals, or even other people that do not reflect the reality of the experience of interacting with the projected-upon thing or individual (see Farah & Helberlein, 2007). In many cases, this may be perfectly harmless or may even be a form of play, which is actually one of the

[b]It should be made clear that this kind of tragic occurrence could still come to pass under the supervision of carebots—especially if the robots are, for instance, not capable of detecting such an infestation. However, the important distinction here, we believe, is that robots are not capable of moral judgments and thus not capable of acting maleficently, whereas a human caregiver knowingly allowing their patients to come to harm is acting with maleficence. This distinction is important to the ensuing arguments. It is true, however, that neutral moral agents such as robots may still exhibit apparently immoral characteristics. We thank an anonymous reviewer for prompting us to make this clear.

capabilities authors utilizing the objective-list approach identify as a means to leading a fulfilling life.

The best response to an objective-list approach may be a case-by-case analysis of carebot use among individuals. The question then becomes "does the use of a carebot by this individual ultimately result in their leading a better life?" If the answer is yes, then the usage of a carebot by that individual is not to be discouraged. Otherwise, the individual in need of care may indeed be better off with a human caregiver. Ultimately, as long as the individual in need of care is sound of mind, the decision of how they wish to receive that care should be up to them.

Ethics of care concerns

The final category of ethical analysis of carebots identified by Vandemeulebroucke and colleagues is arguments related to care-ethical approaches. Ethics of Care was first introduced in the work of Carol Gilligan (1982) and proved to be a defining influence that propelled work in feminist ethics. Gilligan's theory posits that males and females have different voices that guide their moral behaviors. For males, the voice that is dominant is the one that is logical and concerned with ideas of equality, autonomy, justice, and individualism. The voice that is dominant in females is the voice that speaks up on issues of others—those outside of the individual—and is called the "interpersonal voice." Care means safety for others and caring about their issues (Muuss, 1988). Gilligan noted that women more often than men focus on the idea of selfishness and responsibility in their moral thinking (Gilligan, 1982).

According to Vandemeulebroucke and colleagues, "[c]are-ethical approaches start from the particular care relationship between caregivers and care receivers, and progressively widen their scope to include a contextual level and then a political level. They stress that meaningful care relationships consist of 'caring about' and 'caring for' someone" (22). The major concern care ethicists raise is that robots cannot really *care*; thus the care relationship, which is essential for feminine morality, gets disrupted and becomes meaningless. Replacing real caregivers with robots is seen as detrimental for society. Beyond those concerns, issues that have been raised include objectification of elders (objectification in general being a major concern in feminist literature), deception, and social isolation. There is another important dimension of the care-based approach to the ethical assessment of carebots. Namely, some feminist literature has addressed the economic and political context of "feminized jobs" such as caregiving (see Standing, 1989): these jobs tend to be the least respected and paid in any given society. Thus even though the instrumental and economic contribution (in terms of increased efficiency and productivity) of

carebots in aged care may be a good thing, it may contribute to additional de-valuation of the work provided by women.

Thus when considering any given carebot technology on a case-by-case basis, special attention needs to be given to the potential of their widespread use to further marginalize certain groups. It is our position that human caregivers are important and deserve to be compensated for the skilled services they render, and that it would be better that any large-scale implementation of carebot technology does not displace or devalue human caregivers. Considering demographic trends that may lead to a gap between the number of caregivers available and the amount of caregiving services required by the elderly, however, carebot technology may help fill this gap or serve as an extension or augmentation of the services provided by human caregivers. If carebots are used to assist nurses and reduce their work loads, they may lead to an overall reduction in nurse burnout, which has been shown to decrease quality of care (Poghosyan, Clarke, Finlayson, & Aiken, 2010). However, if the caregiving industry utilized carebots to replace nurses completely, we would agree with care ethics critics that it would be unethical to do so.

Toward an ethical consensus on carebots

As is readily apparent, deception, objectification, social isolation, and displacement or marginalization of humans are crucial concerns that are recognized by ethicists. Indeed, these are valid concerns, and some carebot developers and funding bodies in the United States (such as the National Science Foundation) seem to be taking them seriously (consider the example of Pearl, above). Thus, as mentioned previously, most insightful ethical evaluation of carebots cannot be done in general, but on a case-by-case basis. However, there is one general point we would like to stress. Namely, unlike humans, carebots will certainly not be intentionally engaged in elder abuse.[c] The crucial question that needs to be answered then is whether carebots can help prevent elder neglect. Suppose Linda heard about the shocking stories in nursing homes and is hoping to be cared for by carebot technology as opposed to human

[c]The issue remains as to who should be held accountable if mistakes do occur while utilizing carebots and a human subsequently comes to harm. The issue of who should be held responsible for a robot's actions is the subject of ongoing debate, especially with regard to legal issues associated with autonomous cars (Lin, 2016). We believe the conclusion about who is morally responsible for a robot's actions is far from clear, and not something we can establish within the scope of this chapter, but we do thank an anonymous reviewer for constructive comments that prompted us to reflect on this issue.

caregivers. She thinks she is experiencing mild cognitive decline, while in fact she is in the early stages of dementia. If she had access to a carebot like Pearl, would she be better off and able to stay in her home longer (which is her preference)?

We think that she would, and not only would this be beneficial in terms of her health outcomes, it would provide an arrangement more respectful of her autonomy. Namely, even though there is a widespread assumption in the literature that people living with dementia are not capable of maintaining any level of basic autonomy (see, e.g., Dworkin, 1993), which in turn leads to their stigmatization (see Dubljević, 2019), part of the problem of maintaining day-to-day functioning is actually the standard response society has to people living with dementia. Dementia does cause memory issues and learning difficulties. Thus when people with dementia are placed in a completely new nursing home environment, which does not have the familiar cues they are used to, this actually increases their cognitive difficulties (Batsch & Mittelman, 2012). Conversely, having access to a carebot like Pearl, which would provide much needed reminders and facilitate continued living in the familiar environment of their home would not only allow dementia patients to choose to stay at home but would also prolong the time they may maintain relatively intact functioning. But why should we care about an individual's choice in the matter if they have dementia?

Contrary to the traditional view of autonomy, which was heavily predicated on memory and cognitive capacities, newer models of autonomy, which better reflect neuroscientific evidence, place the crucial capacities not on cognition, but on valuing (see Jaworska, 1999) and volition (Dubljević, 2013). According to Jaworska, dementia patients in early stages of the disease progression are in fact autonomous: "[…] in the context of dementia […] , so long as [the person] still holds values, he is capable of self-governance and can form new critical interests" (Jaworska, 1999, p. 134). Since Linda from our example clearly has values, has no issues with volitional disturbances, and merely has cognitive deficits that are remediable with the use of carebots, there is an ethical obligation to respect her autonomy and provide her with the tools to maintain that as long as possible. An argument can be made that anthropomorphic robots might be problematic in cases like Linda's, but nursebots such as Pearl (and perhaps virtual assistants) could offset such concerns and provide help with managing medication use and even financial matters (e.g., reminders to pay the bills, making sure they are in fact paid, etc.). Of course, since dementia is a progressive illness, Linda will eventually need to be cared for in an assisted living facility, but only in the later stages when the disease starts affecting her socio-moral judgment and volitional capacities (see Dubljević, 2020). This does not completely solve the issues of the poor state of affairs in certain nursing homes, but arguably, with

more people (like Linda) being able to stay at home, the facilities with human caregivers would not be as overcrowded and understaffed, and both of these issues contribute to de facto neglect of elders. Moreover, if carebots are used in such assisted living facilities, they could potentially record (and report) instances of elder abuse, thus helping reduce a major social concern.

However, some might not be convinced with the argument from newer conceptions of autonomy and insist that the only people whose autonomy should be respected are those in full control of their cognitive capacities (e.g., memory). In fact, previously we have argued that elders *not suffering from significant cognitive impairment* should be assumed capable of arranging for social activities to provide necessary mental stimulation. Now we wish to extend the argument from autonomy to the elderly in early stages of dementia, which may encounter resistance from skeptics. To those we submit that they are prejudiced toward different forms of cognitive capacities that provide "scaffolding" for human autonomy. Consider Clark and Chalmers' hypothetical example of two individuals in New York who have different ways of instantiating their choices and navigating the world: Otto and Inga. They both hear of an intriguing exhibition at the Museum of Modern Art (MOMA). Inga recalls it is on 53rd street and sets off. Otto, however, suffers from early stage dementia, and as a result, he always carries a notebook in which he records information he thinks he may need. He hears of the exhibition at MOMA, reads the address from his notebook, and sets off in this direction (Clark & Chalmers, 1998). Clark and Chalmers argue that Otto's usage of a notebook to find the exhibit is comparable to Inga's usage of her own memory to do the same.[d] Both Inga and Otto have beliefs about where the MOMA is located, and they both retrieve these beliefs somehow to get there. It is just that Otto's beliefs are not stored in his brain—they are stored externally to his body. Effectively, they both are able to use their beliefs to achieve the same result, and thus exercise their autonomy. Now, this is not to say that cognitive capacities are not relevant for autonomy—they certainly are, but this thought experiment serves to point out that the actual exercise of cognitive capacities is what we need to respect in human beings, whether or not they are elderly and suffering from mild cognitive decline or dementia.

[d]As noted by an anonymous reviewer, in this day and age most people use GPS devices and not notebooks. However, two issues need to be noted. First, elderly people like Otto and Inga in the example may be averse to using new technology and resort to notebooks. Second, the point of the argument is that even low-tech external memory storage (e.g., a notebook) may help increase the autonomy of human agents. Conversely, high-tech applications, such as GPS devices and carebots, could also be seen as uncontroversial external means of augmenting human agency and autonomy. See Bauer and Dubljević (2019) for a related discussion.

Arguably, the use of carebots, if freely chosen by a person like Linda, is autonomous in two ways: it flows from her values and unfettered volition, and it helps her maintain a level of cognitive functioning through externalizing memory storage and retrieval. The fact that Otto uses a low-tech tool to help maintain his cognitive functioning whereas Linda uses carebots makes no difference. If we care for and respect *persons*, then we should care for and respect their choice of cognitive scaffolding. Just as it would be cruel and arbitrary to destroy Otto's notebook or deny him access to it, so it would be cruel and arbitrary to deny people like Linda access to carebot technology, especially if it can help them live independently for longer.

However, the skeptics may point out that people in the early stages of dementia cannot be cognizant of privacy concerns that may arise from the use of carebots and may be easily deceived. It will be remembered that this was the caveat introduced in the principlist literature on autonomy in the ethics of carebots, whereas the deception concern has been raised across ethical theories. To this, we respond that this is not an issue that needs to be solved at the level of every single individual living with dementia, but rather may be addressed with "ethical design" (Baldini, Botterman, Neisse, & Tallacchini, 2018) of carebots. Just like computers may be run in "safe mode" that excludes certain functionalities (especially in terms of access to external networks), carebots designed to work with people living with dementia may be programmed to guard their privacy via "embedded protective technological solutions" (Baldini et al., 2018, 913). Also, nonanthropomorphic design of dementia carebots addresses the issue of deception and guarantees that the "ascription of personhood" to inanimate objects is not on a distinctly higher level from someone caring about their new car.

Conclusion

As populations around the world age and human caregivers become more scarce, carebot technology presents itself as a potential solution to this widening needs gap. While many ethical concerns have been raised about carebot technology, there is no general ethical concern that would lead to anything like a moratorium on carebots. While some ethical concerns may persist, especially when concerning individuals who require more advanced caregiving services (such as those with dementia), carebot technology may at the very least be used to help free up human caregivers to provide more specialized care to those in need of it while allowing those without more advanced caregiving needs to lead an independent lifestyle and care for themselves for longer. In fact, once individual sources of moral unease and concern are addressed with ethically informed design,

the use of carebot technology to extend or even replace human caregivers may be a logical solution for meeting the rising demands for providing care for the elderly. The analysis of presently available carebot technology leads us to conclude that developments in the near future will bring about more functions that may be instrumental in providing adequate elder care and extending the autonomy and independent functioning of elders with or without cognitive deficits. When considering the use of carebots to help care for the elderly, it is important to respect the autonomy of individuals who elect to utilize carebot technology. Like any emerging technology, there may be some initial reservations about carebot technology that will eventually subside as individuals opt into the usage of such technology, and it thus becomes more commonplace.

Acknowledgments

The authors would like to thank two anonymous reviewers for detailed, helpful comments, and the members of the NeuroComputational Ethics Research Group at NC State University for their helpful feedback and discussion (in alphabetical order): Elizabeth Eskander, Dr. Jovan Milojevich, Anirudh Nair, Leila Ouchchy, and Abigail Scheper.

References

Ackerman, E. (2018, September 18). *Moxi prototype from diligent robotics starts helping out in hospitals.* Retrieved from *(2018, September 18).* https://spectrum.ieee.org/automaton/robotics/industrial-robots/moxi-prototype-from-diligent-robotics-starts-helping-out-in-hospitals.

Aminuddin, R., Sharkey, A., & Levita, L. (2016, March). Interaction with the Paro robot may reduce psychophysiological stress responses. In *2016 11th ACM/IEEE international conference on human-robot interaction (HRI)* (pp. 593–594). IEEE.

Angwin, J., Larson, J., Mattu, S., & Kirchner, L. (2016). Machine bias. *ProPublica*, 2016. May 23.

Baldini, G., Botterman, M., Neisse, R., & Tallacchini, M. (2018). Ethical design in the Internet of things. *Science and Engineering Ethics*, *24*, 905–925.

Batsch, N. L., & Mittelman, M. S. (2012). *World Alzheimer report 2012: Overcoming the stigma of dementia.* London: Alzheimer's Disease International.

Bauer, W. A., & Dubljević, V. (2019). AI assistants and the paradox of internal automaticity. *Neuroethics*. https://doi.org/10.1007/s12152-019-09423-6.

Beauchamp, T. L., & Childress, J. F. (2009). *Principles of biomedical ethics* (6th ed.). New York, NY: Oxford University Press.

Camp, J. V. (2019, March 8). *My Jibo is dying and It's breaking my heart. Retrieved from (2019, March 8). https://www.wired.com/story/jibo-is-dying-eulogy/.*

Charova, K., Schaeffer, C., & Garron, L. (2011). *Computers and robots: Decision-makers in an automated world. Retrieved from (2011). https://cs.stanford.edu/people/eroberts/cs201/projects/2010-11/ComputersMakingDecisions/robotic-nurses/index.html.*

Clark, A., & Chalmers, D. (1998). The extended mind. *Analysis*, *58*(1), 719.

Cooper, C., Selwood, A., & Livingston, G. (March 2008). The prevalence of elder abuse and neglect: A systematic review. *Age and Ageing*, *37*(2), 151–160.

Crisp, R. (2017). *Well-being. The Stanford encyclopedia of philosophy. In E. N. Zalta (Ed.), The Stanford encyclopedia of philosophy* (Fall). Retrieved from (2017). https://plato.stanford.edu/archives/fall2017/entries/well-being/.

Dubljević, V. (2013). Autonomy in neuroethics: Political and not metaphysical. *American Journal of Bioethics Neuroscience*, *4*(3), 44–51.

Dubljević, V. (2019). Disease and wellness across the lifespan: A global perspective on mental health burden of dementia. In D. Stein, & I. Singh (Eds.), *Global mental health and neuroethics*. Amsterdam, Netherlands: Elsevier. (In Press).

Dubljević, V. (2020). *The principle of autonomy and behavioral variant frontotemporal dementia.* (Under review).

Dworkin, R. (1993). *Life's dominion: An argument about abortion, euthanasia and individual freedom.* New York: Alfred A. Knopf.

Farah, M. J., & Helberlein, A. S. (2007). Personhood and neuroscience: Naturalizing or nihilating? *American Journal of Bioethics*, *7*(1), 37–48.

Feil-Seifer, D., & Matarić, M. J. (2011). Socially assistive robotics. *IEEE Robotics and Automation Magazine*, *18*(1), 24–31.

Gilligan, C. (1982). *In a different voice.* Cambridge: Harvard University Press.

Harper, S. (2014). Economic and social implications of aging societies. *Science*, *346*(6209), 587–591.

Haselton, T. (2018, January 9). *Here's a smart robot for the elderly that can play videos, chat and more. Retrieved from (2018, January 9). https://www.cnbc.com/2018/01/09/elliq-robot-for-elderly-first-look.html.*

Jaworska, A. (1999). Respecting the margins of agency: Alzheimer's patients and the capacity to value. *Philosophy and Public Affairs*, *28*(2), 105–138.

Lin, P. (2016). Why ethics matters for autonomous cars. In *Autonomous driving* (pp. 69–85). Berlin, Heidelberg: Springer.

McGinn, C., Bourke, E., Murtagh, A., Donovan, C., & Cullinan, M. F. (2019, March). Meeting Stevie: Perceptions of a socially assistive robot by residents and staff in a long-term care facility. In *2019 14th ACM/IEEE international conference on human-robot interaction (HRI)* (pp. 602–603). IEEE.

Muuss, R. (1988). Carol Gilligan's theory of sex differences in the development of moral reasoning during adolescence. *Adolescence*, *23*(89), 229–243.

Pepito, J. A., Locsin, R. C., & Constantino, R. E. (2019). Caring for older persons in a technologically advanced nursing future. *Health*, *11*(5), 439–463.

Phillips, K. (2018, May 1). *She modeled in New York and worked for the navy. At 93, parasites ate her alive at a nursing home. Retrieved from (2018, May 1). https://www.washingtonpost.com/news/to-your-health/wp/2018/05/01/she-modeled-in-new-york-and-worked-for-the-navy-at-93-parasites-ate-her-alive-at-a-nursing-home/.*

Poghosyan, L., Clarke, S. P., Finlayson, M., & Aiken, L. H. (2010). Nurse burnout and quality of care: Cross-national investigation in six countries. *Research in Nursing & Health*, *33*(4), 288–298.

Robotics Today (2015). *Jibo. [online]. Retrieved from(2015). https://www.roboticstoday.com/robots/jibo-description.*

Sharkey, A., & Sharkey, N. (2011). Children, the elderly, and interactive robots. Anthropomorphism and deception in robot care and companionship. *IEEE Robotics and Automation Magazine*, *18*(1), 32–38.

Sparrow, R., & Sparrow, L. (2006). In the hands of machines? The future of aged care. *Minds and Machines*, *16*(2), 141–161.

Standing, G. (1989). Global feminization through flexible labor. *World Development*, *17*(7), 1077–1095.

Szondy, D. (2015, May 2). *Robear robot care bear designed to serve Japan's aging population. Retrieved from (2015, May 2). https://newatlas.com/robear-riken/36219/.*

Vandemeulebroucke, T., de Casterle, B. D., & Gastmans, C. (2018). The use of care robots in aged care: A systematic review of argument-based ethics literature. *Archives of Gerontology and Geriatrics*, *74*, 15–25.

World Health Organization [WHO] (2012). *Dementia: A public health priority.* Geneva: World Health Organization.

Conclusion: Looking to the future for trust in human-robot interaction

Chang S. Nam[a] *and Joseph B. Lyons*[b]

[a]Edward P. Fitts Department of Industrial and Systems Engineering, North Carolina State University, Raleigh, NC, United States, [b]Air Force Research Laboratory, Wright-Patterson Air Force Base, OH, United States

This book was motivated to provide students, researchers, practitioners, engineers, and other scientists who have or do not have formal training in human-robot interaction (HRI) with a synopsis of key findings and theoretical and technical advances in the areas of trust in HRI. This book presents 24 chapters, including an introductory chapter with an emphasis on the evolution of trust in HRI, in the following five main themes which supported and drove the important opportunities for trust research in the context of HRI: multifaceted nature of trust in HRI (Part I); explainable trust in HRI (Part II); trust in human-robot teaming (Part III); cognitive, behavioral, and neural modeling of trust (Part IV); and culture, emotion, and ethics in HRI (Part V).

As evidenced by the contributed chapters in this book, trust in HRI has attracted worldwide research attention and great success has been achieved in recent years; great strides are currently being made to extend HRI trust research. However, it is also true that trust in HRI research still faces many challenges and a lot of additional research needs to be conducted on this topic. Some example research and technology areas include, but are not limited to, the following:

- The development and validation of computational models or frameworks that can lead to novel predictions of the operator's trust of robots in dynamic HRI contexts.
- Research examining the full gamut of robotic form (e.g., appearance) and its influence on human trust and acceptance.
- Research examining situational factors that shape HRI.
- Investigation into issues of trust with respect to ubiquitous co-robots.
- The design, development, implementation, and validation of robotics technology and engineering features for improving trust in HRI.
- The identification of emerging applications, barriers, and actions to advance human-robot interaction.

- The development of standards for human-robot dialog based on HRI research.
- The development of a comprehensive understanding of the role of robot-based intent.
- The development of advanced research methodologies through enhanced inter- and multidisciplinary collaboration with researchers in other fields.

In summary, we hope that the collection of chapters in this book will be able to serve as the next step in the journey of advancing the science of trust in the domain of human-robot interaction.

Index

Note: Page numbers followed by *f* indicate figures and *t* indicate tables.

A

Ability, 7–9, 14, 87, 409–410
  attributes of, 88*t*
  to control, 432
  of system, 4–5
  trustee, 383–384
ABI trust scale, 87, 88*t*
Acceptance observation history, 468
Action-perception system (APS), 479–480
Activity Context, 385–387
Actor-critic (A3C) method, 332–334
Adaptability, 321
Adaptive Parameter EXploration (APEX), 361, 367
Affect-based trust, 267–268
Affect infusion model (AIM), 517–520, 522
Affective trust, 249, 250*t*
Affect priming, 518
Affects. *See also* Emotions
  interpersonal trust
    human-automation interaction, 521–522
    human-human interaction, 520–521
  negative, 518–520
  positive, 518–520
After-action review, 239, 241–242, 251
Agent's mental model, 249–251
Agent transparency, 322–323
Agreeableness, 162–163, 170
AIM. *See* Affect infusion model (AIM)
Air Force Office of Scientific Research, 494, 499–500
Alexa robot, 204–205
Allied technologies, 59
Amazon, 533–534
Amazon Alexa, 554–555
Amazon Mechanical Turk (AMT), 17–19, 75, 344, 466–467
Analogic trust, 249, 250*t*
Analysis of variance (ANOVA)
  one-way
    dispositional robot trust, 504–505
    Rotter Trust Questionnaire, 503
  three-way mixed, 506
Analytic trust, 249, 250*t*
Anthropomorphism, 77, 479–484
  emotional gestures and conversation, 482–484
  kinematics, 481–482
  qualities, 107
  uncanny valley effect, 478–481
Aquinas' Christian teleolog, 540
Area under the trust curve (AUTC)
  proximate human-robot interaction, 372
  qualitative trust models measurement, 451
Argument-based dialogue, 72
Aristotle's moral philosophy, 537–538
Aristotle's teleology, 539, 542–543
Artificial intelligence (AI), 59, 197, 209–210
  advances in, 429
  agent communication, 303–304
  Aristotelian teleological model, 537–538, 543
  Catholic Church and, 540
  challenge, 538–539
  community, 331
  data-driven techniques, 329
  erosion of human judgment, 537
  ethical issues, 542
  in Protestant view, 539, 541
  vs. robot partners, 542
  Roman Catholic view, 539
  technology's herding capacities, 534–535
  willingly/unwillingly partnered with, 534
Asimov's Laws of Robotics, 67, 494
Atari games, 333–334, 336, 341–342
Atari 2600 games, 333–334
Attention to Process, 325
Attribute-based characteristics, 355–356
Augsburg Biosignal Toolbox, 95
Automated technologies, 493–494, 497

Automatic Ground Collision Avoidance System (Auto-GCAS), 63, 301–302
distributed and dynamic team trust, 303–304, 307–308, 314
network in, 303*f*
predispositional trust, 307
reputation of, 315
trustworthiness development, 302
Automatic target recognition (ATR) technique, 306–307
Automation, 197, 233, 235–236, 321–322
bias, 201–202
cross-cultural trust, 497
factors affecting, 324–326
global factors in, 164–165
growth of, 494
human interaction with, 321–322
intelligibility, 325
level of automation, 326
predictability, 325
reliability, 325
reliance, human-autonomy teaming, 263–264
specific factors, 165–166
swift trust, 412–414
transparency, 325
trust in, 163–166
Autonomous Agent Teaming, 69
Autonomous partner (AP), 72–73
Autonomous squad member (ASM), 216–217, 217*f*
Autonomous systems, 61–68
mental models for, 68–69
super-machines/sub-humans, 74
Autonomous vehicle (AV), 32, 40
Autonomy, risk assessment, 431
Autoregressive moving average vector (ARMAV) model, 454–455, 470
bilateral trust and self-confidence, 459–461
dynamic Bayesian networks, 463–464
humans' trust in robots, 455–457
robots' trust in humans, 457–459
Avatar, 524–525

B

Baxter robot, 456
Bayesian belief, 224
Bayesian decision-makers, 76
Bayesian networks, dynamic. *See* Dynamic Bayesian networks (DBNs)
Behavioral change, 40
mixed objective measures, 42
Behavioral measures, human-autonomy teaming, 278–279
Bellman equation, 332
Beneficence, issues of, 560
Benevolence, 7–9, 87, 409–410
attributes of, 88*t*
trustee, 384–385
Bidirectional communications, and transparency, 217–228
Bilateral haptic mixed-initiative tele-operation scheme, 460
Bilateral trust, and self-confidence, 459–461
Binomial logistic regressions, 505–506
Buddy, 515
Business contexts, 98, 98*f*

C

Calibrated trust, 4–5
Carebots, 555–556
dementia, 553, 565–566
early stages, 566
nonanthropomorphic design, 567
ethical and social concerns, 557
care ethicists, 563–564
deontology, 558–559
objective-list approach, 561–563
principlist approaches, 559–561
ethical consensus, 564–567
feminist literature, 563–564
issues of economic justice, 560–561
moratorium on, 567–568
Cartesian revolution, 535–536
Cat5 cables, 99–100
Categorical imperative, 536–537
Category-driven process, 410
Catholic theology, 540
Checklist for Trust in Automation, 277–278
Chen's SAT levels, 239–240
Classical moral philosophy, 535, 537–538
developers, 535–536
ethical challenges, 532
failure reasons, 533
to human identity, 534
natural effects, 535
secular ethics model, 542
value-assigning functions, 532–533
Closed-loop communication, 321
Coactive Design, 389–390
Cognition, 66
abilities and skills, 129–131
emotions and, 516

Cognitive performance, 455
Collaborative task
humans' trust in robots, 457
robots' trust in humans, 457–458, 459*f*
Collectivism, 495
Communication dialogue, 97–99
Communication process, 93
Competence, 7–9, 14–15
Complacency-potential rating scale (CPRS), 171–172
Complexity, 65
Compliance, 91, 451
Comprehensibility, 65
Computational trust models, 450
Computer-based Aerial Port Simulation task, 521
Computer game experience, 437
Computer-mediated communication (CMC), 71
Confidence, human-autonomy teaming, 263–264
Conscientiousness, 161–162, 170–171
Conscious cues, 478–479
Consistency, 8
Constitutive explanations, for transparency, 331
Constrained first-principles filtering approach, 288–289
Constraint violations, 440
Consumer-ready social robots, 515
Context, 386–388
Context factors, 92, 415–416*t*, 416
Context-of-use scripted dialogue, 98
Contextual environmental factors, 148
Controllability
HAIER, 435
multivariate analysis of variance, 438–439
risk budget, 433
surfacing budget, 440, 440*f*
Convolutional neural networks (CNNs), 329
Cooperative inverse reinforcement learning (RL), 334
Core competencies, 63
Cronbach's Alpha, 101, 103, 106
Cross-cultural trust, 494–495
automation, 497
Chinese and American cultures, 497
constructs, 494–495
descriptive and inferential tests, 502–503
dimension, 495–497
experimental task, 499
human-robot research, 497
individualism–collectivism, 495
masculinity–femininity, 496
native and expatriate, 501*t*
obedience, 497–498
participants, 500–502
power distance, 495
procedure, 502
quasi-experimental fashion, 498–499
results, 502–508
robotic peacekeepers (*see* Robotic peacekeepers (RPKs))
time orientation, 496
uncertainty avoidance, 495
Cultural identity, 498
Cultural phenomenon, 531–532. *See also* Cross-cultural trust

D

Data-driven approach, 364
Data-driven clustering approach, 284–285
Debriefing, 241–242
postmission, 242, 253
team, 247–248
Decision-analytical models, 461–462
Deep learning approaches, 393
Deep neural networks
reinforcement learning (*see* Deep reinforcement learning (DRL))
systems transparency based on, 326
Deep Q-network (DQN) model
agent behaviors explanations, 336
Atari games, 333
double, 336–337, 337*f*
Deep reinforcement learning (DRL), 323–324
advantages, 332
computational models, 333–335
object-saliency-based explainability, 329–330, 335
natural language explanation, 341–345, 345*f*
visual explanation, 336–341, 338*f*, 340–341*f*, 345*f*
Pac-Man (*see* Ms. Pac-Man)
reinforcement learning and, 331–335
visual and verbal explanations, 326, 327*f*
Defense Advanced Research Projects Agency (DARPA) Explainable Artificial Intelligence (XAI) program, 393
Degree of Autonomy, 326
Delegation process, 237

Dementia, carebot, 553, 565–566
early stages, 566
nonanthropomorphic design, 567
Deontological ethics, 536
Deontology, 558–559
Desert Survival Problem, 170
Dijkstra planner, 435
Directability, interdependence relationships, 390, 392
Discreetness, 8
Displaced transparency, 237–238, 240
methods for, 240–242
posteriori displacement, 241–242
priori displacement, 241
synchronize mental models, 249–251
time frames for, 240, 240*f*
and trust, 248–251
Displacement, 240, 240*f*
Dispositional trust, 170, 430–431
Hofstede's cultural dimensions, 498
human-autonomy teaming, 267
in-vivo robot *vs.*, 510–511
level of, varied by cultural group, 504–505
measurement, 310–311, 503
personality trait, 493–494
qualitative models, 451
of robots, 502
Schaefer's Trust Perception Scale-HRI, 504–508, 506*f*, 508*f*
Distributed dynamic team trust (D2T2), 302
applications, 314–316
behavioral measures, 311
human-AI-robot teaming, 307, 314–316
interaction-based measures, 313–314
inter-relational trust states, 304–305, 305*f*
measurement techniques, 309–314, 310*f*
multiple artificial agents on team, 306–307
scalable control strategies for multiagent systems, 315–316
sensor-based measurement, 310*f*, 311–312
social network analysis approach, 314
stakeholder types, 303–304
surveys, 310–311
transitivity concept, 304–305, 304*f*
Dorsomedial prefrontal cortex (DMPFC), 96
Double deep Q-network (DDQN) model, 336–337, 337*f*
Dual-process models, human cognition, 199–200
Dyadic Trust Scale measurement, 35–36
Dynamic Bayesian networks (DBNs), 361–364, 462–463, 470
OPTIMo, 463–465
partially observable Markov decision processes, 465–467
Dynamic trust
measurement of, 153–155
model of, 151*f*

E

Efficient explanations, for transparency, 330
Eisenhower, Dwight, 242
Electrocardiography (ECG), 95
Electroencephalography (EEG), 312
ElliQ, 556
Embedded path planning algorithms, 429
Emotional gestures, 482–484
Emotional Stability, 161–162
Emotions, 515–516
affect infusion model, 517–520
affects and (*see* Affects)
aspects of being human, 516–520
to automation, 523–524
characterization, 517
and cognition, 516
human-human interaction, 520–521
social robots, 523
advantages and disadvantages, 525
animals, 523–524
benefits, 524
consumer-ready, 515
designs, 524–525
with people, 515–516
psychological and physiological effects, 523–524
Environmental antecedents, 144
Environmental factors, contextual, 148
Environmental variability, maps, 439–440, 444
Environmental variables, 167
Error-related negativity (ERN), 485–486
Ethical, 7
Ethical dilemmas, 204
Ethical relational risk, 187–188
Ethical risk, 187–188
Ethical risk-taking propensity, 191
Ethical situational risk, 187–188
Ethics
of carebots, 557, 560, 564–567
care ethicists, 563–564
deontology, 558–559
objective-list approach, 561–563
principlist approaches, 559–561
Catholic theology, 540

Ethics *(Continued)*
classical moral philosophy, 535, 537–538
developers, 535–536
ethical challenges, 532
failure reasons, 533
to human identity, 534
natural effects, 535
secular ethics model, 542
value-assigning functions, 532–533
deontological, 536
Kantian ethics, 536–537
secular ethics model, 542
of technology, 546–548
teleological ethics, 537–539
theological considerations, 539
Ethics of Care, 563–564
Ethnicity, 90, 110
Executing teammate (ET), 244–246
Expectations, human-autonomy teaming, 263–264
Experiential trust, 493–494
robotic peacekeepers, 504–505
Experimental task, 499
Explainable artificial intelligence (XAI), 218, 220–223
Explanations
communication, 203–204
deep reinforcement learning, 326, 327*f*
verbal explanations, 326, 327*f*, 342–343, 345*f*
visual explanations, 326, 327*f*, 336–341, 338*f*, 340–341*f*
displaced transparency, 242
during interpersonal interaction, 199–200
manipulative, 204–205
for transparency, 329–331
trust *vs.* ethics, 204–205
Expressions, physiological trust, 95
Extraversion, 162, 168–170

F

Facebook, 533–534
Face-to-face (FtF), 71
Facial action coding system (FACS), 95
Facial expressions, physiological trust, 95
Facial recognition software, 545
Fair, 7
Fairness, 8
Feedback, robot factors, 92
Femininity, cross-cultural trust, 496
Financial relational risk, 185–186
Financial risk, 185–186
Financial risk-taking propensity, 190–191
Financial situational risk, 185–186
First-order lag model, 454–455
Five-factor model (FFM), 161
fMRI, 479–481
Folk model, 450–451
Formal explanations, for transparency, 331
Form-Function Attribution Bias (FFAB), 417–418
Foxx, Anthony, 201–202
F-16 pilot, 72, 73*f*
Fukushima Daiichi nuclear plant, 407–408
Functional explanations, for transparency, 330

G

Galvanic skin response (GSR), 312
Gambling research, 432
Gambling risk-taking, 190
Gaussian distribution, 463–464
Gender, 89
General Data Protection Regulation (GDPR), 559–560
10-Gigabit network hub, 99–100
Gilligan's theory, 563
Global factors
in automation, 164–165
interpersonal trust, 161
in robots, 168–170
Google Home, 554–555
Google Scholar, 28
Graphical user interfaces, designing, 99
Greatest Happiness Principle, 537
Ground truth, 96

H

Halo effect, 70
Health Insurance Portability and Accountability Act (HIPAA), 559–560
Heart rate, physiological trust, 95
Heidegger's notion of enframing, 544–545
Herding process, 534
Heuristic processing strategy, 517–518
High-performance teams
experimental evidence, 321
human-human communication, 237, 242
with well-calibrated trust, 295–296
History-based trust, 493–494
human-autonomy teaming, 269
human-robot survey, 502
robotic peacekeepers, 504, 506

Hofstede's cultural framework. *See* Cross-cultural trust
Honesty, 8–9
HRI Trust Perception Scale, 34–35
Human-agent communication
  idealized, 219*f*
  XAI through, 220–223
Human-agent SMM, through ToM, 223–228
Human-agent teaming, 210–211
Human-AI-robot teaming, 302
  D2T2, 307, 314–316
  microcosm of, 304–305
  problematic artificial agents in, 315
Human-automation interaction, 234, 248, 522
  affect in, 521–522
  canonical problem in, 234–235
  community, 450
  domains, 241
  faulty, 412
  human-robot interaction *vs.*, 355, 358–359
  models, 357–358
  risk associated with, 358
  transparency, 236
  trust in, 4–6
  trust models, 357–358
Human-automation teams, effects for, 416–417
Human–automation trust, commonalities, 383
Human-autonomy interface for exploration of risks (HAIER), 433–435, 444
  controllability, 435
  design features, 434
  experimental interfaces, 434*f*
  observability, 434–435
Human-autonomy teaming, 261
  affect-based trust, 267–268
  automation reliance, 263–264
  complexity, 293
  confidence, 263–264
  construction, 262–281, 267*f*
  control authority, 295
  dispositional trust, 267
  ecological validity, 293
  expectations, 263–264
  history-based trust, 269
  individual differences, 294
  learned trust, 270–271
  measurement tools, 277, 293
    behavioral measures, 278–279
    psychophysiological measures, 279–281
    self-report measures, 277–278
  metrics for, 292–296
  perceived trustworthiness, 268–269
  research designs, 281–282
    experimental paradigm, 282–283
    inter- and intraindividual variability challenges, 287–292, 294
    performance quality, 283–287, 284*f*
    psychophysiological quantities, 289, 290*f*
    trust-based decision state estimation modeling process, 289, 290*f*
  risk and uncertainty, 263–264
  situational trust, 270
  transparency in, 326–331
  trust definition, 262–271, 292
  trust process model, 271–276, 271*f*, 294
    development, 272–274, 273*f*, 292
    trust calibration, 275–276
    trust *vs.* trust-based outcomes, 274–275
  trust propensity, 266–267
  vulnerability, 263–264
Human cognition, dual-process models, 199–200
Human-computer interaction (HCI), 87–89
Human factors, 89–91
  ethnicity, 90
  gender, 89
  role, 91
Human-human communication, 237
  high-performing teams, 237, 242
Human-human domain, 14
Human-human interaction (HHI)
  brain activations, 480–481
  photorefractive keratectomy laser surgery, 532
  trust in, 6–9, 86, 546–548
Human-human (H-H) teams, 409, 411, 414, 415–416*t*
  application, 419
  context factors, 416
  future research, 419–421
  imported information, 418–419
  propensity to trust, 416–417
  surface-level cues, 417–418
Human-human trust, 67
  generalization limitations, 180–181
  model, 381
Humanlike qualities, 91
Human–machine communication, 71–73
Human-machine interactions, 236–237
Human-machine teaming (HMT), 60, 64–68
Human personality, in human-robot interactions, 167–174

Human-related antecedents, 144, 152
experience and abilities, 146
human characteristics, 145
Human-related variables, 167
Human-robot interaction (HRI), 27–28, 145–146, 353, 477–478, 571
advantages and disadvantages, 525
attribute-based characteristics, 355–356
automation literature, 163–166
autoregressive moving average vector model, 455–459
brain activations, 480–481
comparison within papers, 44–48*t*
environment-related factors, 356
experimental design and variables, 105
experimental layout for, 106*f*
experimental manipulations, 31–36
factors that influencing, 452
goal, 353–354
vs. human-automation interaction, 355, 358–359
human personality and trust in, 159, 167–174
human personality in, 167–174
human-robot interaction, 104–108
human-to-human interaction, 101–104
human-to-robot-to-human interaction, 108–112
identifying gaps in, 87–89
interpersonal trust, 160–163
Mayer's definition in, 181
method, 105–106
neural dynamics of trust, 478–479
objectives and hypotheses, 104–105
open questions, 19–20
performance-based characteristics, 355–356
performance-based questions, 31–33
photorefractive keratectomy laser surgery, 531–532
physical, 456–457
proximate interaction (*see* Proximate human-robot interaction (HRI))
relation-based trust questions, 33–34
research, 389
research on robots, 166–167
research problem, 96
results and discussion, 106–108
robot-related factors, 355–356
Schaefer's Trust Perception Scale-HRI, 504–508, 506*f*, 508–509*f*
surveys, 37–38
theoretical framework, 355–356
trust, 9–13, 86–87, 90*f*, 546–548
trust calibration, 353–354, 356
trust definitions in, 28–30
trust questions, 30–38
Human-robot interaction simulation testbed
with HTML front-end, 221*f*
with Unity front-end, 222*f*
Human-robot research, risk in, 184
Human-robot teaming (HRT), 60, 70, 97, 409, 411, 414, 415–416*t*
characteristics, 419
context factors, 416
future research, 419–421
imported information, 418–419
needs, 411
propensity to trust, 416–417
social norms in, 421
surface-level cues, 417–418
Human-robot trust, 199–200
challenges, 413
cross-cultural, 497
domains of risk, 179
ethical quandary, 494
history-based, 502
in human-robot teams, 407, 411
measuring, 13–15
methodology, 115
objective risk, 179
perceived risk, 179
requirements, 202–203
risk and, 180–184
theory of, 115
understanding risk, 192–193
Human-to-human interaction
experimental design and variables, 101
method, 102
objectives and hypotheses, 101
results and discussion, 103–104
Human-to-robot-to-human interaction
experimental design and variables, 108–109
experimental layout for, 111*f*
method, 109
objectives and hypotheses, 108
results and discussion, 110–112
Human-to-robot trust
equation for, 458
self-confidence in, 459–460
Human-unique competencies, 63
Hyperreality, 545–548

I
iCAT robot, 70
Image encoder, 343
Immoral technological process, 544–545
IMPACT paradigm, 214–216, 215*f*
Imported information, 410–411, 415–416*t*, 418–419
Independence, 553–555
Independent agency argument, 542–543
Individual characteristics, 127–129
Individualism, cross-cultural trust, 495
Individual-related factors, 126–131
Industrial robots, 85
  and worker safety, 123–124
Inference-based models, 467
Innate propensity, 382–383
Input-process-output (IPO) models, 61
Instagram, 533–534
Integrity, 7–9, 15, 87, 409–410
  trust components, attributes of, 88*t*
  trustee, 384–385
Intelligence, Surveillance, and Reconnaissance (ISR) scenario, 465
Intelligent agent (IA), 214
Intelligent autonomous system (IAS), 59–61
Intelligent robots, 334
Intelligent systems, 60
  mental models for, 68–69
  super-machines/sub-humans, 74
Intentional frame, 237
Interaction-based measures, 313–314
Interactive team cognition (ITC) theory, 313–314
Interdependence, 380–381
  analysis of potential workflows (how), 399–400
  assessment of potential interdependence (who), 397–399
  Coactive Design, 389–390
  context role, 386–388
  model joint work (what), 396
  relational trust, 389–390
  risk-taking relationship, 381–382, 381*f*
    engage in, 385–386
    situational factors, 385
    trustee factors, 383–385
    trustor characteristics, 382–383
Interdependence analysis (IA) tables, 381, 395–396, 396*f*, 400
Interdependence relationships
  developement, 390–394
  enable partial trust, 394
  encourage opportunistic trust, 394–395
  manage complexities of trust, 395
  resolve uncertainty, 394
International Federation of Robotics report, 123–124
Interpersonal skills, 133
Interpersonal social cognition, 199
Interpersonal trust, 301–302, 408
  affects
    human-automation interaction, 521–522
    human-human interaction, 520–521
  agreeableness, 162–163
  commonalities, 383
  conscientiousness, 162
  conventional models, 408–409
  and distrust, 94
  emotional stability/neuroticism, 161–162
  extraversion, 162
  findings of, 510
  global factors, 161
  measurement, 503
  openness to experience, 163
  and personality, 160–163
  scaffold for, 417–418
  stakeholders, 304
  target of, 493–494
  trustworthiness, 268–269
Interpersonal Trust Scale (ITS), 277
Interpersonal voice, 563
Interpretability, 329–330
Inverse reinforcement learning (RL), 334
Investment risk-taking, 190
In-vivo robot trust, 510–511
Isaac Asimov's Laws of Robotics, 384
Ishiguro, Hiroshi, 540

J
Jibo, 515, 556

K
Kantian ethics, 536–537
Kinematics, anthropomorphism, 481–482
Knowledge-based error, 66
Kolmogorov–Smirnov test, 506–508

L
Language, social factors, 92
Learned trust, 308–309, 430–431
  human-autonomy teaming, 270–271
  qualitative models, 451

Learning-based verbal explanation model, 343, 345*f*
Lee and See's concept
  multitiered model, 249, 250*t*, 251
  performance, 383–384
  process, 385
  purpose, 384
Leung and Cohen's theory, 497
Levene's test, 438, 506–508
Libertarianism, 537
Likert scales, 451
Logistic regression analysis, 440
Logitech H111 headset, 102
Loyalty, 8

M

Machine learning (ML), 197, 323–324, 329
Macro-expressions, 95
Manipulative explanations, 204–205
Map environment variable, 439–440, 444
Markov Decision Process (MDP), 331–332
Masculinity, 496
Matching task, visual explanation, 338–339, 338*f*, 340*f*, 345*f*
Maximum voluntary contraction (MVC), 455
Mechanistic explanations, for transparency, 330
Medical robots, 85
Mentalizing network (MTN), 480–483
Mental Model Reconciliation, 246–247
Mental models, 60–61
  agent's, 249–251
  and design, 68–73
  human–machine communication, 71–73
  individual differences in, 74–75
  for intelligent, autonomous systems, 68–69
  research directions and applications, 77–78
  super-machines/sub-humans, 74
  team cohesion, 69–71
  threat detection task, 75–77
Mental model synchronization, 246–248
  displaced transparency, 249–251
  situation awareness through, 252
MetLife, 533–534
M-2000iA, 478
Microcosm, human-AI-robot system, 304–305
Micro-expressions, 95
Microsoft Kinect V2, 99–100, 102
Military peacekeepers, 498
Mill's theory, 537
Mixed Initiative eXperimental (MIX) testbed, 165
Mixed objective measures
  behavioral change, 42
  following advice, 42–43
  task delegation, 41–42
Mixed papers, 44–48*t*, 49–50
Mixed trust questions, 34–36
Model-based systems, 324
MOMA. *See* Museum of Modern Art (MOMA)
Mood
  affect infusion model, 517–518
  positive *vs.* negative, 518–519
The Moon is a Harsh Mistress (Heinlein), 545–546
Moral characteristics, 9, 15
Moral philosophy. *See* Ethics
Moral side, 9
Moral trust, 3, 6–7, 14
Moral trust expectations, 20–21
Motivational Theory of Human Robot Teamwork, 97
Motives, 8
Moxi, 556
Ms. Pac-Man
  learned behavior, 337–338
  learning-based verbal explanation model, 343
  matching task, 338–339, 338*f*, 340*f*
  prediction task, 339–341, 341*f*
  rule-based verbal explanation model, 342–343
  rules of, 331, 336–337, 341–342
  screenshot, 336–337, 337*f*
Multidimensional conception, 15–19
Multidimensional Measure of Trust (MDMT), 17, 18*f*, 21
Multidimensional psychological attitude, 324–325
Multiple Errands Task (MET), 499
Multivariate analysis of variance (MANOVA), 438–439, 439*t*
Museum of Modern Art (MOMA), 566
Musical instrument digital interface (MIDI), 534

N

NAO robot, 525
NAO robot version 4, 105

NASA Task Load Index, 224, 240–241
National Institute for Occupational Safety and Health (NIOSH), 137–138
Natural language explanation, 341–345
  experimental results, 343–345
  learning-based verbal explanation model, 343, 345*f*
  rule-based verbal explanation model, 342–343, 345*f*
Natural language processing (NLP), 217–218
Negative affects, 518–520
Negative Attitudes to Robots scale (NARS), 75, 77
Negative emotions, 517, 520
Neural dynamics of trust
  anthropomorphism modulation, 479–484
    emotional gestures and conversation, 482–484
    kinematics, 481–482
    uncanny valley effect, 478–481
  brain activations, 480–481
  decision and feedback processing, 484–486
  human-robotic interaction, 478–479
  perfect automation schema, 484
  trust calibration, 484–486
Neural-net techniques, 85–86, 357
Neuro-fuzzy method, 103, 112–113, 114*t*
Neurological basis, of trust, 96
Neuroticism, 161–162, 169–170
NeuroTrust, 467–470
Never Let Me Go (Ishiguro), 544–545
Nicomachean Ethics, 537–538
Nonthreatening, 7
Nonverbal communication, 70–71
Normalization of deviance, 247–248
Norman's seven stages of actions, 432
Nursebot, 557*f*, 565–566

O

Obedience, robotic peacekeepers, 498, 509–511
  rates, 504*f*, 505–506, 507*f*, 508
  regression analyses of, 503–504
Objective-list approach, 561–563
Objective measures
  of trust, 38–43
  types of, 38–39
Object-saliency-based explainability, in DRL, 329–330, 335
  natural language explanation, 341–345
    experimental results, 343–345
    learning-based verbal explanation model, 343, 345*f*
    rule-based verbal explanation model, 342–343, 345*f*
  visual explanation, 336–341
    human experiments, 337–341
    matching task, 338–339, 338*f*, 340*f*, 345*f*
    prediction task, 339–341, 341*f*, 345*f*
Object saliency maps
  natural language explanation, 341–344
  visual explanation, 336–341, 337–338*f*
Observability
  interdependence relationships, 390–394
  Lee and See's design considerations, 391–392
  multivariate analysis of variance, 438–439
  risk budget, 433–435, 443
Observing teammate (OT), 244–246
Occupational Safety and Health Administration (OSHA), 359
O-DDQN, 338–341
Old Dominion University (ODU), 499–500
One-way analysis of variance (ANOVA)
  dispositional robot trust, 504–505
  Rotter Trust Questionnaire, 503
Online Probabilistic Trust Inference Model (OPTIMo)
  dynamic Bayesian networks, 463–465, 470
  proximate human-robot interaction, 361–364, 367–368
OpenFace, 95
Openness, 8
Openness to Experience, 163, 170–171
Opportunistic trust, 394–395
Overtrust, 125, 198, 201–202
  autonomous vehicles, 203
  avoiding inadvertent, 202–203
Oxytocin, 166

P

Pac-Man. *See* Ms. Pac-Man
Paired t-test, 344
Paradigm shifts, 532–533
PARO, 556
Partially observable Markov decision process (POMDP), 198–199, 220–222, 224, 364–366, 465–467
Path length, 437
Path planning algorithms, 435
  embedded, 429
  participants practice, 436

Path planning algorithms *(Continued)*
risk amount during, 429
time horizon, 443, 446
Peacekeeping, in foreign country, 497–498. *See also* Robotic peacekeepers (RPKs)
Pearl, 556, 557*f*, 565–566
Pepper, 515
Perceived relational risk, 182, 184–189
Perceived situational risk, 182–183, 183*f*, 185–189
Perceived trust, 94
Perfect automation schema (PAS), 75, 77, 165, 172–174, 412, 484
Performance
design features and effect on, 444
Lee and See's concept, 383–384
participant, 437
Performance-based characteristics
human-robot interaction, 355–356
proximate human-robot interaction, 359
robot, 359
Performance-based factors, 146–147
Performance-based objective measures
behavioral change, 40
following advice, 40–41
task delegation, 39–40
task intervention, 39
Performance-based papers, 43–49
Performance-based trust, 29–30, 44–48*t*
questions, 31–33
Performance of system, 4–5
Performance relational risk, 186
Performance risk, 186
Performance robots, 523
Performance side, 9
Performance situational risk, 186
Performance trust, 3, 6, 14
Performer (P), 397–399
Perseveration Syndrome, 445
Persistence, 7
Personality, interpersonal trust and, 160–163
Personality trait, dispositional trust, 493–494
Person perception network (PPN), 482–483
Photorefractive keratectomy (PRK) laser surgery, 531
Physical relational risk, 186
Physical risk, 186
Physical risk-taking propensity, 191
Physical situational risk, 186
Physiological trust, 94–95, 99
facial expressions, 95
heart rate, 95
postural gestures, 95
voice, 94–95
Policy, 323–324, 332
Positive affects, 518–520
Positive emotions, 517, 520
Postman's technopoly concept, 534–535
Postmission debriefing, 242, 253
Postural gestures, physiological trust, 95
Power distance, cross-cultural trust, 495
Praat, 94–95
Predictability, 8–9
interdependence relationships, 390, 392–394
trust in automation, 325
Prediction task, visual explanation, 339–341, 341*f*, 345*f*
Predispositional trust, 307–309
Preference, risk-taking relationship model, 383, 386
Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA), 452, 453*f*
Premission planning, 251
Presence, social factors, 92
Principal Components Analysis (PCA), 16
Principlist approaches, carebot, 559–561
Privacy relational risk, 188
Privacy risk, 188
Privacy situational risk, 188
Probabilistic models, human-automation interaction, 357
Process, 388–389
Lee and See's concept, 385
Promise fulfillment, 8
Propensity, 382–383
Propensity to trust, 171–172, 410–411, 415–416*t*, 416–417
human-automation interaction, 521–522
human-autonomy teaming, 266–267
human-human interaction, 520–521
Protestantism, 541–542
Proximate human-robot interaction (HRI), 353, 358
area under the curve measurement, 372
critical factors within, 358–359
opportunities for, 359
optimal teaming during, 354
performance-based, 359
robots engaging in, 354*f*
table-clearing task, 368
trust-aware control

Proximate human-robot interaction (HRI) *(Continued)*
aims, 359–360, 373
challenges, 371–373
existing frameworks, 361–369, 362–363*t*
robot movement behaviors, 369–371, 373
trust aware robot control framework, 361–364, 367
trust-POMDP framework, 364–368, 370
Proximate interaction, 89
Psychological relational risk, 186–187
Psychological risk, 186–187
Psychological situational risk, 186–187
Psychological trust, 94
Psychophysiological measures, 279–281
PsychSim, 221–222, 224
Purpose, Lee and See's concept, 384

Q

Q-learning, 333–334
Qualitative models, 451–453
human-automation interaction, 357
measurement, 451
Qualtircs.com, 344
Quantitative models, 360, 366, 452–453
Q-value function, 332

R

Reallusion Crazytalk software, 105
Reasonable trust, 234
Receiver operating characteristic (ROC) curve, 339–340, 340*f*
Recreational risk-taking propensity, 191
Reeti robot, 107–108
Reeti version 1, 105
Regression models, human-automation interaction, 357
Reinforcement learning (RL), 331–335
challenges, 333
deep neural networks in (*see* Deep reinforcement learning (DRL))
Markov decision process, 331–332
Relational trust, 389–390
Relation-based objective measures, 41
Relation-based papers, 49
Relation-based trust, 29–30, 44–48*t*
questions, 33–34
Relative expected loss (REL) score, 461
Reliability, 7, 9, 14–15, 146–147, 150–151
of performance, 4–5
trust in automation, 325
Reliable, 14
Reliance Intentions scale, 75–76
Remote interaction, 89
Research needs, 137–138
Right to explanation, 324
Risk
domains of, 179, 185–191
in human-robot research, 184
perceived relational risk, 182, 184–189
perceived situational risk, 182–183, 183*f*, 185–189
and trust, 180–184
types of, 183–184
understanding, 192–193
Risk-aware autonomy, 430, 432–433
Risk budget
concept of, 432–433, 444
logistic regression analysis, 440
observability and controllability, 433
Risk perception, 429–431, 437
component, 432–433
human development of, 431
influence diagram, 430–431, 430*f*
vs. leg risk, 442
mean rating, 437
metrics, 441
mode trend, 442–443, 442–443*f*
Risk-seeking behaviors, 432
Risk-taking choice, 13–14
Risk-taking propensity, 184, 189–191
Risk-taking relationship (RTR), 381–382, 381*f*
engage in, 385–386, 394
situational factors, 385
trustee factors, 383–385
trustor characteristics, 382–383
Risk-taking tendencies, 182
Robear, 556
Robot attributes, 147–148
Robot creatures, 201
Robot factors
compliance, 91
feedback, 92
humanlike, 91
transparency, 91
Robot-generated explanations, 201
automation bias, 201–202
human in age, 199–200
insights for improving, 203–204
reconnaissance task, 198–199
utility evaluation, 199

Robotic peacekeepers (RPKs), 494
average self-reported trust rating, 507*f*, 508, 509*f*
cross-cultural trust (*see* Cross-cultural trust)
dichotomous obedience, 508
experiential trust, 504–505
experimental environment, featuring interaction with, 500*f*
history-based trust, 504, 506
in-vivo trust, 510–511
Multiple Errands Task, 499
obedience, 498, 509–511
rates, 504*f*, 505–506, 507*f*, 508
regression analyses of, 503–504
participants, 502
quasi-experimental fashion, 498–499
Schaefer's Trust Perception Scale-HRI, 506, 508, 509*f*
Robotic swarms, control strategies, 315–316
Robotic systems, 408
Robotic work environments, design of, 133–135
Robot movement behaviors, 369–371, 373
Robot Operating System (ROS), 99–100, 100*f*, 109
Robot performance, 456
linear function, 455
partially observable Markov decision process, 466
Robot-related antecedents, 144
contextual environmental factors, 148
performance-based factors, 146–147
robot attributes, 147–148
Robot-related variables, 167
RoboTrust, 467–470
Robots, 452
for complex duties, 494
design of, 133–135
dispositional trust of, 502, 504–508, 506*f*, 508*f*
global factors in, 168–170
in human-machine teams complements, 408
humans' trust in, 455–457
motion planning scheme, 457
performance-based characteristics, 359
reputation, 462
research on, 166–167
for society (*see* Social robots)
swift trust, 414–419
trust in, 143–144
types of, 523
Robot Threat Assessment (RoTA) scale, 74–77
Robot-to-human trust, equation for, 457–458
Robot trust model, human-robot teams, 409, 411, 414, 415–416*t*
characteristics, 419
context factors, 416
imported information, 418–419
needs, 411
propensity to trust, 416–417
social norms in, 421
surface-level cues, 417–418
Rock-paper-scissors game, 413
Rotter Trust Questionnaire, 503, 503*f*
Rotter Trust Scale, 503–504
RPKs. *See* Robotic peacekeepers (RPKs)
Rule-based error, 66
Rule-based verbal explanation model, 342–343, 345*f*

S

Schaefer Human-Robot Trust Questionnaire, 509–510
Schaefer's Trust Perception Scale-HRI
dispositional robot trust score, 504–508, 506*f*, 508*f*
robotic peacekeepers trust ratings on, 506, 508, 509*f*
Scopus, 28
Scoring, 99
Secular ethics model, 542–546
Security relational risk, 188–189
Security risk, 188–189
Security situational risk, 188–189
Self-confidence, 5, 433, 442–443
bilateral trust, 459–461
human-robot interaction, 356
rating, 437
risk evaluation, 441
Self-regulation, 68
Self-report measures
human-autonomy teaming, 277–278
quantitative models, 452–453
Semantic space, trust in, 16
Seminal trust model, 301–302
Sensor-based measurement, of trust, 310*f*, 311–312
Sensor error, 429
Shared mental models (SMMs), 62, 72, 210–211, 210*f*, 313, 321
Simulable competencies, 63

Simulated robot, 494, 497–498
Sincerity, 7–8
Situational factors, human-robotic interaction, 478–479
Situational trust, 308–309, 430–431
  human-autonomy teaming, 270
  qualitative models, 451
Situation awareness (SA), 209–210
  errors in, 245
  levels of, 243–245
  supervisor's, 243–244
  through mental model synchronization, 252
Situation awareness-based transparency (SAT) scale, 211–217, 212–213*f*, 238–239, 239*f*
  displaced transparency, 251
  observing and executing teammate, 244–245
Situation Awareness Rating Scale, 224
Skill-based error, 66
Skype, 102
Social agency, 66
Social categorization mechanisms, 416–417
Social cognition, interpersonal, 199
Social concerns, 557–564
Social factors
  language, 92
  presence, 92
  teamwork, 93
Social hierarchy, 495
Social interaction, 89
Social media, 533–534, 540
Social network analysis approach, 314
Social norms, in human-robot teams, 421
Social relational risk, 187
Social risk, 187
Social risk-taking propensity, 191
Social robots, 420, 449, 470–471, 523
  advantages and disadvantages, 525
  animals, 523–524
  benefits, 524
  consumer-ready, 515
  designs, 524–525
  emotions (*see* Emotions)
  incorrect trust calibration, 322
  neural correlates of trust, 480–481
  with people, 515–516
  psychological and physiological effects, 523–524
Social situational risk, 187
Soliloquy process, 246–247
Spearman rank-order correlations, 441, 441*t*
Specific factors, in robots, 171–174
Stakeholders
  distributed dynamic team trust and, 303–304
  types, 301–302
  views on others' trust, 306
Stereotyping, 202–203
Stevie, 556
Stochastic filtering method, 288–289
Subconscious cues, 478–479
Subjective factors, 441–443
Substantive processing strategy, 518
Supervisory control, 237, 444
Supervisory methods, machine learning, 323–324
Surface budget, 433–434, 437, 440, 440*f*
Surface-level cues, 410–411, 415–416*t*, 417–418
Surgeon-robot partnering, 544
Suspicion propensity, 174
Swift trust, 234, 409–411
  future research, 419–421
  human-human teams, 409, 411, 414, 415–416*t*
    application, 419
    context factors, 416
    imported information, 418–419
    propensity to trust, 416–417
    surface-level cues, 417–418
  human-robot teams, 409, 411, 414, 415–416*t*
    characteristics, 419
    context factors, 416
    imported information, 418–419
    needs, 411
    propensity to trust, 416–417
    surface-level cues, 417–418
  in machines, 412–414
  in robots, 414–419, 415–416*t*

T

Table-clearing task, proximate HRI, 368
Task delegation
  mixed objective measures, 41–42
  performance-based objective measures, 39–40
Task intervention, 39
Team cognition, 313–314
Team cohesion, designing for, 69–71
Team debriefing, 247–248
Teaming, 66

Teammate-likeness, 68–69
Team orientation, 321
Team processes
  human-machine teaming, 64–68
  supporting effective human teams, 61–62
  and trust, 62–64
  and trust in intelligent, 61–68
Teams, 61–62
Team's effectiveness, 407–408
Team trust, 301
  Auto-GCAS, 301–302, 303*f*, 307
  communication, 313–314
  distributed dynamic (*see* Distributed dynamic team trust (D2T2))
  network properties, 309
Teamwork, 321
  social factors, 93
Technocracy, 534–535
Technology Acceptance Model (TAM), 412
Technology ethics, 546–548
Technopoly, 534–535
Telecommunication feedback, 92
Teleological explanations, for transparency, 330
Teleology
  Aristotle's, 539, 542–543
  ethics, 537–539
Tesla, 387–388, 393
Theology, 542
  Catholicism, 540
  considerations, 539
Theory of Mind (ToM), 199, 210–211, 223–228, 331
Three-way mixed ANOVA, 506
Time, and trust, 148–153
Time horizon, 431–432
  control, 432, 444
Time loss relational risk, 187
Time loss risk, 187
Time loss situational risk, 187
Time orientation, cross-cultural trust, 496
Time series models, 357
Traffic Collision Avoidance System (TCAS^{c}), 398–399
Transactive Memory Systems (TMS), 72
Transcranial magnetic stimulation (TMS), 481–482
Transparency, 7, 146–147, 322–323
  during automation interactions, 234
  challenge of, 236–237
  displaced (*see* Displaced transparency)
  explainability, 329–330
  explanations types for, 330–331
  goal of, 322–323
  in human-autonomy teaming, 326–331
  intent/purpose, 328
  interpretability, 329–330
  performance, 326–328
  problem of, 234–236
  process, 328–329
  robot factors, 91
  situation assessment stage, 328
  trust in automation, 325
Transparency information, 238–240
  needed during mission for supervisor awareness, 246*f*
  thought experiment, 243–246, 243*f*
Transparent robots
  explanations (*see* Explanations)
  trust calibration process, 198–199
Trust, 4, 234, 380–381, 450–451 See also *specific trusts*
  in automation (*see* Automation)
  changes in expectations, 17–19
  component of, 182–184
  concept of, 477
  consensus rates of sorting task, 18*f*
  defining, 86–87, 180–181
  designing, 93–96
  determining and modeling, 89–93
  dimensions, 11*t*
  displaced transparency and, 248–251
  dynamics module, 364–365
  experimental paradigms, 52
  formalization of, 51–52
  frequencyanalysis, 10*t*
  general trust, 94
  in human-automation interaction, 4–6
  in human-autonomy teaming, 326–331
  in human-human interaction, 6–9, 86
  in human-robot interaction, 9–13, 86–87, 90*f*
  in intelligent, 61–68
  measurement of, 15–19
  multidimensional conception, 15–19
  neurological basis, 96
  objective measures of, 38–43
  physiological trust, 94–95
  predictors of, 75–77
  as process, 388–389
  psychological trust, 94
  questions, 30–38, 53
  in robots, 143–144

Trust *(Continued)*
in semantic space, 16
sorting trust words, 17
team processes and, 62–64
vs. trust-based outcomes, 274–275
types of, 265
Trust antecedents, categories of, 144–148
Trust aware conservative control (TACtiC), 464
Trust-aware control, proximate HRI
aims, 359–360, 373
challenges, 371–373
existing frameworks, 361–369, 362–363*t*
robot movement behaviors, 369–371, 373
trust aware robot control framework, 361–364, 367
trust-POMDP framework, 364–368, 370
Trust aware robot control framework (TACtiC), 361–364, 367
Trust calibration, 379–380, 398–399
human-autonomy teaming, 275–276
human-robot interaction, 353–354, 356
incorrect, 322
Lee and See's three methods, 249, 250*t*
neural dynamics of trust, 484–486
process, 198–199
Trustee
characteristics, 520–521
factors, 383–385
Trust fall exercise, 3
Trust-free model, 365
Trust in Automation Scale, 33–34
Trust of entirety, 372
Trustor characteristics, 382–383
Trustor-trustee interaction, 520–521
Trust Perception Scale-HRI (TPS-HRI), 277–278
Trust-POMDP framework, 364–368, 370
Trust rating, 437
Trust transitivity, 304–305, 304*f*
Trustworthiness, 409–410
with human, 478–479
human-autonomy teaming, 268–269
interpersonal trust, 268–269
Twitter, 533–534
2×2 mixed factorial design, 435–436

U

Uncanny valley effect, 478–481
Uncertainty avoidance, 495
Uninhabited aerial vehicles (UAVs), 186
United States, population by age group, 553, 554*f*
United States Transportation Secretary, 201–202
Unmanned ground vehicle (UGV), 216
Unmanned underwater vehicle (UUV), 433–434, 445
Unsupervised methods, machine learning, 323–324
Usability and Trust Survey, 277–278
US Air Force, 307–308
Use case scenarios, 92
Utilitarianism, 537, 541–542

V

Valence, 517
Verbal decoder, 343
Verbal explanations, deep reinforcement learning, 326, 327*f*
learning-based model, 343, 345*f*
rule-based model, 342–343, 345*f*
Virginia Modeling, Analysis and Simulation Center (VMASC), 499–500
Virtual Avatar, 107–108, 110*f*
Visual explanation, deep reinforcement learning, 326, 327*f*, 336–341
human experiments, 337–341
matching task, 338–339, 338*f*, 340*f*
prediction task, 339–341, 341*f*, 345*f*
Voice
communication, 71
physiological trust, 94–95
Vulnerability, human-autonomy teaming, 263–264

W

White-coat effect, 94
Worker safety, industrial robots and, 123–124
Worker's trust in robots
changes over time, 131–133
important for workplace safety, 125–126
individual-related factors impacting, 126–131
Worker training, 135–136
Working memory, 129
Workplace safety
implications and recommendations for, 133–138
worker trust important for, 125–126
World Trade Center Disaster, 407–408

Y

Yerkes-Dodson Law, 455, 460, 470–471

TRUST IN HUMAN-ROBOT INTERACTION

EDITED BY CHANG S. NAM AND JOSEPH B. LYONS

Despite the dire prognostications of science fiction of robotic and artificial intelligence systems replacing humans, future robotic systems are intended to partner with humans to accomplish operational goals. This notion of partnering raises an interesting set of challenges for human-robot interaction (HRI). *Trust in Human-Robot Interaction* addresses the gamut of factors that influence trust of robotic systems. The book presents the theory, fundamentals, techniques, and diverse applications of the behavioral, cognitive, and neural mechanisms of trust in human-robot interaction, covering topics such as individual differences, transparency, communication, physical design, privacy, and ethics.

Key Features:

- Presents a repository of the open questions and challenges in trust in HRI
- Includes contributions from many disciplines participating in HRI research—psychology, neuroscience, sociology, engineering, and computer science
- Examines human information processing as a foundation for understanding HRI
- Details the methods and techniques used to test and quantify trust in HRI

ELSEVIER

ACADEMIC PRESS
An imprint of Elsevier
elsevier.com/books-and-journals

ISBN 978-0-12-819472-0
9 780128 194720